



US012409188B2

(12) **United States Patent**
Freier et al.(10) **Patent No.: US 12,409,188 B2**(45) **Date of Patent: Sep. 9, 2025**(54) **MODULATORS OF HSD17B13 EXPRESSION**(71) Applicant: **Ionis Pharmaceuticals, Inc.**, Carlsbad, CA (US)(72) Inventors: **Susan M. Freier**, San Diego, CA (US);
Susan F. Murray, San Diego, CA (US)(73) Assignee: **Ionis Pharmaceuticals, Inc.**, Carlsbad, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/334,632**(22) Filed: **Jun. 14, 2023**(65) **Prior Publication Data**

US 2023/0310483 A1 Oct. 5, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/416,108, filed as application No. PCT/US2019/068051 on Dec. 20, 2019.

(60) Provisional application No. 62/827,524, filed on Apr. 1, 2019, provisional application No. 62/825,581, filed on Mar. 28, 2019, provisional application No. 62/783,680, filed on Dec. 21, 2018.

(51) **Int. Cl.****A61K 31/7088** (2006.01)**A61P 1/16** (2006.01)**C12N 15/113** (2010.01)(52) **U.S. Cl.**CPC **A61K 31/7088** (2013.01); **A61P 1/16** (2018.01); **C12N 15/1137** (2013.01)(58) **Field of Classification Search**CPC **A61K 31/7088**; **A61P 1/16**; **C12N 15/1137**
See application file for complete search history.(56) **References Cited****U.S. PATENT DOCUMENTS**

3,687,808 A 8/1972 Merigan, Jr. et al.
4,751,219 A 6/1988 Kempen
4,845,205 A 7/1989 Huynh Dinh et al.
5,034,506 A 7/1991 Summerton et al.
5,130,302 A 7/1992 Spielvogel et al.
5,134,066 A 7/1992 Rogers et al.
5,166,315 A 11/1992 Summerton et al.
5,175,273 A 12/1992 Bischofberger et al.
5,185,444 A 2/1993 Summerton et al.
5,367,066 A 11/1994 Urdea et al.
5,432,272 A 7/1995 Benner
5,434,257 A 7/1995 Matteucci et al.
5,457,187 A 10/1995 Gmeiner et al.
5,457,191 A 10/1995 Cook et al.
5,459,255 A 10/1995 Cook et al.
5,484,908 A 1/1996 Froehler et al.
5,502,177 A 3/1996 Matteucci et al.
5,525,711 A 6/1996 Hawkins et al.
5,552,540 A 9/1996 Haralambidis
5,587,469 A 12/1996 Cook et al.

5,587,470 A 12/1996 Cook et al.
5,594,121 A 1/1997 Froehler et al.
5,596,091 A 1/1997 Switzer
5,614,617 A 3/1997 Cook et al.
5,645,985 A 7/1997 Froehler et al.
5,681,941 A 10/1997 Cook et al.
5,698,685 A 12/1997 Summerton et al.
5,750,692 A 5/1998 Cook et al.
5,763,588 A 6/1998 Matteucci et al.
5,808,027 A 9/1998 Cook et al.
5,811,534 A 9/1998 Cook et al.
5,830,653 A 11/1998 Froehler et al.
5,859,221 A 1/1999 Cook et al.
5,948,903 A 9/1999 Cook et al.
5,994,517 A 11/1999 Ts'o et al.
6,005,087 A 12/1999 Cook et al.
6,005,096 A 12/1999 Matteucci et al.
6,166,199 A 12/2000 Cook et al.
6,268,490 B1 7/2001 Imanishi et al.
6,300,319 B1 10/2001 Manoharan
6,383,812 B1 5/2002 Chen et al.
6,525,031 B2 2/2003 Manoharan
6,525,191 B1 2/2003 Ramasamy
6,531,584 B1 3/2003 Cook et al.
6,620,916 B1 9/2003 Takahara et al.
6,660,720 B2 12/2003 Manoharan
6,670,461 B1 12/2003 Wengel et al.
6,673,661 B1 1/2004 Liu et al.
6,770,748 B2 8/2004 Imanishi et al.
6,794,499 B2 9/2004 Wengel et al.
6,906,182 B2 6/2005 Ts'o et al.
6,908,903 B1 6/2005 Theodore et al.
7,034,133 B2 4/2006 Wengel et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 103520724 A 1/2014
CN 103520724 B 5/2016

(Continued)

OTHER PUBLICATIONS

Manoharan M., et al., "Lipidic Nucleic Acids," Tetrahedron Letters, 1995, vol. 36, No. 21, pp. 3651-3654.
Manoharan M., et al., "Oligonucleotide Conjugates: Alteration of the Pharmacokinetic Properties of Antisense Agents," Nucleosides & Nucleotides, 1995, vol. 14, No. 3-5, pp. 969-973.
Manoharan M., "Oligonucleotide Conjugates as Potential Antisense Drugs with Improved Uptake, Biodistribution, Targeted Delivery, and Mechanism of Action", Antisense & Nucleic Acid Drug Development, 2002, vol. 12, pp. 103-128.
Merwin J R., et al., "Targeted Delivery of DNA Using YEE(GaINAcAH)₃, a Synthetic Glycopeptide Ligand for the Asialoglycoprotein Receptor", Bioconjug Chem, 1994, vol. 5, No. 6, pp. 612-620.

(Continued)

Primary Examiner — J. E. Angell

(74) Attorney, Agent, or Firm — Sheppard Mullin Richter & Hampton LLP

(57) **ABSTRACT**

Methods, compounds, and compositions useful for inhibiting HSD17B13 expression are provided. Such compounds, compositions, and methods are useful for treating, preventing, or ameliorating a disease associated with HSD 17B 13.

16 Claims, No Drawings**Specification includes a Sequence Listing.**

(56)

References Cited**U.S. PATENT DOCUMENTS**

7,053,207 B2 5/2006 Wengel
 7,262,177 B2 8/2007 Ts'o et al.
 7,399,845 B2 7/2008 Seth et al.
 7,427,672 B2 9/2008 Imanishi et al.
 7,491,805 B2 2/2009 Vargeese et al.
 7,547,684 B2 6/2009 Seth et al.
 7,569,686 B1 8/2009 Bhat et al.
 7,572,582 B2 8/2009 Wengel et al.
 7,582,744 B2 9/2009 Manoharan et al.
 7,666,854 B2 2/2010 Seth et al.
 7,696,345 B2 4/2010 Allerson et al.
 7,723,509 B2 5/2010 Manoharan et al.
 7,741,457 B2 6/2010 Seth et al.
 7,750,131 B2 7/2010 Seth et al.
 7,851,615 B2 12/2010 Manoharan et al.
 7,875,733 B2 1/2011 Bhat et al.
 7,939,677 B2 5/2011 Bhat et al.
 8,022,193 B2 9/2011 Seth et al.
 8,030,467 B2 10/2011 Seth et al.
 8,034,909 B2 10/2011 Wengel et al.
 8,080,644 B2 12/2011 Wengel et al.
 8,088,746 B2 1/2012 Seth et al.
 8,088,904 B2 1/2012 Swayze et al.
 8,106,022 B2 1/2012 Manoharan et al.
 8,124,745 B2 2/2012 Allerson et al.
 8,137,695 B2 3/2012 Rozema et al.
 8,153,365 B2 4/2012 Wengel et al.
 8,158,601 B2 4/2012 Chen et al.
 8,268,980 B2 9/2012 Seth et al.
 8,278,283 B2 10/2012 Seth et al.
 8,278,425 B2 10/2012 Prakash et al.
 8,278,426 B2 10/2012 Seth et al.
 8,313,772 B2 11/2012 Rozema et al.
 8,344,125 B2 1/2013 Manoharan et al.
 8,349,308 B2 1/2013 Yurkovetskiy et al.
 8,404,862 B2 3/2013 Manoharan et al.
 8,435,491 B2 5/2013 Wang et al.
 8,440,803 B2 5/2013 Swayze et al.
 8,450,467 B2 5/2013 Manoharan et al.
 8,501,805 B2 8/2013 Seth et al.
 8,501,930 B2 8/2013 Rozema et al.
 8,530,640 B2 9/2013 Seth et al.
 8,541,548 B2 9/2013 Rozema et al.
 8,546,556 B2 10/2013 Seth et al.
 8,552,163 B2 10/2013 Lee et al.
 RE44,779 E 2/2014 Imanishi et al.
 9,005,906 B2 4/2015 Swayze et al.
 9,012,421 B2 4/2015 Migawa et al.
 2003/0077829 A1 4/2003 MacLachlan
 2003/0119724 A1 6/2003 Ts'O et al.
 2003/0158403 A1 8/2003 Manoharan et al.
 2003/0175906 A1 9/2003 Manoharan et al.
 2005/0164235 A1 7/2005 Manoharan et al.
 2006/0148740 A1 7/2006 Platenburg
 2006/0183886 A1 8/2006 Tso et al.
 2008/0039618 A1 2/2008 Allerson et al.
 2008/0108801 A1 5/2008 Manoharan et al.
 2008/0206869 A1 8/2008 Smith et al.
 2008/0281041 A1 11/2008 Rozema et al.
 2008/0281044 A1 11/2008 Monahan et al.
 2009/0203132 A1 8/2009 Swayze et al.
 2009/0203135 A1 8/2009 Forst et al.
 2009/0286973 A1 11/2009 Manoharan et al.
 2010/0240730 A1 9/2010 Beigelman et al.
 2011/0097264 A1 4/2011 Wang et al.
 2011/0097265 A1 4/2011 Wang et al.
 2011/0123520 A1 5/2011 Manoharan et al.
 2011/0207799 A1 8/2011 Rozema et al.
 2011/0269814 A1 11/2011 Manoharan et al.
 2012/0035115 A1 2/2012 Manoharan et al.
 2012/0095075 A1 4/2012 Manoharan et al.
 2012/0101148 A1 4/2012 Aking et al.
 2012/0128760 A1 5/2012 Manoharan et al.
 2012/0136042 A1 5/2012 Manoharan et al.
 2012/0157509 A1 6/2012 Hadwiger et al.

2012/0165393 A1 6/2012 Rozema et al.
 2012/0230938 A1 9/2012 Rozema et al.
 2013/0004427 A1 1/2013 El-Sayed et al.
 2013/0109817 A1 5/2013 Yurkovetskiy
 2013/0121954 A1 5/2013 Wakefield et al.
 2013/0130378 A1 5/2013 Manoharan et al.
 2013/0178512 A1 7/2013 Manoharan et al.
 2013/0203836 A1 8/2013 Rajeev et al.
 2013/0236968 A1 9/2013 Manoharan et al.
 2015/0191727 A1 7/2015 Migawa et al.
 2018/0216104 A1 8/2018 Abul-Husn et al.
 2022/0273691 A1 9/2022 Freier et al.

FOREIGN PATENT DOCUMENTS

CN 107208092 A 9/2017
 WO 9720563 A1 6/1997
 WO 9746098 A1 12/1997
 WO 9813381 A1 4/1998
 WO 9914226 A2 3/1999
 WO 0063364 A2 10/2000
 WO 0243771 A2 6/2002
 WO 2004024757 A2 3/2004
 WO 2004101619 A1 11/2004
 WO 2004106356 A1 12/2004
 WO 2007134181 A2 11/2007
 WO 2008098788 A2 8/2008
 WO 2008101157 A1 8/2008
 WO 2009073809 A2 6/2009
 WO 2009082607 A2 7/2009
 WO 2009126933 A2 10/2009
 WO 2009134487 A2 11/2009
 WO 2010054406 A1 5/2010
 WO 2010088537 A2 8/2010
 WO 2010129709 A1 11/2010
 WO 2010144740 A1 12/2010
 WO 2010148013 A2 12/2010
 WO 2011038356 A2 3/2011
 WO 2011100131 A2 8/2011
 WO 2011120053 A1 9/2011
 WO 2011163121 A1 12/2011
 WO 2012037254 A1 3/2012
 WO 2012068187 A1 5/2012
 WO 2012083046 A2 6/2012
 WO 2012083185 A2 6/2012
 WO 2012089352 A1 7/2012
 WO 2012089602 A1 7/2012
 WO 2012177947 A2 12/2012
 WO 2013033230 A1 3/2013
 WO 2013075035 A1 5/2013
 WO 2013165816 A2 11/2013
 WO 2013166121 A1 11/2013
 WO 2014179620 A1 11/2014
 WO 2015106128 A2 7/2015
 WO 2017011276 A1 1/2017
 WO 2017015555 A1 1/2017
 WO 2017136758 A1 8/2017
 WO 2018/136702 A1 7/2018
 WO 2018/136758 A1 7/2018
 WO 2018190970 A1 10/2018
 WO 2019183329 A1 9/2019
 WO 2020/061177 A1 3/2020
 WO 2020132564 A1 6/2020
 WO 2023/039076 A1 3/2023
 WO 2023/091644 A2 5/2023

OTHER PUBLICATIONS

Mishra R.K., et al., "Improved Leishmanicidal Effect of Phosphorotioate Antisense Oligonucleotides by LDL-Mediated Delivery," *Biochimica et Biophysica Acta*, 1995, vol. 1264, No. 2, pp. 229-237.
 Nishina K., et al., "Efficient In Vivo Delivery of siRNA to the Liver by Conjugation of α -Tocopherol", *Molecular Therapy*, vol. 16, No. 4, Apr. 2008, pp. 734-740.
 Nishina T., et al., "Chimeric Antisense Oligonucleotide Conjugated to α -Tocopherol", *Molecular Therapy Nucleic Acids*, 2015, 4, e220, pp. 1-10.

(56)

References Cited

OTHER PUBLICATIONS

- Oberhauser B., et al., "Effective Incorporation of 2'-O-Methyl-Oligoribonucleotides into Liposomes and Enhanced Cell Association through Modification with Thiocholesterol," *Nucleic Acids Research*, 1992, vol. 20, No. 3, pp. 533-538.
- Oka N., et al., "An Oxazaphospholidine Approach for the Stereocontrolled Synthesis of Oligonucleoside Phosphorothioates", *J Am Chem Soc*, Jun. 14, 2003, vol. 125, No. 27, pp. 8307-8317.
- Orum H., et al., "Locked Nucleic Acids: A Promising Molecular Family for Gene-Function Analysis and Antisense Drug Development," *Current Opinion in Molecular Therapeutics*, 2001, vol. 3, No. 3, pp. 239-243.
- Pavia A A., et al., "Synthetic TN Glycopeptide Related to Human Glycophorin AM", High-field Proton and Carbon-13 Nuclear Magnetic Resonance Study, *Int J Pep Protein Res*, 1983, vol. 22, pp. 539-548.
- Pujol A M., et al., "A Sulfur Tripod Glycoconjugate that Releases a High-Affinity Copper Chelator in Hepatocytes", *Angewandte Chem. Int. Ed.*, 2012, vol. 51, pp. 7445-7448.
- Rajur S B., et al., "Covalent Protein-Oligonucleotide Conjugates for Efficient Delivery of Antisense Molecules", *Bioconjugate Chem.*, 1997, vol. 8, pp. 935-940.
- Ravi Kanth V V., et al., "Genetics of Non-Alcoholic Fatty Liver Disease: From Susceptibility and Nutrient Interactions to Management", *World Journal of Hepatology*, Jul. 18, 2016, vol. 8, Issue 20, pp. 827-837.
- Rensen P C N., et al., "Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asialoglycoprotein Receptor", *J. Med. Chem.*, 2004, vol. 47, doi:10.1021/jm049481d, pp. 5798-5808, XP002551237.
- Rensen P C N., et al., "Determination of the Upper Size Limit for Uptake and Processing of Ligands by the Asialoglycoprotein Receptor on Hepatocytes in Vitro and in Vivo", *J. Biol. Chem.*, Oct. 5, 2001, vol. 276, No. 40, pp. 37577-37584.
- Rensen P C N., et al., "Stimulation of Liver-Directed Cholesterol Flux in Mice by Novel N-Acetylgalactosamine-Terminated Glycolipids with High Affinity for the Asialoglycoprotein Receptor", *Arterioscler Thromb Vase Biol*, 2006, vol. 26, pp. 169-175.
- Saison-Behmoaras T., et al., "Short Modified Antisense Oligonucleotides Directed Against Ha-ras Point Mutation Induce Selective Cleavage of the mRNA and Inhibit T24 Cells Proliferation," *The EMBO Journal*, 1991, vol. 10, No. 5, pp. 1111-1118.
- Sanghvi et al., "Carbohydrate Modifications in Antisense Research", *ACS Symposium Series* vol. 580, Chapters 3 and 4, 1994, pp. 40-65.
- Sanghvi Y.S., "Heterocyclic Base Modifications in Nucleic Acids and Their Applications in Antisense Oligonucleotides," *Antisense Research and Applications*, Chapter 15, 1993, pp. 273-288.
- Banyal A J., "NASH: A Global Health Problem", *Hepatology Research*, 2011, vol. 41, pp. 670-674.
- Sato M., et al., "Glycoinsulins: Dendritic Sialyloligosaccharide-Displaying Insulins Showing a Prolonged Blood-Sugar-Lowering Activity", *J. Am. Chem. Soc.*, 2004, vol. 126, No. 43, pp. 14013-14022.
- Seth P.P., et al., "Short Antisense Oligonucleotides with Novel 2'-4' Conformationally Restricted Nucleoside Analogues Show Improved Potency Without Increased Toxicity in Animals," *Journal of Medicinal Chemistry*, 2009, vol. 52, No. 1, pp. 10-13.
- Shea R G., et al., "Synthesis, Hybridization Properties and Antiviral Activity of Lipid-Oligodeoxynucleotide Conjugates", *Nucleic Acids Research*, vol. 18, No. 13, 1990, pp. 3777-3783.
- Singh S.K., et al., "LNA (Locked Nucleic Acids): Synthesis and High-Affinity Nucleic Acid Recognition," *Chemical Communications*, 1998, vol. 4, pp. 455-456.
- Singh S.K., et al., "Synthesis of 2'-Amino-LNA: A Novel Conformationally Restricted High-Affinity Oligonucleotide Analogue with a Handle," *Journal of Organic Chemistry*, 1998, vol. 63, No. 26, pp. 10035-10039.
- Slidregt L A J M., et al., "Design and Synthesis of Novel Amphiphilic Dendritic Galactosides for Selective Targeting of Liposomes to the Hepatic Asialoglycoprotein Receptor", *J. Med. Chem.*, 1999, vol. 42, pp. 609-618.
- Srivastava P., et al., "Five- and Six-Membered Conformationally Locked 2',4'-Carbocyclic ribo-Thymidines: Synthesis, Structure, and Biochemical Studies", *J. Am. Chem. Soc.*, 2007, vol. 129, No. 26, pp. 8362-8379.
- Su W., et al., "Role of HSD17B13 in the liver physiology and pathophysiology", *Molecular and Cellular Endocrinology*, 2018, pp. 1-7.
- Svinarchuk F.P., et al., "Inhibition of HIV Proliferation in MT-4 Cells by Antisense Oligonucleotide Conjugated to Lipophilic Groups," *Biochimie*, 1993, vol. 75, No. 1-2, pp. 49-54.
- Tomiya N., et al., "Liver-Targeting of Primaquine-(Poly-γ-Glutamic Acid) and its Degradation in Rat Hepatocytes", *Bioorganic & Medicinal Chemistry*, 2013, vol. 21, pp. 5275-5281.
- Toyokuni T., et al., "Synthetic Vaccines: I. Synthesis of Multivalent Tn Antigen Cluster-Lysyllysine Conjugates", *Tetrahedron Lett*, 1990, vol. 31, No. 19, pp. 2673-2676.
- Valentijn A R P M., et al., "Solid-Phase Synthesis of Lysine-Based Cluster Galactosides with High Affinity for the Asialoglycoprotein Receptor", *Tetrahedron*, 1997, vol. 53, No. 2, pp. 759-770.
- Van Rossenberg SMW., et al., "Stable Polyplexes Based on Arginine-Containing Oligopeptides for in Vivo Gene Delivery", *Gene Therapy*, 2004, vol. 11, pp. 457-464.
- Wahlestedt C., et al., "Potent and Nontoxic Antisense Oligonucleotides Containing Locked Nucleic Acids," *Proceedings of the National Academy of Science*, May 9, 2000, vol. 97, No. 10, pp. 5633-5638.
- Wan W B., et al., "Synthesis, Biophysical Properties and Biological Activity of Second Generation Antisense Oligonucleotides Containing Chiral Phosphorothioate Linkages", *Nucleic Acids Research*, Nov. 14, 2014, vol. 42, No. 22, pp. 13456-13468.
- Westerlind U., et al., "Ligands of the Asialoglycoprotein Receptor for Targeted Gene Delivery, Part 1: Synthesis of and Binding Studies with Biotinylated Cluster Glycosides Containing N-Acetylgalactosamine", *Glycoconjugate Journal*, 2004, vol. 21, pp. 227-241.
- Woolf T.M., et al., "Specificity of Antisense Oligonucleotides in Vivo," *Proceedings of the National Academy of Sciences of the United States of America (PNAS)*, Aug. 1992, vol. 89, No. 16, pp. 7305-7309.
- Zhang J., et al., "PowerBLAST: A New Network BLAST Application for Interactive or Automated Sequence Analysis and Annotation", *Genome Research*, 1997, vol. 7, pp. 649-656.
- Zhang X., et al., "Omic Studies Reveal the Pathogenic Lipid Droplet Proteins in Non-Alcoholic Fatty Liver Disease", *Protein Cell*, 2017, vol. 8, No. 1, pp. 4-13.
- Zhou C., et al., "Fine Tuning of Electrostatics around the Internucleotide Phosphate through Incorporation of Modified 2',4'-Carbocyclic-LNAs and -ENAs Leads to Significant Modulation of Antisense Properties", *J. Org. Chem*, 2009, vol. 74, No. 1, pp. 118-134.
- European Search Report for EP Patent Application No. EP 19 77 2339, dated Jan. 21, 2022.
- International Search Report for International Application No. PCT/US2019/23333, dated Jun. 6, 2019.
- International Search Report for International Application No. PCT/US2019/68051, dated May 14, 2020.
- Su et al., "Liver X receptor α induces 17 β -hydroxysteroid dehydrogenase-13 expression through SREBP-1c", *Am. J. Phys. Endocrin. Metab.*, 312(4):E357-E367 (Apr. 2017).
- Su et al., "Comparative proteomic study reveals 17 β -HSD13 as a pathogenic protein in nonalcoholic fatty liver disease" *PNAS.*, 111(31):11437-11442 (Jul. 2014).
- Wang M. et al., "Inhibition of LXR α /SREBP-1c-Mediated Hepatic Steatosis by Jiang-Zhi Granule", *Evid. Based Compl. Alt. Med.*, 2013:1-10 (2013).
- Abul-Husn N.S., et al., "A Protein-Truncating HSD17B13 Variant and Protection from Chronic Liver Disease", *New Englnd Journal of Mdeicine*, Mar. 22, 2018, vol. 378, No. 12, pp. 1096-1106.

(56)

References Cited

OTHER PUBLICATIONS

- Adam M., et al., "Hydroxysteroid (17 β) Dehydrogenase 13 Deficiency Triggers Hepatic Steatosis and Inflammation in Mice", *The FASEB Journal*, 2018, vol. 32, No. 6, pp. 3434-3447.
- Albaek N., et al., "Analogues of a Locked Nucleic Acid with Three Carbon 2', 4'-Linkages: Synthesis by Ring Closing Metathesis and Influence of Nucleic Acid Duplex Stability and Structure", *J. Org. Chem.* 2006, vol. 71, No. 20, pp. 7731-7740.
- Altschul S.F., et al., "Basic Local Alignment Search Tool", *Journal of Molecular Biology*, Oct. 1990, vol. 215 (3), Elsevier, Netherlands, pp. 403-410.
- Biessen E A L., et al., "Novel Hepatotrophic Prodrugs of the Antiviral Nucleoside 9-(2-Phosphonylmethoxyethyl) Adenine with Improved Pharmacokinetics and Antiviral Activity", *FASEB J.*, Sep. 2000, vol. 14, pp. 1784-1792.
- Biessen E A L., et al., "The Cholesterol Derivative of a Triantennary Galactoside with High Affinity for the Hepatic Asialoglycoprotein Receptor: A Potent Cholesterol Lowering Agent", *J. Med. Chem.* 1995, vol. 38, No. 11, pp. 1846-1852.
- Biessen E A L., et al., "Synthesis of Cluster Galactosides with High Affinity for the Hepatic Asialoglycoprotein Receptor", *J. Med. Chem.* 1995, vol. 38, No. 9, pp. 1538-1546.
- Braasch D A., et al., "Novel Antisense and Peptide Nucleic Acid Strategies for Controlling Gene Expression", *Biochemistry*, Apr. 9, 2002, vol. 41, No. 14, pp. 4503-4510.
- Braasch D.A., et al., "Locked Nucleic Acid (LNA): Fine-Tuning the Recognition of DNA and RNA," *Chemistry and Biology*, 2001, vol. 8, pp. 1-7.
- Connolly D T., et al., "Binding and Endocytosis of Cluster Glycosides by Rabbit Hepatocytes", *J Biol Chem*, Jan. 25, 1982, vol. 257, No. 2, pp. 939-945.
- Crooke S T., "Antisense Drug Technology", Second Edition, CRC Press, 2008, Chapters 1-28, 414 Pages.
- Crooke S.T., et al., "Pharmacokinetic Properties of Several Novel Oligonucleotide Analogs in Mice," *The Journal of Pharmacology and Experimental Therapeutics*, 1996, vol. 277, No. 2, pp. 923-937.
- Duff R J., et al., "Intrabody Tissue-Specific Delivery of Antisense Conjugates in Animals: Ligand-Linker-Antisense Oligomer Conjugates", *Methods in Enzymology*, 1999, vol. 313, pp. 297-321.
- Egli M., et al., "Synthesis, Improved Antisense Activity and Structural Rationale for the Divergent RNA Affinities of 3'-Fluoro Hexitol Nucleic Acid (FHNA and Ara-FHNA) Modified Oligonucleotides," *Journal of the American Chemical Society*, Oct. 19, 2011, vol. 133, No. 41, pp. 16642-16649.
- Elayadi A.N., et al., "Application of PNA and LNA Oligomers to Chemotherapy," *Current Opinion in Investigational Drugs*, 2001, vol. 2, pp. 558-561.
- Engelisch U., et al., "Chemically Modified Oligonucleotides as Probes and Inhibitors," *Angewandte Chemie*, Jun. 1991, vol. 30, No. 6, pp. 613-729.
- Freier S.M., et al., "The Ups and Downs of Nucleic Acid Duplex Stability: Structure-Stability Studies on Chemically-Modified DNA: RNA Duplexes," *Nucleic Acids Research*, 1997, vol. 25, No. 22, pp. 4429-4443.
- Frieden M., et al., "Expanding the Design Horizon of Antisense Oligonucleotides with Alpha-L-LNA," *Nucleic Acids Research*, 2003, vol. 31, No. 21, pp. 6365-6372.
- Gautschi O., et al., "Activity of a Novel bcl-2/bcl-xL-Bispecific Antisense Oligonucleotide Against Tumors of Diverse Histologic Origins," *Journal of the National Cancer Institute*, Mar. 21, 2001, vol. 93, No. 6, pp. 463-471.
- Hashimoto E., et al., "Prevalence, Gender, Ethnic Variations, and Prognosis of NASH", *J Gastroenterol*, 2011, vol. 46, pp. 63-69.
- International Preliminary Report on Patentability for International Application No. PCT/US2019/068051, dated Jul. 1, 2021, 8 Pages.
- Jayaprakash K N., et al., "Non-Nucleoside Building Blocks for Copper-Assisted and Copper-Free Click Chemistry for the Efficient Synthesis of RNA Conjugates", *Organic Letters*, 2010, vol. 12, No. 23, pp. 5410-5413.
- Kabanov A.V., et al., "A New Class of Antivirals: Antisense Oligonucleotides Combined with a Hydrophobic Substituent Effectively Inhibit Influenza Virus Reproduction and Synthesis of Virus-Specific Proteins in MDCK Cells," *FEBS Letters*, Jan. 1990, vol. 259, No. 2, pp. 327-330.
- Kahali B., et al., "Insights from Genome-Wide Association Analyses of Nonalcoholic Fatty Liver Disease", *Semin Liver Dis* Nov. 2015, vol. 35, No. 4, 375-391, 29 Pages.
- Kato K., et al., "N-Acetylgalactosamine Incorporation into a Peptide Containing Consecutive Threonine Residues by UDP-N-Acetyl-D-Galactosaminide:Polypeptide N-Acetylgalactosaminyltransferases", *Glycobiology*, 2001, vol. 11, No. 10, pp. 821-829.
- Khorev O., et al., "Trivalent, Gal/GalNAc-Containing Ligands Designed for the Asialoglycoprotein Receptor", *Bioorganic & Medicinal Chemistry*, 2008, vol. 16, pp. 5216-5231.
- Kim J M., et al., "Oligomeric Glycopeptidomimetics Bearing the Cancer Related TN-Antigen", *Tetrahedron Letters*, 1997, vol. 38, No. 20, pp. 3487-3490.
- Kornilova A Y., et al., "Development of a Fluorescence Polarization Binding Assay for Asialoglycoprotein Receptor", *Analytical Biochemistry*, 2012, vol. 425, pp. 43-46.
- Koshkin A.A., et al., "LNA (Locked Nucleic Acids): Synthesis of the Adenine, Cytosine, Guanine, 5-Methylcytosine, Thymine and Uracil Bicyclonucleoside Monomers, Oligomerisation and Unprecedented Nucleic Acid Recognition," *Tetrahedron*, 1998, vol. 54, No. 14, pp. 3607-3630.
- Kristiansen M N B., et al., "Molecular Characterization of Microvesicular and Macrovesicular Steatosis Shows Widespread Differences in Metabolic Pathways", *Lipids*, Feb. 5, 2019, vol. 54, pp. 109-115.
- Kroschwitz J.I., "Concise Encyclopedia of Polymer Science and Engineering," John Wiley & Sons, NY, 1990, pp. 858-859.
- Kumar R., et al., "The First Analogues of LNA (Locked Nucleic Acids): Phosphorothioate-LNA and 2'-Thio-LNA," *Bioorganic and Medicinal Chemistry Letters*, 1998, vol. 8, No. 16, pp. 2219-2222.
- Kumar V., et al., "Design, Synthesis, Biophysical and Primer Extension Studies of Novel Acyclic Butyl Nucleic Acid (BuNA) †", *Organic Biomolecular Chemistry*, Jul. 9, 2013, vol. 11, pp. 5853-5865.
- Lee D J., et al., "Synthesis of Multivalent Neoglyconjugates of MUC1 by the Conjugation of Carbohydrate-Centered, Triazole-Linked Glycoclusters to MUC1 Peptides using Click Chemistry", *J Org Chem*, 2012, vol. 77, pp. 7564-7571.
- Lee M-R., et al., "Protein Microarrays to Study Carbohydrate-Recognition Events", *Bioorg Med Chem Lett*, 2006, vol. 16, No. 19, pp. 5132-5135.
- Lee R T., et al., "Preparation of Cluster Glycosides of N-Acetylgalactosamine that have Subnanomolar Binding Constants Towards the Mammalian Hepatic Gal/GalNAc-specific Receptor", *Glycoconjugate J.*, 1987, vol. 4, pp. 317-328.
- Lee R T., et al., "Facile Synthesis of a High-Affinity Ligand for Mammalian Hepatic Lectin Containing Three Terminal N-Acetylgalactosamine Residues", *Bioconjugate Chem.*, 1997, vol. 8, No. 5, pp. 762-765.
- Lee R T., et al., "New and More Efficient Multivalent Glyco-Ligands for Asialoglycoprotein Receptor of Mammalian Hepatocytes", *Bioorganic & Medicinal Chemistry*, 2011, vol. 19, pp. 2494-2500.
- Lee R T., et al., "New Synthetic Cluster Ligands for Galactose/N-Acetylgalactosamine-Specific Lectin of Mammalian Liver", *Biochem*, 1984, vol. 23, pp. 4255-4261.
- Lee R T., et al., "Synthesis of Peptide-Based Trivalent Scaffold for Preparation of Cluster Glycosides", *Methods in Enzymology*, 2003, vol. 362, pp. 38-43.
- Lee Y C., "Synthesis of Some Cluster Glycosides Suitable for Attachment to Proteins or Solid Matrices", *Carbohydrate Research*, 1978, vol. 67, pp. 509-514.
- Letsinger R L., et al., "Cholesteryl-Conjugated Oligonucleotides: Synthesis, Properties, and Activity as Inhibitors of Replication of Human Immunodeficiency Virus in Cell Culture", *Proc. Natl. Acad. Sci. USA*, Sep. 1989, vol. 86, pp. 6553-6556.
- Leumann C J., "DNA Analogues: From Supramolecular Principles to Biological Properties", *Bioorganic & Medicinal Chemistry*, 2002, vol. 10, pp. 841-854.

(56)

References Cited

OTHER PUBLICATIONS

Liu S., et al., "Molecular Cloning and Expression Analysis of a New Gene for Short-Chain Dehydrogenase/Reductase 9", *Acta Biochimica Polonica*, 2007, vol. 54, No. 1, pp. 213-218.

Maier L.J., et al., "Comparative Hybrid Arrest By Tandem Antisense Oligodeoxyribonucleotides or Oligodeoxy-Ribonucleoside Methylphosphonates in A Cell-Free System," *Nucleic Acids Research*, 1988, vol. 16, No. 8, pp. 3341-3358.

Maier M. A., et al., "Synthesis of Antisense Oligonucleotides Conjugated to a Multivalent Carbohydrate Cluster for Cellular Targeting", *Bioconjugate Chemistry*, 2003, vol. 14, doi:10.1021/bc020028v, pp. 18-29, XP002510288.

Maierhofer C., et al., "Probing Multivalent Carbohydrate-Lectin Interactions by an Enzyme-Linked Lectin Assay Employing Covalently Immobilized Carbohydrates", *Bioorganic & Medicinal Chemistry*, 2007, vol. 15, pp. 7661-7676.

Manoharan M., et al., "Chemical Modifications to Improve Uptake and Bioavailability of Antisense Oligonucleotides," *Annals of the New York Academy of Sciences*, 1992, vol. 660, pp. 306-309.

Manoharan M., et al., "Cholic Acid-Oligonucleotide Conjugates for Antisense Applications," *Bioorganic and Medicinal Chemistry Letters*, 1994, vol. 4, No. 8, pp. 1053-1060.

Manoharan M., et al., "Introduction of a Lipophilic Thioether Tether in the Minor Groove of Nucleic Acids for Antisense Applications," *Bioorganic and Medicinal Chemistry Letters*, 1993, vol. 3, No. 12, pp. 2765-2770.

Caldwell S., et al., "Cryptogenic Cirrhosis: What We Are Missing ?", *Curr Gastroenterol Rep*, Jan. 27, 2010, pp. 40-48.

Chan J.H.P., et al., "Antisense Oligonucleotides: from Design to Therapeutic Application", *Clinical and Experimental Pharmacology and Physiology*, 2006, vol. 33, No. 5-6, pp. 533-540, DOI: doi.org/10.1111/j.1440-1681.2006.04403.x.

Shemesh C.S., et al., "Assessment of the Drug Interaction Potential of Unconjugated and GalNAc3-Conjugated 2'-MOE-ASOs", *Molecular Therapy—Nucleic Acids*, vol. 9, 2017, pp. 34-47, DOI: 10.1016/j.omtn.2017.08.012.

Supplementary European Search Report in European Patent Application No. 19900290.8, dated Apr. 28, 2023, 5 Pages.

Ma et al., "Hsd17b13 Deficiency Does Not Protect Mice From Obesogenic Diet Injury" *Hepatology* (2021) 73: 1701-1716.

Zhaochun "Practical Clinical Hepatology" People's Military Medical Press (2015) p. 379.

1

MODULATORS OF HSD17B13 EXPRESSION**SEQUENCE LISTING**

The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled BIOL0350WSEQ_ST25.txt created Dec. 13, 2019, which is 640 kb in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD

The present embodiments provide methods, compounds, and compositions useful for inhibiting hydroxysteroid 17-beta dehydrogenase 13 (LOC345275; 17-beta hydroxysteroid dehydrogenase; HSD17B13; HSD17013) expression, and in certain instances, reducing the amount of HSD17B13 protein in a cell or animal, which can be useful for treating, preventing, or ameliorating a disease associated with HSD17B13.

BACKGROUND

Non-alcoholic fatty liver disease (NAFLD) covers a spectrum of liver disease from steatosis to nonalcoholic steatohepatitis (NASH) and cirrhosis. NAFLD is defined as fat accumulation in the liver exceeding 5% by weight, in the absence of significant alcohol consumption, steatogenic medication, or hereditary disorders (Kotronen et al, *Arterioscler Thromb. Vasc. Biol.* 2008, 28: 27-38).

Non-alcoholic steatohepatitis (NASH) is NAFLD with signs of inflammation and hepatic injury. NASH is defined histologically by macrovesicular steatosis, hepatocellular ballooning, and lobular inflammatory infiltrates (Sanyal, *Hepatol. Res.* 2011, 41: 670-4). NASH is estimated to affect 2-3% of the general population. In the presence of other pathologies, such as obesity or diabetes, the estimated prevalence increases to 7% and 62% respectively (Hashimoto et al, *J. Gastroenterol.* 2011, 46(1): 63-69).

SUMMARY

Certain embodiments provided herein are compounds and methods for reducing the amount or activity of HSD17B13 mRNA, and in certain embodiments, reducing the amount or activity of HSD17B13 protein in a cell or animal. In certain embodiments, the animal has a liver disease. In certain embodiments, the disease is NASH. In certain embodiments, the disease is alcoholic steatohepatitis (ASH). In certain embodiments, the disease is NAFLD. In certain embodiments, the disease is hepatic steatosis. In certain embodiments, the disease is cirrhosis. In certain embodiments, the disease is hepatocellular carcinoma. In certain embodiments, the disease is alcoholic liver disease. In certain embodiments, the disease is HCV hepatitis. In certain embodiments, the disease is chronic hepatitis. In certain embodiments, the disease is hereditary hemochromatosis. In certain embodiments, the disease is primary sclerosing cholangitis. Certain compounds provided herein are directed to compounds and compositions that reduce liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an animal.

Certain embodiments provided herein are directed to potent and tolerable compounds and compositions useful for inhibiting HSD17B13 expression, which can be useful for

2

treating, preventing, ameliorating, or slowing progression of liver diseases. Certain embodiments provided herein are directed to compounds and compositions that are more potent or have greater therapeutic value than compounds publicly disclosed.

DETAILED DESCRIPTION

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the embodiments, as claimed. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of "or" means "and/or" unless stated otherwise. Furthermore, the use of the term "including" as well as other forms, such as "includes" and "included", is not limiting.

The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, treatises, and GenBank and NCBI reference sequence records are hereby expressly incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

It is understood that the sequence set forth in each SEQ ID NO in the examples contained herein is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Compounds described by ION number indicate a combination of nucleobase sequence, chemical modification, and motif.

Definitions

Unless otherwise indicated, the following terms have the following meanings:

"2'-deoxyfuranosyl sugar moiety" or "2'-deoxyfuranosyl sugar" means a furanosyl sugar moiety having two hydrogens at the 2'-position. 2'-deoxyfuranosyl sugar moieties may be unmodified or modified and may be substituted at positions other than the 2'-position or unsubstituted. A β -D-2'-deoxyribose sugar moiety in the context of an oligonucleotide is an unsubstituted, unmodified 2'-deoxyfuranosyl and is found in naturally occurring deoxyribonucleic acids (DNA).

"2'-deoxynucleoside" means a nucleoside comprising 2'-H(H) furanosyl sugar moiety, as found in naturally occurring deoxyribonucleic acids (DNA). In certain embodiments, a 2'-deoxynucleoside may comprise a modified nucleobase or may comprise an RNA nucleobase (uracil).

"2'-O-methoxyethyl" (also 2'-MOE) refers to a 2'-O(CH₂)₂—OCH₃) in the place of the 2'-OH group of a ribosyl ring. A 2'-O-methoxyethyl modified sugar is a modified sugar.

"2'-MOE nucleoside" (also 2'-O-methoxyethyl nucleoside) means a nucleoside comprising a 2'-MOE modified sugar moiety.

"2'-substituted nucleoside" or "2'-modified nucleoside" means a nucleoside comprising a 2'-substituted or 2'-modified sugar moiety. As used herein, "2'-substituted" or "2'-modified" in reference to a sugar moiety means a sugar moiety comprising at least one 2'-substituent group other than H or OH.

“3' target site” refers to the nucleotide of a target nucleic acid which is complementary to the 3'-most nucleotide of a particular compound.

“5' target site” refers to the nucleotide of a target nucleic acid which is complementary to the 5'-most nucleotide of a particular compound.

“5-methylcytosine” means a cytosine with a methyl group attached to the 5 position.

“About” means within 10% of a value. For example, if it is stated, “the compounds affected about 70% inhibition of HSD17B13”, it is implied that HSD17B13 levels are inhibited within a range of 60/c and 80%.

“Administration” or “administering” refers to routes of introducing a compound or composition provided herein to an individual to perform its intended function. An example of a route of administration that can be used includes, but is not limited to parenteral administration, such as subcutaneous, intravenous, or intramuscular injection or infusion.

“Administered concomitantly” or “co-administration” means administration of two or more compounds in any manner in which the pharmacological effects of both are manifest in the patient. Concomitant administration does not require that both compounds be administered in a single pharmaceutical composition, in the same dosage form, by the same route of administration, or at the same time. The effects of both compounds need not manifest themselves at the same time. The effects need only be overlapping for a period of time and need not be coextensive. Concomitant administration or co-administration encompasses administration in parallel or sequentially.

“Amelioration” refers to an improvement or lessening of at least one indicator, sign, or symptom of an associated disease, disorder, or condition. In certain embodiments, amelioration includes a delay or slowing in the progression or severity of one or more indicators of a condition or disease. The progression or severity of indicators may be determined by subjective or objective measures, which are known to those skilled in the art.

“Animal” refers to a human or non-human animal, including, but not limited to, mice, rats, rabbits, dogs, cats, pigs, and non-human primates, including, but not limited to, monkeys and chimpanzees.

“Antisense activity” means any detectable and/or measurable activity attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid compared to target nucleic acid levels or target protein levels in the absence of the antisense compound to the target.

“Antisense compound” means a compound comprising an oligonucleotide and optionally one or more additional features, such as a conjugate group or terminal group. Examples of antisense compounds include single-stranded and double-stranded compounds, such as, oligonucleotides, ribozymes, siRNAs, shRNAs, ssRNAs, and occupancy-based compounds.

“Antisense inhibition” means reduction of target nucleic acid levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels in the absence of the antisense compound.

“Antisense mechanisms” are all those mechanisms involving hybridization of a compound with target nucleic acid, wherein the outcome or effect of the hybridization is either target degradation or target occupancy with concomi-

tant stalling of the cellular machinery involving, for example, transcription or splicing.

“Antisense oligonucleotide” means an oligonucleotide having a nucleobase sequence that is complementary to a target nucleic acid or region or segment thereof. In certain embodiments, an antisense oligonucleotide is specifically hybridizable to a target nucleic acid or region or segment thereof.

“Bicyclic nucleoside” or “BNA” means a nucleoside comprising a bicyclic sugar moiety. “Bicyclic sugar” or “bicyclic sugar moiety” means a modified sugar moiety comprising two rings, wherein the second ring is formed via a bridge connecting two of the atoms in the first ring thereby forming a bicyclic structure. In certain embodiments, the first ring of the bicyclic sugar moiety is a furanosyl moiety. In certain embodiments, the bicyclic sugar moiety does not comprise a furanosyl moiety.

“Branching group” means a group of atoms having at least 3 positions that are capable of forming covalent linkages to at least 3 groups. In certain embodiments, a branching group provides a plurality of reactive sites for connecting tethered ligands to an oligonucleotide via a conjugate linker and/or a cleavable moiety.

“Cell-targeting moiety” means a conjugate group or portion of a conjugate group that is capable of binding to a particular cell type or particular cell types.

“cEt” or “constrained ethyl” means a ribosyl bicyclic sugar moiety wherein the second ring of the bicyclic sugar is formed via a bridge connecting the 4'-carbon and the 2'-carbon, wherein the bridge has the formula: 4'-CH(CH₃)—O-2', and wherein the methyl group of the bridge is in the S configuration.

“cEt nucleoside” means a nucleoside comprising a cEt modified sugar moiety.

“Chemical modification” in a compound describes the substitutions or changes through chemical reaction, of any of the units in the compound relative to the original state of such unit. “Modified nucleoside” means a nucleoside having, independently, a modified sugar moiety and/or modified nucleobase. “Modified oligonucleotide” means an oligonucleotide comprising at least one modified internucleoside linkage, a modified sugar, and/or a modified nucleobase.

“Chemically distinct region” refers to a region of a compound that is in some way chemically different than another region of the same compound. For example, a region having 2'-O-methoxyethyl nucleotides is chemically distinct from a region having nucleotides without 2'-O-methoxyethyl modifications.

“Chimeric antisense compounds” means antisense compounds that have at least 2 chemically distinct regions, each position having a plurality of subunits.

“Cleavable bond” means any chemical bond capable of being split. In certain embodiments, a cleavable bond is selected from among: an amide, a polyamide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, a di-sulfide, or a peptide.

“Cleavable moiety” means a bond or group of atoms that is cleaved under physiological conditions, for example, inside a cell, an animal, or a human.

“Complementary” in reference to an oligonucleotide means the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to the following pairs: adenine (A) and thymine (T), adenine

(A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (^mC) and guanine (G) unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. By contrast, “fully complementary” or “100% complementary” in reference to oligonucleotides means that such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches.

“Conjugate group” means a group of atoms that is attached to an oligonucleotide. Conjugate groups include a conjugate moiety and a conjugate linker that attaches the conjugate moiety to the oligonucleotide.

“Conjugate linker” means a group of atoms comprising at least one bond that connects a conjugate moiety to an oligonucleotide.

“Conjugate moiety” means a group of atoms that is attached to an oligonucleotide via a conjugate linker.

“Contiguous” in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, “contiguous nucleobases” means nucleobases that are immediately adjacent to each other in a sequence.

“Designing” or “Designed to” refer to the process of designing a compound that specifically hybridizes with a selected nucleic acid molecule.

“Diluent” means an ingredient in a composition that lacks pharmacological activity, but is pharmaceutically necessary or desirable. For example, the diluent in an injected composition can be a liquid, e.g. saline solution.

“Differently modified” means chemical modifications or chemical substituents that are different from one another, including absence of modifications. Thus, for example, a MOE nucleoside and an unmodified DNA nucleoside are “differently modified,” even though the DNA nucleoside is unmodified. Likewise, DNA and RNA are “differently modified,” even though both are naturally-occurring unmodified nucleosides.

Nucleosides that are the same but for comprising different nucleobases are not differently modified. For example, a nucleoside comprising a 2'-OMe modified sugar and an unmodified adenine nucleobase and a nucleoside comprising a 2'-OMe modified sugar and an unmodified thymine nucleobase are not differently modified.

“Dose” means a specified quantity of a compound or pharmaceutical agent provided in a single administration, or in a specified time period. In certain embodiments, a dose may be administered in two or more boluses, tablets, or injections. For example, in certain embodiments, where subcutaneous administration is desired, the desired dose may require a volume not easily accommodated by a single injection. In such embodiments, two or more injections may be used to achieve the desired dose. In certain embodiments, a dose may be administered in two or more injections to minimize injection site reaction in an individual. In other embodiments, the compound or pharmaceutical agent is administered by infusion over an extended period of time or continuously. Doses may be stated as the amount of pharmaceutical agent per hour, day, week or month.

“Dosing regimen” is a combination of doses designed to achieve one or more desired effects.

“Double-stranded antisense compound” means an antisense compound comprising two oligomeric compounds that are complementary to each other and form a duplex, and wherein one of the two said oligomeric compounds comprises an oligonucleotide.

“Effective amount” means the amount of compound sufficient to effectuate a desired physiological outcome in an individual in need of the compound. The effective amount may vary among individuals depending on the health and physical condition of the individual to be treated, the taxonomic group of the individuals to be treated, the formulation of the composition, assessment of the individual’s medical condition, and other relevant factors.

“Efficacy” means the ability to produce a desired effect.

“Expression” includes all the functions by which a gene’s coded information is converted into structures present and operating in a cell. Such structures include, but are not limited to, the products of transcription and translation.

“Gapmer” means an oligonucleotide comprising an internal region having a plurality of nucleosides that support RNase H cleavage positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as the “gap” and the external regions may be referred to as the “wings.”

“HSD17B13” means any nucleic acid or protein of HSD17B13. “HSD17B13 nucleic acid” means any nucleic acid encoding HSD17B13. For example, in certain embodiments, a HSD17B13 nucleic acid includes a DNA sequence encoding HSD17B13, an RNA sequence transcribed from DNA encoding HSD17B13 (including genomic DNA comprising introns and exons), and an mRNA sequence encoding HSD17B13. “HSD17B13 mRNA” means an mRNA encoding a HSD17B13 protein. The target may be referred to in either upper or lower case.

“HSD17B13 specific inhibitor” refers to any agent capable of specifically inhibiting HSD17B13 RNA and/or HSD17B13 protein expression or activity at the molecular level. For example, HSD17B13 specific inhibitors include nucleic acids (including antisense compounds), peptides, antibodies, small molecules, and other agents capable of inhibiting the expression of HSD17B13 RNA and/or HSD17B13 protein.

“Hybridization” means the annealing of oligonucleotides and/or nucleic acids. While not limited to a particular mechanism, the most common mechanism of hybridization involves hydrogen bonding, which may be Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense compound and a nucleic acid target. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an oligonucleotide and a nucleic acid target.

“Immediately adjacent” means there are no intervening elements between the immediately adjacent elements of the same kind (e.g. no intervening nucleobases between the immediately adjacent nucleobases).

“Individual” means a human or non-human animal selected for treatment or therapy.

“Inhibiting the expression or activity” refers to a reduction or blockade of the expression or activity relative to the expression of activity in an untreated or control sample and does not necessarily indicate a total elimination of expression or activity.

“Internucleoside linkage” means a group or bond that forms a covalent linkage between adjacent nucleosides in an oligonucleotide. “Modified internucleoside linkage” means any internucleoside linkage other than a naturally occurring, phosphate internucleoside linkage. Non-phosphate linkages are referred to herein as modified internucleoside linkages.

“Lengthened oligonucleotides” are those that have one or more additional nucleosides relative to an oligonucleotide disclosed herein, e.g. a parent oligonucleotide.

“Linked nucleosides” means adjacent nucleosides linked together by an internucleoside linkage.

“Linker-nucleoside” means a nucleoside that links an oligonucleotide to a conjugate moiety. Linker-nucleosides are located within the conjugate linker of a compound. Linker-nucleosides are not considered part of the oligonucleotide portion of a compound even if they are contiguous with the oligonucleotide.

“Mismatch” or “non-complementary” means a nucleobase of a first oligonucleotide that is not complementary to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned. For example, nucleobases including but not limited to a universal nucleobase, inosine, and hypoxanthine, are capable of hybridizing with at least one nucleobase but are still mismatched or non-complementary with respect to nucleobase to which it hybridized. As another example, a nucleobase of a first oligonucleotide that is not capable of hybridizing to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned is a mismatch or non-complementary nucleobase.

“Modulating” refers to changing or adjusting a feature in a cell, tissue, organ or organism. For example, modulating HSD17B13 RNA can mean to increase or decrease the level of HSD17B13 RNA and/or HSD17B13 protein in a cell, tissue, organ or organism. A “modulator” effects the change in the cell, tissue, organ or organism. For example, a HSD17B13 compound can be a modulator that decreases the amount of HSD17B13 RNA and/or HSD17B13 protein in a cell, tissue, organ or organism.

“MOE” means methoxyethyl.

“Monomer” refers to a single unit of an oligomer. Monomers include, but are not limited to, nucleosides and nucleotides.

“Motif” means the pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages, in an oligonucleotide.

“Natural” or “naturally occurring” means found in nature.

“Non-bicyclic modified sugar” or “non-bicyclic modified sugar moiety” means a modified sugar moiety that comprises a modification, such as a substituent, that does not form a bridge between two atoms of the sugar to form a second ring.

“Nucleic acid” refers to molecules composed of monomeric nucleotides. A nucleic acid includes, but is not limited to, ribonucleic acids (RNA), deoxyribonucleic acids (DNA), single-stranded nucleic acids, and double-stranded nucleic acids.

“Nucleobase” means a heterocyclic moiety capable of pairing with a base of another nucleic acid. As used herein a “naturally occurring nucleobase” is adenine (A), thymine (T), cytosine (C), uracil (U), and guanine (G). A “modified nucleobase” is a naturally occurring nucleobase that is chemically modified. A “universal base” or “universal nucleobase” is a nucleobase other than a naturally occurring nucleobase and modified nucleobase, and is capable of pairing with any nucleobase.

“Nucleobase sequence” means the order of contiguous nucleobases in a nucleic acid or oligonucleotide independent of any sugar or internucleoside linkage.

“Nucleoside” means a compound comprising a nucleobase and a sugar moiety. The nucleobase and sugar moiety are each, independently, unmodified or modified. “Modified

nucleoside” means a nucleoside comprising a modified nucleobase and/or a modified sugar moiety. Modified nucleosides include abasic nucleosides, which lack a nucleobase.

“Oligomeric compound” means a compound comprising a single oligonucleotide and, optionally, one or more additional features, such as a conjugate group or terminal group.

“Oligonucleotide” means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another. Unless otherwise indicated, oligonucleotides consist of 8-80 linked nucleosides. “Modified oligonucleotide” means an oligonucleotide, wherein at least one sugar, nucleobase, or internucleoside linkage is modified. “Unmodified oligonucleotide” means an oligonucleotide that does not comprise any sugar, nucleobase, or internucleoside modification.

“Parent oligonucleotide” means an oligonucleotide whose sequence is used as the basis of design for more oligonucleotides of similar sequence but with different lengths, motifs, and/or chemistries. The newly designed oligonucleotides may have the same or overlapping sequence as the parent oligonucleotide.

“Parenteral administration” means administration through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g. intrathecal or intracerebroventricular administration.

“Pharmaceutically acceptable carrier or diluent” means any substance suitable for use in administering to an individual. For example, a pharmaceutically acceptable carrier can be a sterile aqueous solution, such as PBS or water-for-injection.

“Pharmaceutically acceptable salts” means physiologically and pharmaceutically acceptable salts of compounds, such as oligomeric compounds or oligonucleotides, i.e., salts that retain the desired biological activity of the parent compound and do not impart undesired toxicological effects thereto.

“Pharmaceutical agent” means a compound that provides a therapeutic benefit when administered to an individual.

“Pharmaceutical composition” means a mixture of substances suitable for administering to an individual. For example, a pharmaceutical composition may comprise one or more compounds or salt thereof and a sterile aqueous solution.

“Phosphorothioate linkage” means a modified phosphate linkage in which one of the non-bridging oxygen atoms is replaced with a sulfur atom. A phosphorothioate internucleoside linkage is a modified internucleoside linkage.

“Phosphorus moiety” means a group of atoms comprising a phosphorus atom. In certain embodiments, a phosphorus moiety comprises a mono-, di-, or tri-phosphate, or phosphorothioate.

“Portion” means a defined number of contiguous (i.e., linked) nucleobases of a nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an oligomeric compound.

“Prevent” refers to delaying or forestalling the onset, development or progression of a disease, disorder, or condition for a period of time from minutes to indefinitely.

“Prodrug” means a compound in a form outside the body which, when administered to an individual, is metabolized to another form within the body or cells thereof. In certain

embodiments, the metabolized form is the active, or more active, form of the compound (e.g., drug). Typically conversion of a prodrug within the body is facilitated by the action of an enzyme(s) (e.g., endogenous or viral enzyme) or chemical(s) present in cells or tissues, and/or by physiologic conditions.

“Reduce” means to bring down to a smaller extent, size, amount, or number.

“RefSeq No.” is a unique combination of letters and numbers assigned to a sequence to indicate the sequence is for a particular target transcript (e.g., target gene). Such sequence and information about the target gene (collectively, the gene record) can be found in a genetic sequence database. Genetic sequence-databases include the NCBI Reference Sequence database, GenBank, the European Nucleotide Archive, and the DNA Data Bank of Japan (the latter three forming the International Nucleotide Sequence Database Collaboration or INSDC).

“Region” is defined as a portion of the target nucleic acid having at least one identifiable structure, function, or characteristic.

“RNAi compound” means an antisense compound that acts, at least in part, through RISC or Ago2, but not through RNase H, to modulate a target nucleic acid and/or protein encoded by a target nucleic acid. RNAi compounds include, but are not limited to double-stranded siRNA, single-stranded RNA (ssRNA), and microRNA, including microRNA mimics.

“Segments” are defined as smaller or sub-portions of regions within a nucleic acid.

“Side effects” means physiological disease and/or conditions attributable to a treatment other than the desired effects. In certain embodiments, side effects include injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, myopathies, and malaise. For example, increased aminotransferase levels in serum may indicate liver toxicity or liver function abnormality. For example, increased bilirubin may indicate liver toxicity or liver function abnormality.

“Single-stranded” in reference to a compound means the compound has only one oligonucleotide.

“Self-complementary” means an oligonucleotide that at least partially hybridizes to itself. A compound consisting of one oligonucleotide, wherein the oligonucleotide of the compound is self-complementary, is a single-stranded compound. A single-stranded compound may be capable of binding to a complementary compound to form a duplex.

“Sites” are defined as unique nucleobase positions within a target nucleic acid.

“Specifically hybridizable” refers to an oligonucleotide having a sufficient degree of complementarity between the oligonucleotide and a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids. In certain embodiments, specific hybridization occurs under physiological conditions.

“Specifically inhibit” with reference to a target nucleic acid means to reduce or block expression of the target nucleic acid while exhibiting fewer, minimal, or no effects on non-target nucleic acids. Reduction does not necessarily indicate a total elimination of the target nucleic acid’s expression.

“Standard cell assay” means assay(s) described in the Examples and reasonable variations thereof.

“Standard in vivo experiment” means the procedure(s) described in the Example(s) and reasonable variations thereof.

“Stereorandom chiral center” in the context of a population of molecules of identical molecular formula means a chiral center having a random stereochemical configuration. For example, in a population of molecules comprising a stereorandom chiral center, the number of molecules having the (S) configuration of the stereorandom chiral center may be but is not necessarily the same as the number of molecules having the (R) configuration of the stereorandom chiral center. The stereochemical configuration of a chiral center is considered random when it is the result of a synthetic method that is not designed to control the stereochemical configuration. In certain embodiments, a stereorandom chiral center is a stereorandom phosphorothioate internucleoside linkage.

“Sugar moiety” means an unmodified sugar moiety or a modified sugar moiety. As used herein, “unmodified sugar moiety” means a p-D-ribose moiety, as found in naturally occurring RNA, or a β-D-2'-deoxyribose moiety as found in naturally occurring DNA. As used herein, “modified sugar moiety” or “modified sugar” means a sugar surrogate or a furanosyl sugar moiety other than a β-D-ribose or a β-D-2'-deoxyribose. Modified furanosyl sugar moieties may be modified or substituted at a certain position(s) of the sugar moiety, substituted, or unsubstituted, and they may or may not have a stereoconfiguration other than β-D-ribose. Modified furanosyl sugar moieties include bicyclic sugars and non-bicyclic sugars.

“Sugar surrogate” means a modified sugar moiety that does not comprise a furanosyl or tetrahydrofuran ring (is not a “furanosyl sugar moiety”) and that can link a nucleobase to another group, such as an internucleoside linkage, conjugate group, or terminal group in an oligonucleotide. Modified nucleosides comprising sugar surrogates can be incorporated into one or more positions within an oligonucleotide and such oligonucleotides are capable of hybridizing to complementary oligomeric compounds or nucleic acids.

“Synergy” or “synergize” refers to an effect of a combination that is greater than additive of the effects of each component alone at the same doses.

“Target gene” refers to a gene encoding a target.

“Targeting” means the specific hybridization of a compound to a target nucleic acid in order to induce a desired effect.

“Target nucleic acid,” “target RNA,” “target RNA transcript” and “nucleic acid target” all mean a nucleic acid capable of being targeted by compounds described herein.

“Target region” means a portion of a target nucleic acid to which one or more compounds is targeted.

“Target segment” means the sequence of nucleotides of a target nucleic acid to which a compound is targeted. “5' target site” refers to the 5'-most nucleotide of a target segment. “3' target site” refers to the 3'-most nucleotide of a target segment.

“Terminal group” means a chemical group or group of atoms that is covalently linked to a terminus of an oligonucleotide.

“Therapeutically effective amount” means an amount of a compound, pharmaceutical agent, or composition that provides a therapeutic benefit to an individual.

“Treat” refers to administering a compound or pharmaceutical composition to an animal in order to effect an alteration or improvement of a disease, disorder, or condition in the animal.

Certain Embodiments

Certain embodiments provide methods, compounds and compositions for inhibiting HSD17B13 expression.

11

Certain embodiments provide compounds targeted to a HSD17B13 nucleic acid. In certain embodiments, the HSD17B13 nucleic acid has the sequence set forth in RefSeq or GENBANK Accession No. NM_178135.4 (incorporated by reference, disclosed herein as SEQ ID NO: 1), the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000 (incorporated by reference, disclosed herein as SEQ ID NO: 2), the sequence listed as SEQ ID NO: 3, or GENBANK Accession No. NM_001136230.2 (incorporated by reference, disclosed herein as SEQ ID NO: 4). In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 9 to 80 linked nucleosides and having a nucleobase sequence comprising at least 9 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 10 to 80 linked nucleosides and having a nucleobase sequence comprising at least 10 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 11 to 80 linked nucleosides and having a nucleobase sequence comprising at least 11 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 11 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 80 linked nucleosides and having a nucleobase sequence comprising at least 12 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 12 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 to 80 linked

12

nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of the nucleobase sequence of any one of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded.

In certain embodiments, the compound is double-stranded.

Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 3095-3369, 3371-7564, 7565-7672, 7673-8777, 8778-8909, 8910-10401, 10402-10508, 10509-12040, 12041-12178, 12179-15641, 15642-15758, 15759-20692, or 20693-22212 of SEQ ID NO: 2, wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

In certain embodiments, compounds target within nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183,

3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 14800-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-

17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2.

In certain embodiments, compounds have at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-

15

11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

In certain embodiments, compounds have a nucleobase sequence complementary to nucleobases 1323-1338, 1600-1615, 1627-1642, or 1782-1797 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

In certain embodiments, compounds have a nucleobase sequence comprising a portion of at least 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobases complementary within an equal length portion of nucleobases 1323-1338, 1600-1615, 1627-1642, or 1782-1797 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

16

In certain embodiments, compounds have a nucleobase sequence complementary to nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2.

In certain embodiments, compounds have a nucleobase sequence comprising a portion of at least 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobases complementary within an equal length portion of nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

In certain embodiments, any of the foregoing modified oligonucleotides has at least one modified internucleoside linkage, at least one modified sugar, and/or at least one modified nucleobase.

In certain embodiments, at least one nucleoside of any of the foregoing modified oligonucleotides comprises a modified sugar. In certain embodiments, the modified sugar comprises a 2'-O-methoxyethyl group. In certain embodiments, the modified sugar is a bicyclic sugar, such as a 4'-CH(CH₃)-O-2' group, a 4'-CH₂-O-2' group, or a 4'-(CH₂)₂-O-2' group.

In certain embodiments, at least one internucleoside linkage of the modified oligonucleotide comprises a modified internucleoside linkage, such as a phosphorothioate internucleoside linkage.

In certain embodiments, at least one nucleobase of any of the foregoing modified oligonucleotides comprises a modified nucleobase, such as 5-methylcytosine.

In certain embodiments, any of the foregoing modified oligonucleotides has:

- a gap segment consisting of linked deoxynucleosides;
- a 5' wing segment consisting of linked nucleosides; and
- a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide consists of 12 to 30 linked nucleosides and has a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 8-2896. In certain embodiments, the modified oligonucleotide is 16 to 30 linked nucleosides in length having a nucleobase sequence comprising the nucleobase sequence recited in any one of SEQ ID NOs: 8-2896. In certain embodiments, the modified oligonucleotide is 16 linked nucleosides in length having a nucleobase sequence consisting of the sequence recited in any one of SEQ ID NOs: 8-2896.

17

In certain embodiments, a compound comprises or consists of a modified oligonucleotide consisting of 16 to 80 linked nucleobases and having a nucleobase sequence comprising the nucleobase sequence recited in any one of 8-2896, wherein the modified oligonucleotide has:

- a gap segment consisting of linked 2'-deoxynucleosides;
- a 5' wing segment consisting of linked nucleosides; and
- a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides

In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, and 1595, wherein the modified oligonucleotide has:

- a gap segment consisting of ten linked 2'-deoxynucleosides;
- a 5' wing segment consisting of three linked nucleosides; and
- a 3' wing segment consisting of three linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 819, wherein the modified oligonucleotide has:

- a gap segment consisting of nine linked 2'-deoxynucleosides;
- a 5' wing segment consisting of three linked nucleosides; and
- a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt sugar; wherein the sugar residues of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases

18

in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 1595, wherein the modified oligonucleotide has:

- a gap segment consisting of ten linked nucleosides;
- a 5' wing segment consisting of three linked nucleosides; and
- a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

In any of the foregoing embodiments, the compound or oligonucleotide can be at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% complementary to a nucleic acid encoding HSD17B13.

In any of the foregoing embodiments, the compound can be single-stranded. In certain embodiments, the compound comprises deoxyribonucleotides. In certain embodiments, the compound is double-stranded. In certain embodiments, the compound is double-stranded and comprises ribonucleotides. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

In any of the foregoing embodiments, the compound can consist of 8 to 80, 10 to 30, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked nucleosides. In certain embodiments, the compound comprises or consists of an oligonucleotide.

In certain embodiments, a compound comprises a modified oligonucleotide described herein and a conjugate group. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 5' end of the modified oligonucleotide. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 3' end of the modified oligonucleotide. In certain embodiments, the conjugate group comprises at least one N-Acetylgalactosamine (GalNAc), at least two N-Acetylgalactosamines (GalNAcs), or at least three N-Acetylgalactosamines (GalNAcs).

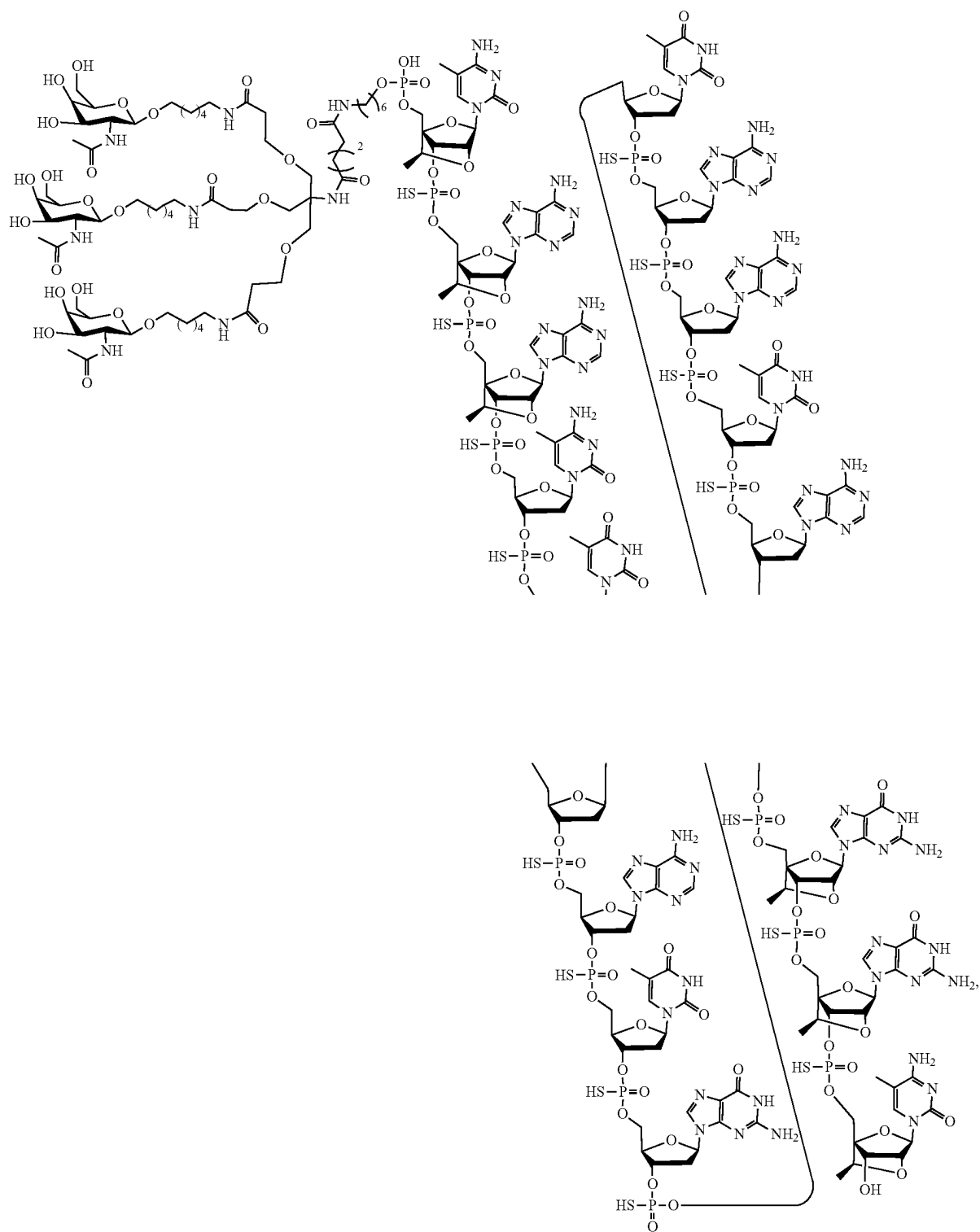
In certain embodiments, compounds or compositions provided herein comprise a pharmaceutically acceptable salt of the modified oligonucleotide. In certain embodiments, the salt is a sodium salt. In certain embodiments, the salt is a potassium salt.

19

In certain embodiments, a compound provided herein is according to the following chemical structure:

20

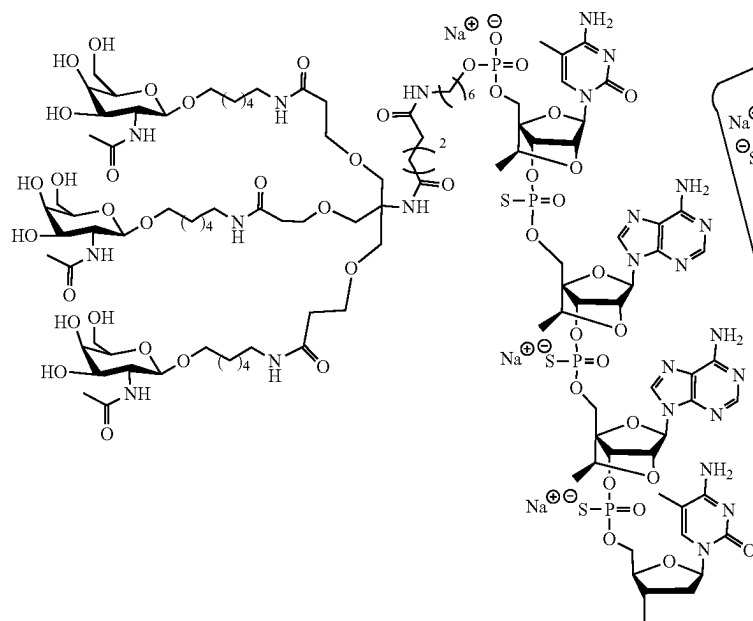
(SEQ ID NO: 286)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

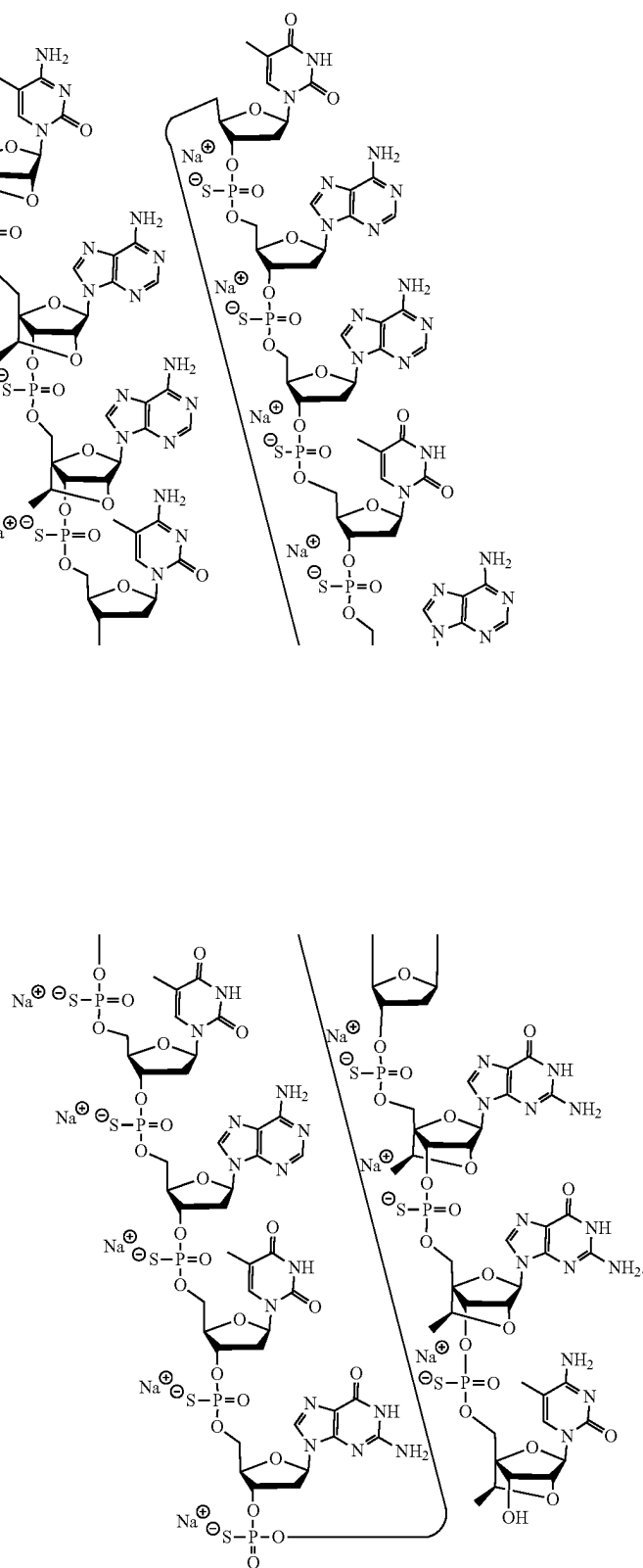
21

In certain embodiments, a compound provided herein is according to the following chemical structure:



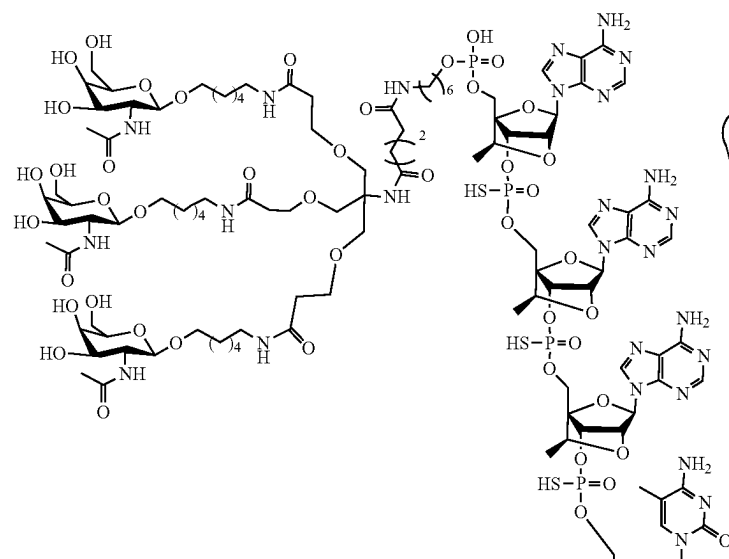
22

(SEQ ID NO: 286)



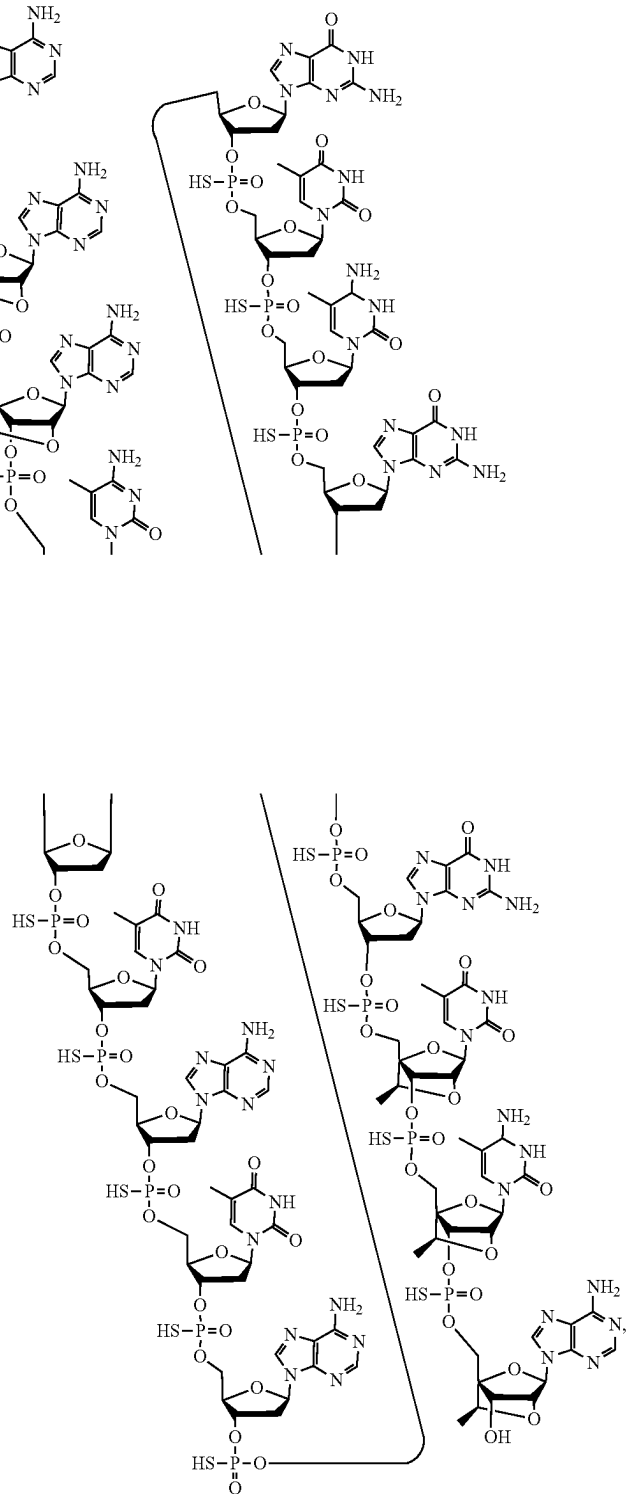
23

In certain embodiments, a compound provided herein is according to the following chemical structure:



24

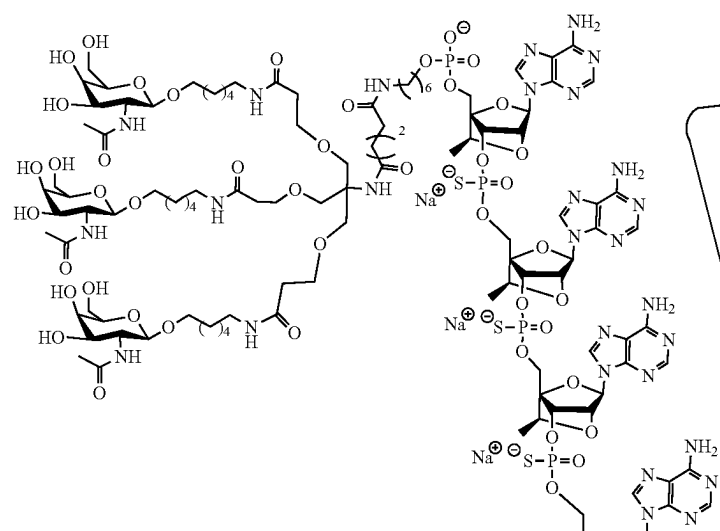
(SEQ ID NO: 817)



or a salt thereof. In certain embodiments, the compound is
a sodium salt. In certain embodiments, the compound is a
potassium salt.

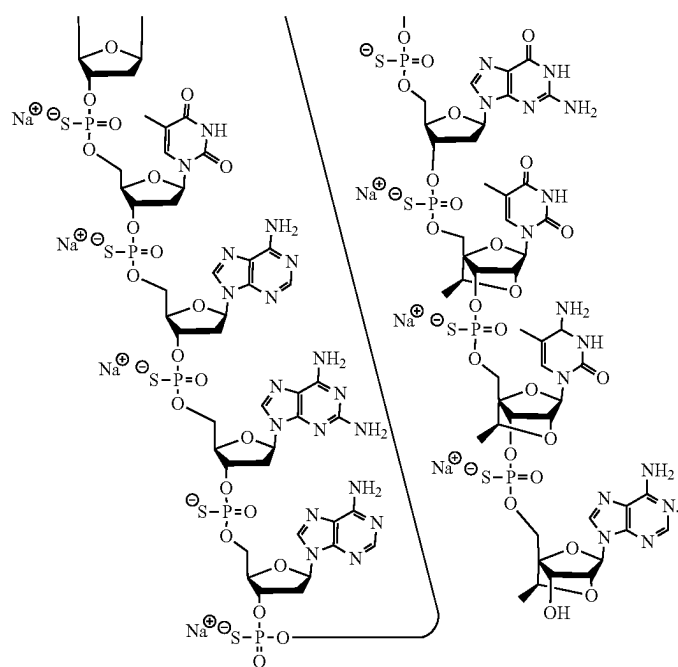
25

In certain embodiments, a compound provided herein is according to the following chemical structure:



26

(SEQ ID NO: 817)

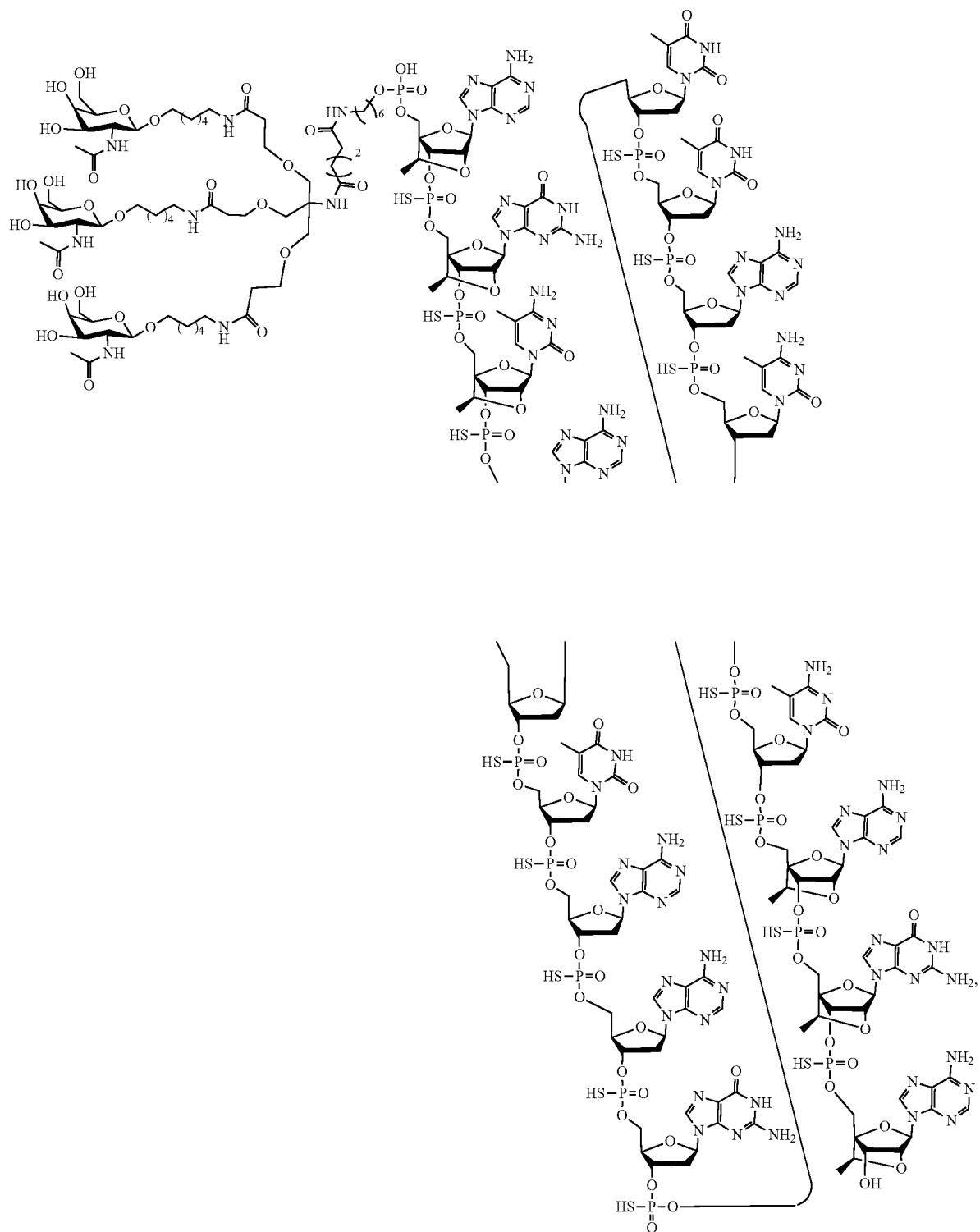


27

In certain embodiments, a compound provided herein is according to the following chemical structure:

28

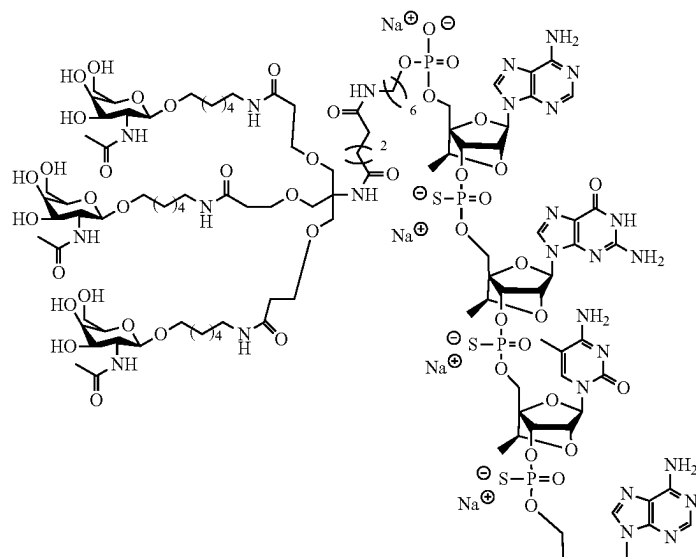
(SEQ ID NO: 983)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

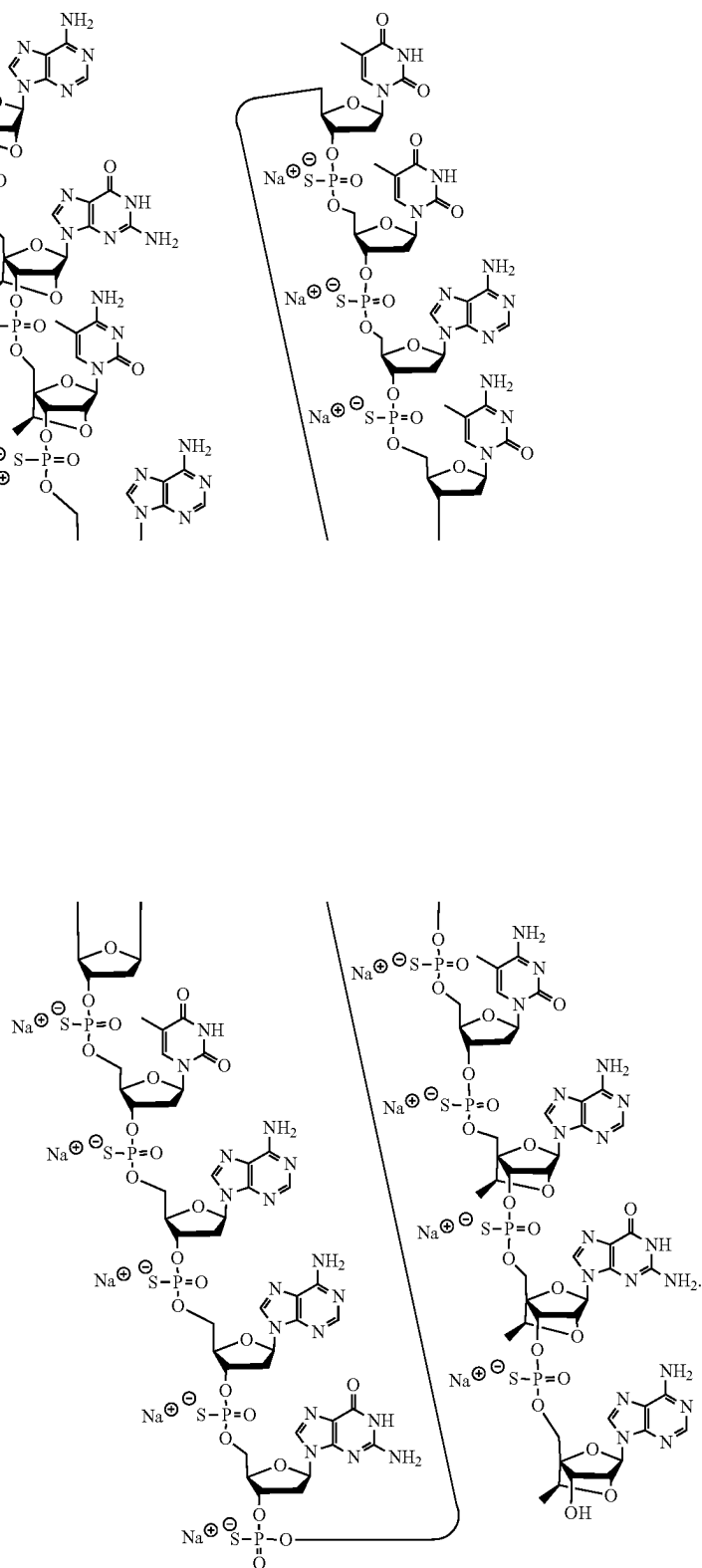
29

In certain embodiments, a compound provided herein is according to the following chemical structure:



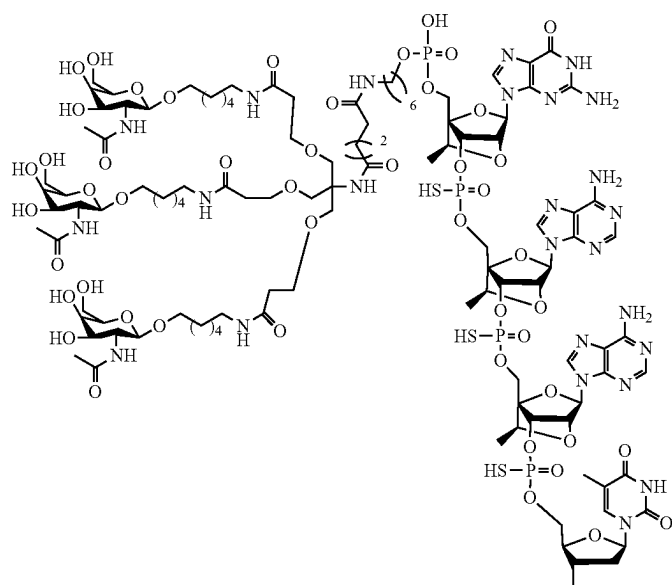
30

(SEQ ID NO: 983)



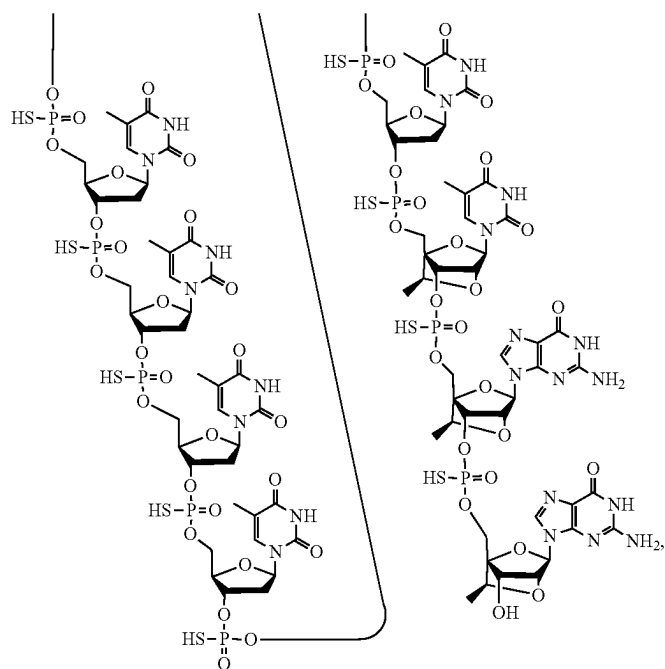
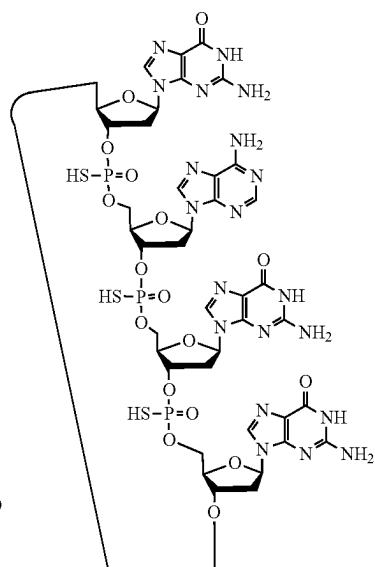
31

In certain embodiments, a compound provided herein is according to the following chemical structure:



32

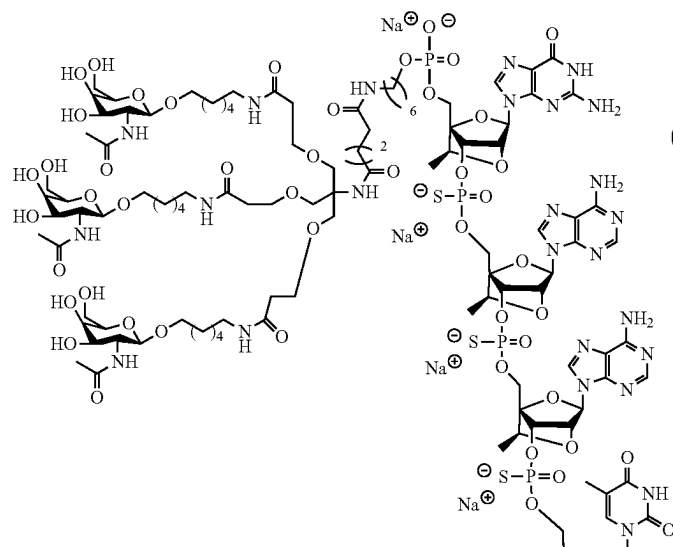
(SEQ ID NO: 1215)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

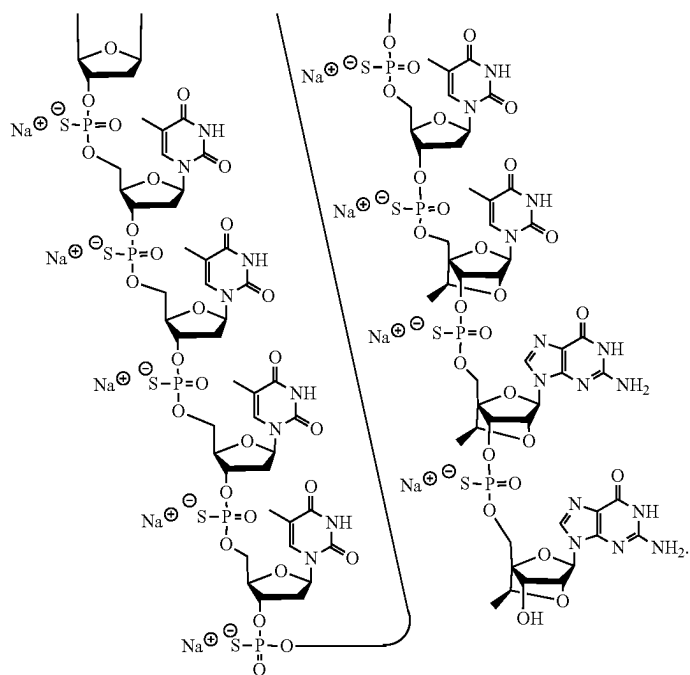
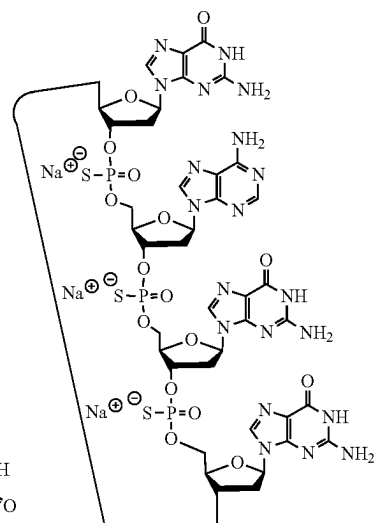
33

In certain embodiments, a compound provided herein is according to the following chemical structure:



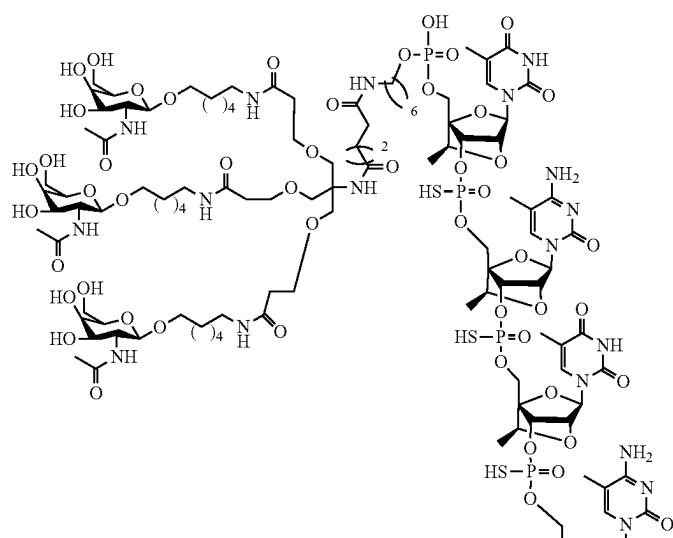
34

(SEQ ID NO: 1215)



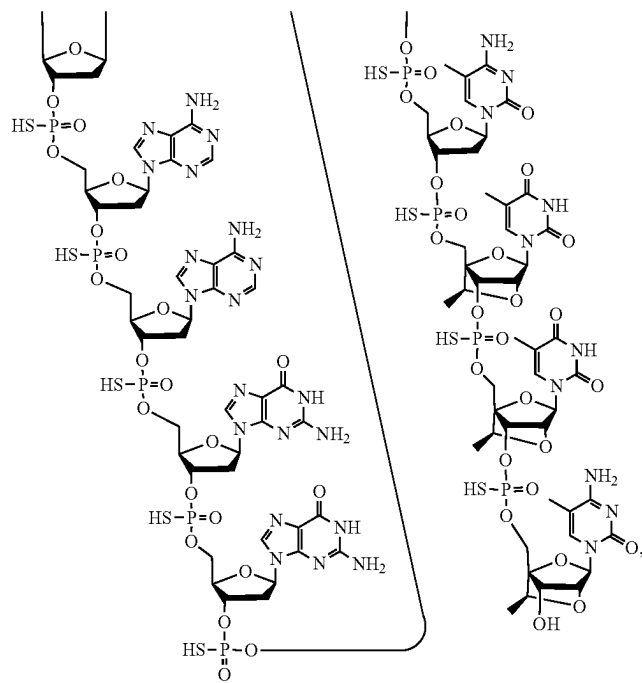
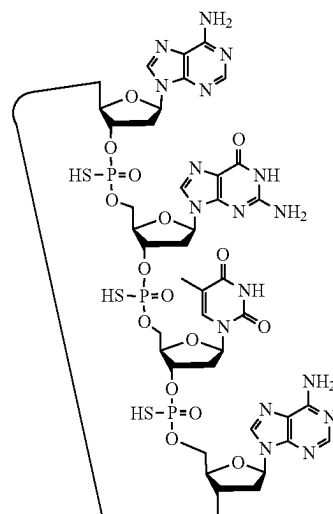
35

In certain embodiments, a compound provided herein is according to the following chemical structure:



36

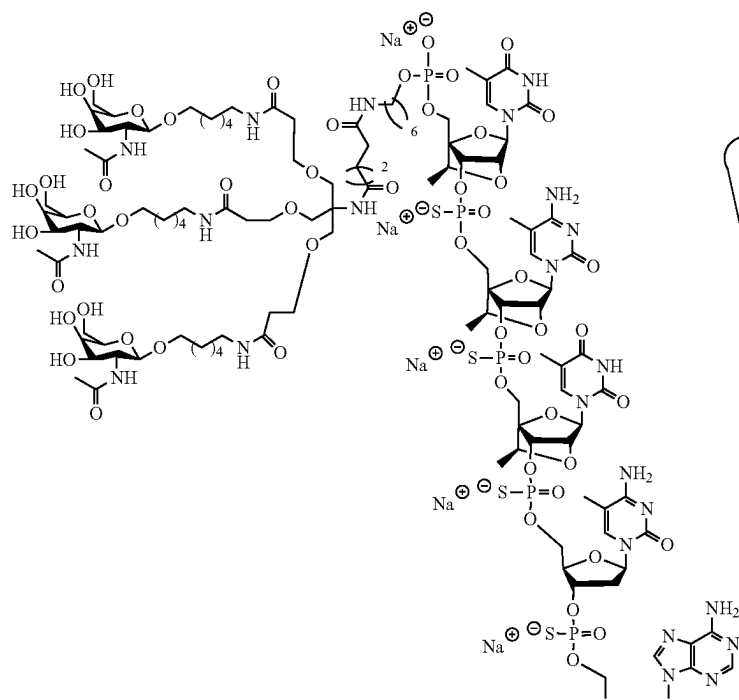
(SEQ ID NO: 747)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

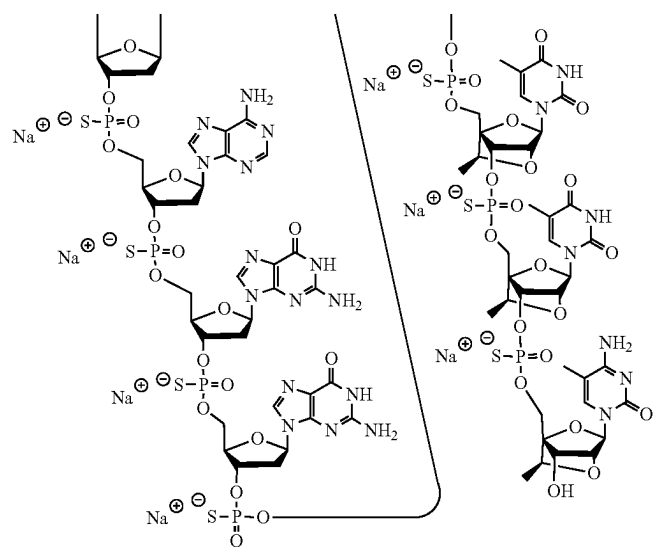
37

In certain embodiments, a compound provided herein is according to the following chemical structure:



38

(SEQ ID NO: 747)

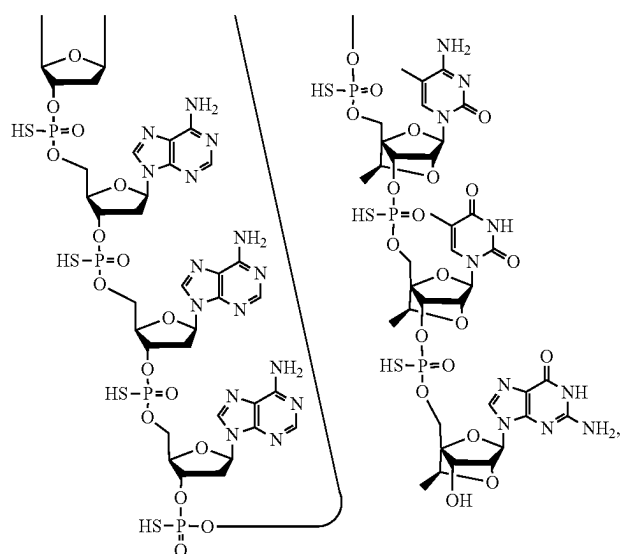
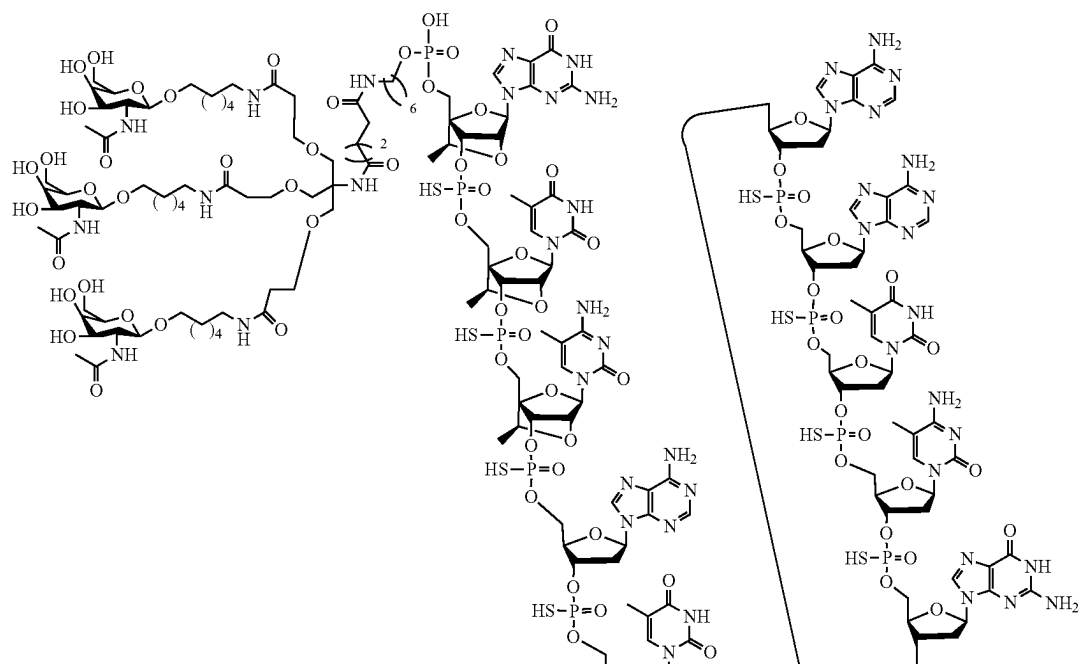


39

In certain embodiments, a compound provided herein is according to the following chemical structure:

40

(SEQ ID NO: 43)



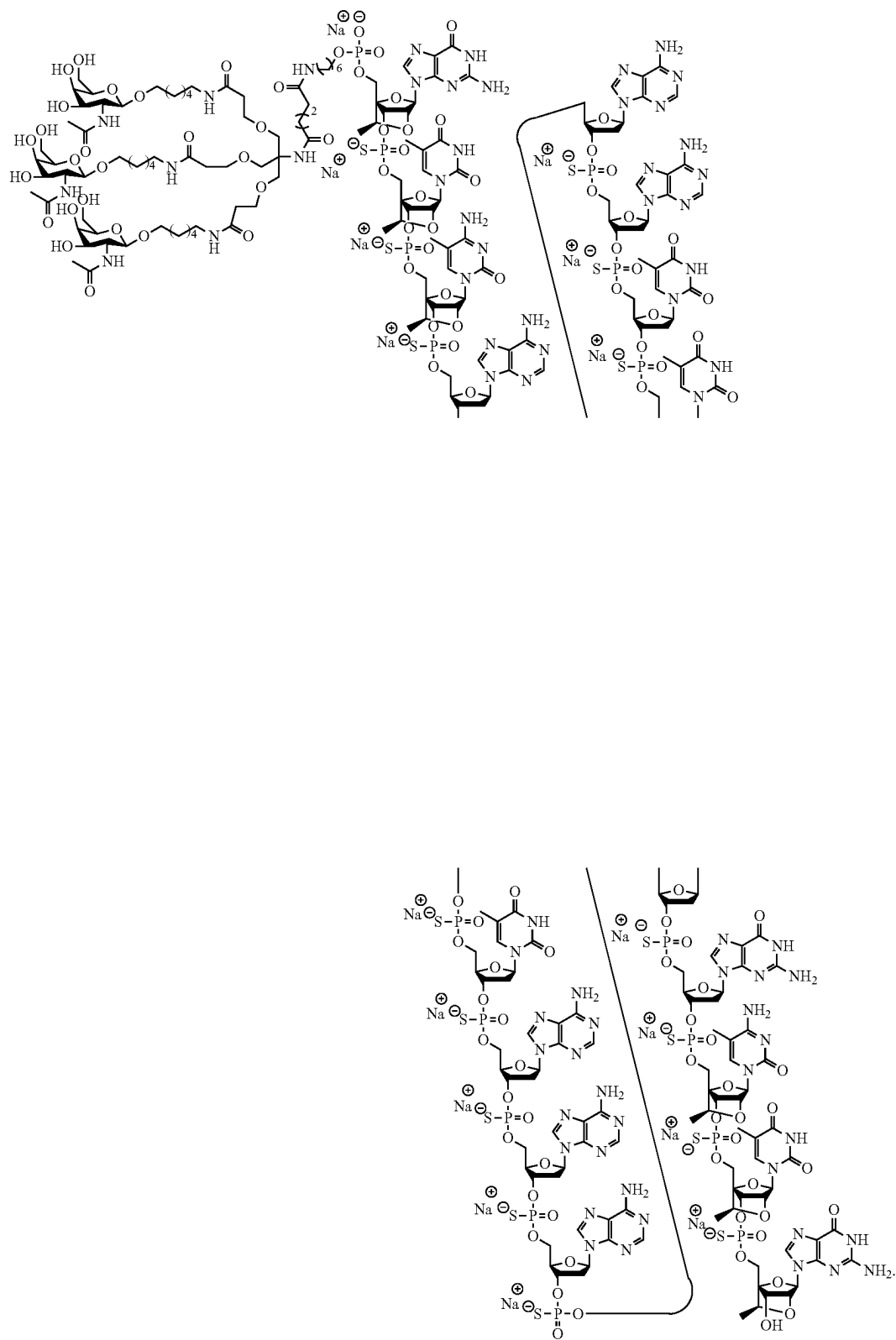
or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

41

In certain embodiments, a compound provided herein is according to the following chemical structure:

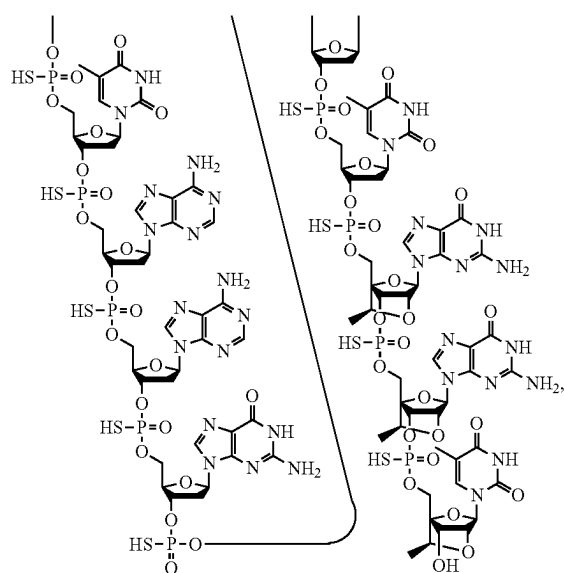
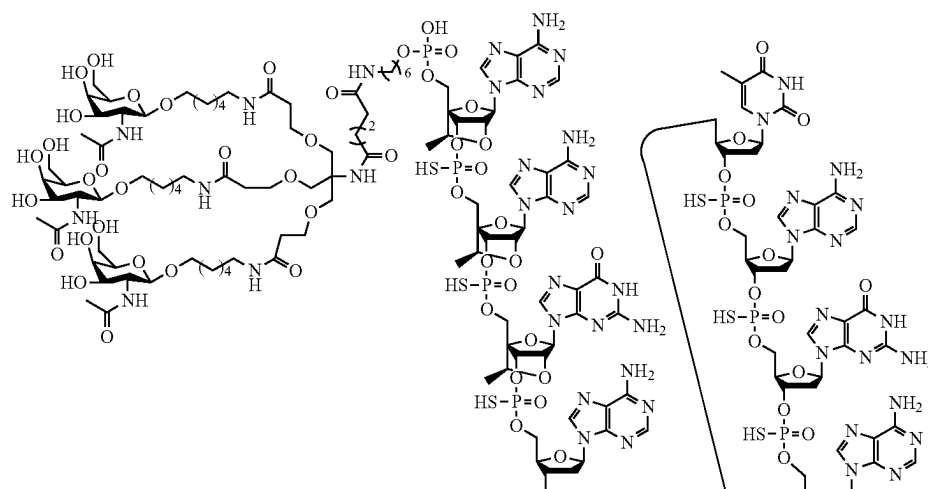
42

(SEQ ID NO: 43)



In certain embodiments, a compound provided herein is according to the following chemical structure:

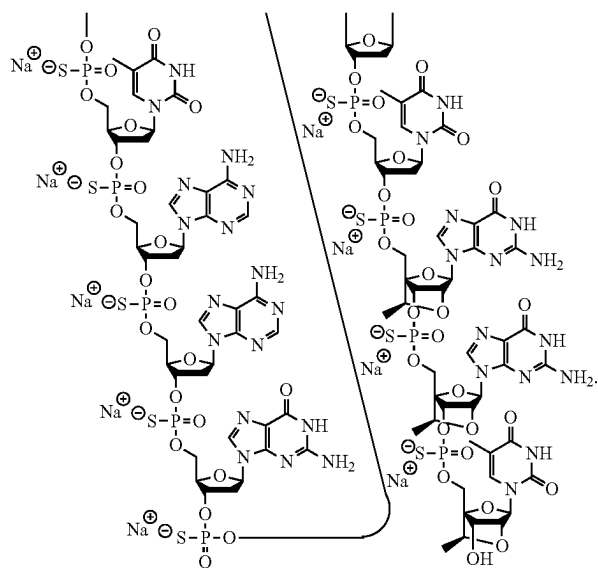
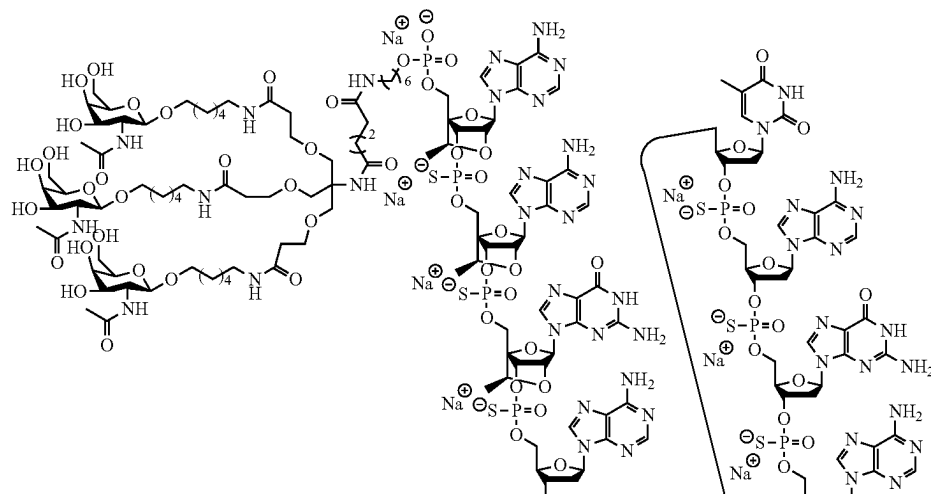
(SEQ ID NO: 355)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

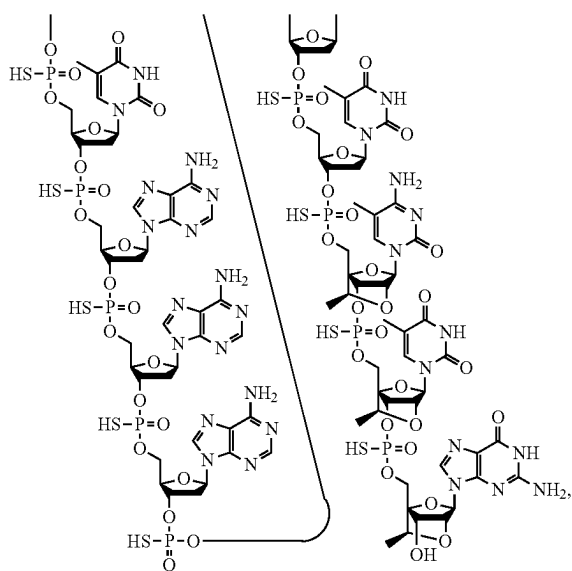
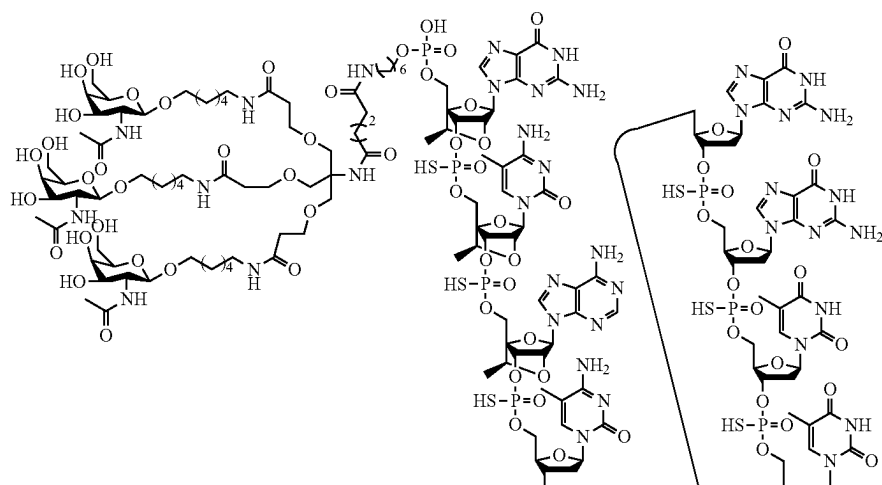
In certain embodiments a compound provided herein is according to the following chemical structure:

(SEQ ID NO: 355)



In certain embodiments, a compound provided herein is according to the following chemical structure:

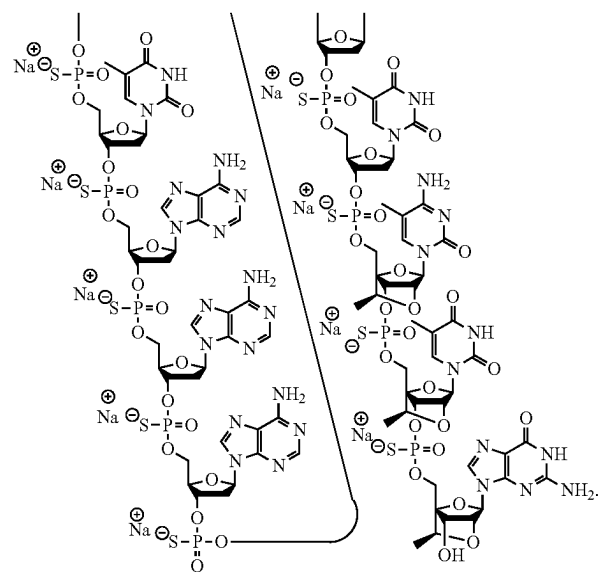
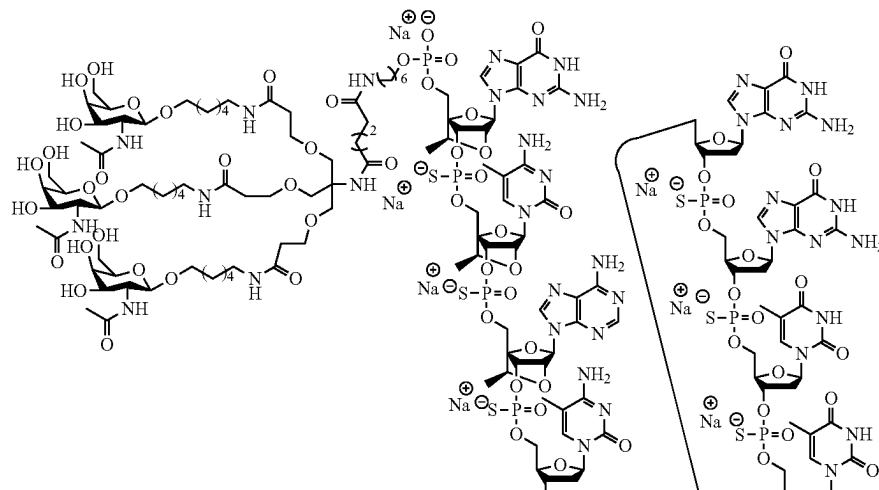
(SEQ ID NO: 1602)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

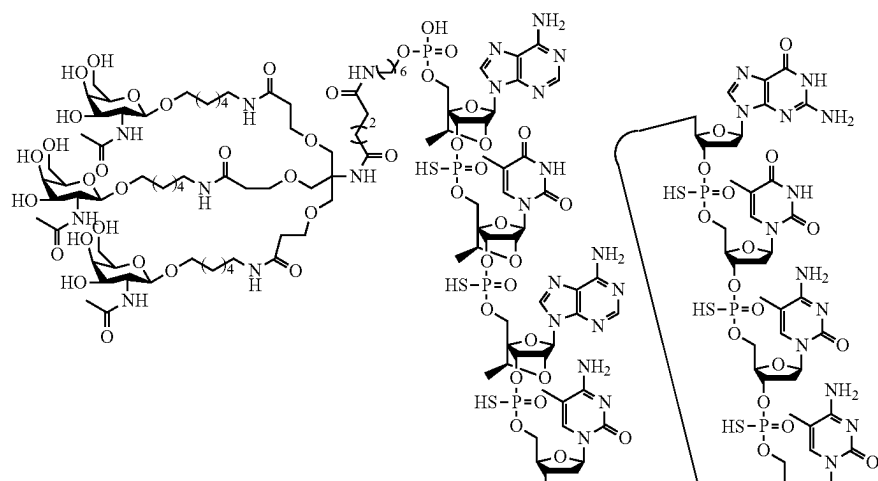
In certain embodiments, a compound provided herein is according to the following chemical structure:

(SEQ ID NO: 1602)



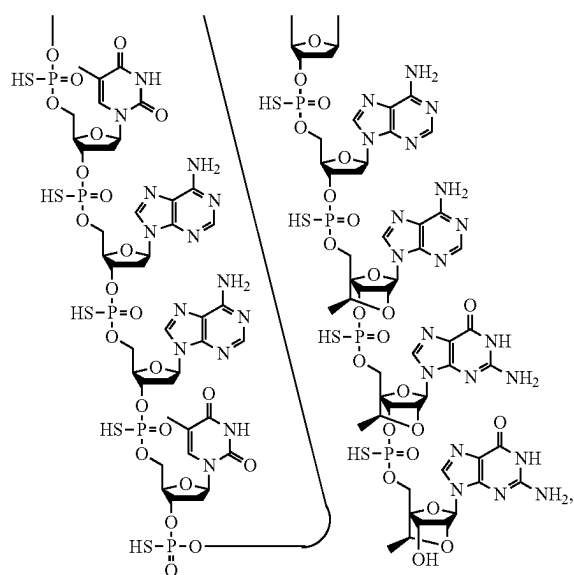
51

In certain embodiments, a compound provided herein k
according to the following chemical structure:



52

(SEQ ID NO: 201)



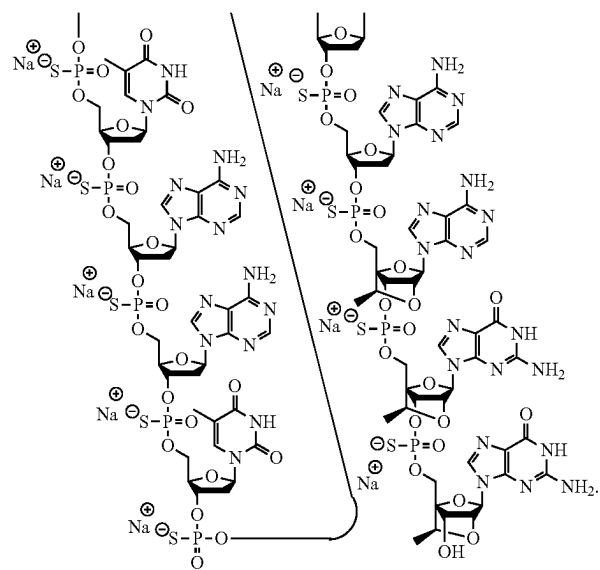
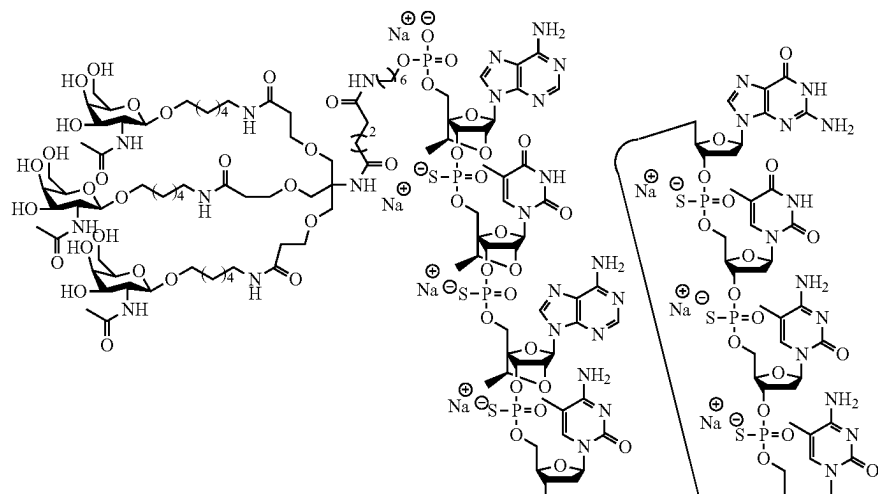
or a salt thereof. In certain embodiments, the compound is 65
a sodium salt. In certain embodiments, the compound is a
potassium salt.

53

In certain embodiments, a compound provided herein is according to the following chemical structure:

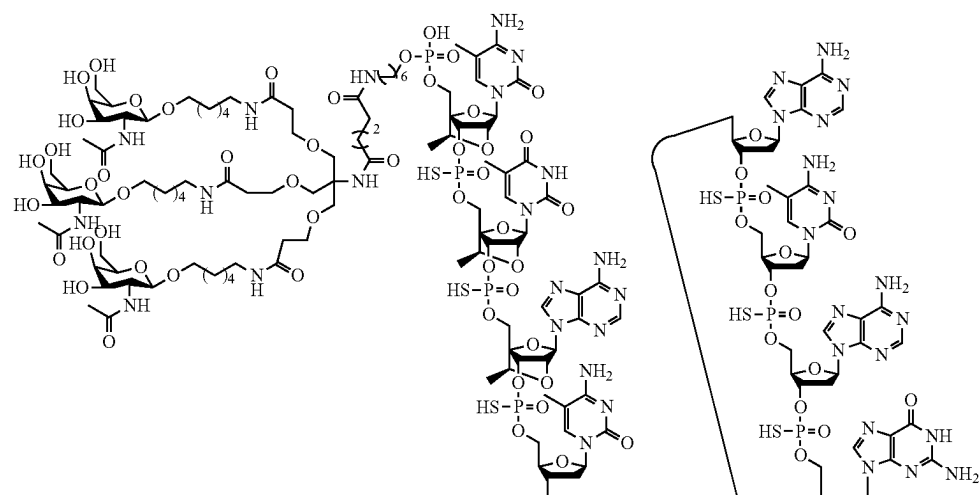
54

(SEQ ID NO: 201)



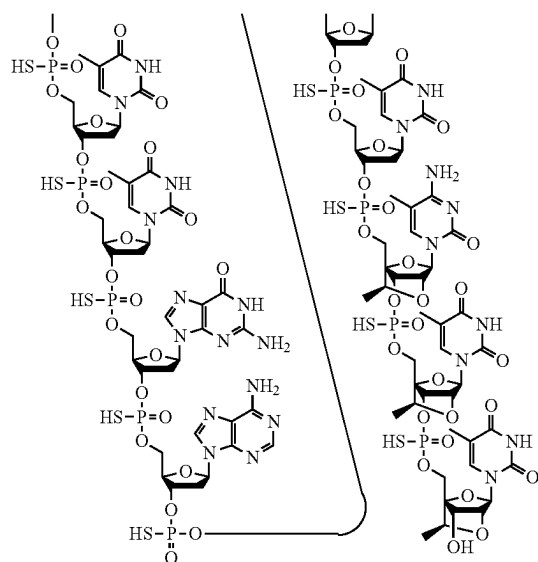
55

In certain embodiments, a compound provided herein is according to the following chemical structure:



56

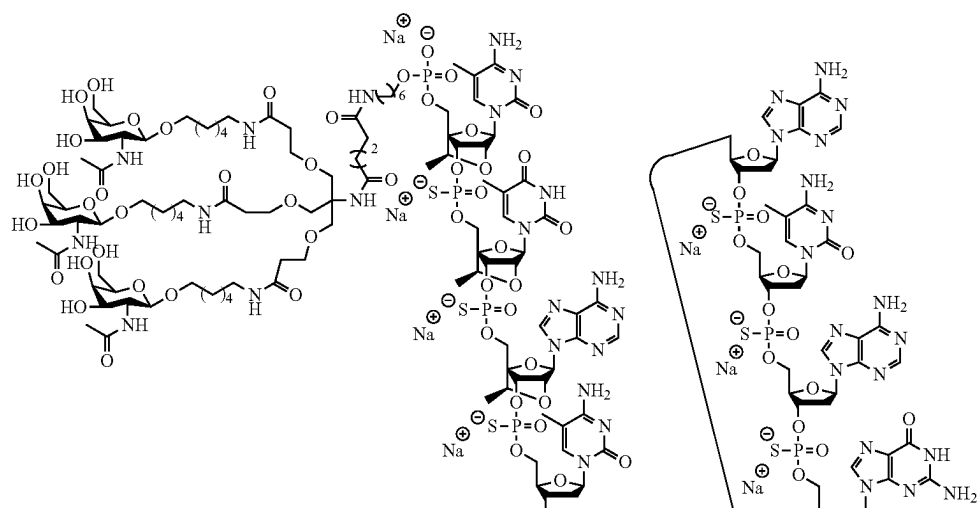
(SEQ ID NO: 734)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

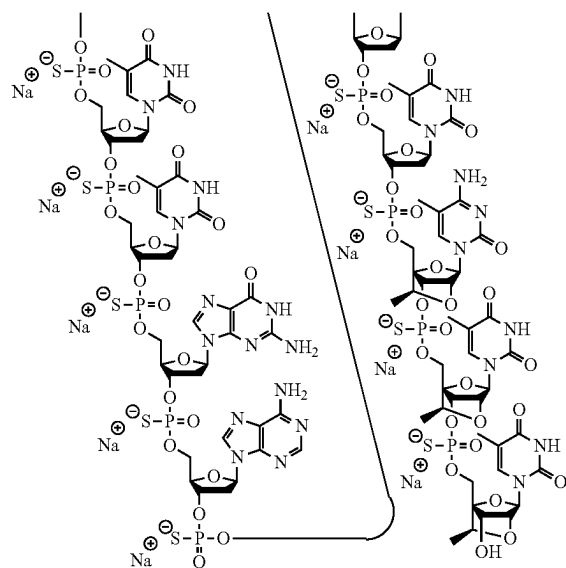
57

In certain embodiments, a compound provided herein is according to the following chemical structure:



58

(SEQ ID NO: 734)

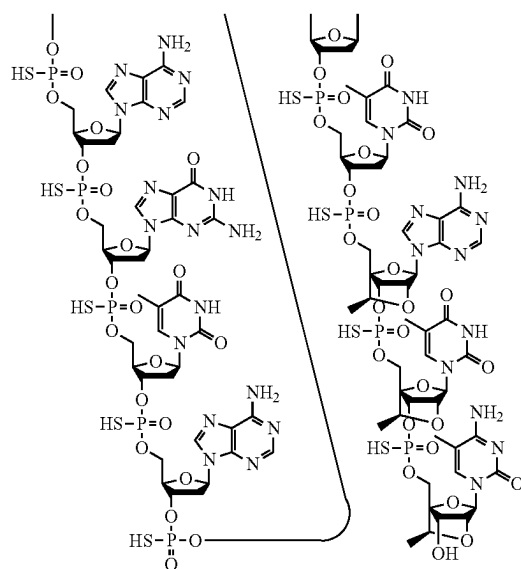
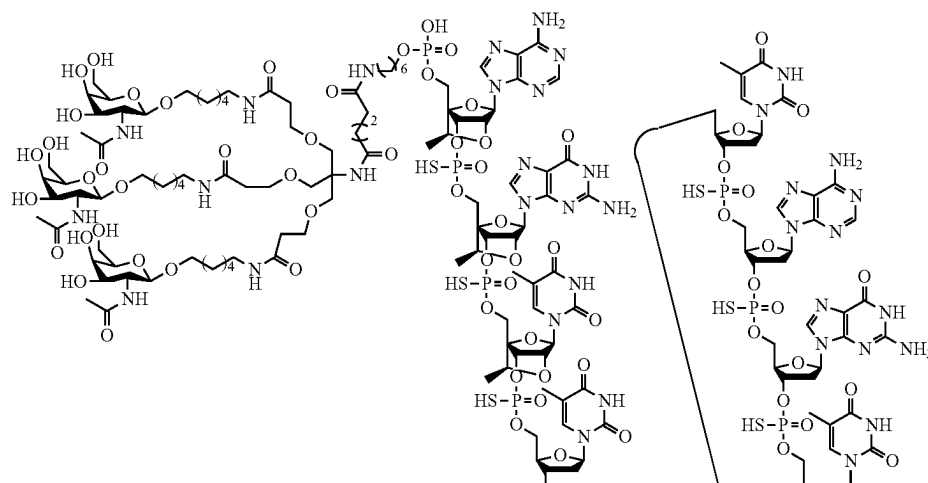


59

In certain embodiments, a compound provided herein is according to the following chemical structure:

60

(SEQ ID NO: 1249)



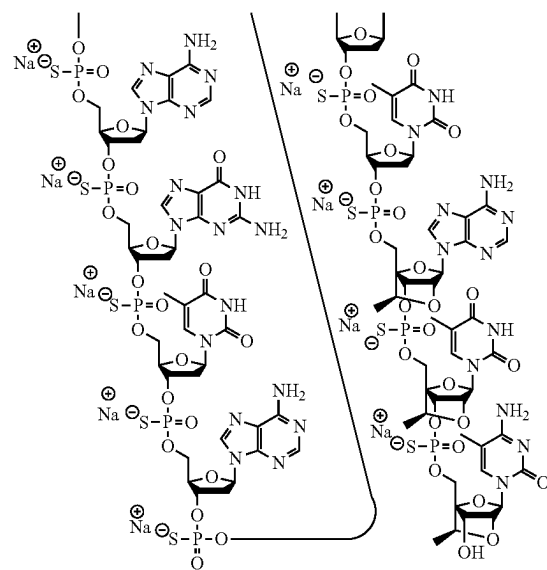
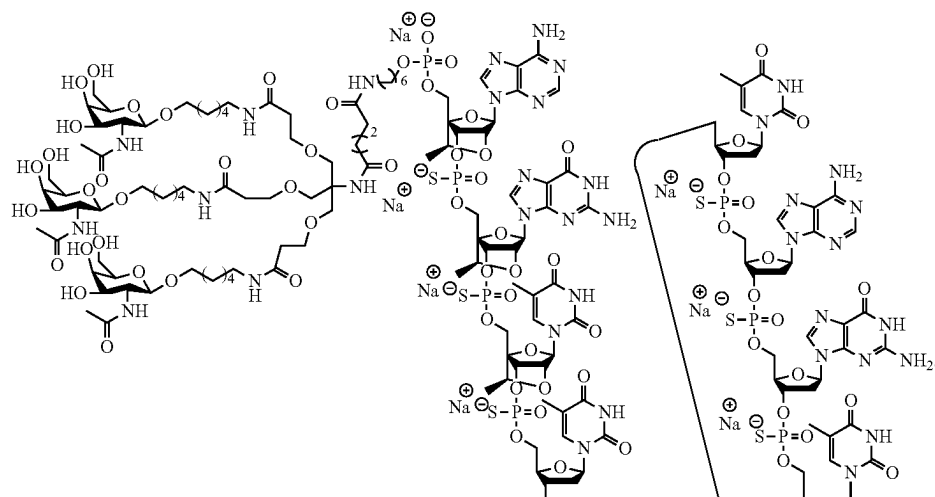
or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

61

In certain embodiments, a compound provided herein is according to the following chemical structure:

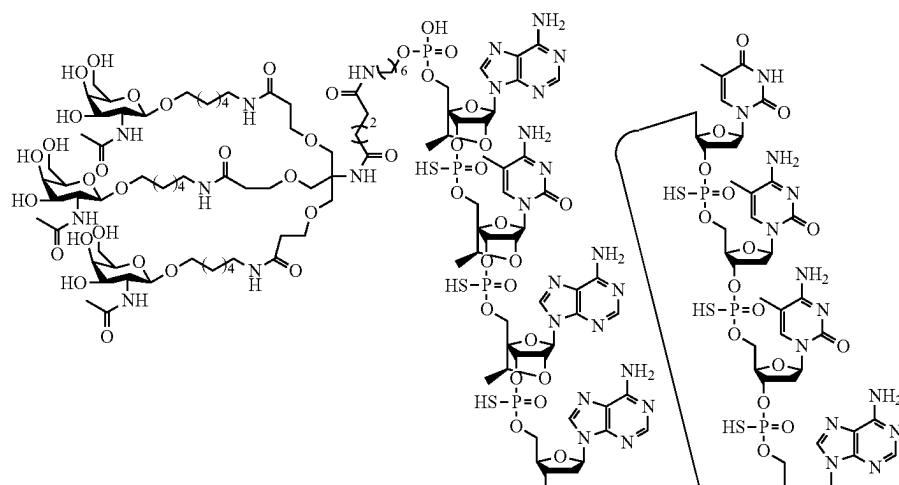
62

(SEQ ID NO: 1249)



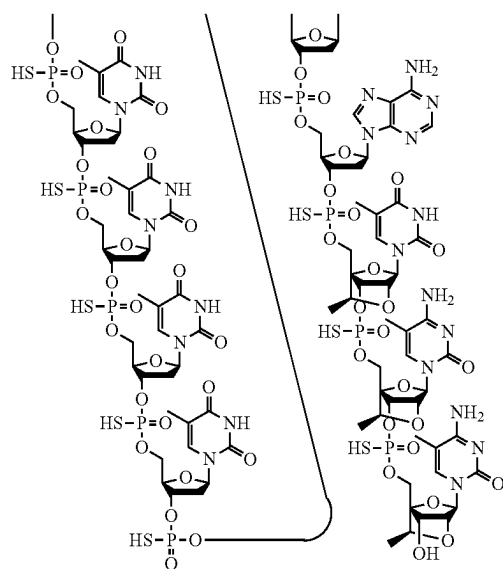
63

In certain embodiments, a compound provided herein is according to the following chemical structure:



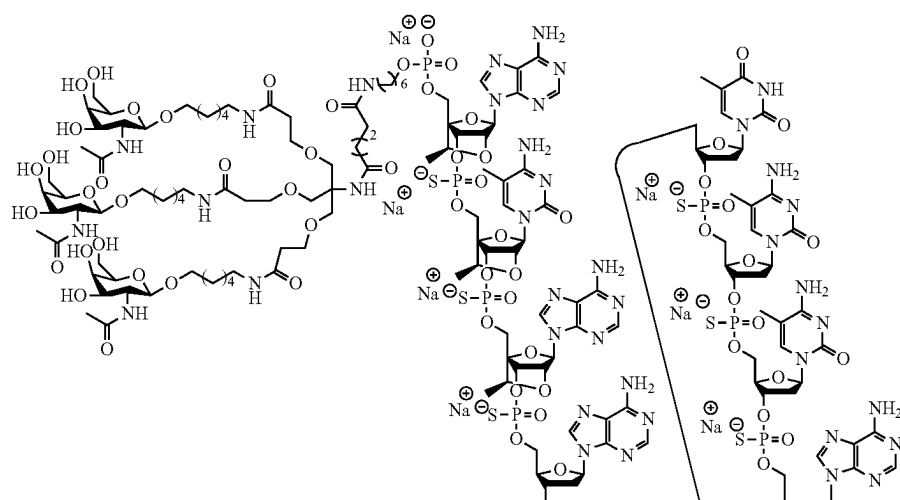
64

(SEQ ID NO: 513)

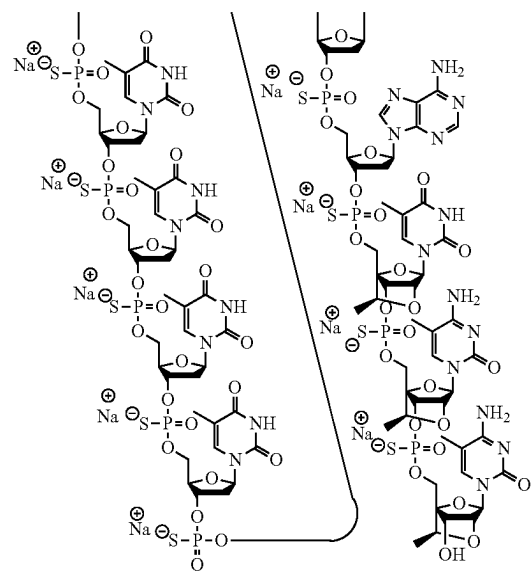


or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

In certain embodiments, a compound provided herein is according to the following chemical structure:



(SEQ ID NO: 513)

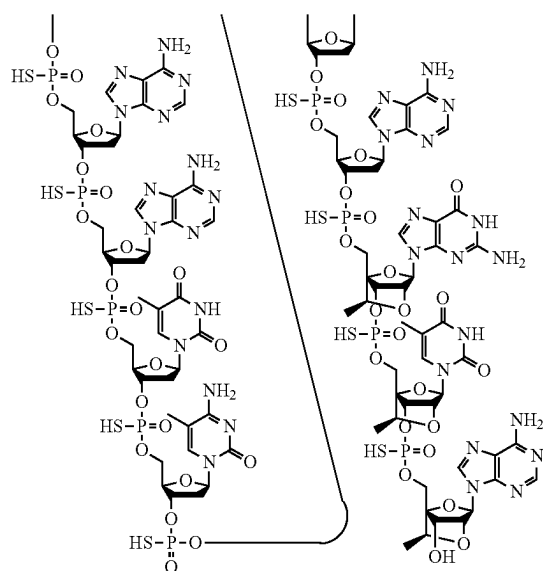
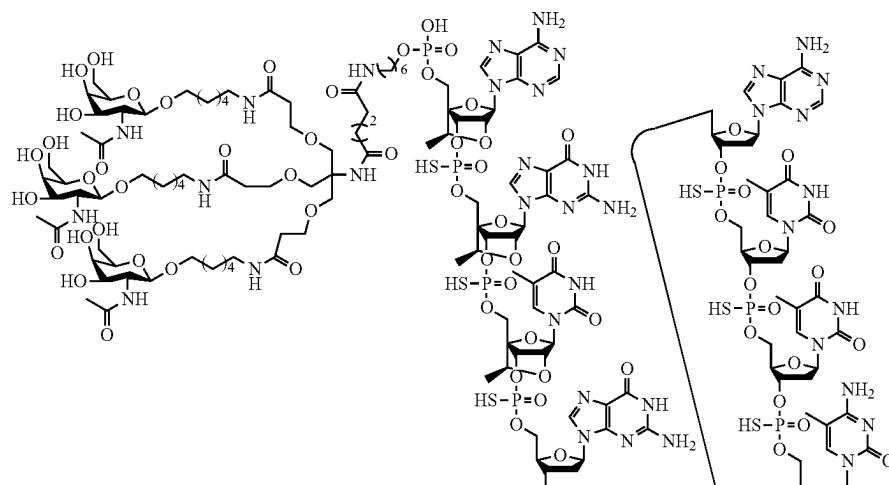


67

In certain embodiments, a compound provided herein is according to the following chemical structure:

68

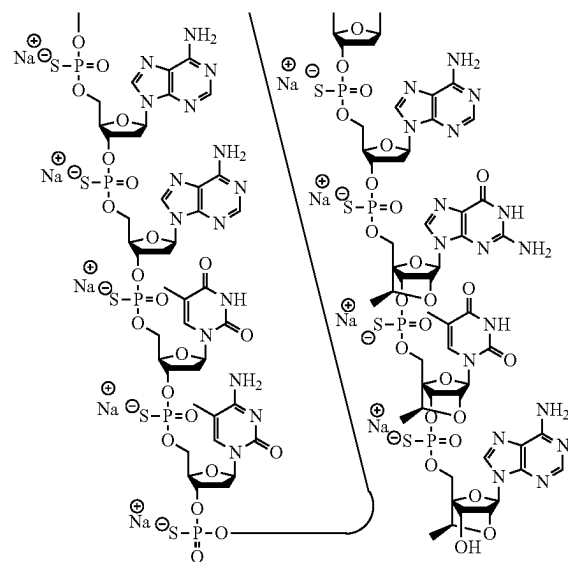
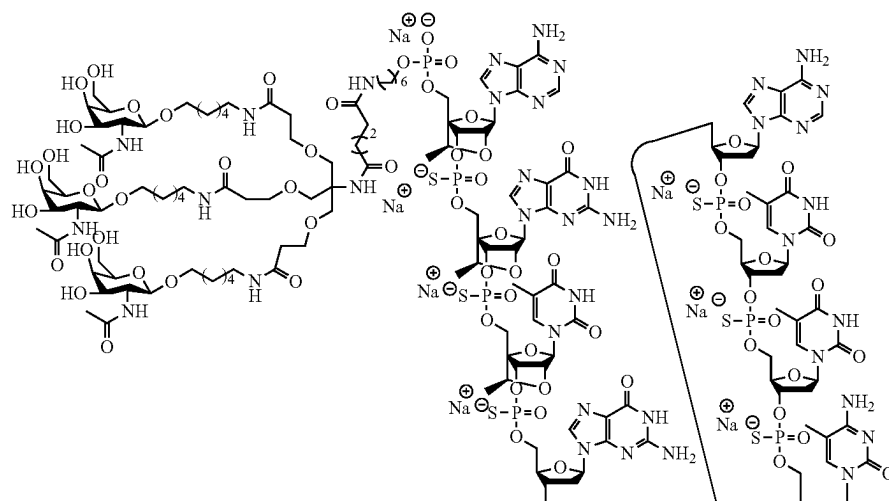
(SEQ ID NO: 1449)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

In certain embodiments, a compound provided herein is according to the following chemical structure:

(SEQ ID NO: 1449)

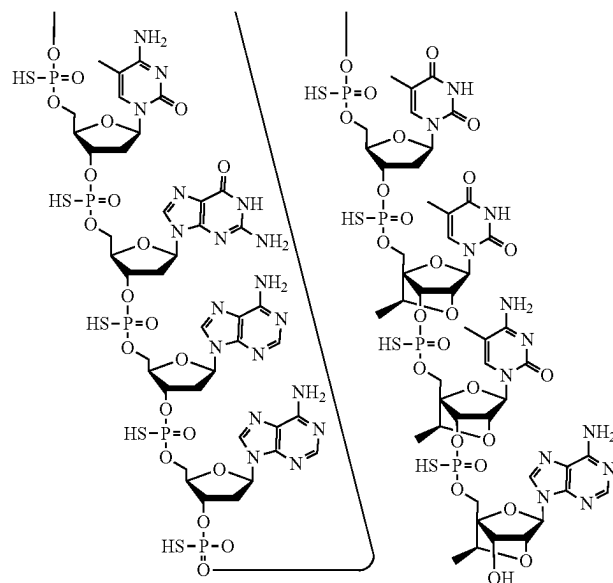
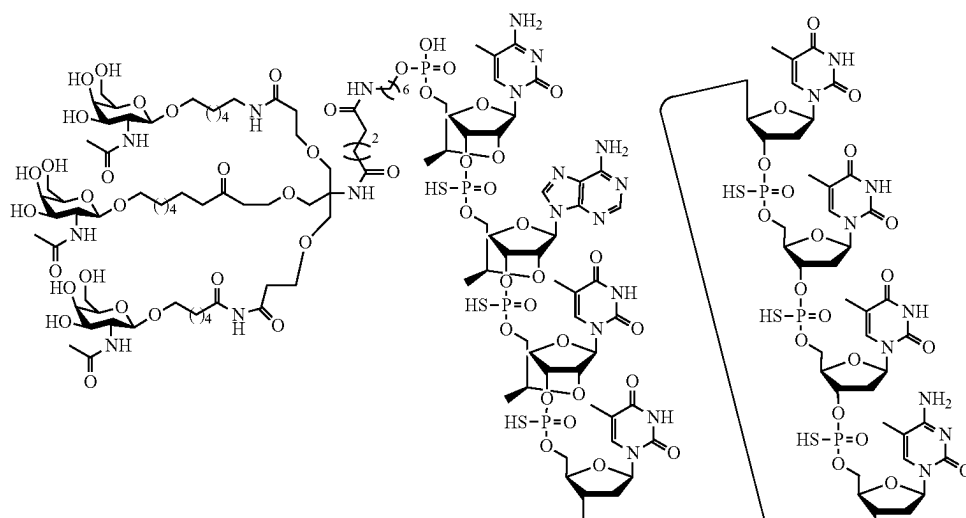


71

In certain embodiments, a compound provided herein is according to the following chemical structure:

72

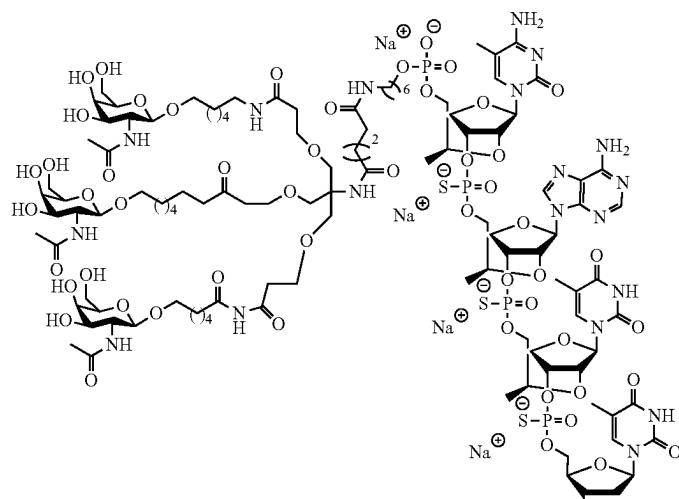
(SEQ ID NO: 1448)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

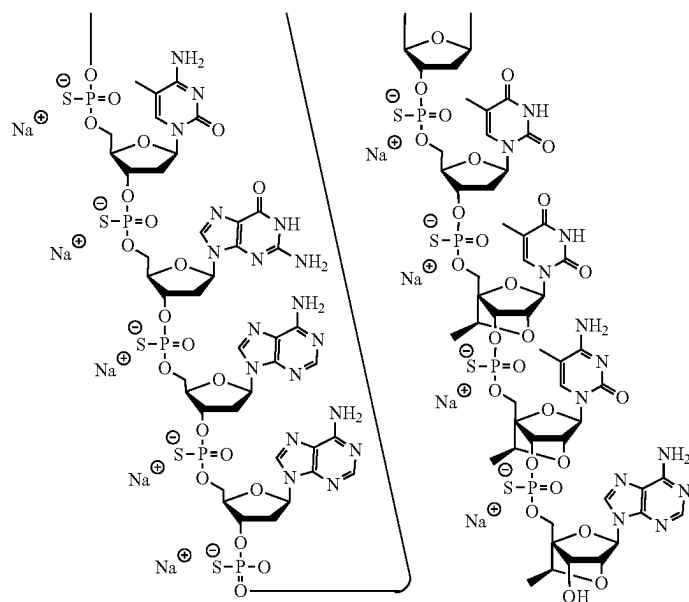
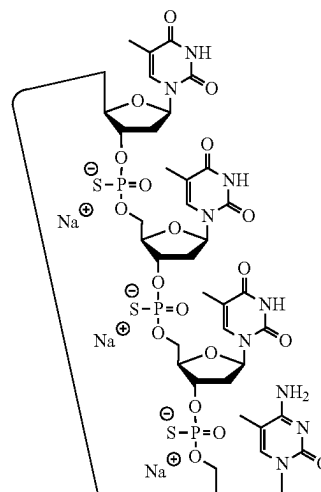
73

In certain embodiments, a compound provided herein is according to the following chemical structure:



74

(SEQ ID NO: 1448)

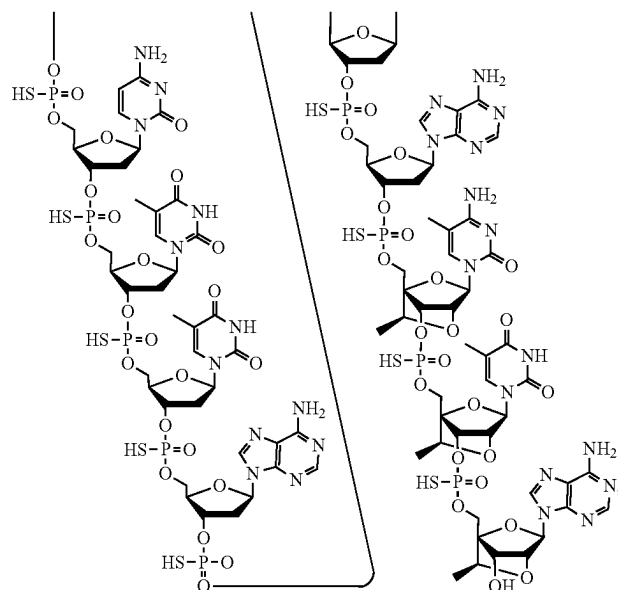
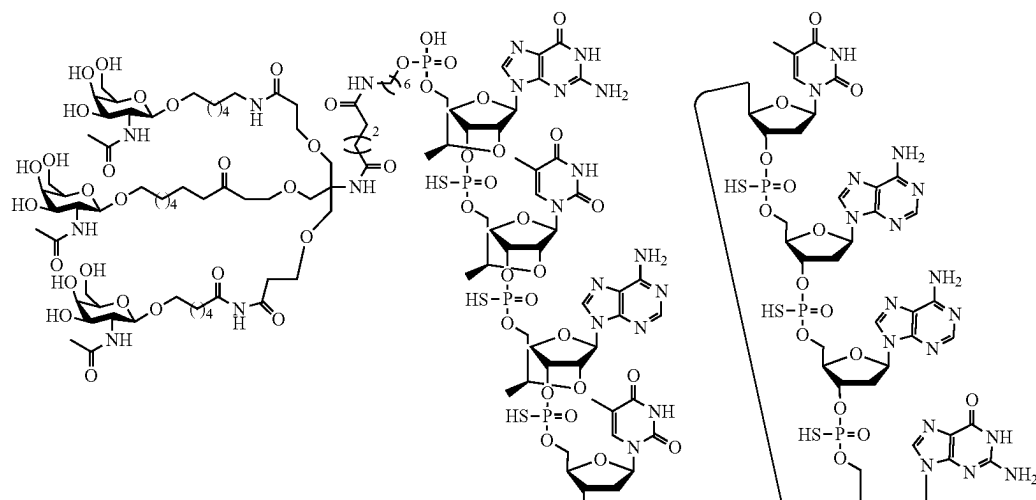


75

In certain embodiments, a compound provided herein is according to the following chemical structure:

76

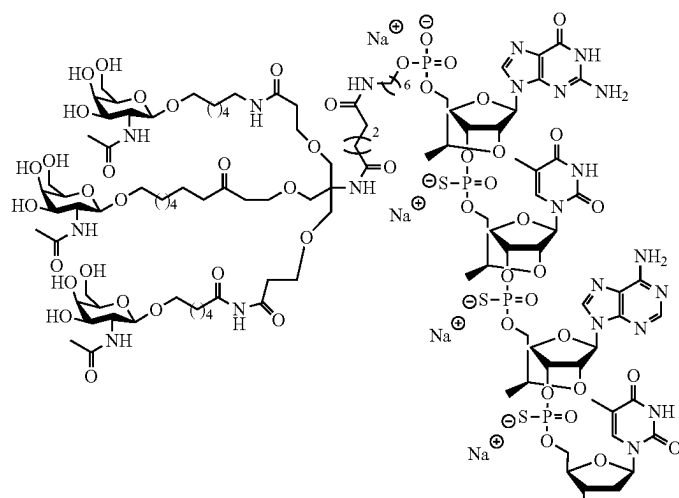
(SEQ ID NO: 1595)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

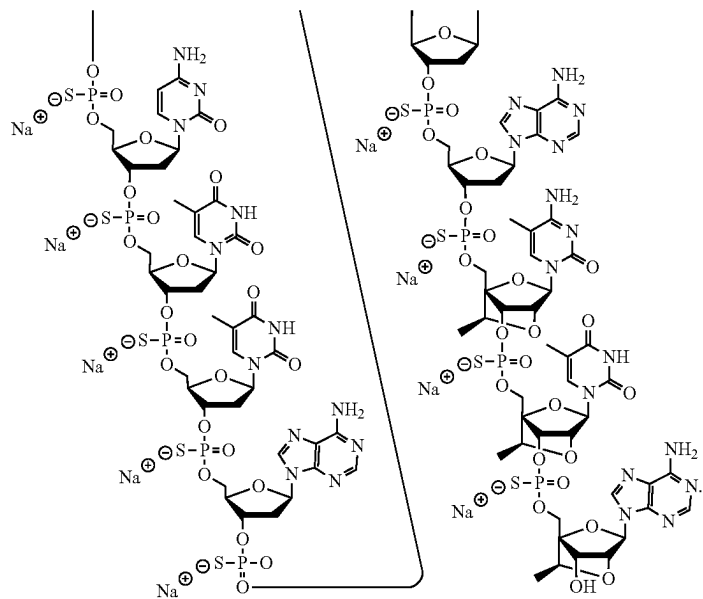
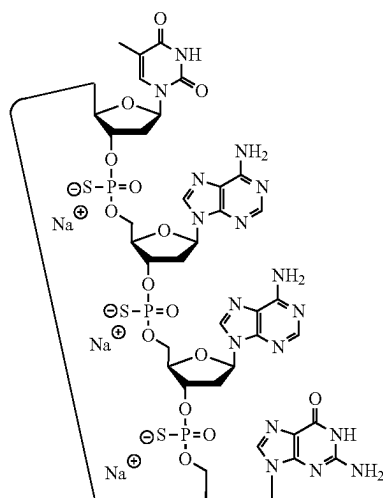
77

In certain embodiments, a compound provided herein is according to the following chemical structure:



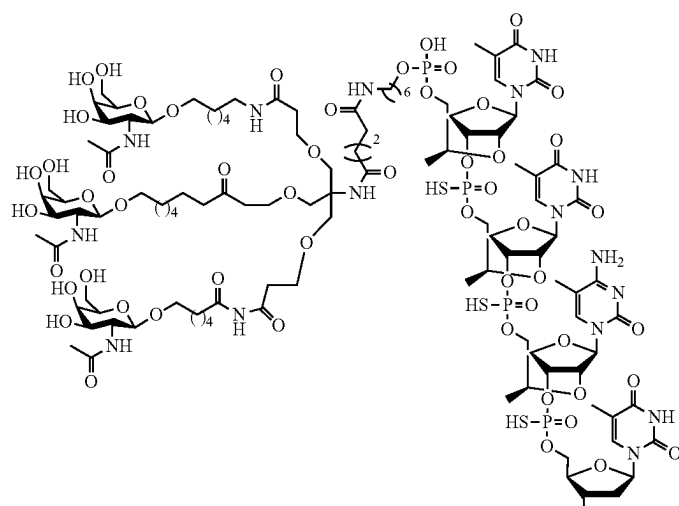
78

(SEQ ID NO: 1595)



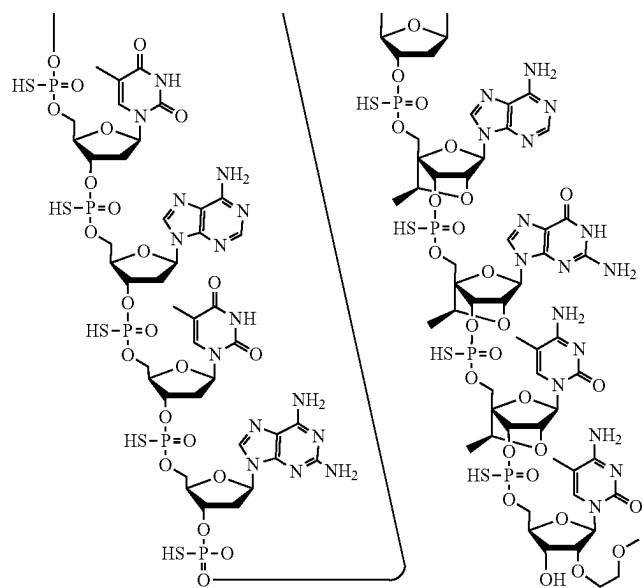
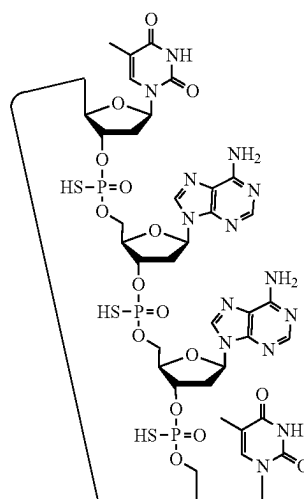
79

In certain embodiments, a compound provided herein is according to the following chemical structure:



80

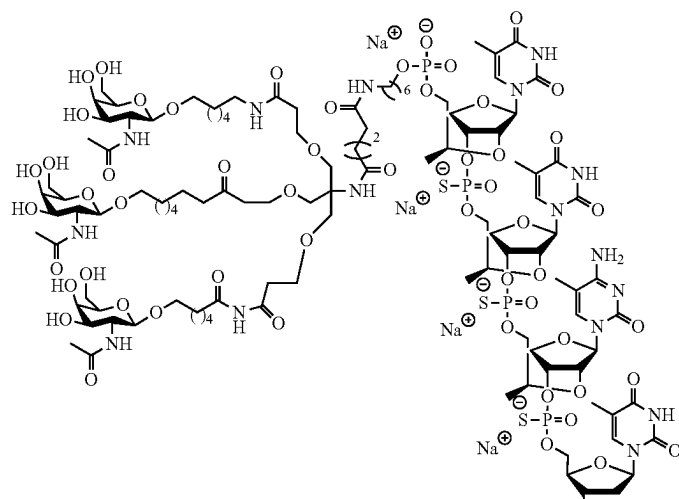
(SEQ ID NO: 819)



or a salt thereof. In certain embodiments, the compound is
a sodium salt. In certain embodiments, the compound is a
potassium salt.

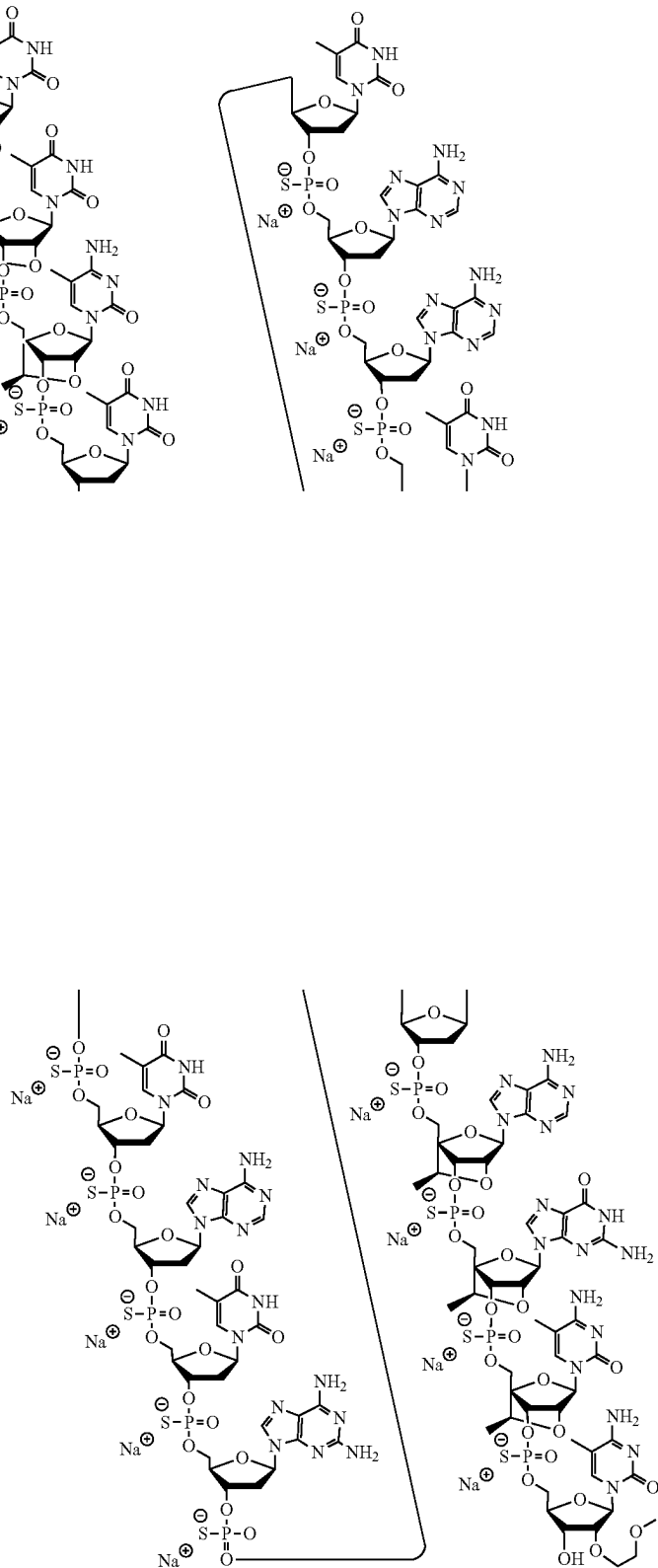
81

In certain embodiments, a compound provided herein is according to the following chemical structure:



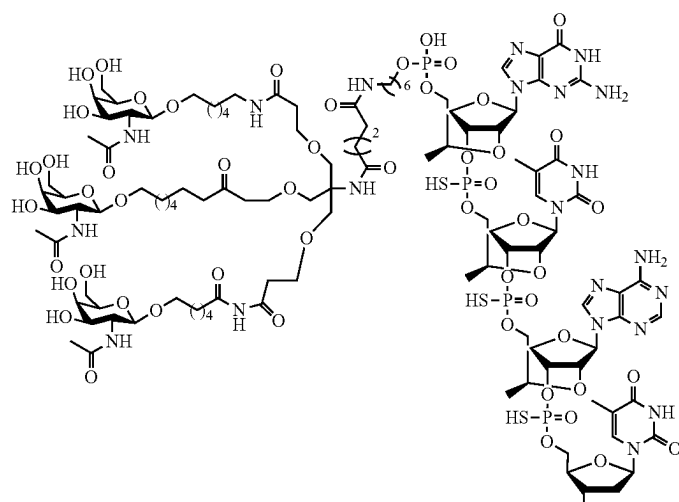
82

(SEQ ID NO: 819)



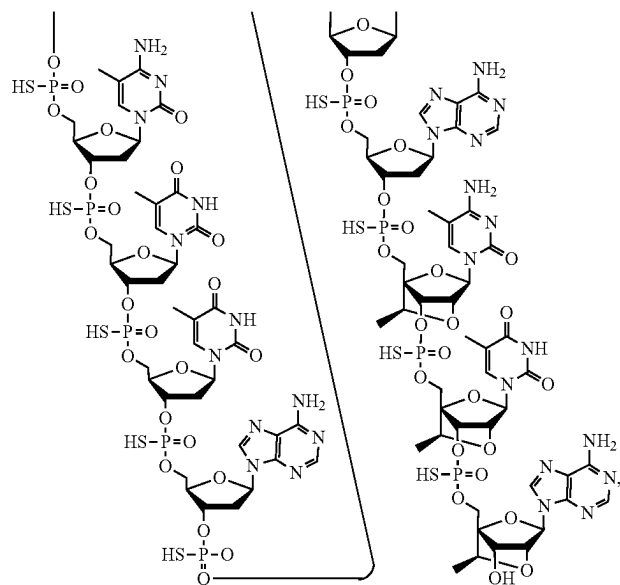
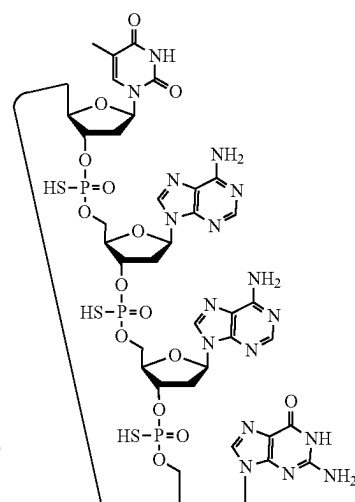
83

In certain embodiments, a compound provided herein is according to the following chemical structure:



84

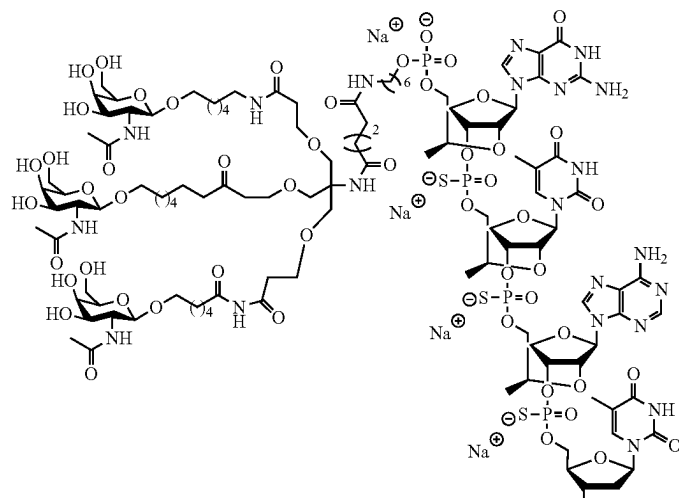
(SEQ ID NO: 1595)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

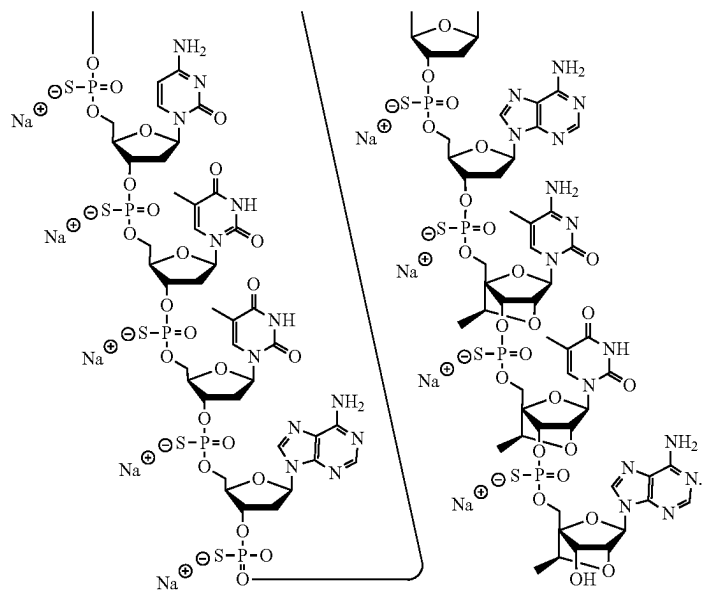
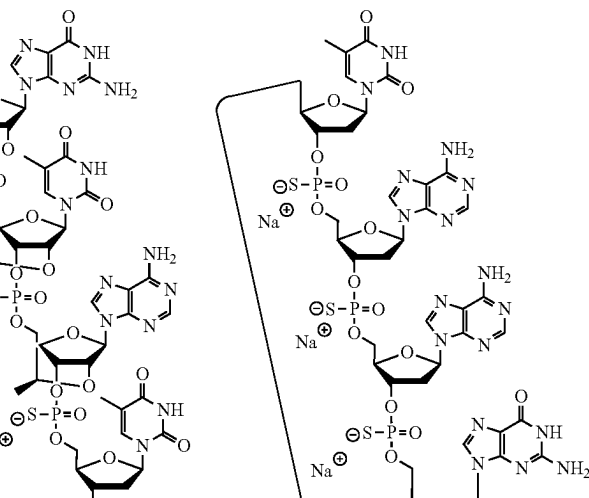
85

In certain embodiments, a compound provided herein is according to the following chemical structure:

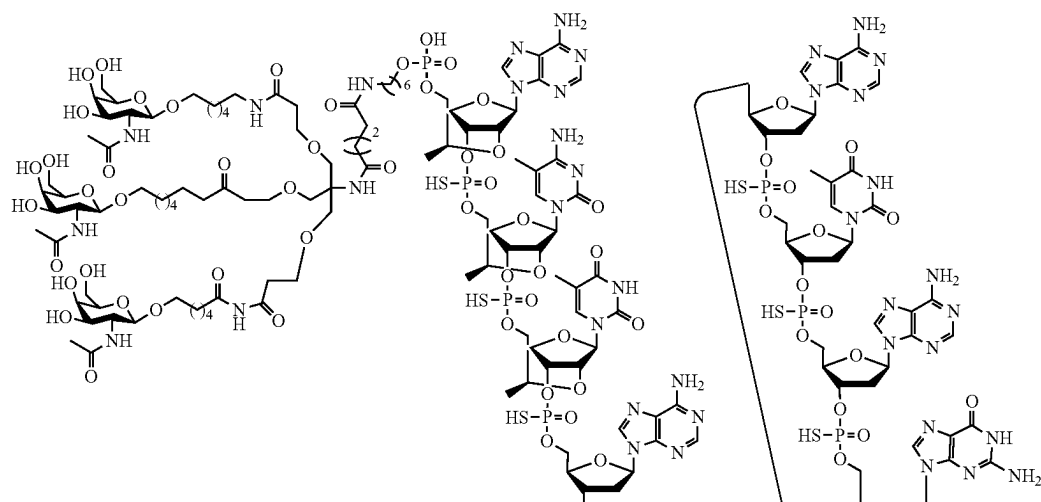


86

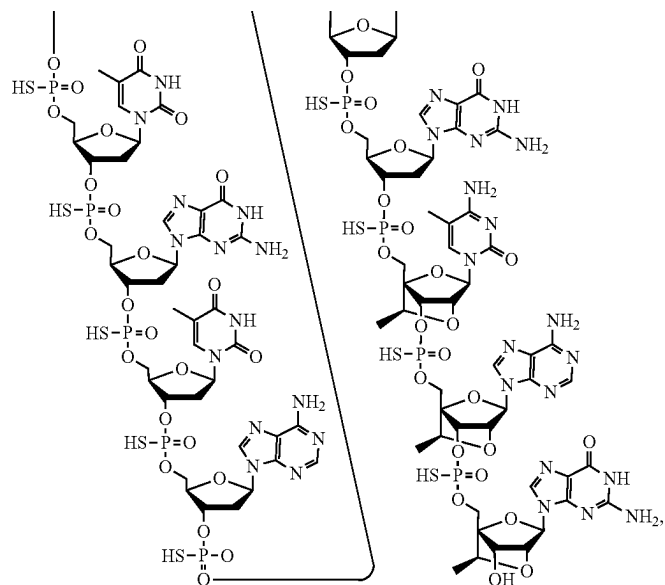
(SEQ ID NO: 1595)



In certain embodiments, a compound provided herein is according to the following chemical structure:



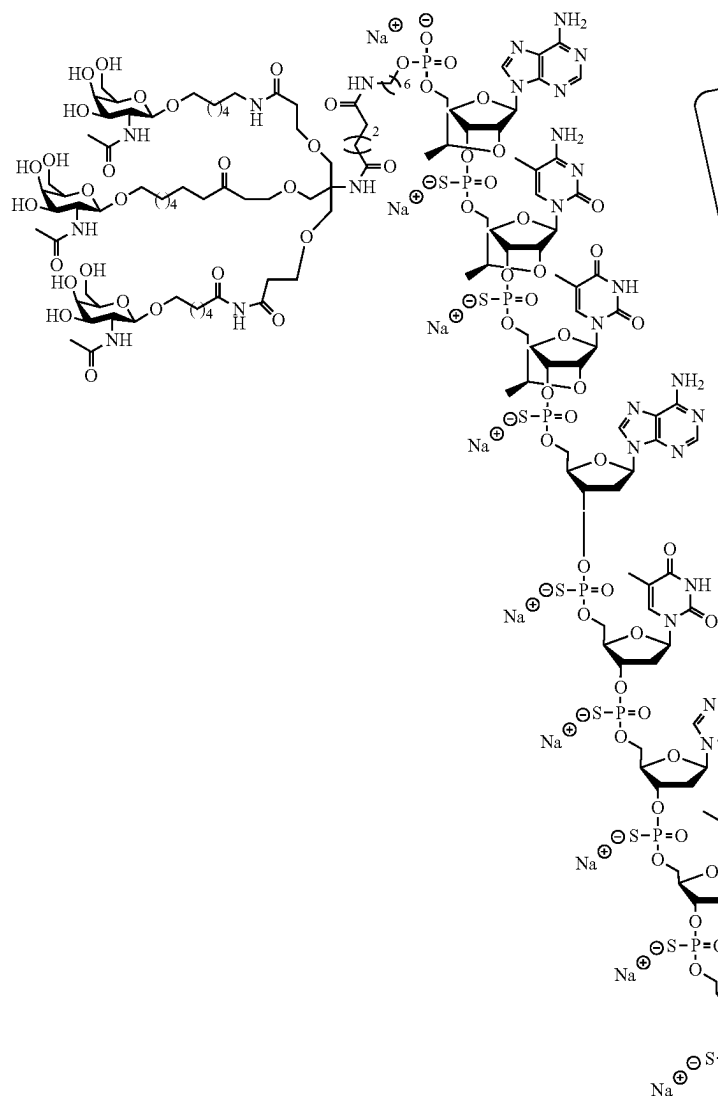
(SEQ ID NO: 208)



or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

89

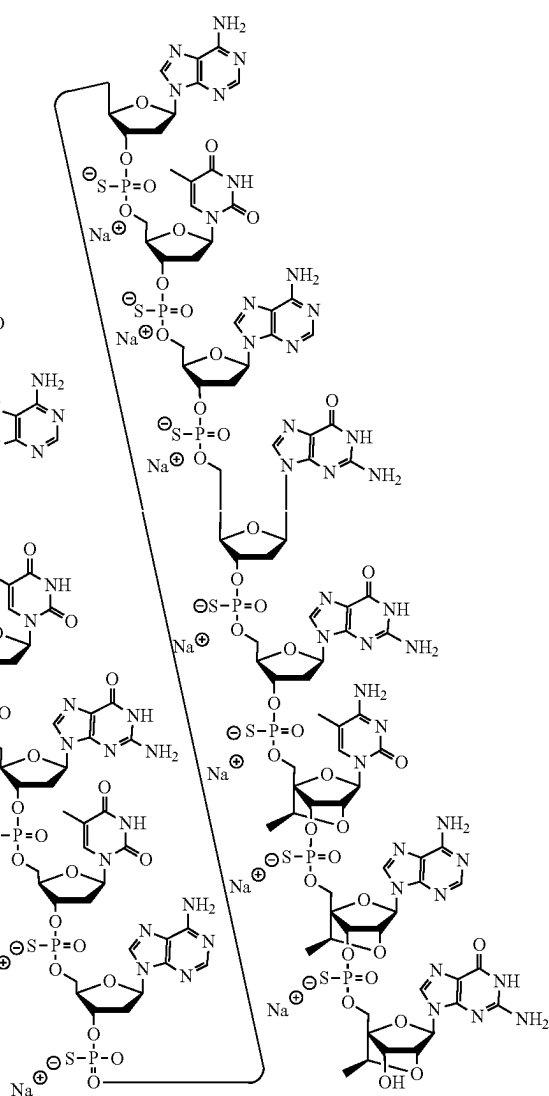
In certain embodiments, a compound provided herein is according to the following chemical structure:



90

acid of a compound followed by the term “or a salt thereof” expressly includes all such forms that may be fully or

(SEQ ID NO: 208)



Under certain conditions, certain compounds disclosed herein act as acids. Although such compounds may be drawn or described in protonated (free acid) form, or ionized and in association with a cation (salt) form, aqueous solutions of such compounds exist in equilibrium among such forms. For example, a phosphate linkage of an oligonucleotide in aqueous solution exists in equilibrium among free acid, anion and salt forms. Unless otherwise indicated, compounds described herein are intended to include all such forms. Moreover, certain oligonucleotides have several such linkages, each of which is in equilibrium. Thus, oligonucleotides in solution exist in an ensemble of forms at multiple positions all at equilibrium. The term “oligonucleotide” is intended to include all such forms. Drawn structures necessarily depict a single form. Nevertheless, unless otherwise indicated, such drawings are likewise intended to include corresponding forms. Herein, a structure depicting the free

partially protonated/de-protonated/in association with a cation. In certain instances, one or more specific cation is identified.

In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with sodium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with potassium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in PBS. In certain embodiments, modified oligonucleotides or oligomeric compounds are in water. In certain such embodiments, the pH of the solution is adjusted with NaOH and/or HCl to achieve a desired pH. In certain embodiments, the compounds or compositions as described herein are active by virtue of having at least one of an in vitro IC_{50} of less than 2 μ M, less than 1.5 μ M, less than 1 M, less than 0.9 μ M, less than 0.8 μ M, less than 0.7 M, less than 0.6 μ M, less than 0.5 M, less than 0.4 μ M, less

than 0.3 μ M, less than 0.2 μ M, less than 0.1 μ M, less than 0.05 μ M, less than 0.04 μ M, less than 0.03 μ M, less than 0.02 μ M, or less than 0.01 μ M.

In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having at least one of an increase in alanine transaminase (ALT) or aspartate transaminase (AST) value of no more than 4 fold, 3 fold, or 2 fold over control animals, or an increase in liver, spleen, or kidney weight of no more than 30%, 20%, 15%, 12%, 10%, 5%, or 2% compared to control animals. In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having no increase of ALT or AST over control animals. In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having no increase in liver, spleen, or kidney weight over control animals.

Certain embodiments provide a composition comprising the compound of any of the aforementioned embodiments or any pharmaceutically acceptable salt thereof and at least one of a pharmaceutically acceptable carrier or diluent. In certain embodiments, the composition has a viscosity less than about 40 centipoise (cP), less than about 30 centipoise (cP), less than about 20 centipoise (cP), less than about 15 centipoise (cP), or less than about 10 centipoise (cP). In certain embodiments, the composition having any of the aforementioned viscosities comprises a compound provided herein at a concentration of about 100 mg/mL, about 125 mg/mL, about 150 mg/mL, about 175 mg/mL, about 200 mg/mL, about 225 mg/mL, about 250 mg/mL, about 275 mg/mL, or about 300 mg/mL. In certain embodiments, the composition having any of the aforementioned viscosities and/or compound concentrations has a temperature of room temperature, or about 20° C., about 21° C., about 22° C., about 23° C., about 24° C., about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., or about 30° C.

Non-Limiting Numbered Embodiments Include:

Embodiment 1: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, or at least 12 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, or at least 12 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896.

Embodiment 2: Certain embodiments provide a compound comprising modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. Certain embodiments provide a compound comprising modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896.

Embodiment 3: Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 8-2896.

Embodiment 4: Certain embodiments provide a compound comprising modified oligonucleotide consisting of 8 to 80 linked nucleosides, or consisting of 12 to 30 linked nucleosides, wherein the modified oligonucle-

otide has a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-

14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 85/c, at least 90/c, at least 95%, or 100% complementary to SEQ ID NO: 2.

Embodiment 5: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence complementary within nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498,

7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032,

20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of SEQ ID NO: 2, and wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2.

Embodiment 6: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases 100% complementary to an equal length portion of nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 1.

Embodiment 7: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence complementary within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of SEQ ID NO: 1, and wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 1.

Embodiment 8: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16 contiguous nucleobases 100% complementary to an equal length portion of nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of SEQ ID NO: 2, and wherein the nucleobase

sequence of the modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2.

Embodiment 9: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819.

Embodiment 10: Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NO: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819.

Embodiment 11: Certain embodiments provide the compound of embodiments 1-6, wherein the oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to SEQ ID NO: 2 over the entire length of the oligonucleotide.

Embodiment 12: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises at least one modification selected from at least one modified internucleoside linkage, at least one modified nucleoside comprising a modified sugar moiety, and at least one modified nucleoside comprising a modified nucleobase.

Embodiment 13: Certain embodiments provide the compound of embodiment 12, wherein the modified internucleoside linkage is a phosphorothioate internucleoside linkage.

Embodiment 14: Certain embodiments provide the compound of embodiments 12 or 13, wherein at least one nucleoside of the modified oligonucleotide is a 2'-substituted nucleoside or a bicyclic nucleoside.

Embodiment 15: Certain embodiments provide the compound of embodiment 14, wherein the bicyclic sugar moiety of the bicyclic nucleoside is selected from the group consisting of: 4'-(CH₂)—O-2' (LNA); 4'-(CH₂)₂—O-2' (ENA); and 4'-CH(CH₃)—O-2' (cEt).

Embodiment 16: Certain embodiments provide the compound of any one of embodiments 12-15, wherein the modified nucleoside is 2'-O-methoxyethyl nucleoside.

Embodiment 17: Certain embodiments provide the compound of any one of embodiments 12-16, wherein the modified nucleobase is a 5-methylcytosine.

Embodiment 18: Certain embodiments provide the compound of any one of embodiments 1-17, wherein the modified oligonucleotide has:

a gap segment consisting of linked 2'-deoxynucleosides;

a 5' wing segment consisting of linked nucleosides; and

a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

Embodiment 19: Certain embodiments provide the compound of any one of embodiment 1-18, wherein the

97

modified sugar moiety of each nucleoside of each wing segment is selected from 2'-O-Me, 2'-MOE, and cEt sugar moieties.

Embodiment 20: Certain embodiments provide the compound of any of one embodiment 1-19, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of three linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt nucleoside; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 21: Certain embodiments provide the compound of any of one embodiment 1-20, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of three linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt nucleoside; wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 22: Certain embodiments provide the compound of any of the preceding embodiments, wherein the modified oligonucleotide comprises:

a gap segment consisting of nine linked 2'-deoxynucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt nucleoside; wherein the sugar moieties of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 23: Certain embodiments provide the compound of any of the preceding embodiments, wherein the modified oligonucleotide comprises:

a gap segment consisting of nine linked 2'-deoxynucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each

98

nucleoside of the 5' wing segment comprises a cEt nucleoside; wherein each sugar moiety of the nucleosides of the 3' wing segment are selected from cEt and 2'-MOE sugar moieties; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 24: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked nucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt nucleoside; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 25: Embodiment 24: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked nucleosides;

a 5' wing segment consisting of three linked nucleosides; and

a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the nucleosides of the gap segment are selected from deoxynucleosides, cEt nucleosides and 2'-MOE nucleosides, wherein each nucleoside of each of the wing segments comprises a cEt nucleoside; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 26: Certain embodiments provide the compound of any one of embodiments 1-25, wherein the compound is single-stranded.

Embodiment 27: Certain embodiments provide the compound of any one of embodiments 1-25, wherein the modified oligonucleotide is hybridized to a second oligonucleotide to form a double-stranded antisense compound.

Embodiment 28: Certain embodiments provide the compound of any one of embodiments 1-27, wherein at least one of the nucleotides of the modified oligonucleotide is a ribonucleotide.

Embodiment 29: Certain embodiments provide the compound of any one of embodiments 1-27, wherein at least one of the nucleotides of the modified oligonucleotide is a deoxyribonucleotide.

Embodiment 30: Certain embodiments provide the compound of any one of embodiments 1-23, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides.

Embodiment 31: Certain embodiments provide the compound of any one of embodiments 1-30, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides.

Embodiment 32: Certain embodiments provide the compound of any one of embodiments 1-30, wherein the modified oligonucleotide consists of 16 to 30 linked nucleosides.

Embodiment 33: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence selected from any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819, wherein the modified oligonucleotide comprises:

- a gap segment consisting of linked deoxynucleosides;
- a 5' wing segment consisting of linked nucleosides; and
- a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

Embodiment 34: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence selected from any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, and 1595, wherein the modified oligonucleotide comprises

- a gap segment consisting of ten linked deoxynucleosides;
- a 5' wing segment consisting of three linked nucleosides; and
- a 3' wing segment consisting of three linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein the 5' wing segment and the 3' wing segment comprise cEt sugars; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

Embodiment 35: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence consisting of SEQ ID NO: 819, wherein the modified oligonucleotide comprises:

- a gap segment consisting of nine linked 2'-deoxynucleosides;
- a 5' wing segment consisting of three linked nucleosides; and
- a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each

nucleoside of the 5' wing segment comprises a cEt sugar; wherein the sugar residues of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 36: Certain embodiments provide a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence consisting of SEQ ID NO: 1595, wherein the modified oligonucleotide comprises:

- a gap segment consisting of ten linked nucleosides;

- a 5' wing segment consisting of three linked nucleosides; and

- a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine.

Embodiment 37: Certain embodiments provide the compound of any preceding embodiment, comprising a conjugate group.

Embodiment 38: Certain embodiments provide the compound of embodiment 37, wherein the conjugate group comprises a GalNAc cluster comprising 1-3 GalNAc ligands.

Embodiment 39: Certain embodiments provide the compound of embodiment 37 or embodiment 38, wherein the conjugate group comprises a conjugate linker consisting of a single bond.

Embodiment 40: Certain embodiments provide the compound of any one of embodiments 37-39, wherein the conjugate group comprises a cleavable linker.

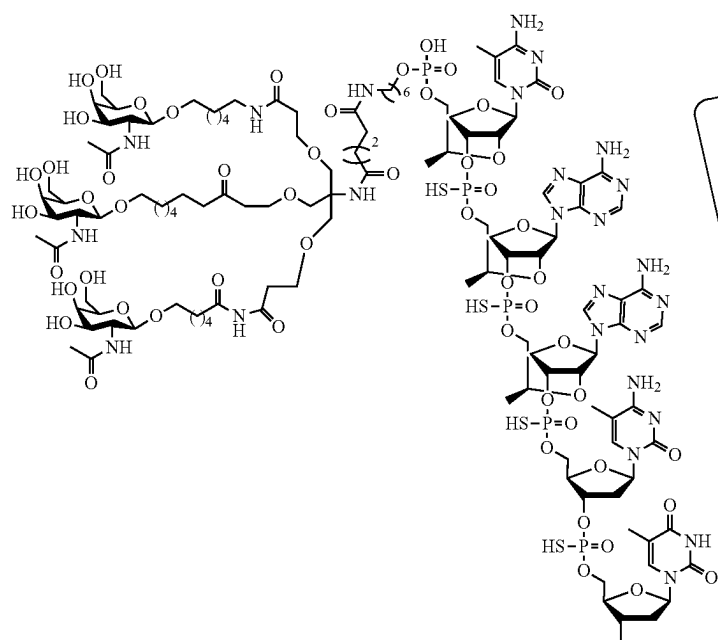
Embodiment 41: Certain embodiments provide the compound of any one of embodiments 37-40, wherein the conjugate group comprises a conjugate linker comprising 1-3 linker-nucleosides.

Embodiment 42: Certain embodiments provide the compound of any one of embodiments 37-41, wherein the conjugate group is attached to the modified oligonucleotide at the 5'-end of the modified oligonucleotide.

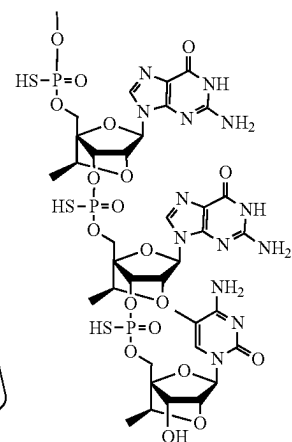
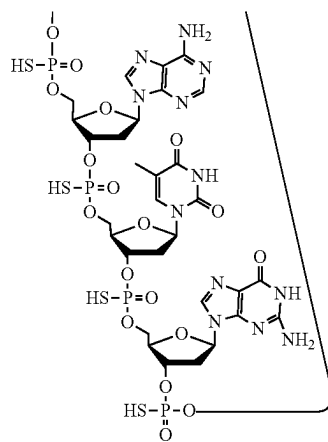
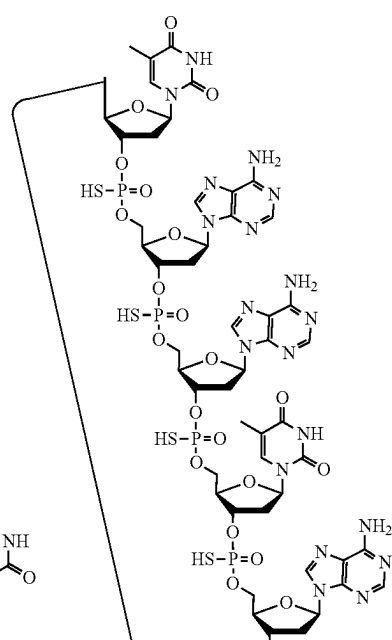
Embodiment 42: Certain embodiments provide the compound of any one of embodiments 37-41, wherein the conjugate group is attached to the modified oligonucleotide at the 3'-end of the modified oligonucleotide.

Embodiment 43: Certain embodiments provide the compound having the following chemical structure or salt thereof:

101

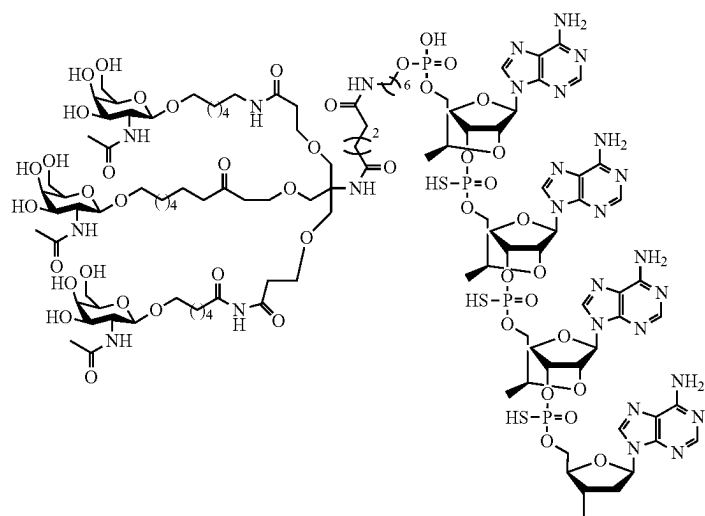


102

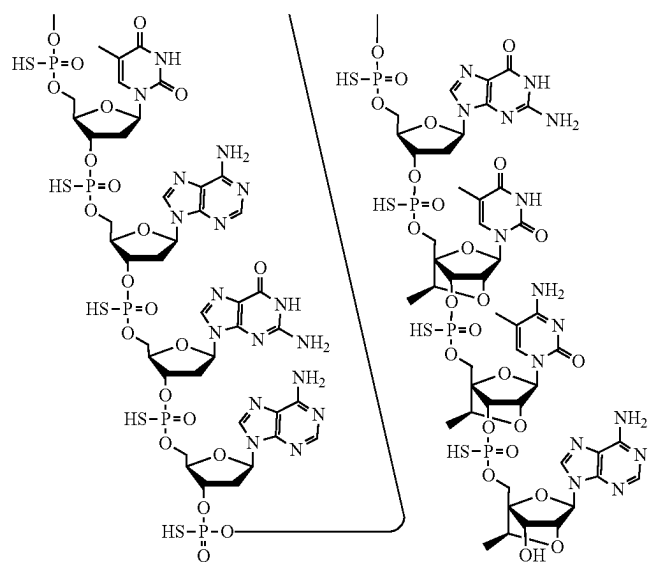
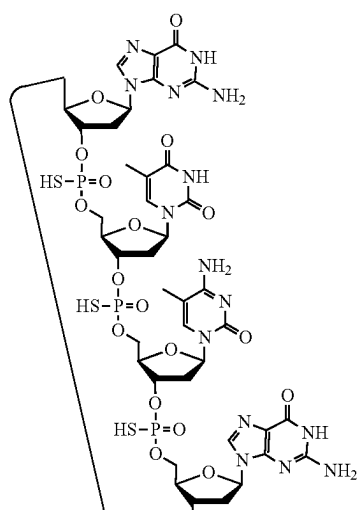


Embodiment 44: Certain embodiments provide the com- 65
 pound having the following chemical structure or salt
 thereof:

103

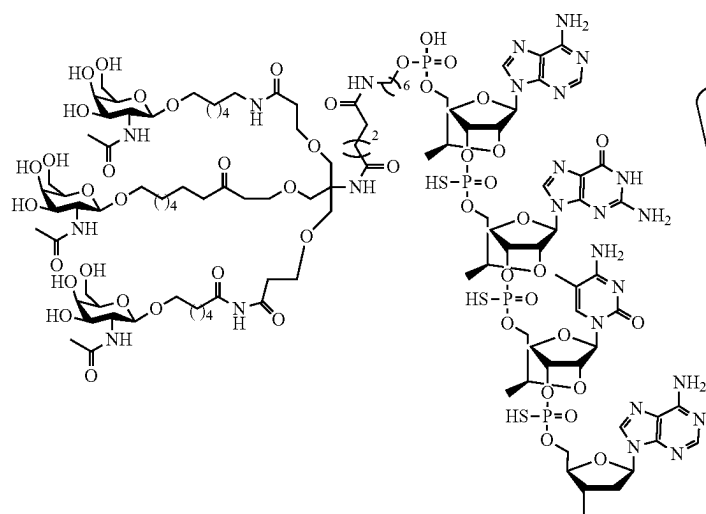


104

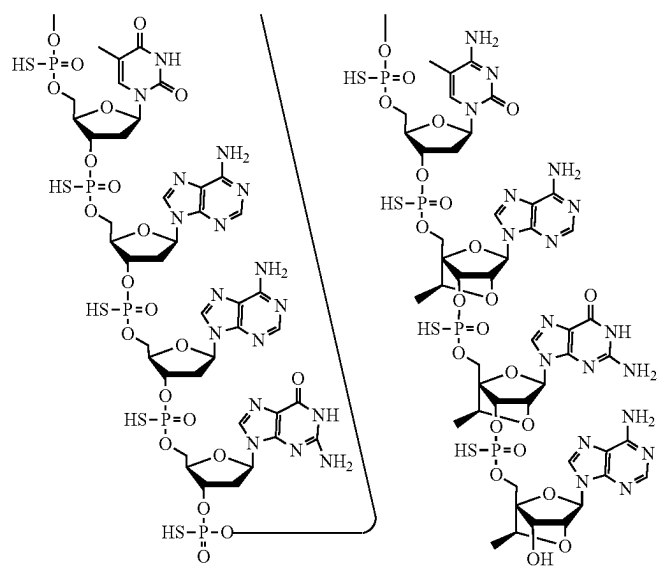
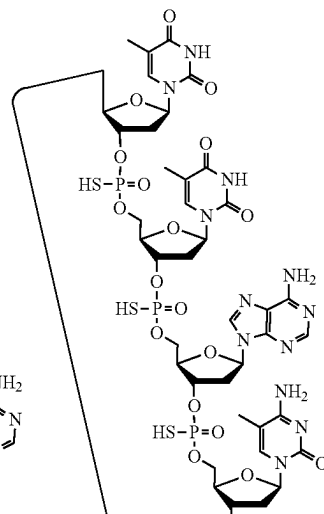


Embodiment 45: Certain embodiments provide the compound having the following chemical structure or salt thereof:

105

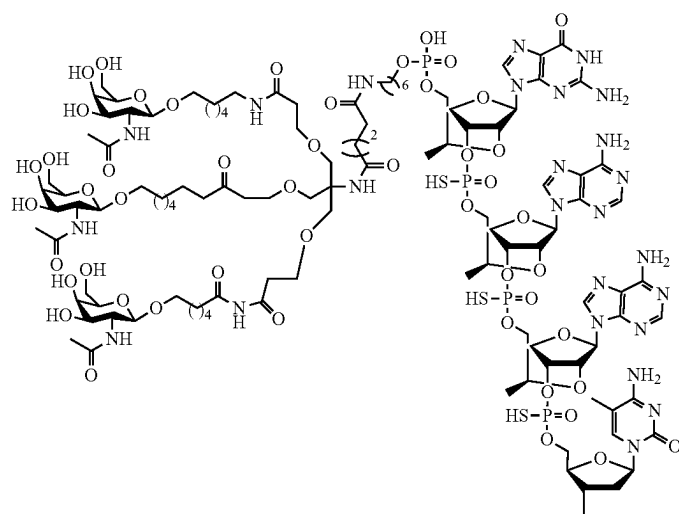


106

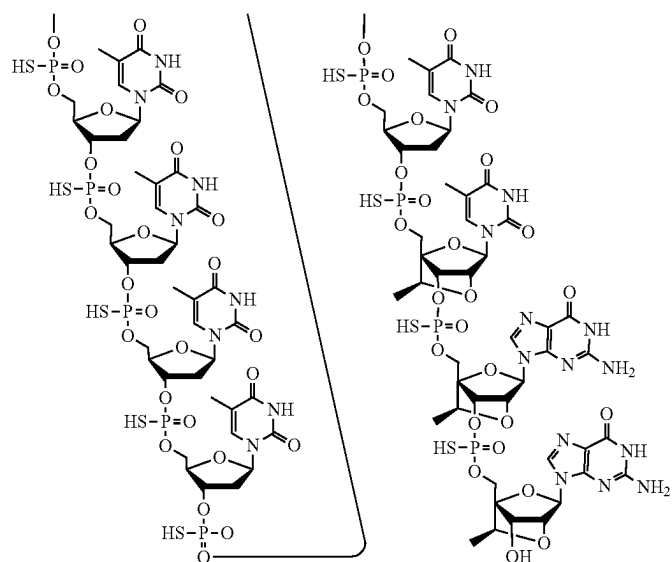
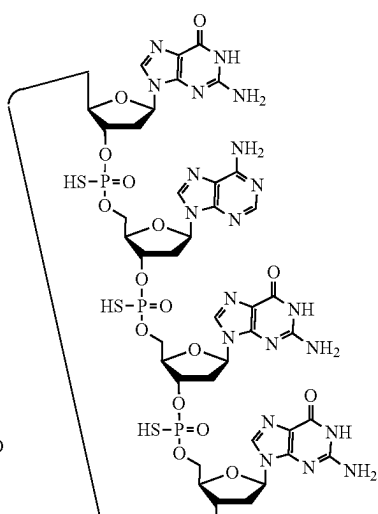


Embodiment 46: Certain embodiments provide the compound having the following chemical structure or salt thereof:

107

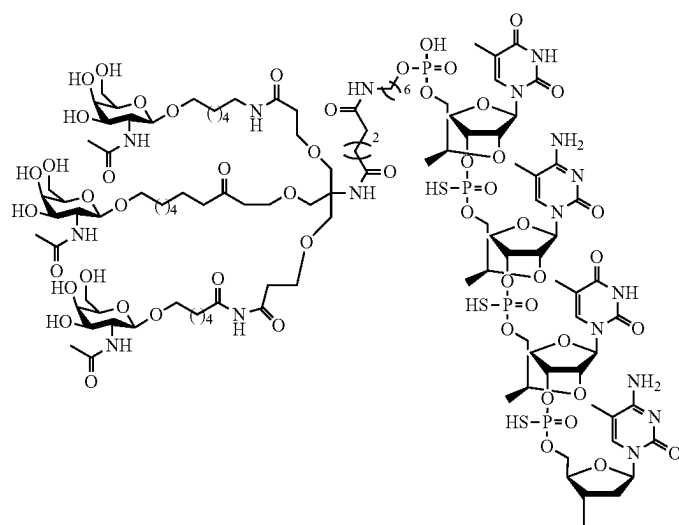


108

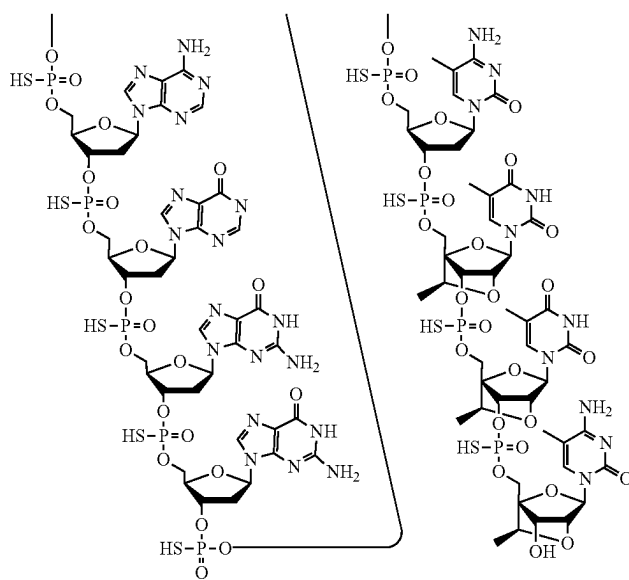
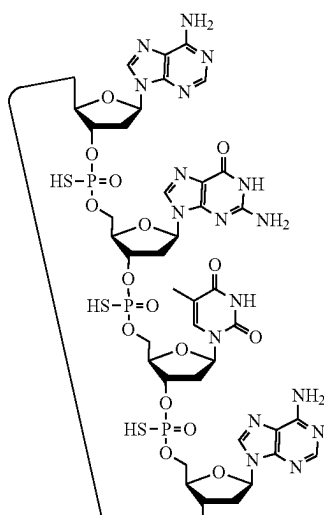


Embodiment 47: Certain embodiments provide the compound having the following chemical structure or salt thereof:

109

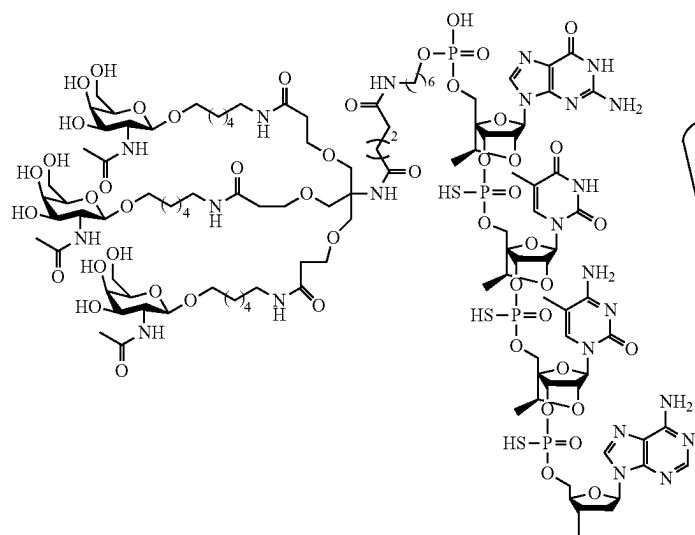


110

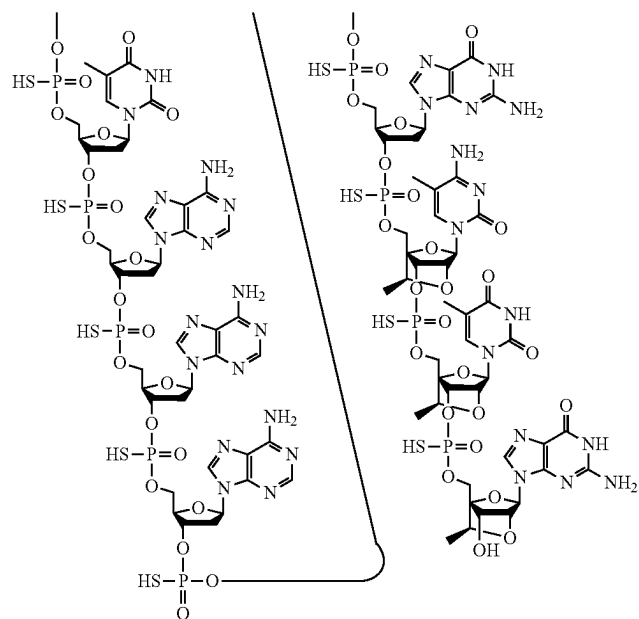
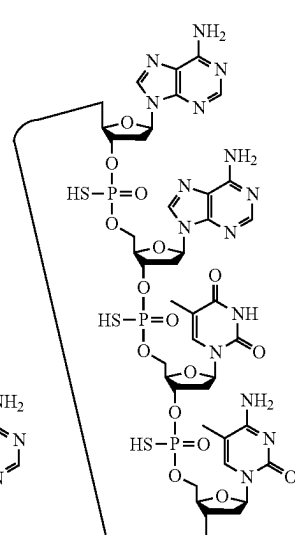


Embodiment 48: Certain embodiments provide the compound having the following chemical structure or salt thereof:

111

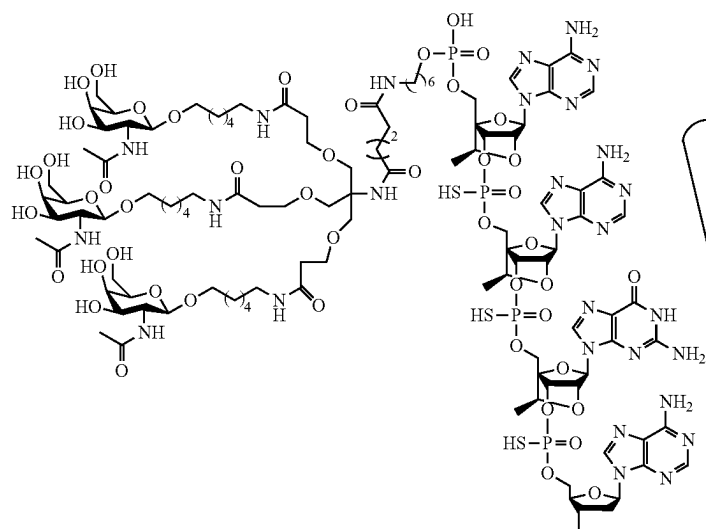


112

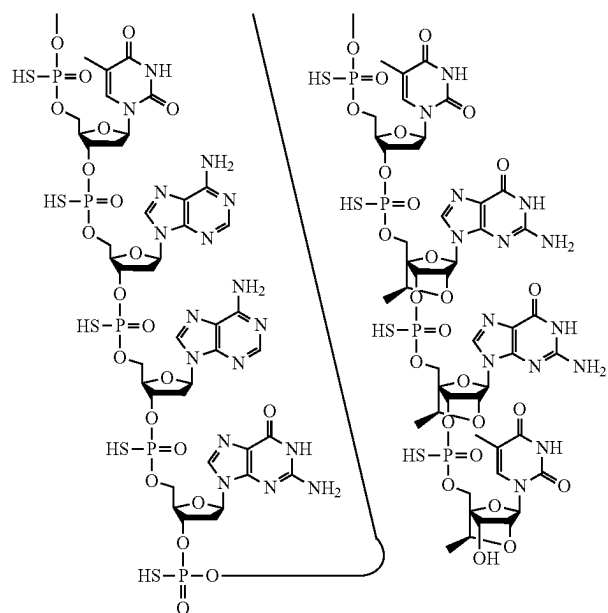
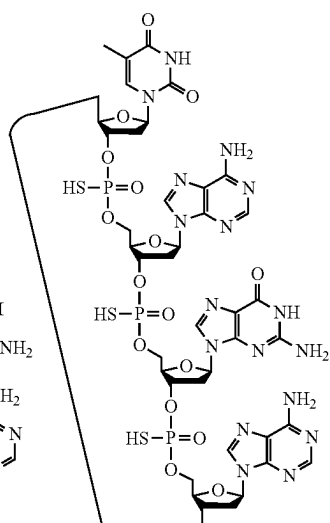


Embodiment 49: Certain embodiments provide the compound having the following chemical structure or salt thereof:

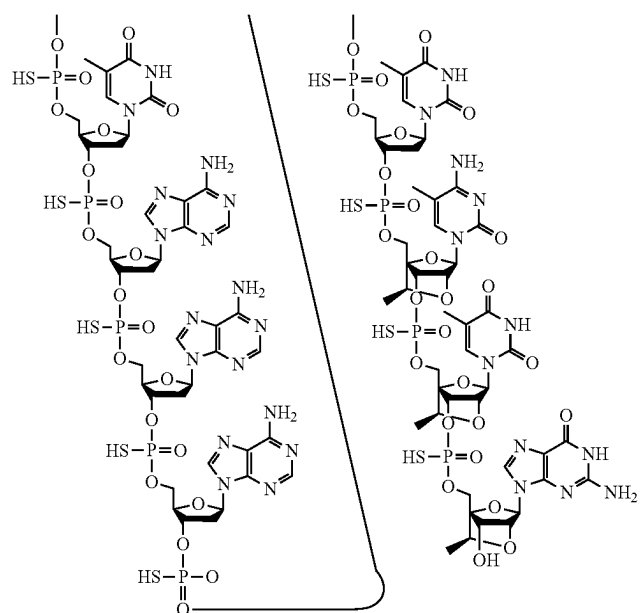
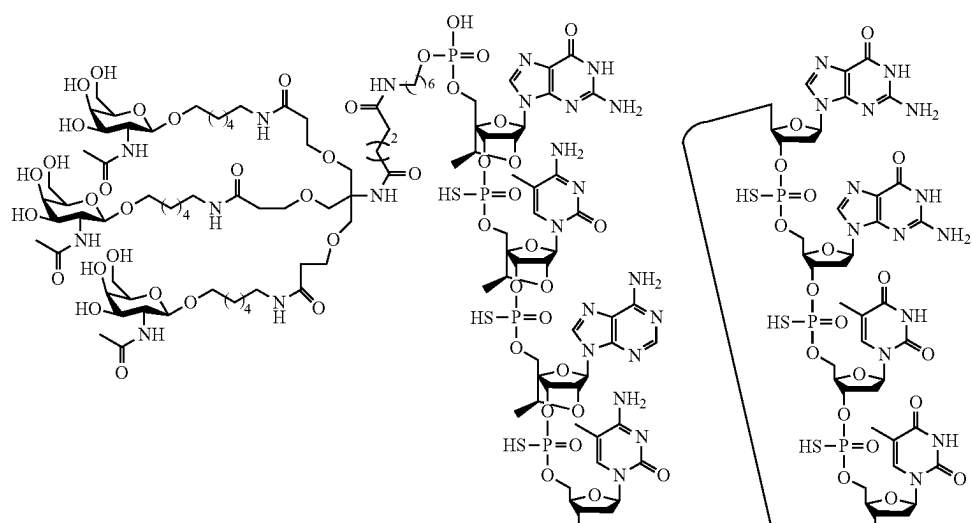
113



114

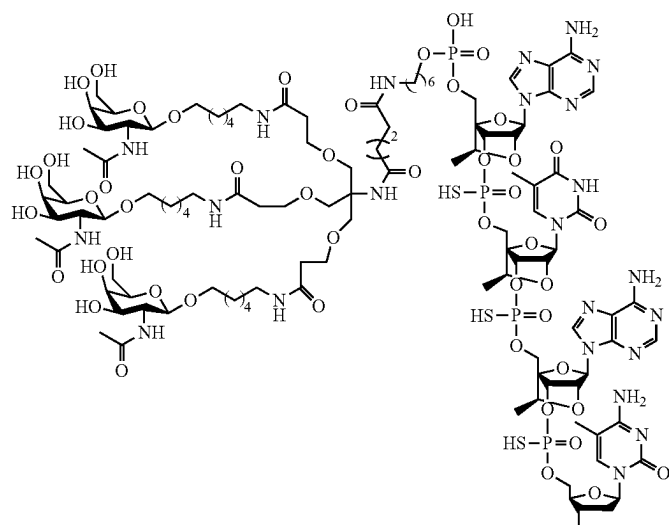


Embodiment 50: Certain embodiments provide the compound having the following chemical structure or salt thereof:

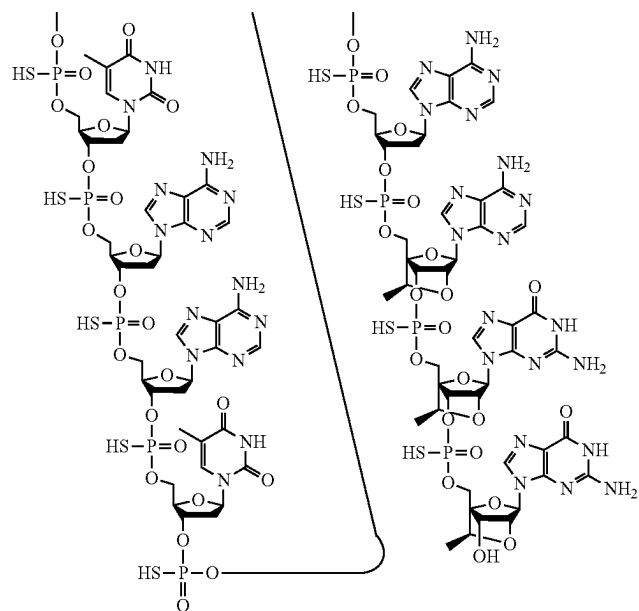
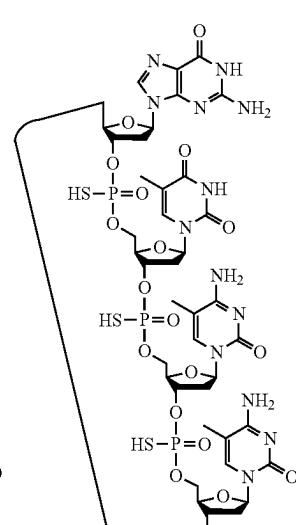


Embodiment 51: Certain embodiments provide the com- 65
pound having the following chemical structure or salt
thereof:

117

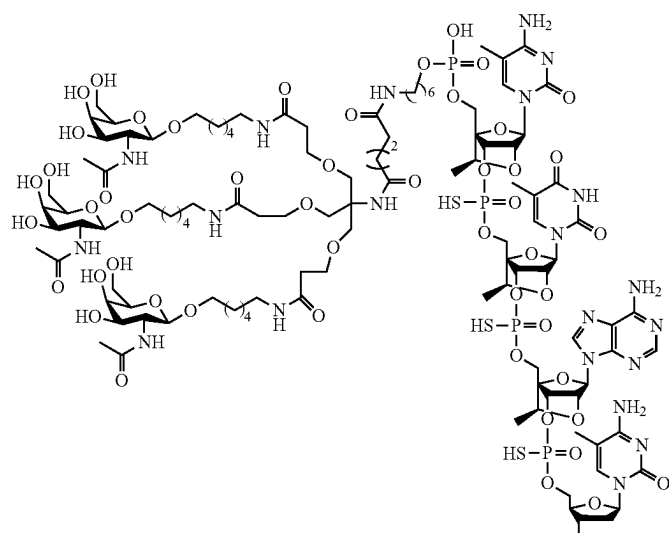


118

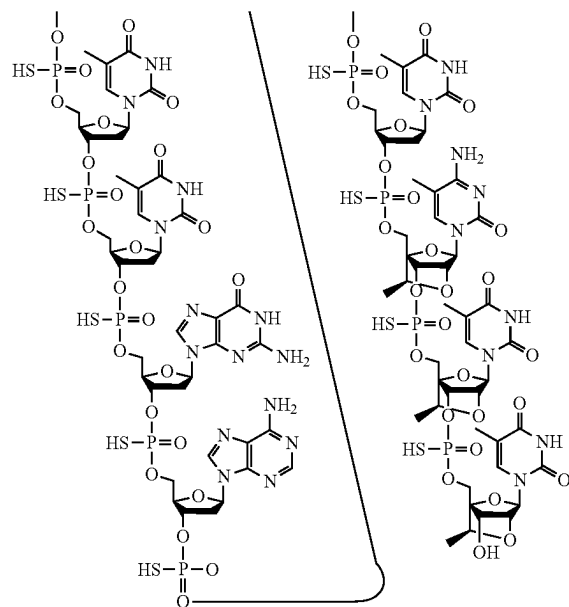
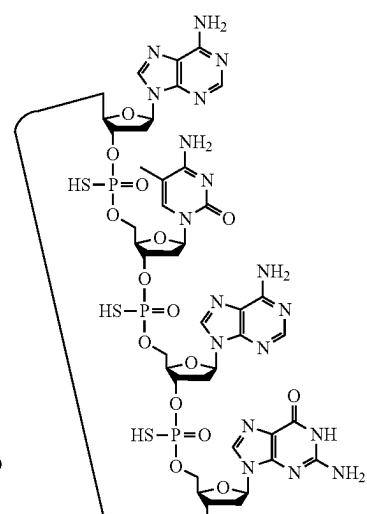


Embodiment 52: Certain embodiments provide the compound having the following chemical structure or salt thereof:

119

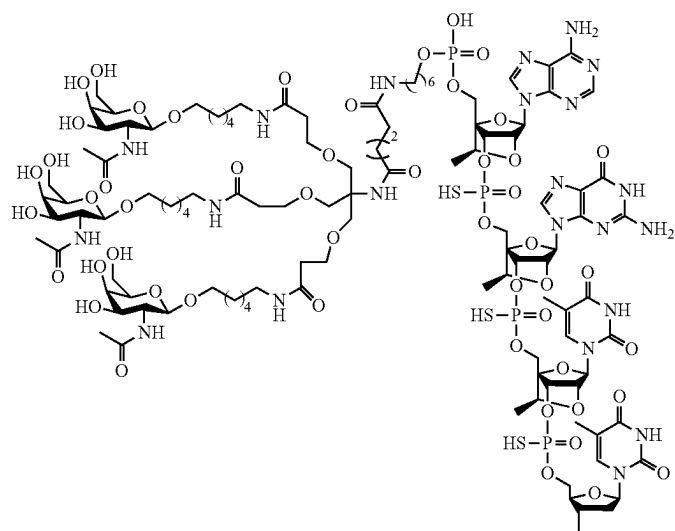


120

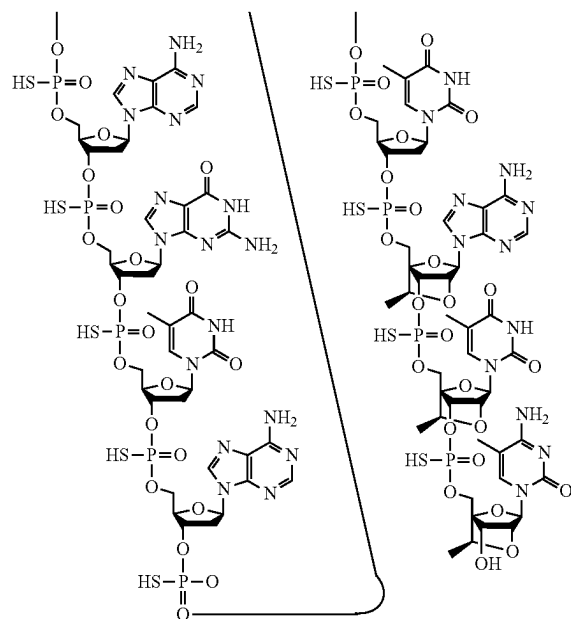
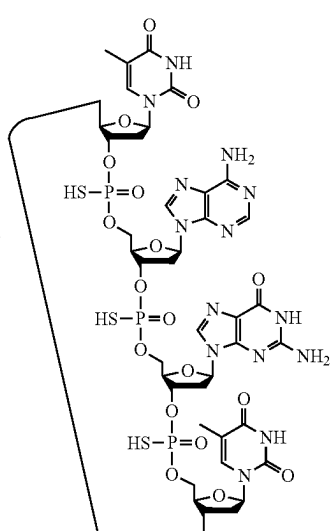


Embodiment 53: Certain embodiments provide the com- 65
 pound having the following chemical structure or salt
 thereof:

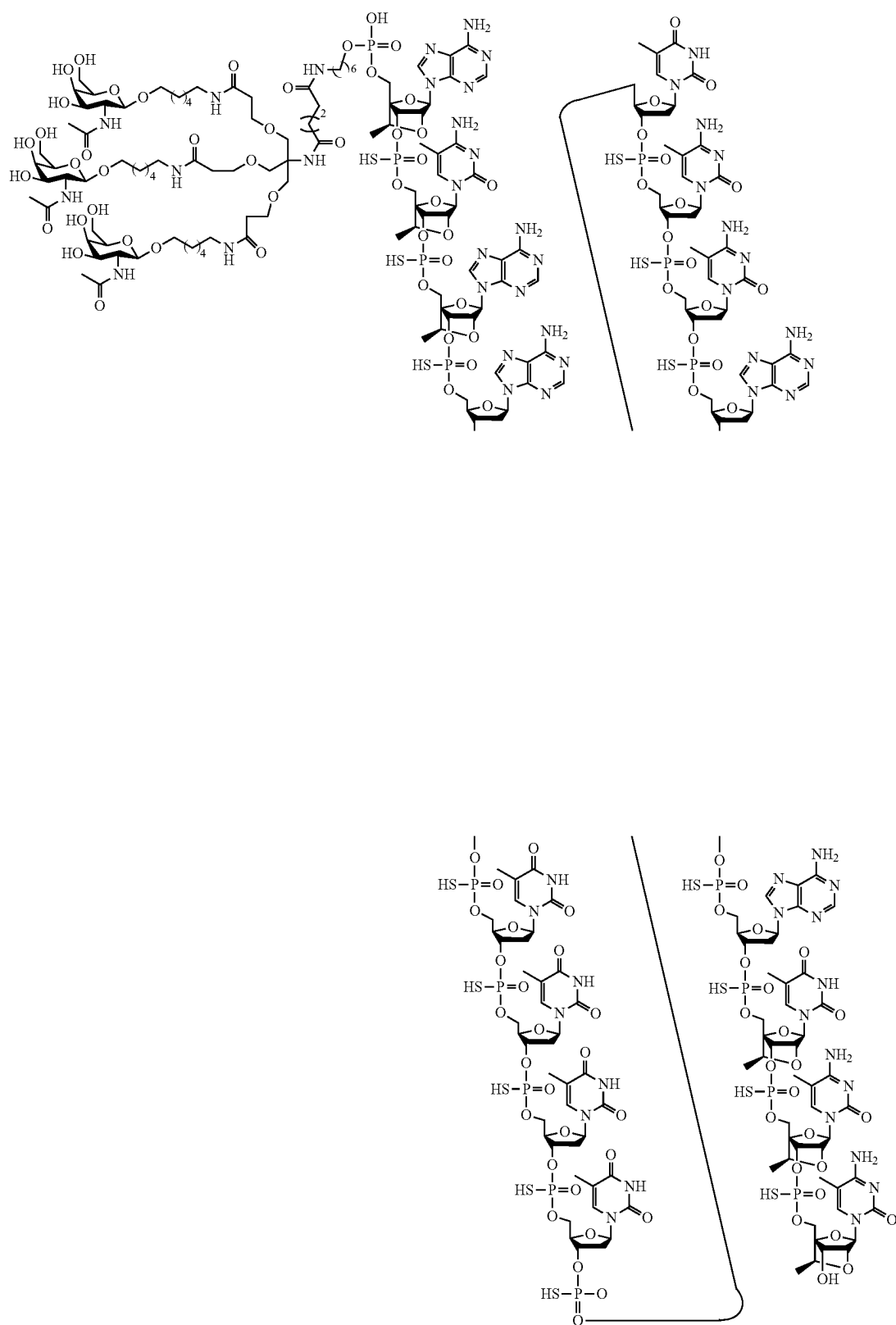
121



122

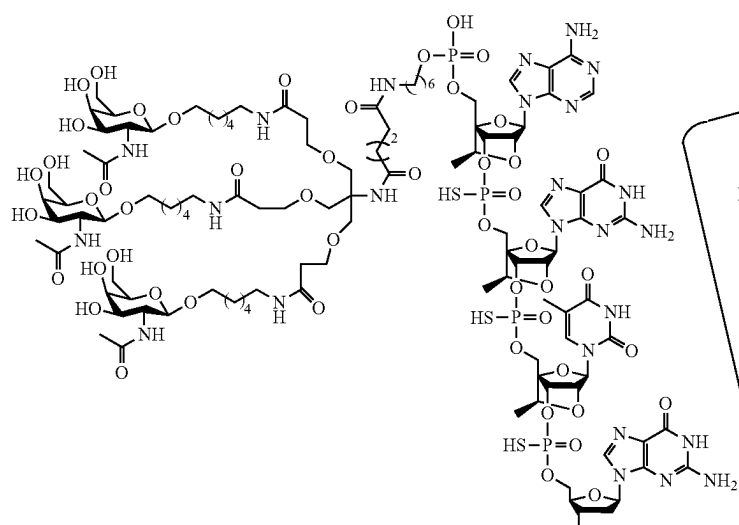


Embodiment 54: Certain embodiments provide the compound having the following chemical structure or salt thereof:

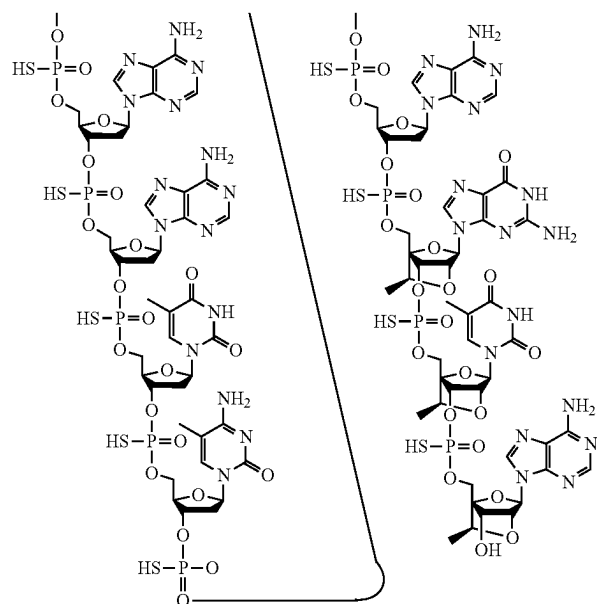
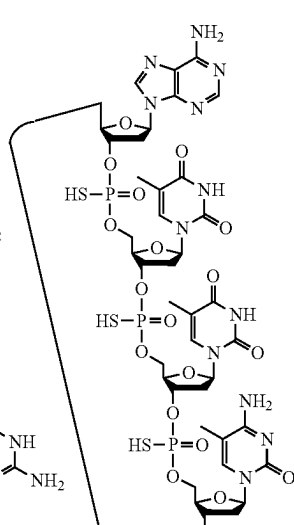


Embodiment 55: Certain embodiments provide the com- 65
pound having the following chemical structure or salt
thereof:

125

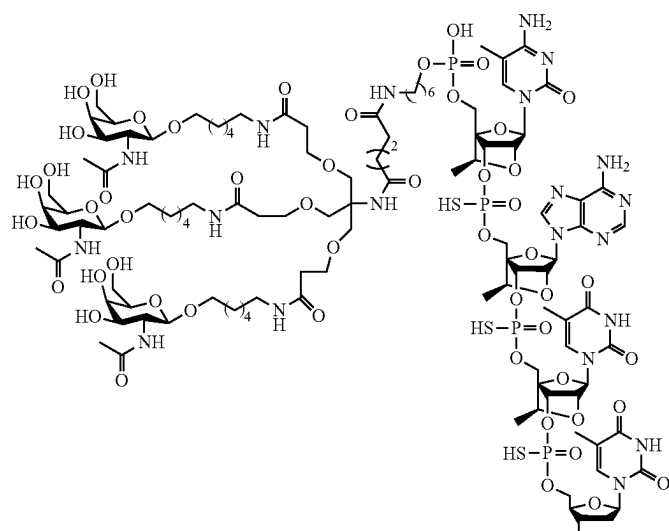


126

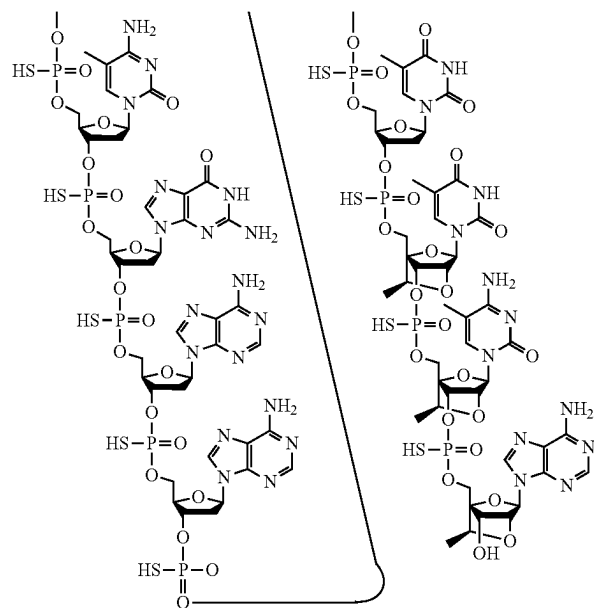
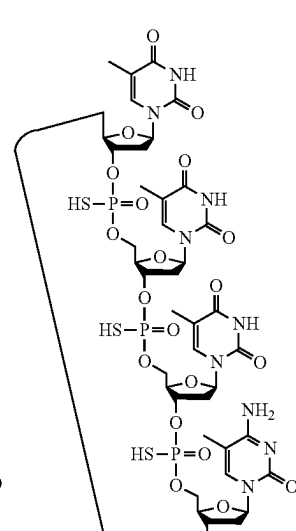


Embodiment 56: Certain embodiments provide the compound having the following chemical structure or salt thereof:

127

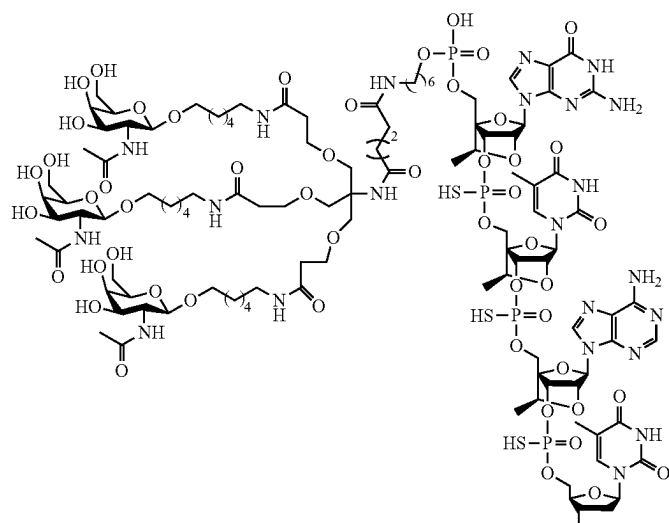


128

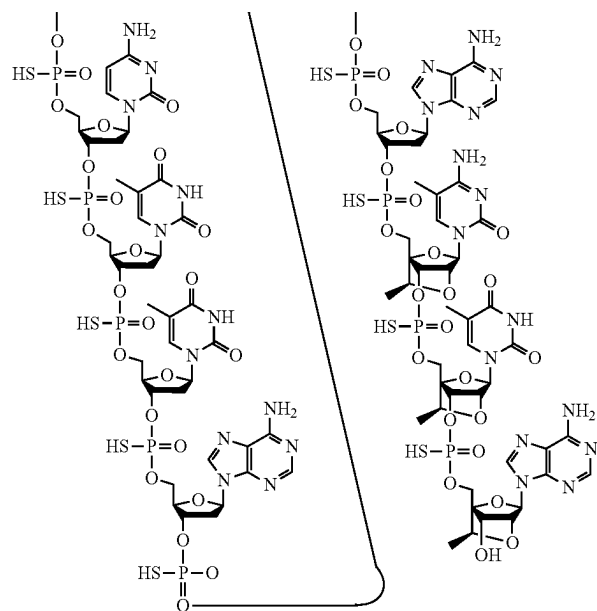
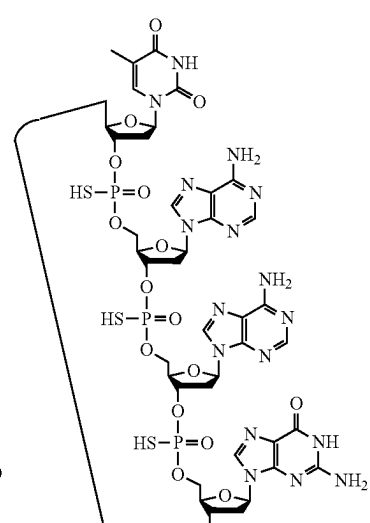


Embodiment 57: Certain embodiments provide the compound having the following chemical structure or salt thereof:

129

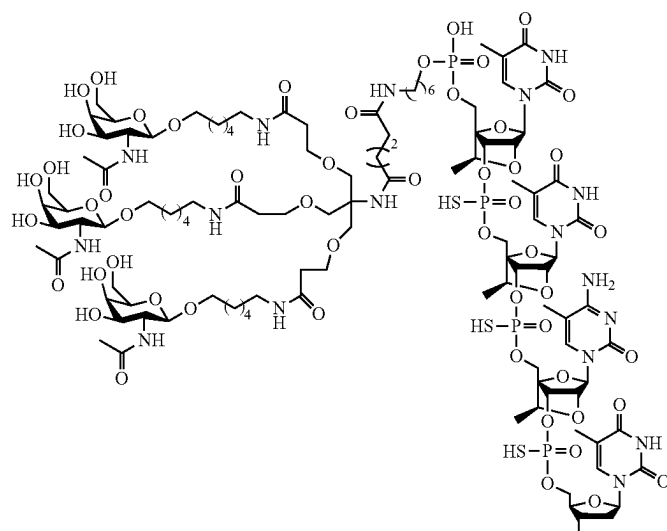


130

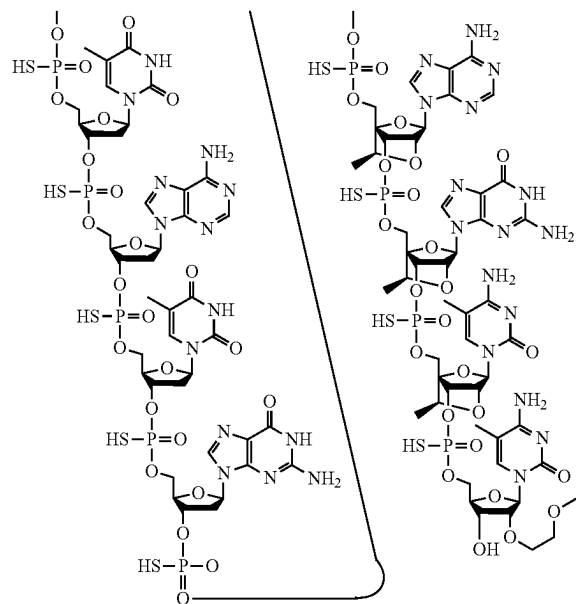
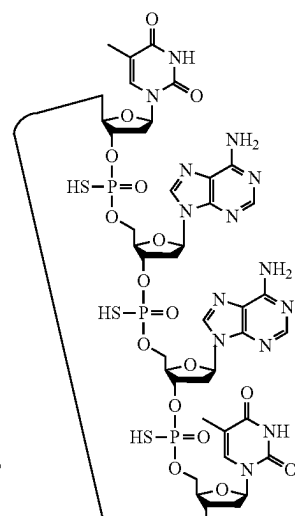


Embodiment 58: Certain embodiments provide the compound having the following chemical structure or salt thereof:

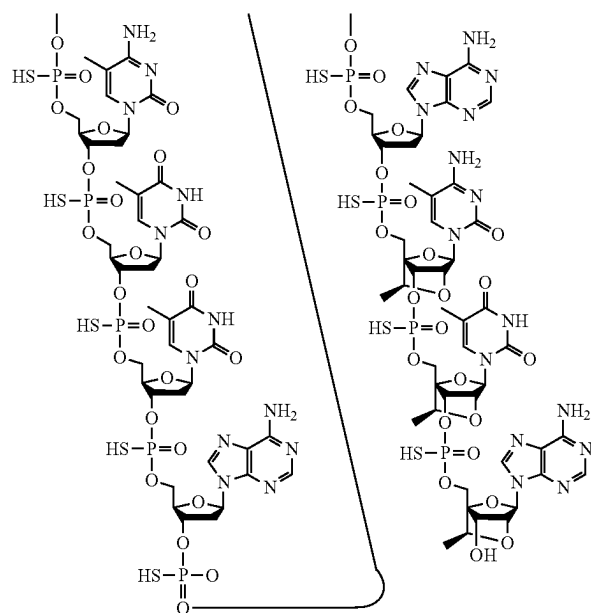
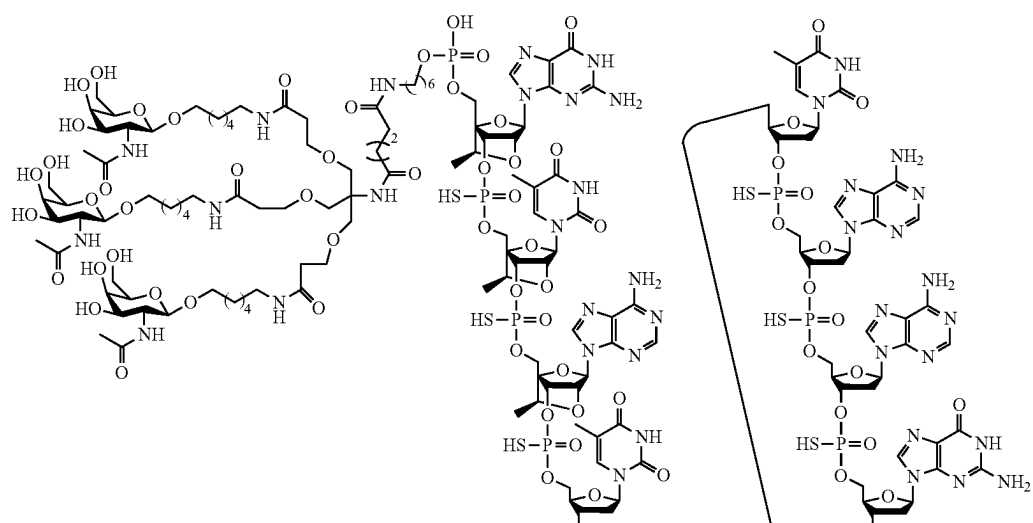
131



132

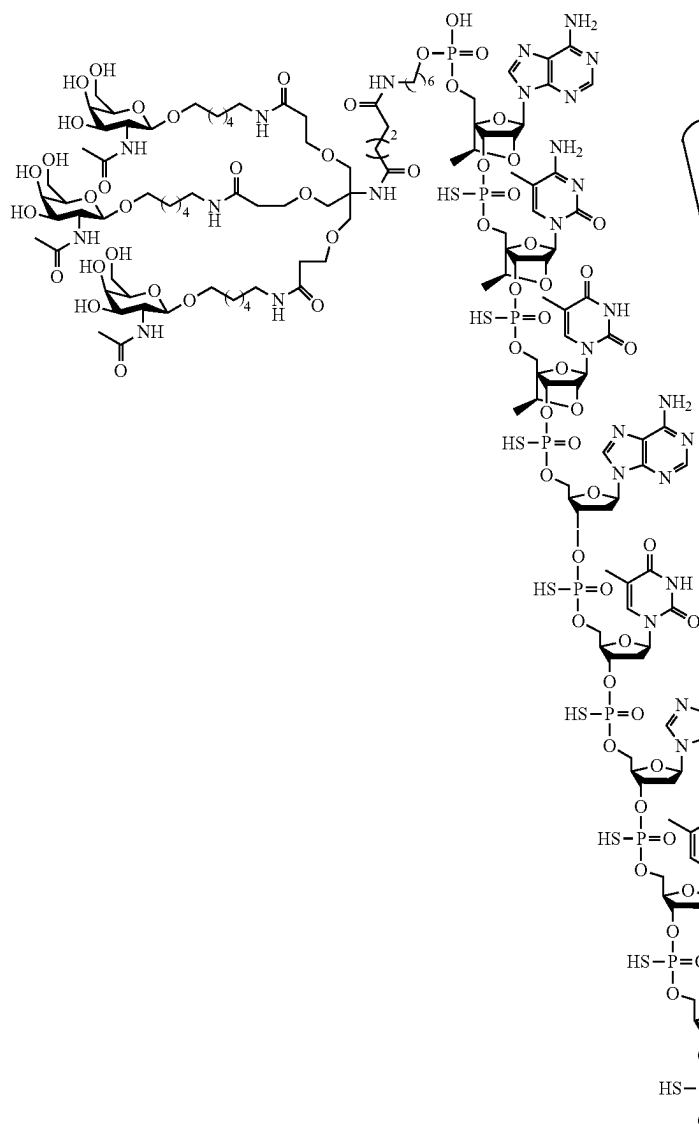


Embodiment 59: Certain embodiments provide the compound having the following chemical structure or salt thereof:

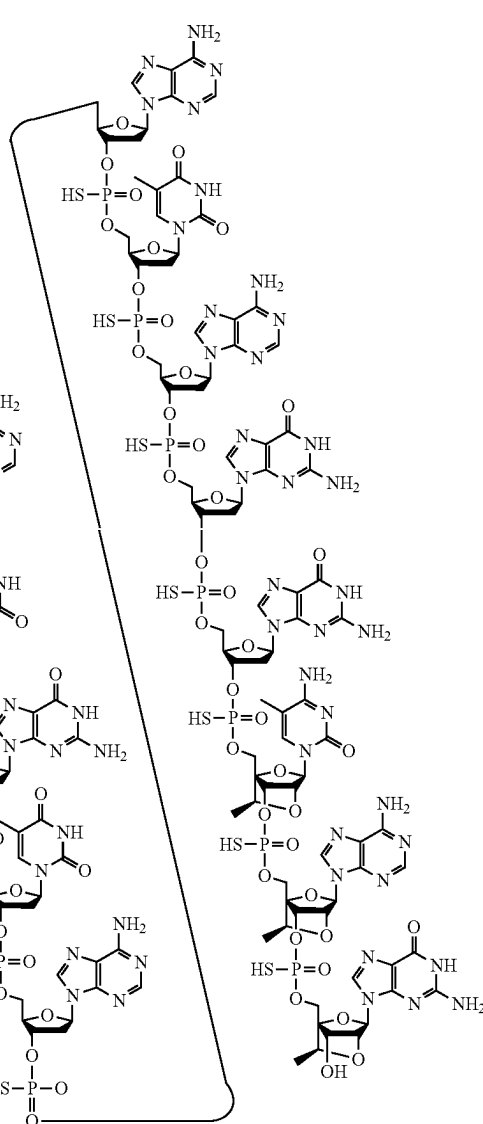


Embodiment 60: Certain embodiments provide the com- 65
pound having the following chemical structure or salt
thereof:

135



136



45

Embodiment 61: Certain embodiments provide the compound of any one of the embodiments 43-60, wherein the compound is a sodium salt.

Embodiment 62: Certain embodiments provide the compound of any one of the embodiments 43-60, wherein the compound is a potassium salt.

Embodiment 63: Certain embodiments provide the chiroly enriched population of the compound of any one of embodiments 1-62, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having a particular stereochemical configuration.

Embodiment 64: Certain embodiments provide the chiroly enriched population of embodiment 63, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having the (Sp) configuration.

Embodiment 65: Certain embodiments provide the chiroly enriched population of embodiment 63 or embodi-

ment 64, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having the (Rp) configuration.

Embodiment 66: Certain embodiments provide the chiroly enriched population of embodiment 63, wherein the population is enriched for compounds having a modified oligonucleotide having a particular, independently selected stereochemical configuration at each phosphorothioate internucleoside linkage.

Embodiment 67: Certain embodiments provide the chiroly enriched population of embodiment 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having the (Sp) configuration at each phosphorothioate internucleoside linkage.

Embodiment 68: Certain embodiments provide the chiroly enriched population of embodiment 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having the (Rp) configuration at each phosphorothioate internucleoside linkage.

137

Embodiment 69: Certain embodiments provide the chirally enriched population of embodiment 63 or 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having at least 3 contiguous phosphorothioate internucleoside linkages in the Sp-Sp-Rp configurations, in the 5' to 3' direction. 5

Embodiment 70: Certain embodiments provide a population of compounds having a modified oligonucleotide of any of embodiments 69, wherein all of the phosphorothioate internucleoside linkages of the modified oligonucleotide are stereorandom. 10

Embodiment 71: Certain embodiments provide a composition comprising the compound of any one of embodiments 1-70 and a pharmaceutically acceptable carrier. 15

Embodiment 72: Certain embodiments provide a composition comprising a compound of any preceding embodiment, for use in therapy.

Embodiment 73: Certain embodiments provide a method of treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual comprising administering to the individual a compound complementary to HSD17B13, thereby treating, preventing, or ameliorating the disease. 20

Embodiment 74: Certain embodiments provide a method of administering the compound of embodiments 1-70 or composition of embodiment 71 or embodiment 72 to an individual. 25

Embodiment 75: Certain embodiments provide the method of embodiments 73 or 74, wherein the individual has a liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. 30

Embodiment 76: Certain embodiments provide the method of embodiment 73 or embodiment 74, wherein the compound is an antisense compound. 35

Embodiment 77: Certain embodiments provide the method of any of embodiments 73-76, wherein administering the compound ameliorates liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in the individual. 40

Embodiment 78: Certain embodiments provide a method of inhibiting expression of HSD17B13 in a cell comprising contacting the cell with a compound comprising a modified oligonucleotide complementary to HSD17B13, thereby inhibiting expression of HSD17B13 in the cell. 45

Embodiment 79: Certain embodiments provide the method of embodiment 78, wherein the cell is in the liver of an individual.

Embodiment 80: Certain embodiments provide the method of embodiment 79, wherein the individual has, or is at risk of having, liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. 50

Embodiment 81: Certain embodiments provide a method of ameliorating, reducing or inhibiting liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an individual, comprising administering a compound comprising a modified oligonucleotide complementary 65

138

to HSD17B13 to the individual, thereby reducing or inhibiting liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in the individual.

Embodiment 82: Certain embodiments provide the method of embodiment 81, wherein the individual has, or is at risk of having, liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

Embodiment 83: Certain embodiments provide the method of any one of embodiments 78-82, wherein the compound is an antisense compound.

Embodiment 84: Certain embodiments provide the method of any one of embodiments 78-83, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72.

Embodiment 85: Certain embodiments provide the method of embodiment 83 or embodiment 84, wherein the compound or composition is administered parenterally.

Embodiment 86: Certain embodiments provide the use of a compound comprising a modified oligonucleotide complementary to HSD17B13 for treating, preventing, or ameliorating a disease associated with HSD17B13.

Embodiment 87: Certain embodiments provide the use of embodiment 86, wherein the disease is liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

Embodiment 88: Certain embodiments provide the use of embodiments 86 or 87, wherein the compound is an antisense compound.

Embodiment 89: Certain embodiments provide the use of any one of embodiments 86-88, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72.

Embodiment 90: Certain embodiments provide the use of a compound comprising a modified oligonucleotide complementary to HSD17B13 in the manufacture of a medicament for treating, preventing, or ameliorating a disease associated with HSD17B13.

Embodiment 91: Certain embodiments provide the use of embodiment 90, wherein the disease is liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

Embodiment 92: Certain embodiments provide the use of embodiments 90 or 91, wherein the compound is an antisense compound.

Embodiment 93: Certain embodiments provide the use of any one of embodiments 90-92, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72.

Certain Indications

Certain embodiments provided herein relate to methods of inhibiting HSD17B13 expression, which can be useful for treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual, by administration of a compound that targets HSD17B13. In certain embodiments, the compound can be a HSD17B13 specific inhibitor. In

certain embodiments, the compound can be an antisense compound, an oligomeric compound, or an oligonucleotide targeted to HSD17B13.

Examples of diseases associated with HSD17B13 treatable, preventable, and/or ameliorable with the methods provided herein include liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, non-alcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Certain compounds provided herein are directed to compounds and compositions that reduce liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an animal.

In certain embodiments, a method of treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby treating, preventing, or ameliorating the disease. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13. In certain embodiments, the disease is a liver disease. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an animal.

In certain embodiments, a method of treating, preventing, or ameliorating liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an animal comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby treating, preventing, or ameliorating liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase

sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13.

In certain embodiments, a method of inhibiting expression of HSD17B13 in an individual having, or at risk of having, a disease associated with HSD17B13 comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby inhibiting expression of HSD17B13 in the individual. In certain embodiments, administering the compound inhibits expression of HSD17B13 in the liver. In certain embodiments, the disease is a liver disease. In certain embodiments, the individual has, or is at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, non-alcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the individual has, or is at risk of having, liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation.

In certain embodiments, a method of inhibiting expression of HSD17B13 in a cell comprises contacting the cell with a compound comprising a HSD17B13 specific inhibitor, thereby inhibiting expression of HSD17B13 in the cell. In certain embodiments, the cell is a hepatocyte. In certain embodiments, the cell is in the liver. In certain embodiments, the cell is in the liver of an individual who has, or is at risk of having, liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to

HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

In certain embodiments, a method of reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, a disease associated with HSD17B13 comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in the individual. In certain embodiments, the individual has, or is at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13.

Certain embodiments are drawn to a compound comprising a HSD17B13 specific inhibitor for use in treating a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least

8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally.

Certain embodiments are drawn to a compound comprising a HSD17B13 specific inhibitor for use in reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the manufacture or preparation of a medicament for treating a disease associated with HSD17B13. Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the preparation of a medicament for treating a disease associated with HSD17B13. In certain embodiments, the disease is a liver disease. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of

SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing

embodiments, the compound can be an antisense compound or oligomeric compound.

Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the manufacture or preparation of a medicament for reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Certain

embodiments are drawn to use of a compound comprising a HSD17B13 specific inhibitor for the preparation of a medicament for treating a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain

embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing

embodiments, the compound can be an antisense compound or oligomeric compound.

In any of the foregoing methods or uses, the compound can be targeted to HSD17B13. In certain embodiments, the compound comprises or consists of a modified oligonucleotide, for example, a modified oligonucleotide consisting of 8 to 80 linked nucleosides, 10 to 30 linked nucleosides, 12 to 30 linked nucleosides, or 20 linked nucleosides. In certain embodiments, the modified oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4. In certain embodiments, the modified oligonucleotide comprises at least one modified internucleoside linkage, at least one modified sugar and/or at least one modified nucleobase. In certain embodiments, the modified internucleoside linkage is a phosphorothioate internucleoside linkage, the modified sugar is a bicyclic sugar or a 2'-O-methoxyethyl modified sugar, and the modified nucleobase is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide comprises a gap segment consisting of linked deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting

of linked nucleosides, wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

In any of the foregoing embodiments, the modified oligonucleotide consists of 12 to 30, 15 to 30, 15 to 25, 15 to 24, 16 to 24, 17 to 24, 18 to 24, 19 to 24, 20 to 24, 19 to 22, 20 to 22, 16 to 20, or 16 or 20 linked nucleosides. In certain embodiments, the modified oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4.

In any of the foregoing methods or uses, the compound comprises or consists of a modified oligonucleotide consisting of 16 to 30 linked nucleosides and having a nucleobase sequence comprising any one of SEQ ID NOs: 8-2896, wherein the modified oligonucleotide has:

- a gap segment consisting of linked 2'-deoxynucleosides;
- a 5' wing segment consisting of linked nucleosides; and
- a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide is 16-30 linked nucleosides in length. In certain embodiments, the modified oligonucleotide is 16 linked nucleosides in length.

In any of the foregoing methods or uses, the compound can be administered parenterally. For example, in certain embodiments the compound can be administered through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g. intrathecal or intracerebroventricular administration.

Certain Compounds

In certain embodiments, compounds described herein can be antisense compounds. In certain embodiments, the antisense compound comprises or consists of an oligomeric compound. In certain embodiments, the oligomeric compound comprises a modified oligonucleotide. In certain embodiments, the modified oligonucleotide has a nucleobase sequence complementary to that of a target nucleic acid.

In certain embodiments, a compound described herein comprises or consists of a modified oligonucleotide. In certain embodiments, the modified oligonucleotide has a nucleobase sequence complementary to that of a target nucleic acid.

In certain embodiments, a compound or antisense compound is single-stranded. Such a single-stranded compound or antisense compound comprises or consists of an oligomeric compound. In certain embodiments, such an oligomeric compound comprises or consists of an oligonucleotide and optionally a conjugate group. In certain embodiments, the oligonucleotide is an antisense oligonucleotide. In certain embodiments, the oligonucleotide is modified. In certain embodiments, the oligonucleotide of a single-stranded antisense compound or oligomeric compound comprises a self-complementary nucleobase sequence.

In certain embodiments, compounds are double-stranded. Such double-stranded compounds comprise a first modified oligonucleotide having a region complementary to a target nucleic acid and a second modified oligonucleotide having a region complementary to the first modified oligonucleotide. In certain embodiments, the modified oligonucleotide

is an RNA oligonucleotide. In such embodiments, the thymine nucleobase in the modified oligonucleotide is replaced by a uracil nucleobase. In certain embodiments, compound comprises a conjugate group. In certain embodiments, one of the modified oligonucleotides is conjugated. In certain embodiments, both the modified oligonucleotides are conjugated. In certain embodiments, the first modified oligonucleotide is conjugated. In certain embodiments, the second modified oligonucleotide is conjugated. In certain embodiments, the first modified oligonucleotide is 16-30 linked nucleosides in length and the second modified oligonucleotide is 16-30 linked nucleosides in length. In certain embodiments, one of the modified oligonucleotides has a nucleobase sequence comprising at least 8 contiguous nucleobases of any of SEQ ID NOs: 8-2035.

In certain embodiments, antisense compounds are double-stranded. Such double-stranded antisense compounds comprise a first oligomeric compound having a region complementary to a target nucleic acid and a second oligomeric compound having a region complementary to the first oligomeric compound. The first oligomeric compound of such double stranded antisense compounds typically comprises or consists of a modified oligonucleotide and optionally a conjugate group. The oligonucleotide of the second oligomeric compound of such a double-stranded antisense compound may be modified or unmodified. Either or both oligomeric compounds of a double-stranded antisense compound may comprise a conjugate group. The oligomeric compounds of double-stranded antisense compounds may include non-complementary overhanging nucleosides.

Examples of single-stranded and double-stranded compounds include, but are not limited to, oligonucleotides, siRNAs, microRNA targeting oligonucleotides, and single-stranded RNAi compounds, such as small hairpin RNAs (shRNAs), single-stranded siRNAs (ssRNAs), and microRNA mimics.

In certain embodiments, a compound described herein has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

In certain embodiments, a compound described herein comprises an oligonucleotide 12 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 12 to 22 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 14 to 30 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 14 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 15 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 15 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 18 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 18 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 20 to 30 linked subunits in length. In other words, such oligonucleotides are 12 to 30

linked subunits, 14 to 30 linked subunits, 14 to 20 subunits, 15 to 30 subunits, 15 to 20 subunits, 16 to 30 subunits, 16 to 20 subunits, 17 to 30 subunits, 17 to 20 subunits, 18 to 30 subunits, 18 to 20 subunits, or 20 to 30 subunits in length, respectively. In certain embodiments, a compound described herein comprises an oligonucleotide 14 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 18 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 19 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 20 linked subunits in length. In other embodiments, a compound described herein comprises an oligonucleotide 8 to 80, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked subunits. In certain such embodiments, the compound described herein comprises an oligonucleotide 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 linked subunits in length, or a range defined by any two of the above values. In some embodiments the linked subunits are nucleotides, nucleosides, or nucleobases.

In certain embodiments, the compound may further comprise additional features or elements, such as a conjugate group, that are attached to the oligonucleotide. In certain embodiments, such compounds are antisense compounds. In certain embodiments, such compounds are oligomeric compounds. In embodiments where a conjugate group comprises a nucleoside (i.e. a nucleoside that links the conjugate group to the oligonucleotide), the nucleoside of the conjugate group is not counted in the length of the oligonucleotide.

In certain embodiments, compounds may be shortened or truncated. For example, a single subunit may be deleted from the 5' end (5' truncation), or alternatively from the 3' end (3' truncation). A shortened or truncated compound targeted to a HSD17B13 nucleic acid may have two subunits deleted from the 5' end, or alternatively, may have two subunits deleted from the 3' end of the compound. Alternatively, the deleted nucleosides may be dispersed throughout the compound.

When a single additional subunit is present in a lengthened compound, the additional subunit may be located at the 5' or 3' end of the compound. When two or more additional subunits are present, the added subunits may be adjacent to each other, for example, in a compound having two subunits added to the 5' end (5' addition), or alternatively, to the 3' end (3' addition) of the compound. Alternatively, the added subunits may be dispersed throughout the compound.

It is possible to increase or decrease the length of a compound, such as an oligonucleotide, and/or introduce mismatch bases without eliminating activity (Woolf et al. *Proc. Natl. Acad. Sci. USA* 1992, 89:7305-7309; Gautschi et al. *J. Natl. Cancer Inst. March* 2001, 93:463-471; Maher and Dolnick *Nuc. Acid. Res.* 1998, 16:3341-3358). However, seemingly small changes in oligonucleotide sequence, chemistry and motif can make large differences in one or more of the many properties required for clinical development (Seth et al. *J. Med. Chem.* 2009, 52, 10; Egli et al. *J. Am. Chem. Soc.* 2011, 133, 16642).

In certain embodiments, compounds described herein are interfering RNA compounds (RNAi), which include double-stranded RNA compounds (also referred to as short-inter-

fering RNA or siRNA) and single-stranded RNAi compounds (or ssRNA). Such compounds work at least in part through the RISC pathway to degrade and/or sequester a target nucleic acid (thus, include microRNA/microRNA-mimic compounds). As used herein, the term siRNA is meant to be equivalent to other terms used to describe nucleic acid molecules that are capable of mediating sequence-specific RNAi, for example, short interfering RNA (siRNA), double-stranded RNA (dsRNA), micro-RNA (miRNA), short hairpin RNA (shRNA), short interfering oligonucleotide, short interfering nucleic acid, short interfering modified oligonucleotide, chemically modified siRNA, post-transcriptional gene silencing RNA (ptgsRNA), and others. In addition, as used herein, the term "RNAi" is meant to be equivalent to other terms used to describe sequence-specific RNA interference, such as post transcriptional gene silencing, translational inhibition, or epigenetics.

In certain embodiments, a compound described herein can comprise any of the oligonucleotide sequences targeted to HSD17B13 described herein. In certain embodiments, the compound can be double-stranded. In certain embodiments, the compound comprises a first strand comprising at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion of any one of SEQ ID NOs: 8-2035 and a second strand. In certain embodiments, the compound comprises a first strand comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2035 and a second strand. In certain embodiments, the compound comprises ribonucleotides in which the first strand has uracil (U) in place of thymine (T) in any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises (i) a first strand comprising a nucleobase sequence complementary to the site on HSD17B13 to which any of SEQ ID NOs: 8-2035 is targeted, and (ii) a second strand. In certain embodiments, the compound comprises one or more modified nucleotides in which the 2' position of the sugar contains a halogen (such as a fluorine group; 2'-F) or contains an alkoxy group (such as a methoxy group; 2'-OMe). In certain embodiments, the compound comprises at least one 2'-F sugar modification and at least one 2'-OMe sugar modification. In certain embodiments, the at least one 2'-F sugar modification and at least one 2'-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the dsRNA compound. In certain embodiments, the compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramidate, phosphorothioate, and phosphorodithioate linkages. The compounds may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the compound contains one or two capped strands, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000.

In certain embodiments, the first strand of the compound is an siRNA guide strand and the second strand of the compound is an siRNA passenger strand. In certain embodiments, the second strand of the compound is complementary to the first strand. In certain embodiments, each strand of the compound is 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides in length. In certain embodiments, the first or second strand of the compound can comprise a conjugate group.

In certain embodiments, a compound described herein can comprise any of the oligonucleotide sequences targeted to HSD17B13 described herein. In certain embodiments, the

compound is single stranded. In certain embodiments, such a compound is a single-stranded RNAi (ssRNAi) compound. In certain embodiments, the compound comprises at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion of any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises the nucleobase sequence of any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises ribonucleotides in which uracil (U) is in place of thymine (T) in any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises a nucleobase sequence complementary to the site on HSD17B13 to which any of SEQ ID NOs: 8-2035 is targeted. In certain embodiments, the compound comprises one or more modified nucleotides in which the 2' position in the sugar contains a halogen (such as fluorine group; 2'-F) or contains an alkoxy group (such as a methoxy group; 2'-OMe). In certain embodiments, the compound comprises at least one 2'-F sugar modification and at least one 2'-OMe sugar modification. In certain embodiments, the at least one 2'-F sugar modification and at least one 2'-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the compound. In certain embodiments, the compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramidate, phosphorothioate, and phosphorodithioate linkages. The compounds may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the compound contains a capped strand, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000. In certain embodiments, the compound consists of 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides. In certain embodiments, the compound can comprise a conjugate group.

Certain Mechanisms

In certain embodiments, compounds described herein comprise or consist of modified oligonucleotides. In certain embodiments, compounds described herein are antisense compounds. In certain embodiments, compounds comprise oligomeric compounds. In certain embodiments, compounds described herein are capable of hybridizing to a target nucleic acid, resulting in at least one antisense activity. In certain embodiments, compounds described herein selectively affect one or more target nucleic acid. Such compounds comprise a nucleobase sequence that hybridizes to one or more target nucleic acid, resulting in one or more desired antisense activity and does not hybridize to one or more non-target nucleic acid or does not hybridize to one or more non-target nucleic acid in such a way that results in a significant undesired antisense activity.

In certain antisense activities, hybridization of a compound described herein to a target nucleic acid results in recruitment of a protein that cleaves the target nucleic acid. For example, certain compounds described herein result in RNase H mediated cleavage of the target nucleic acid. RNase H is a cellular endonuclease that cleaves the RNA strand of an RNA:DNA duplex. The DNA in such an RNA:DNA duplex need not be unmodified DNA. In certain embodiments, compounds described herein are sufficiently "DNA-like" to elicit RNase H activity. Further, in certain embodiments, one or more non-DNA-like nucleoside in the gap of a gapmer is tolerated.

In certain antisense activities, compounds described herein or a portion of the compound is loaded into an RNA-induced silencing complex (RISC), ultimately result-

ing in cleavage of the target nucleic acid. For example, certain compounds described herein result in cleavage of the target nucleic acid by Argonaute. Compounds that are loaded into RISC are RNAi compounds. RNAi compounds may be double-stranded (siRNA) or single-stranded (ssRNA).

In certain embodiments, hybridization of compounds described herein to a target nucleic acid does not result in recruitment of a protein that cleaves that target nucleic acid. In certain such embodiments, hybridization of the compound to the target nucleic acid results in alteration of splicing of the target nucleic acid. In certain embodiments, hybridization of the compound to a target nucleic acid results in inhibition of a binding interaction between the target nucleic acid and a protein or other nucleic acid. In certain such embodiments, hybridization of the compound to a target nucleic acid results in alteration of translation of the target nucleic acid.

Antisense activities may be observed directly or indirectly. In certain embodiments, observation or detection of an antisense activity involves observation or detection of a change in an amount of a target nucleic acid or protein encoded by such target nucleic acid, a change in the ratio of splice variants of a nucleic acid or protein, and/or a phenotypic change in a cell or animal.

Target Nucleic Acids, Target Regions and Nucleotide Sequences

In certain embodiments, compounds described herein comprise or consist of an oligonucleotide comprising a region that is complementary to a target nucleic acid. In certain embodiments, the target nucleic acid is an endogenous RNA molecule. In certain embodiments, the target nucleic acid encodes a protein. In certain such embodiments, the target nucleic acid is selected from an mRNA and a pre-mRNA, including intronic, exonic and untranslated regions. In certain embodiments, the target RNA is an mRNA. In certain embodiments, the target nucleic acid is a pre-mRNA. In certain such embodiments, the target region is entirely within an intron. In certain embodiments, the target region spans an intron/exon junction. In certain embodiments, the target region is at least 50% within an intron.

Nucleotide sequences that encode HSD17B13 include, without limitation, the following: RefSeq or GENBANK Accession No. NM_178135.4 (incorporated by reference, disclosed herein as SEQ ID NO: 1), the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000 (incorporated by reference, disclosed herein as SEQ ID NO: 2), the sequence listed as SEQ ID NO: 3, or GENBANK Accession No. NM_001136230.2 (incorporated by reference, disclosed herein as SEQ ID NO: 4).

Hybridization

In some embodiments, hybridization occurs between a compound disclosed herein and a HSD17B13 nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.

Hybridization can occur under varying conditions. Hybridization conditions are sequence-dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.

Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in

the art. In certain embodiments, the compounds provided herein are specifically hybridizable with a HSD17B13 nucleic acid.

Complementarity

An oligonucleotide is said to be complementary to another nucleic acid when the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to the following pairs: adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (mC) and guanine (G), unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. An oligonucleotide is fully complementary or 100% complementary when such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches.

In certain embodiments, compounds described herein comprise or consist of modified oligonucleotides. In certain embodiments, compounds described herein are antisense compounds. In certain embodiments, compounds comprise oligomeric compounds. Non-complementary nucleobases between a compound and a HSD17B13 nucleic acid may be tolerated provided that the compound remains able to specifically hybridize to a target nucleic acid. Moreover, a compound may hybridize over one or more segments of a HSD17B13 nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).

In certain embodiments, the compounds provided herein, or a specified portion thereof are at least, or are up to 70%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% complementary to a HSD17B13 nucleic acid, a target region, target segment, or specified portion thereof. In certain embodiments, the compounds provided herein, or a specified portion thereof, are 70% to 75%, 75% to 80%, 80% to 85%, 85% to 90%, 90% to 95%, 95% to 100%, or any number in between these ranges, complementary to a HSD17B13 nucleic acid, a target region, target segment, or specified portion thereof. Percent complementarity of a compound with a target nucleic acid can be determined using routine methods.

For example, a compound in which 18 of 20 nucleobases of the compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining non-complementary nucleobases may be clustered or interspersed with complementary nucleobases and need not be contiguous to each other or to complementary nucleobases. As such, a compound which is 18 nucleobases in length having four non-complementary nucleobases which are flanked by two regions of complete complementarity with the target nucleic acid would have 77.8% overall complementarity with the target nucleic acid. Percent complementarity of a compound with a region of a target nucleic acid can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs known in the art (Altschul et al., *J. Mol. Biol.*, 1990, 215, 403-410; Zhang and Madden, *Genome Res.*, 1997, 7, 649-656). Percent homology, sequence identity or complementarity, can be determined by, for example, the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madi-

son Wis.), using default settings, which uses the algorithm of Smith and Waterman (Adv. Appl. Math., 1981, 2, 482 489).

In certain embodiments, compounds described herein, or specified portions thereof, are fully complementary (i.e. 100% complementary) to a target nucleic acid, or specified portion thereof. For example, a compound may be fully complementary to a HSD17B13 nucleic acid, or a target region, or a target segment or target sequence thereof. As used herein, "fully complementary" means each nucleobase of a compound is complementary to the corresponding nucleobase of a target nucleic acid. For example, a 20 nucleobase compound is fully complementary to a target sequence that is 400 nucleobases long, so long as there is a corresponding 20 nucleobase portion of the target nucleic acid that is fully complementary to the compound. "Fully complementary" can also be used in reference to a specified portion of the first and/or the second nucleic acid. For example, a 20 nucleobase portion of a 30 nucleobase compound can be "fully complementary" to a target sequence that is 400 nucleobases long. The 20 nucleobase portion of the 30 nucleobase compound is fully complementary to the target sequence if the target sequence has a corresponding 20 nucleobase portion wherein each nucleobase is complementary to the 20 nucleobase portion of the compound. At the same time, the entire 30 nucleobase compound may or may not be fully complementary to the target sequence, depending on whether the remaining 10 nucleobases of the compound are also complementary to the target sequence.

In certain embodiments, compounds described herein comprise one or more mismatched nucleobases relative to the target nucleic acid. In certain such embodiments, anti-sense activity against the target is reduced by such mismatch, but activity against a non-target is reduced by a greater amount. Thus, in certain such embodiments, selectivity of the compound is improved. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, or 8 from the 5'-end of the gap region. In certain such embodiments, the mismatch is at position 9, 8, 7, 6, 5, 4, 3, 2, 1 from the 3'-end of the gap region. In certain such embodiments, the mismatch is at position 1, 2, 3, or 4 from the 5'-end of the wing region. In certain such embodiments, the mismatch is at position 4, 3, 2, or 1 from the 3'-end of the wing region. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide not having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 5'-end of the oligonucleotide. In certain such embodiments, the mismatch is at position, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 3'-end of the oligonucleotide.

The location of a non-complementary nucleobase may be at the 5' end or 3' end of the compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e. linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer oligonucleotide.

In certain embodiments, compounds described herein that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a HSD17B13 nucleic acid, or specified portion thereof.

In certain embodiments, compounds described herein that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21,

22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a HSD17B13 nucleic acid, or specified portion thereof.

In certain embodiments, compounds described herein also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e. linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of a compound. In certain embodiments, the compounds are complementary to at least an 8 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 9 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 10 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least an 11 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 12 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 13 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 14 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 15 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 16 nucleobase portion of a target segment. Also contemplated are compounds that are complementary to at least a 9, 10, 17, 18, 19, 20, or more nucleobase portion of a target segment, or a range defined by any two of these values.

Identity

The compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NOs, or compound represented by a specific ION number, or portion thereof. In certain embodiments, compounds described herein are antisense compounds or oligomeric compounds. In certain embodiments, compounds described herein are modified oligonucleotides. As used herein, a compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the compounds described herein as well as compounds having non-identical bases relative to the compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the compound. Percent identity of an compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

In certain embodiments, compounds described herein, or portions thereof, are, or are at least, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to one or more of the compounds or SEQ ID NOs, or a portion thereof, disclosed herein. In certain embodiments, compounds described herein are about 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical, or any percentage between such values, to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific ION number, or portion thereof, in which the compounds comprise an oligonucleotide having one or more mismatched nucleobases.

In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 5'-end of the oligonucleotide. In certain such embodiments, the mismatch is at position, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 3'-end of the oligonucleotide.

In certain embodiments, compounds described herein comprise or consist of antisense compounds. In certain embodiments, a portion of the antisense compound is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

In certain embodiments, compounds described herein comprise or consist of oligonucleotides. In certain embodiments, a portion of the oligonucleotide is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

Certain Modified Compounds

In certain embodiments, compounds described herein comprise or consist of oligonucleotides consisting of linked nucleosides. Oligonucleotides may be unmodified oligonucleotides (RNA or DNA) or may be modified oligonucleotides. Modified oligonucleotides comprise at least one modification relative to unmodified RNA or DNA (i.e., comprise at least one modified nucleoside (comprising a modified sugar moiety and/or a modified nucleobase) and/or at least one modified internucleoside linkage).

A. Modified Nucleosides

Modified nucleosides comprise a modified sugar moiety or a modified nucleobase or both a modified sugar moiety and a modified nucleobase.

1. Modified Sugar Moieties

In certain embodiments, sugar moieties are non-bicyclic modified sugar moieties. In certain embodiments, modified sugar moieties are bicyclic or tricyclic sugar moieties. In certain embodiments, modified sugar moieties are sugar surrogates. Such sugar surrogates may comprise one or more substitutions corresponding to those of other types of modified sugar moieties.

In certain embodiments, modified sugar moieties are non-bicyclic modified furanosyl sugar moieties comprising one or more acyclic substituent, including, but not limited, to substituents at the 2', 4', and/or 5' positions. In certain embodiments, the furanosyl sugar moiety is a ribosyl sugar moiety. In certain embodiments, one or more acyclic substituent of non-bicyclic modified sugar moieties is branched. Examples of 2'-substituent groups suitable for non-bicyclic modified sugar moieties include but are not limited to: 2'-F, 2'-OCH₃ ("OMe" or "O-methyl"), and 2'-O(CH₂)₂OCH₃ ("MOE"). In certain embodiments, 2'-substituent groups are selected from among: halo, allyl, amino, azido, SH, CN, OCN, CF₃, OCF₃, O—C₁-C₁₀ alkoxy, O—C₁-C₁₀ substituted alkoxy, O—C₁-C₁₀ alkyl, O—C₁-C₁₀ substituted alkyl, S-alkyl, N(R_m)-alkyl, O-alkenyl, S-alkenyl, N(R_m)-alkenyl, O-alkynyl, S-alkynyl, N(R_m)-alkynyl, O-alkylenyl-O-alkyl, alkynyl, alkaryl, aralkyl, O-alkaryl, O-aralkyl, O(CH₂)₂SCH₃, O(CH₂)₂ON(R_m)(R_n) or OCH₂C(=O)—N(R_m)(R_n), where each R_m and R_n is, independently, H, an amino protecting group, or substituted or unsubstituted C₁-C₁₀ alkyl, and the 2'-substituent groups described in Cook et al., U.S. Pat. No. 6,531,584; Cook et al., U.S. Pat. No. 5,859,221; and Cook et al., U.S. Pat. No. 6,005,087. Certain

dently selected from among: hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro (NO₂), thiol, thioalkoxy, thioalkyl, halogen, alkyl, aryl, alkenyl and alkynyl. Examples of 4'-substituent groups suitable for linearly non-bicyclic modified sugar moieties include, but are not limited to, alkoxy (e.g., methoxy), alkyl, and those described in Manoharan et al., WO 2015/106128. Examples of 5'-substituent groups suitable for non-bicyclic modified sugar moieties include, but are not limited to: 5'-methyl (R or S), 5'-vinyl, and 5'-methoxy. In certain embodiments, non-bicyclic modified sugars comprise more than one non-bridging sugar substituent, for example, 2'-F-5'-methyl sugar moieties and the modified sugar moieties and modified nucleosides described in Migawa et al., WO 2008/101157 and Rajeev et al., US2013/0203836.

In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, NH₂, N₃, OCF₃, OCH₃, O(CH₂)₃NH₂, CH₂CH=CH₂, OCH₂CH=CH₂, OCH₂CH₂OCH₃, O(CH₂)₂SCH₃, O(CH₂)₂ON(R_m)(R_n), O(CH₂)₂O(CH₂)₂N(CH₃)₂, and N-substituted acetamide (OCH₂C(=O)—N(R_m)(R_n)), where each R_m and R_n is, independently, H, an amino protecting group, or substituted or unsubstituted C₁-C₁₀ alkyl.

In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, OCF₃, OCH₃, OCH₂CH₂OCH₃, O(CH₂)₂SCH₃, O(CH₂)₂ON(CH₃)₂, O(CH₂)₂O(CH₂)₂N(CH₃)₂, and OCH₂C(=O)—N(H)CH₃ ("NMA").

In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, OCH₃, and OCH₂CH₂OCH₃.

Nucleosides comprising modified sugar moieties, such as non-bicyclic modified sugar moieties, are referred to by the position(s) of the substitution(s) on the sugar moiety of the nucleoside. For example, nucleosides comprising 2'-substituted or 2'-modified sugar moieties are referred to as 2'-substituted nucleosides or 2'-modified nucleosides.

Certain modified sugar moieties comprise a bridging sugar substituent that forms a second ring resulting in a bicyclic sugar moiety. In certain such embodiments, the bicyclic sugar moiety comprises a bridge between the 4' and the 2' furanose ring atoms. In certain such embodiments, the furanose ring is a ribose ring. Examples of such 4' to 2' bridging sugar substituents include, but are not limited to: 4'-CH₂-2', 4'-(CH₂)₂-2', 4'-(CH₂)₃-2', 4'-CH₂-O-2' ("LNA"), 4'-CH₂-S-2', 4'-(CH₂)₂-O-2' ("ENA"), 4'-CH(CH₃)-O-2' (referred to as "constrained ethyl" or "cEt" when in the S configuration), 4'-CH₂-O-CH₂-2', 4'-CH₂-N(R)-2', 4'-CH(CH₂OCH₃)-O-2' ("constrained MOE" or "cMOE") and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 7,399,845; Bhat et al., U.S. Pat. No. 7,569,686; Swayze et al., U.S. Pat. No. 7,741,457; and Swayze et al., U.S. Pat. No. 8,022,193), 4'-C(CH₃)(CH₃)-O-2' and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 8,278,283), 4'-CH₂-N(OCH₃)-2' and analogs thereof (see, e.g., Prakash et al., U.S. Pat. No. 8,278,425), 4'-CH₂-O-N(CH₃)-2' (see, e.g., Allerson et al., U.S. Pat. No. 7,696,345 and Allerson et al., U.S. Pat. No. 8,124,745), 4'-CH₂-C(H)(CH₃)-2' (see, e.g., Zhou, et al., *J. Org. Chem.*, 2009, 74, 118-134), 4'-CH₂-C(=CH₂)-2' and analogs thereof (see e.g., Seth et al., U.S. Pat. No. 8,278,426), 4'-C(R_aR_b)-N(R)-O-2', 4'-C(R_aR_b)-O-N(R)-2', 4'-CH₂-O-N(R)-2', and 4'-CH₂-N(R)-O-2', wherein each R, R_a, and R_b is,

155

independently, H, a protecting group, or C₁-C₁₂ alkyl (see, e.g. Imanishi et al., U.S. Pat. No. 7,427,672).

In certain embodiments, such 4' to 2' bridges independently comprise from 1 to 4 linked groups independently selected from: $-\text{[C(R}_a\text{)(R}_b\text{)]}_n-$, $-\text{[C(R}_a\text{)(R}_b\text{)]}_n\text{O}-$, $-\text{C(R}_a\text{)=C(R}_b\text{)-}$, $-\text{C(R}_a\text{)=N-}$, $-\text{C(=NR}_a\text{)-}$, $-\text{C(=O)-}$, $-\text{C(=S)-}$, $-\text{O-}$, $-\text{Si(R}_a\text{)}_2-$, $-\text{S(=O)}_x-$, and $-\text{N(R}_a\text{)-}$;

wherein:

x is 0, 1, or 2;

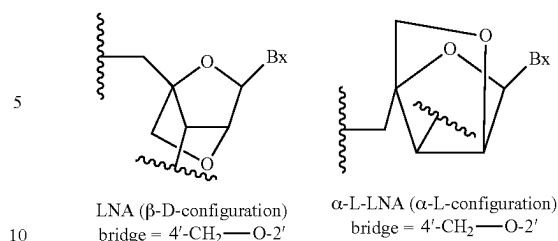
n is 1, 2, 3, or 4;

each R_a and R_b is, independently, H, a protecting group, hydroxyl, C₁-C₁₂ alkyl, substituted C₁-C₁₂ alkyl, C₂-C₁₂ alkenyl, substituted C₂-C₁₂ alkenyl, C₂-C₁₂ alkynyl, substituted C₂-C₁₂ alkynyl, C₅-C₂₀ aryl, substituted C₅-C₂₀ aryl, heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C₅-C₇ alicyclic radical, substituted C₅-C₇ alicyclic radical, halogen, OJ₁, NJ₁J₂, SJ₁, N₃, COOJ₁, acyl (C(=O)-H), substituted acyl, CN, sulfonyl (S(=O)₂-J₁), or sulfoxyl (S(=O)-J₁); and each J₁ and J₂ is, independently, H, C₁-C₁₂ alkyl, substituted C₁-C₁₂ alkyl, C₂-C₁₂ alkenyl, substituted C₂-C₁₂ alkenyl, C₂-C₁₂ alkynyl, substituted C₂-C₁₂ alkynyl, C₅-C₂₀ aryl, substituted C₅-C₂₀ aryl, acyl (C(=O)-H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C₁-C₁₂ aminoalkyl, substituted C₁-C₁₂ aminoalkyl, or a protecting group.

Additional bicyclic sugar moieties are known in the art, see, for example: Freier et al., *Nucleic Acids Research*, 1997, 25(22), 4429-4443; Albaek et al., *J. Org. Chem.*, 2006, 71, 7731-7740; Singh et al., *Chem. Commun.*, 1998, 4, 455-456; Koshkin et al., *Tetrahedron*, 1998, 54, 3607-3630; Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A.*, 2000, 97, 5633-5638; Kumar et al., *Bioorg. Med. Chem. Lett.*, 1998, 8, 2219-2222; Singh et al., *J. Org. Chem.*, 1998, 63, 10035-10039; Srivastava et al., *J. Am. Chem. Soc.*, 2007, 129, 8362-8379; Elayadi et al., *Curr. Opinion Inven. Drugs*, 2001, 2, 558-561; Braasch et al., *Chem. Biol.*, 2001, 8, 1-7; Orum et al., *Curr. Opinion Mol. Ther.* 2001, 3, 239-243; Wengel et al., U.S. Pat. No. 7,053,207; Imanishi et al., U.S. Pat. No. 6,268,490; Imanishi et al. U.S. Pat. No. 6,770,748; Imanishi et al., U.S. RE44,779; Wengel et al., U.S. Pat. No. 6,794,499; Wengel et al., U.S. Pat. No. 6,670,461; Wengel et al., U.S. Pat. No. 7,034,133; Wengel et al., U.S. Pat. No. 8,080,644; Wengel et al., U.S. Pat. No. 8,034,909; Wengel et al., U.S. Pat. No. 8,153,365; Wengel et al., U.S. Pat. No. 7,572,582; and Ramasamy et al., U.S. Pat. No. 6,525,191; Torsten et al., WO 2004/106356; Wengel et al., WO 1999/014226; Seth et al., WO 2007/134181; Seth et al., U.S. Pat. No. 7,547,684; Seth et al., U.S. Pat. No. 7,666,854; Seth et al., U.S. Pat. No. 8,088,746; Seth et al., U.S. Pat. No. 7,750,131; Seth et al., U.S. Pat. No. 8,030,467; Seth et al., U.S. Pat. No. 8,268,980; Seth et al., U.S. Pat. No. 8,546,556; Seth et al., U.S. Pat. No. 8,530,640; Migawa et al., U.S. Pat. No. 9,012,421; Seth et al., U.S. Pat. No. 8,501,805; Allerson et al., US2008/0039618; and Migawa et al., US2015/0191727.

In certain embodiments, bicyclic sugar moieties and nucleosides incorporating such bicyclic sugar moieties are further defined by isomeric configuration. For example, an LNA nucleoside (described herein) may be in the α-L configuration or in the β-D configuration.

156

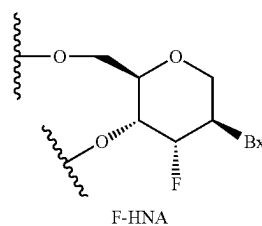


α-L-methyleneoxy (4'-CH₂-O-2') or α-L-LNA bicyclic nucleosides have been incorporated into oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research*, 2003, 21, 6365-6372). Herein, general descriptions of bicyclic nucleosides include both isomeric configurations. When the positions of specific bicyclic nucleosides (e.g., LNA or cEt) are identified in exemplified embodiments herein, they are in the β-D configuration, unless otherwise specified.

In certain embodiments, modified sugar moieties comprise one or more non-bridging sugar substituent and one or more bridging sugar substituent (e.g., 5'-substituted and 4'-2' bridged sugars).

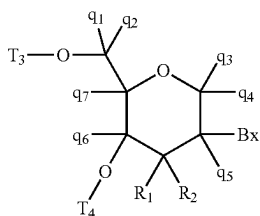
In certain embodiments, modified sugar moieties are sugar surrogates. In certain such embodiments, the oxygen atom of the sugar moiety is replaced, e.g., with a sulfur, carbon or nitrogen atom. In certain such embodiments, such modified sugar moieties also comprise bridging and/or non-bridging substituents as described herein. For example, certain sugar surrogates comprise a 4'-sulfur atom and a substitution at the 2'-position (see, e.g., Bhat et al., U.S. Pat. No. 7,875,733 and Bhat et al., U.S. Pat. No. 7,939,677) and/or the 5' position.

In certain embodiments, sugar surrogates comprise rings having other than 5 atoms. For example, in certain embodiments, a sugar surrogate comprises a six-membered tetrahydropyran ("THP"). Such tetrahydropyrans may be further modified or substituted. Nucleosides comprising such modified tetrahydropyrans include, but are not limited to, hexitol nucleic acid ("HNA"), anitol nucleic acid ("ANA"), manitol nucleic acid ("MNA") (see e.g., Leumann, C.J. *Bioorg. & Med. Chem.* 2002, 10, 841-854), fluoro HNA:



("F-HNA", see e.g., Swayze et al., U.S. Pat. No. 8,088,904; Swayze et al., U.S. Pat. No. 8,440,803; and Swayze et al., U.S. Pat. No. 9,005,906) F-HNA can also be referred to as a F-THP or 3'-fluoro tetrahydropyran, and nucleosides comprising additional modified THP compounds having the formula:

157



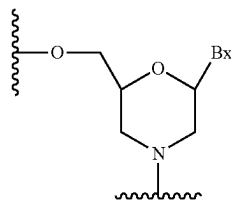
wherein, independently, for each of said modified THP nucleoside:

Bx is a nucleobase moiety;

T₃ and T₄ are each, independently, an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide or one of T₃ and T₄ is an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide and the other of T₃ and T₄ is H, a hydroxyl protecting group, a linked conjugate group, or a 5' or 3'-terminal group; q₁, q₂, q₃, q₄, q₅, q₆ and q₇ are each, independently, H, C₁-C₆ alkyl, substituted C₁-C₆ alkyl, C₂-C₆ alkenyl, substituted C₂-C₆ alkenyl, C₂-C₆ alkynyl, or substituted C₂-C₆ alkynyl; and each of R₁ and R₂ is independently selected from among: hydrogen, halogen, substituted or unsubstituted alkoxy, NJ₁J₂, SJ₁, N₃, OC(=X)J₁, OC(=X)NJ₁J₂, NJ₃C(=X)NJ₁J₂, and CN, wherein X is O, S or NJ₁, and each J₁, J₂, and J₃ is, independently, H or C₁-C₆ alkyl.

In certain embodiments, modified THP nucleosides are provided wherein q₁, q₂, q₃, q₄, q₅, q₆ and q₇ are each H. In certain embodiments, at least one of q₁, q₂, q₃, q₄, q₅, q₆ and q₇ is other than H. In certain embodiments, at least one of q₁, q₂, q₃, q₄, q₅, q₆ and q₇ is methyl. In certain embodiments, modified THP nucleosides are provided wherein one of R₁ and R₂ is F. In certain embodiments, R₁ is F and R₂ is H, in certain embodiments, R₁ is methoxy and R₂ is H, and in certain embodiments, R₁ is methoxyethoxy and R₂ is H.

In certain embodiments, sugar surrogates comprise rings having more than 5 atoms and more than one heteroatom. For example, nucleosides comprising morpholino sugar moieties and their use in oligonucleotides have been reported (see, e.g., Braasch et al., *Biochemistry*, 2002, 41, 4503-4510 and Summerton et al., U.S. Pat. No. 5,698,685; Summerton et al., U.S. Pat. No. 5,166,315; Summerton et al., U.S. Pat. No. 5,185,444; and Summerton et al., U.S. Pat. No. 5,034,506. As used here, the term "morpholino" means a sugar surrogate having the following structure:



In certain embodiments, morpholinos may be modified, for example, by adding or altering various substituent groups from the above morpholino structure. Such sugar surrogates are referred to herein as "modified morpholinos."

In certain embodiments, sugar surrogates comprise acyclic moieties. Examples of nucleosides and oligonucleotides comprising such acyclic sugar surrogates include, but are

158

not limited to: peptide nucleic acid ("PNA"), acyclic butyl nucleic acid (see, e.g., Kumar et al., *Org. Biomol. Chem.*, 2013, 11, 5853-5865), and nucleosides and oligonucleotides described in Manoharan et al., US2013/130378.

Many other bicyclic and tricyclic sugar and sugar surrogate ring systems are known in the art that can be used in modified nucleosides.

In certain embodiments, modified oligonucleotides comprise a 2'-OMe modified nucleoside in the gap of a cEt gapmer. In certain embodiments, the 2'-OMe modified nucleoside is at position 2 or 3 of the gap. In certain embodiments, the 2'-OMe modified nucleoside is at position 2 of the gap. In certain embodiments, modified oligonucleotides have a sugar motif kkkdmdddddskkk, wherein each k represents a cEt nucleoside, each d represents a 2'-β-D-deoxyribosyl nucleoside, and each m represents a nucleoside comprising a 2'-OMe sugar moiety ("2'-OMe nucleoside").

In certain embodiments, modified oligonucleotides have a formula A-B-C, wherein A is a 5'-region, B is a central region, and C is a 3'-region. In certain embodiments, A and C each consist of 1-5 linked nucleosides and B consists of 7-11 linked nucleosides. In certain embodiments, A and C consist of 3 linked cEt nucleosides. In certain embodiments, the second nucleoside of the central region comprises a 2'-OMe modified sugar moiety, and each of the remaining central region nucleosides comprises a 2'-p-D-deoxyribosyl sugar moiety.

In certain embodiments, modified oligonucleotides comprise a neutral phosphonate linkage at one or more positions. In certain embodiments, modified oligonucleotides comprise a methoxypropyl phosphonate linkage at one or more positions. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer, and the internucleoside linkage motif comprises at least one methoxypropyl phosphonate linkage in the gap. In certain embodiments, the methoxypropyl phosphonate linkage is between the second and third nucleosides in the gap. In certain embodiments, the methoxypropyl phosphonate linkage is between the third and fourth nucleosides in the gap.

2. Modified Nucleobases

Nucleobase (or base) modifications or substitutions are structurally distinguishable from, yet functionally interchangeable with, naturally occurring or synthetic unmodified nucleobases. Both natural and modified nucleobases are capable of participating in hydrogen bonding. Such nucleobase modifications can impart nuclease stability, binding affinity or some other beneficial biological property to antisense compounds.

In certain embodiments, compounds described herein comprise modified oligonucleotides. In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising an unmodified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleosides that does not comprise a nucleobase, referred to as an abasic nucleoside.

In certain embodiments, modified nucleobases are selected from: 5-substituted pyrimidines, 6-azapyrimidines, alkyl or alkynyl substituted pyrimidines, alkyl substituted purines, and N-2, N-6 and O-6 substituted purines. In certain embodiments, modified nucleobases are selected from: 2-aminopropyladenine, 5-hydroxymethyl cytosine, 5-methylcytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-N-methylguanine, 6-N-methyladenine, 2-propyladenine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-propynyl

(C≡C—CH₃) uracil, 5-propynylcytosine, 6-azouracil, 6-azocytosine, 6-azothymine, 5-ribosyluracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl, 8-aza and other 8-substituted purines, 5-halo, particularly, 5-bromo, 5-trifluoromethyl, 5-halouracil, and 5-halocytosine, 7-methylguanine, 7-methyladenine, 2-F-adenine, 2-aminoadenine, 7-deazaguanine, 7-deazaadenine, 3-deazaguanine, 3-deazaadenine, 6-N-benzoyladenine, 2-N-isobutyrylguanine, 4-N-benzoylcytosine, 4-N-benzoyluracil, 5-methyl 4-N-benzoylcytosine, 5-methyl 4-N-benzoyluracil, universal bases, hydrophobic bases, promiscuous bases, size-expanded bases, and fluorinated bases. Further modified nucleobases include tricyclic pyrimidines, such as 1,3-diazaphenoxazine-2-one, 1,3-diazaphenothiazine-2-one, and 9-(2-aminoethoxy)-1,3-diazaphenoxazine-2-one (G-clamp). Modified nucleobases may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example, 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in Merigan et al., U.S. Pat. No. 3,687,808, those disclosed in The Concise Encyclopedia Of Polymer Science And Engineering, Kroschwitz, J. I., Ed., John Wiley & Sons, 1990, 858-859; Englisch et al., Angewandte Chemie, International Edition, 1991, 30, 613; Sanghvi, Y. S., Chapter 15, Antisense Research and Applications, Crooke, S. T. and Lebleu, B., Eds., CRC Press, 1993, 273-288; and those disclosed in Chapters 6 and 15, Antisense Drug Technology, Crooke S. T., Ed., CRC Press, 2008, 163-166 and 442-443.

Publications that teach the preparation of certain of the above noted modified nucleobases, as well as other modified nucleobases include without limitation, Manoharan et al., US2003/0158403, Manoharan et al., US2003/0175906; Dinh et al., U.S. Pat. No. 4,845,205; Spielvogel et al., U.S. Pat. No. 5,130,302; Rogers et al., U.S. Pat. No. 5,134,066; Bischofberger et al., U.S. Pat. No. 5,175,273; Urdea et al., U.S. Pat. No. 5,367,066; Benner et al., U.S. Pat. No. 5,432,272; Matteucci et al., U.S. Pat. No. 5,434,257; Gmeiner et al., U.S. Pat. No. 5,457,187; Cook et al., U.S. Pat. No. 5,459,255; Froehler et al., U.S. Pat. No. 5,484,908; Matteucci et al., U.S. Pat. No. 5,502,177; Hawkins et al., U.S. Pat. No. 5,525,711; Haralambidis et al., U.S. Pat. No. 5,552,540; Cook et al., U.S. Pat. No. 5,587,469; Froehler et al., U.S. Pat. No. 5,594,121; Switzer et al., U.S. Pat. No. 5,596,091; Cook et al., U.S. Pat. No. 5,614,617; Froehler et al., U.S. Pat. No. 5,645,985; Cook et al., U.S. Pat. No. 5,681,941; Cook et al., U.S. Pat. No. 5,811,534; Cook et al., U.S. Pat. No. 5,750,692; Cook et al., U.S. Pat. No. 5,948,903; Cook et al., U.S. Pat. No. 5,587,470; Cook et al., U.S. Pat. No. 5,457,191; Matteucci et al., U.S. Pat. No. 5,763,588; Froehler et al., U.S. Pat. No. 5,830,653; Cook et al., U.S. Pat. No. 5,808,027; Cook et al., U.S. Pat. No. 6,166,199; and Matteucci et al., U.S. Pat. No. 6,005,096.

In certain embodiments, compounds targeted to a HSD17B13 nucleic acid comprise one or more modified nucleobases. In certain embodiments, the modified nucleobase is 5-methylcytosine. In certain embodiments, each cytosine is a 5-methylcytosine.

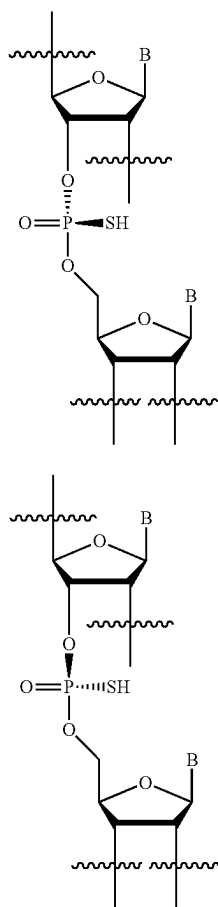
Modified Internucleoside Linkages

The naturally occurring internucleoside linkage of RNA and DNA is a Y to 5 phosphodiester linkage. In certain

embodiments, compounds described herein having one or more modified, i.e. non-naturally occurring, internucleoside linkages are often selected over compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

Representative internucleoside linkages having a chiral center include but are not limited to alkylphosphonates and phosphorothioates. Modified oligonucleotides comprising internucleoside linkages having a chiral center can be prepared as populations of modified oligonucleotides comprising stereorandom internucleoside linkages, or as populations of modified oligonucleotides comprising phosphorothioate linkages in particular stereochemical configurations. In certain embodiments, populations of modified oligonucleotides comprise phosphorothioate internucleoside linkages wherein all of the phosphorothioate internucleoside linkages are stereorandom. Such modified oligonucleotides can be generated using synthetic methods that result in random selection of the stereochemical configuration of each phosphorothioate linkage. Nonetheless, as is well understood by those of skill in the art, each individual phosphorothioate of each individual oligonucleotide molecule has a defined stereoconfiguration. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising one or more particular phosphorothioate internucleoside linkages in a particular, independently selected stereochemical configuration. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 65% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 70% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 80% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 90% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 99% of the molecules in the population. Such chirally enriched populations of modified oligonucleotides can be generated using synthetic methods known in the art, e.g., methods described in Oka et al., *JACS* 125, 8307 (2003), Wan et al. *Nuc. Acid. Res.* 42, 13456 (2014), and WO 2017/015555. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one indicated phosphorothioate in the (Sp) configuration. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one phosphorothioate in the (Rp) configuration. In certain embodiments, modified oligonucleotides comprising (Rp) and/or (Sp) phosphorothioates comprise one or more of the following formulas, respectively, wherein "B" indicates a nucleobase:

161



Unless otherwise indicated, chiral internucleoside linkages of modified oligonucleotides described herein can be stereorandom or in a particular stereochemical configuration.

In certain embodiments, compounds targeted to a HSD17B13 nucleic acid comprise one or more modified internucleoside linkages. In certain embodiments, the modified internucleoside linkages are phosphorothioate linkages. In certain embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage.

In certain embodiments, compounds described herein comprise oligonucleotides. Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiester, phosphotriester, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

In certain embodiments, nucleosides of modified oligonucleotides may be linked together using any internucleoside linkage. The two main classes of internucleoside linking groups are defined by the presence or absence of a phosphorus atom. Representative phosphorus-containing internucleoside linkages include, but are not limited to, phosphates, which contain a phosphodiester bond ("P=O") (also referred to as unmodified or naturally occurring linkages), phosphotriesters, methylphosphonates, phosphoramidates,

162

and phosphorothioates ("P=S"), and phosphorodithioates ("HS=P=S"). Representative non-phosphorus containing internucleoside linking groups include, but are not limited to, methylenemethylimino ($-\text{CH}_2-\text{N}(\text{CH}_3)-\text{O}-\text{CH}_2-$), thiodiester, thionocarbamate ($-\text{O}-\text{C}(=\text{O})(\text{NH})-\text{S}-$), siloxane ($-\text{O}-\text{SiH}_2-\text{O}-$), and N,N'-dimethylhydrazine ($-\text{CH}_2-\text{N}(\text{CH}_3)-\text{N}(\text{CH}_3)-$). Modified internucleoside linkages, compared to naturally occurring phosphate linkages, can be used to alter, typically increase, nuclease resistance of the oligonucleotide. In certain embodiments, internucleoside linkages having a chiral atom can be prepared as a racemic mixture, or as separate enantiomers. Representative chiral internucleoside linkages include, but are not limited to, alkylphosphonates and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing internucleoside linkages are well known to those skilled in the art.

Neutral internucleoside linkages include, without limitation, phosphotriesters, methylphosphonates, MMI ($3'-\text{CH}_2-\text{N}(\text{CH}_3)-\text{O}-5'$), amide-3 ($3'-\text{CH}_2-\text{C}(=\text{O})-\text{N}(\text{H})-5'$), amide-4 ($3'-\text{CH}_2-\text{N}(\text{H})-\text{C}(=\text{O})-5'$), formacetal ($3'-\text{O}-\text{CH}_2-\text{O}-5'$), methoxypropyl, and thioformacetal ($3'-\text{S}-\text{CH}_2-\text{O}-5'$). Further neutral internucleoside linkages include nonionic linkages comprising siloxane (dialkylsiloxane), carboxylate ester, carboxamide, sulfide, sulfonate ester and amides (See, for example: Carbohydrate Modifications in Antisense Research; Y. S. Sanghvi and P. D. Cook, Eds., ACS Symposium Series 580; Chapters 3 and 4, 40-65). Further neutral internucleoside linkages include nonionic linkages comprising mixed N, O, S and CH_2 component parts.

In certain embodiments, oligonucleotides comprise modified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or modified internucleoside linkage motif. In certain embodiments, internucleoside linkages are arranged in a gapped motif. In such embodiments, the internucleoside linkages in each of two wing regions are different from the internucleoside linkages in the gap region. In certain embodiments, the internucleoside linkages in the wings are phosphodiester and the internucleoside linkages in the gap are phosphorothioate. The nucleoside motif is independently selected, so such oligonucleotides having a gapped internucleoside linkage motif may or may not have a gapped nucleoside motif and, if it does have a gapped nucleoside motif, the wing and gap lengths may or may not be the same.

In certain embodiments, oligonucleotides comprise a region having an alternating internucleoside linkage motif. In certain embodiments, oligonucleotides comprise a region of uniformly modified internucleoside linkages. In certain such embodiments, the oligonucleotide comprises a region that is uniformly linked by phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide is uniformly linked by phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate and at least one internucleoside linkage is phosphorothioate.

In certain embodiments, the oligonucleotide comprises at least 6 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 8 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 10 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least

6 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 8 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 10 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 12 consecutive phosphorothioate internucleoside linkages. In certain such embodiments, at least one such block is located at the 3' end of the oligonucleotide. In certain such embodiments, at least one such block is located within 3 nucleosides of the 3' end of the oligonucleotide.

In certain embodiments, oligonucleotides comprise one or more methylphosphonate linkages. In certain embodiments, oligonucleotides having a gapmer nucleoside motif comprise a linkage motif comprising all phosphorothioate linkages except for one or two methylphosphonate linkages. In certain embodiments, one methylphosphonate linkage is in the central gap of an oligonucleotide having a gapmer nucleoside motif.

In certain embodiments, it is desirable to arrange the number of phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages to maintain nuclease resistance. In certain embodiments, it is desirable to arrange the number and position of phosphorothioate internucleoside linkages and the number and position of phosphodiester internucleoside linkages to maintain nuclease resistance. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased while still maintaining nuclease resistance. In certain embodiments, it is desirable to decrease the number of phosphorothioate internucleoside linkages while retaining nuclease resistance. In certain embodiments, it is desirable to increase the number of phosphodiester internucleoside linkages while retaining nuclease resistance.

3. Certain Motifs

In certain embodiments, compounds described herein comprise oligonucleotides. Oligonucleotides can have a motif, e.g. a pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified sugar. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more modified internucleoside linkage. In such embodiments, the modified, unmodified, and differently modified sugar moieties, nucleobases, and/or internucleoside linkages of a modified oligonucleotide define a pattern or motif. In certain embodiments, the patterns of sugar moieties, nucleobases, and internucleoside linkages are each independent of one another. Thus, a modified oligonucleotide may be described by its sugar motif, nucleobase motif and/or internucleoside linkage motif (as used herein, nucleobase motif describes the modifications to the nucleobases independent of the sequence of nucleobases).

a. Certain Sugar Motifs

In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise one or more type of modified sugar and/or unmodified sugar moiety arranged along the oligo-

nucleotide or region thereof in a defined pattern or sugar motif. In certain instances, such sugar motifs include, but are not limited to, any of the sugar modifications discussed herein.

In certain embodiments, modified oligonucleotides comprise or consist of a region having a gapmer motif, which comprises two external regions or "wings" and a central or internal region or "gap". The three regions of a gapmer motif (the 5'-wing, the gap, and the 3'-wing) form a contiguous sequence of nucleosides, wherein at least some of the sugar moieties of the nucleosides of each of the wings differ from at least some of the sugar moieties of the nucleosides of the gap. Specifically, at least the sugar moieties of the nucleosides of each wing that are closest to the gap (the 3'-most nucleoside of the 5'-wing and the 5'-most nucleoside of the 3'-wing) differ from the sugar moiety of the neighboring gap nucleosides, thus defining the boundary between the wings and the gap (i.e., the wing/gap junction). In certain embodiments, the sugar moieties within the gap are the same as one another. In certain embodiments, the gap includes one or more nucleosides having a sugar moiety that differs from the sugar moiety of one or more other nucleosides of the gap. In certain embodiments, the sugar motifs of the two wings are the same as one another (symmetric gapmer). In certain embodiments, the sugar motif of the 5-wing differs from the sugar motif of the 3-wing (asymmetric gapmer).

In certain embodiments, the wings of a gapmer comprise 1-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 2-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 3-5 nucleosides. In certain embodiments, the nucleosides of a gapmer are all modified nucleosides.

In certain embodiments, the gap of a gapmer comprises 7-12 nucleosides. In certain embodiments, the gap of a gapmer comprises 7-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 8-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 10 nucleosides. In certain embodiments, each nucleoside of the gap of a gapmer is an unmodified 2'-deoxy nucleoside.

In certain embodiments, the gapmer is a deoxy gapmer. In such embodiments, the nucleosides on the gap side of each wing/gap junction are unmodified 2'-deoxy nucleosides and the nucleosides on the wing sides of each wing/gap junction are modified nucleosides. In certain such embodiments, each nucleoside of the gap is an unmodified 2'-deoxy nucleoside. In certain such embodiments, each nucleoside of each wing is a modified nucleoside.

In certain embodiments, a modified oligonucleotide has a fully modified sugar motif wherein each nucleoside of the modified oligonucleotide comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif wherein each nucleoside of the region comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif, wherein each nucleoside within the fully modified region comprises the same modified sugar moiety, referred to herein as a uniformly modified sugar motif. In certain embodiments, a fully modified oligonucleotide is a uniformly modified oligonucleotide. In certain embodiments, each nucleoside of a uniformly modified oligonucleotide comprises the same 2'-modification.

b. Certain Nucleobase Motifs

In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise modified and/or unmodified nucleobases arranged along the oligonucleotide or region thereof

in a defined pattern or motif. In certain embodiments, each nucleobase is modified. In certain embodiments, none of the nucleobases are modified. In certain embodiments, each purine or each pyrimidine is modified. In certain embodiments, each adenine is modified. In certain embodiments, each guanine is modified. In certain embodiments, each thymine is modified. In certain embodiments, each uracil is modified. In certain embodiments, each cytosine is modified. In certain embodiments, some or all of the cytosine nucleobases in a modified oligonucleotide are 5-methylcytosines.

In certain embodiments, modified oligonucleotides comprise a block of modified nucleobases. In certain such embodiments, the block is at the 3'-end of the oligonucleotide. In certain embodiments, the block is within 3 nucleosides of the 3'-end of the oligonucleotide. In certain embodiments, the block is at the 5'-end of the oligonucleotide. In certain embodiments, the block is within 3 nucleosides of the 5'-end of the oligonucleotide.

In certain embodiments, oligonucleotides having a gapmer motif comprise a nucleoside comprising a modified nucleobase. In certain such embodiments, one nucleoside comprising a modified nucleobase is in the central gap of an oligonucleotide having a gapmer motif. In certain such embodiments, the sugar moiety of said nucleoside is a 2'-deoxyribose moiety. In certain embodiments, the modified nucleobase is selected from: a 2-thiopyrimidine and a 5-propynepyrimidine.

c. Certain Internucleoside Linkage Motifs

In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise modified and/or unmodified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or motif. In certain embodiments, essentially each internucleoside linking group is a phosphate internucleoside linkage (P=O). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is a phosphorothioate (P=S). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is independently selected from a phosphorothioate and phosphate internucleoside linkage. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer and the internucleoside linkages within the gap are all modified. In certain such embodiments, some or all of the internucleoside linkages in the wings are unmodified phosphate linkages. In certain embodiments, the terminal internucleoside linkages are modified. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer, and the internucleoside linkage motif comprises at least one phosphodiester internucleoside linkage in at least one wing, wherein the at least one phosphodiester linkage is not a terminal internucleoside linkage, and the remaining internucleoside linkages are phosphorothioate internucleoside linkages. In certain such embodiments, all of the phosphorothioate linkages are stereorandom. In certain embodiments, all of the phosphorothioate linkages in the wings are (Sp) phosphorothioates, and the gap comprises at least one Sp, Sp, Rp motif. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising such internucleoside linkage motifs.

4. Certain Modified Oligonucleotides

In certain embodiments, compounds described herein comprise modified oligonucleotides. In certain embodiments, the above modifications (sugar, nucleobase, internucleoside linkage) are incorporated into a modified oligonucleotide. In certain embodiments, modified

oligonucleotides are characterized by their modification, motifs, and overall lengths. In certain embodiments, such parameters are each independent of one another. Thus, unless otherwise indicated, each internucleoside linkage of an oligonucleotide having a gapmer sugar motif may be modified or unmodified and may or may not follow the gapmer modification pattern of the sugar modifications. For example, the internucleoside linkages within the wing regions of a sugar gapmer may be the same or different from one another and may be the same or different from the internucleoside linkages of the gap region of the sugar motif. Likewise, such gapmer oligonucleotides may comprise one or more modified nucleobases independent of the gapmer pattern of the sugar modifications. Furthermore, in certain instances, an oligonucleotide is described by an overall length or range and by lengths or length ranges of two or more regions (e.g., a regions of nucleosides having specified sugar modifications). In such circumstances, it may be possible to select numbers for each range that result in an oligonucleotide having an overall length falling outside the specified range. In such circumstances, both elements must be satisfied. For example, in certain embodiments, a modified oligonucleotide consists of 15-20 linked nucleosides and has a sugar motif consisting of three regions, A, B, and C, wherein region A consists of 2-6 linked nucleosides having a specified sugar motif, region B consists of 6-10 linked nucleosides having a specified sugar motif, and region C consists of 2-6 linked nucleosides having a specified sugar motif. Such embodiments do not include modified oligonucleotides where A and C each consist of 6 linked nucleosides and B consists of 10 linked nucleosides (even though those numbers of nucleosides are permitted within the requirements for A, B, and C) because the overall length of such oligonucleotide will be 22, which exceeds the upper limit of the overall length of the modified oligonucleotide (20). Herein, if a description of an oligonucleotide is silent with respect to one or more parameters, such parameter is not limited. Thus, a modified oligonucleotide described only as having a gapmer sugar motif without further description may have any length, internucleoside linkage motif, and nucleobase motif. Unless otherwise indicated, all modifications are independent of nucleobase sequence.

Certain Conjugated Compounds

In certain embodiments, the compounds described herein comprise or consist of an oligonucleotide (modified or unmodified) and, optionally, one or more conjugate groups and/or terminal groups. Conjugate groups consist of one or more conjugate moiety and a conjugate linker which links the conjugate moiety to the oligonucleotide. Conjugate groups may be attached to either or both ends of an oligonucleotide and/or at any internal position. In certain embodiments, conjugate groups are attached to the 2'-position of a nucleoside of a modified oligonucleotide. In certain embodiments, conjugate groups that are attached to either or both ends of an oligonucleotide are terminal groups. In certain such embodiments, conjugate groups or terminal groups are attached at the 3' and/or 5'-end of oligonucleotides. In certain such embodiments, conjugate groups (or terminal groups) are attached at the 3'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 3'-end of oligonucleotides. In certain embodiments, conjugate groups (or terminal groups) are attached at the 5'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 5'-end of oligonucleotides.

In certain embodiments, the oligonucleotide is modified. In certain embodiments, the oligonucleotide of a compound has a nucleobase sequence that is complementary to a target

nucleic acid. In certain embodiments, oligonucleotides are complementary to a messenger RNA (mRNA). In certain embodiments, oligonucleotides are complementary to a pre-mRNA. In certain embodiments, oligonucleotides are complementary to a sense transcript.

Examples of terminal groups include but are not limited to conjugate groups, capping groups, phosphate moieties, protecting groups, modified or unmodified nucleosides, and two or more nucleosides that are independently modified or unmodified.

In certain embodiments, oligonucleotides are covalently attached to one or more conjugate groups. In certain embodiments, conjugate groups modify one or more properties of the attached oligonucleotide, including but not limited to pharmacodynamics, pharmacokinetics, stability, binding, absorption, tissue distribution, cellular distribution, cellular uptake, charge and clearance. In certain embodiments, conjugate groups impart a new property on the attached oligonucleotide, e.g., fluorophores or reporter groups that enable detection of the oligonucleotide. Certain conjugate groups and conjugate moieties have been described previously, for example: cholesterol moiety (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86, 6553-6556), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4, 1053-1060), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N. Y. Acad. Sci.*, 1992, 660, 306-309; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3, 2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20, 533-538), an aliphatic chain, e.g., do-decan-diol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10, 1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259, 327-330; Svinarchuk et al., *Biochimie*, 1993, 75, 49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36, 3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18, 3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14, 969-973), or adamantane acetic acid a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264, 229-237), an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Cooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277, 923-937), a tocopherol group (Nishina et al., *Molecular Therapy Nucleic Acids*, 2015, 4, e220; and Nishina et al., *Molecular Therapy*, 2008, 16, 734-740), or a GalNAc cluster (e.g., WO2014/179620).

1. Conjugate Moieties

Conjugate moieties include, without limitation, intercalators, reporter molecules, polyamines, polyamides, peptides, carbohydrates (e.g., GalNAc), vitamin moieties, polyethylene glycols, thioethers, polyethers, cholesterol, thiocholesterols, cholic acid moieties, folate, lipids, phospholipids, biotin, phenazine, phenanthridine, anthraquinone, adamantane, acridine, fluoresceins, rhodamines, coumarins, fluorophores, and dyes.

In certain embodiments, a conjugate moiety comprises an active drug substance, for example, aspirin, warfarin, phenylbutazone, ibuprofen, suprofen, fen-bufen, ketoprofen, (S)-(+)-pranoprofen, carprofen, danylsarcosine, 2,3,5-triiodobenzoic acid, fingolimod, flufenamic acid, folinic acid, a benzothiadiazide, chlorothiazide, a diazepine, indomethacin, a barbiturate, a cephalosporin, a sulfa drug, an antidiabetic, an antibacterial or an antibiotic.

2. Conjugate Linkers

Conjugate moieties are attached to oligonucleotides through conjugate linkers. In certain compounds, the conjugate linker is a single chemical bond (i.e., the conjugate moiety is attached directly to an oligonucleotide through a

single bond). In certain compounds, a conjugate moiety is attached to an oligonucleotide via a more complex conjugate linker comprising one or more conjugate linker moieties, which are sub-units making up a conjugate linker. In certain embodiments, the conjugate linker comprises a chain structure, such as a hydrocarbyl chain, or an oligomer of repeating units such as ethylene glycol, nucleosides, or amino acid units.

In certain embodiments, a conjugate linker comprises one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether, and hydroxylamino. In certain such embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and amide groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and ether groups. In certain embodiments, the conjugate linker comprises at least one phosphorus moiety. In certain embodiments, the conjugate linker comprises at least one phosphate group. In certain embodiments, the conjugate linker includes at least one neutral linking group.

In certain embodiments, conjugate linkers, including the conjugate linkers described above, are bifunctional linking moieties, e.g., those known in the art to be useful for attaching conjugate groups to parent compounds, such as the oligonucleotides provided herein. In general, a bifunctional linking moiety comprises at least two functional groups. One of the functional groups is selected to bind to a particular site on a parent compound and the other is selected to bind to a conjugate group. Examples of functional groups used in a bifunctional linking moiety include but are not limited to electrophiles for reacting with nucleophilic groups and nucleophiles for reacting with electrophilic groups. In certain embodiments, bifunctional linking moieties comprise one or more groups selected from amino, hydroxyl, carboxylic acid, thiol, alkyl, alkenyl, and alkynyl.

Examples of conjugate linkers include but are not limited to pyrrolidine, 8-amino-3,6-dioxaoctanoic acid (ADO), succinimidyl 4-(N-maleimidomethyl) cyclohexane-1-carboxylate (SMCC) and 6-aminohexanoic acid (AHEX or AHA). Other conjugate linkers include but are not limited to substituted or unsubstituted C₁-C₁₀ alkyl, substituted or unsubstituted C₂-C₁₀ alkenyl or substituted or unsubstituted C₂-C₁₀ alkynyl, wherein a nonlimiting list of preferred substituent groups includes hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro, thiol, thioalkoxy, halogen, alkyl, aryl, alkenyl and alkynyl.

In certain embodiments, conjugate linkers comprise 1-10 linker-nucleosides. In certain embodiments, such linker-nucleosides are modified nucleosides. In certain embodiments such linker-nucleosides comprise a modified sugar moiety. In certain embodiments, linker-nucleosides are unmodified. In certain embodiments, linker-nucleosides comprise an optionally protected heterocyclic base selected from a purine, substituted purine, pyrimidine or substituted pyrimidine. In certain embodiments, a cleavable moiety is a nucleoside selected from uracil, thymine, cytosine, 4-N-benzoylcytosine, 5-methylcytosine, 4-N-benzoyl-5-methylcytosine, adenine, 6-N-benzoyladenine, guanine and 2-N-isobutyrylguanine. It is typically desirable for linker-nucleosides to be cleaved from the compound after it reaches a target tissue. Accordingly, linker-nucleosides are typically linked to one another and to the remainder of the compound through cleavable bonds. In certain embodiments, such cleavable bonds are phosphodiester bonds.

169

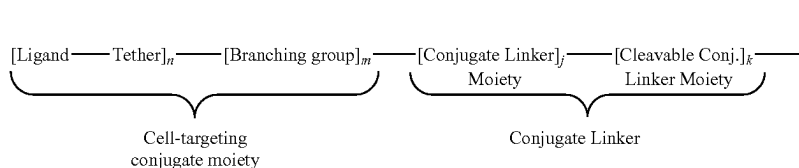
Herein, linker-nucleosides are not considered to be part of the oligonucleotide. Accordingly, in embodiments in which an compound comprises an oligonucleotide consisting of a specified number or range of linked nucleosides and/or a specified percent complementarity to a reference nucleic acid and the compound also comprises a conjugate group comprising a conjugate linker comprising linker-nucleosides, those linker-nucleosides are not counted toward the length of the oligonucleotide and are not used in determining the percent complementarity of the oligonucleotide for the reference nucleic acid. For example, a compound may comprise (1) a modified oligonucleotide consisting of 8-30 nucleosides and (2) a conjugate group comprising 1-10 linker-nucleosides that are contiguous with the nucleosides of the modified oligonucleotide. The total number of contiguous linked nucleosides in such a compound is more than 30. Alternatively, a compound may comprise a modified

170

In certain embodiments, a cleavable moiety comprises or consists of one or more linker-nucleosides. In certain such embodiments, the one or more linker-nucleosides are linked to one another and/or to the remainder of the compound through cleavable bonds. In certain embodiments, such cleavable bonds are unmodified phosphodiester bonds. In certain embodiments, a cleavable moiety is 2'-deoxy nucleoside that is attached to either the 3' or 5'-terminal nucleoside of an oligonucleotide by a phosphate internucleoside linkage and covalently attached to the remainder of the conjugate linker or conjugate moiety by a phosphate or phosphorothioate linkage. In certain such embodiments, the cleavable moiety is 2'-deoxyadenosine.

3. Certain Cell-Targeting Conjugate Moieties

In certain embodiments, a conjugate group comprises a cell-targeting conjugate moiety. In certain embodiments, a conjugate group has the general formula:



oligonucleotide consisting of 8-30 nucleosides and no conjugate group. The total number of contiguous linked nucleosides in such a compound is no more than 30. Unless otherwise indicated conjugate linkers comprise no more than 10 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 5 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 3 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 2 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 1 linker-nucleoside.

In certain embodiments, it is desirable for a conjugate group to be cleaved from the oligonucleotide. For example, in certain circumstances compounds comprising a particular conjugate moiety are better taken up by a particular cell type, but once the compound has been taken up, it is desirable that the conjugate group be cleaved to release the unconjugated or parent oligonucleotide. Thus, certain conjugate linkers may comprise one or more cleavable moieties. In certain embodiments, a cleavable moiety is a cleavable bond. In certain embodiments, a cleavable moiety is a group of atoms comprising at least one cleavable bond. In certain embodiments, a cleavable moiety comprises a group of atoms having one, two, three, four, or more than four cleavable bonds. In certain embodiments, a cleavable moiety is selectively cleaved inside a cell or subcellular compartment, such as a lysosome. In certain embodiments, a cleavable moiety is selectively cleaved by endogenous enzymes, such as nucleases.

In certain embodiments, a cleavable bond is selected from among: an amide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, or a disulfide. In certain embodiments, a cleavable bond is one or both of the esters of a phosphodiester. In certain embodiments, a cleavable moiety comprises a phosphate or phosphodiester. In certain embodiments, the cleavable moiety is a phosphate linkage between an oligonucleotide and a conjugate moiety or conjugate group.

wherein n is from 1 to about 3, m is 0 when n is 1, m is 1 when n is 2 or greater, j is 1 or 0, and k is 1 or 0.

In certain embodiments, n is 1, j is 1 and k is 0. In certain embodiments, n is 1, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 2, j is 1 and k is 0. In certain embodiments, n is 2, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 3, j is 1 and k is 0. In certain embodiments, n is 3, j is 0 and k is 1. In certain embodiments, n is 3, j is 1 and k is 1.

In certain embodiments, conjugate groups comprise cell-targeting moieties that have at least one tethered ligand. In certain embodiments, cell-targeting moieties comprise two tethered ligands covalently attached to a branching group. In certain embodiments, cell-targeting moieties comprise three tethered ligands covalently attached to a branching group.

In certain embodiments, the cell-targeting moiety comprises a branching group comprising one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain such embodiments, the branching group comprises a branched aliphatic group comprising groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl and ether groups. In certain embodiments, the branching group comprises a mono or polycyclic ring system.

In certain embodiments, each tether of a cell-targeting moiety comprises one or more groups selected from alkyl, substituted alkyl, ether, thioether, disulfide, amino, oxo, amide, phosphodiester, and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, thioether, disulfide, amino, oxo, amide,

171

and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, phosphodiester, ether, amino, oxo, and amide, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, amino, oxo, and amid, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, amino, and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and phosphodiester, in any combination. In certain embodiments, each tether comprises at least one phosphorus linking group or neutral linking group. In certain embodiments, each tether comprises a chain from about 6 to about 20 atoms in length. In certain embodiments, each tether comprises a chain from about 10 to about 18 atoms in length. In certain embodiments, each tether comprises about 10 atoms in chain length.

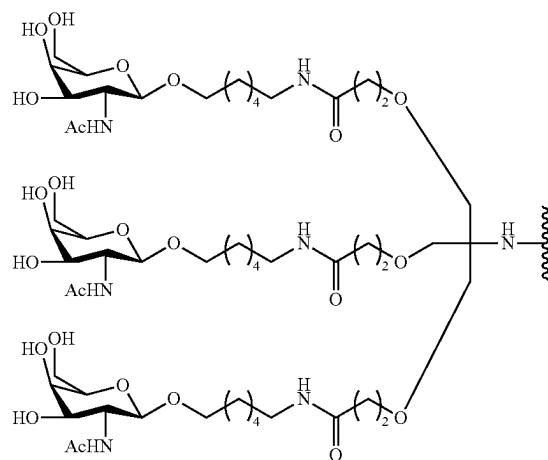
In certain embodiments, each ligand of a cell-targeting moiety has an affinity for at least one type of receptor on a target cell. In certain embodiments, each ligand has an affinity for at least one type of receptor on the surface of a mammalian liver cell. In certain embodiments, each ligand has an affinity for the hepatic asialoglycoprotein receptor (ASGP-R). In certain embodiments, each ligand is a carbohydrate. In certain embodiments, each ligand is, independently selected from galactose, N-acetyl galactoseamine (GalNAc), mannose, glucose, glucoseamine and fucose. In certain embodiments, each ligand is N-acetyl galactoseamine (GalNAc). In certain embodiments, the cell-targeting moiety comprises 3 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 2 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 1 GalNAc ligand.

In certain embodiments, each ligand of a cell-targeting moiety is a carbohydrate, carbohydrate derivative, modified carbohydrate, polysaccharide, modified polysaccharide, or polysaccharide derivative. In certain such embodiments, the conjugate group comprises a carbohydrate cluster (see, e.g., Maier et al., "Synthesis of Antisense Oligonucleotides Conjugated to a Multivalent Carbohydrate Cluster for Cellular Targeting," *Bioconjugate Chemistry*, 2003, 14, 18-29 or Rensen et al., "Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asialoglycoprotein Receptor," *J. Med. Chem.* 2004, 47, 5798-5808). In certain such embodiments, each ligand is an amino sugar or a thio sugar. For example, amino sugars may be selected from any number of compounds known in the art, such as sialic acid, α -D-galactosamine, β -muramic acid, 2-deoxy-2-methylamino-L-glucopyranose, 4,6-dideoxy-4-formamido-2,3-di-O-methyl-D-mannopyranose, 2-deoxy-2-sulfoamino-D-glucopyranose

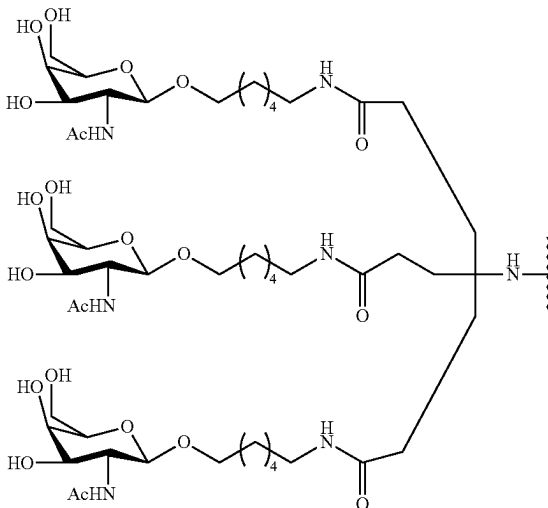
172

and N-sulfo-D-glucosamine, and N-glycoloyl- α -neuraminic acid. For example, thio sugars may be selected from 5-Thio- β -D-glucopyranose, methyl 2,3,4-tri-O-acetyl-1-thio-6-O-trityl- α -D-glucopyranoside, 4-thio- β -D-galactopyranose, and ethyl 3,4,6,7-tetra-O-acetyl-2-deoxy-1,5-dithio- α -D-glucopyranoside.

In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:



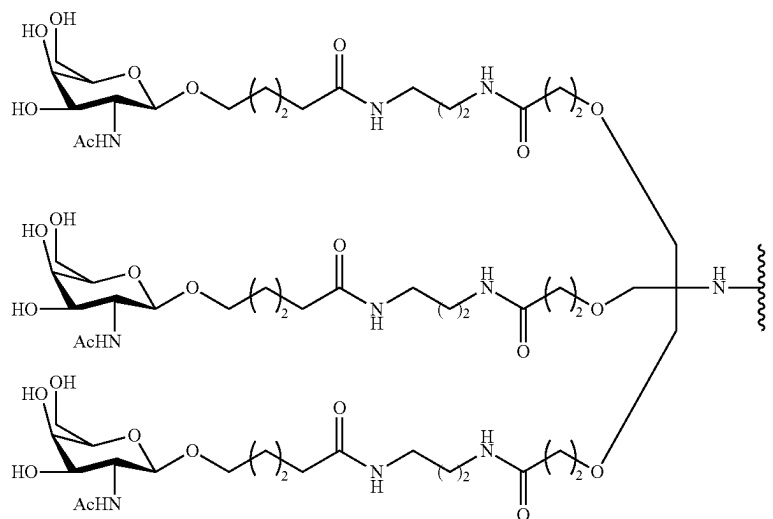
In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:



173

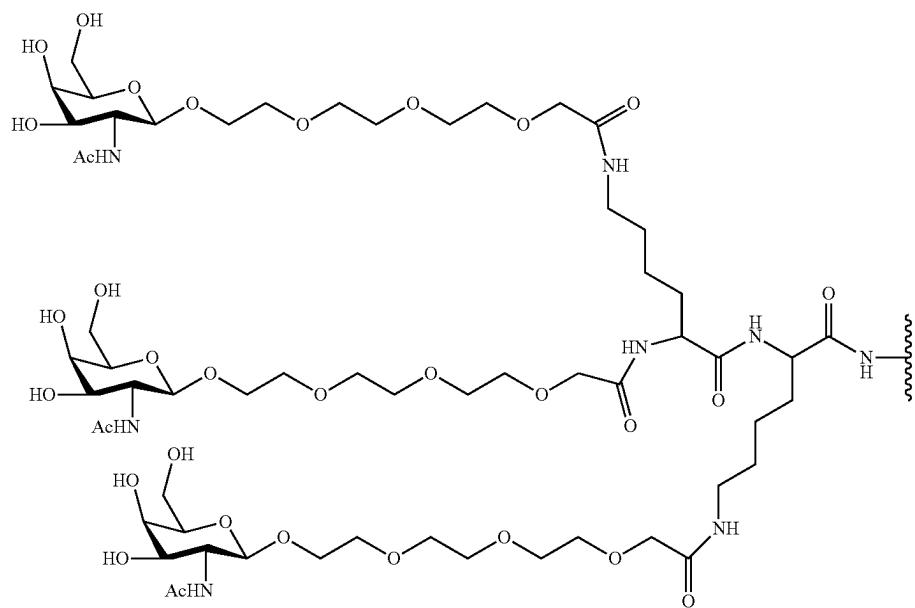
174

In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:



In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

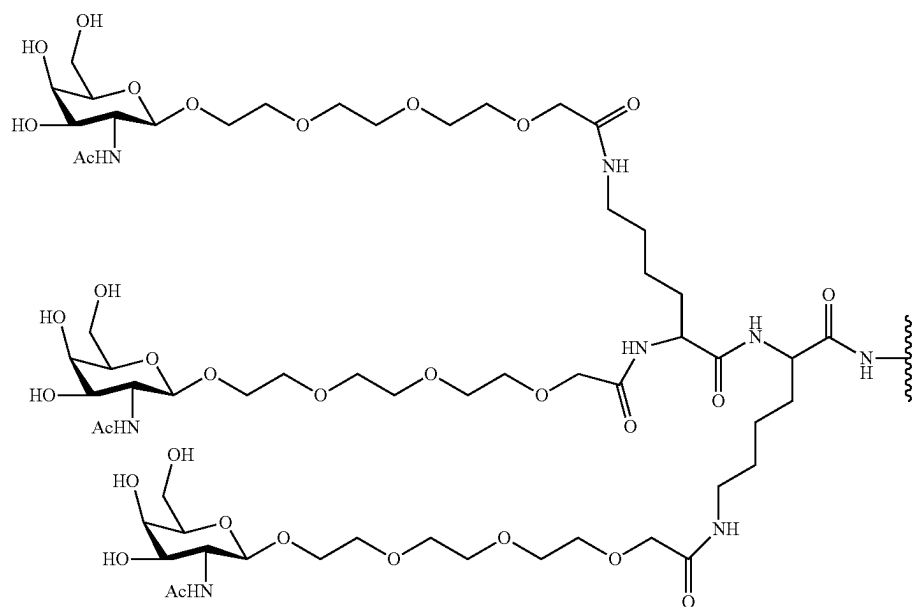
35



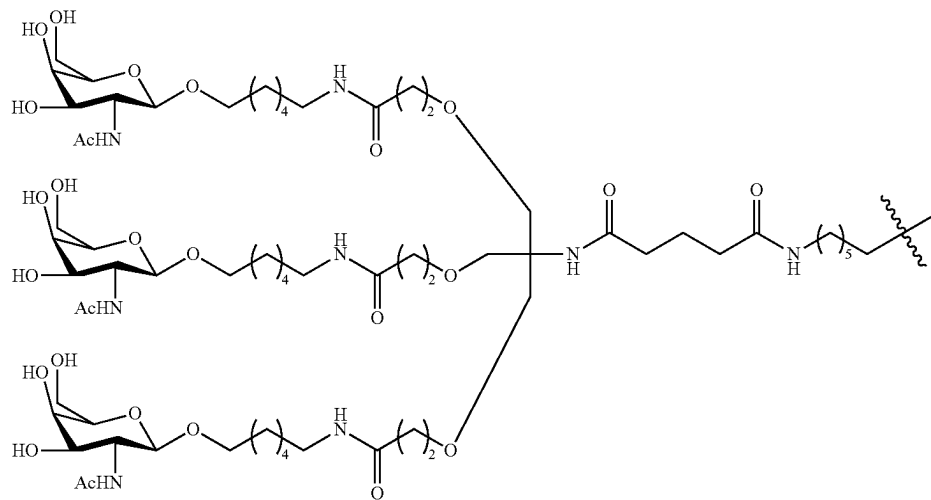
175

176

In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

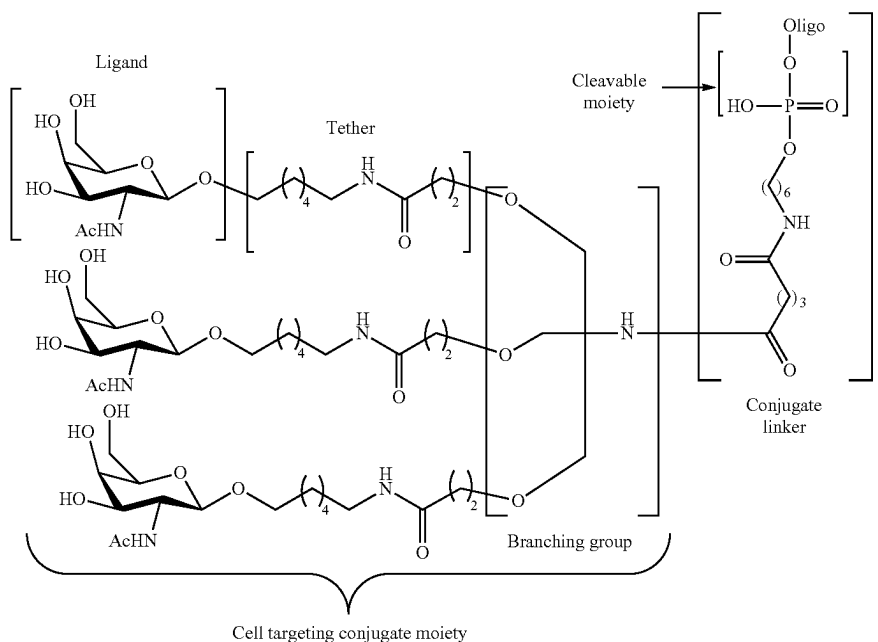


In certain embodiments, compounds comprise a conjugate group described herein as "LICA-1". LICA-1 has the formula:



177

In certain embodiments, compounds described herein comprise LICA-1 and a cleavable moiety within the conjugate linker have the formula:



178

2008, 16, 5216-5231; Lee et al., *Bioorg Med Chem*, 2011, 19, 2494-2500; Kornilova et al., *Analyt Biochem*, 2012, 425, 43-46; Pujol et al., *Angew Chemie Int Ed Engl*, 2012, 51,

wherein oligo is an oligonucleotide.

Representative United States patents, United States patent application publications, international patent application publications, and other publications that teach the preparation of certain of the above noted conjugate groups, compounds comprising conjugate groups, tethers, conjugate linkers, branching groups, ligands, cleavable moieties as well as other modifications include without limitation, U.S. Pat. Nos. 5,994,517, 6,300,319, 6,660,720, 6,906,182, 7,262,177, 7,491,805, 8,106,022, 7,723,509, US 2006/0148740, US 2011/0123520, WO 2013/033230 and WO 2012/037254, Biessen et al., *J. Med. Chem.* 1995, 38, 1846-1852, Lee et al., *Bioorganic & Medicinal Chemistry* 2011, 19, 2494-2500, Rensen et al., *J. Biol. Chem.* 2001, 276, 37577-37584, Rensen et al., *J. Med. Chem.* 2004, 47, 5798-5808, Slidregt et al., *J. Med. Chem.* 1999, 42, 609-618, and Valentijn et al., *Tetrahedron*, 1997, 53, 759-770.

In certain embodiments, modified oligonucleotides comprise a gapmer or fully modified sugar motif and a conjugate group comprising at least one, two, or three GalNAc ligands. In certain embodiments, compounds comprise a conjugate group found in any of the following references: Lee, *Carbohydr Res*, 1978, 67, 509-514; Connolly et al., *J Biol Chem*, 1982, 257, 939-945; Pavia et al., *Int J Pep Protein Res*, 1983, 22, 539-548; Lee et al., *Biochem*, 1984, 23, 4255-4261; Lee et al., *Glycoconjugate J*, 1987, 4, 317-328; Toyokuni et al., *Tetrahedron Lett*, 1990, 31, 2673-2676; Biessen et al., *J Med Chem*, 1995, 38, 1538-1546; Valentijn et al., *Tetrahedron*, 1997, 53, 759-770; Kim et al., *Tetrahedron Lett*, 1997, 38, 3487-3490; Lee et al., *Bioconjug Chem*, 1997, 8, 762-765; Kato et al., *Glycobiol*, 2001, 11, 821-829; Rensen et al., *J Biol Chem*, 2001, 276, 37577-37584; Lee et al., *Methods Enzymol*, 2003, 362, 38-43; Westerlind et al., *Glycoconj J*, 2004, 21, 227-241; Lee et al., *Bioorg Med Chem Lett*, 2006, 16(19), 5132-5135; Maierhofer et al., *Bioorg Med Chem*, 2007, 15, 7661-7676; Khorev et al., *Bioorg Med Chem*,

7445-7448; Biessen et al., *J Med Chem*, 1995, 38, 1846-1852; Slidregt et al., *J Med Chem*, 1999, 42, 609-618; Rensen et al., *J Med Chem*, 2004, 47, 5798-5808; Rensen et al., *Arterioscler Thromb Vasc Biol*, 2006, 26, 169-175; van Rossenberg et al., *Gene Ther*, 2004, 11, 457-464; Sato et al., *J Am Chem Soc*, 2004, 126, 14013-14022; Lee et al., *J Org Chem*, 2012, 77, 7564-7571; Biessen et al., *FASEB J*, 2000, 14, 1784-1792; Rajur et al., *Bioconjug Chem*, 1997, 8, 935-940; Duff et al., *Methods Enzymol*, 2000, 313, 297-321; Maier et al., *Bioconjug Chem*, 2003, 14, 18-29; Jayaprakash et al., *Org Lett*, 2010, 12, 5410-5413; Manoharan, *Antisense Nucleic Acid Drug Dev*, 2002, 12, 103-128; Merwin et al., *Bioconjug Chem*, 1994, 5, 612-620; Tomiya et al., *Bioorg Med Chem*, 2013, 21, 5275-5281; International applications WO1998/013381; WO2011/038356; WO1997/046098; WO2008/098788; WO2004/101619; WO2012/037254; WO2011/120053; WO2011/100131; WO2011/163121; WO2012/177947; WO2013/033230; WO2013/075035; WO2012/083185; WO2012/083046; WO2009/082607; WO2009/134487; WO2010/144740; WO2010/148013; WO1997/020563; WO2010/088537; WO2002/043771; WO2010/129709; WO2012/068187; WO2009/126933; WO2004/024757; WO2010/054406; WO2012/089352; WO2012/089602; WO2013/166121; WO2013/165816; U.S. Pat. Nos. 4,751,219; 8,552,163; 6,908,903; 7,262,177; 5,994,517; 6,300,319; 8,106,022; 7,491,805; 7,491,805; 7,582,744; 8,137,695; 6,383,812; 6,525,031; 6,660,720; 7,723,509; 8,541,548; 8,344,125; 8,313,772; 8,349,308; 8,450,467; 8,501,930; 8,158,601; 7,262,177; 6,906,182; 6,620,916; 8,435,491; 8,404,862; 7,851,615; Published U.S. Patent Application Publications US2011/0097264; US2011/0097265; US2013/0004427; US2005/0164235; US2006/0148740; US2008/0281044; US2010/0240730; US2003/0119724; US2006/0183886; US2008/0206869; US2011/0269814; US2009/0286973; US2011/0207799; US2012/0136042; US2012/0165393; US2008/0281041; US2009/

0203135; US2012/0035115; US2012/0095075; US2012/0101148; US2012/0128760; US2012/0157509; US2012/0230938; US2013/0109817; US2013/0121954; US2013/0178512; US2013/0236968; US2011/0123520; US2003/0077829; US2008/0108801; and US2009/0203132.

In certain embodiments, compounds are single-stranded. In certain embodiments, compounds are double-stranded. Compositions and Methods for Formulating Pharmaceutical Compositions

Compounds described herein may be admixed with pharmaceutically acceptable active or inert substances for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

Certain embodiments provide pharmaceutical compositions comprising one or more compounds or a salt thereof. In certain embodiments, the compounds are antisense compounds or oligomeric compounds. In certain embodiments, the compounds comprise or consist of a modified oligonucleotide. In certain such embodiments, the pharmaceutical composition comprises a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutical composition comprises a sterile saline solution and one or more compound. In certain embodiments, such pharmaceutical composition consists of a sterile saline solution and one or more compound. In certain embodiments, the sterile saline is pharmaceutical grade saline. In certain embodiments, a pharmaceutical composition comprises one or more compound and sterile water. In certain embodiments, a pharmaceutical composition consists of one compound and sterile water. In certain embodiments, the sterile water is pharmaceutical grade water. In certain embodiments, a pharmaceutical composition comprises one or more compounds and phosphate-buffered saline (PBS). In certain embodiments, a pharmaceutical composition consists of one or more compound and sterile PBS. In certain embodiments, the sterile PBS is pharmaceutical grade PBS. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

A compound described herein targeted to HSD17B13 nucleic acid can be utilized in pharmaceutical compositions by combining the compound with a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutically acceptable diluent is water, such as sterile water suitable for injection. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising a compound targeted to HSD17B13 nucleic acid and a pharmaceutically acceptable diluent. In certain embodiments, the pharmaceutically acceptable diluent is water. In certain embodiments, the compound comprises or consists of a modified oligonucleotide provided herein.

Pharmaceutical compositions comprising compounds provided herein encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. In certain embodiments, the compounds are antisense compounds or oligomeric compounds. In certain embodiments, the compound comprises or consists of a modified oligonucleotide. Accordingly, for example, the disclosure is also drawn to pharmaceutically acceptable salts of compounds, prodrugs,

pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

A prodrug can include the incorporation of additional nucleosides at one or both ends of a compound which are cleaved by endogenous nucleases within the body, to form the active compound.

In certain embodiments, the compounds or compositions further comprise a pharmaceutically acceptable carrier or diluent.

EXAMPLES

The Examples below describe the screening process to identify compounds targeted to HSD17B13.

Non-Limiting Disclosure and Incorporation by Reference

Although the sequence listing accompanying this filing identifies each sequence as either "RNA" or "DNA" as required, in reality, those sequences may be modified with any combination of chemical modifications. One of skill in the art will readily appreciate that such designation as "RNA" or "DNA" to describe modified oligonucleotides is, in certain instances, arbitrary. For example, an oligonucleotide comprising a nucleoside comprising a 2'-OH sugar moiety and a thymine base could be described as a DNA having a modified sugar (2'-OH for the natural 2'-H of DNA) or as an RNA having a modified base (thymine (methylated uracil) for natural uracil of RNA).

Accordingly, nucleic acid sequences provided herein, including, but not limited to, those in the sequence listing, are intended to encompass nucleic acids containing any combination of natural or modified RNA and/or DNA, including, but not limited to, such nucleic acids having modified nucleobases. By way of further example and without limitation, an oligonucleotide having the nucleobase sequence "ATCGATCG" encompasses any oligonucleotides having such nucleobase sequence, whether modified or unmodified, including, but not limited to, such compounds comprising RNA bases, such as those having sequence "AUCGAUCG" and those having some DNA bases and some RNA bases such as "AUCGATCG" and compounds having other modified nucleobases, such as "AT^mCGAUCG," wherein ^mC indicates a cytosine base comprising a methyl group at the 5-position.

Certain compounds described herein (e.g. modified oligonucleotides) have one or more asymmetric centers and thus give rise to enantiomers, diastereomers, and other stereoisomeric configurations that may be defined, in terms of absolute stereochemistry, as (R) or (S), as α or β , such as for sugar anomers, or as (D) or (L), such as for amino acids, etc. Compounds provided herein that are drawn or described as having certain stereoisomeric configurations include only the indicated compounds. Compounds provided herein that are drawn or described with undefined stereochemistry include all such possible isomers, including their stereorandom and optically pure forms. Likewise, all tautomeric forms of the compounds provided herein are included unless otherwise indicated. Unless otherwise indicated, oligomeric compounds and modified oligonucleotides described herein are intended to include corresponding salt forms.

Compounds described herein include variations in which one or more atoms are replaced with a non-radioactive isotope or radioactive isotope of the indicated element. For example, compounds herein that comprise hydrogen atoms encompass all possible deuterium substitutions for each of the ¹H hydrogen atoms. Isotopic substitutions encompassed

181

by the compounds herein include, but are not limited to: ^2H or ^3H in place of ^1H , ^{13}C or ^{14}C in place of ^{12}C , ^{15}N in place of ^{14}N , ^{17}O or ^{18}O in place of ^{16}O , and ^{33}S , ^{34}S , ^{35}S , or ^{36}S in place of ^{32}S .

While certain compounds, compositions and methods described herein have been described with specificity in accordance with certain embodiments, the following examples serve only to illustrate the compounds described herein and are not intended to limit the same. Each of the references recited in the present application is incorporated herein by reference in its entirety.

Example 1: Antisense Inhibition of Human HSD17B13 in HepaRG Cells by cEt Gapmers

Modified oligonucleotides complementary to an HSD17B13 nucleic acid were synthesized and tested for their effect on HSD17B13 RNA levels in vitro. The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each separate experiment are presented in tables below.

The modified oligonucleotides were all 3-10-3 cEt gapmers (i.e. had a central gap segment often 2'-deoxynucleosides flanked on each side by wing segments each comprising three cEt modified nucleosides). The internucleoside linkages throughout each modified oligonucleotide were phosphorothioate (P=S) linkages. All cytosine nucleobases throughout each modified oligonucleotide were 5-methylcytosines.

"Start site" indicates the 5'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. "Stop site" indicates the 3'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. As shown in the Tables below, the modified

182

oligonucleotide are complementary to either the human HSD17B13 mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_178135.4) or to the human HSD17B13 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000) or to both. 'N/A' indicates that the modified oligonucleotide is not complementary to that particular target sequence with 100% complementarity.

Cultured HepaRG cells at a density of 20,000 cells per well were transfected using electroporation with 1,000 nM of modified oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and HSD17B13 RNA levels were measured by quantitative real-time RTPCR. Human primer probe set RTS43553 (forward sequence AGACTACAGAAGTTTCTTCTGAAC, designated herein as SEQ ID NO: 5; reverse sequence CATCTCTGGCTGGAGCTTATIT, designated herein as SEQ ID NO: 6; probe sequence TTTGAAGCAGTGGTTGGCCACAAA, designated herein as SEQ ID NO: 7) was used to measure RNA levels. HSD17B13 RNA levels were normalized to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of HSD17B13 relative to untreated control cells. As used herein, a value of '0' indicates that treatment with the modified oligonucleotide did not inhibit HSD17B13 mRNA levels. An asterisk (*) indicates that the modified oligonucleotide is complementary to the target transcript within the amplicon region of the primer probe set and so, the associated data is not reliable. In such instances, additional using alternate primer probe sets must be performed to accurately assess the potency and efficacy of such modified oligonucleotides.

TABLE 1

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245068	1	16	3095	3110	GGAGGAGGTACTGTCT	44	8
1245094	59	74	3153	3168	GATGTTTCATGGCTTTG	81	9
1245120	111	126	3205	3220	ACTCCAAGTAGGAGTA	0	10
1245146	181	196	3275	3290	GCTCCAGTAATGAGAA	34	11
1245172	251	266	3345	3360	ATCCACAGAACCAAT	34	12
1245198	299	314	7588	7603	TAGTTTTCGGCACTCA	35	13
1245224	359	374	7648	7663	AGAGCGATAGATCTCT	8	14
1245250	402	417	8796	8811	CCACGATTGTTACATC	42	15
1245276	432	447	8826	8841	CGGCTGGATATACTGT	35	16
1245302	473	488	8867	8882	AAATGTCTTGGTAATC	30	17
1245328	515	530	10401	10416	AAGTGCTTTTGTGATC	41	18
1245354	561	576	10447	10462	CCACTGTGACGATGTG	0	19
1245380	614	629	N/A	N/A	GCTGGAACAATATGGG	36	20
1245406	670	685	12088	12103	CCCAAGGCCTGAAGTT	61	21

TABLE 1-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245432	759	774	15640	15655	ATACAGGCCATAATCT	0	22
1245458	790	805	15671	15686	ATCAGACTTCTTACGA	45	23
1245484	842	857	15723	15738	ATTGATATACGATGGA	31	24
1245510*	907	922	20722	20737	TTCTGCATACGATTTA	83	25
1245536*	973	988	20788	20803	TCTGGCTGGAGCTTAT	85	26
1245562	1014	1029	20829	20844	CATTGATTCGAACTA	40	27
1245588	1088	1103	20903	20918	TTGACTGCTGCTAGTG	30	28
1245614	1260	1275	21075	21090	GGTAGCTTTTGTCCAC	66	29
1245640	1314	1329	21129	21144	CAGTCTTAAACCTTCC	70	30
1245666	1341	1356	21156	21171	TGGCTACAGATTGGAA	55	31
1245692	1385	1400	21200	21215	TTAGCTGTGCACTCAT	61	32
1245718	1423	1438	21238	21253	CCAGGTTGAGATAAAG	49	33
1245744	1499	1514	21314	21329	AGAGTTGCCACCGTTT	76	34
1245770	1558	1573	21373	21388	CACTTTTGGTGGACTT	21	35
1245796	1617	1632	21432	21447	CGGTCACCTTTCATAA	69	36
1245822	1710	1725	21525	21540	ATCTCTGGGACCAAGG	72	37
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	55	38
1245848	1772	1787	21587	21602	ATAGCCAGTACAGTTC	65	39
1245874	2177	2192	21992	22007	GGAGTCGATTATTTT	41	40
1245900	2244	2259	22059	22074	GTCCATGCAAAAGCAT	62	41
1245926	N/A	N/A	3405	3420	AATACATATCTTACTC	37	42
1245952	N/A	N/A	3552	3567	GTCATAAAAATCGCTG	72	43
1245978	N/A	N/A	3948	3963	GAAAAGCCTGACTCAC	48	44
1246004	N/A	N/A	4517	4532	CATCAAGCCCTTTTCA	23	45
1246030	N/A	N/A	5078	5093	CTAAAGGAGATCTGAG	47	46
1246056	N/A	N/A	5222	5237	TAGCTTAAACTCCAAT	60	47
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	87	48
1246082	N/A	N/A	5817	5832	CTCAAAGTCAGTATCC	70	49
1246108	N/A	N/A	5989	6004	GGCAAGGATACCTGAA	67	50
1246134	N/A	N/A	6233	6248	CTCATTTATGTACCAA	87	51
1246160	N/A	N/A	6474	6489	TATGTAATAGGCAGTA	62	52
1246186	N/A	N/A	7482	7497	GACAATTAACATTCGG	69	53
1246212	N/A	N/A	7951	7966	CTAGGAAGTGATCCAT	66	54
1246238	N/A	N/A	8243	8258	CTAATAACTACATGAC	18	55
1246264	N/A	N/A	8694	8709	TCCAAAGATGAGAGTC	17	56
1246290	N/A	N/A	9002	9017	GCAAAGATCTGGCCAG	22	57

TABLE 1-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 1	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
	Start Site	1 Stop Site	Start Site	2 Stop Site			
1246316	N/A	N/A	9348	9363	TCCAAAAGTGTCTCG	31	58
1246342	N/A	N/A	9518	9533	CTACACTAATATTGAG	20	59
1246368	N/A	N/A	9715	9730	AGTGAACATACATTGT	48	60
1246394	N/A	N/A	9981	9996	ATTAAGACAGTTGAGT	34	61
1246420	N/A	N/A	10211	10226	ATTTATTGGTATGGTA	46	62
1246446	N/A	N/A	10730	10745	CTGTAATTGGCTCTTG	28	63
1246472	N/A	N/A	11261	11276	TCTTATCTTGGGCACC	44	64
1246498	N/A	N/A	11742	11757	TCCATTGAATCTTCAA	57	65
1246524	N/A	N/A	11985	12000	GAAACCAATCCTCAGC	3	66
1246550	N/A	N/A	12421	12436	TGGTAACGGTGATCAA	74	67
1246576	N/A	N/A	12624	12639	TTAAGCAGATGGCTTA	0	68
1246602	N/A	N/A	12761	12776	ACAGAGAATTGTTTAG	39	69
1246628	N/A	N/A	13605	13620	AGACAATACAGGATAG	15	70
1246654	N/A	N/A	13862	13877	CCTTAGGAAAGCTCAT	29	71
1246680	N/A	N/A	14041	14056	ATCAATGCCTTAGCCC	38	72
1246706	N/A	N/A	14199	14214	TATTATGTGATTGAGT	86	73
1246732	N/A	N/A	14399	14414	ATTCATAACATAGGC	36	74
1246758	N/A	N/A	14730	14745	ACTCTAAATACCCCTTG	16	75
1246784	N/A	N/A	15316	15331	GGATTAATCATGGGAC	44	76
1246810	N/A	N/A	15405	15420	TCATAGCTCACTTAGT	0	77
1246836	N/A	N/A	15775	15790	TTTAGTATTGGGTGT	21	78
1246862	N/A	N/A	16407	16422	CAATTGCTCTATAGAT	15	79
1246888	N/A	N/A	17767	17782	AGCATATTCATTTGGC	61	80
1246914	N/A	N/A	18033	18048	AGTTTATATGGATTG	67	81
1246940	N/A	N/A	18816	18831	CTAGTAATTGCATCTG	57	82
1246966	N/A	N/A	19445	19460	CTGGGATAGTGGAGGA	21	83
1246992	N/A	N/A	19686	19701	GCTAAAAGCTCACCAA	26	84
1247018	N/A	N/A	19902	19917	GATACCCAGGTTGCTT	53	85
1247044	N/A	N/A	20186	20201	ATTAGAAGTCAGCCCA	42	86
1247070	N/A	N/A	20401	20416	GCTATAGTAATTGCTA	33	87

TABLE 2

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245069	2	17	3096	3111	GGGAGGAGGTACTGTC	35	88
1245095	61	76	3155	3170	ATGATGTTTCATGGCTT	72	89
1245121	112	127	3206	3221	GACTCCAAGTAGGAGT	0	90
1245147	182	197	3276	3291	AGCTCCAGTAATGAGA	26	91
1245173	263	278	N/A	N/A	GCGCTTATTAATATCC	3	92
1245199	300	315	7589	7604	CTAGTTTTCGGCACTC	33	93
1245225	360	375	7649	7664	GAGAGCGATAGATCTC	1	94
1245251	403	418	8797	8812	ACCACGATTGTTACAT	34	95
1245277	433	448	8827	8842	TCGGCTGGATATACTG	23	96
1245303	476	491	8870	8885	CTCAAATGTCTTGGTA	29	97
1245329	516	531	10402	10417	GAAGTGCTTTTGTGAT	34	98
1245355	562	577	10448	10463	GCCACTGTGACGATGT	9	99
1245381	615	630	N/A	N/A	TGCTGGAACAATATGG	36	100
1245407	671	686	12089	12104	TCCAAGGCCTGAAGT	43	101
1245433	760	775	15641	15656	AATACAGGCCATAATC	16	102
1245459	791	806	15672	15687	TATCAGACTTCTTACG	59	103
1245485	843	858	15724	15739	TATTGATATACGATGG	16	104
1245511*	908	923	20723	20738	ATTCTGCATACGATTT	89	105
1245537*	974	989	20789	20804	CTCTGGCTGGAGCTTA	86	106
1245563	1015	1030	20830	20845	GCATTGATTCGAAACT	66	107
1245589	1089	1104	20904	20919	TTTGACTGCTGCTAGT	21	108
1245615	1261	1276	21076	21091	AGGTAGCTTTTGTCCA	56	109
1245641	1315	1330	21130	21145	ACAGTCTTAAACCTTC	76	110
1245667	1342	1357	21157	21172	ATGGCTACAGATTGGA	72	111
1245693	1386	1401	21201	21216	CTTAGCTGTGCACTCA	60	112
1245719	1425	1440	21240	21255	GTCCAGGTTGAGATAA	60	113
1245745	1500	1515	21315	21330	TAGAGTTGCACCGTTT	58	114
1245771	1559	1574	21374	21389	CCACTTTTGGTGGACT	17	115
1245797	1618	1633	21433	21448	TCGGTCACCTTTCATA	64	116
1245823	1711	1726	21526	21541	CATCTCTGGACCAAG	51	117
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	46	38
1245849	1773	1788	21588	21603	AATAGCCAGTACAGTT	46	118
1245875	2178	2193	21993	22008	GGGAGTCGGATTATTT	33	119
1245901	2245	2260	22060	22075	AGTCCATGCAAAAGCA	77	120
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	90	121
1245953	N/A	N/A	3553	3568	TGTCATAAAATCGCT	57	122

TABLE 2-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245979	N/A	N/A	3962	3977	TACTAATGTCCAAGGA	77	123
1246005	N/A	N/A	4529	4544	AAGGTTAGATTTCATC	80	124
1246031	N/A	N/A	5079	5094	CCTAAAGGAGATCTGA	48	125
1246057	N/A	N/A	5225	5240	CTTTAGCTTAACTCC	69	126
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	84	48
1246083	N/A	N/A	5833	5848	GATGTTAGGACCCAGT	64	127
1246109	N/A	N/A	5990	6005	AGGCAAGGATACCTGA	42	128
1246135	N/A	N/A	6298	6313	GGAGATATAACATTAC	70	129
1246161	N/A	N/A	6475	6490	CTATGTAATAGGCAGT	73	130
1246187	N/A	N/A	7483	7498	AGACAATTAACATTCG	62	131
1246213	N/A	N/A	7952	7967	ACTAGGAAGTGATCCA	58	132
1246239	N/A	N/A	8244	8259	CCTAATAACTACATGA	3	133
1246265	N/A	N/A	8695	8710	ATCCAAAGATGAGAGT	5	134
1246291	N/A	N/A	9003	9018	GGCAAAGATCTGGCCA	10	135
1246317	N/A	N/A	9349	9364	GTCCAAAAGTGTCCTC	35	136
1246343	N/A	N/A	9526	9541	CCTAGATTCTACACTA	14	137
1246369	N/A	N/A	9718	9733	GATAGTGAACATACAT	54	138
1246395	N/A	N/A	9983	9998	GTATTAAGACAGTTGA	45	139
1246421	N/A	N/A	10212	10227	AATTTATTGGTATGGT	30	140
1246447	N/A	N/A	10831	10846	GTAACGACTCTGTAA	8	141
1246473	N/A	N/A	11308	11323	GCACAAGCACACTGTA	7	142
1246499	N/A	N/A	11758	11773	CTACATGTAAGGTTTT	11	143
1246525	N/A	N/A	12000	12015	TAGCTAAGGGAGAGTG	9	144
1246551	N/A	N/A	12429	12444	AGACAGGGTGGTAACG	51	145
1246577	N/A	N/A	12625	12640	GTTAAGCAGATGGCTT	41	146
1246603	N/A	N/A	12765	12780	CTCAACAGAGAATTGT	29	147
1246629	N/A	N/A	13612	13627	GCTTTGAAGACAATAC	42	148
1246655	N/A	N/A	13863	13878	CCCTTAGGAAAGCTCA	37	149
1246681	N/A	N/A	14042	14057	TATCAATGCCTTAGCC	15	150
1246707	N/A	N/A	14200	14215	TTATTATGTGATTGAG	23	151
1246733	N/A	N/A	14432	14447	ACTGAAGGCTGTGTAC	14	152
1246759	N/A	N/A	14731	14746	AACTCTAAATACCCTT	28	153
1246785	N/A	N/A	15317	15332	AGGATTAATCATGGGA	37	154
1246811	N/A	N/A	15407	15422	TTTCATAGCTCACTTA	16	155
1246837	N/A	N/A	15787	15802	CTCTATGGTGTTTTA	32	156
1246863	N/A	N/A	16416	16431	AATACTCCCAATTGC	0	157

TABLE 2-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246889	N/A	N/A	17780	17795	AGAAGATATTATCAGC	64	158
1246915	N/A	N/A	18087	18102	CACCAATGCAGTTTGT	42	159
1246941	N/A	N/A	18817	18832	GCTAGTAATTGCATCT	47	160
1246967	N/A	N/A	19474	19489	GCTATAACTGGAAGGA	58	161
1246993	N/A	N/A	19687	19702	TGCTAAAAGCTCACCA	49	162
1247019	N/A	N/A	19906	19921	TTAAGATACCCAGGTT	37	163
1247045	N/A	N/A	20187	20202	TATTAGAAGTCAGCCC	27	164
1247071	N/A	N/A	20402	20417	GGCTATAGTAATTGCT	13	165

TABLE 3

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245070	5	20	3099	3114	CTAGGGAGGAGTACT	29	166
1245096	62	77	3156	3171	GATGATGTTTCATGGCT	86	167
1245122	113	128	3207	3222	CGACTCCAAGTAGGAG	10	168
1245148	200	215	3294	3309	CCTGCCTATTCCATGC	24	169
1245174	264	279	N/A	N/A	CGCGCTTATTAATATC	6	170
1245200	301	316	7590	7605	CCTAGTTTTTCGGCACT	62	171
1245226	361	376	7650	7665	AGAGAGCGATAGATCT	15	172
1245252	404	419	8798	8813	CACCACGATTGTTACA	36	173
1245278	434	449	8828	8843	ATCGGCTGGATATACT	10	174
1245304	477	492	8871	8886	CCTCAAATGCTCTGGT	6	175
1245330	524	539	10410	10425	CGATGGAAGAAGTGCT	3	176
1245356	563	578	10449	10464	AGCCACTGTGACGATG	10	177
1245382	616	631	N/A	N/A	TTGCTGGAACAATATG	20	178
1245408	688	703	12106	12121	GTTTTGATACCAGTTT	67	179
1245434	761	776	15642	15657	CAATACAGGCCATAAT	0	180
1245460	792	807	15673	15688	CTATCAGACTTCTTAC	39	181
1245486	844	859	15725	15740	ATATTGATATACGATG	0	182
1245512*	909	924	20724	20739	TATTCTGCATACGATT	81	183
1245538*	975	990	20790	20805	TCTCTGGCTGGAGCTT	90	184
1245564	1016	1031	20831	20846	AGCATTGATTCGAAAC	70	185
1245590	1090	1105	20905	20920	GTTTGACTGCTGCTAG	58	186
1245616	1262	1277	21077	21092	GAGGTAGCTTTTGTCC	59	187

TABLE 3-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt capmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245642	1316	1331	21131	21146	AACAGTCTTAAACCTT	82	188
1245668	1343	1358	21158	21173	CATGGCTACAGATTGG	54	189
1245694	1387	1402	21202	21217	TCTTAGCTGTGCACTC	69	190
1245720	1426	1441	21241	21256	TGTCCAGGTTGAGATA	48	191
1245746	1501	1516	21316	21331	ATAGAGTTGCACCGTT	57	192
1245772	1560	1575	21375	21390	TCCACTTTTGGTGGAC	16	193
1245798	1619	1634	21434	21449	GTCGGTCACCTTTCAT	85	194
1245824	1712	1727	21527	21542	ACATCTCTGGGACCAA	78	195
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	64	38
1245850	1774	1789	21589	21604	TAATAGCCAGTACAGT	64	196
1245876	2179	2194	21994	22009	TGGGAGTCGGATTATT	44	197
1245902	2246	2261	22061	22076	TAGTCCATGCAAAAGC	66	198
1245928	N/A	N/A	3434	3449	AAGTAGATGGTAAGTC	85	199
1245954	N/A	N/A	3554	3569	ATGTCATAAAATCGC	53	200
1245980	N/A	N/A	3963	3978	ATACTAATGTCCAAGG	89	201
1246006	N/A	N/A	4535	4550	GATTTGAAGGTTAGAT	47	202
1246032	N/A	N/A	5080	5095	GCCTAAAGGAGATCTG	67	203
1246058	N/A	N/A	5239	5254	TAGCAAAACACTTGCT	21	204
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	87	48
1246084	N/A	N/A	5834	5849	GGATGTTAGGACCCAG	81	205
1246110	N/A	N/A	6035	6050	ATACTTTTCCGTCTCA	90	206
1246136	N/A	N/A	6303	6318	CTCTAGGAGATATAAC	49	207
1246162	N/A	N/A	6476	6491	ACTATGTAATAGGCAG	85	208
1246188	N/A	N/A	7681	7696	AAATTTGTGAACCTGC	26	209
1246214	N/A	N/A	7953	7968	AACTAGGAAGTGATCC	58	210
1246240	N/A	N/A	8245	8260	ACCTAATAACTACATG	0	211
1246266	N/A	N/A	8703	8718	CCAATATAATCCAAAG	28	212
1246292	N/A	N/A	9006	9021	TAAGGCAAGATCTGG	46	213
1246318	N/A	N/A	9350	9365	CGTCCAAAAGTGTCTT	53	214
1246344	N/A	N/A	9527	9542	CCCTAGATTCTACACT	22	215
1246370	N/A	N/A	9719	9734	GGATAGTGAACATACA	62	216
1246396	N/A	N/A	9984	9999	TGTATTAAGACAGTTG	34	217
1246422	N/A	N/A	10335	10350	CCAAATCAGGTAGTTT	13	218
1246448	N/A	N/A	10832	10847	AGTAAACGACTCTGTA	33	219
1246474	N/A	N/A	11385	11400	TTTtagTCAGGTAGAG	3	220
1246500	N/A	N/A	11761	11776	ATCCTACATGTAAGGT	24	221

TABLE 3-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1 and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246526	N/A	N/A	12002	12017	GTTAGCTAAGGGAGAG	18	222
1246552	N/A	N/A	12433	12448	TTAAAGACAGGGTGGT	21	223
1246578	N/A	N/A	12626	12641	TGTTAAGCAGATGGCT	39	224
1246604	N/A	N/A	12777	12792	AATGAGGCGGCACTCA	20	225
1246630	N/A	N/A	13639	13654	TGACAATGTGCAGCTC	64	226
1246656	N/A	N/A	13864	13879	ACCCTTAGGAAAGCTC	52	227
1246682	N/A	N/A	14043	14058	ATATCAATGCCTTAGC	32	228
1246708	N/A	N/A	14202	14217	GATTATTATGTGATTG	39	229
1246734	N/A	N/A	14435	14450	TATACTGAAGGCTGTG	44	230
1246760	N/A	N/A	14800	14815	GAATTGCTCACCCTTT	24	231
1246786	N/A	N/A	15318	15333	TAGGATTAATCATGGG	47	232
1246812	N/A	N/A	15418	15433	CAGTAGGTGTGTTTCA	47	233
1246838	N/A	N/A	15793	15808	AAAAAGCTCTATTGGT	0	234
1246864	N/A	N/A	16424	16439	GATATGTCAATACTCC	59	235
1246890	N/A	N/A	17857	17872	CATTGAAGTCTATAC	21	236
1246916	N/A	N/A	18091	18106	GATACCAATGCAGT	73	237
1246942	N/A	N/A	18818	18833	TGCTAGTAATTGCATC	17	238
1246968	N/A	N/A	19475	19490	GGCTATAACTGGAAGG	52	239
1246994	N/A	N/A	19688	19703	ATGCTAAAAGCTCACC	50	240
1247020	N/A	N/A	19907	19922	CTTAAGATACCCAGGT	36	241
1247046	N/A	N/A	20188	20203	TTATTAGAAGTCAGCC	36	242
1247072	N/A	N/A	20403	20418	TGGCTATAGTAATTGC	34	243

TABLE 4

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245071	6	21	3100	3115	CCTAGGGAGGAGGTAC	34	244
1245097	63	78	3157	3172	GGATGATGTTTCATGGC	82	245
1245123	114	129	3208	3223	ACGACTCCAAGTAGGA	24	246
1245149	202	217	3296	3311	TGCCTGCCTATTCCAT	40	247
1245175	265	280	N/A	N/A	CCGCGCTTATTAATAT	21	248
1245201	302	317	7591	7606	GCCTAGTTTTTCGGCAC	12	249
1245227	362	377	7651	7666	TAGAGAGCGATAGATC	22	250

TABLE 4-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245253	405	420	8799	8814	TCACCACGATTGTTAC	34	251
1245279	435	450	8829	8844	GATCGGCTGGATATAC	7	252
1245305	478	493	8872	8887	ACCTCAAATGTCTTGG	17	253
1245331	525	540	10411	10426	TCGATGGAAGAAGTGC	34	254
1245357	566	581	10452	10467	TGAAGCCACTGTGACG	49	255
1245383	623	638	12041	12056	GGCAAATTTGCTGGAA	54	256
1245409	689	704	12107	12122	GGTTTTGATACCAGTT	58	257
1245435	762	777	15643	15658	CCAATACAGGCCATAA	38	258
1245461	793	808	15674	15689	TCTATCAGACTTCTTA	49	259
1245487	845	860	15726	15741	GATATTGATATACGAT	2	260
1245513*	910	925	20725	20740	ATATTCTGCATACGAT	36	261
1245539*	977	992	20792	20807	CATCTCTGGCTGGAGC	79	262
1245565	1017	1032	20832	20847	CAGCATTGATTCGAAA	69	263
1245591	1091	1106	20906	20921	CGTTTGACTGCTGCTA	58	264
1245617	1263	1278	21078	21093	GGAGGTAGCTTTTGTC	43	265
1245643	1317	1332	21132	21147	GAACAGTCTTAAACCT	64	266
1245669	1344	1359	21159	21174	GCATGGCTACAGATTG	56	267
1245695	1388	1403	21203	21218	CTCTTAGCTGTGCACT	53	268
1245721	1427	1442	21242	21257	ATGTCCAGGTTGAGAT	55	269
1245747	1502	1517	21317	21332	AATAGAGTTGCACCGT	48	270
1245773	1561	1576	21376	21391	GTCCACTTTTGGTGGA	14	271
1245799	1620	1635	21435	21450	AGTCGGTCACCTTTCA	66	272
1245825	1713	1728	21528	21543	AACATCTCTGGACCA	74	273
1245851	1775	1790	21590	21605	GTAATAGCCAGTACAG	77	274
1245877	2180	2195	21995	22010	GTGGGAGTCGGATTAT	37	275
1245903	2251	2266	22066	22081	GAGGATAGTCCATGCA	59	276
1245929	N/A	N/A	3437	3452	GATAAGTAGATGGTAA	59	277
1245955	N/A	N/A	3573	3588	TTCTATCAACCTGCAC	62	278
1245981	N/A	N/A	3964	3979	AATACTAATGTCCAAG	65	279
1246007	N/A	N/A	4572	4587	ATATAGCCCTTTCCCC	26	280
1246033	N/A	N/A	5081	5096	CGCCTAAAGGAGATCT	59	281
1246059	N/A	N/A	5241	5256	GTTAGCAAAACACTTG	71	282
1246085	N/A	N/A	5840	5855	AAACATGGATGTTAGG	56	283
1246111	N/A	N/A	6036	6051	TATACTTTTCCGTCTC	71	284
1246137	N/A	N/A	6305	6320	TACTCTAGGAGATATA	33	285
1246163	N/A	N/A	6478	6493	CAACTATGTAATAGGC	79	286

TABLE 4-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt qampers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246189	N/A	N/A	7684	7699	AAGAAATTTGTGAACC	21	287
1246215	N/A	N/A	7973	7988	AGTTAATAGGACTAAA	13	288
1246241	N/A	N/A	8246	8261	AACCTAATAACTACAT	0	289
1246267	N/A	N/A	8705	8720	GACCAATATAATCCAA	34	290
1246293	N/A	N/A	9009	9024	GATTAAGGCAAAGATC	25	291
1246319	N/A	N/A	9373	9388	CTAGAACACTTGCCCTC	25	292
1246345	N/A	N/A	9540	9555	GAATCCAGATCTGCCC	29	293
1246371	N/A	N/A	9721	9736	GAGGATAGTGAACATA	50	294
1246397	N/A	N/A	9985	10000	CTGTATTAAGACAGTT	41	295
1246423	N/A	N/A	10336	10351	TCCAAATCAGGTAGTT	22	296
1246449	N/A	N/A	10833	10848	GAGTAAACGACTCTGT	43	297
1246475	N/A	N/A	11386	11401	CTTTTAGTCAGGTAGA	33	298
1246501	N/A	N/A	11780	11795	GCTTAATGTTTCAGTTT	50	299
1246527	N/A	N/A	12009	12024	AAGCATTGTTAGCTAA	11	300
1246553	N/A	N/A	12465	12480	ATTTATTTGCTGGTCC	47	301
1246579	N/A	N/A	12647	12662	TATACTAGGATTAGAA	2	302
1246605	N/A	N/A	12778	12793	AAATGAGGCGGCACTC	20	303
1246631	N/A	N/A	13704	13719	AATGTAAGAAGCCACG	26	304
1246657	N/A	N/A	13881	13896	CCCAAGAGTGGCAGGA	29	305
1246683	N/A	N/A	14044	14059	CATATCAATGCCTTAG	29	306
1246709	N/A	N/A	14205	14220	GCAGATTATTATGTGA	59	307
1246735	N/A	N/A	14436	14451	TTTACTGAAGGCTGT	23	308
1246761	N/A	N/A	14916	14931	ATACATTAGCAAGCTA	69	309
1246787	N/A	N/A	15319	15334	CTAGGATTAATCATGG	65	310
1246813	N/A	N/A	15420	15435	TCCAGTAGGTGTGTTT	47	311
1246839	N/A	N/A	16116	16131	CTCTATTGGGCCAGGC	46	312
1246865	N/A	N/A	16461	16476	GATATTATGTTCTTGG	59	313
1246891	N/A	N/A	17909	17924	GGAAATTGTTGCTGTT	46	314
1246917	N/A	N/A	18097	18112	GAAATTGATACACCAA	67	315
1246943	N/A	N/A	18819	18834	ATGCTAGTAATTGCAT	16	316
1246969	N/A	N/A	19484	19499	AGAATTAAGGGCTATA	45	317
1246995	N/A	N/A	19689	19704	GATGCTAAAAGCTCAC	46	318
1247021	N/A	N/A	19908	19923	TCTTAAGATACCCAGG	58	319
1247047	N/A	N/A	20189	20204	GTTATTAGAAGTCAGC	56	320
1247073	N/A	N/A	20415	20430	AATTATGCCTTGTGGC	19	321

TABLE 5

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245072	7	22	3101	3116	TCCTAGGGAGGAGGTA	33	322
1245098	64	79	3158	3173	AGGATGATGTTTCATGG	88	323
1245124	115	130	3209	3224	AACGACTCCAAGTAGG	27	324
1245150	205	220	3299	3314	GTCTGCCTGCCTATTC	53	325
1245176	266	281	N/A	N/A	ACCGCGCTTATTAATA	4	326
1245202	303	318	7592	7607	CGCCTAGTTTTTCGGCA	23	327
1245228	363	378	7652	7667	TTAGAGAGCGATAGAT	23	328
1245254	406	421	8800	8815	TTCACCACGATTGTTA	40	329
1245280	436	451	8830	8845	AGATCGGCTGGATATA	27	330
1245306	479	494	8873	8888	GACCTCAAATGTCTTG	12	331
1245332	526	541	10412	10427	ATCGATGGAAGAAGTG	39	332
1245358	573	588	10459	10474	CGCACACTGAAGCCAC	56	333
1245384	624	639	12042	12057	CGGCAAATTTGCTGGA	29	334
1245410	690	705	12108	12123	AGGTTTTGATACCAGT	60	335
1245436	763	778	15644	15659	TCCAATACAGGCCATA	62	336
1245462	795	810	15676	15691	CATCTATCAGACTTCT	38	337
1245488	846	861	15727	15742	AGATATTGATATACGA	20	338
1245514*	911	926	20726	20741	AATATTCTGCATACGA	70	339
1245540*	978	993	20793	20808	ACATCTCTGGCTGGAG	85	340
1245566	1018	1033	20833	20848	GCAGCATTGATTCGAA	63	341
1245592	1105	1120	20920	20935	TAATTAATCTTGTTTCG	63	342
1245618	1264	1279	21079	21094	GGGAGGTAGCTTTTGT	10	343
1245644	1318	1333	21133	21148	TGAACAGTCTTAAACC	56	344
1245670	1345	1360	21160	21175	GGCATGGCTACAGATT	69	345
1245696	1389	1404	21204	21219	TCTCTTAGCTGTGCAC	79	346
1245722	1428	1443	21243	21258	TATGTCCAGGTTGAGA	64	347
1245748	1503	1518	21318	21333	GAATAGAGTTGCACCG	55	348
1245774	1562	1577	21377	21392	GGTCCACTTTTGGTGG	13	349
1245800	1621	1636	21436	21451	GAGTCGGTCACCTTTC	82	350
1245826	1715	1730	21530	21545	TAAACATCTCTGGGAC	51	351
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	58	38
1245852	1776	1791	21591	21606	TGTAATAGCCAGTACA	46	352
1245878	2181	2196	21996	22011	AGTGGGAGTCGGATTA	50	353
1245904	2252	2267	22067	22082	AGAGGATAGTCCATGC	74	354
1245930	N/A	N/A	3439	3454	AAGATAAGTAGATGGT	79	355
1245956	N/A	N/A	3602	3617	AGCTTGGAGGAGACT	69	356

TABLE 5-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245982	N/A	N/A	3999	4014	GTTAATGTAGTGTTTA	77	357
1246008	N/A	N/A	4573	4588	AATATAGCCCTTTCCC	44	358
1246034	N/A	N/A	5082	5097	TCGCCTAAAGGAGATC	70	359
1246060	N/A	N/A	5267	5282	TGCAAAATGTGATGCC	69	360
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	90	48
1246086	N/A	N/A	5841	5856	CAAACATGGATGTTAG	49	361
1246112	N/A	N/A	6037	6052	TTATACTTTTCCGTCT	61	362
1246138	N/A	N/A	6323	6338	ATAGGGAGACCATGCT	64	363
1246164	N/A	N/A	6567	6582	AAAACATTGCTCTTCG	59	364
1246190	N/A	N/A	7751	7766	TAAATCAGGCAGTCAT	35	365
1246216	N/A	N/A	7974	7989	AAGTTAATAGGACTAA	17	366
1246242	N/A	N/A	8248	8263	ATAACCTAATAACTAC	26	367
1246268	N/A	N/A	8714	8729	AAGTTTTGGGACCAAT	11	368
1246294	N/A	N/A	9012	9027	CTTGATTAAGGCAAAG	38	369
1246320	N/A	N/A	9375	9390	ATCTAGAACACTTGCC	27	370
1246346	N/A	N/A	9588	9603	CTGGTAAAGGTAAGGG	58	371
1246372	N/A	N/A	9722	9737	AGAGGATAGTGAACAT	59	372
1246398	N/A	N/A	9986	10001	ACTGTATTAAGACAGT	20	373
1246424	N/A	N/A	10337	10352	TTCCAAATCAGGTAGT	12	374
1246450	N/A	N/A	10834	10849	TGAGTAAACGACTCTG	23	375
1246476	N/A	N/A	11387	11402	ACTTTTAGTCAGGTAG	26	376
1246502	N/A	N/A	11781	11796	GGCTTAATGTTTCAGTT	54	377
1246528	N/A	N/A	12011	12026	TTAAGCATTGTTAGCT	16	378
1246554	N/A	N/A	12466	12481	AATTTATTGCTGGTC	27	379
1246580	N/A	N/A	12648	12663	ATATACTAGGATTAGA	33	380
1246606	N/A	N/A	12779	12794	CAAATGAGGCGGCACT	32	381
1246632	N/A	N/A	13707	13722	GCAAAATGTAAGAAGCC	4	382
1246658	N/A	N/A	13894	13909	CTATCATGCCTTCCCC	1	383
1246684	N/A	N/A	14046	14061	TACATATCAATGCCTT	38	384
1246710	N/A	N/A	14206	14221	TGCAGATTATTATGTG	38	385
1246736	N/A	N/A	14437	14452	CTTATACTGAAGGCTG	48	386
1246762	N/A	N/A	14934	14949	CATTATAAGCTAACTA	39	387
1246788	N/A	N/A	15321	15336	GGCTAGGATTAATCAT	44	388
1246814	N/A	N/A	15421	15436	ATCCAGTAGGTGTGTT	41	389
1246840	N/A	N/A	16123	16138	ATAATAGCTCTATTGG	46	390
1246866	N/A	N/A	16533	16548	CAAGACTTAAACACCA	55	391

TABLE 5-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246892	N/A	N/A	17910	17925	GGGAAATTGTTGCTGT	42	392
1246918	N/A	N/A	18099	18114	CTGAAATTGATACACC	67	393
1246944	N/A	N/A	18827	18842	AAACCTTCATGCTAGT	27	394
1246970	N/A	N/A	19485	19500	TAGAATTAGGGCTAT	31	395
1246996	N/A	N/A	19699	19714	GATTAATCTTGATGCT	26	396
1247022	N/A	N/A	19909	19924	ATCTTAAGATACCCAG	70	397
1247048	N/A	N/A	20192	20207	GCAGTTATTAGAAGTC	76	398
1247074	N/A	N/A	20419	20434	GTAAAATTATGCCTTG	50	399

TABLE 6

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245073	8	23	3102	3117	GTCCTAGGGAGGAGGT	36	400
1245099	65	80	3159	3174	TAGGATGATGTTTCATG	72	401
1245125	116	131	3210	3225	CAACGACTCCAAGTAG	0	402
1245151	206	221	3300	3315	AGTCTGCCTGCCTATT	47	403
1245177	267	282	N/A	N/A	CACCGCGCTTATTAAT	7	404
1245203	304	319	7593	7608	ACGCCTAGTTTTTCGGC	7	405
1245229	364	379	7653	7668	TTTAGAGAGCGATAGA	13	406
1245255	407	422	8801	8816	ATTCACCACGATTGTT	0	407
1245281	437	452	8831	8846	AAGATCGGCTGGATAT	1	408
1245307	481	496	8875	8890	TTGACCTCAAATGTCT	0	409
1245333	527	542	10413	10428	CATCGATGGAAGAAGT	16	410
1245359	574	589	10460	10475	CCGCACACTGAAGCCA	34	411
1245385	625	640	12043	12058	GCGCAAAATTTGCTGG	0	412
1245411	691	706	12109	12124	GAGGTTTTGATACCAG	59	413
1245437	764	779	15645	15660	CTCCAATACAGGCCAT	20	414
1245463	796	811	15677	15692	CCATCTATCAGACTTC	78	415
1245489	847	862	15728	15743	AAGATATTGATATACG	45	416
1245515*	912	927	20727	20742	GAATATTCTGCATACG	68	417
1245541*	979	994	20794	20809	TACATCTCTGGCTGGA	86	418
1245567	1019	1034	20834	20849	TGCAGCATTGATTCTGA	47	419
1245593	1106	1121	20921	20936	GTAATTAATCTTGTTTC	52	420
1245619	1265	1280	21080	21095	AGGGAGGTAGCTTTTG	46	421

TABLE 6-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245645	1319	1334	21134	21149	TTGAACAGTCTTAAAC	33	422
1245671	1346	1361	21161	21176	TGGCATGGCTACAGAT	57	423
1245697	1390	1405	21205	21220	ATCTCTTAGCTGTGCA	63	424
1245723	1429	1444	21244	21259	ATATGTCCAGGTTGAG	61	425
1245749	1504	1519	21319	21334	AGAATAGAGTTGCACC	70	426
1245775	1563	1578	21378	21393	GGGTCCACTTTTGGTG	13	427
1245801	1622	1637	21437	21452	AGAGTCGGTCACCTTT	71	428
1245827	1718	1733	21533	21548	GTCTAAACATCTCTGG	58	429
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	57	38
1245853	1777	1792	21592	21607	ATGTAATAGCCAGTAC	52	430
1245879	2182	2197	21997	22012	TAGTGGGAGTCGGATT	54	431
1245905	2253	2268	22068	22083	AAGAGGATAGTCCATG	65	432
1245931	N/A	N/A	3474	3489	AGGGAATTTATCAAAC	47	433
1245957	N/A	N/A	3610	3625	CATTTAGCAGCTTGGGA	88	434
1245983	N/A	N/A	4002	4017	GTTGTTAATGTAGTGT	77	435
1246009	N/A	N/A	4641	4656	ATACGACTTCCTTCTA	44	436
1246035	N/A	N/A	5104	5119	GAAAACTCAGCCAGCA	78	437
1246061	N/A	N/A	5278	5293	AGCTAGACAATTGCAA	30	438
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	87	48
1246087	N/A	N/A	5842	5857	CCAAACATGGATGTTA	43	439
1246113	N/A	N/A	6038	6053	ATTATACTTTTCCGTC	75	440
1246139	N/A	N/A	6327	6342	TTTCATAGGGAGACCA	71	441
1246165	N/A	N/A	6607	6622	CTAAAGCAGGCTACTT	32	442
1246191	N/A	N/A	7752	7767	TTAATCAGGCAGTCA	45	443
1246217	N/A	N/A	7975	7990	AAAGTTAATAGGACTA	12	444
1246243	N/A	N/A	8249	8264	CATAACCTAATAACTA	10	445
1246269	N/A	N/A	8717	8732	TCAAAGTTTGGGACC	42	446
1246295	N/A	N/A	9080	9095	TCCACTTGAATCTGT	32	447
1246321	N/A	N/A	9376	9391	GATCTAGAACACTTGC	27	448
1246347	N/A	N/A	9589	9604	GCTGGTAAAGGTAAGG	65	449
1246373	N/A	N/A	9751	9766	CTCCAAATCCCAACC	31	450
1246399	N/A	N/A	9998	10013	AAATCTTGTGTAACGT	41	451
1246425	N/A	N/A	10359	10374	GACAATGACATAGACG	0	452
1246451	N/A	N/A	10835	10850	CTGAGTAACGACTCT	33	453
1246477	N/A	N/A	11389	11404	CAACTTTTAGTCAGGT	66	454
1246503	N/A	N/A	11782	11797	TGGCTTAATGTTTCAGT	50	455

TABLE 6-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246529	N/A	N/A	12012	12027	ATTAAGCATTTGTTAGC	0	456
1246555	N/A	N/A	12467	12482	CAATTTATTGCTGGT	31	457
1246581	N/A	N/A	12649	12664	TATATACTAGGATTAG	10	458
1246607	N/A	N/A	12781	12796	CTCAAATGAGGCGGCA	56	459
1246633	N/A	N/A	13711	13726	CGTAGCAAATGTAAGA	0	460
1246659	N/A	N/A	13896	13911	ATCTATCATGCCTTCC	6	461
1246685	N/A	N/A	14047	14062	TTACATATCAATGCCT	12	462
1246711	N/A	N/A	14243	14258	GTTAAACTTCATTCC	5	463
1246737	N/A	N/A	14438	14453	CCTTATACTGAAGGCT	41	464
1246763	N/A	N/A	14935	14950	TCATTATAAGCTAACT	0	465
1246789	N/A	N/A	15328	15343	GAGGAATGGCTAGGAT	47	466
1246815	N/A	N/A	15424	15439	CATATCCAGTAGGTGT	37	467
1246841	N/A	N/A	16128	16143	CCATAATAATAGCTCT	48	468
1246867	N/A	N/A	16755	16770	AAAGTTAGTTGGGCGA	26	469
1246893	N/A	N/A	17911	17926	TGGGAAATTGTTGCTG	38	470
1246919	N/A	N/A	18100	18115	ACTGAAATTGATACAC	49	471
1246945	N/A	N/A	18828	18843	GAAACCTTCATGCTAG	31	472
1246971	N/A	N/A	19486	19501	CTAGAATTAAGGGCTA	42	473
1246997	N/A	N/A	19700	19715	TGATTAATCTTGATGC	15	474
1247023	N/A	N/A	19910	19925	CATCTTAAGATACCCA	76	475
1247049	N/A	N/A	20248	20263	CCTAAACAAACACTAT	8	476
1247075	N/A	N/A	20422	20437	ACAGTAAATTATGCC	73	477

TABLE 7

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245074	10	25	3104	3119	TAGTCCTAGGGAGGAG	68	478
1245100	66	81	3160	3175	CTAGGATGATGTTTCAT	47	479
1245126	117	132	3211	3226	CCAACGACTCCAAGTA	20	480
1245152	207	222	3301	3316	TAGTCTGCCTGCCTAT	44	481
1245178	268	283	N/A	N/A	ACACCGCGCTTATTAA	7	482
1245204	316	331	7605	7620	GCATGCGCAGTGACGC	36	483
1245230	365	380	7654	7669	ATTTAGAGAGCGATAG	14	484

TABLE 7-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245256	408	423	8802	8817	TATTCACCACGATTGT	2	485
1245282	438	453	8832	8847	GAAGATCGGCTGGATA	16	486
1245308	482	497	8876	8891	GTTGACCTCAAATGTC	0	487
1245334	528	543	10414	10429	TCATCGATGGAAGAAG	24	488
1245360	575	590	10461	10476	GCCGCACACTGAAGCC	25	489
1245386	626	641	12044	12059	AGCGGCAAATTTGCTG	5	490
1245412	692	707	12110	12125	TGAGGTTTTGATACCA	63	491
1245438	765	780	15646	15661	TCTCCAATACAGGCCA	21	492
1245464	797	812	15678	15693	TCCATCTATCAGACTT	42	493
1245490*	858	873	15739	15754	GTAGTCTCAGAAAGAT	45	494
1245516*	926	941	20741	20756	CACTGCTTCAAATTGA	64	495
1245542*	980	995	20795	20810	ATACATCTCTGGCTGG	88	496
1245568	1020	1035	20835	20850	TTGCAGCATTGATTCG	59	497
1245594	1107	1122	20922	20937	GGTAATTAATCTTGTT	58	498
1245620	1266	1281	21081	21096	TAGGGAGGTAGCTTTT	46	499
1245646	1320	1335	21135	21150	CTTGAACAGTCTTAAA	39	500
1245672	1347	1362	21162	21177	GTGGCATGGCTACAGA	56	501
1245698	1391	1406	21206	21221	GATCTCTTAGCTGTGC	64	502
1245724	1430	1445	21245	21260	AATATGTCCAGGTTGA	64	503
1245750	1505	1520	21320	21335	CAGAAATAGAGTTGCAC	50	504
1245776	1564	1579	21379	21394	AGGGTCCACTTTTGGT	11	505
1245802	1623	1638	21438	21453	TAGAGTCGGTCACCTT	65	506
1245828	1719	1734	21534	21549	TGTCTAAACATCTCTG	58	507
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	51	38
1245854	1778	1793	21593	21608	TATGTAATAGCCAGTA	50	508
1245880	2183	2198	21998	22013	GTAGTGGGAGTCGGAT	48	509
1245906	2254	2269	22069	22084	CAAGAGGATAGTCCAT	58	510
1245932	N/A	N/A	3487	3502	AGACTAAGGGACCAGG	80	511
1245958	N/A	N/A	3611	3626	TCATTTAGCAGCTTGG	79	512
1245984	N/A	N/A	4123	4138	ACAATTTTCCAATCC	86	513
1246010	N/A	N/A	4642	4657	AATACGACTTCCTTCT	26	514
1246036	N/A	N/A	5105	5120	TGAAAACCTCAGCCAGC	77	515
1246062	N/A	N/A	5287	5302	TTATAACTGAGCTAGA	8	516
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	89	48
1246088	N/A	N/A	5843	5858	CCCAAACATGGATGTT	72	517

TABLE 7-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246114	N/A	N/A	6039	6054	AATTATACTTTTCCGT	66	518
1246140	N/A	N/A	6328	6343	CTTTCATAGGGAGACC	73	519
1246166	N/A	N/A	6608	6623	TCTAAAGCAGGCTACT	32	520
1246192	N/A	N/A	7757	7772	CCAATTTAAATCAGGC	73	521
1246218	N/A	N/A	7976	7991	CAAAGTTAATAGGACT	50	522
1246244	N/A	N/A	8280	8295	GTCTAAAGTGCTGGTT	39	523
1246270	N/A	N/A	8747	8762	GAACACTTTCCTGGG	0	524
1246296	N/A	N/A	9110	9125	ACCAAGTAGCTTACAT	2	525
1246322	N/A	N/A	9377	9392	GGATCTAGAACACTTG	32	526
1246348	N/A	N/A	9598	9613	ATGTGAAGAGCTGGTA	42	527
1246374	N/A	N/A	9761	9776	CCCAAGCTACCTCCAA	15	528
1246400	N/A	N/A	10001	10016	GTGAAATCTTGTTAA	44	529
1246426	N/A	N/A	10360	10375	AGACAATGACATAGAC	14	530
1246452	N/A	N/A	10836	10851	ACTGAGTAAACGACTC	45	531
1246478	N/A	N/A	11428	11443	CCCTTTATGTCTTTGG	35	532
1246504	N/A	N/A	11790	11805	CTATTGATTGGCTTAA	31	533
1246530	N/A	N/A	12038	12053	AAATTTGCTGGAAC TG	15	534
1246556	N/A	N/A	12478	12493	AGCTAGATACTCAATT	2	535
1246582	N/A	N/A	12652	12667	GAATATATACTAGGAT	5	536
1246608	N/A	N/A	12782	12797	CCTCAAATGAGGCGGC	47	537
1246634	N/A	N/A	13730	13745	ATTAAGTACTGTGAGA	1	538
1246660	N/A	N/A	13897	13912	CATCTATCATGCCTTC	18	539
1246686	N/A	N/A	14048	14063	ATTACATATCAATGCC	36	540
1246712	N/A	N/A	14250	14265	AGCGAATGTTAAACT	16	541
1246738	N/A	N/A	14439	14454	TCCTTATACTGAAGGC	22	542
1246764	N/A	N/A	14936	14951	CTCATTATAAGCTAAC	13	543
1246790	N/A	N/A	15350	15365	AGTAGTGGAGCCAGAC	39	544
1246816	N/A	N/A	15425	15440	TCATATCCAGTAGGTG	32	545
1246842	N/A	N/A	16129	16144	TCCATAATAATAGCTC	69	546
1246868	N/A	N/A	16905	16920	CGTCAAAAACCGTCAA	20	547
1246894	N/A	N/A	17923	17938	GAGTATTTCTCTGGG	23	548
1246920	N/A	N/A	18506	18521	ACTAATTTAGTCAACT	18	549
1246946	N/A	N/A	18846	18861	TTTAACTACCCTCACA	0	550
1246972	N/A	N/A	19487	19502	TCTAGAATTAAGGGCT	45	551
1246998	N/A	N/A	19701	19716	CTGATTAATCTTGATG	32	552
1247024	N/A	N/A	19911	19926	GCATCTTAAGATACCC	82	553

TABLE 7-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1247050	N/A	N/A	20249	20264	GCCTAAACAAACACTA	37	554
1247076	N/A	N/A	20481	20496	CATAGTGGACTTCATT	34	555

TABLE 8

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245075	11	26	3105	3120	GTAGTCCTAGGGAGGA	77	556
1245101	67	82	3161	3176	TCTAGGATGATGTTCA	64	557
1245127	118	133	3212	3227	ACCAACGACTCCAAGT	16	558
1245153	208	223	3302	3317	GTAGTCTGCCTGCCTA	68	559
1245179	269	284	N/A	N/A	CACACCGCGCTTATTA	11	560
1245205	317	332	7606	7621	CGCATGCGCAGTGACG	46	561
1245231	366	381	7655	7670	GATTTAGAGAGCGATA	44	562
1245257	409	424	8803	8818	TTATTACCACGATTG	40	563
1245283	439	454	8833	8848	AGAAGATCGGCTGGAT	29	564
1245309	483	498	8877	8892	TGTTGACCTCAAATGT	14	565
1245335	529	544	10415	10430	ATCATCGATGGAAGAA	30	566
1245361	576	591	10462	10477	GGCCGCACACTGAAGC	9	567
1245387	632	647	12050	12065	GCCAACAGCGGCAAAT	30	568
1245413	693	708	12111	12126	ATGAGGTTTTGATACC	47	569
1245439	766	781	15647	15662	GTCTCCAATACAGGCC	47	570
1245465	798	813	15679	15694	TTCCATCTATCAGACT	33	571
1245491*	866	881	N/A	N/A	AAACTTCTGTAGTCTC	67	572
1245517*	927	942	20742	20757	CCACTGCTTCAAATTG	77	573
1245543*	981	996	20796	20811	CATACATCTCTGGCTG	69	574
1245569	1021	1036	20836	20851	TTTGACGATTGATTC	53	575
1245595	1109	1124	20924	20939	CAGGTAATTAATCTTG	54	576
1245621	1267	1282	21082	21097	TTAGGGAGGTAGCTTT	54	577
1245647	1321	1336	21136	21151	ACTTGAACAGTCTTAA	63	578
1245673	1348	1363	21163	21178	TGTGGCATGGCTACAG	24	579
1245699	1392	1407	21207	21222	TGATCTCTTAGCTGTG	59	580
1245725	1431	1446	21246	21261	AAATATGTCCAGTTG	68	581

TABLE 8-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1245751	1506	1521	21321	21336	CCAGAATAGAGTTGCA	62	582
1245777	1565	1580	21380	21395	GAGGGTCCACTTTTGG	23	583
1245803	1624	1639	21439	21454	ATAGAGTCGGTCACCT	60	584
1245829	1721	1736	21536	21551	ATTGTCTAAACATCTC	84	585
1245855	1779	1794	21594	21609	TTATGTAATAGCCAGT	77	586
1245881	2184	2199	21999	22014	TGTAGTGGGAGTCGGA	62	587
1245907	2255	2270	22070	22085	ACAAGAGGATAGTCCA	64	588
1245933	N/A	N/A	3488	3503	CAGACTAAGGGACCAG	75	589
1245959	N/A	N/A	3612	3627	CTCATTTAGCAGCTTG	70	590
1245985	N/A	N/A	4136	4151	CAAGGAGTACTTCACA	66	591
1246011	N/A	N/A	4643	4658	TAATACGACTTCCTTC	32	592
1246037	N/A	N/A	5125	5140	GCAAAAGGAGCCAGTT	57	593
1246063	N/A	N/A	5290	5305	GTTTTATAACTGAGCT	84	594
1246089	N/A	N/A	5871	5886	CTAAACCCTGGAGCAG	42	595
1246115	N/A	N/A	6040	6055	AAATTATACCTTTCCG	85	596
1246141	N/A	N/A	6337	6352	GAAGAAGGACTTTCAT	40	597
1246167	N/A	N/A	6609	6624	CTCTAAGCAGGCTAC	51	598
1246193	N/A	N/A	7758	7773	TCCAATTTAAATCAGG	61	599
1246219	N/A	N/A	7977	7992	GCAAAGTTAATAGGAC	52	600
1246245	N/A	N/A	8282	8297	TGGTCTAAAGTGCTGG	61	601
1246271	N/A	N/A	8895	8910	CCCAAAATGTCCTAG	18	602
1246297	N/A	N/A	9111	9126	GACCAAGTAGCTTACA	28	603
1246323	N/A	N/A	9379	9394	ATGGATCTAGAACACT	43	604
1246349	N/A	N/A	9600	9615	ATATGTGAAGAGCTGG	51	605
1246375	N/A	N/A	9762	9777	ACCCAAGCTACCTCCA	9	606
1246401	N/A	N/A	10002	10017	AGTGAAATCTTGTGTA	41	607
1246427	N/A	N/A	10380	10395	GATCATTCGAGAAAGA	23	608
1246453	N/A	N/A	10849	10864	ACATATGGGTACACT	19	609
1246479	N/A	N/A	11468	11483	GCAAATACAGCAGTAC	56	610
1246505	N/A	N/A	11791	11806	TCTATTGATTGGCTTA	39	611
1246531	N/A	N/A	12039	12054	CAAATTTGCTGGAACT	39	612
1246557	N/A	N/A	12482	12497	AAGGAGCTAGATACTC	19	613
1246583	N/A	N/A	12654	12669	CAGAATATATACTAGG	55	614
1246609	N/A	N/A	12795	12810	TAAGAGTCAGTATCCT	52	615
1246635	N/A	N/A	13731	13746	TATTAAGTACTGTGAG	0	616
1246661	N/A	N/A	13915	13930	CATTATCCTTACTCCC	16	617

TABLE 8-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1246687	N/A	N/A	14049	14064	CATTACATATCAATGC	29	618
1246713	N/A	N/A	14262	14277	AGTTAGGGAGCCAGCG	38	619
1246739	N/A	N/A	14465	14480	CTGGAGGTATGTCATA	23	620
1246765	N/A	N/A	15022	15037	TGATAAAATAGGGTGC	44	621
1246791	N/A	N/A	15351	15366	GAGTAGTGGAGCCAGA	44	622
1246817	N/A	N/A	15427	15442	TTTCATATCCAGTAGG	36	623
1246843	N/A	N/A	16189	16204	AATAACTGTTCTCCCC	20	624
1246869	N/A	N/A	16956	16971	GACAACAAACAATGGG	39	625
1246895	N/A	N/A	17924	17939	AGAGTATTTCTCTGG	32	626
1246921	N/A	N/A	18507	18522	CACTAATTTAGTCAAC	7	627
1246947	N/A	N/A	18847	18862	TTTTAACTACCCTCAC	2	628
1246973	N/A	N/A	19488	19503	ATCTAGAATTAAGGGC	41	629
1246999	N/A	N/A	19702	19717	GCTGATTAATCTTGAT	48	630
1247025	N/A	N/A	19921	19936	TCCAAGGAGTGCATCT	46	631
1247051	N/A	N/A	20356	20371	TTATTGCCTGAACACA	58	632
1247077	N/A	N/A	20482	20497	TCATAGTGGACTTCAT	36	633

TABLE 9

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1245076	12	27	3106	3121	TGTAGTCCTAGGGAGG	86	634
1245102	68	83	3162	3177	TTCTAGGATGATGTTT	72	635
1245128	119	134	3213	3228	CACCAACGACTCCAAG	16	636
1245154	211	226	3305	3320	TAAGTAGTCTGCCTGC	19	637
1245180	271	286	N/A	N/A	TCCACACCGCCTTAT	32	638
1245206	318	333	7607	7622	ACGCATGCGCAGTGAC	52	639
1245232	367	382	7656	7671	TGATTTAGAGAGCGAT	35	640
1245258	410	425	8804	8819	ATTATTCACCACGATT	29	641
1245284	440	455	8834	8849	GAGAAGATCGGCTGGA	6	642
1245310	484	499	8878	8893	ATGTTGACCTCAAATG	0	643
1245336	530	545	10416	10431	CATCATCGATGGAAGA	14	644
1245362	577	592	10463	10478	TGGCCGCACACTGAAG	0	645

TABLE 9-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245388	638	653	12056	12071	GTGAAAGCCAACAGCG	62	646
1245414	696	711	12114	12129	GACATGAGGTTTGGAT	52	647
1245440	767	782	15648	15663	TGTCTCCAATACAGGC	39	648
1245466	799	814	15680	15695	ATTCCATCTATCAGAC	7	649
1245492*	867	882	N/A	N/A	GAAACTTCTGTAGTCT	86	650
1245518*	931	946	20746	20761	CCAACCACTGCTTCAA	98	651
1245544*	982	997	20797	20812	GCATACATCTCTGGCT	27	652
1245570	1023	1038	20838	20853	GCTTTGCAGCATTGAT	61	653
1245596	1110	1125	20925	20940	ACAGGTAATTAATCTT	39	654
1245622	1268	1283	21083	21098	TTAGGGAGGTAGCTT	24	655
1245648	1322	1337	21137	21152	TACTTGAACAGTCTTA	48	656
1245674	1349	1364	21164	21179	CTGTGGCATGGCTACA	66	657
1245700	1393	1408	21208	21223	TTGATCTCTTAGCTGT	74	658
1245726	1461	1476	21276	21291	CTAGGGAAATCTTTCA	48	659
1245752	1507	1522	21322	21337	TCCAGAATAGAGTTGC	46	660
1245778	1566	1581	21381	21396	AGAGGGTCCACTTTTG	34	661
1245804	1625	1640	21440	21455	AATAGAGTCGGTCACC	58	662
1245830	1728	1743	21543	21558	GCCTAAAATTGTCTAA	32	663
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	49	38
1245856	1780	1795	21595	21610	CTTATGTAATAGCCAG	74	664
1245882	2185	2200	22000	22015	ATGTAGTGGGAGTCGG	72	665
1245908	2256	2271	22071	22086	AACAAGAGGATAGTCC	54	666
1245934	N/A	N/A	3489	3504	ACAGACTAAGGGACCA	68	667
1245960	N/A	N/A	3613	3628	CCTCATTTAGCAGCTT	81	668
1245986	N/A	N/A	4138	4153	CTCAAGGAGTACTTCA	82	669
1246012	N/A	N/A	4644	4659	TTAATACGACTTCCTT	35	670
1246038	N/A	N/A	5126	5141	GGCAAAGGAGCCAGT	43	671
1246064	N/A	N/A	5655	5670	GTTACCAGAGCATTCA	73	672
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	89	48
1246090	N/A	N/A	5872	5887	GCTAAACCCTGGAGCA	5	673
1246116	N/A	N/A	6089	6104	CAAGTTTTCACCTCAG	93	674
1246142	N/A	N/A	6357	6372	GGGAAAGAAGTCTCCT	13	675
1246168	N/A	N/A	6610	6625	TCTCTAAAGCAGGCTA	60	676
1246194	N/A	N/A	7770	7785	TAGCAGCTAAAATCCA	66	677
1246220	N/A	N/A	7978	7993	TGCAAAGTTAATAGGA	43	678
1246246	N/A	N/A	8487	8502	CTATATAAATTCTGAC	0	679

TABLE 9-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246272	N/A	N/A	8939	8954	TAAGAGAAGGTGCACA	12	680
1246298	N/A	N/A	9130	9145	GATACTGGTGGACAAG	13	681
1246324	N/A	N/A	9410	9425	GTATAGCTGCATTCT	35	682
1246350	N/A	N/A	9601	9616	CATATGTGAAGAGCTG	25	683
1246376	N/A	N/A	9776	9791	AACTAGAACTCCCAAC	0	684
1246402	N/A	N/A	10025	10040	GGCTTTATCATTCTAA	20	685
1246428	N/A	N/A	10384	10399	TAAGGATCATTGCAGA	0	686
1246454	N/A	N/A	10855	10870	CCATTTACATATGGTG	28	687
1246480	N/A	N/A	11469	11484	GGCAAATACAGCAGTA	64	688
1246506	N/A	N/A	11792	11807	CTCTATTGATTGGCTT	65	689
1246532	N/A	N/A	12168	12183	CTTACCTTGTGCTTGG	34	690
1246558	N/A	N/A	12486	12501	GACTAAGGAGCTAGAT	25	691
1246584	N/A	N/A	12655	12670	GCAGAATATATACTAG	55	692
1246610	N/A	N/A	12799	12814	GCTGTAAGAGTCAGTA	89	693
1246636	N/A	N/A	13733	13748	CTTATTAAGTACTGTG	0	694
1246662	N/A	N/A	13916	13931	CCATTATCCTTACTCC	27	695
1246688	N/A	N/A	14089	14104	CTACATGGATCCAACA	5	696
1246714	N/A	N/A	14266	14281	GAGGAGTTAGGGAGCC	36	697
1246740	N/A	N/A	14469	14484	ACAACCTGGAGGTATGT	0	698
1246766	N/A	N/A	15023	15038	TTGATAAAATAGGGTG	46	699
1246792	N/A	N/A	15353	15368	AGGAGTAGTGGAGCCA	16	700
1246818	N/A	N/A	15443	15458	TATCAGAACTTATAC	0	701
1246844	N/A	N/A	16191	16206	GAAATAACTGTTCTCC	32	702
1246870	N/A	N/A	16969	16984	TCCAATATATACTGAC	72	703
1246896	N/A	N/A	17930	17945	GCTGAGAGGTATTTTC	51	704
1246922	N/A	N/A	18508	18523	ACACTAATTTAGTCAA	16	705
1246948	N/A	N/A	18848	18863	CTTTTAACTACCCTCA	16	706
1246974	N/A	N/A	19490	19505	CCATCTAGAATTAAGG	0	707
1247000	N/A	N/A	19731	19746	ACTGAAGGTCTGAGCT	20	708
1247026	N/A	N/A	19922	19937	CTCCAAGGAGTGCATC	53	709
1247052	N/A	N/A	20359	20374	AAATTATTGCCTGAAC	32	710
1247078	N/A	N/A	20484	20499	CTTCATAGTGGACTTC	79	711

TABLE 10

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245077	13	28	3107	3122	GTGTAGTCCTAGGGAG	67	712
1245103	71	86	3165	3180	GATTTCTAGGATGATG	57	713
1245129	120	135	3214	3229	TCACCAACGACTCCAA	19	714
1245155	212	227	3306	3321	ATAAGTAGTCTGCCTG	43	715
1245181	272	287	N/A	N/A	CTCCACACCGCGCTTA	7	716
1245207	319	334	7608	7623	TACGCATGCGCAGTGA	60	717
1245233	368	383	7657	7672	CTGATTTAGAGAGCGA	61	718
1245259	411	426	8805	8820	CATTATTCAACACGAT	33	719
1245285	441	456	8835	8850	TGAGAAGATCGGCTGG	37	720
1245311	485	500	8879	8894	GATGTTGACCTCAAAT	0	721
1245337	531	546	10417	10432	CCATCATCGATGGAAG	8	722
1245363	578	593	10464	10479	GTGGCCGCACACTGAA	41	723
1245389	642	657	12060	12075	CTCTGTGAAAGCCAAC	50	724
1245415	698	713	12116	12131	GAGACATGAGGTTTGT	77	725
1245441	768	783	15649	15664	CTGTCTCCAATACAGG	29	726
1245467	801	816	15682	15697	GTATTCCATCTATCAG	52	727
1245493*	877	892	N/A	N/A	CGTTCAGGAAGAAACT	70	728
1245519*	932	947	20747	20762	GCCAACCACTGCTTCA	97	729
1245545*	983	998	20798	20813	TGCATACATCTCTGGC	53	730
1245571	1024	1039	20839	20854	AGCTTTGCAGCATTGA	62	731
1245597	1111	1126	20926	20941	GACAGGTAATTAATCT	60	732
1245623	1269	1284	21084	21099	TTTtagggaggtagct	39	733
1245649	1323	1338	21138	21153	CTACTTGAACAGTCTT	80	734
1245675	1350	1365	21165	21180	TCTGTGGCATGGCTAC	67	735
1245701	1394	1409	21209	21224	CTTGATCTCTTAGCTG	51	736
1245727	1462	1477	21277	21292	GCTAGGGAAATCTTTC	66	737
1245753	1508	1523	21323	21338	GTCCAGAATAGAGTTG	60	738
1245779	1567	1582	21382	21397	TAGAGGGTCCACTTTT	36	739
1245805	1626	1641	21441	21456	AAATAGAGTCGGTCAC	59	740
1245831	1729	1744	21544	21559	AGCCTAAAATTGTCTA	53	741
1245857	1781	1796	21596	21611	TCTTATGTAATAGCCA	77	742
1245883	2186	2201	22001	22016	GATGTAGTGGGAGTCG	67	743
1245909	2257	2272	22072	22087	AAACAAGAGGATAGTC	48	744
1245935	N/A	N/A	3492	3507	TATACAGACTAAGGGA	24	745
1245961	N/A	N/A	3629	3644	GATAATATGTGACTTG	68	746
1245987	N/A	N/A	4139	4154	TCTCAAGGAGTACTTC	80	747

TABLE 10-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246013	N/A	N/A	4645	4660	TTTAATACGACTTCCT	72	748
1246039	N/A	N/A	5156	5171	CAAACGCTACTGTGT	57	749
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	91	48
1246091	N/A	N/A	5875	5890	TAGGCTAAACCCTGGA	64	750
1246117	N/A	N/A	6134	6149	CTCTAATGTTTCTTGA	53	751
1246143	N/A	N/A	6422	6437	TAAACTGTATGCCTCT	49	752
1246169	N/A	N/A	6622	6637	CAGTAGGCAGTTTCTC	72	753
1246195	N/A	N/A	7776	7791	TTCAGTTAGCAGCTAA	72	754
1246221	N/A	N/A	7979	7994	CTGCAAAGTTAATAGG	57	755
1246247	N/A	N/A	8529	8544	GTTTATCAGTAAAGTG	64	756
1246273	N/A	N/A	8940	8955	TTAAGAGAAGGTGCAC	38	757
1246299	N/A	N/A	9146	9161	GCCAAGCTTTGTGTCA	64	758
1246325	N/A	N/A	9413	9428	TCTGTATAGCTGCATT	47	759
1246351	N/A	N/A	9603	9618	CACATATGTGAAGAGC	71	760
1246377	N/A	N/A	9780	9795	ATAGAACTAGAACTCC	22	761
1246403	N/A	N/A	10058	10073	GATAATTTGGTGTCTAT	59	762
1246429	N/A	N/A	10385	10400	TTAAGGATCATTCAG	14	763
1246455	N/A	N/A	10931	10946	CATAAGGCCGATTCAT	32	764
1246481	N/A	N/A	11482	11497	GGAAAGGATATTTGGC	58	765
1246507	N/A	N/A	11793	11808	ACTCTATTGATTGGCT	57	766
1246533	N/A	N/A	12178	12193	TGATTTTGACCTTACC	11	767
1246559	N/A	N/A	12487	12502	TGACTAAGGAGCTAGA	31	768
1246585	N/A	N/A	12685	12700	ATTAAGCTTAGTATAT	10	769
1246611	N/A	N/A	12810	12825	AGCTAACTCAGGCTGT	14	770
1246637	N/A	N/A	13738	13753	GCATACTTATTAAGTA	32	771
1246663	N/A	N/A	13917	13932	TCCATTATCCTTACTC	15	772
1246689	N/A	N/A	14090	14105	TCTACATGGATCCAAC	26	773
1246715	N/A	N/A	14280	14295	TGTAAAGGCTGGGTGA	26	774
1246741	N/A	N/A	14473	14488	TATGACAACTGGAGGT	14	775
1246767	N/A	N/A	15024	15039	GTTGATAAAATAGGGT	42	776
1246793	N/A	N/A	15354	15369	AAGGAGTAGTGGAGCC	38	777
1246819	N/A	N/A	15445	15460	GTTATCAGAACTTAT	31	778
1246845	N/A	N/A	16219	16234	TTATAGACTGGGTAGG	51	779
1246871	N/A	N/A	17017	17032	GTTTAGCAATTTAGCA	50	780
1246897	N/A	N/A	17956	17971	ATTTAGAACTGCTCC	21	781

TABLE 10-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246923	N/A	N/A	18521	18536	GCTTAGTACCAAGACA	62	782
1246949	N/A	N/A	18849	18864	CCTTTTAACTACCCTC	22	783
1246975	N/A	N/A	19533	19548	AGGCTAAAATGGTCAT	36	784
1247001	N/A	N/A	19736	19751	CTAAAACTGAAGTCT	40	785
1247027	N/A	N/A	19936	19951	TAGGAACATTCCTCT	24	786
1247053	N/A	N/A	20362	20377	TCTAAATTATTGCCTG	43	787
1247079	N/A	N/A	20485	20500	ACTTCATAGTGGACTT	61	788

TABLE 11

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245078	14	29	3108	3123	TGTGTAGTCCTAGGGA	50	789
1245104	72	87	3166	3181	GGATTTC TAGGATGAT	58	790
1245130	121	136	3215	3230	TTCACCAACGACTCCA	62	791
1245156	213	228	3307	3322	CATAAGTAGTCTGCCT	31	792
1245182	273	288	N/A	N/A	CCTCCACACCGCGCTT	16	793
1245208	320	335	7609	7624	ATACGCATGCGCAGTG	38	794
1245234	369	384	7658	7673	CCTGATTTAGAGAGCG	42	795
1245260	412	427	8806	8821	GCATTATTCACCACGA	59	796
1245286	442	457	8836	8851	CTGAGAAGATCGGCTG	37	797
1245312	486	501	8880	8895	GGATGTTGACCTCAA	5	798
1245338	532	547	10418	10433	TCCATCATCGATGGAA	31	799
1245364	579	594	10465	10480	CGTGGCCGCACACTGA	42	800
1245390	645	660	12063	12078	GACCTCTGTGAAAGCC	46	801
1245416	715	730	12133	12148	TTCACAAAAAC TGGGC	37	802
1245442	769	784	15650	15665	TCTGTCTCCAATACAG	13	803
1245468	802	817	15683	15698	AGTATTCATCTATCA	62	804
1245494*	891	906	20706	20721	AAATCGCTGAGGCGCG	89	805
1245520*	933	948	20748	20763	GGCCAACCACTGCTTC	93	806
1245546*	984	999	20799	20814	ATGCATACATCTCTGG	53	807
1245572	1025	1040	20840	20855	AAGCTTTGCAGCATTG	59	808
1245598	1112	1127	20927	20942	AGACAGGTAATTAATC	29	809
1245624	1270	1285	21085	21100	CTTTTAGGGAGGTAGC	62	810

TABLE 11-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 3 Start Site	SEQ ID NO: 4 Start Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245650	1324	1339	21139	21154	GCTACTTGAACAGTCT	66	811
1245676	1351	1366	21166	21181	TTCTGTGGCATGGCTA	78	812
1245702	1395	1410	21210	21225	ACTTGATCTCTTAGCT	59	813
1245728	1463	1478	21278	21293	GGCTAGGGAAATCTTT	57	814
1245754	1509	1524	21324	21339	AGTCCAGAATAGAGTT	58	815
1245780	1568	1583	21383	21398	ATAGAGGGTCCACTTT	21	816
1245806	1627	1642	21442	21457	AAAATAGAGTCGGTCA	71	817
1245832	1730	1745	21545	21560	GAGCCTAAAATTGTCT	49	818
1245858	1782	1797	21597	21612	TTCTTATGTAATAGCC	71	819
1245884	2187	2202	22002	22017	TGATGTAGTGGGAGTC	65	820
1245910	2349	2364	22164	22179	CTACAAGAGTTATTT	31	821
1245936	N/A	N/A	3493	3508	ATATACAGACTAAGGG	24	822
1245962	N/A	N/A	3671	3686	AAGTTATTATCCCCAC	60	823
1245988	N/A	N/A	4220	4235	AACTAAACATGACAGC	72	824
1246014	N/A	N/A	4646	4661	TTTAAATACGACTTCC	57	825
1246040	N/A	N/A	5164	5179	ATAAGAGCCAACTGC	49	826
1246066	N/A	N/A	5660	5675	CATAAGTTACCAGAGC	80	827
1246092	N/A	N/A	5878	5893	CTGTAGGCTAAACCT	47	828
1246118	N/A	N/A	6135	6150	GCTCTAATGTTTCTTG	62	829
1246144	N/A	N/A	6423	6438	GTAAACTGTATGCCTC	74	830
1246170	N/A	N/A	6624	6639	TACAGTAGGCAGTTTC	60	831
1246196	N/A	N/A	7777	7792	CTTCAGTTAGCAGCTA	50	832
1246222	N/A	N/A	8039	8054	ACTACATTAACACCAA	46	833
1246248	N/A	N/A	8534	8549	TGCAAGTTTATCAGTA	70	834
1246274	N/A	N/A	8941	8956	CTTAAGAGAAGGTGCA	43	835
1246300	N/A	N/A	9188	9203	CTTAACAAACCTCCAC	3	836
1246326	N/A	N/A	9424	9439	ATTATATGAGGTCTGT	59	837
1246352	N/A	N/A	9604	9619	TCACATATGTGAAGAG	27	838
1246378	N/A	N/A	9781	9796	CATAGAACTAGAACTC	19	839
1246404	N/A	N/A	10059	10074	TGATAATTTGGTGTCA	3	840
1246430	N/A	N/A	10386	10401	CTTAAGGATCATTGCA	29	841
1246456	N/A	N/A	10971	10986	GTAATCTTGAGGCAGG	51	842
1246482	N/A	N/A	11511	11526	TATTATGAGGATCTGG	44	843
1246508	N/A	N/A	11853	11868	AGCAGAATTGTGAACG	67	844
1246534	N/A	N/A	12194	12209	CATACCCATTCTAACT	6	845

TABLE 11-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246560	N/A	N/A	12488	12503	TTGACTAAGGAGCTAG	20	846
1246586	N/A	N/A	12686	12701	CATTAAGCTTAGTATA	19	847
1246612	N/A	N/A	12813	12828	CATAGCTAACTCAGGC	29	848
1246638	N/A	N/A	13748	13763	CTTACATATTGCATAC	35	849
1246664	N/A	N/A	13951	13966	CTAAGTTAGCCCCAG	15	850
1246690	N/A	N/A	14091	14106	ATCTACATGGATCCAA	12	851
1246716	N/A	N/A	14283	14298	GAATGTAAAGGCTGGG	42	852
1246742	N/A	N/A	14477	14492	GGAGTATGACAACTGG	28	853
1246768	N/A	N/A	15170	15185	CGTTTAAATCAGAGCA	54	854
1246794	N/A	N/A	15355	15370	TAAGGAGTAGTGGAGC	18	855
1246820	N/A	N/A	15492	15507	ATTAATGCCACCCTAC	6	856
1246846	N/A	N/A	16223	16238	CTTTTATAGACTGGG	67	857
1246872	N/A	N/A	17019	17034	AGGTTTAGCAATTTAG	54	858
1246898	N/A	N/A	17985	18000	ATAGTTATCTTCTCAC	48	859
1246924	N/A	N/A	18585	18600	TATTAATATGACTTGC	25	860
1246950	N/A	N/A	18850	18865	GCCTTTTAACTACCCT	0	861
1246976	N/A	N/A	19534	19549	TAGGCTAAAATGGTCA	59	862
1247002	N/A	N/A	19739	19754	GATCTAAACTGAAGG	44	863
1247028	N/A	N/A	19970	19985	AATTCATACAGTTCG	66	864
1247054	N/A	N/A	20363	20378	GTCTAAATTATTGCCT	45	865
1247080	N/A	N/A	20489	20504	CATAACTTCATAGTGG	31	866

TABLE 12

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245079	15	30	3109	3124	TTGTGTAGTCCTAGGG	64	867
1245105	75	90	3169	3184	GAAGGATTTCTAGGAT	71	868
1245131	122	137	3216	3231	CTTACCAACGACTCC	8	869
1245157	214	229	3308	3323	TCATAAGTAGTCTGCC	62	870
1245183	274	289	7563	7578	TCCTCCACACCGCGCT	20	871
1245209	321	336	7610	7625	CATACGCATGCGCAGT	60	872
1245235	370	385	7659	7674	ACCTGATTTAGAGAGC	65	873
1245261	413	428	8807	8822	AGCATTATTACCACG	53	874

TABLE 12-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245287	443	458	8837	8852	GCTGAGAAGATCGGCT	10	875
1245313	487	502	8881	8896	AGGATGTTGACCTCAA	4	876
1245339	533	548	10419	10434	CTCCATCATCGATGGA	0	877
1245365	593	608	10479	10494	GTAAGGAATCCCTTCG	22	878
1245391	646	661	12064	12079	AGACCTCTGTGAAAGC	50	879
1245417	723	738	12141	12156	ACCCAGTATTCACAAA	10	880
1245443	770	785	15651	15666	ATCTGTCTCCAATACA	55	881
1245469	803	818	15684	15699	AAGTATTCCATCTATC	52	882
1245495*	892	907	20707	20722	AAAATCGCTGAGGCGC	97	883
1245521*	934	949	20749	20764	TGGCCAACCACTGCTT	98	884
1245547*	985	1000	20800	20815	CATGCATACATCTCTG	55	885
1245573	1029	1044	20844	20859	AATAAAGCTTTGCAGC	26	886
1245599	1117	1132	20932	20947	CAGGAAGACAGGTAAT	60	887
1245625	1271	1286	21086	21101	ACTTTTAGGGAGGTAG	20	888
1245651	1325	1340	21140	21155	TGCTACTTGAACAGTC	81	889
1245677	1353	1368	21168	21183	TATTCTGTGGCATGGC	71	890
1245703	1396	1411	21211	21226	AACTTGATCTCTTAGC	64	891
1245729	1464	1479	21279	21294	AGGCTAGGGAAATCTT	62	892
1245755	1510	1525	21325	21340	AAGTCCAGAATAGAGT	60	893
1245781	1569	1584	21384	21399	TATAGAGGGTCCACTT	17	894
1245807	1628	1643	21443	21458	TAAAATAGAGTCGGTC	81	895
1245833	1731	1746	21546	21561	TGAGCCTAAAATTGTC	30	896
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	57	38
1245859	1784	1799	21599	21614	GTTTCTTATGTAATAG	46	897
1245885	2188	2203	22003	22018	TTGATGTAGTGGGAGT	68	898
1245911	2350	2365	22165	22180	ACTACAAGAGGTATT	46	899
1245937	N/A	N/A	3494	3509	CATATACAGACTAAGG	57	900
1245963	N/A	N/A	3672	3687	TAAGTTATTATCCCCA	70	901
1245989	N/A	N/A	4249	4264	CTCTAAGTGTAAGAGA	28	902
1246015	N/A	N/A	4647	4662	TTTTTAATACGACTTC	52	903
1246041	N/A	N/A	5165	5180	CATAAGAGCCAACTG	27	904
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	90	48
1246067	N/A	N/A	5661	5676	GCATAAGTTACCAGAG	79	905
1246093	N/A	N/A	5889	5904	GTATTTTCGCTACTGTA	76	906
1246119	N/A	N/A	6149	6164	CCTAAGATAAAATTGC	18	907

TABLE 12-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246145	N/A	N/A	6425	6440	TGGTAAACTGTATGCC	76	908
1246171	N/A	N/A	6625	6640	GTACAGTAGGCAGTTT	50	909
1246197	N/A	N/A	7782	7797	AACTACTTCAGTTAGC	53	910
1246223	N/A	N/A	8043	8058	GAAGACTACATTAACA	36	911
1246249	N/A	N/A	8562	8577	AAATTTTCAGTACCCAG	46	912
1246275	N/A	N/A	8942	8957	TCTTAAGAGAAGGTGC	39	913
1246301	N/A	N/A	9189	9204	CCTTAACAAACCTCCA	27	914
1246327	N/A	N/A	9426	9441	CAATTATATGAGGTCT	36	915
1246353	N/A	N/A	9610	9625	GCCAATTCACATATGT	34	916
1246379	N/A	N/A	9887	9902	TATATGTTGATGTTAC	34	917
1246405	N/A	N/A	10060	10075	ATGATAAATTTGGTGTC	55	918
1246431	N/A	N/A	10387	10402	TCTTAAGGATCATTGC	0	919
1246457	N/A	N/A	10973	10988	TTGTAATCTTGAGGCA	27	920
1246483	N/A	N/A	11512	11527	TTATTATGAGGATCTG	50	921
1246509	N/A	N/A	11854	11869	AAGCAGAATTGTGAAC	18	922
1246535	N/A	N/A	12201	12216	CATACCACATACCCAT	0	923
1246561	N/A	N/A	12489	12504	CTTGACTAAGGAGCTA	24	924
1246587	N/A	N/A	12687	12702	TCATTAAGCTTAGTAT	31	925
1246613	N/A	N/A	12814	12829	ACATAGCTAACTCAGG	38	926
1246639	N/A	N/A	13763	13778	GTACTATAGTATTTAC	27	927
1246665	N/A	N/A	13952	13967	TCTAAGTTAGCCCCCA	36	928
1246691	N/A	N/A	14092	14107	CATCTACATGGATCCA	63	929
1246717	N/A	N/A	14316	14331	TTAAGATGTCCAGGCA	35	930
1246743	N/A	N/A	14479	14494	AAGGAGTATGACAACT	28	931
1246769	N/A	N/A	15211	15226	CATAAAGCTGGATTGT	0	932
1246795	N/A	N/A	15356	15371	GTAAGGAGTAGTGGAG	37	933
1246821	N/A	N/A	15493	15508	CATTAATGCCACCCTA	8	934
1246847	N/A	N/A	16254	16269	AGTAGATTTGGTAGAG	67	935
1246873	N/A	N/A	17041	17056	GCTTATCGAGTTTAGC	17	936
1246899	N/A	N/A	17987	18002	GTATAGTTATCTTCTC	59	937
1246925	N/A	N/A	18662	18677	ATACTATTCTCCAAC	25	938
1246951	N/A	N/A	18871	18886	ATAAGAAGAGAGCTCA	14	939
1246977	N/A	N/A	19544	19559	GTTTAGCTGATAGGCT	47	940
1247003	N/A	N/A	19749	19764	TATGAGTAAAGATCTA	10	941
1247029	N/A	N/A	20004	20019	GTAATCAGTTTTCTC	79	942

TABLE 12-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1247055	N/A	N/A	20373	20388	GTAAGGTAAAGTCTAA	37	943
1247081	N/A	N/A	20490	20505	ACATAACTTCATAGTG	27	944

TABLE 13

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245080	16	31	3110	3125	CTTGTGTAGTCCTAGG	71	945
1245106	91	106	3185	3200	ATGGTGATCAGAAGCA	85	946
1245132	123	138	3217	3232	ACTTCACCAACGACTC	15	947
1245158	215	230	3309	3324	TTCATAAGTAGTCTGTC	64	948
1245184	275	290	7564	7579	TTCTCCACACCGCGC	47	949
1245210	322	337	7611	7626	ACATACGCATGCGCAG	20	950
1245236	371	386	7660	7675	CACCTGATTAGAGAG	11	951
1245262	416	431	8810	8825	CCCAGCATTATTCACC	19	952
1245288	444	459	8838	8853	TGCTGAGAAGATCGGC	7	953
1245314	488	503	8882	8897	TAGGATGTTGACCTCA	19	954
1245340	534	549	10420	10435	TCTCCATCATCGATGG	24	955
1245366	594	609	10480	10495	GGTAAGGAATCCCTTC	43	956
1245392	649	664	12067	12082	GTCAGACCTCTGTGAA	34	957
1245418	724	739	12142	12157	AACCCAGTATTCACAA	0	958
1245444	771	786	15652	15667	CATCTGTCTCCAATAC	32	959
1245470	804	819	15685	15700	TAAGTATTCATCTAT	31	960
1245496*	893	908	20708	20723	TAAAATCGCTGAGGCG	87	961
1245522*	935	950	20750	20765	GTGGCCAACCACTGCT	99	962
1245548*	986	1001	20801	20816	TCATGCATACATCTCT	70	963
1245574	1055	1070	20870	20885	ATATTATCAGGACTGA	47	964
1245600	1136	1151	20951	20966	CGTAAATATTCTTGAG	61	965
1245626	1272	1287	21087	21102	TACTTTTAGGGAGGTA	28	966
1245652	1326	1341	21141	21156	ATGCTACTTGAACAGT	47	967
1245678	1354	1369	21169	21184	ATATTCTGTGGCATGG	69	968
1245704	1403	1418	21218	21233	CTGCTGAACTTGATC	50	969
1245730	1465	1480	21280	21295	GAGGCTAGGGAAATCT	56	970

TABLE 13-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245756	1516	1531	21331	21346	GTAATAAAGTCCAGAA	66	971
1245782	1570	1585	21385	21400	ATATAGAGGGTCCACT	18	972
1245808	1629	1644	21444	21459	TTAAATAGAGTCGGT	65	973
1245834	1732	1747	21547	21562	TTGAGCCTAAAATTGT	36	974
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	58	38
1245860	1792	1807	21607	21622	GGTCCATTGTTTCTTA	36	975
1245886	2189	2204	22004	22019	CTTGATGTAGTGGGAG	83	976
1245912	2351	2366	22166	22181	AACTACAAGAGGTTAT	57	977
1245938	N/A	N/A	3495	3510	ACATATACAGACTAAG	46	978
1245964	N/A	N/A	3673	3688	CTAAGTTATTATCCCC	70	979
1245990	N/A	N/A	4296	4311	GTATAGTATTCTTCAC	81	980
1246016	N/A	N/A	4649	4664	CATTTTAAATACGACT	57	981
1246042	N/A	N/A	5166	5181	GCATAAGAGCCAAACT	67	982
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	87	48
1246068	N/A	N/A	5662	5677	AGCATAAGTTACCAGA	87	983
1246094	N/A	N/A	5891	5906	TTGTATTTCGCTACTG	58	984
1246120	N/A	N/A	6170	6185	CCTAAATTCAAAGACC	64	985
1246146	N/A	N/A	6429	6444	GTAATGGTAAACTGTA	70	986
1246172	N/A	N/A	6737	6752	TGAGATTACCCCTGGC	53	987
1246198	N/A	N/A	7785	7800	CATAACTACTTCAGTT	3	988
1246224	N/A	N/A	8121	8136	CTTAGTTCATTGTGAT	69	989
1246250	N/A	N/A	8564	8579	CTAAATTCAGTACCC	33	990
1246276	N/A	N/A	8943	8958	ATCTTAAGAGAAGGTG	43	991
1246302	N/A	N/A	9190	9205	CCCTTAACAAACCTCC	43	992
1246328	N/A	N/A	9429	9444	AACCAATTATATGAGG	38	993
1246354	N/A	N/A	9632	9647	GTGTAGTTTGGTGGGC	33	994
1246380	N/A	N/A	9908	9923	CCTATTTGTAATATTG	0	995
1246406	N/A	N/A	10062	10077	TTATGATAATTGGTG	0	996
1246432	N/A	N/A	10388	10403	ATCTTAAGGATCATTG	7	997
1246458	N/A	N/A	10976	10991	GCATTGTAATCTTGAG	35	998
1246484	N/A	N/A	11513	11528	TTTATTATGAGGATCT	43	999
1246510	N/A	N/A	11856	11871	TTAAGCAGAATTGTGA	7	1000
1246536	N/A	N/A	12203	12218	ATCATACCACATACCC	45	1001
1246562	N/A	N/A	12501	12516	GCAGAAATTCACCTTG	74	1002
1246588	N/A	N/A	12690	12705	GAATCATTAAGCTTAG	63	1003

TABLE 13-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246614	N/A	N/A	12816	12831	CCACATAGCTAACTCA	53	1004
1246640	N/A	N/A	13768	13783	CAATAGTACTATAGTA	14	1005
1246666	N/A	N/A	13953	13968	CTCTAAGTTAGCCCCC	38	1006
1246692	N/A	N/A	14126	14141	AATTACTCCGTTCTGTC	6	1007
1246718	N/A	N/A	14317	14332	ATTAAGATGTCCAGGC	31	1008
1246744	N/A	N/A	14480	14495	CAAGGAGTATGACAAC	6	1009
1246770	N/A	N/A	15213	15228	TCCATAAAGCTGGATT	13	1010
1246796	N/A	N/A	15358	15373	TGGTAAGGAGTAGTGG	34	1011
1246822	N/A	N/A	15494	15509	GCATTAATGCCACCCT	36	1012
1246848	N/A	N/A	16255	16270	CAGTAGATTGGTAGA	49	1013
1246874	N/A	N/A	17042	17057	AGCTTATCGAGTTTAG	54	1014
1246900	N/A	N/A	17988	18003	AGTATAGTTATCTTCT	67	1015
1246926	N/A	N/A	18663	18678	GATACTATTCTCCAAC	44	1016
1246952	N/A	N/A	18872	18887	GATAAGAAGAGAGCTC	36	1017
1246978	N/A	N/A	19549	19564	ATATTGTTTAGCTGAT	14	1018
1247004	N/A	N/A	19751	19766	GATATGAGTAAAGATC	23	1019
1247030	N/A	N/A	20006	20021	GAGTAATCAGTTTTC	65	1020
1247056	N/A	N/A	20374	20389	AGTAAGGTAAAGTCTA	55	1021
1247082	N/A	N/A	20492	20507	TGACATAACTTCATAG	51	1022

TABLE 14

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245081	17	32	3111	3126	CCTTGTTAGTCCTAG	67	1023
1245107	92	107	3186	3201	GATGGTGATCAGAAGC	52	1024
1245133	124	139	3218	3233	AACTTCACCAACGACT	0	1025
1245159	216	231	3310	3325	ATTCATAAGTAGTCTG	35	1026
1245185	276	291	7565	7580	TTTCCTCCACACCGCG	18	1027
1245211	323	338	7612	7627	CACATACGCATGCGCA	59	1028
1245237	372	387	7661	7676	TCACCTGATTTAGAGA	13	1029
1245263	418	433	8812	8827	GTCCCAGCATTATTCA	21	1030
1245289	445	460	8839	8854	GTGCTGAGAAGATCGG	0	1031
1245315	489	504	8883	8898	CTAGGATGTTGACCTC	0	1032

TABLE 14-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1245341	535	550	10421	10436	CTCTCCATCATCGATG	0	1033
1245367	595	610	10481	10496	AGGTAAGGAATCCCTT	0	1034
1245393	650	665	12068	12083	TGTCAGACCTCTGTGA	0	1035
1245419	725	740	12143	12158	GAACCCAGTATTCACA	0	1036
1245445	777	792	15658	15673	CGACTTCATCTGTCTC	19	1037
1245471	805	820	15686	15701	GTAAGTATTCATCTA	45	1038
1245497*	894	909	20709	20724	TTAAATCGCTGAGGC	67	1039
1245523*	936	951	20751	20766	TGTGGCCAACCACTGC	75	1040
1245549*	987	1002	20802	20817	ATCATGCATACATCTC	62	1041
1245575	1075	1090	20890	20905	GTGCCAACAATGTT	43	1042
1245601	1150	1165	20965	20980	CCTATGAAAACTACG	36	1043
1245627	1277	1292	21092	21107	GTATTTACTTTTAGGG	61	1044
1245653	1327	1342	21142	21157	AATGCTACTTGAACAG	54	1045
1245679	1355	1370	21170	21185	GATATTCTGTGGCATG	46	1046
1245705	1404	1419	21219	21234	CCTGCTGAACTTGAT	19	1047
1245731	1466	1481	21281	21296	AGAGGCTAGGGAATC	63	1048
1245757	1517	1532	21332	21347	AGTAATAAGTCCAGA	43	1049
1245783	1571	1586	21386	21401	AATATAGAGGGTCCAC	7	1050
1245809	1630	1645	21445	21460	TTTAAATAGAGTCGG	69	1051
1245835	1733	1748	21548	21563	TTTGAGCCTAAAATTG	26	1052
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	47	38
1245861	1793	1808	21608	21623	GGGTCCATTGTTTCTT	21	1053
1245887	2190	2205	22005	22020	TCTTGATGTAGTGGGA	44	1054
1245913	2352	2367	22167	22182	TAACTACAAGAGGTTA	22	1055
1245939	N/A	N/A	3507	3522	AACTAACCTGACACAT	20	1056
1245965	N/A	N/A	3674	3689	CCTAAGTTATTATCCC	18	1057
1245991	N/A	N/A	4360	4375	ATACATCAACTCCTCT	64	1058
1246017	N/A	N/A	4650	4665	TCATTTTAAATACGAC	48	1059
1246043	N/A	N/A	5167	5182	TGCATAAGAGCCAAAC	56	1060
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	84	48
1246069	N/A	N/A	5663	5678	GAGCATAAGTTACCAG	61	1061
1246095	N/A	N/A	5916	5931	CTTGAATGAGTGGTCT	64	1062
1246121	N/A	N/A	6184	6199	CAATTTACTTTTCAGCC	71	1063
1246147	N/A	N/A	6437	6452	AACTAGTGGTAATGGT	32	1064
1246173	N/A	N/A	6741	6756	GCAATGAGATTACCCC	83	1065

TABLE 14-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246199	N/A	N/A	7786	7801	ACATAACTACTTCAGT	28	1066
1246225	N/A	N/A	8122	8137	GCTTAGTTCATTGTGA	45	1067
1246251	N/A	N/A	8587	8602	GGGATAAAGTGTCTT	47	1068
1246277	N/A	N/A	8944	8959	TATCTTAAGAGAAGGT	29	1069
1246303	N/A	N/A	9191	9206	ACCCTTAACAAACCTC	36	1070
1246329	N/A	N/A	9430	9445	TAACCAATTATATGAG	0	1071
1246355	N/A	N/A	9636	9651	CTCCGTGTAGTTGGT	28	1072
1246381	N/A	N/A	9922	9937	GAATTCACATCTCCC	38	1073
1246407	N/A	N/A	10063	10078	CTTATGATAATTTGGT	2	1074
1246433	N/A	N/A	10389	10404	GATCTTAAGGATCATT	0	1075
1246459	N/A	N/A	11000	11015	TAAATCCAAATCACTC	39	1076
1246485	N/A	N/A	11514	11529	TTTTATTATGAGGATC	18	1077
1246511	N/A	N/A	11857	11872	ATTAAGCAGAATTGTG	8	1078
1246537	N/A	N/A	12205	12220	TTATCATACACATAC	19	1079
1246563	N/A	N/A	12512	12527	CAAGAGTTCCTGCAGA	17	1080
1246589	N/A	N/A	12692	12707	TCGAATCATTAAGCTT	34	1081
1246615	N/A	N/A	12817	12832	ACCACATAGCTAACTC	50	1082
1246641	N/A	N/A	13769	13784	GCAATAGTACTATAGT	39	1083
1246667	N/A	N/A	13954	13969	CCTCTAAGTTAGCCCC	97	1084
1246693	N/A	N/A	14127	14142	AAATTAAGTCCGTTCTG	28	1085
1246719	N/A	N/A	14321	14336	TCAAATTAAGATGTCC	6	1086
1246745	N/A	N/A	14499	14514	TGAGAATTAAGAGGG	25	1087
1246771	N/A	N/A	15217	15232	GTATTCATAAAGCTG	10	1088
1246797	N/A	N/A	15359	15374	ATGGTAAGGAGTAGTG	17	1089
1246823	N/A	N/A	15495	15510	TGCATTAATGCCACCC	22	1090
1246849	N/A	N/A	16257	16272	GACAGTAGATTGGTA	34	1091
1246875	N/A	N/A	17043	17058	AAGCTTATCGAGTTTA	26	1092
1246901	N/A	N/A	17989	18004	TAGTATAGTTATCTTC	28	1093
1246927	N/A	N/A	18666	18681	TGGGATACTATTCTCC	18	1094
1246953	N/A	N/A	18873	18888	GGATAAGAAGAGAGCT	46	1095
1246979	N/A	N/A	19550	19565	GATATTGTTTAGCTGA	59	1096
1247005	N/A	N/A	19753	19768	CTGATATGAGTAAAGA	21	1097
1247031	N/A	N/A	20008	20023	GAGAGTAATCAGTTTT	27	1098
1247057	N/A	N/A	20375	20390	AAGTAAGGTAAAGTCT	20	1099
1247083	N/A	N/A	20571	20586	TTTAACCTTCCTCCGT	16	1100

TABLE 15

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245082	18	33	3112	3127	TCCTTGTTAGTCTCTA	61	1101
1245108	93	108	3187	3202	TGATGGTGATCAGAAG	0	1102
1245134	125	140	3219	3234	AAACTTCACCAACGAC	27	1103
1245160	217	232	3311	3326	AATTCATAAGTAGTCT	28	1104
1245186	277	292	7566	7581	GTTTCCTCCACACCGC	45	1105
1245212	324	339	7613	7628	CCACATACGCATGCGC	68	1106
1245238	373	388	N/A	N/A	TTCACCTGATTTAGAG	9	1107
1245264	419	434	8813	8828	TGTCCCAGCATTATTC	31	1108
1245290	446	461	8840	8855	GGTGCTGAGAAGATCG	33	1109
1245316	490	505	8884	8899	CCTAGGATGTTGACCT	3	1110
1245342	536	551	10422	10437	TCTCTCCATCATCGAT	24	1111
1245368	596	611	10482	10497	GAGGTAAGGAATCCCT	27	1112
1245394	651	666	12069	12084	ATGTCAGACCTCTGTG	19	1113
1245420	726	741	12144	12159	TGAACCCAGTATTAC	10	1114
1245446	778	793	15659	15674	ACGACTTCATCTGTCT	8	1115
1245472	806	821	15687	15702	GGTAAGTATTCATCT	12	1116
1245498*	895	910	20710	20725	TTTAAATCGCTGAGG	24	1117
1245524*	937	952	20752	20767	TTGTGGCCAACCACTG	83	1118
1245550*	988	1003	20803	20818	TATCATGCATACATCT	45	1119
1245576	1076	1091	20891	20906	AGTGCCAAACCAATGT	62	1120
1245602	1154	1169	20969	20984	CAGACCTATGAAAAAC	26	1121
1245628	1296	1311	21111	21126	GTGTAATAAGTTCTC	61	1122
1245654	1328	1343	21143	21158	GAATGCTACTTGAACA	44	1123
1245680	1356	1371	21171	21186	TGATATTCTGTGGCAT	69	1124
1245706	1405	1420	21220	21235	GCCTGCTGAAACTTGA	18	1125
1245732	1483	1498	21298	21313	GGGCTAATGAAAAGG	14	1126
1245758	1520	1535	21335	21350	TCAAGTAATAAAGTCC	49	1127
1245784	1572	1587	21387	21402	AAATATAGAGGGTCCA	0	1128
1245810	1631	1646	21446	21461	ATTTAAATAGAGTCG	24	1129
1245836	1738	1753	21553	21568	TAATTTTGTAGCCTAA	41	1130
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	55	38
1245862	1794	1809	21609	21624	TGGGTCCATTGTTTCT	35	1131
1245888	2191	2206	22006	22021	GTCTTGATGTAGTGGG	67	1132
1245914	2353	2368	22168	22183	ATAACTACAAGAGGTT	64	1133
1245940	N/A	N/A	3508	3523	TAACTAACCTGACACA	57	1134
1245966	N/A	N/A	3675	3690	CCCTAAGTTATTATCC	25	1135

TABLE 15-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1245992	N/A	N/A	4361	4376	TATACATCAACTCCTC	39	1136
1246018	N/A	N/A	4652	4667	AGTCATTTTAAATACG	13	1137
1246044	N/A	N/A	5168	5183	TTGCATAAGAGCCAAA	46	1138
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	75	48
1246070	N/A	N/A	5736	5751	TAGTTGGTAGCTTGCT	66	1139
1246096	N/A	N/A	5918	5933	ACCTTGAATGAGTGGT	26	1140
1246122	N/A	N/A	6185	6200	GCAATTTACTTTCAGC	66	1141
1246148	N/A	N/A	6441	6456	TCTAAACTAGTGGTAA	24	1142
1246174	N/A	N/A	6743	6758	TTGCAATGAGATTACC	29	1143
1246200	N/A	N/A	7787	7802	GACATAACTACTTCAG	66	1144
1246226	N/A	N/A	8133	8148	TTTGTTAGAGTGCTTA	80	1145
1246252	N/A	N/A	8596	8611	CATAAAGGTGGGATAA	23	1146
1246278	N/A	N/A	8945	8960	GTATCTTAAGAGAAGG	30	1147
1246304	N/A	N/A	9208	9223	AAAGATGTACTACTGAC	57	1148
1246330	N/A	N/A	9431	9446	CTAACCAATTATATGA	21	1149
1246356	N/A	N/A	9648	9663	AATGTATGAGGTCTCC	65	1150
1246382	N/A	N/A	9926	9941	GATTGAATTCATATC	0	1151
1246408	N/A	N/A	10064	10079	CCTTATGATAATTGG	29	1152
1246434	N/A	N/A	10390	10405	TGATCTTAAGGATCAT	0	1153
1246460	N/A	N/A	11030	11045	CATTCTAACTACTAT	11	1154
1246486	N/A	N/A	11542	11557	GACAATGGTTGCCACT	63	1155
1246512	N/A	N/A	11858	11873	GATTAAAGCAGAATTGT	37	1156
1246538	N/A	N/A	12209	12224	CAATTTATCATACCAC	37	1157
1246564	N/A	N/A	12513	12528	ACAAGAGTTCTTGCAG	29	1158
1246590	N/A	N/A	12693	12708	TTCGAATCATTAAGCT	63	1159
1246616	N/A	N/A	12862	12877	CACCATGAAGGACTCC	48	1160
1246642	N/A	N/A	13770	13785	TGCAATAGTACTATAG	24	1161
1246668	N/A	N/A	13975	13990	ATTTAGCATTGCCTGT	10	1162
1246694	N/A	N/A	14128	14143	TAAATTACTCCGTTCT	36	1163
1246720	N/A	N/A	14362	14377	GTGCAAGATGATGACA	19	1164
1246746	N/A	N/A	14541	14556	GTATTTGAGATTCACC	41	1165
1246772	N/A	N/A	15244	15259	GCTAAGTATACTTTCT	37	1166
1246798	N/A	N/A	15360	15375	GATGGTAAGGAGTAGT	28	1167
1246824	N/A	N/A	15496	15511	CTGCATTAAATGCCACC	67	1168
1246850	N/A	N/A	16272	16287	CAAAGTTTGGGCAGAG	44	1169

TABLE 15-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1246876	N/A	N/A	17046	17061	AAAAAGCTTATCGAGT	8	1170
1246902	N/A	N/A	17991	18006	GTTAGTATAGTTATCT	66	1171
1246928	N/A	N/A	18671	18686	TTTACTGGGATACTAT	25	1172
1246954	N/A	N/A	18901	18916	AGGGAAGATCTGGCTG	4	1173
1246980	N/A	N/A	19563	19578	AGCTATTGTCTTTGAT	39	1174
1247006	N/A	N/A	19760	19775	CTCTTATCTGATATGA	7	1175
1247032	N/A	N/A	20037	20052	GCATTAAACTCTCAT	35	1176
1247058	N/A	N/A	20376	20391	CAAGTAAGGTAAAGTC	33	1177
1247084	N/A	N/A	20572	20587	GTTTAACTTTCCTCCG	30	1178

TABLE 16

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
1245083	19	34	3113	3128	GTCCTTGTGTAGTCCT	62	1179
1245109	98	113	3192	3207	GTAGATGATGGTGATC	36	1180
1245135	126	141	3220	3235	AAAACCTCACCAACGA	29	1181
1245161	240	255	3334	3349	CCAATATGCTCTGTCTG	65	1182
1245187	278	293	7567	7582	AGTTTCCTCCACACCG	28	1183
1245213	325	340	7614	7629	ACCACATACGCATGCG	45	1184
1245239	374	389	N/A	N/A	CTTCACTGATTTAGA	7	1185
1245265	421	436	8815	8830	ACTGTCCCAGCATTAT	40	1186
1245291	452	467	8846	8861	ATCCTTGGTGCTGAGA	13	1187
1245317	491	506	8885	8900	TCCTAGGATGTTGACC	0	1188
1245343	537	552	10423	10438	TTCTCTCCATCATCGA	35	1189
1245369	597	612	10483	10498	TGAGGTAAGGAATCCC	2	1190
1245395	652	667	12070	12085	GATGTCAGACCTCTGT	6	1191
1245421	727	742	12145	12160	GTGAACCCAGTATTCA	1	1192
1245447	779	794	15660	15675	TACGACTTCATCTGTC	29	1193
1245473	807	822	15688	15703	TGGTAAGTATTCATC	26	1194
1245499*	896	911	20711	20726	ATTTAAATCGCTGAG	30	1195
1245525*	938	953	20753	20768	TTTGTGGCCAACCACT	88	1196
1245551*	989	1004	20804	20819	TTATCATGCATACATC	52	1197
1245577	1077	1092	20892	20907	TAGTGCCAAACCAATG	46	1198

TABLE 16-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245603	1180	1195	20995	21010	GTTTTTAAGAGGCATG	63	1199
1245629	1298	1313	21113	21128	CTGTGTAATAAGTTC	38	1200
1245655	1329	1344	21144	21159	GGAATGCTACTTGAAC	63	1201
1245681	1357	1372	21172	21187	TTGATATTCTGTGGCA	87	1202
1245707	1406	1421	21221	21236	TGCCTGCTGAACTTG	31	1203
1245733	1488	1503	21303	21318	GTTTTGGGCTAATGAA	54	1204
1245759	1538	1553	21353	21368	GTTATACAGAAGACAG	54	1205
1245785	1573	1588	21388	21403	GAAATATAGAGGGTCC	22	1206
1245811	1654	1669	21469	21484	GGGCTAGAACTTAAAA	35	1207
1245837	1742	1757	21557	21572	GCTTTAATTTTGTGAGC	26	1208
1245863	1795	1810	21610	21625	TTGGGTCCATTGTTTC	44	1209
1245889	2192	2207	22007	22022	AGTCTTGATGTAGTGG	63	1210
1245915	2354	2369	22169	22184	TATAACTACAAGAGGT	46	1211
1245941	N/A	N/A	3509	3524	CTAACTAACCTGACAC	23	1212
1245967	N/A	N/A	3703	3718	TAATTTTCAGATCCCG	75	1213
1245993	N/A	N/A	4362	4377	ATATACATCAACTCCT	43	1214
1246019	N/A	N/A	4991	5006	GAATTTTGTGAGTTGG	77	1215
1246045	N/A	N/A	5173	5188	TTAGATTGCATAAGAG	61	1216
1246071	N/A	N/A	5738	5753	TTTAGTTGGTAGCTTG	63	1217
1246097	N/A	N/A	5919	5934	AACCTTGAATGAGTGG	50	1218
1246123	N/A	N/A	6204	6219	CATAGGACATGGAGAC	69	1219
1246149	N/A	N/A	6443	6458	ATTCTAACTAGTGGT	39	1220
1246175	N/A	N/A	6784	6799	CTCCTTATTGTTAGA	26	1221
1246201	N/A	N/A	7788	7803	TGACATAACTACTTCA	22	1222
1246227	N/A	N/A	8137	8152	AAAGTTTGTAGAGTG	75	1223
1246253	N/A	N/A	8597	8612	TCATAAAGGTGGGATA	40	1224
1246279	N/A	N/A	8959	8974	GTTATAAGTTTCATGT	21	1225
1246305	N/A	N/A	9215	9230	GCATTGAAAAGATGTA	0	1226
1246331	N/A	N/A	9476	9491	TCTATACAAAGGCAGT	53	1227
1246357	N/A	N/A	9649	9664	TAATGTATGAGGTCTC	51	1228
1246383	N/A	N/A	9928	9943	ATGATTGAATTCACCTA	33	1229
1246409	N/A	N/A	10112	10127	TAAACAATATTAGGG	0	1230
1246435	N/A	N/A	10494	10509	CCAATATGGGATGAGG	60	1231
1246461	N/A	N/A	11074	11089	TTACAACCTGGTTTCA	25	1232
1246487	N/A	N/A	11543	11558	TGACAATGGTTGCCAC	40	1233
1246513	N/A	N/A	11859	11874	TGATTAAGCAGAATTG	57	1234

TABLE 16-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246539	N/A	N/A	12210	12225	TCAATTTATCATACCA	26	1235
1246565	N/A	N/A	12514	12529	GACAAGAGTCTTGCA	22	1236
1246591	N/A	N/A	12694	12709	TTTCGAATCATTAGC	30	1237
1246617	N/A	N/A	12875	12890	ACCTATGGTCTAACAC	14	1238
1246643	N/A	N/A	13771	13786	TTGCAATAGTACTATA	3	1239
1246669	N/A	N/A	13976	13991	TATTTAGCATTGCCTG	12	1240
1246695	N/A	N/A	14129	14144	CTAAATTACTCCGTTC	20	1241
1246721	N/A	N/A	14377	14392	TAGAATTGTCTGTTAG	9	1242
1246747	N/A	N/A	14542	14557	CGTATTGAGATTACAC	59	1243
1246773	N/A	N/A	15245	15260	AGCTAAGTATACTTTC	18	1244
1246799	N/A	N/A	15361	15376	AGATGGTAAGGAGTAG	26	1245
1246825	N/A	N/A	15511	15526	ATATTCCAGGATTGTC	26	1246
1246851	N/A	N/A	16273	16288	GCAAAGTTTGGGCAGA	50	1247
1246877	N/A	N/A	17047	17062	AAAAAAGCTTATCGAG	7	1248
1246903	N/A	N/A	17992	18007	AGTTAGTATAGTTATC	74	1249
1246929	N/A	N/A	18695	18710	AAAGACTTTGAGACTC	68	1250
1246955	N/A	N/A	18925	18940	GAAAGATGGAATGAGC	54	1251
1246981	N/A	N/A	19588	19603	AGGTAAACTAAAGTAC	6	1252
1247007	N/A	N/A	19778	19793	CATACCACTCTTCTCA	24	1253
1247033	N/A	N/A	20039	20054	ATGCATTAAACTCTC	12	1254
1247059	N/A	N/A	20377	20392	ACAAGTAAGGTAAAGT	38	1255
1247085	N/A	N/A	20589	20604	AGAGTTATCTGGTTTG	56	1256

TABLE 17

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245084	20	35	3114	3129	AGTCCTTGTTAGTCC	64	1257
1245110	100	115	3194	3209	GAGTAGATGATGGTGA	28	1258
1245136	127	142	3221	3236	AAAAACTTCACCAACG	8	1259
1245162	241	256	3335	3350	ACCAATATGCTCTGTC	67	1260
1245188	289	304	7578	7593	CACTCAGCTGCAGTTT	13	1261
1245214	326	341	7615	7630	TACCACATACGCATGC	33	1262
1245240	375	390	N/A	N/A	TCTTCACCTGATTAG	24	1263
1245266	422	437	8816	8831	TACTGTCCCAGCATTA	21	1264

TABLE 17-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245292	453	468	8847	8862	CATCCTTGGTGCTGAG	12	1265
1245318	492	507	8886	8901	GTCCTAGGATGTTGAC	1	1266
1245344	538	553	10424	10439	TTTCTCTCCATCATCG	27	1267
1245370	598	613	10484	10499	ATGAGGTAAGGAATCC	0	1268
1245396	654	669	12072	12087	CTGATGTCAGACCTCT	35	1269
1245422	728	743	12146	12161	GGTGAACCCAGTATTC	7	1270
1245448	780	795	15661	15676	TTACGACTTCATCTGT	36	1271
1245474	810	825	15691	15706	TATTGGTAAGTATTC	67	1272
1245500*	897	912	20712	20727	GATTTAAATCGCTGA	49	1273
1245526*	939	954	20754	20769	TTTTGTGGCCAACCAC	77	1274
1245552*	992	1007	20807	20822	TCATTATCATGCATAC	67	1275
1245578	1078	1093	20893	20908	CTAGTGCCAAACCAAT	49	1276
1245604	1183	1198	20998	21013	GAAGTTTTTAAGAGGC	64	1277
1245630	1299	1314	21114	21129	CCTGTGTAAATAAGTT	18	1278
1245656	1331	1346	21146	21161	TTGGAATGCTACTTGA	72	1279
1245682	1358	1373	21173	21188	GTGATATTCTGTGGC	80	1280
1245708	1408	1423	21223	21238	GCTGCCTGCTGAAACT	38	1281
1245734	1489	1504	21304	21319	CGTTTTGGGCTAATGA	48	1282
1245760	1540	1555	21355	21370	GAGTTATACAGAAGAC	61	1283
1245786	1574	1589	21389	21404	GGAAATATAGAGGGTC	50	1284
1245812	1668	1683	21483	21498	AAAAGGTTATCATGGG	49	1285
1245838	1750	1765	21565	21580	CTGTGTTAGCTTTAAT	67	1286
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	61	38
1245864	1796	1811	21611	21626	CTTGGGTCCATTGTTT	44	1287
1245890	2193	2208	22008	22023	TAGTCTTGATGTAGTG	55	1288
1245916	2355	2370	22170	22185	TTATAACTACAAGAGG	68	1289
1245942	N/A	N/A	3511	3526	ATCTAACTAACCTGAC	0	1290
1245968	N/A	N/A	3704	3719	GTAATTTTCAGATCCC	82	1291
1245994	N/A	N/A	4363	4378	TATATACATCAACTCC	61	1292
1246020	N/A	N/A	5029	5044	GCTTAAATGTTTAACC	67	1293
1246046	N/A	N/A	5175	5190	GTTTAGATTGCATAAG	46	1294
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	88	48
1246072	N/A	N/A	5739	5754	ATTTAGTTGGTAGCTT	35	1295
1246098	N/A	N/A	5920	5935	AAACCTGAATGAGTG	37	1296
1246124	N/A	N/A	6205	6220	CCATAGGACATGGAGA	68	1297
1246150	N/A	N/A	6444	6459	AATTCTAACTAGTGG	31	1298

TABLE 17-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246176	N/A	N/A	6795 7104	6810 7119	TAAAAATTGGCCTCCT	29	1299
1246202	N/A	N/A	7804	7819	ATATATCCTTCCTTGG	20	1300
1246228	N/A	N/A	8139	8154	GTAAAGTTTGTTAGAG	54	1301
1246254	N/A	N/A	8598	8613	GTCATAAAGGTGGGAT	28	1302
1246280	N/A	N/A	8960	8975	CGTTATAAGTTTCATG	33	1303
1246306	N/A	N/A	9256	9271	AATCATGGTGTGGAGG	11	1304
1246332	N/A	N/A	9497	9512	GGAAGATTAATCATAA	32	1305
1246358	N/A	N/A	9650	9665	CTAATGTATGAGGTCT	46	1306
1246384	N/A	N/A	9937	9952	TAATGAATCATGATTG	2	1307
1246410	N/A	N/A	10145	10160	GAATCAAATAGATCCT	46	1308
1246436	N/A	N/A	10495	10510	ACCAATATGGGATGAG	24	1309
1246462	N/A	N/A	11078	11093	ATAATTACAACCTGGT	44	1310
1246488	N/A	N/A	11547 11572	11562 11587	CTCTTGACAATGGTTG	50	1311
1246514	N/A	N/A	11861	11876	CATGATTAAGCAGAAT	42	1312
1246540	N/A	N/A	12211	12226	ATCAATTTATCATAACC	11	1313
1246566	N/A	N/A	12515	12530	AGACAAGAGTTCTTGC	30	1314
1246592	N/A	N/A	12701	12716	AAATTGGTTTCGAATC	32	1315
1246618	N/A	N/A	12889	12904	CTTTAGTCAACAGTAC	12	1316
1246644	N/A	N/A	13783	13798	AACAATCAGGAGTTGC	31	1317
1246670	N/A	N/A	13997	14012	CTCTATAAAATCAATC	0	1318
1246696	N/A	N/A	14130	14145	TCTAAATTACTCCGTT	15	1319
1246722	N/A	N/A	14378	14393	GTAGAATTGTCTGTTA	51	1320
1246748	N/A	N/A	14556	14571	TCCGGTAAGAATTTTCG	63	1321
1246774	N/A	N/A	15246	15261	TAGCTAAGTATACTTT	10	1322
1246800	N/A	N/A	15364	15379	ATTAGATGGTAAGGAG	4	1323
1246826	N/A	N/A	15570	15585	GTAAATGATGGCAGGA	34	1324
1246852	N/A	N/A	16274	16289	AGCAAAGTTTGGGCAG	44	1325
1246878	N/A	N/A	17077	17092	AAATTAGAGGCCCAGG	43	1326
1246904	N/A	N/A	17994	18009	GAAGTTAGTATAGTTA	59	1327
1246930	N/A	N/A	18732	18747	CTAGTTACAAACCACC	43	1328
1246956	N/A	N/A	18958	18973	AGATTTACTTCCTTGT	41	1329
1246982	N/A	N/A	19590	19605	TAAGGTAACTAAAGT	9	1330
1247008	N/A	N/A	19781	19796	ATACATACCACTCTTC	18	1331
1247034	N/A	N/A	20087	20102	AGTAACCTTTATTAACG	12	1332

TABLE 17-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1247060	N/A	N/A	20381	20396	AATCACAAGTAAGGTA	52	1333
1247086	N/A	N/A	20599	20614	GAATACGCTGAGAGTT	11	1334

TABLE 18

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1245085	21	36	3115	3130	CAGTCCTTGTGTAGTC	70	1335
1245111	101	116	3195	3210	GGAGTAGATGATGGTG	31	1336
1245137	154	169	3248	3263	GCCACAGATTTCTCC	49	1337
1245163	242	257	3336	3351	AACCAATATGCTCTGT	51	1338
1245189	290	305	7579	7594	GCACTCAGCTGCAGTT	29	1339
1245215	327	342	7616	7631	CTACCACATACGCATG	25	1340
1245241	392	407	8786	8801	TACATCACCCTTCT	37	1341
1245267	423	438	8817	8832	ATACTGTCCCAGCATT	32	1342
1245293	454	469	8848	8863	TCATCCTTGGTGCTGA	12	1343
1245319	493	508	8887	8902	TGTCCTAGGATGTTGA	41	1344
1245345	548	563	10434	10449	GTGGCCATGATTCTC	40	1345
1245371	599	614	10485	10500	GATGAGGTAAGGAATC	0	1346
1245397	655	670	12073	12088	TCTGATGTCAGACCTC	42	1347
1245423	729	744	12147	12162	TGGTGAACCCAGTATT	0	1348
1245449	781	796	15662	15677	CTTACGACTTCATCTG	38	1349
1245475	812	827	15693	15708	CTTATTGGTAAGTATT	22	1350
1245501*	898	913	20713	20728	CGATTTAAAATCGCTG	40	1351
1245527*	940	955	20755	20770	ATTTTGTGGCCAACCA	59	1352
1245553	996	1011	20811	20826	CATATCATTATCATGC	66	1353
1245579	1079	1094	20894	20909	GCTAGTGCCAAACCAA	70	1354
1245605	1192	1207	21007	21022	GTAAGCACAGAAGTTT	55	1355
1245631	1300	1315	21115	21130	CCCTGTGTAAATAAGT	41	1356
1245657	1332	1347	21147	21162	ATTGGAATGCTACTTG	53	1357
1245683	1359	1374	21174	21189	TGTTGATATTCTGTGG	68	1358
1245709	1412	1427	21227	21242	TAAAGCTGCCTGCTGA	44	1359
1245735	1490	1505	21305	21320	CCGTTTGGGCTAATG	58	1360
1245761	1545	1560	21360	21375	CTTCAGAGTTATACAG	35	1361
1245787	1575	1590	21390	21405	AGGAAATATAGAGGGT	54	1362

TABLE 18-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245813	1670	1685	21485	21500	GAAAAAGGTTATCATG	31	1363
1245839	1751	1766	21566	21581	CCTGTGTTAGCTTTAA	67	1364
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	58	38
1245865	1797	1812	21612	21627	TCTTGGGTCCATTGTT	60	1365
1245891	2194	2209	22009	22024	TTAGTCTTGATGTAGT	61	1366
1245917	2378	2393	22193	22208	GTTTAGAAGTCAAACG	0	1367
1245943	N/A	N/A	3512	3527	CATCTAACTAACCTGA	0	1368
1245969	N/A	N/A	3750	3765	TAAAGGTTTCTGTTGC	66	1369
1245995	N/A	N/A	4405	4420	ATTCGAATTCTTCAA	65	1370
1246021	N/A	N/A	5042	5057	GAATCATTCACTAGCT	65	1371
1246047	N/A	N/A	5177	5192	CAGTTTAGATTGCATA	60	1372
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	85	48
1246073	N/A	N/A	5740	5755	GATTTAGTTGGTAGCT	63	1373
1246099	N/A	N/A	5921	5936	CAACCTTGAATGAGT	53	1374
1246125	N/A	N/A	6206	6221	ACCATAGGACATGGAG	62	1375
1246151	N/A	N/A	6445	6460	TAATTCTAACTAGTG	11	1376
1246177	N/A	N/A	6796	6811	ATAAAAATTGGCCTCC	41	1377
1246203	N/A	N/A	7828	7843	TGGATTACATACAAGA	68	1378
1246229	N/A	N/A	8140	8155	TGTAAAGTTTGTTAGA	35	1379
1246255	N/A	N/A	8599	8614	TGTCATAAAGGTGGGA	48	1380
1246281	N/A	N/A	8961	8976	CCGTTATAAGTTTCAT	54	1381
1246307	N/A	N/A	9257	9272	AAATCATGGTGTGGAG	1	1382
1246333	N/A	N/A	9499	9514	CTGGAAGATTAATCAT	31	1383
1246359	N/A	N/A	9652	9667	AGCTAATGTATGAGGT	56	1384
1246385	N/A	N/A	9938	9953	CTAATGAATCATGATT	20	1385
1246411	N/A	N/A	10180	10195	CTAATCTCAGTACTCC	53	1386
1246437	N/A	N/A	10496	10511	TACCAATATGGGATGA	1	1387
1246463	N/A	N/A	11079	11094	TATAATTACAACCTGG	46	1388
1246489	N/A	N/A	11548 11573	11563 11588	TCTCTTGACAATGGTT	70	1389
1246515	N/A	N/A	11865	11880	CTTACATGATTAAGCA	44	1390
1246541	N/A	N/A	12238	12253	GCCTAAACATTTCTCA	2	1391
1246567	N/A	N/A	12516	12531	GAGACAAGAGTTCTTG	19	1392
1246593	N/A	N/A	12703	12718	AAAAATTGGTTTCGAA	13	1393
1246619	N/A	N/A	13551	13566	CTCTCTAAAGGGCTGC	55	1394
1246645	N/A	N/A	13813	13828	TTTATTGGGCCCAAT	18	1395

TABLE 18-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246671	N/A	N/A	14007	14022	ATCTAGCAACCTCTAT	12	1396
1246697	N/A	N/A	14131	14146	TTCTAAATTACTCCGT	53	1397
1246723	N/A	N/A	14380	14395	GTGTAGAATTGTCTGT	42	1398
1246749	N/A	N/A	14557	14572	TTCCGGTAAGAATTTTC	33	1399
1246775	N/A	N/A	15248	15263	TATAGCTAAGTATACT	23	1400
1246801	N/A	N/A	15365	15380	CATTAGATGGTAAGGA	32	1401
1246827	N/A	N/A	15571	15586	AGTAAATGATGGCAGG	40	1402
1246853	N/A	N/A	16287	16302	TAGAATAGTCTTCAGC	70	1403
1246879	N/A	N/A	17135	17150	AATAGCCATGTAGCTA	28	1404
1246905	N/A	N/A	17995	18010	TGAAGTTAGTATAGTT	43	1405
1246931	N/A	N/A	18733	18748	CCTAGTTACAAACCAC	51	1406
1246957	N/A	N/A	18962	18977	TGCAAGATTACTTCC	45	1407
1246983	N/A	N/A	19598	19613	CTCTATAATAAGGTAA	0	1408
1247009	N/A	N/A	19782	19797	GATACATACCACTCTT	52	1409
1247035	N/A	N/A	20110	20125	CAAGACATTCTAGCCT	58	1410
1247061	N/A	N/A	20383	20398	GTAATCACAAGTAAGG	70	1411
1247087	N/A	N/A	20600	20615	AGAATACGCTGAGAGT	56	1412

TABLE 19

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245086	22	37	3116	3131	TCAGTCCTTGTGTAGT	72	1413
1245112	103	118	3197	3212	TAGGAGTAGATGATGG	15	1414
1245138	166	181	3260	3275	ACAATCTCCCCAGCCA	67	1415
1245164	243	258	3337	3352	GAACCAATATGCTCTG	61	1416
1245190	291	306	7580	7595	GGCACTCAGCTGCAGT	49	1417
1245216	328	343	7617	7632	TCTACCACATACGCAT	55	1418
1245242	393	408	8787	8802	TTACATCACCACTTC	37	1419
1245268	424	439	8818	8833	TATACTGTCCCAGCAT	40	1420
1245294	456	471	8850	8865	CTTCATCCTTGGTGCT	30	1421
1245320	494	509	8888	8903	ATGTCCTAGGATGTTG	50	1422
1245346	551	566	10437	10452	GATGTGGCCATGATTT	9	1423
1245372	606	621	10492	10507	AATATGGGATGAGGTA	53	1424
1245398	656	671	12074	12089	TTCTGATGTCAGACCT	42	1425

TABLE 19-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245424	730	745	12148	12163	TTGGTGAACCCAGTAT	31	1426
1245450	782	797	15663	15678	TCTTACGACTTCATCT	34	1427
1245476	813	828	15694	15709	TCTTATTGGTAAGTAT	25	1428
1245502*	899	914	20714	20729	ACGATTTAAATCGCT	44	1429
1245528*	941	956	20756	20771	GATTTTGTGGCCAACC	91	1430
1245554	1006	1021	20821	20836	CGAACTATTTCATATC	66	1431
1245580	1080	1095	20895	20910	TGCTAGTGCCAAACCA	78	1432
1245606	1198	1213	21013	21028	GTTTATGTAAGCACAG	67	1433
1245632	1301	1316	21116	21131	TCCCTGTGTAAATAAG	35	1434
1245658	1333	1348	21148	21163	GATTGGAATGCTACTT	58	1435
1245684	1364	1379	21179	21194	GTTCTTGTGATATTC	84	1436
1245710	1414	1429	21229	21244	GATAAAGCTGCCTGCT	42	1437
1245736	1491	1506	21306	21321	ACCGTTTGGGCTAAT	78	1438
1245762	1547	1562	21362	21377	GACTTCAGAGTTATAC	16	1439
1245788	1589	1604	21404	21419	GACTATAAAAGGGAG	40	1440
1245814	1702	1717	21517	21532	GACCAAGGATATATGA	48	1441
1245840	1752	1767	21567	21582	TCCTGTGTTAGCTTTA	56	1442
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	57	38
1245866	1798	1813	21613	21628	CTCTTGGGTCCATTGT	40	1443
1245892	2195	2210	22010	22025	ATTAGTCTTGATGTAG	67	1444
1245918	2379	2394	22194	22209	AGTTTAGAAGTCAAAC	23	1445
1245944	N/A	N/A	3530	3545	TCTTAATTACATTCCC	64	1446
1245970	N/A	N/A	3751	3766	CTAAAGGTTTCTGTTG	34	1447
1245996	N/A	N/A	4406	4421	CATTCGAATTTCTTCA	77	1448
1246022	N/A	N/A	5045	5060	AGTGAATCATTGAGTA	86	1449
1246048	N/A	N/A	5178	5193	ACAGTTTAGATTGCAT	84	1450
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	82	48
1246074	N/A	N/A	5741	5756	TGATTTAGTTGGTAGC	79	1451
1246100	N/A	N/A	5923	5938	GGCAAACCTTGAATGA	61	1452
1246126	N/A	N/A	6207	6222	AACCATAGGACATGGA	71	1453
1246152	N/A	N/A	6446	6461	GTAATTCTAACTAGT	39	1454
1246178	N/A	N/A	6914	6929	CTCGGAAGTTTAGAC	47	1455
1246204	N/A	N/A	7829	7844	GTGGATTACATACAAG	57	1456
1246230	N/A	N/A	8141	8156	ATGTAAAGTTTGTAG	12	1457
1246256	N/A	N/A	8610	8625	CAAGGGAACACTGTCA	58	1458
1246282	N/A	N/A	8962	8977	TCCGTTATAAGTTTCA	65	1459

TABLE 19-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246308	N/A	N/A	9260	9275	ACAAAATCATGGTGTG	26	1460
1246334	N/A	N/A	9510	9525	ATATTGAGGCACTGGA	53	1461
1246360	N/A	N/A	9653	9668	TAGCTAATGTATGAGG	48	1462
1246386	N/A	N/A	9939	9954	ACTAATGAATCATGAT	39	1463
1246412	N/A	N/A	10181	10196	ACTAATCTCAGTACTC	45	1464
1246438	N/A	N/A	10498	10513	CTTACCAATATGGGAT	11	1465
1246464	N/A	N/A	11155	11170	TGGAAACATGCTTGGC	62	1466
1246490	N/A	N/A	11549 11574	11564 11589	TTCTCTTGACAATGGT	67	1467
1246516	N/A	N/A	11892	11907	GTATAAATGGCAATTC	56	1468
1246542	N/A	N/A	12251	12266	TATTAGTTGGCCTGCC	14	1469
1246568	N/A	N/A	12598	12613	TCTTATCCACACCTTC	31	1470
1246594	N/A	N/A	12704	12719	AAAAAATTGGTTTCGA	20	1471
1246620	N/A	N/A	13574	13589	CTATCAGAACAGTTCA	48	1472
1246646	N/A	N/A	13814	13829	TTTTTATTGGGCCCAA	51	1473
1246672	N/A	N/A	14008	14023	AATCTAGCAACCTCTA	22	1474
1246698	N/A	N/A	14132	14147	TTTCTAAATTACTCCG	54	1475
1246724	N/A	N/A	14382	14397	TAGTGTAGAATTGTCT	46	1476
1246750	N/A	N/A	14600	14615	AGAAATCTAAGAACCC	25	1477
1246776	N/A	N/A	15252	15267	ATTCTATAGCTAAGTA	23	1478
1246802	N/A	N/A	15366	15381	GCATTAGATGGTAAGG	64	1479
1246828	N/A	N/A	15633	15648	CCATAATCTGTGATTA	9	1480
1246854	N/A	N/A	16289	16304	GTTAGAATAGTCTTCA	46	1481
1246880	N/A	N/A	17136	17151	GAATAGCCATGTAGCT	21	1482
1246906	N/A	N/A	17996	18011	ATGAAGTTAGTATAGT	55	1483
1246932	N/A	N/A	18734	18749	GCCTAGTTACAAACCA	56	1484
1246958	N/A	N/A	18968	18983	GACTAATGCAAGATTT	55	1485
1246984	N/A	N/A	19599	19614	ACTCTATAATAAGGTA	42	1486
1247010	N/A	N/A	19785	19800	ACTGATACATAACCACT	52	1487
1247036	N/A	N/A	20112	20127	ATCAAGACATTCTAGC	61	1488
1247062	N/A	N/A	20385	20400	TAGTAATCACAAGTAA	30	1489
1247088	N/A	N/A	20601	20616	TAGAATACGCTGAGAG	47	1490

TABLE 20

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245087	23	38	3117	3132	TTCAGTCCTTGTGTAG	45	1491
1245113	104	119	3198	3213	GTAGGAGTAGATGATG	7	1492
1245139	167	182	3261	3276	AACAATCTCCCCAGCC	59	1493
1245165	244	259	3338	3353	AGAACCAATATGCTCT	38	1494
1245191	292	307	7581	7596	CGGCACTCAGCTGCAG	55	1495
1245217	329	344	7618	7633	GTCTACCACATACGCA	66	1496
1245243	394	409	8788	8803	GTTACATCACCCTT	41	1497
1245269	425	440	8819	8834	ATATACTGTCCCAGCA	43	1498
1245295	457	472	8851	8866	TCTTCATCCTTGGTGC	48	1499
1245321	495	510	8889	8904	AATGTCCTAGGATGTT	70	1500
1245347	552	567	10438	10453	CGATGTGGCCATGATT	44	1501
1245373	607	622	10493	10508	CAATATGGGATGAGGT	62	1502
1245399	657	672	12075	12090	GTTCTGATGTCAGACC	55	1503
1245425	731	746	12149	12164	TTTGGTGAACCCAGTA	34	1504
1245451	783	798	15664	15679	TTCTTACGACTTCATC	47	1505
1245477	835	850	15716	15731	TACGATGGAACAAAAA	20	1506
1245503*	900	915	20715	20730	TACGATTTAAATCGC	0	1507
1245529*	942	957	20757	20772	TGATTTTGTGGCCAAC	72	1508
1245555	1007	1022	20822	20837	TCGAACTATTCATAT	64	1509
1245581	1081	1096	20896	20911	CTGCTAGTGCCAAACC	74	1510
1245607	1202	1217	21017	21032	GTATGTTTATGTAAGC	80	1511
1245633	1305	1320	21120	21135	ACCTTCCCTGTGTAAA	60	1512
1245659	1334	1349	21149	21164	AGATTGGAATGCTACT	63	1513
1245685	1366	1381	21181	21196	GTGTTCTTGTGATAT	81	1514
1245711	1415	1430	21230	21245	AGATAAAGCTGCCTGC	48	1515
1245737	1492	1507	21307	21322	CACCGTTTTGGGCTAA	85	1516
1245763	1548	1563	21363	21378	GGACTTCAGAGTTATA	36	1517
1245789	1599	1614	21414	21429	TATCTTATAAGACTAT	43	1518
1245815	1703	1718	21518	21533	GGACCAAGGATATATG	56	1519
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	67	38
1245867	1799	1814	21614	21629	TCTCTGGGTCCATTG	63	1520
1245893	2196	2211	22011	22026	GATTAGTCTTGATGTA	65	1521
1245919	N/A	N/A	3356	3371	ACCTTATTAATATCCC	71	1522
1245945	N/A	N/A	3532	3547	GTTCTTAATTACATTTC	64	1523
1245971	N/A	N/A	3753	3768	CACTAAAGGTTTCTGT	56	1524
1245997	N/A	N/A	4407	4422	TCATTCGAATTTCTTC	76	1525

TABLE 20-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	87	1526
1246049	N/A	N/A	5179	5194	AACAGTTTAGATTGCA	71	1527
1246075	N/A	N/A	5742	5757	CTGATTAGTTGGTAG	78	1528
1246101	N/A	N/A	5928	5943	TTTAGGGCAAACCTTG	67	1529
1246127	N/A	N/A	6208	6223	TAACCATAGGACATGG	70	1530
1246153	N/A	N/A	6447	6462	AGTAATCTAAACTAG	31	1531
1246179	N/A	N/A	7101	7116	AAATTGGCCTCCTGGG	44	1532
1246205	N/A	N/A	7846	7861	ATAATGGGCAGAGCTG	53	1533
1246231	N/A	N/A	8165	8180	TTATTGTAGGGATTGA	63	1534
1246257	N/A	N/A	8612	8627	GCCAAGGGAACACTGT	61	1535
1246283	N/A	N/A	8975	8990	GAAGTATGTGAACCTC	56	1536
1246309	N/A	N/A	9312	9327	ATTACTTCTGATGTCC	65	1537
1246335	N/A	N/A	9511	9526	AATATTGAGGCACTGG	54	1538
1246361	N/A	N/A	9656	9671	AGGTAGCTAATGTATG	53	1539
1246387	N/A	N/A	9940	9955	CACTAATGAATCATGA	25	1540
1246413	N/A	N/A	10182	10197	CACTAATCTCAGTACT	48	1541
1246439	N/A	N/A	10501	10516	ATACTTACCAATATGG	0	1542
1246465	N/A	N/A	11156	11171	ATGGAACATGCTTGG	48	1543
1246491	N/A	N/A	11565	11580	CAATGGTTGATAGTTT	47	1544
1246517	N/A	N/A	11894	11909	GAGTATAAATGGCAAT	53	1545
1246543	N/A	N/A	12252	12267	CTATTAGTTGGCCTGC	32	1546
1246569	N/A	N/A	12600	12615	GTTCTTATCCACACCT	57	1547
1246595	N/A	N/A	12705	12720	TAAAAAATGGTTTCG	22	1548
1246621	N/A	N/A	13575	13590	ACTATCAGAACAGTTC	62	1549
1246647	N/A	N/A	13815	13830	GTTTTTATTGGGCCCA	67	1550
1246673	N/A	N/A	14009	14024	CAATCTAGCAACCTCT	51	1551
1246699	N/A	N/A	14166	14181	AGCAAGCCAACAGAGA	40	1552
1246725	N/A	N/A	14384	14399	CTTAGTGTAGAATTGT	56	1553
1246751	N/A	N/A	14614	14629	AATTATTTCTTGCGAG	16	1554
1246777	N/A	N/A	15253	15268	AATTCTATAGCTAAGT	9	1555
1246803	N/A	N/A	15367	15382	GGCATTAGATGGTAAG	62	1556
1246829	N/A	N/A	15634	15649	GCCATAATCTGTGATT	0	1557
1246855	N/A	N/A	16290	16305	AGTTAGAATAGCTTTC	67	1558
1246881	N/A	N/A	17139	17154	ATTGAATAGCCATGTA	39	1559
1246907	N/A	N/A	17997	18012	TATGAAGTTAGTATAG	30	1560
1246933	N/A	N/A	18746	18761	TATATACCTGTTGCCT	55	1561

TABLE 20-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246959	N/A	N/A	19388	19403	ATGAGTAGGCAACTGA	61	1562
1246985	N/A	N/A	19608	19623	CTATTATGCACTCTAT	32	1563
1247011	N/A	N/A	19786	19801	CACTGATACATAACCAC	64	1564
1247037	N/A	N/A	20114	20129	GCATCAAGACATTCTA	72	1565
1247063	N/A	N/A	20387	20402	TATAGTAATCACAAGT	53	1566
1247089	N/A	N/A	20602	20617	TTAGAATACGCTGAGA	48	1567

TABLE 21

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245088	24	39	3118	3133	GTTCACTCCTTGTGTA	38	1568
1245114	105	120	3199	3214	AGTAGGAGTAGATGAT	0	1569
1245140	168	183	3262	3277	GAACAATCTCCCCAGC	63	1570
1245166	245	260	3339	3354	CAGAACCAATATGCTC	56	1571
1245192	293	308	7582	7597	TCGGCACTCAGCTGCA	52	1572
1245218	330	345	7619	7634	AGTCTACCACATACGC	65	1573
1245244	395	410	8789	8804	TGTTACATCACCCTACT	35	1574
1245270	426	441	8820	8835	GATATACTGTCCCAGC	34	1575
1245296	458	473	8852	8867	CTCTTCATCCTTGGTG	25	1576
1245322	496	511	8890	8905	AAATGTCCTAGGATGT	36	1577
1245348	553	568	10439	10454	ACGATGTGGCCATGAT	52	1578
1245374	608	623	N/A	N/A	ACAATATGGGATGAGG	71	1579
1245400	658	673	12076	12091	AGTTCTGATGTCAGAC	40	1580
1245426	749	764	N/A	N/A	TAATCTTGTGCTTGGA	54	1581
1245452	784	799	15665	15680	CTTCTTACGACTTCAT	45	1582
1245478	836	851	15717	15732	ATACGATGGAACAAAA	27	1583
1245504*	901	916	20716	20731	ATACGATTTAAAATCG	17	1584
1245530*	966	981	20781	20796	GGAGCTTATTTATTCA	38	1585
1245556	1008	1023	20823	20838	TTCGAACTATTTCATA	44	1586
1245582	1082	1097	20897	20912	GCTGCTAGTGCCAAAC	67	1587
1245608	1250	1265	21065	21080	GTCCACCTTTAAATGG	33	1588
1245634	1306	1321	21121	21136	AACCTTCCCTGTGTAA	38	1589
1245660	1335	1350	21150	21165	CAGATTGGAATGCTAC	74	1590
1245686	1367	1382	21182	21197	TGTGTTCTTGTGATA	69	1591

TABLE 21-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245712	1416	1431	21231	21246	GAGATAAAGCTGCCTG	67	1592
1245738	1493	1508	21308	21323	GCACCGTTTTGGGCTA	74	1593
1245764	1551	1566	21366	21381	GGTGGACTTCAGAGTT	26	1594
1245790	1600	1615	21415	21430	GTATCTTATAAGACTA	81	1595
1245816	1704	1719	21519	21534	GGGACCAAGGATATAT	17	1596
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	54	38
1245842	1766	1781	21581	21596	AGTACAGTTCCTTTTC	69	1597
1245868	1800	1815	21615	21630	TTCTCTGGGTCCATT	44	1598
1245894	2197	2212	22012	22027	AGATTAGTCTTGATGT	63	1599
1245920	N/A	N/A	N/A	N/A	CACCTTATTAATATCC	34	1600
1245946	N/A	N/A	3537	3552	GCTTAGTTCTTAATTA	28	1601
1245972	N/A	N/A	3754	3769	GCACTAAAGTTTCTG	75	1602
1245998	N/A	N/A	4408	4423	TTCATTCTGAATTCTT	71	1603
1246024	N/A	N/A	5051	5066	CCTTAGAGTGAATCAT	65	1604
1246050	N/A	N/A	5180	5195	CAACAGTTTAGATTGC	66	1605
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	91	48
1246076	N/A	N/A	5743	5758	ACTGATTTAGTTGGTA	55	1606
1246102	N/A	N/A	5929	5944	ATTTAGGGCAAACCTT	35	1607
1246128	N/A	N/A	6209	6224	ATAACCATAGGACATG	70	1608
1246154	N/A	N/A	6453	6468	CTAAGTAGTAATTCTA	35	1609
1246180	N/A	N/A	7103	7118	AAAAATTGGCCTCCTG	41	1610
1246206	N/A	N/A	7847	7862	AATAATGGGCAGAGCT	43	1611
1246232	N/A	N/A	8166	8181	GTTATTGTAGGGATTG	69	1612
1246258	N/A	N/A	8643	8658	CAAGACAGACTGTTGA	39	1613
1246284	N/A	N/A	8976	8991	AGAAGTATGTGAAGTC	41	1614
1246310	N/A	N/A	9315	9330	GTAATTACTTCTGATG	41	1615
1246336	N/A	N/A	9512	9527	TAATATTGAGGCACTG	33	1616
1246362	N/A	N/A	9659	9674	TATAGGTAGCTAATGT	31	1617
1246388	N/A	N/A	9941	9956	ACACTAATGAATCATG	7	1618
1246414	N/A	N/A	10183	10198	ACACTAATCTCAGTAC	6	1619
1246440	N/A	N/A	10502	10517	GATACTTACCAATATG	22	1620
1246466	N/A	N/A	11210	11225	CTGTAGGTTTTGCTTC	27	1621
1246492	N/A	N/A	11567	11582	GACAATGGTTGATAGT	67	1622
1246518	N/A	N/A	11925	11940	GCCAAAAATCAGCCAC	11	1623
1246544	N/A	N/A	12253	12268	TCTATTAGTTGGCCTG	26	1624
1246570	N/A	N/A	12609	12624	AATTTTCGGGTTCTTA	35	1625

TABLE 21-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246596	N/A	N/A	12710	12725	TCCAGTAAAAAATTGG	12	1626
1246622	N/A	N/A	13577	13592	CCACTATCAGAACAGT	31	1627
1246648	N/A	N/A	13841	13856	ACGAAGCCTGAATGCC	52	1628
1246674	N/A	N/A	14010	14025	ACAATCTAGCAACCTC	44	1629
1246700	N/A	N/A	14167	14182	TAGCAAGCCAACAGAG	21	1630
1246726	N/A	N/A	14390	14405	CATAGGCTTAGTGTAG	38	1631
1246752	N/A	N/A	14615	14630	GAATTATTTCTTGCGA	61	1632
1246778	N/A	N/A	15279	15294	TCAAAGCCTGTTGGAT	35	1633
1246804	N/A	N/A	15368	15383	AGGCATTAGATGGTAA	51	1634
1246830	N/A	N/A	15750	15765	TACTTACTTCTGTAGT	12	1635
1246856	N/A	N/A	16291	16306	TAGTTAGAATAGTCTT	23	1636
1246882	N/A	N/A	17143	17158	GTAAATTGAATAGCCA	55	1637
1246908	N/A	N/A	17999	18014	CTTATGAAGTTAGTAT	43	1638
1246934	N/A	N/A	18747	18762	ATATATACCTGTTGCC	28	1639
1246960	N/A	N/A	19389	19404	GATGAGTAGGCAACTG	42	1640
1246986	N/A	N/A	19609	19624	TCTATTATGCACTCTA	55	1641
1247012	N/A	N/A	19815	19830	GTTCAAATTGGATGCA	68	1642
1247038	N/A	N/A	20124	20139	TCTGATTACAGCATCA	48	1643
1247064	N/A	N/A	20388	20403	CTATAGTAATCACAAG	35	1644
1247090	N/A	N/A	20603	20618	TTTAGAATACGCTGAG	44	1645

TABLE 22

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245089	25	40	3119	3134	GGTTCAGTCCTTGTGT	57	1646
1245115	106	121	3200	3215	AAGTAGGAGTAGATGA	0	1647
1245141	171	186	3265	3280	TGAGAACAATCTCCCC	69	1648
1245167	246	261	3340	3355	ACAGAACCAATATGCT	40	1649
1245193	294	309	7583	7598	TTCGGCACTCAGCTGC	41	1650
1245219	354	369	7643	7658	GATAGATCTCTTCTCT	0	1651
1245245	396	411	8790	8805	TTGTTACATCACCCAC	34	1652
1245271	427	442	8821	8836	GGATATACTGTCCCAG	9	1653
1245297	464	479	8858	8873	GGTAATCTCTTCATCC	9	1654

TABLE 22-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245323	497	512	8891	8906	AAAATGTCCTAGGATG	0	1655
1245349	554	569	10440	10455	GACGATGTGGCCATGA	42	1656
1245375	609	624	N/A	N/A	AACAATATGGGATGAG	39	1657
1245401	659	674	12077	12092	AAGTTCTGATGTCAGA	7	1658
1245427	750	765	N/A	N/A	ATAATCTTGTGCTTGG	36	1659
1245453	785	800	15666	15681	ACTTCTTACGACTTCA	43	1660
1245479	837	852	15718	15733	TATACGATGGAACAAA	4	1661
1245505*	902	917	20717	20732	CATACGATTAAAATC	13	1662
1245531*	967	982	20782	20797	TGGAGCTTATTTATTC	39	1663
1245557	1009	1024	20824	20839	ATTCGAACTATTCAT	24	1664
1245583	1083	1098	20898	20913	TGCTGCTAGTGCCAAA	58	1665
1245609	1251	1266	21066	21081	TGTCCACCTTTAAATG	29	1666
1245635	1307	1322	21122	21137	AAACCTTCCCTGTGTA	34	1667
1245661	1336	1351	21151	21166	ACAGATTGGAATGCTA	50	1668
1245687	1379	1394	21194	21209	GTGCACTATTCTGTG	6	1669
1245713	1418	1433	21233	21248	TTGAGATAAAGCTGCC	49	1670
1245739	1494	1509	21309	21324	TGCACCGTTTGGGCT	36	1671
1245765	1552	1567	21367	21382	TGGTGGACTTCAGAGT	17	1672
1245791	1601	1616	21416	21431	TGTATCTTATAAGACT	45	1673
1245817	1705	1720	21520	21535	TGGGACCAAGGATATA	20	1674
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	48	38
1245843	1767	1782	21582	21597	CAGTACAGTTCCTTTT	43	1675
1245869	1801	1816	21616	21631	CTTCTCTTGGGTCCAT	51	1676
1245895	2198	2213	22013	22028	AAGATTAGTCTTGATG	25	1677
1245921	N/A	N/A	N/A	N/A	TCACCTTATTAATATC	0	1678
1245947	N/A	N/A	3546	3561	AAAATCGCTGCTTAGT	35	1679
1245973	N/A	N/A	3755	3770	GGCACTAAAGGTTTCT	62	1680
1245999	N/A	N/A	4409	4424	GTTCAATTCGAATTTCT	76	1681
1246025	N/A	N/A	5053	5068	AGCCTTAGAGTGAATC	74	1682
1246051	N/A	N/A	5187	5202	CCCAACGCAACAGTTT	66	1683
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	84	48
1246077	N/A	N/A	5759	5774	AATAGGAGAGTCTTTC	47	1684
1246103	N/A	N/A	5930	5945	TATTTAGGGCAAACCT	40	1685
1246129	N/A	N/A	6210	6225	CATAACCATAGGACAT	41	1686
1246155	N/A	N/A	6454	6469	GCTAAGTAGTAATTCT	57	1687
1246181	N/A	N/A	7119	7134	GATTAGTTAACCTTTT	58	1688

TABLE 22-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246207	N/A	N/A	7848	7863	TAATAATGGGCAGAGC	30	1689
1246233	N/A	N/A	8171	8186	ACAGAGTTATTGTAGG	58	1690
1246259	N/A	N/A	8670	8685	ATTGTATGAGGTCTCT	49	1691
1246285	N/A	N/A	8977	8992	CAGAAGTATGTGAACT	22	1692
1246311	N/A	N/A	9329	9344	CTATATACATCCAAGT	6	1693
1246337	N/A	N/A	9513	9528	CTAATATTGAGGCACT	51	1694
1246363	N/A	N/A	9661	9676	GCTATAGGTAGCTAAT	20	1695
1246389	N/A	N/A	9942	9957	CACACTAATGAATCAT	30	1696
1246415	N/A	N/A	10185	10200	CAACACTAATCTCAGT	4	1697
1246441	N/A	N/A	10520	10535	ATAACATGGCTGGCAT	14	1698
1246467	N/A	N/A	11235	11250	CATAGACATGTGTTCA	52	1699
1246493	N/A	N/A	11568	11583	TGACAATGGTTGATAG	16	1700
1246519	N/A	N/A	11953	11968	GTATCATTGGCTTAA	29	1701
1246545	N/A	N/A	12254	12269	TTCTATTAGTTGGCCT	28	1702
1246571	N/A	N/A	12610	12625	TAATTTTCGGGTCTT	40	1703
1246597	N/A	N/A	12723	12738	GGATTAATCCCTTCC	10	1704
1246623	N/A	N/A	13594	13609	GATAGAGCACATGGCC	20	1705
1246649	N/A	N/A	13842	13857	CACGAAGCCTGAATGC	24	1706
1246675	N/A	N/A	14011	14026	CACAATCTAGCAACCT	40	1707
1246701	N/A	N/A	14170	14185	GAATAGCAAGCCAACA	64	1708
1246727	N/A	N/A	14391	14406	ACATAGGCTTAGTGTA	0	1709
1246753	N/A	N/A	14616	14631	CGAATTATTTCTTGCG	58	1710
1246779	N/A	N/A	15280	15295	CTCAAAGCCTGTTGGA	10	1711
1246805	N/A	N/A	15380	15395	AAAATGGTGGCGAGGC	0	1712
1246831*	N/A	N/A	15751	15766	GTACTTACTTCTGTAG	0	1713
1246857	N/A	N/A	16292	16307	TTAGTTAGAATAGTCT	8	1714
1246883	N/A	N/A	17144	17159	AGTAAATTGAATAGCC	39	1715
1246909	N/A	N/A	18000	18015	GCTTATGAAGTTAGTA	67	1716
1246935	N/A	N/A	18798	18813	CGTAAAATTGTGTCTC	43	1717
1246961	N/A	N/A	19397	19412	GCATAAGAGATGAGTA	33	1718
1246987	N/A	N/A	19610	19625	ATCTATTATGCACTCT	24	1719
1247013	N/A	N/A	19819	19834	CATAGTTCAAATTGGA	39	1720
1247039	N/A	N/A	20138	20153	GGGAAAAAGTGTGTTC	20	1721
1247065	N/A	N/A	20395	20410	GTAATTGCTATAGTAA	33	1722
1247091	N/A	N/A	20604	20619	ATTAGAATACGCTGA	19	1723

TABLE 23

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245090	26	41	3120	3135	TGGTTCAGTCCTTGTG	50	1724
1245116	107	122	3201	3216	CAAGTAGGAGTAGATG	5	1725
1245142	172	187	3266	3281	ATGAGAACAATCTCCC	65	1726
1245168	247	262	3341	3356	CACAGAACCAATATGC	54	1727
1245194	295	310	7584	7599	TTTCGGCACTCAGCTG	29	1728
1245220	355	370	7644	7659	CGATAGATCTCTTCTC	15	1729
1245246	398	413	8792	8807	GATTGTTACATCACCC	65	1730
1245272	428	443	8822	8837	TGGATATACTGTCCCA	24	1731
1245298	465	480	8859	8874	TGGTAATCTCTTCATC	23	1732
1245324	500	515	8894	8909	CCAAAAATGTCCTAGG	38	1733
1245350	557	572	10443	10458	TGTGACGATGTGGCCA	42	1734
1245376	610	625	N/A	N/A	GAACAATATGGGATGA	26	1735
1245402	660	675	12078	12093	GAAGTTCCTGATGTCAG	0	1736
1245428	751	766	N/A	N/A	CATAATCTTGCTCTTG	37	1737
1245454	786	801	15667	15682	GACTTCTTACGACTTC	38	1738
1245480	838	853	15719	15734	ATATACGATGGAACAA	0	1739
1245506*	903	918	20718	20733	GCATACGATTTAAAT	0	1740
1245532*	969	984	20784	20799	GCTGGAGCTTATTTAT	16	1741
1245558	1010	1025	20825	20840	GATTCGAACTATTCA	57	1742
1245584	1084	1099	20899	20914	CTGCTGCTAGTGCCAA	57	1743
1245610	1256	1271	21071	21086	GCTTTTGTCCACCTTT	68	1744
1245636	1308	1323	21123	21138	TAAACCTTCCCTGTGT	17	1745
1245662	1337	1352	21152	21167	TACAGATTGGAATGCT	46	1746
1245688	1380	1395	21195	21210	TGTGCACTCATTCTGT	18	1747
1245714	1419	1434	21234	21249	GTTGAGATAAAGCTGC	54	1748
1245740	1495	1510	21310	21325	TTGCACCGTTTTGGGC	45	1749
1245766	1553	1568	21368	21383	TTGGTGGACTTCAGAG	8	1750
1245792	1602	1617	21417	21432	ATGTATCTTATAAGAC	0	1751
1245818	1706	1721	21521	21536	CTGGGACCAAGGATAT	32	1752
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	50	38
1245844	1768	1783	21583	21598	CCAGTACAGTTCCTTT	66	1753
1245870	1802	1817	21617	21632	TCTTCTCTTGGGTCCA	59	1754
1245896	2199	2214	22014	22029	CAAGATTAGTCTTGAT	27	1755
1245922	N/A	N/A	3357	3372	TACCTTATTAATATCC	54	1756
1245948	N/A	N/A	3547	3562	AAAAATCGCTGCTTAG	25	1757
1245974	N/A	N/A	3834	3849	GACTACGGACTGGCAA	60	1758

TABLE 23-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246000	N/A	N/A	4414	4429	AGAAAGTTCATTCGAA	61	1759
1246026	N/A	N/A	5057	5072	TTACAGCCTTAGAGTG	21	1760
1246052	N/A	N/A	5200	5215	TTGAACCGTATTCCC	69	1761
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	77	48
1246078	N/A	N/A	5760	5775	GAATAGGAGAGTCTTT	58	1762
1246104	N/A	N/A	5955	5970	ATTGAGATGACAGTGG	70	1763
1246130	N/A	N/A	6211	6226	CCATAACCATAGGACA	63	1764
1246156	N/A	N/A	6455	6470	TGCTAAGTAGTAATTC	32	1765
1246182	N/A	N/A	7121	7136	AAGATTAGTTAACCTT	24	1766
1246208	N/A	N/A	7851	7866	CAATAATAATGGGCAG	43	1767
1246234	N/A	N/A	8205	8220	GAGTCATAATTTGCTT	57	1768
1246260	N/A	N/A	8672	8687	CTATTGTATGAGGTCT	42	1769
1246286	N/A	N/A	8978	8993	CCAGAAGTATGTGAAC	8	1770
1246312	N/A	N/A	9330	9345	CCTATATACATCCAAG	53	1771
1246338	N/A	N/A	9514	9529	ACTAATATTGAGGCAC	19	1772
1246364	N/A	N/A	9662	9677	AGCTATAGGTAGCTAA	0	1773
1246390	N/A	N/A	9943	9958	CCACACTAATGAATCA	43	1774
1246416	N/A	N/A	10202	10217	TATGGTAAGCCCATG	30	1775
1246442	N/A	N/A	10522	10537	ATATAACATGGCTGGC	29	1776
1246468	N/A	N/A	11236	11251	ACATAGACATGTGTTT	35	1777
1246494	N/A	N/A	11590	11605	CTCAATAACTGAGTTA	4	1778
1246520	N/A	N/A	11962	11977	GAGAAGGCGGTATCAT	48	1779
1246546	N/A	N/A	12255	12270	CTTCTATTAGTTGGCC	11	1780
1246572	N/A	N/A	12611	12626	TTAATTTTCGGTTCT	31	1781
1246598	N/A	N/A	12724	12739	AGGATTAATTCCTTTC	19	1782
1246624	N/A	N/A	13595	13610	GGATAGAGCACATGGC	33	1783
1246650	N/A	N/A	13845	13860	GTACACGAAGCCTGAA	42	1784
1246676	N/A	N/A	14012	14027	TCACAATCTAGCAACC	4	1785
1246702	N/A	N/A	14171	14186	TGAATAGCAAGCCAAC	4	1786
1246728	N/A	N/A	14393	14408	AAACATAGGCTTAGTG	16	1787
1246754	N/A	N/A	14724	14739	AATACCCTTGGTGGAG	16	1788
1246780	N/A	N/A	15312	15327	TAATCATGGGACAGGA	15	1789
1246806	N/A	N/A	15381	15396	CAAAATGGTGGCGAGG	10	1790
1246832	N/A	N/A	15771	15786	GTATTTGGGTGTTCTG	36	1791
1246858	N/A	N/A	16293	16308	TTTAGTTAGAATAGTC	15	1792
1246884	N/A	N/A	17227	17242	GTACAAATTTATGCCA	45	1793

TABLE 23-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246910	N/A	N/A	18001	18016	GGCTTATGAAGTTAGT	59	1794
1246936	N/A	N/A	18812	18827	TAATTGCATCTGCTCG	46	1795
1246962	N/A	N/A	19398	19413	TGCATAAGAGATGAGT	19	1796
1246988	N/A	N/A	19611	19626	TATCTATTATGCACTC	48	1797
1247014	N/A	N/A	19820	19835	TCATAGTTCAAATTGG	47	1798
1247040	N/A	N/A	20172	20187	CAATTATGAATCTGCA	25	1799
1247066	N/A	N/A	20396	20411	AGTAATTGCTATAGTA	48	1800
1247092	N/A	N/A	20605	20620	CATTTAGAATACGCTG	25	1801

TABLE 24

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245091	27	42	3121	3136	CTGGTTCAGTCCTTGT	63	1802
1245117	108	123	3202	3217	CCAAGTAGGAGTAGAT	19	1803
1245143	175	190	3269	3284	GTAATGAGAACAATCT	48	1804
1245169	248	263	3342	3357	CCACAGAACCAATATG	49	1805
1245195	296	311	7585	7600	TTTTCGGCACTCAGCT	13	1806
1245221	356	371	7645	7660	GCGATAGATCTCTTCT	24	1807
1245247	399	414	8793	8808	CGATTGTTACATCACC	47	1808
1245273	429	444	8823	8838	CTGGATATACTGTCCC	20	1809
1245299	469	484	8863	8878	GTCTTGGTAATCTCTT	37	1810
1245325	511	526	N/A	N/A	GCTTTTGTGATCCAAA	43	1811
1245351	558	573	10444	10459	CTGTGACGATGTGGCC	28	1812
1245377	611	626	N/A	N/A	GGAACAATATGGGATG	36	1813
1245403	665	680	12083	12098	GGCCTGAAGTTCTGAT	0	1814
1245429	752	767	N/A	N/A	CCATAATCTTGTGCTT	32	1815
1245455	787	802	15668	15683	AGACTTCTTACGACTT	41	1816
1245481	839	854	15720	15735	GATATACGATGGAACA	46	1817
1245507*	904	919	20719	20734	TGCATACGATTAAAA	18	1818
1245533*	970	985	20785	20800	GGCTGGAGCTTATTTA	57	1819
1245559	1011	1026	20826	20841	TGATTCGAACTATTC	29	1820
1245585	1085	1100	20900	20915	ACTGCTGCTAGTGCCA	47	1821
1245611	1257	1272	21072	21087	AGCTTTTGTCCACCTT	57	1822
1245637	1310	1325	21125	21140	CTTAAACCTTCCCTGT	22	1823

TABLE 24-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245663	1338	1353	21153	21168	CTACAGATTGGAATGC	52	1824
1245689	1381	1396	21196	21211	CTGTGCACTCATTCTG	51	1825
1245715	1420	1435	21235	21250	GGTTGAGATAAAGCTG	76	1826
1245741	1496	1511	21311	21326	GTTGCACCGTTTGGG	56	1827
1245767	1554	1569	21369	21384	TTTGGTGGACTTCAGA	16	1828
1245793	1613	1628	21428	21443	CACCTTTCATAATGTA	59	1829
1245819	1707	1722	21522	21537	TCTGGGACCAAGGATA	30	1830
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	50	38
1245845	1769	1784	21584	21599	GCCAGTACAGTTCCTT	61	1831
1245871	2174	2189	21989	22004	GTCGGATTATTTTTC	61	1832
1245897	2200	2215	22015	22030	ACAAGATTAGTCTTGA	45	1833
1245923	N/A	N/A	3368	3383	GATGTATACATTACCT	67	1834
1245949	N/A	N/A	3549	3564	ATAAAAATCGCTGCTT	11	1835
1245975	N/A	N/A	3877	3892	ATCATATGCTAAGTGC	73	1836
1246001	N/A	N/A	4415	4430	CAGAAAGTTCATTGCA	67	1837
1246027	N/A	N/A	5061	5076	TACATTACAGCCTTAG	75	1838
1246053	N/A	N/A	5204	5219	GTTTTTTGAACCGTAT	71	1839
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	74	48
1246079	N/A	N/A	5761	5776	AGAATAGGAGAGTCTT	64	1840
1246105	N/A	N/A	5957	5972	GTATTGAGATGACAGT	68	1841
1246131	N/A	N/A	6212	6227	CCCATAACCATAGGAC	47	1842
1246157	N/A	N/A	6464	6479	GCAGTAAAGTGCTAAG	83	1843
1246183	N/A	N/A	7122	7137	GAAGATTAGTTAACCT	13	1844
1246209	N/A	N/A	7938	7953	CATATTCCTGATGAT	14	1845
1246235	N/A	N/A	8238	8253	AACTACATGACCTGGG	27	1846
1246261	N/A	N/A	8674	8689	TTCTATTGTATGAGGT	0	1847
1246287	N/A	N/A	8980	8995	GTCCAGAAGTATGTGA	9	1848
1246313	N/A	N/A	9331	9346	TCCTATATACATCCAA	45	1849
1246339	N/A	N/A	9515	9530	CACTAATATTGAGGCA	31	1850
1246365	N/A	N/A	9663	9678	CAGCTATAGGTAGCTA	0	1851
1246391	N/A	N/A	9966	9981	TAGTGTAAGCTGAGAG	75	1852
1246417	N/A	N/A	10203	10218	GTATGGTAAGCCCAT	36	1853
1246443	N/A	N/A	10523	10538	TATATAACATGGCTGG	8	1854
1246469	N/A	N/A	11238	11253	TGACATAGACATGTGT	42	1855
1246495	N/A	N/A	11621	11636	CTGTAGGACTCTGCTC	48	1856
1246521	N/A	N/A	11963	11978	TGAGAAGCGGTATCA	20	1857

TABLE 24-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246547	N/A	N/A	12337	12352	CCCTAATGTGTTTTCC	25	1858
1246573	N/A	N/A	12612	12627	CTTAATTTTCGGGTTC	37	1859
1246599	N/A	N/A	12725	12740	TAGGATTAATCCCTT	0	1860
1246625	N/A	N/A	13601	13616	AATACAGGATAGAGCA	0	1861
1246651	N/A	N/A	13859	13874	TAGGAAAGCTCATGGT	27	1862
1246677	N/A	N/A	14018	14033	GGAAATTCACAATCTA	22	1863
1246703	N/A	N/A	14172	14187	TTGAATAGCAAGCCAA	31	1864
1246729	N/A	N/A	14394	14409	TAAACATAGGCTTAGT	34	1865
1246755	N/A	N/A	14726	14741	TAAATACCCCTTGGTGG	7	1866
1246781	N/A	N/A	15313	15328	TTAATCATGGGACAGG	27	1867
1246807	N/A	N/A	15382	15397	TCAAAATGGTGGCGAG	31	1868
1246833	N/A	N/A	15772	15787	AGTATTTGGGTGTTCT	17	1869
1246859	N/A	N/A	16327	16342	TCACTATATTGCATTC	35	1870
1246885	N/A	N/A	17239	17254	AGGGAATGTTATGTAC	49	1871
1246911	N/A	N/A	18012	18027	AAAAGGCCCAAGGCTT	30	1872
1246937	N/A	N/A	18813	18828	GTAATTGCATCTGCTC	51	1873
1246963	N/A	N/A	19399	19414	ATGCATAAGAGATGAG	30	1874
1246989	N/A	N/A	19649	19664	CTATGCAAGCCTTCAC	19	1875
1247015	N/A	N/A	19827	19842	GTAATATTCATAGTTC	49	1876
1247041	N/A	N/A	20173	20188	CCAATTATGAATCTGC	61	1877
1247067	N/A	N/A	20397	20412	TAGTAATTGCTATAGT	7	1878
1247093	N/A	N/A	20606	20621	TCATTTAGAATACGCT	20	1879

TABLE 25

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245092	31	46	3125	3140	CCTTCTGGTTCAGTCC	72	1880
1245118	109	124	3203	3218	TCCAAGTAGGAGTAGA	24	1881
1245144	176	191	3270	3285	AGTAATGAGAACAATC	45	1882
1245170	249	264	3343	3358	CCCACAGAACCAATAT	16	1883
1245196	297	312	7586	7601	GTTTTCGGCACTCAGC	51	1884
1245222	357	372	7646	7661	AGCGATAGATCTCTTC	31	1885
1245248	400	415	8794	8809	ACGATTGTTACATCAC	53	1886

TABLE 25-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245274	430	445	8824	8839	GCTGGATATACTGTCC	36	1887
1245300	471	486	8865	8880	ATGTCTTGTAATCTC	23	1888
1245326	513	528	N/A	N/A	GTGCTTTTGTGATCCA	62	1889
1245352	559	574	10445	10460	ACTGTGACGATGTGGC	40	1890
1245378	612	627	N/A	N/A	TGGAACAATATGGGAT	46	1891
1245404	667	682	12085	12100	AAGGCCTGAAGTTCTG	5	1892
1245430	755	770	N/A	N/A	AGGCCATAATCTTGTG	33	1893
1245456	788	803	15669	15684	CAGACTTCTTACGACT	33	1894
1245482	840	855	15721	15736	TGATATACGATGGAAC	13	1895
1245508*	905	920	20720	20735	CTGCATACGATTTAAA	32	1896
1245534*	971	986	20786	20801	TGGCTGGAGCTTATTT	87	1897
1245560	1012	1027	20827	20842	TTGATTCGAACTATT	55	1898
1245586	1086	1101	20901	20916	GACTGCTGCTAGTGCC	67	1899
1245612	1258	1273	21073	21088	TAGCTTTTGTCCACCT	60	1900
1245638	1312	1327	21127	21142	GTCTTAAACCTTCCCT	63	1901
1245664	1339	1354	21154	21169	GCTACAGATTGGAATG	47	1902
1245690	1383	1398	21198	21213	AGCTGTGCACTCATTC	57	1903
1245716	1421	1436	21236	21251	AGGTTGAGATAAAGCT	74	1904
1245742	1497	1512	21312	21327	AGTTGCACCGTTTTGG	55	1905
1245768	1556	1571	21371	21386	CTTTTGGTGGACTTCA	10	1906
1245794	1615	1630	21430	21445	GTCACCTTTCATAATG	45	1907
1245820	1708	1723	21523	21538	CTCTGGGACCAAGGAT	51	1908
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	55	38
1245846	1770	1785	21585	21600	AGCCAGTACAGTTCCT	58	1909
1245872	2175	2190	21990	22005	AGTCGGATTATTTTTT	42	1910
1245898	2201	2216	22016	22031	AACAAGATTAGTCTTG	40	1911
1245924	N/A	N/A	3401	3416	CATATCTTACTCTGTG	47	1912
1245950	N/A	N/A	3550	3565	CATAAAAATCGCTGCT	45	1913
1245976	N/A	N/A	3882	3897	CGTTTATCATATGCTA	88	1914
1246002	N/A	N/A	4416	4431	CCAGAAAGTTCATTCTG	81	1915
1246028	N/A	N/A	5062	5077	TTACATTACAGCCTTA	67	1916
1246054	N/A	N/A	5218	5233	TTAAACTCCAATGTGT	36	1917
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	82	48
1246080	N/A	N/A	5800	5815	CTCAATAGCCTTTTG	64	1918
1246106	N/A	N/A	5979	5994	CCTGAAAAGCCCTATT	0	1919
1246132	N/A	N/A	6222	6237	ACCAAACCTTGCCCAT	46	1920

TABLE 25-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1246158	N/A	N/A	6465	6480	GGCAGTAAAGTGCTAA	62	1921
1246184	N/A	N/A	7123	7138	TGAAGATTAGTTAACC	21	1922
1246210	N/A	N/A	7941	7956	ATCCATATTCCTGAT	26	1923
1246236	N/A	N/A	8239	8254	TAACTACATGACCTGG	37	1924
1246262	N/A	N/A	8688	8703	GATGAGAGTCTTCTT	26	1925
1246288	N/A	N/A	9000	9015	AAAGATCTGGCCAGTT	3	1926
1246314	N/A	N/A	9332	9347	ATCCTATATACATCCA	76	1927
1246340	N/A	N/A	9516	9531	ACACTAATATTGAGGC	68	1928
1246366	N/A	N/A	9712	9727	GAACATACATTGTTGC	58	1929
1246392	N/A	N/A	9972	9987	GTTGAGTAGTGAAGC	53	1930
1246418	N/A	N/A	10209	10224	TTATTGGTATGGTAAG	14	1931
1246444	N/A	N/A	10728	10743	GTAATTGGCTCTTGAG	44	1932
1246470	N/A	N/A	11239	11254	ATGACATAGACATGTG	23	1933
1246496	N/A	N/A	11652	11667	CTACAAAAGTGGTATC	0	1934
1246522	N/A	N/A	11982	11997	ACCAATCCTCAGCTTT	11	1935
1246548	N/A	N/A	12353	12368	GATCATTCCTGCTAGT	35	1936
1246574	N/A	N/A	12613	12628	GCTTAATTTTCGGGTT	36	1937
1246600	N/A	N/A	12726	12741	TTAGGATTAATTCCT	14	1938
1246626	N/A	N/A	13603	13618	ACAATACAGGATAGAG	23	1939
1246652	N/A	N/A	13860	13875	TTAGGAAAGCTCATGG	26	1940
1246678	N/A	N/A	14026	14041	CTTACTAAGGAAATTC	19	1941
1246704	N/A	N/A	14174	14189	CCTTGAATAGCAAGCC	61	1942
1246730	N/A	N/A	14395	14410	ATAAACATAGGCTTAG	41	1943
1246756	N/A	N/A	14728	14743	TCTAAATACCTTGGT	26	1944
1246782	N/A	N/A	15314	15329	ATTAATCATGGGACAG	20	1945
1246808	N/A	N/A	15383	15398	ATCAAAATGGTGGCGA	20	1946
1246834	N/A	N/A	15773	15788	TAGTATTTGGGTGTTT	27	1947
1246860	N/A	N/A	16377	16392	ACCAAAGTTCAGCAG	31	1948
1246886	N/A	N/A	17675	17690	CAAAACCTTTGTTGCC	14	1949
1246912	N/A	N/A	18015	18030	TCAAAAAGGCCAAGG	46	1950
1246938	N/A	N/A	18814	18829	AGTAATTGCATCTGCT	42	1951
1246964	N/A	N/A	19404	19419	CAGGAATGCATAAGAG	33	1952
1246990	N/A	N/A	19662	19677	ACCAGAATCCATTCTA	60	1953
1247016	N/A	N/A	19860	19875	AATCAATCTGTCTGAA	34	1954
1247042	N/A	N/A	20174	20189	CCCAATTATGAATCTG	62	1955

TABLE 25-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1247068	N/A	N/A	20398	20413	ATAGTAATTGCTATAG	36	1956
1247094	N/A	N/A	20607	20622	GTCATTAGAATACGC	46	1957

TABLE 26

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1245093	58	73	3152	3167	ATGTTTCATGGCTTTGC	73	1958
1245119	110	125	3204	3219	CTCCAAGTAGGAGTAG	5	1959
1245145	180	195	3274	3289	CTCCAGTAATGAGAAC	40	1960
1245171	250	265	3344	3359	TCCCACAGAACCAATA	24	1961
1245197	298	313	7587	7602	AGTTTTCGGCACTCAG	16	1962
1245223	358	373	7647	7662	GAGCGATAGATCTCTT	0	1963
1245249	401	416	8795	8810	CACGATTGTTACATCA	37	1964
1245275	431	446	8825	8840	GGCTGGATATACTGTC	18	1965
1245301	472	487	8866	8881	AATGTCTTGGTAATCT	20	1966
1245327	514	529	N/A	N/A	AGTGCTTTTGTGATCC	25	1967
1245353	560	575	10446	10461	CACTGTGACGATGTGG	10	1968
1245379	613	628	N/A	N/A	CTGGAACAATATGGGA	38	1969
1245405	669	684	12087	12102	CCAAGGCCTGAAGTTC	20	1970
1245431	758	773	N/A	N/A	TACAGGCCATAATCTT	16	1971
1245457	789	804	15670	15685	TCAGACTTCTTACGAC	36	1972
1245483	841	856	15722	15737	TTGATATACGATGGAA	38	1973
1245509*	906	921	20721	20736	TCTGCATACGATTAA	44	1974
1245535*	972	987	20787	20802	CTGGCTGGAGCTTATT	82	1975
1245561	1013	1028	20828	20843	ATTGATTCGAACTAT	22	1976
1245587	1087	1102	20902	20917	TGACTGCTGCTAGTGC	50	1977
1245613	1259	1274	21074	21089	GTAGCTTTTGTCCACC	63	1978
1245639	1313	1328	21128	21143	AGTCTTAAACCTTCCC	77	1979
1245665	1340	1355	21155	21170	GGCTACAGATTGGAAT	23	1980
1245691	1384	1399	21199	21214	TAGCTGTGCACTCATT	40	1981
1245717	1422	1437	21237	21252	CAGGTTGAGATAAAGC	49	1982
1245743	1498	1513	21313	21328	GAGTTGCACCGTTTGT	48	1983
1245769	1557	1572	21372	21387	ACTTTTGGTGGACTTC	0	1984
1245795	1616	1631	21431	21446	GGTCACCTTTCATAAT	57	1985

TABLE 26-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	HSD17B13 (% Inhibition)	SEQ ID NO
1245821	1709	1724	21524	21539	TCTCTGGGACCAAGGA	56	1986
1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC	44	38
1245847	1771	1786	21586	21601	TAGCCAGTACAGTTCC	52	1987
1245873	2176	2191	21991	22006	GAGTCGGATTATTTT	44	1988
1245899	2204	2219	22019	22034	CAAAACAAGATTAGTC	18	1989
1245925	N/A	N/A	3403	3418	TACATATCTTACTCTG	40	1990
1245951	N/A	N/A	3551	3566	TCATAAAAAATCGCTGC	26	1991
1245977	N/A	N/A	3902	3917	GTTTATCATGTGCCAC	83	1992
1246003	N/A	N/A	4434	4449	GTAAATTCTTGCTATC	40	1993
1246029	N/A	N/A	5063	5078	GTTACATTACAGCCTT	71	1994
1246055	N/A	N/A	5221	5236	AGCTTAAACTCCAATG	40	1995
1246065	N/A	N/A	5659	5674	ATAAGTTACCAGAGCA	80	48
1246081	N/A	N/A	5816	5831	TCAAAGTCAGTATCCC	79	1996
1246107	N/A	N/A	5983	5998	GATACCTGAAAAGCCC	49	1997
1246133	N/A	N/A	6223	6238	TACCAAACCTGCCCAT	33	1998
1246159	N/A	N/A	6473	6488	ATGTAATAGGCAGTAA	33	1999
1246185	N/A	N/A	7481	7496	ACAATTAACATTCGGC	77	2000
1246211	N/A	N/A	7950	7965	TAGGAAGTGATCCATA	30	2001
1246237	N/A	N/A	8240	8255	ATAACTACATGACCTG	29	2002
1246263	N/A	N/A	8693	8708	CCAAAGATGAGAGTCT	19	2003
1246289	N/A	N/A	9001	9016	CAAAGATCTGGCCAGT	1	2004
1246315	N/A	N/A	9346	9361	CAAAAGTGTCCTCGAT	5	2005
1246341	N/A	N/A	9517	9532	TACACTAATATTGAGG	30	2006
1246367	N/A	N/A	9713	9728	TGAACATACATTGTTG	29	2007
1246393	N/A	N/A	9976	9991	GACAGTTGAGTAGTGT	46	2008
1246419	N/A	N/A	10210	10225	TTTATTGGTATGGTAA	25	2009
1246445	N/A	N/A	10729	10744	TGTAATTGGCTCTTGA	0	2010
1246471	N/A	N/A	11248	11263	ACCAAGGCCATGACAT	17	2011
1246497	N/A	N/A	11653	11668	TCTACAAAACCTGGTAT	9	2012
1246523	N/A	N/A	11983	11998	AACCAATCCTCAGCTT	0	2013
1246549	N/A	N/A	12411	12426	GATCAAATGTATGTGC	25	2014
1246575	N/A	N/A	12614	12629	GGCTTAATTTTCGGGT	17	2015
1246601	N/A	N/A	12727	12742	TTTAGGATTAATCCC	16	2016
1246627	N/A	N/A	13604	13619	GACAATACAGGATAGA	43	2017
1246653	N/A	N/A	13861	13876	CTTAGGAAAGCTCATG	0	2018
1246679	N/A	N/A	14027	14042	CCTTACTAAGGAAATT	28	2019

TABLE 26-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	HSD17B13 (%) Inhibition	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1246705	N/A	N/A	14194	14209	TGTGATTGAGTTCTCC	62	2020
1246731	N/A	N/A	14397	14412	TCATAAACATAGGCTT	58	2021
1246757	N/A	N/A	14729	14744	CTCTAAATACCCCTTG	7	2022
1246783	N/A	N/A	15315	15330	GATTAATCATGGGACA	9	2023
1246809	N/A	N/A	15384	15399	TATCAAAATGGTGGCG	18	2024
1246835	N/A	N/A	15774	15789	TTAGTATTTGGGTGTT	29	2025
1246861	N/A	N/A	16382	16397	TCCCAACCAAACTCC	82	2026
1246887	N/A	N/A	17766	17781	GCATATTCATTGGCC	6	2027
1246913	N/A	N/A	18018	18033	GTTTCAAAAAGCCCA	39	2028
1246939	N/A	N/A	18815	18830	TAGTAATGCATCTGC	51	2029
1246965	N/A	N/A	19436	19451	TGGAGGAAAGCTTCAA	10	2030
1246991	N/A	N/A	19675	19690	ACCAAGAGACACCACC	41	2031
1247017	N/A	N/A	19889	19904	CTTACGACAGGTCATC	39	2032
1247043	N/A	N/A	20185	20200	TTAGAAGTCAGCCCAA	37	2033
1247069	N/A	N/A	20400	20415	CTATAGTAATTGCTAT	0	2034
1247095	N/A	N/A	20608	20623	GGTCATTAGAATACG	39	2035

Example 2: Dose-Dependent Inhibition of Human
HSD17B13 in HepaRG Cells by cEt Gapmers

Certain modified oligonucleotides described in the studies above exhibiting significant in vitro inhibition of HSD17B13 RNA were selected and tested at various doses in HepaRG cells.

The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with modified oligonucleotides diluted to different concentrations as specified in the Tables below. After a treatment period of approximately 24 hours, HSD17B13 RNA levels were measured as previously described using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total RNA, as measured by RIBOGREEN®. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells.

The half maximal inhibitory concentration (IC₅₀) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

TABLE 27

Multi-dose assay of modified oligonucleotides in HepaRG cells							
40	ION	% Inhibition					IC50
		No.	6 nM	32 nM	160 nM	800 nM	
45	1245651	0	6	24	57	82	0.6
	1245927	0	19	55	80	90	0.2
	1245930	0	0	30	77	86	0.4
	1245957	8	8	49	77	87	0.2
	1245984	12	14	62	78	90	0.1
	1246065	0	10	46	65	89	0.3
50	1246066	0	13	35	58	76	0.5
	1246068	0	5	30	61	83	0.5
	1246110	3	12	44	82	87	0.2
	1246115	8	14	45	75	89	0.2
	1246116	6	17	48	72	83	0.2
	1246134	3	21	55	79	91	0.2
	1246163	2	15	43	75	92	0.2

TABLE 28

Multi-dose assay of modified oligonucleotides in HepaRG cells							
60	ION	% Inhibition					IC50
	No.	6 nM	32 nM	160 nM	800 nM	4000 nM	(μM)
65	1245681	0	12	38	72	90	0.3
	1245790	1	8	37	67	93	0.3
	1245968	15	23	49	75	93	0.2
	1245976	0	16	49	73	88	0.2

TABLE 28-continued

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% Inhibition					IC50 (μ M)
	6 nM	32 nM	160 nM	800 nM	4000 nM	
1245977	0	10	41	75	77	0.3
1245999	1	14	48	82	90	0.2
1246022	13	24	56	83	93	0.1
1246023	13	24	66	90	93	0.1
1246104	0	0	35	67	87	0.4
1246157	0	22	40	75	84	0.3
1246226	11	7	40	71	88	0.3
1246667	89	95	97	98	100	<0.006
1246853	0	2	11	57	82	0.7
1246861	4	0	12	31	67	2.6

Example 3: Antisense Inhibition of Human HSD17B13 in HepaRG Cells by cEt Gapmers

Additional modified oligonucleotides complementary to an HSD17B13 nucleic acid were synthesized and tested for their effect on HSD17B13 RNA levels in vitro. The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below.

Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with 1,000 nM of modified oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and HSD17B13 RNA levels were measured by quantitative real-time RTPCR. Human primer probe set RTS43553 was

used to measure RNA levels. HSD17B13 RNA levels were adjusted according to total RNA content, as measured by RIBOGREEN® S. Results are presented as percent inhibition of HSD17B13 relative to untreated control cells. As used herein, a value of '0' indicates that treatment with the modified oligonucleotide did not inhibit HSD17B13 mRNA levels. An asterisk (*) indicates that the modified oligonucleotide is complementary to the target transcript within the amplicon region of the primer probe set and so, the associated data is not reliable. In such instances, additional using alternate primer probe sets must be performed to accurately assess the potency and efficacy of such modified oligonucleotides.

The modified oligonucleotides described in the tables below were 3-10-3 cEt gapmers. The internucleoside linkages throughout each modified oligonucleotide were phosphorothioate (P=S) linkages. All cytosine residues throughout each modified oligonucleotide were 5-methylcytosines.

"Start site" indicates the 5'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. "Stop site" indicates the 3'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. As shown in the Tables below, the modified oligonucleotide are complementary to either the human HSD17B13 mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_178135.4) or to the human HSD17B13 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000) or to both. 'N/A' indicates that the modified oligonucleotide is not complementary to that particular target sequence with 100% complementarity.

TABLE 29

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	% Inhibition	SEQ ID NO	
	Start Site	Stop Site	Start Site	Stop Site				
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	91	121	
1246023	N/A	N/A	5046	5061	GAGTGAATCATTTCAGT	89	1526	
1340071	N/A	N/A	20191	20206	CAGTTATTAGAAGTCA	85	2036	
1340085	N/A	N/A	20588	20603	GAGTTATCTGGTTTGC	84	2037	
1340086	N/A	N/A	9335	9350	TCGATCCTATATACAT	40	2038	
1340096	N/A	N/A	19711	19726	GCTGAAACTGCTGATT	38	2039	
1340099	1763	1778	21578	21593	ACAGTTCCTTTTCCTG	58	2040	
1340104	N/A	N/A	10500	10515	TACTTACCAATATGGG	13	2041	
1340120	N/A	N/A	11585	11600	TAAGTGAATTATTCTC	41	2042	
1340158	N/A	N/A	6440	6455	CTAAACTAGTGGTAAT	23	2043	
1340170	N/A	N/A	9657	9672	TAGGTAGCTAATGTAT	34	2044	
1340194	N/A	N/A	11077	11092	TAATTACAACCTGGTT	52	2045	
1340201	N/A	N/A	14279	14294	GTAAGGCTGGGTGAG	41	2046	
1340204	N/A	N/A	12202	12217	TCATACCACATACCCA	41	2047	
1340209	N/A	N/A	18520	18535	CTTAGTACCAAGACAC	60	2048	
1340231	N/A	N/A	16134	16149	GATGCTCCATAATAAT	42	2049	

TABLE 29-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340239	N/A	N/A	19912	19927	TGCATCTTAAGATACC	81	2050
1340247	N/A	N/A	16286	16301	AGAATAGTCTTCAGCA	80	2051
1340256	N/A	N/A	19532	19547	GGCTAAAATGGTCATC	75	2052
1340276	N/A	N/A	19394	19409	TAAGAGATGAGTAGGC	46	2053
1340278	N/A	N/A	15250	15265	TCTATAGCTAAGTATA	0	2054
1340279	N/A	N/A	17802	17817	TATTCTTATGTCATCC	75	2055
1340290	809	824	15690	15705	ATTGGTAAGTATTCCA	50	2056
1340305	1177	1192	20992	21007	TTTAAGAGGCATGAAA	40	2057
1340308	N/A	N/A	20111	20126	TCAAGACATTCTAGCC	79	2058
1340330	N/A	N/A	12476	12491	CTAGATACTCAATTTA	30	2059
1340360	N/A	N/A	12800	12815	GGCTGTAAGAGTCAGT	88	2060
1340368	N/A	N/A	13827	13842	CCTACCAAGAGGGTTT	34	2061
1340380	N/A	N/A	12005	12020	ATTGTTAGCTAAGGGA	22	2062
1340390	N/A	N/A	14175	14190	GCCTTGAATAGCAAGC	32	2063
1340405	N/A	N/A	13598	13613	ACAGGATAGAGCACAT	44	2064
1340412	N/A	N/A	14914	14929	ACATTAGCAAGCTAAG	52	2065
1340423	N/A	N/A	9538	9553	ATCCAGATCTGCCCTA	39	2066
1340427	N/A	N/A	8535	8550	TTGCAAGTTTATCAGT	66	2067
1340432	N/A	N/A	4225	4240	GTTTCAACTAAACATG	65	2068
1340438	N/A	N/A	3479	3494	GGACCAGGGAATTTAT	55	2069
1340439	1255	1270	21070	21085	CTTTTGTCCACCTTTA	76	2070
1340448	N/A	N/A	5672	5687	GTCAGTAGAGAGCATA	85	2071
1340453	N/A	N/A	6084	6099	TTTCACCTCAGGTGAC	52	2072
1340459	N/A	N/A	5083	5098	CTCGCCTAAAGGAGAT	56	2073
1340465	N/A	N/A	5895	5910	TCCTTTGTATTTGCT	68	2074
1340479	N/A	N/A	6785	6800	CCTCCTTATTTGTTAG	41	2075
1340491	N/A	N/A	3699	3714	TTTCAGATCCCGTTCT	45	2076
1340525	N/A	N/A	9965	9980	AGTGTAAGCTGAGAGT	71	2077
1340542	N/A	N/A	5186	5201	CCAACGCAACAGTTTA	77	2078
1340546	N/A	N/A	5647	5662	AGCATTTCATCAGATGT	83	2079
1340556	N/A	N/A	5268	5283	TTGCAAAATGTGATGC	75	2080
1340563	N/A	N/A	3990	4005	GTGTTTACAAGTAAGA	79	2081
1340567	N/A	N/A	6260	6275	AGATGGGCAAGGCCAC	59	2082
1340576	N/A	N/A	11499	11514	CTGGAGAAGAGTTTAC	38	2083
1340580	N/A	N/A	8641	8656	AGACAGACTGTTGAGC	74	2084
1340597	N/A	N/A	18036	18051	CATAGTTTATATGGAT	56	2085

TABLE 29-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340601	N/A	N/A	18138	18153	TGTTATCTCAAGTCAG	81	2086
1340613	N/A	N/A	5832	5847	ATGTTAGGACCCAGTC	60	2087
1340661	N/A	N/A	9201	9216	TACACTGACAACCCCTT	56	2088
1340665	N/A	N/A	14601	14616	GAGAAATCTAAGAACC	42	2089
1340667	N/A	N/A	9936	9951	AATGAATCATGATTGA	33	2090
1340669	N/A	N/A	10000	10015	TGAAATCTTGTGTAAC	43	2091
1340684	N/A	N/A	8125	8140	AGTGCTTAGTTCATTG	84	2092
1340685	N/A	N/A	16385	16400	CAATCCCAACCAAACT	49	2093
1340689	N/A	N/A	3600	3615	CTTGGAAGGAGACTGG	60	2094
1340697	N/A	N/A	4661	4676	CCGCCCTTAAGTCATT	53	2095
1340704	N/A	N/A	6329	6344	ACTTTCATAGGGAGAC	80	2096
1340705	74	89	3168	3183	AAGGATTCTAGGATG	71	2097
1340722	N/A	N/A	5737	5752	TTAGTTGGTAGCTTGC	72	2098
1340767	N/A	N/A	9441	9456	TCCCACAAAATAACC	24	2099
1340773	N/A	N/A	9773	9788	TAGAACTCCCAACCCA	44	2100
1340776	N/A	N/A	6732	6747	TTACCCCTGGCTTTTC	26	2101
1340785	N/A	N/A	7781	7796	ACTACTTCAGTTAGCA	79	2102
1340792	N/A	N/A	9115	9130	GACAGACCAAGTAGCT	39	2103
1340812	N/A	N/A	10264	10279	CTTCCAAGCATTCCAT	57	2104
1340813	N/A	N/A	7526	7541	CCGAAAAAAGTGGAGG	31	2105
1340836	N/A	N/A	18694	18709	AAGACTTTGAGACTCT	62	2106
1340850	N/A	N/A	8744	8759	CACTTTCAGTGGGTGT	7	2107
1340852	1424	1439	21239	21254	TCCAGGTTGAGATAAA	70	2108
1340872	1614	1629	21429	21444	TCACCTTTCATAATGT	63	2109
1340878	N/A	N/A	12627	12642	TTGTTAAGCAGATGGC	84	2110
1340889	N/A	N/A	6597	6612	CTACTTTCAAACCTTG	57	2111
1340903	N/A	N/A	20390	20405	TGCTATAGTAATCACA	65	2112

TABLE 30

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	84	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	89	1526
1340078	N/A	N/A	16295	16310	CCTTTAGTTAGAATAG	37	2113

TABLE 30-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340094	N/A	N/A	19547	19562	ATTGTTTAGCTGATAG	61	2114
1340113	N/A	N/A	12667	12682	TGTTAACCTGCAGCAG	39	2115
1340122	N/A	N/A	14192	14207	TGATTGAGTTCTCCAC	48	2116
1340126	N/A	N/A	19779	19794	ACATACCACTCTTCTC	35	2117
1340130	N/A	N/A	17954	17969	TTAGAAACTGCTCCTC	48	2118
1340146	N/A	N/A	8132	8147	TTGTTAGAGTGCTTAG	76	2119
1340149	N/A	N/A	11545	11560	CTTGACAATGGTTGCC	58	2120
1340152	N/A	N/A	10138	10153	ATAGATCCTTTTTTGG	6	2121
1340154	N/A	N/A	6284	6299	ACATTCTTCAGGTGTT	88	2122
1340156	N/A	N/A	20194	20209	TTGCAGTTATTAGAAG	13	2123
1340179	N/A	N/A	14315	14330	TAAGATGTCCAGGCAT	17	2124
1340197	N/A	N/A	10554	10569	GAAAGTGAATGCTCCC	71	2125
1340220	N/A	N/A	14611	14626	TATTTCTTGCGAGAAA	30	2126
1340241	N/A	N/A	16220	16235	TTTATAGACTGGGTAG	59	2127
1340255	1783	1798	21598	21613	TTTCTTATGTAATAGC	76	2128
1340260	1714	1729	21529	21544	AAACATCTCTGGGACC	76	2129
1340277	N/A	N/A	19914	19929	AGTGCATCTTAAGATA	57	2130
1340291	N/A	N/A	18540	18555	GAAACTCACTTCCTAC	56	2131
1340297	N/A	N/A	14917	14932	AATACATTAGCAAGCT	45	2132
1340304	697	712	12115	12130	AGACATGAGGTTTGA	59	2133
1340310	N/A	N/A	13638	13653	GACAATGTGCAGCTCT	67	2134
1340321	N/A	N/A	6606	6621	TAAAGCAGGCTACTTT	9	2135
1340323	N/A	N/A	19396	19411	CATAAGAGATGAGTAG	44	2136
1340342	N/A	N/A	5279	5294	GAGCTAGACAATTGCA	73	2137
1340350	N/A	N/A	9716	9731	TAGTGAACATACATTG	23	2138
1340361	N/A	N/A	5935	5950	GCTGCTATTTAGGGCA	53	2139
1340364	N/A	N/A	5678	5693	TTCTAGGTCAGTAGAG	58	2140
1340369	1433	1448	21248	21263	TAAATATGTCCAGGT	75	2141
1340385	N/A	N/A	4389	4404	AGACCAAAAAGATGGC	80	2142
1340395	N/A	N/A	11794	11809	AACTCTATTGATTGGC	72	2143
1340416	N/A	N/A	12496	12511	AATTCACCTTGACTAA	9	2144
1340419	N/A	N/A	12410	12425	ATCAAATGTATGTGCT	73	2145
1340443	N/A	N/A	11117	11132	CTACAGAGGAGTTTGC	27	2146
1340445	N/A	N/A	5847	5862	ACTCCCCAAACATGGA	48	2147
1340458	N/A	N/A	7834	7849	GCTGAGTGGATTACAT	44	2148
1340469	N/A	N/A	6456	6471	GTGCTAAGTAGTAATT	58	2149

TABLE 30-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340476	815	830	15696	15711	TTTCTTATTGGTAAGT	44	2150
1340512	N/A	N/A	6736	6751	GAGATTACCCCTGGCT	72	2151
1340522	N/A	N/A	6915	6930	ACTCGGGAAGTTTAGA	26	2152
1340533	N/A	N/A	9456	9471	GTGATATTTCATCT	55	2153
1340543	N/A	N/A	20115	20130	AGCATCAAGACATTCT	79	2154
1340549	N/A	N/A	9968	9983	AGTAGTGAAGCTGAG	88	2155
1340552	N/A	N/A	12802	12817	CAGGCTGTAAGAGTCA	56	2156
1340555	N/A	N/A	5158	5173	GCCAACTGCTACTGT	73	2157
1340558	N/A	N/A	15423	15438	ATATCCAGTAGGTGTG	55	2158
1340606	N/A	N/A	18046	18061	CAAGTTTGTCATAGT	78	2159
1340614	1311	1326	21126	21141	TCTTAAACCTTCCCTG	21	2160
1340651	N/A	N/A	9344	9359	AAAGTGCTCCTCGATCC	36	2161
1340671	N/A	N/A	10293	10308	TGAGAGAACTTATACA	32	2162
1340673	N/A	N/A	5191	5206	TATCCCAACGCAACA	62	2163
1340674	N/A	N/A	4663	4678	ACCCGCCCTTAAGTCA	48	2164
1340679	N/A	N/A	5758	5773	ATAGGAGAGTCTTCA	77	2165
1340681	N/A	N/A	8547	8562	GCACATCATGTTTTCG	44	2166
1340683	N/A	N/A	9947	9962	TACACCACACTAATGA	26	2167
1340696	N/A	N/A	8656	8671	CTCAGAGTTCAGGCAA	67	2168
1340713	N/A	N/A	6191	6206	GACAGAGCAATTTACT	55	2169
1340734	N/A	N/A	8964	8979	ACTCCGTATAAGTTT	42	2170
1340747	218	233	3312	3327	AAATTCTAAGTAGTC	39	2171
1340757	N/A	N/A	18143	18158	CCTTATGTTATCTCAA	67	2172
1340761	N/A	N/A	4068	4083	CCTTACCAGAATTAC	63	2173
1340762	N/A	N/A	5651	5666	CCAGAGCATTTCATCAG	83	2174
1340768	N/A	N/A	9210	9225	GAAAAGATGTACACTG	56	2175
1340770	N/A	N/A	3505	3520	CTAACCTGACACATAT	18	2176
1340771	1179	1194	20994	21009	TTTTTAAGAGGCATGA	60	2177
1340775	N/A	N/A	9593	9608	AAGAGCTGGTAAAGGT	47	2178
1340798	N/A	N/A	3619	3634	GACTTGCCTCATTTAG	69	2179
1340802	N/A	N/A	9793	9808	CCTATGCAAATTCATA	34	2180
1340803	N/A	N/A	6333	6348	AAGGACTTTCATAGGG	71	2181
1340824	N/A	N/A	20421	20436	CAGTAAAATTATGCCT	38	2182
1340831	N/A	N/A	3702	3717	AATTTTCAGATCCCGT	74	2183
1340833	N/A	N/A	9117	9132	AAGACAGACCAAGTAG	16	2184
1340837	N/A	N/A	13955	13970	ACCTCTAAGTTAGCCC	28	2185

TABLE 30-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	% Inhibition	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1340845	N/A	N/A	18697	18712	CAAAAGACTTTGAGAC	22	2186
1340863	N/A	N/A	20598	20613	AATACGCTGAGAGTTA	48	2187
1340874	N/A	N/A	7528	7543	TTCCGAAAAAGTGGGA	13	2188
1340891	N/A	N/A	16895	16910	CGTCAAATAGGGCTGG	53	2189

TABLE 31

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	% Inhibition	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	91	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	83	1526
1340082	N/A	N/A	16329	16344	CTTCACTATATTGCAT	50	2190
1340083	N/A	N/A	20116	20131	CAGCATCAAGACATTC	72	2191
1340092	N/A	N/A	16221	16236	TTTTATAGACTGGGTA	53	2192
1340098	N/A	N/A	19818	19833	ATAGTTCAAATTGGAT	48	2193
1340125	N/A	N/A	15778	15793	TGTTTTAGTATTGGG	48	2194
1340131	N/A	N/A	18720	18735	CACCATGACAGCTTCA	65	2195
1340141	N/A	N/A	11213	11228	CTTCTGTAGGTTTTC	60	2196
1340142	N/A	N/A	17984	17999	TAGTTATCTTCTCACT	52	2197
1340148	2240	2255	22055	22070	ATGCAAAAGCATTCTA	50	2198
1340164	N/A	N/A	11546	11561	TCTTGACAATGGTTGC	77	2199
1340183	N/A	N/A	11795	11810	TAACTCTATTGATTGG	51	2200
1340187	N/A	N/A	20628	20643	GACAACAGAGTTCTGT	9	2201
1340192	N/A	N/A	12498	12513	GAAATTCACCTTGACT	41	2202
1340206	1330	1345	21145	21160	TGGAATGCTACTTGAA	85	2203
1340208	N/A	N/A	9969	9984	GAGTAGTGTAAGCTGA	80	2204
1340215	N/A	N/A	12416	12431	ACGGTGATCAATGTA	53	2205
1340221	N/A	N/A	16900	16915	AAAACCGTCAAATAGG	28	2206
1340230	N/A	N/A	18658	18673	TATTCTCCAACCTCAGG	54	2207
1340235	N/A	N/A	18145	18160	CTCCTTATGTTATCTC	57	2208
1340245	N/A	N/A	15446	15461	TGTTATCAGAACTTA	44	2209
1340270	N/A	N/A	7550	7565	GCTGTAATTAGGAAGA	24	2210
1340274	N/A	N/A	10584	10599	AGAACCAGGACTCTCC	67	2211
1340275	219	234	3313	3328	CAAATTCATAAGTAGT	76	2212

TABLE 31-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340284	1467	1482	21282	21297	AAGAGGCTAGGGAAAT	66	2213
1340292	N/A	N/A	19417	19432	GACAACTGATGTCCAG	12	2214
1340312	N/A	N/A	20423	20438	GACAGTAAAATTATGC	57	2215
1340314	N/A	N/A	18075	18090	TTGTACCCTGTTCTCA	64	2216
1340320	N/A	N/A	14318	14333	AATTAAGATGTCCAGG	13	2217
1340324	N/A	N/A	9950	9965	TTCTACACCACACTAA	12	2218
1340329	1181	1196	20996	21011	AGTTTTTAAGAGGCAT	75	2219
1340337	N/A	N/A	13659	13674	TCACAAAGTCCCTGAC	11	2220
1340353	N/A	N/A	7125	7140	CATGAAGATTAGTTAA	11	2221
1340355	N/A	N/A	8657	8672	TCTCAGAGTTCAGGCA	54	2222
1340356	N/A	N/A	5680	5695	GTTTCTAGGTCAGTAG	73	2223
1340366	N/A	N/A	6457	6472	AGTGCTAAGTAGTAAT	71	2224
1340367	N/A	N/A	5870	5885	TAAACCCTGGAGCAGC	66	2225
1340370	N/A	N/A	5159	5174	AGCCAAACTGCTACTG	78	2226
1340426	N/A	N/A	9597	9612	TGTGAAGAGCTGGTAA	41	2227
1340472	N/A	N/A	6335	6350	AGAAGGACTTTCATAG	46	2228
1340475	N/A	N/A	8136	8151	AAGTTTGTAGAGTGC	74	2229
1340484	N/A	N/A	12709	12724	CCAGTAAAAAATTGGT	23	2230
1340503	N/A	N/A	14613	14628	ATTATTTCTTGCGAGA	60	2231
1340514	N/A	N/A	9345	9360	AAAAGTGTCTCGATC	17	2232
1340515	N/A	N/A	5762	5777	AAGAATAGGAGAGTCT	70	2233
1340539	N/A	N/A	3721	3736	GACTTTTGTGTTGTAGC	74	2234
1340560	N/A	N/A	6285	6300	TACATTCTTCAGGTGT	61	2235
1340577	N/A	N/A	5280	5295	TGAGCTAGACAATTGC	78	2236
1340603	699	714	12117	12132	AGAGACATGAGGTTTT	77	2237
1340604	N/A	N/A	20195	20210	ATTGCAGTTATTAGAA	42	2238
1340617	N/A	N/A	3516	3531	CCTTCATCTAACTAAC	30	2239
1340636	N/A	N/A	12804	12819	CTCAGGCTGTAAGAGT	21	2240
1340647	N/A	N/A	5195	5210	ACCGTATTCCCAACGC	90	2241
1340659	N/A	N/A	8965	8980	AACTCCGTTATAAGTT	11	2242
1340664	N/A	N/A	3676	3691	CCCCTAAGTTATTATC	33	2243
1340690	N/A	N/A	9726	9741	TGTGAGAGGATAGTGA	50	2244
1340724	N/A	N/A	4862	4877	AACTGAAACGATCCCTC	52	2245
1340725	N/A	N/A	9466	9481	GGCAGTTGAAGTTGAT	46	2246
1340727	N/A	N/A	8589	8604	GTGGGATAAACTGTTTC	52	2247
1340746	N/A	N/A	19564	19579	TAGCTATTGTCTTTGA	63	2248

TABLE 31-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340781	N/A	N/A	13956	13971	TACCTCTAAGTTAGCC	28	2249
1340789	N/A	N/A	14193	14208	GTGATTGAGTTCTCCA	80	2250
1340790	N/A	N/A	9118	9133	CAAGACAGACCAAGTA	27	2251
1340830	N/A	N/A	7835	7850	AGCTGAGTGGATTACA	29	2252
1340834	N/A	N/A	6203	6218	ATAGGACATGGAGACA	82	2253
1340835	N/A	N/A	5653	5668	TACCAGAGCATTATC	55	2254
1340838	N/A	N/A	10294	10309	TTGAGAGAACTTATAC	21	2255
1340839	N/A	N/A	19915	19930	GAGTGCATCTTAAGAT	43	2256
1340841	N/A	N/A	4069	4084	GCCTTACCAGAATTTA	75	2257
1340864	N/A	N/A	9796	9811	AATCCTATGCAAATTC	42	2258
1340865	1716	1731	21531	21546	CTAAACATCTCTGGGA	62	2259
1340870	N/A	N/A	4402	4417	CGAATTTCTTCAAAGA	55	2260
1340875	N/A	N/A	5974	5989	AAAGCCCTATTCTTCG	65	2261
1340876	N/A	N/A	6626	6641	TGTACAGTAGGCAGTT	75	2262
1340883	N/A	N/A	9224	9239	CAAGTGACAGCATTGA	48	2263
1340884	N/A	N/A	6738	6753	ATGAGATTACCCCTGG	62	2264
1340885	N/A	N/A	10184	10199	AACACTAATCTCAGTA	15	2265
1340892	N/A	N/A	14918	14933	TAATACATTAGCAAGC	33	2266

TABLE 32

Inhibition of HSD17B13 RNA by 3-10-3 cEt qapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	59	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTGAGT	78	1526
1340075	N/A	N/A	13957	13972	CTACCTCTAAGTTAGC	6	2267
1340077	1468	1483	21283	21298	GAAGAGGCTAGGGAAA	44	2268
1340093	N/A	N/A	18744	18759	TATACCTGTTGCCTAG	11	2269
1340108	N/A	N/A	10848	10863	CATATGGTGACTACTG	15	2270
1340129	N/A	N/A	16904	16919	GTCAAAAACCGTCAAA	39	2271
1340144	254	269	3348	3363	AATATCCACAGAACC	60	2272
1340151	N/A	N/A	11796	11811	ATAACTCTATTGATTG	39	2273
1340157	N/A	N/A	9147	9162	TGCCAAGCTTTGTGTC	38	2274
1340163	N/A	N/A	14617	14632	CCGAATTATTTCTTGC	60	2275
1340166	N/A	N/A	4412	4427	AAAGTTCATTGCAATT	63	2276

TABLE 32-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340168	N/A	N/A	18086	18101	ACCAATGCAGTTTGTA	75	2277
1340169	N/A	N/A	6460	6475	TAAAGTGCTAAGTAGT	50	2278
1340171	N/A	N/A	11553	11568	GTTTTTCTCTTGACAA	59	2279
1340176	N/A	N/A	16117	16132	GCTCTATTGGGCCAGG	47	2280
1340181	N/A	N/A	18176	18191	CGCGAGATGATGTTGC	53	2281
1340211	N/A	N/A	18661	18676	TACTATTCTCCAACCTC	47	2282
1340226	1182	1197	20997	21012	AAGTTTTTAAGAGGCA	77	2283
1340233	N/A	N/A	12499	12514	AGAAATTCACCTTGAC	39	2284
1340268	N/A	N/A	16224	16239	TCTTTTATAGACTGG	82	2285
1340273	N/A	N/A	11237	11252	GACATAGACATGTGTT	26	2286
1340283	1717	1732	21532	21547	TCTAAACATCTCTGGG	40	2287
1340286	N/A	N/A	19846	19861	AACCTATGTGTGCACC	73	2288
1340327	N/A	N/A	12418	12433	TAACGGTGATCAAATG	17	2289
1340332	N/A	N/A	10295	10310	TTTGAGAGAACTTATA	0	2290
1340334	N/A	N/A	8973	8988	AGTATGTGAACCTCCGT	81	2291
1340335	N/A	N/A	5654	5669	TTACCAGAGCATTTCAT	50	2292
1340352	N/A	N/A	9970	9985	TGAGTAGTGTAAGCTG	62	2293
1340375	N/A	N/A	19444	19459	TGGGATAGTGGAGGAA	13	2294
1340377	N/A	N/A	17986	18001	TATAGTTATCTTCTCA	46	2295
1340386	N/A	N/A	14919	14934	ATAATACATTAGCAAG	7	2296
1340387	N/A	N/A	5873	5888	GGCTAAACCCTGGAGC	36	2297
1340388	N/A	N/A	8164	8179	TATTGTAGGGATTGAA	40	2298
1340397	N/A	N/A	3741	3756	CTGTTGCAAGTTCTTC	85	2299
1340398	N/A	N/A	13688	13703	TTTACTCACTTCTGGT	30	2300
1340415	N/A	N/A	12815	12830	CACATAGCTAACTCAG	42	2301
1340430	N/A	N/A	5197	5212	GAACCGTATTCCTAAC	72	2302
1340433	N/A	N/A	9467	9482	AGGCAGTTGAAGTTGA	47	2303
1340446	N/A	N/A	15491	15506	TTAATGCCACCCTACC	19	2304
1340450	N/A	N/A	5712	5727	GATTGCACAAACCACA	73	2305
1340451	N/A	N/A	6301	6316	CTAGGAGATATAACAT	40	2306
1340452	N/A	N/A	5978	5993	CTGAAAAGCCCTATTC	11	2307
1340454	N/A	N/A	7680	7695	AATTTGTGAACCTGCA	20	2308
1340468	N/A	N/A	20197	20212	ACATTGCAGTTATTAG	42	2309
1340471	N/A	N/A	14385	14400	GCTTAGTGTAAGATTG	16	2310
1340478	N/A	N/A	6740	6755	CAATGAGATTACCCCT	16	2311
1340489	N/A	N/A	6388	6403	GATGATTCTGACTCAT	55	2312

TABLE 32-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	% Inhibition	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1340498	N/A	N/A	20470	20485	TCATTAGGAAAAGTGT	16	2313
1340521	N/A	N/A	8593	8608	AAAGGTGGGATAAACT	33	2314
1340534	N/A	N/A	10187	10202	GCCAACTAATCTCA	40	2315
1340545	N/A	N/A	20117	20132	ACAGCATCAAGACATT	53	2316
1340547	N/A	N/A	4864	4879	TGAACTGAAACGATCC	58	2317
1340554	N/A	N/A	9225	9240	ACAAGTGACAGCATTG	44	2318
1340566	700	715	12118	12133	CAGAGACATGAGGTTT	61	2319
1340574	N/A	N/A	9599	9614	TATGTGAAGAGCTGGT	63	2320
1340578	N/A	N/A	3678	3693	ATCCCCTAAGTTATTA	13	2321
1340585	N/A	N/A	5782	5797	GGATTGCAGAAGAGTA	76	2322
1340610	N/A	N/A	20629	20644	TGACAACAGAGTTCTG	11	2323
1340628	N/A	N/A	19585	19600	TAACTAAAGTACTTG	1	2324
1340631	N/A	N/A	4071	4086	TGGCCTTACCAGAATT	46	2325
1340634	N/A	N/A	7127	7142	GACATGAAGATTAGTT	3	2326
1340638	N/A	N/A	9727	9742	CTGTGAGAGGATAGTG	33	2327
1340645	1352	1367	21167	21182	ATTCTGTGGCATGGCT	69	2328
1340668	2242	2257	22057	22072	CCATGCAAAAGCATTC	72	2329
1340701	N/A	N/A	3544	3559	AATCGCTGCTTAGTTC	41	2330
1340740	N/A	N/A	19999	20014	CAGTTTTCTCATGAT	65	2331
1340777	N/A	N/A	14195	14210	ATGTGATTGAGTTCTC	36	2332
1340783	N/A	N/A	9915	9930	CTATCTCCCTATTTGT	15	2333
1340821	N/A	N/A	7942	7957	GATCCATATTCCTGA	74	2334
1340822	N/A	N/A	5281	5296	CTGAGCTAGACAATTG	56	2335
1340832	N/A	N/A	8668	8683	TGTATGAGGTCTCTCA	66	2336
1340873	N/A	N/A	6221	6236	CCAAACTTGCCATAA	45	2337
1340879	N/A	N/A	5161	5176	AGAGCCAACTGCTAC	55	2338
1340894	N/A	N/A	9372	9387	TAGAACACTTGCCTCA	31	2339
1340895	N/A	N/A	9953	9968	GAGTTCTACACCACAC	76	2340
1340901	N/A	N/A	16332	16347	CTACTTCACTATATTG	33	2341
1340902	N/A	N/A	6628	6643	GCTGTACAGTAGGCAG	66	2342
1340910	N/A	N/A	12722	12737	GATTAATTCCTTCCA	30	2343

TABLE 33

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	85	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCACT	79	1526
1340100	N/A	N/A	19467	19482	CTGGAAGGAGGATTCA	39	2344
1340117	N/A	N/A	12419	12434	GTAACGGTGATCAAAT	34	2345
1340119	N/A	N/A	9971	9986	TTGAGTAGTGAAGCT	52	2346
1340123	N/A	N/A	9602	9617	ACATATGTGAAGAGCT	66	2347
1340127	1185	1200	21000	21015	CAGAAGTTTTTAAGAG	30	2348
1340135	N/A	N/A	19847	19862	GAACCTATGTGTGCAC	53	2349
1340145	N/A	N/A	6391	6406	CTTGATGATCTGACT	57	2350
1340189	N/A	N/A	20479	20494	TAGTGGACTTCATTAG	52	2351
1340190	N/A	N/A	14921	14936	CTATAATACATTAGCA	3	2352
1340193	N/A	N/A	14618	14633	CCCGAATTATTTCTTG	34	2353
1340196	N/A	N/A	16225	16240	CTCTTTTATAGACTG	63	2354
1340205	N/A	N/A	19586	19601	GTAAACTAAAGTACTT	0	2355
1340212	N/A	N/A	10930	10945	ATAAGGCCGATTCATG	38	2356
1340218	N/A	N/A	20001	20016	ATCAGTTTTCCTCATG	48	2357
1340228	N/A	N/A	20361	20376	CTAAATTATTGCCTGA	39	2358
1340229*	N/A	N/A	20685	20700	GAAGAAACCTGTACAA	85	2359
1340234	N/A	N/A	16337	16352	ATTAACACTTCACTA	14	2360
1340238	N/A	N/A	11855	11870	TAAGCAGAATTGTGAA	35	2361
1340244	N/A	N/A	16968	16983	CCAATATATACTGACA	68	2362
1340246	N/A	N/A	13959	13974	GTCTACCTCTAAGTTA	3	2363
1340258	N/A	N/A	11570	11585	CTTGACAATGGTTGAT	54	2364
1340266	N/A	N/A	18748	18763	AATATATACCTGTTGC	38	2365
1340271	N/A	N/A	17990	18005	TTAGTATAGTTATCTT	37	2366
1340309	N/A	N/A	14196	14211	TATGTGATTGAGTTCT	28	2367
1340311	N/A	N/A	18178	18193	CTCGCGAGATGATGTT	40	2368
1340339	N/A	N/A	6467	6482	TAGGCAGTAAAGTGCT	48	2369
1340345	N/A	N/A	18092	18107	TGATACACCAATGCAG	44	2370
1340348	N/A	N/A	5283	5298	AACTGAGCTAGACAAT	44	2371
1340349	N/A	N/A	6225	6240	TGTACCAAACCTTGCCC	53	2372
1340354	N/A	N/A	9148	9163	ATGCCAAGCTTTGTGT	50	2373
1340357	N/A	N/A	3375	3390	GTTGGAAGATGTATAC	53	2374
1340402	N/A	N/A	9958	9973	GCTGAGAGTTCTACAC	45	2375
1340406	N/A	N/A	8594	8609	TAAAGGTGGGATAAAC	0	2376
1340428	N/A	N/A	9921	9936	AATTCATCTCTCCCT	10	2377

TABLE 33-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340440	1360	1375	21175	21190	TTGTTGATATTCTGTG	52	2378
1340449	N/A	N/A	5198	5213	TGAACCGTATTCCCAA	76	2379
1340466	N/A	N/A	6638	6653	TTGGTTGGAGGCTGTA	60	2380
1340492	N/A	N/A	15497	15512	GCTGCATTAATGCCAC	46	2381
1340501	N/A	N/A	6742	6757	TGCAATGAGATTACCC	54	2382
1340504	N/A	N/A	12500	12515	CAGAAATTCACCTTGA	28	2383
1340505	N/A	N/A	6302	6317	TCTAGGAGATATAACA	48	2384
1340506	N/A	N/A	12758	12773	GAGAATTGTTTAGTTC	47	2385
1340548	N/A	N/A	5715	5730	CCAGATTGCACAAACC	72	2386
1340553	N/A	N/A	10188	10203	TGCCAACACTAATCTC	30	2387
1340571	N/A	N/A	9498	9513	TGGAAGATTAATCATA	12	2388
1340582	2243	2258	22058	22073	TCCATGCAAAGCATT	62	2389
1340626	N/A	N/A	5657	5672	AAGTTACCAGAGCATT	70	2390
1340653	N/A	N/A	5984	5999	GGATACCTGAAAAGCC	68	2391
1340655	701	716	12119	12134	GCAGAGACATGAGGTT	55	2392
1340657	N/A	N/A	4076	4091	TTTCATGGCCTTACCA	52	2393
1340678	N/A	N/A	7130	7145	TTGGACATGAAGATTA	18	2394
1340692	N/A	N/A	3756	3771	AGGCACTAAGGTTTC	61	2395
1340693	N/A	N/A	4476	4491	GGACAAATGCCTGAGA	52	2396
1340695	N/A	N/A	13689	13704	GTTTACTCACTTCTGG	29	2397
1340716	N/A	N/A	8168	8183	GAGTTATTGTAGGGAT	69	2398
1340723	N/A	N/A	8669	8684	TTGTATGAGGTCTCTC	50	2399
1340750	N/A	N/A	7682	7697	GAAATTTGTGAACCTG	32	2400
1340759	N/A	N/A	9755	9770	CTACCTCCAAATCCCC	27	2401
1340763	N/A	N/A	7947	7962	GAAGTGATCCATATTC	44	2402
1340786	N/A	N/A	12890	12905	TCTTTAGTCAACAGTA	68	2403
1340788	N/A	N/A	18667	18682	CTGGGATACTATTCTC	57	2404
1340800	N/A	N/A	14388	14403	TAGGCTTAGTGTAGAA	38	2405
1340805	N/A	N/A	9319	9334	CCAAGTAATTACTTCT	68	2406
1340807	1471	1486	21286	21301	AAGGAAGAGGCTAGGG	57	2407
1340808	N/A	N/A	5804	5819	TCCCCTCAAATAGCCT	60	2408
1340823	N/A	N/A	20119	20134	TTACAGCATCAAGACA	46	2409
1340842	N/A	N/A	16124	16139	AATAATAGCTCTATTG	0	2410
1340846	N/A	N/A	8981	8996	TGTCCAGAAGTATGTG	0	2411
1340857	N/A	N/A	5162	5177	AAGAGCCAACTGCTA	63	2412
1340859	N/A	N/A	3568	3583	TCAACCTGCACACCAT	72	2413

TABLE 33-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340893	N/A	N/A	5876	5891	GTAGGCTAAACCCTGG	62	2414
1340896	N/A	N/A	11290	11305	ATGTTTTGATCCAGGG	46	2415
1340898	1720	1735	21535	21550	TTGTCTAAACATCTCT	60	2416
1340899	N/A	N/A	3679	3694	CATCCCCTAAGTTATT	24	2417
1340904	N/A	N/A	9381	9396	CCATGGATCTAGAACA	15	2418
1340905	N/A	N/A	10345	10360	CGTTAGGTTTCCAAAT	0	2419
1340908	N/A	N/A	5048	5063	TAGAGTGAATCATTCA	75	2420

TABLE 34

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	88	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	86	1526
1340080	N/A	N/A	10970	10985	TAATCTTGAGGCAGGA	17	2421
1340088	N/A	N/A	12420	12435	GGTAACGGTGATCAAA	51	2422
1340111	N/A	N/A	9519	9534	TCTACACTAATATTGA	21	2423
1340121	N/A	N/A	14619	14634	CCCCGAATTATTTCTT	35	2424
1340132	N/A	N/A	16379	16394	CAACCAAACCTCCAGC	72	2425
1340133	N/A	N/A	12502	12517	TGCAGAAATTCACCTT	54	2426
1340160	N/A	N/A	8675	8690	CTTCTATTGTATGAGG	40	2427
1340175	N/A	N/A	11866	11881	TCTTACATGATTAAGC	35	2428
1340198	N/A	N/A	15019	15034	TAAAATAGGGTGCAGT	58	2429
1340199	N/A	N/A	20120	20135	ATTACAGCATCAAGAC	63	2430
1340200	N/A	N/A	14197	14212	TTATGTGATTGAGTTC	34	2431
1340222	734	749	12152	12167	ATTTTGGTGAACCCA	8	2432
1340243	N/A	N/A	13690	13705	CGTTTACTCACTTCTG	21	2433
1340259	N/A	N/A	16126	16141	ATAATAATAGCTCTAT	1	2434
1340261	1197	1212	21012	21027	TTTATGTAAGCACAGA	70	2435
1340264	2247	2262	22062	22077	ATAGTCCATGCAAAAG	43	2436
1340269	1371	1386	21186	21201	ATTCTGTGTTCTTGTT	67	2437
1340280	N/A	N/A	17993	18008	AAGTTAGTATAGTTAT	20	2438
1340303	N/A	N/A	17014	17029	TAGCAATTAGCAAGA	54	2439
1340307	N/A	N/A	11571	11586	TCTTGACAATGGTTGA	75	2440
1340318	N/A	N/A	19607	19622	TATTATGCACCTCTATA	9	2441

TABLE 34-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340338	N/A	N/A	18672	18687	TTTACTGGGATACTA	27	2442
1340347	N/A	N/A	9924	9939	TTGAATTCACATATCTC	45	2443
1340358	N/A	N/A	5199	5214	TTGAACCGTATTCCCA	76	2444
1340363	N/A	N/A	9975	9990	ACAGTTGAGTAGTGTA	72	2445
1340396	N/A	N/A	20002	20017	AATCAGTTTTCCTCAT	53	2446
1340400	N/A	N/A	20483	20498	TTCATAGTGGACTTCA	51	2447
1340418	N/A	N/A	20378	20393	CACAAGTAAGGTAAAG	54	2448
1340425	N/A	N/A	6747	6762	CTAGTTGCAATGAGAT	65	2449
1340431	N/A	N/A	10199	10214	GGTAAGCCCCATGCCA	38	2450
1340437	N/A	N/A	11307	11322	CACAAGCACACTGTAA	2	2451
1340455	N/A	N/A	5716	5731	GCCAGATTGCACAAAC	60	2452
1340462	N/A	N/A	16262	16277	GCAGAGACAGTAGATT	63	2453
1340470	N/A	N/A	6228	6243	TTATGTACCAAACCTTG	53	2454
1340493	N/A	N/A	5985	6000	AGGATACCTGAAAAGC	81	2455
1340508	N/A	N/A	14392	14407	AACATAGGCTTAGTGT	18	2456
1340510	N/A	N/A	4103	4118	ATCCTGTAAACACTTC	79	2457
1340513	N/A	N/A	7949	7964	AGGAAGTGATCCATAT	60	2458
1340518	N/A	N/A	9004	9019	AGGCAAAGATCTGGCC	7	2459
1340519	N/A	N/A	9386	9401	GCACCCCATGGATCTA	35	2460
1340524	1544	1559	21359	21374	TTCAGAGTTATACAGA	66	2461
1340527	N/A	N/A	3680	3695	CCATCCCCTAAGTTAT	14	2462
1340544	1022	1037	20837	20852	CTTTGCAGCATTGATT	47	2463
1340561	N/A	N/A	3782	3797	GAAGTGTGTGCTTGT	77	2464
1340570	N/A	N/A	19887	19902	TACGACAGGTCATCTT	58	2465
1340572	N/A	N/A	6426	6441	ATGGTAAACTGTATGC	78	2466
1340575	N/A	N/A	9606	9621	ATTCACATATGTGAAG	0	2467
1340588	N/A	N/A	4481	4496	TTTTTGGACAAATGCC	50	2468
1340609	N/A	N/A	7683	7698	AGAAATTGTGAACCT	39	2469
1340620	N/A	N/A	18963	18978	ATGCAAGATTACTTC	38	2470
1340640	N/A	N/A	5049	5064	TTAGAGTGAATCATTC	69	2471
1340642	N/A	N/A	6308	6323	TGCTACTCTAGGAGAT	60	2472
1340648	N/A	N/A	5809	5824	CAGTATCCCCTCAAAT	53	2473
1340650	1722	1737	21537	21552	AATTGCTAAACATCT	62	2474
1340670	N/A	N/A	8170	8185	CAGAGTTATTGTAGGG	83	2475
1340680	N/A	N/A	5169	5184	ATTGCATAAGAGCCAA	67	2476
1340687	519	534	10405	10420	GAAGAAGTGCTTTTGT	50	2477

TABLE 34-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340691	N/A	N/A	9757	9772	AGCTACCTCCAAATTC	0	2478
1340700	N/A	N/A	14077	14092	AACAACCCAGCCTCGG	20	2479
1340711	N/A	N/A	9959	9974	AGCTGAGAGTTCTACA	67	2480
1340717	N/A	N/A	3569	3584	ATCAACCTGCACACCA	63	2481
1340742	N/A	N/A	19472	19487	TATAACTGGAAGGAGG	44	2482
1340744	N/A	N/A	12780	12795	TCAAATGAGGCGGCAC	46	2483
1340749	N/A	N/A	6710	6725	GGAATGCTCATATTAA	19	2484
1340753	N/A	N/A	15498	15513	TGCTGCATTAATGCCA	15	2485
1340756	N/A	N/A	5284	5299	TAAGTGAGCTAGACAA	53	2486
1340765	N/A	N/A	9152	9167	GTACATGCCAAGCTTT	61	2487
1340778	N/A	N/A	6468	6483	ATAGGCAGTAAAGTGC	56	2488
1340828	N/A	N/A	8600	8615	CTGTCATAAAGTGCG	59	2489
1340840	N/A	N/A	9320	9335	TCCAAGTAATTACTTC	72	2490
1340843	N/A	N/A	18505	18520	CTAATTTAGTCAACTT	40	2491
1340848	N/A	N/A	5658	5673	TAAGTTACCAGAGCAT	77	2492
1340861	N/A	N/A	3429	3444	GATGGTAAGTCAAATA	85	2493
1340871	N/A	N/A	7426	7441	TTGGCCGAGGAGGCG	3	2494
1340909	N/A	N/A	13548	13563	TCTAAAGGGCTGCTCT	33	2495
1340913	N/A	N/A	5883	5898	CGCTACTGTAGGCTAA	68	2496
1340916	N/A	N/A	18094	18109	ATTGATACACCAATGC	66	2497

TABLE 35

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	84	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTTCAGT	81	1526
1340081	1028	1043	20843	20858	ATAAAGCTTTGCAGCA	54	2498
1340090	N/A	N/A	9961	9976	TAAGCTGAGAGTTCTA	64	2499
1340106	N/A	N/A	19473	19488	CTATAACTGGAAGGAG	43	2500
1340110	N/A	N/A	17228	17243	TGTACAAATTTATGCC	67	2501
1340114	565	580	10451	10466	GAAGCCACTGTGACGA	58	2502
1340118	N/A	N/A	9608	9623	CAATTCACATATGTGA	11	2503
1340128	N/A	N/A	14461	14476	AGGTATGTCATACTTC	5	2504

TABLE 35-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340143	N/A	N/A	15021	15036	GATAAAATAGGGTGCA	52	2505
1340150	N/A	N/A	15501	15516	ATTGCTGCATTAATG	20	2506
1340165	N/A	N/A	14198	14213	ATTATGTGATTGAGTT	46	2507
1340177	N/A	N/A	3682	3697	GCCCATCCCCTAAGTT	33	2508
1340185	N/A	N/A	14621	14636	TGCCCCGAATTATTTC	3	2509
1340191	N/A	N/A	19890	19905	GCTTACGACAGGTCAT	73	2510
1340195	N/A	N/A	16127	16142	CATAATAATAGCTCTA	41	2511
1340210	N/A	N/A	18969	18984	TGACTAATGCAAGATT	45	2512
1340223	N/A	N/A	13729	13744	TTAAGTACTGTGAGAC	35	2513
1340254	N/A	N/A	13563	13578	GTTCAACTGAACCTCT	55	2514
1340265	N/A	N/A	11897	11912	TGAGAGTATAAATGGC	67	2515
1340272	N/A	N/A	11576	11591	TATTCTCTTGACAATG	47	2516
1340294	N/A	N/A	19651	19666	TTCTATGCAAGCCTTC	51	2517
1340300	N/A	N/A	17998	18013	TTATGAAGTTAGTATA	31	2518
1340302	N/A	N/A	18509	18524	GACACTAATTTAGTCA	11	2519
1340319	2248	2263	22063	22078	GATAGTCCATGCAAAA	64	2520
1340322	735	750	12153	12168	GATTTTGGTGAACCC	30	2521
1340336	N/A	N/A	9925	9940	ATTGAATTCATCTCT	31	2522
1340372	N/A	N/A	5988	6003	GCAAGGATACCTGAAA	67	2523
1340373	N/A	N/A	6309	6324	CTGCTACTCTAGGAGA	66	2524
1340379	N/A	N/A	18095	18110	AATTGATACACCAATG	55	2525
1340382	N/A	N/A	3570	3585	TATCAACCTGCACACC	62	2526
1340384	N/A	N/A	18673	18688	ATTTTACTGGGATACT	49	2527
1340394	N/A	N/A	12503	12518	TTGCAGAAATTCACCT	36	2528
1340401	N/A	N/A	5664	5679	AGAGCATAAGTTACCA	88	2529
1340404	N/A	N/A	9977	9992	AGACAGTTGAGTAGTG	63	2530
1340414	N/A	N/A	7954	7969	AAACTAGGAAGTGATC	39	2531
1340422	N/A	N/A	5202	5217	TTTTTGAACCGTATTC	56	2532
1340429	N/A	N/A	16275	16290	CAGCAAAGTTTGGGCA	37	2533
1340434	N/A	N/A	20121	20136	GATTACAGCATCAAGA	52	2534
1340435	N/A	N/A	3784	3799	TAGAACTGTGTTGCTT	78	2535
1340482	1199	1214	21014	21029	TGTTTATGTAAGCACA	74	2536
1340496	N/A	N/A	8605	8620	GAACACTGTCATAAAG	43	2537
1340499	N/A	N/A	20380	20395	ATCACAAAGTAAGGTAA	62	2538
1340500	N/A	N/A	16380	16395	CCAACCAAACTTCCAG	70	2539
1340526	N/A	N/A	10974	10989	ATTGTAATCTTGAGGC	39	2540

TABLE 35-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340529	N/A	N/A	9158	9173	CTCCTGGTACATGCCA	47	2541
1340538	N/A	N/A	6428	6443	TAATGGTAAACTGTAT	32	2542
1340584	N/A	N/A	6748	6763	CCTAGTTGCAATGAGA	77	2543
1340587	N/A	N/A	12422	12437	GTGGTAACGGTGATCA	49	2544
1340590	N/A	N/A	6711	6726	GGGAATGCTCATATTA	18	2545
1340594	N/A	N/A	9057	9072	TGTCAGAGACTCTGTG	39	2546
1340607	N/A	N/A	7708	7723	CGTCCTAGGAAACAAA	45	2547
1340619	N/A	N/A	8236	8251	CTACATGACCTGGGTC	59	2548
1340630	N/A	N/A	4133	4148	GGAGTACTTCACAATT	79	2549
1340646	N/A	N/A	5170	5185	GATTGCATAAGAGCCA	82	2550
1340654	N/A	N/A	5810	5825	TCAGTATCCCCCAAA	64	2551
1340677	N/A	N/A	8706	8721	GGACCAATATAATCCA	20	2552
1340706	N/A	N/A	9323	9338	ACATCCAAGTAATTAC	56	2553
1340731	N/A	N/A	5717	5732	TGCCAGATTGCACAAA	54	2554
1340737	N/A	N/A	20486	20501	AACTTCATAGTGGACT	68	2555
1340741	N/A	N/A	14165	14180	GCAAGCCAACAGAGAG	52	2556
1340748	N/A	N/A	11384	11399	TTTAGTCAGGTAGAGT	30	2557
1340751	N/A	N/A	9411	9426	TGTATAGCTGCATTTT	37	2558
1340772	N/A	N/A	5886	5901	TTTCGCTACTGTAGGC	62	2559
1340780	N/A	N/A	5054	5069	CAGCCTTAGAGTGAAT	82	2560
1340782	N/A	N/A	4537	4552	GAGATTTGAAGGTTAG	82	2561
1340804	N/A	N/A	20005	20020	AGTAATCAGTTTTCCT	65	2562
1340811	1546	1561	21361	21376	ACTTCAGAGTTATACA	35	2563
1340815	N/A	N/A	7473	7488	CATTCGGCTAGGCGCG	46	2564
1340816	N/A	N/A	9759	9774	CAAGCTACCTCCAAAT	25	2565
1340817	1372	1387	21187	21202	CATTCTGTGTTCTTGT	60	2566
1340820	N/A	N/A	3432	3447	GTAGATGGTAAGTCAA	86	2567
1340826	1723	1738	21538	21553	AAATTGTCTAAACATC	24	2568
1340847	N/A	N/A	6246	6261	ACTGGATGGATTCTCTC	83	2569
1340854	N/A	N/A	10201	10216	ATGGTAAGCCCCATGC	22	2570
1340855	N/A	N/A	6471	6486	GTAATAGGCAGTAAAG	77	2571
1340866	N/A	N/A	12794	12809	AAGAGTCAGTATCCTC	63	2572
1340877	N/A	N/A	9521	9536	ATTCTACACTAATATT	0	2573
1340880	N/A	N/A	5285	5300	ATAACTGAGCTAGACA	71	2574

TABLE 36

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	89	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCACT	91	1526
1340087	N/A	N/A	11578	11593	GTTATTCTCTTGACAA	42	2575
1340089	N/A	N/A	11388	11403	AACTTTTAGTCAGGTA	49	2576
1340116	N/A	N/A	16381	16396	CCCAACCAAACTTCCA	62	2577
1340124	603	618	10489	10504	ATGGGATGAGGTAAGG	46	2578
1340136	N/A	N/A	13782	13797	ACAATCAGGAGTTGCA	62	2579
1340153	N/A	N/A	5223	5238	TTAGCTTAAACTCCAA	78	2580
1340162	N/A	N/A	12504	12519	CTTGCAGAAATTCACC	40	2581
1340188	N/A	N/A	19489	19504	CATCTAGAATTAAGGG	53	2582
1340203	N/A	N/A	18690	18705	CTTTGAGACTCTTGTT	61	2583
1340207	N/A	N/A	18002	18017	AGGCTTATGAAGTTAG	57	2584
1340216	N/A	N/A	18971	18986	GTTGACTAATGCAAGA	59	2585
1340225	N/A	N/A	18096	18111	AAATTGATACACCAAT	14	2586
1340227	N/A	N/A	19891	19906	TGCTTACGACAGGTCA	81	2587
1340232	N/A	N/A	19656	19671	ATCCATTCTATGCAAG	46	2588
1340240	1397	1412	21212	21227	AAACTGATCTCTTAG	61	2589
1340248	1726	1741	21541	21556	CTAAAATTGTCTAAAC	35	2590
1340262	N/A	N/A	12423	12438	GGTGGTAACGGTGATC	31	2591
1340281	N/A	N/A	20122	20137	TGATTACAGCATCAAG	69	2592
1340285	2249	2264	22064	22079	GGATAGTCCATGCAAA	64	2593
1340293	1597	1612	21412	21427	TCTTATAAGACTATAA	17	2594
1340299	N/A	N/A	14905	14920	AGCTAAGAGACACTTC	41	2595
1340313	N/A	N/A	13568	13583	GAACAGTTCAACTGAA	10	2596
1340326	1054	1069	20869	20884	TATTATCAGGACTGAA	63	2597
1340328	N/A	N/A	20382	20397	TAATCACAAGTAAGGT	54	2598
1340331	N/A	N/A	8707	8722	GGGACCAATATAATCC	31	2599
1340351	N/A	N/A	9194	9209	ACAACCCTTAACAAAC	26	2600
1340371	N/A	N/A	6479	6494	CCAATATGTAATAGG	68	2601
1340378	N/A	N/A	9982	9997	TATTAAGACAGTTGAG	44	2602
1340407	N/A	N/A	4140	4155	TTCTCAAGGAGTACTT	76	2603
1340417	N/A	N/A	9422	9437	TATATGAGGTCTGTAT	42	2604
1340436	N/A	N/A	9765	9780	CCAACCAAGCTACCT	15	2605
1340441	N/A	N/A	5812	5827	AGTCAGTATCCCCTCA	82	2606
1340442	N/A	N/A	5718	5733	CTGCCAGATTGCACAA	44	2607
1340485	N/A	N/A	5058	5073	ATTACAGCCTTAGAGT	56	2608

TABLE 36-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340486	N/A	N/A	3571	3586	CTATCAACCTGCACAC	39	2609
1340487	N/A	N/A	6749	6764	GCCTAGTTGCAATGAG	61	2610
1340509	N/A	N/A	7476	7491	TAACATTCGGCTAGGC	52	2611
1340511	N/A	N/A	3933	3948	CTGCGCACATATTTTA	92	2612
1340516	N/A	N/A	5667	5682	TAGAGAGCATAAGTTA	59	2613
1340523	N/A	N/A	9927	9942	TGATTGAATTCACTAT	N.D.	2614
1340536	N/A	N/A	11924	11939	CCAAAAATCAGCCACT	0	2615
1340550	N/A	N/A	7768	7783	GCAGCTAAATCCAAT	70	2616
1340557	N/A	N/A	17672	17687	AACCTTTGTTGCCTGG	36	2617
1340559	N/A	N/A	14462	14477	GAGGTATGTCATACTT	29	2618
1340562	N/A	N/A	15510	15525	TATTCAGGATTGCT	43	2619
1340564	N/A	N/A	3435	3450	TAAGTAGATGGTAAGT	74	2620
1340581	N/A	N/A	8613	8628	GGCCAAGGGAACACTG	36	2621
1340589	1200	1215	21015	21030	ATGTTTATGTAAGCAC	70	2622
1340591	N/A	N/A	3693	3708	ATCCCGTTCTTGCCCA	30	2623
1340593	N/A	N/A	12169	12184	CCTTACCTTGTGCTTG	20	2624
1340595	N/A	N/A	8017	8032	ATCTCACAAGGGAAT	73	2625
1340596	N/A	N/A	6435	6450	CTAGTGGTAATGGTAA	62	2626
1340602	N/A	N/A	10206	10221	TTGGTATGGTAAGCCC	62	2627
1340605	N/A	N/A	5286	5301	TATACTGAGCTAGAC	53	2628
1340621	N/A	N/A	9327	9342	ATATACATCCAAGTAA	18	2629
1340622	N/A	N/A	6247	6262	CACTGGATGGATTCT	72	2630
1340641	N/A	N/A	5176	5191	AGTTTAGATTGCATAA	50	2631
1340644	N/A	N/A	16282	16297	TAGTCTTCAGCAAAGT	49	2632
1340652	N/A	N/A	14168	14183	ATAGCAAGCCAACAGA	37	2633
1340675	N/A	N/A	14201	14216	ATTATTATGTGATTGA	39	2634
1340682	N/A	N/A	9525	9540	CTAGATTCTACACTAA	17	2635
1340688	N/A	N/A	9629	9644	TAGTTTGGTGGGCATG	28	2636
1340699	N/A	N/A	20487	20502	TAACTTCATAGTGGAC	76	2637
1340707	N/A	N/A	16130	16145	CTCCATAATAATAGCT	59	2638
1340710	N/A	N/A	9962	9977	GTAAGCTGAGAGTTCT	49	2639
1340736	N/A	N/A	4640	4655	TACGACTTCCTCTAA	67	2640
1340774	N/A	N/A	15212	15227	CCATAAAGCTGGATTG	40	2641
1340779	N/A	N/A	5992	6007	AGAGGCAAGGATACCT	78	2642
1340797	N/A	N/A	18517	18532	AGTACCAAGACACTAA	66	2643
1340814	N/A	N/A	8237	8252	ACTACATGACCTGGGT	54	2644

TABLE 36-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340818	N/A	N/A	9077	9092	ACTTGAATTCTGTGTC	52	2645
1340849	N/A	N/A	6316	6331	GACCATGCTGCTACTC	74	2646
1340853	N/A	N/A	10978	10993	CAGCATTGTAATCTTG	43	2647
1340858	N/A	N/A	6715	6730	GCCAGGGAATGCTCAT	38	2648
1340860	N/A	N/A	12796	12811	GTAAGAGTCAGTATCC	47	2649
1340886	N/A	N/A	5888	5903	TATTCGCTACTGTAG	48	2650
1340887	N/A	N/A	20009	20024	AGAGAGTAATCAGTTT	39	2651

TABLE 37

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	83	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	76	1526
1340073	N/A	N/A	12797	12812	TGTAAGAGTCAGTATC	43	2652
1340074	1203	1218	21018	21033	AGTATGTTTATGTAAG	48	2653
1340084	N/A	N/A	18692	18707	GACTTTGAGACTCTTG	65	2654
1340091	N/A	N/A	9333	9348	GATCCTATATACATCC	49	2655
1340101	N/A	N/A	11391	11406	TGCAACTTTTAGTCAG	41	2656
1340103	N/A	N/A	16383	16398	ATCCCAACCAAACCTTC	15	2657
1340140	605	620	10491	10506	ATATGGGATGAGGTAA	37	2658
1340159	N/A	N/A	6029	6044	TTCCGTCTCACAATCT	79	2659
1340174	N/A	N/A	13571	13586	TCAGAACAGTTCAACT	30	2660
1340178	1598	1613	21413	21428	ATCTTATAAGACTATA	32	2661
1340202	N/A	N/A	19658	19673	GAATCCATTCTATGCA	47	2662
1340217	N/A	N/A	11579	11594	AGTTATTCTCTTGACA	48	2663
1340236	N/A	N/A	20491	20506	GACATAACTTCATAGT	39	2664
1340237	N/A	N/A	16131	16146	GCTCCATAATAATAGC	21	2665
1340250	N/A	N/A	12170	12185	ACCTTACCTTGTGCTT	0	2666
1340252	N/A	N/A	18518	18533	TAGTACCAAGACACTA	27	2667
1340253	N/A	N/A	18003	18018	AAGGCTTATGAAGTTA	61	2668
1340317	N/A	N/A	18098	18113	TGAAATTGATACACCA	58	2669
1340376	N/A	N/A	5304	5319	CGGCCTATTCTTCTGT	25	2670
1340389	N/A	N/A	7478	7493	ATTAACATTCGGCTAG	17	2671
1340392	N/A	N/A	19498	19513	AATATCAGCCATCTAG	24	2672

TABLE 37-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340410	1074	1089	20889	20904	TGCCAAACCAATGTTT	32	2673
1340413	N/A	N/A	9195	9210	GACAACCCCTTAACAAA	24	2674
1340447	N/A	N/A	3436	3451	ATAAGTAGATGGTAAG	58	2675
1340461	N/A	N/A	17676	17691	TCAAAACCTTTGTTGC	6	2676
1340463	N/A	N/A	6716	6731	TGCCAGGGAATGCTCA	23	2677
1340464	N/A	N/A	9647	9662	ATGTATGAGGTCTCCG	63	2678
1340477	N/A	N/A	13824	13839	ACCAAGAGGGTTTTTA	7	2679
1340495	N/A	N/A	4648	4663	ATTTTAAATACGACTT	35	2680
1340497	N/A	N/A	9963	9978	TGTAAGCTGAGAGTTC	45	2681
1340507	N/A	N/A	9766	9781	CCCAACCCAAGCTACC	22	2682
1340517	N/A	N/A	18983	18998	CACTTTGGTATTGTTG	53	2683
1340528	N/A	N/A	8636	8651	GACTGTTGAGCTCAAA	52	2684
1340537	N/A	N/A	5669	5684	AGTAGAGAGCATAAGT	52	2685
1340540	N/A	N/A	8040	8055	GACTACATTAACACCA	58	2686
1340541	N/A	N/A	6438	6453	AAACTAGTGGTAATGG	47	2687
1340568	N/A	N/A	20123	20138	CTGATTACAGCATCAA	40	2688
1340573	N/A	N/A	6762	6777	CTTGAGAGTGATTGCC	64	2689
1340579	2250	2265	22065	22080	AGGATAGTCCATGCAA	64	2690
1340592	N/A	N/A	10218	10233	GATACAAATTTATTGG	33	2691
1340598	N/A	N/A	5720	5735	CACTGCCAGATTGCAC	38	2692
1340608	N/A	N/A	3575	3590	CTTCTATCAACCTGC	68	2693
1340615	N/A	N/A	14475	14490	AGTATGACAACCTGGAG	15	2694
1340616	N/A	N/A	20046	20061	CGTAACCATGCATTAA	68	2695
1340618	N/A	N/A	6248	6263	CCACTGGATGGATTTC	68	2696
1340624	N/A	N/A	7772	7787	GTTAGCAGCTAAAATC	45	2697
1340635	N/A	N/A	5814	5829	AAAGTCAGTATCCCCT	77	2698
1340649	N/A	N/A	4222	4237	TCAACTAAACATGACA	77	2699
1340660	N/A	N/A	9933	9948	GAATCATGATTGAATT	39	2700
1340672	N/A	N/A	3695	3710	AGATCCCGTTCTTGCC	68	2701
1340676	N/A	N/A	5182	5197	CGCAACAGTTTAGATT	57	2702
1340703	N/A	N/A	14911	14926	TTAGCAAGCTAAGAGA	42	2703
1340708	N/A	N/A	12424	12439	GGGTGGTAACGGTGAT	46	2704
1340712	N/A	N/A	3952	3967	CAAGGAAAAGCCTGAC	59	2705
1340715	N/A	N/A	8256	8271	CTGGTTTCATAACCTA	9	2706
1340718	N/A	N/A	9423	9438	TTATATGAGGTCTGTA	32	2707
1340730	N/A	N/A	8709	8724	TTGGGACCAATATAAT	18	2708

TABLE 37-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340738	N/A	N/A	11965	11980	TGTGAGAAGCGGTAT	20	2709
1340743	N/A	N/A	5890	5905	TGTATTTGCTACTGT	76	2710
1340754	N/A	N/A	16284	16299	AATAGTCTTCAGCAAA	48	2711
1340758	N/A	N/A	19904	19919	AAGATACCCAGGTTGC	59	2712
1340784	N/A	N/A	5226	5241	GCTTTAGCTTAACTC	72	2713
1340787	N/A	N/A	9081	9096	GTCCACTTGAATTCTG	18	2714
1340799	N/A	N/A	5066	5081	TGAGTTACATTACAGC	90	2715
1340825	1748	1763	21563	21578	GTGTTAGCTTTAATTT	48	2716
1340844	1398	1413	21213	21228	GAAACTTGATCTCTTA	64	2717
1340851	N/A	N/A	9528	9543	GCCCTAGATTCTACAC	15	2718
1340868	N/A	N/A	15214	15229	TTCCATAAAGCTGGAT	5	2719
1340869	N/A	N/A	11073	11088	TACAACCTGGTTTCAT	25	2720
1340881	N/A	N/A	6321	6336	AGGGAGACCATGCTGC	79	2721
1340888	N/A	N/A	6480	6495	ACCAACTATGTAATAG	47	2722
1340890	N/A	N/A	14204	14219	CAGATTATTATGTGAT	51	2723
1340900	N/A	N/A	9989	10004	GTAAGTGTATTAAGAC	54	2724
1340906	N/A	N/A	12506	12521	TTCTTGCAGAAATTC	42	2725
1340912	N/A	N/A	14169	14184	AATAGCAAGCCAACAG	9	2726
1340914	N/A	N/A	20384	20399	AGTAATCACAAGTAAG	54	2727
1340915	794	809	15675	15690	ATCTATCAGACTTCTT	47	2728

TABLE 38

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	84	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	91	1526
1340076	N/A	N/A	12426	12441	CAGGGTGGTAACGGTG	55	2729
1340079	N/A	N/A	12191	12206	ACCCATTCTAACTTGA	0	2730
1340105	N/A	N/A	13825	13840	TACCAAGAGGGTTTTT	17	2731
1340107	N/A	N/A	12798	12813	CTGTAAGAGTCAGTAT	42	2732
1340112	N/A	N/A	20493	20508	TTGACATAACTTCATA	15	2733
1340115	N/A	N/A	12003	12018	TGTTAGCTAAGGGAGA	0	2734
1340139	N/A	N/A	17768	17783	CAGCATATTCATTTGG	49	2735

TABLE 38-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340147	N/A	N/A	5831	5846	TGTTAGGACCCAGTCT	51	2736
1340161	N/A	N/A	6764	6779	TTCTTGAGAGTGATTG	52	2737
1340167	N/A	N/A	4659	4674	GCCCTTAAGTCATTTT	25	2738
1340173	N/A	N/A	11075	11090	ATTACAACCTGGTTTC	50	2739
1340184	N/A	N/A	14913	14928	CATTAGCAAGCTAAGA	0	2740
1340186	N/A	N/A	13576	13591	CACTATCAGAACAGTT	24	2741
1340214	N/A	N/A	14173	14188	CTTGAATAGCAAGCCA	25	2742
1340219	N/A	N/A	18519	18534	TTAGTACCAAGACACT	41	2743
1340224	N/A	N/A	20109	20124	AAGACATTCTAGCCTG	63	2744
1340242	N/A	N/A	12616	12631	ATGGCTTAATTTTCGG	33	2745
1340251	N/A	N/A	20190	20205	AGTTATTAGAAGTCAG	54	2746
1340257	N/A	N/A	20386	20401	ATAGTAATCACAAGTA	56	2747
1340267	1749	1764	21564	21579	TGTGTTAGCTTTAATT	38	2748
1340296	N/A	N/A	14476	14491	GAGTATGACAACTGGA	23	2749
1340298	808	823	15689	15704	TTGGTAAGTATCCAT	20	2750
1340316	1175	1190	20990	21005	TAAGAGGCATGAAAGG	52	2751
1340325	N/A	N/A	8713	8728	AGTTTTGGGACCAATA	31	2752
1340343	N/A	N/A	11580	11595	GAGTTATTCTCTTGAC	57	2753
1340344	N/A	N/A	7479	7494	AATTAACATTCGGCTA	38	2754
1340365	N/A	N/A	16384	16399	AATCCCAACCAAACTT	23	2755
1340374	N/A	N/A	19673	19688	CAAGAGACACCACCAG	50	2756
1340381	N/A	N/A	6439	6454	TAAACTAGTGGTAATG	14	2757
1340383	N/A	N/A	5894	5909	CCTTTGTATTTTCGCTA	47	2758
1340408	N/A	N/A	6326	6341	TTCATAGGGAGACCAT	65	2759
1340421	N/A	N/A	9964	9979	GTGTAAGCTGAGAGTT	60	2760
1340424	N/A	N/A	10499	10514	ACTTACCAATATGGGA	0	2761
1340457	N/A	N/A	15247	15262	ATAGCTAAGTATACTT	0	2762
1340480	N/A	N/A	18102	18117	ATACTGAAATTGATAC	11	2763
1340481	N/A	N/A	9651	9666	GCTAATGTATGAGGTC	78	2764
1340483	N/A	N/A	9112	9127	AGACCAAGTAGCTTAC	42	2765
1340488	N/A	N/A	5070	5085	GATCTGAGTTACATTA	69	2766
1340502	N/A	N/A	9999	10014	GAAATCTTGTGTAAC	69	2767
1340535	N/A	N/A	9198	9213	ACTGACAACCCTTAAC	10	2768
1340551	N/A	N/A	16285	16300	GAATAGTCTTCAGCAA	51	2769
1340569	N/A	N/A	8050	8065	TTCTACAGAAGACTAC	40	2770
1340583	N/A	N/A	9432	9447	ACTAACCAATTATATG	6	2771

TABLE 38-continued

Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	% Inhibition	SEQ ID NO
1340611	N/A	N/A	16132	16147	TGCTCCATAATAATAG	35	2772
1340623	1603	1618	21418	21433	AATGTATCTTATAAGA	10	2773
1340632	N/A	N/A	18998	19013	AGTCATATTGAAAATC	39	2774
1340637	N/A	N/A	7773	7788	AGTTAGCAGCTAAAAT	33	2775
1340639	N/A	N/A	5185	5200	CAACGCAACAGTTTAG	52	2776
1340656	N/A	N/A	10258	10273	AGCATTCATGATTAA	47	2777
1340658	N/A	N/A	18005	18020	CCAAGGCTTATGAAGT	27	2778
1340662	N/A	N/A	8279	8294	TCTAAAGTGCTGGTTG	34	2779
1340663	N/A	N/A	6482	6497	GCACCACTATGTAAT	53	2780
1340666	N/A	N/A	3697	3712	TCAGATCCCGTCTTG	59	2781
1340702	N/A	N/A	3584	3599	CTCCTGAGTCTTCTA	23	2782
1340714	N/A	N/A	5227	5242	TGCTTTAGCTTAACT	51	2783
1340720	N/A	N/A	19531	19546	GCTAAAATGGTCATCT	40	2784
1340721	2356	2371	22171	22186	TTTATACTACAAGAG	38	2785
1340732	N/A	N/A	8639	8654	ACAGACTGTTGAGCTC	53	2786
1340733	1254	1269	21069	21084	TTTTGTCCACCTTTAA	68	2787
1340735	N/A	N/A	9334	9349	CGATCCTATATACATC	33	2788
1340745	N/A	N/A	9767	9782	TCCCAACCCAAGCTAC	8	2789
1340760	N/A	N/A	19905	19920	TAAGATACCCAGGTTG	39	2790
1340764	N/A	N/A	3478	3493	GACCAGGGAATTTATC	41	2791
1340766	N/A	N/A	9934	9949	TGAATCATGATTGAAT	38	2792
1340793	N/A	N/A	5646	5661	GCATTATCAGATGTT	89	2793
1340794	N/A	N/A	6717	6732	CTGCCAGGGAATGCTC	23	2794
1340796	N/A	N/A	6033	6048	ACTTTTCCGTCTCACA	89	2795
1340806	N/A	N/A	5671	5686	TCAGTAGAGAGCATAA	52	2796
1340809	1417	1432	21232	21247	TGAGATAAAGCTGCCT	63	2797
1340827	N/A	N/A	5732	5747	TGGTAGCTTGCTCACT	71	2798
1340829	N/A	N/A	6249	6264	GCCACTGGATGGATT	36	2799
1340856	N/A	N/A	11392	11407	CTGCAACTTTTAGTCA	21	2800
1340862	N/A	N/A	9530	9545	CTGCCCTAGATTCTAC	33	2801
1340867	N/A	N/A	14263	14278	GAGTTAGGGAGCCAGC	39	2802
1340882	N/A	N/A	4223	4238	TTCAACTAAACATGAC	55	2803
1340897	N/A	N/A	18693	18708	AGACTTTGAGACTCTT	57	2804
1340911	N/A	N/A	3988	4003	GTTTACAAGTAAGAAC	42	2805

355

Modified oligonucleotides complementary to an HSD17B13 nucleic acid with other chemistry modifications were synthesized and tested for their effect on HSD17B13 RNA levels in vitro, as described above. The Chemistry Notation column in the Tables below specifies the chemistry of each modified oligonucleotide; wherein subscript 'd'

356

represents a 2'-β-D-deoxyribosyl sugar moiety, subscript 'e' represents a 2'-MOE sugar moiety, subscript 'y' represents a 2'-O-methyl sugar moiety, subscript 'k' represents a cEt modified sugar moiety, subscript 's' represents a phosphorothioate internucleoside linkage, and superscript 'm' before the cytosine residue represents a 5-methyl cytosine.

TABLE 39

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ks}T_{ds}G_{ds}G_{ds}T_{ds}A_{ds}P_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	78	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}A_{ks}G_{ks}T_k$	78	1526
1341171	700	715	12118	12133	CAGAGACATGAGGTTT	$^mC_{ks}A_{ks}G_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}G_{ds}P_{ds}G_{ks}G_{es}T_{ks}T_{es}T_k$	61	2319
1341175	794	809	15675	15690	ATCTATCAGACTTCTT	$A_{ks}T_k^mC_{ds}T_{ds}A_{ds}T_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ks}T_{es}^mC_{ks}T_{es}T_k$	37	2728
1341179	606	621	10492	10507	AATATGGGATGAGGTA	$A_{ks}A_{ks}T_{ds}A_{ds}T_{ds}G_{ds}G_{ds}G_{ds}A_{ds}T_{ds}G_{ds}T_{ks}G_{es}T_{ks}T_{es}A_k$	52	1424
1341180	1080	1095	20895	20910	TGCTAGTGCCAAACCA	$T_{ks}G_{ks}^mC_{ds}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}A_{es}^mC_{ks}^mC_{es}A_k$	58	1432
1341189	1019	1034	20834	20849	TGCAGCATTGATTCTGA	$T_{ks}G_{ks}^mC_{ds}A_{ds}G_{ds}T_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}G_{ds}T_{ks}T_{es}^mC_{ks}G_{es}A_k$	42	419
1341190	1258	1273	21073	21088	TAGCTTTTGTCCACCT	$T_{ks}A_{ks}G_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}T_{ds}G_{ds}T_{ds}^mC_{ds}^mC_{ks}A_{es}^mC_{ks}^mC_{es}T_k$	71	1900
1341193	1202	1217	21017	21032	GTATGTTTATGTAAGC	$G_{ks}T_{ks}A_{ds}T_{ds}G_{ds}T_{ds}T_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ks}A_{es}A_{ks}G_{es}^mC_k$	72	1511
1341198	1084	1099	20899	20914	CTGCTGCTAGTGCCAA	$^mC_{ks}T_{ks}G_{ks}^mC_{ds}T_{ds}G_{ds}^mC_{ds}T_{ds}A_{ds}G_{ds}T_{ds}G_{ks}^mC_{es}^mC_{ks}A_{es}A_k$	44	1743
1341200	1316	1331	21131	21146	AACAGTCTTAAACCTT	$A_{ks}A_{ks}^mC_{ds}A_{ds}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ks}A_{ks}^mC_{es}^mC_{ks}T_{es}T_k$	37	188
1341213	1324	1339	21139	21154	GCTACTTGAACAGTCT	$G_{ks}^mC_{ks}T_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}A_{ds}^mC_{ds}A_{ks}G_{es}T_{ks}^mC_{es}T_k$	64	811
1341217	1312	1327	21127	21142	GTCTTAAACCTTCCCT	$G_{ks}T_{ks}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}A_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ks}^mC_{es}^mC_{ks}^mC_{es}T_k$	28	1901
1341228	1333	1348	21148	21163	GATTGGAATGCTACTT	$G_{ks}A_{ks}T_{ds}T_{ds}G_{ds}G_{ds}A_{ds}T_{ds}T_{ds}G_{ds}^mC_{ds}T_{ks}A_{es}^mC_{ks}T_{es}T_k$	53	1435
1341234	1356	1371	21171	21186	TGATATTCTGTGGCAT	$T_{ks}G_{ks}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ks}G_{es}^mC_{ks}A_{es}T_k$	46	1124
1341236	1352	1367	21167	21182	ATTCTGTGGCATGGCT	$A_{ks}T_{ks}T_{ds}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}G_{ds}^mC_{ds}A_{ds}T_{ks}G_{es}G_{ks}^mC_{es}T_k$	55	2328
1341239	1340	1355	21155	21170	GGCTACAGATTGGAAT	$G_{ks}G_{ks}^mC_{ds}T_{ds}A_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}T_{ds}T_{ds}G_{ks}G_{es}A_{ks}A_{es}T_k$	73	1980
1341241	1387	1402	21202	21217	TCTTAGCTGTGCACTC	$T_{ks}^mC_{ks}T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}^mC_{ks}A_{es}^mC_{ks}T_{es}^mC_k$	50	190
1341247	1364	1379	21179	21194	GTTCTTGTGATATTC	$G_{ks}T_{ks}T_{ds}^mC_{ds}T_{ds}T_{ds}G_{ds}T_{ds}T_{ds}G_{ds}A_{ds}T_{ks}A_{es}T_{ks}T_{es}^mC_k$	66	1436
1341250	1393	1408	21208	21223	TTGATCTCTTAGCTGT	$T_{ks}T_{ks}G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}P_{ds}G_{ks}^mC_{es}T_{ks}G_{es}T_k$	38	658
1341258	1420	1435	21235	21250	GGTTGAGATAAAGCTG	$G_{ks}G_{ks}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}A_{ds}A_{ks}G_{es}^mC_{ks}T_{es}G_k$	60	1826

TABLE 39-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% Inhibition	SEQ ID NO
1341263	1489	1504	21304	21319	CGTTTTGGGCTAATGA	${}^mC_{ks}G_{ks}T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}$ ${}^mC_{ds}T_{ds}A_{ks}A_{es}T_{ks}G_{es}A_k$	37	1282
1341267	1493	1508	21308	21323	GCACCGTTTTGGGCTA	$G_{ks}{}^mC_{ks}A_{ds}{}^mC_{ds}{}^mC_{ds}G_{ds}T_{ds}T_{ds}$ $T_{ds}T_{ds}G_{ds}G_{ks}G_{es}{}^mC_{ks}T_{es}A_k$	49	1593
1341274	1497	1512	21312	21327	AGTTGCACCGTTTTGG	$A_{ks}G_{ks}T_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}$ $C_{ds}G_{ds}T_{ds}T_{ks}T_{es}T_{ks}G_{es}G_k$	41	1905
1341280	1622	1637	21437	21452	AGAGTCGGTCACCTTT	$A_{ks}G_{ks}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ds}$ $T_{ds}{}^mC_{ds}A_{ds}C_{ks}C_{es}T_{ks}T_{es}T_k$	61	428
1341281	1630	1645	21445	21460	TTTAAATAGAGTCGG	$T_{ks}T_{ks}T_{ds}A_{ds}A_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}A_{ds}G_{ks}T_{es}{}^mC_{ks}G_{es}G_k$	43	1051
1341287	1618	1633	21433	21448	TCGGTCACCTTTTCATA	$T_{ks}{}^mC_{ks}G_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}$ ${}^mC_{ds}T_{ds}T_{ds}T_{ks}{}^mC_{es}A_{ks}T_{es}A_k$	63	116
1341290	1626	1641	21441	21456	AAATAGAGTCGGTCAC	$A_{ks}A_{ks}A_{ds}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}$ ${}^mC_{ds}G_{ds}G_{ks}T_{es}{}^mC_{ks}A_{es}{}^mC_k$	50	740
1341296	1714	1729	21529	21544	AAACATCTCTGGGACC	$A_{ks}A_{ks}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}$ $C_{ds}T_{ds}G_{ds}G_{ks}G_{es}A_{ks}{}^mC_{es}{}^mC_k$	36	2129
1341297	1710	1725	21525	21540	ATCTCTGGGACCAAGG	$A_{ks}T_{ks}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}G_{ds}G$ $dsA_{ds}{}^mC_{ds}{}^mC_{ks}A_{es}A_{es}G_k$	65	37
1341303	1768	1783	21583	21598	CCAGTACAGTTCCTTT	${}^mC_{ks}{}^mC_{ks}A_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}{}^mC_{ks}{}^mC_{es}T_{ks}T_{es}T_k$	75	1753
1341305	1750	1765	21565	21580	CTGTGTAGCTTTAAT	${}^mC_{ks}T_{ks}G_{ds}T_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}$ ${}^mC_{ds}T_{ds}T_{ks}T_{es}A_{ks}A_{es}T_k$	47	1286
1341313	1778	1793	21593	21608	TATGTAATAGCCAGTA	$T_{ks}A_{ks}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}{}^mC_{ds}{}^mC_{ks}A_{es}G_{ks}T_{es}A_k$	78	508
1341315	1782	1797	21597	21612	TTCTTATGTAATAGCC	$T_{ks}T_{ks}{}^mC_{ds}T_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}T_{ds}T_{ks}A_{es}G_{ks}{}^mC_{es}{}^mC_k$	78	819
1341325	2247	2262	22062	22077	ATAGTCCATGCAAAAG	$A_{ks}T_{ks}A_{ds}G_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}$ $T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ks}A_{es}A_{ks}A_{es}G_k$	31	2436
1341328	2193	2208	22008	22023	TAGTCTTGATGTAGTG	$T_{ks}A_{ks}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $T_{ds}G_{ds}T_{ks}A_{es}G_{ks}T_{es}G_k$	43	1288
1341339	N/A	N/A	9330	9345	CCTATATACATCCAAG	${}^mC_{ks}{}^mC_{ks}T_{ds}A_{ds}T_{ds}A_{ds}T_{ds}A_{ds}$ $C_{ds}T_{ds}T_{ds}{}^mC_{ks}{}^mC_{es}A_{ks}A_{es}G_k$	61	1771
1341343	N/A	N/A	11571	11586	TCTTGACAATGGTTGA	$T_{ks}{}^mC_{ks}T_{ds}T_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}T_{ds}G_{ds}G_{ks}T_{es}T_{ks}G_{es}A_k$	40	2440
1341344	N/A	N/A	11855	11870	TAAGCAGAATTGTGAA	$T_{ks}A_{ks}A_{ds}G_{ds}{}^mC_{ds}A_{ds}G_{ds}A_{ds}$ $A_{ds}T_{ds}T_{ds}G_{ks}T_{es}G_{ks}A_{es}A_k$	29	2361
1341347	N/A	N/A	9964	9979	GTGTAAGCTGAGAGTT	$G_{ks}T_{ks}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}{}^mC_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ks}A_{es}G_{ks}T_{es}T_k$	57	2760
1341351	N/A	N/A	12423	12438	GGTGGTAACGGTGATC	$G_{ks}G_{ks}T_{ds}G_{ds}G_{ds}T_{ds}A_{ds}A_{ds}$ $C_{ds}G_{ds}G_{ds}T_{ks}G_{es}A_{ks}T_{es}{}^mC_k$	39	2591
1341354	N/A	N/A	14914	14929	ACATTAGCAAGCTAAG	$A_{ks}{}^mC_{ks}A_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}$ $A_{ds}A_{ds}G_{ds}{}^mC_{ks}T_{es}A_{ks}A_{es}G_k$	23	2065
1341360	N/A	N/A	15494	15509	GCATTAATGCCACCCCT	$G_{ks}{}^mC_{ks}A_{ds}T_{ds}T_{ds}A_{ds}A_{ds}T_{ds}G_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ks}{}^mC_{es}{}^mC_{ks}{}^mC_{es}T_k$	24	1012
1341361	N/A	N/A	16287	16302	TAGAATAGTCTTCAGC	$T_{ks}A_{ks}G_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}T_{ks}{}^mC_{es}A_{ks}G_{es}{}^mC_k$	50	1403
1341375	N/A	N/A	17992	18007	AGTTAGTATAGTTATC	$A_{ks}G_{ks}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ks}T_{es}A_{ks}T_{es}{}^mC_k$	47	1249

TABLE 39-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% Inhibition	SEQ ID NO
1341377	N/A	N/A	16380	16395	CCAACCAAACTTCCAG	${}^mC_{ks}{}^mC_{ks}A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}P_{ds}{}^mC_{ds}T_{ds}T_{ks}{}^mC_{es}{}^mC_{ks}A_{es}G_k$	15	2539
1341382	N/A	N/A	19908	19923	TCTTAAGATACCCAGG	$T_{ks}{}^mC_{ks}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}T_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{es}A_{ks}G_{es}G_k$	35	319
1341387	N/A	N/A	18693	18708	AGACTTTGAGACTCTT	$A_{ks}G_{ks}A_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}P_{ds}{}^mC_{ks}T_{es}{}^mC_{ks}T_{es}T_k$	55	2804
1341389	N/A	N/A	19912	19927	TGCATCTTAAGATACC	$T_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $A_{ds}P_{ds}G_{ds}G_{ds}T_{es}A_{ks}{}^mC_{es}{}^mC_k$	24	2050
1341393	N/A	N/A	20381	20396	AATCACAAGTAAGGTA	$A_{ks}A_{ks}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}A_{ds}A_{ds}$ $G_{ds}T_{ds}A_{ds}P_{ks}G_{es}G_{ks}T_{es}A_k$	59	1333
1341396	N/A	N/A	20116	20131	CAGCATCAAGACATTC	${}^mC_{ks}A_{ks}G_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}{}^mC_{ks}{}^mC_{es}T_{ks}{}^mC_{es}{}^mC_k$	62	2191
1341406	N/A	N/A	12503	12518	TTGCAGAAATTCACCT	$T_{ks}T_{ks}G_{ds}{}^mC_{ds}A_{ds}G_{ds}A_{ds}A_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ks}{}^mC_{es}{}^mC_{ks}{}^mC_{es}T_k$	46	2528
1341407	N/A	N/A	20482	20497	TCATAGTGGACTTCAT	$T_{ks}{}^mC_{ks}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}$ $G_{ds}P_{ds}{}^mC_{ds}T_{ks}T_{es}{}^mC_{ks}A_{es}T_k$	48	633
1341412	N/A	N/A	14196	14211	TATGTGATTGAGTTCT	$T_{ks}A_{ks}T_{ds}G_{ds}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}$ $G_{ds}P_{ds}G_{ks}T_{es}T_{ks}{}^mC_{es}T_k$	23	2367
1341415	N/A	N/A	13952	13967	TCTAAGTTAGCCCCCA	$T_{ks}{}^mC_{ks}T_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}$ $G_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{es}{}^mC_{ks}{}^mC_{es}A_k$	19	928
1341417	N/A	N/A	14170	14185	GAATAGCAAGCCAACA	$G_{ks}A_{ks}A_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}{}^mC_{ds}{}^mC_{ks}A_{es}A_{ks}{}^mC_{es}A_k$	27	1708
1341532	805	820	15686	15701	GTAAGTATTCCATCTA	$G_{ks}T_{ks}A_{ds}G_{ds}G_{ds}T_{ds}A_{ds}T_{ds}T_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ks}T_{es}{}^mC_{ks}T_{es}A_k$	53	1038
1341534	613	628	N/A	N/A	CTGGAACAATATGGGA	${}^mC_{ks}T_{ks}G_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}T_{ds}A_{ds}T_{ks}G_{es}G_{ks}G_{es}A_k$	33	1969
1341538	1091	1106	20906	20921	CGTTTGACTGCTGCTA	${}^mC_{ks}G_{ks}T_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ds}$ $T_{ds}G_{ds}{}^mC_{ds}T_{ks}G_{es}{}^mC_{ks}T_{es}P_k$	48	264
1341541	1320	1335	21135	21150	CTTGAACAGTCTTAAA	${}^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}T_{ds}{}^mC_{ds}T_{ks}T_{es}A_{ks}A_{es}A_k$	38	500
1341543	1308	1323	21123	21138	TAAACCTTCCCTGTGT	$T_{ks}A_{ks}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $C_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}G_{es}T_{ks}G_{es}T_k$	9	1745
1341547	1076	1091	20891	20906	AGTGCCAAACCAATGT	$A_{ks}G_{ks}T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}A_{es}T_{ks}G_{es}T_k$	71	1120
1341551	1347	1362	21162	21177	GTGCATGGCTACAGA	$G_{ks}T_{ks}G_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}$ $G_{ds}{}^mC_{ds}T_{ds}A_{ks}{}^mC_{es}A_{ks}G_{es}P_k$	59	501
1341555	1328	1343	21143	21158	GAATGCTACTTGAACA	$G_{ks}A_{ks}A_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}$ $C_{ds}T_{ds}T_{ds}G_{ks}A_{es}A_{ks}{}^mC_{es}A_k$	48	1123
1341564	1425	1440	21240	21255	GTCCAGGTTGAGATAA	$G_{ks}T_{ks}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ks}A_{es}T_{ks}A_{es}A_k$	36	113
1341565	1461	1476	21276	21291	CTAGGGAAATCTTTCA	${}^mC_{ks}T_{ks}A_{ds}G_{ds}G_{ds}A_{ds}A_{ds}$ $A_{ds}T_{ds}{}^mC_{ds}T_{ks}T_{es}T_{ks}{}^mC_{es}A_k$	46	659
1341567	1415	1430	21230	21245	AGATAAAGCTGCCTGC	$A_{ks}G_{ks}A_{ds}T_{ds}A_{ds}A_{ds}G_{ds}{}^mC_{ds}$ $C_{ds}T_{ds}G_{ds}{}^mC_{ks}{}^mC_{es}T_{ks}G_{es}{}^mC_k$	30	1515
1341568	1718	1733	21533	21548	GTCTAACATCTCTGG	$G_{ks}T_{ks}{}^mC_{ds}T_{ds}A_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}{}^mC_{ds}T_{ks}{}^mC_{es}T_{ks}G_{es}G_k$	50	429
1341569	1504	1519	21319	21334	AGAATAGAGTTGCACC	$A_{ks}G_{ks}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ks}{}^mC_{es}P_{ks}{}^mC_{es}{}^mC_k$	59	426

TABLE 39-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% Inhibition	SEQ ID NO
1341575	1614	1629	21429	21444	TCACCTTTCATAATGT	$T_{ks}^m C_{ks} A_{ds}^m C_{ds}^m C_{ds} T_{ds} T_{ds} T_{ds}$ $^m C_{ds} A_{ds} T_{ds} A_{ks} A_{es} T_{ks} G_{es} T_k$	31	2109
1341578	2186	2201	22001	22016	GATGTAGTGGGAGTCG	$G_{ks} A_{ks} T_{ds} G_{ds} T_{ds} A_{ds} G_{ds} T_{ds} G_{ds}$ $G_{ds} G_{ds} A_{ks} G_{es} T_{ks} ^m C_{es} G_k$	38	743
1341580	2257	2272	22072	22087	AAACAAGAGGATAGTC	$A_{ks} A_{ks} A_{ds}^m C_{ds} A_{ds} A_{ds} G_{ds} A_{ds}$ $G_{ds} G_{ds} A_{ds} T_{ks} A_{es} G_{ks} T_{es} ^m C_k$	52	744
1341585	1774	1789	21589	21604	TAATAGCCAGTACAGT	$T_{ks} A_{ks} A_{ds} T_{ds} A_{ds} G_{ds} ^m C_{ds} ^m C_{ds}$ $A_{ds} G_{ds} T_{ds} A_{ks} ^m C_{es} A_{ks} G_{es} T_k$	44	196
1341588	N/A	N/A	9511	9526	AATATTGAGGCACTGG	$A_{ks} A_{ks} T_{ds} A_{ds} T_{ds} T_{ds} G_{ds} A_{ds} G_{ds}$ $G_{ds} ^m C_{ds} A_{ks} ^m C_{es} T_{ks} G_{es} G_k$	29	1538
1341596	N/A	N/A	9598	9613	ATGTGAAGAGCTGGTA	$A_{ks} T_{ks} G_{ds} T_{ds} G_{ds} A_{ds} A_{ds} G_{ds} A_{ds}$ $G_{ds} ^m C_{ds} T_{ks} G_{es} G_{ks} T_{es} A_k$	54	527
1341603	N/A	N/A	16134	16149	GATGCTCCATAATAAT	$G_{ks} A_{ks} T_{ds} G_{ds} ^m C_{ds} T_{ds} ^m C_{ds} ^m C_{ds}$ $A_{ds} T_{ds} A_{ds} A_{ks} T_{es} A_{ks} T_k$	17	2049
1341604	N/A	N/A	16124	16139	AATAATAGCTCTATTG	$A_k A_{ks} T_{ds} A_{ds} T_{ds} A_{ds} G_{ds} ^m$ $C_{ds} T_{ds} ^m C_{ds} T_{ks} T_{es} T_{ks} T_{es} G_k$	2	2410
1341614	N/A	N/A	17987	18002	GTATAGTTATCTTCTC	$G_{ks} T_{ks} A_{ds} T_{ds} A_{ds} G_{ds} T_{ds} T_{ds} A_{ds}$ $T_{ds} ^m C_{ds} T_{ks} T_{es} ^m C_{ks} T_{es} ^m C_k$	49	937
1341615	N/A	N/A	17997	18012	TATGAAGTTAGTATAG	$T_{ks} A_{ks} T_{ds} G_{ds} A_{ds} A_{ds} G_{ds} T_{ds} T_{ds}$ $A_{ds} G_{ds} T_{ks} A_{es} T_{ks} A_{es} G_k$	47	1560

TABLE 40

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% Inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAAGTCA	$A_{ks} G_{ks} T_{ks} A_{ds} G_{ds} A_{ds} T_{ds} G_{ds} G_{ds}$ $T_{ds} A_{ds} G_{ds} T_{ks} ^m C_{ks} A_k$	76	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks} A_{ks} G_{ks} T_{ds} G_{ds} A_{ds} A_{ds} T_{ds} ^m C_{ds}$ $A_{ds} T_{ds} T_{ds} ^m C_{ds} A_{ks} G_{ks} T_k$	84	1526
1341176	796	811	15677	15692	CCATCTATCAGACTTC	$^m C_{ks} ^m C_{ks} A_{ds} T_{ds} ^m C_{ds} T_{ds} A_{ds} T_{ds}$ $^m C_{ds} A_{ds} G_{ds} A_{ks} ^m C_{es} T_{ks} T_{es} ^m C_k$	52	415
1341182	808	823	15689	15704	TTGGTAAGTATTCCAT	$T_{ks} T_{ks} G_{ds} G_{ds} T_{ds} A_{ds} A_{ds} G_{ds} T_{ds}$ $A_{ds} T_{ds} T_{ks} ^m C_{es} ^m C_{ks} A_{es} T_k$	13	2750
1341183	1077	1092	20892	20907	TAGTGCCAAACCAATG	$T_{ks} A_{ks} G_{ds} T_{ds} G_{ds} ^m C_{ds} ^m C_{ds} A_{ds}$ $A_{ds} A_{ds} ^m C_{ds} ^m C_{ks} A_{es} A_{ks} T_{es} G_k$	66	1198
1341187	1081	1096	20896	20911	CTGCTAGTGCCAAACC	$^m C_{ks} T_{ks} G_{ds} ^m C_{ds} T_{ds} A_{ds} G_{ds} T_{ds} G_{ds}$ $^m C_{ks} A_{ks} A_{ks} A_{es} A_{ks} ^m C_{es} ^m C_k$	51	1510
1341191	1085	1100	20900	20915	ACTGCTGCTAGTGCCA	$A_{ks} ^m C_{ks} T_{ds} G_{ds} ^m C_{ds} T_{ds} G_{ds} ^m C_{ds}$ $T_{ds} A_{ds} G_{ds} T_{ks} G_{es} ^m C_{ks} ^m C_{es} A_k$	39	1821
1341202	1259	1274	21074	21089	GTAGCTTTTGTCCACC	$G_{ks} T_{ks} A_{ds} G_{ds} ^m C_{ds} T_{ds} T_{ds} T_{ds} T_{ds}$ $G_{ds} T_{ds} ^m C_{ks} ^m C_{es} A_{ks} ^m C_{es} ^m C_k$	34	1978
1341204	1313	1328	21128	21143	AGTCTTAAACCTTCCC	$A_{ks} G_{ks} T_{ds} ^m C_{ds} T_{ds} T_{ds} A_{ds} A_{ds} A_{ds}$ $^m C_{ds} ^m C_{ds} T_{ks} T_{es} ^m C_{ks} ^m C_{es} ^m C_k$	43	1979
1341211	1325	1340	21140	21155	TGCTACTTGAACAGTC	$T_{ks} G_{ks} ^m C_{ds} T_{ds} A_{ds} ^m C_{ds} T_{ds} T_{ds} G_{ds}$ $A_{ds} A_{ds} ^m C_{ks} A_{es} G_{ks} T_{es} ^m C_k$	64	889

TABLE 40 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341214	1321	1336	21136	21151	ACTTGACAGTCTTAA	$A_{ks}^m C_{ks} T_{ds} T_{ds} G_{ds} A_{ds} A_{ds}^m C_{ds}$ $A_{ds} G_{ds} T_{ds}^m C_{ks} T_{es} T_{ks} A_{es} A_k$	60	578
1341218	1317	1332	21132	21147	GAACAGTCTTAAACCT	$G_{ks} A_{ks} A_{ds}^m C_{ds} A_{ds} G_{ds} T_{ds}^m C_{ds}$ $T_{ds} T_{ds} A_{ds} A_{ks}^m C_{ks}^m C_{es} T_k$	63	266
1341222	1342	1357	21157	21172	ATGGCTACAGATTGGA	$A_{ks} T_{ks} G_{ds} G_{ds}^m C_{ds} T_{ds} A_{ds}^m C_{ds}$ $A_{ds} G_{ds} A_{ds} T_{ks} T_{es} G_{ks} G_{es} T_k$	57	111
1341226	1329	1344	21144	21159	GGAATGCTACTTGAAC	$G_{ks} G_{ks} A_{ds} A_{ds} T_{ds} G_{ds}^m C_{ds} T_{ds} A_{ds}$ $s^m C_{ds} T_{ds} T_{ks} G_{es} A_{ks} T_{es}^m C_k$	49	1201
1341227	1335	1350	21150	21165	CAGATTGGAATGCTAC	$^m C_{ks} A_{ks} G_{ds} A_{ds} T_{ds} T_{ds} G_{ds} G_{ds} A_{ds}$ $s^m T_{ds} T_{ds} G_{ks}^m C_{es} T_{ks} T_{es}^m C_k$	57	1590
1341237	1353	1368	21168	21183	TATTCTGTGGCATGGC	$T_{ks} A_{ks} T_{ds} T_{ds}^m C_{ds} T_{ds} G_{ds} T_{ds} G_{ds}$ $G_{ds}^m C_{ds} A_{ks} T_{es} G_{ks} G_{es}^m C_k$	57	890
1341242	1366	1381	21181	21196	GTGTTCTTGTGATAT	$G_{ks} T_{ks} G_{ds} T_{ds} T_{ds}^m C_{ds} T_{ds} T_{ds} G_{ds}$ $T_{ds} T_{ds} C T_{ks} A_{es} T_{ks} A_e T_k$	63	1514
1341248	1357	1372	21172	21187	TTGATATTCTGTGGCA	$T_{ks} T_{ks} G_{ds} A_{ds} T_{ds} A_{ds} T_{ds} T_{ds}^m C_{ds}$ $T_{ds} G_{ds} T_{ks} G_{es} G_{ks}^m C_{es} A_k$	55	1202
1341254	1421	1436	21236	21251	AGGTGAGATAAAGCT	$A_{ks} G_{ks} G_{ds} T_{ds} T_{ds} G_{ds} A_{ds} G_{ds} A_{ds}$ $T_{ds} A_{ds} A_{ks} A_{es} G_{ks}^m C_{es} T_k$	60	1904
1341262	1490	1505	21305	21320	CCGTTTTGGGCTAATG	$^m C_{ks}^m C_{ks} G_{ds} T_{ds} T_{ds} T_{ds} T_{ds} G_{ds} G_{ds}$ $ds G_{ds}^m C_{ds} T_{ks} A_{es} T_{ks} T_{es} G_k$	72	1360
1341265	1494	1509	21309	21324	TGCACCGTTTTGGGCT	$T_{ks} G_{ks}^m C_{ds} A_{ds}^m C_{ds}^m C_{ds} G_{ds} T_{ds}$ $T_{ds} T_{ds} T_{ds} G_{ks} G_{es}^m C_{es} T_k$	29	1671
1341269	1464	1479	21279	21294	AGGCTAGGGAATCTT	$A_{ks} G_{ks} G_{ds}^m C_{ds} T_{ds} A_{ds} G_{ds} G_{ds} G_d$ $s^m A_{ds} A_{ds} T_{es}^m C_{ks} T_{es} T_k$	45	892
1341275	1619	1634	21434	21449	GTCGGTCACCTTTTCAT	$G_{ks} T_{ks}^m C_{ds} G_{ds} G_{ds} T_{ds}^m C_{ds} A_{ds}^m$ $C_{ds}^m C_{ds} T_{ds} T_{ks} T_{es}^m C_{ks} A_{es} T_k$	57	194
1341277	1598	1613	21413	21428	ATCTTATAAGACTATA	$A_{ks} T_{ks}^m C_{ds} T_{ds} T_{ds} A_{ds} T_{ds} A_{ds} A_{ds}$ $G_{ds} A_{ds}^m C_{ks} T_{es} A_{ks} T_{es} A_k$	30	2661
1341284	1627	1642	21442	21457	AAAATAGAGTCGGTCA	$A_{ks} A_{ks} A_{ds} A_{ds} T_{ds} A_{ds} G_{ds} A_{ds} G_{ds}$ $T_{ds}^m C_{ds} G_{ks} G_{es} T_{ks}^m C_{es} A_k$	53	817
1341286	1623	1638	21438	21453	TAGAGTCGGTCACCTT	$T_{ks} A_{ks} G_{ds} A_{ds} G_{ds} T_{ds}^m C_{ds} G_{ds} G_{ds}$ $T_{ds}^m C_{ds} A_{ks}^m C_{es}^m C_{ks} T_{es} T_k$	48	506
1341291	1711	1726	21526	21541	CATCTCTGGGACCAAG	$^m C_{ks} A_{ks} T_{ds}^m C_{ds} T_{ds}^m C_{ds} T_{ds} G_{ds}$ $G_{ds} G_{ds} A_{ds}^m C_{ks}^m C_{es} A_{ks} G_k$	44	117
1341292	1715	1730	21530	21545	TAAACATCTCTGGGAC	$T_{ks} A_{ks} A_{ds} A_{ds}^m C_{ds} A_{ds} T_{ds}^m C_{ds} T$ $ds^m C_{ds} T_{ds} G_{ks} G_{es} G_{ks} A_{es}^m C_k$	39	351
1341300	1719	1734	21534	21549	TGTCTAAACATCTCTG	$T_{ks} G_{ks} T_{ds}^m C_{ds} T_{ds} A_{ds} A_{ds} A_{ds}^m C_{ds}$ $ds A_{ds} T_{ds}^m C_{ks} T_{es}^m C_{ks} T_{es} G_k$	59	507
1341302	1770	1785	21585	21600	AGCCAGTACAGTTCTCT	$A_{ks} G_{ks}^m C_{ds}^m C_{ds} A_{ds} G_{ds} T_{ds} A_{ds}^m$ $C_{ds} A_{ds} G_{ds} T_{ks} T_{es}^m C_{ks}^m C_{es} T_k$	49	1909
1341307	1751	1766	21566	21581	CCTGTGTAGCTTTTAA	$^m C_{ks}^m C_{ks} T_{ds} G_{ds} T_{ds} G_{ds} T_{ds} T_{ds} A_{ds}$ $ds G_{ds}^m C_{ds} T_{ks} T_{es} T_{ks} A_{es} A_k$	62	1364
1341310	1783	1798	21598	21613	TTTCTTATGTAATAGC	$T_{ks} T_{ks} T_{ds}^m C_{ds} T_{ds} T_{ds} T_{ds} T_{ds} G_{ds}$ $T_{ds} A_{ds} A_{ks} T_{es} A_{ks} G_{es}^m C_k$	60	2128
1341314	1779	1794	21594	21609	TTATGTAATAGCCAGT	$T_{ks} T_{ks} A_{ds} T_{ds} G_{ds} T_{ds} A_{ds} A_{ds} T_{ds}$ $A_{ds} G_{ds}^m C_{ks}^m C_{es} A_{ks} G_{es} T_k$	63	586
1341316	1775	1790	21590	21605	GTAATAGCCAGTACAG	$G_{ks} T_{ks} A_{ds} A_{ds} T_{ds} A_{ds} G_{ds}^m C_{ds}^m$ $C_{ds} A_{ds} G_{ds} T_{ks} A_{es}^m C_{ks} A_{es} G_k$	47	274

TABLE 40 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341329	2187	2202	22002	22017	TGATGTAGTGGGAGTC	$T_{ks}G_{ks}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}G_{ds}G_{ks}A_{es}G_{ks}T_{es}C_k$	46	820
1341332	N/A	N/A	9514	9529	ACTAATATTGAGGCAC	$A_{ks}C_{ks}T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ks}C_{es}C_{ks}A_{es}C_k$	27	1772
1341334	2250	2265	22065	22080	AGGATAGTCCATGCAA	$A_{ks}G_{ks}G_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $sC_{ds}A_{ds}T_{ks}C_{es}C_{ks}A_{es}A_k$	46	2690
1341337	N/A	N/A	9332	9347	ATCCTATATACATCCA	$A_{ks}T_{ks}C_{ds}C_{ds}T_{ds}A_{ds}T_{ds}A_{ds}T_{ds}$ $dsA_{ds}C_{ds}A_{ks}T_{es}C_{ks}C_{es}A_k$	65	1927
1341342	N/A	N/A	9966	9981	TAGTGTAAAGCTGAGAG	$T_{ks}A_{ks}G_{ds}T_{ds}G_{ds}T_{ds}A_{ds}G_{ds}$ $C_{ds}T_{ds}G_{ks}A_{es}G_{ks}A_{es}G_k$	68	1852
1341346	N/A	N/A	9601	9616	CATATGTGAAGAGCTG	$C_{ks}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}A_d$ $sA_{ds}G_{ds}A_{ks}G_{es}C_{ks}T_{es}G_k$	35	683
1341348	N/A	N/A	11548 11573	11563 11588	TCTCTTGACAATGGTT	$T_{ks}C_{ks}T_{ds}C_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $C_{ds}A_{ds}A_{ds}T_{ks}G_{es}G_{ks}T_{es}T_k$	70 100	1389
1341350	N/A	N/A	14916	14931	ATACATTAGCAAGCTA	$A_{ks}T_{ks}A_{ds}C_{ds}A_{ds}T_{ds}T_{ds}A_{ds}G_d$ $sC_{ds}A_{ds}A_{ks}C_{es}C_{ks}T_{es}A_k$	22	309
1341358	N/A	N/A	14197	14212	TTATGTGATTGAGTTC	$T_{ks}T_{ks}A_{ds}T_{ds}G_{ds}T_{ds}G_{ds}T_{ds}T$ $dsG_{ds}A_{ks}G_{es}T_{ks}T_{es}C_k$	21	2431
1341363	N/A	N/A	15496	15511	CTGCATTAATGCCACC	$C_{ks}T_{ks}G_{ds}C_{ds}A_{ds}T_{ds}T_{ds}A_{ds}A$ $dsT_{ds}G_{ds}C_{ks}C_{es}A_{ks}C_{es}C_k$	6	1168
1341364	N/A	N/A	16127	16142	CATAATAATAGCTCTA	$C_{ks}A_{ks}T_{ds}A_{ds}T_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}C_{ks}T_{es}T_{ks}C_{es}A_k$	13	2511
1341365	N/A	N/A	16289	16304	GTTAGAATAGTCTTCA	$G_{ks}T_{ks}T_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}C_{ks}T_{es}T_{ks}C_{es}A_k$	45	1481
1341370	N/A	N/A	16382	16397	TCCCAACCAACTTCC	$T_{ks}C_{ks}C_{ds}C_{ds}A_{ds}A_{ds}C_{ds}$ $C_{ds}A_{ds}A_{ds}T_{ks}C_{ks}T_{es}T_{ks}C_{es}$ C_k	27	2026
1341371	N/A	N/A	18095	18110	AATTGATACACCAATG	$A_{ks}A_{ks}T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}C_{ds}$ $A_{ds}C_{ds}C_{ks}A_{es}A_{ks}T_{es}G_k$	50	2525
1341376	N/A	N/A	17994	18009	GAAGTTAGTATAGTTA	$G_{ks}A_{ks}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ds}T_{ds}A_{ks}G_{es}T_{ks}T_{es}A_k$	56	1327
1341379	N/A	N/A	17989	18004	TAGTATAGTTATCTTC	$T_{ks}A_{ks}G_{ds}T_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}A_{ds}T_{ks}C_{es}T_{ks}T_{es}C_k$	34	1093
1341380	N/A	N/A	19909	19924	ATCTTAAGATACCCAG	$A_{ks}T_{ks}C_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_d$ $sT_{ds}A_{ds}C_{ks}C_{es}C_{ks}A_{es}G_k$	45	397
1341384	N/A	N/A	19913	19928	GTGCATCTTAAGATAC	$G_{ks}T_{ks}G_{ds}C_{ds}A_{ds}T_{ds}C_{ds}T_{ds}$ $dsA_{ds}A_{ks}G_{ks}A_{es}T_{ks}A_{es}C_k$	20	2806
1341388	N/A	N/A	18695	18710	AAAGACTTTGAGACTC	$A_{ks}A_{ks}A_{ds}G_{ds}A_{ds}C_{ds}T_{ds}T_{ds}$ $sG_{ds}A_{ds}G_{ks}A_{es}C_{ks}T_{es}C_k$	45	1250
1341391	N/A	N/A	20383	20398	GTAATCACAAGTAAGG	$G_{ks}T_{ks}A_{ds}T_{ds}C_{ds}A_{ds}C_{ds}$ $A_{ds}A_{ds}G_{ds}T_{ks}A_{es}A_{ks}G_{es}G_k$	72	1411
1341399	N/A	N/A	20190	20205	AGTTATTAGAAGTCAG	$A_{ks}G_{ks}T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}G_{ds}$ $A_{ds}A_{ds}G_{ks}T_{es}C_{ks}A_{es}G_k$	42	2746
1341401	N/A	N/A	20484	20499	CTTCATAGTGGACTTC	$C_{ks}T_{ks}T_{ds}C_{ds}A_{ds}T_{ds}G_{ds}T$ $dsG_{ds}G_{ks}A_{ks}C_{es}T_{ks}T_{es}C_k$	45	711
1341410	N/A	N/A	13954	13969	CCTCTAAGTTAGCCCC	$C_{ks}C_{ks}T_{ds}C_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}A_{ds}G_{ks}C_{es}C_{ks}C_{es}C_k$	20	1084
1341418	N/A	N/A	12797	12812	TGTAAGAGTCAGTATC	$T_{ks}G_{ks}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}$ $C_{ds}A_{ds}G_{ks}T_{es}A_{ks}T_{es}C_k$	35	2652

TABLE 40 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound 1	SEQ ID NO: Start Site	SEQ ID NO: Stop Site	SEQ ID NO: Start Site	SEQ ID NO: Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341419	N/A	N/A	14172	14187	TTGAATAGCAAGCCAA	$T_{ks}T_{ks}G_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $A_{ds}A_{ds}G_{ks}^mC_{es}^mC_{ks}^mC_{es}^m$	27	1864
1341529	693	708	12111	12126	ATGAGGTTTTGATAACC	$A_{ks}T_{ks}G_{ds}A_{ds}G_{ds}T_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ks}T_{es}A_{ks}^mC_{es}^mC_k$	51	569
1341533	1022	1037	20837	20852	CTTTGCAGCATTGATT	$^mC_{ks}T_{ks}T_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}G_{ds}^m$ $C_{ds}A_{ds}T_{ds}T_{ks}G_{es}A_{ks}T_{es}T_k$	49	2463
1341540	1309	1324	21124	21139	TTAAACCTTCCTGTG	$T_{ks}T_{ks}A_{ds}A_{ds}A_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ds}$ $^mC_{ds}^mC_{ds}^mC_{ks}T_{es}G_{ks}T_{es}G_k$	0	2807
1341545	1175	1190	20990	21005	TAAGAGGCATGAAAGG	$T_{ks}A_{ks}A_{ds}G_{ds}A_{ds}G_{ds}G_{ds}^mC_{ds}A_d$ $s^sT_{ds}G_{ds}A_{ks}A_{es}A_{ks}G_{es}G_k$	54	2751
1341546	1251	1266	21066	21081	TGTCCACCTTTAAATG	$T_{ks}G_{ks}T_{ds}^mC_{ds}^mC_{ds}A_{ds}^mC_{ds}^mC_d$ $s^sT_{ds}T_{ds}T_{ds}A_{ks}A_{es}A_{ks}T_{es}G_k$	27	1666
1341552	1348	1363	21163	21178	TGTGGCATGGCTACAG	$T_{ks}G_{ks}T_{ds}G_{ds}G_{ds}^mC_{ds}A_{ds}T_{ds}G_d$ $s^sG_{ds}^mC_{ds}T_{ks}A_{es}^mC_{ks}A_{es}G_k$	22	579
1341554	1394	1409	21209	21224	CTTGATCTCTTAGCTG	$^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^m$ $C_{ds}T_{ds}T_{ds}A_{ks}G_{es}^mC_{ks}T_{es}G_k$	49	736
1341558	1388	1403	21203	21218	CTCTTAGCTGTGCACT	$^mC_{ks}T_{ks}^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $T_{ds}G_{ds}T_{ds}G_{ks}^mC_{es}A_{ks}^mC_{es}T_k$	56	268
1341560	1498	1513	21313	21328	GAGTTGCACCGTTTTG	$G_{ks}A_{ks}G_{ds}T_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}^m$ $C_{ds}^mC_{ds}G_{ds}T_{ks}T_{es}T_{ks}T_{es}G_k$	61	1983
1341563	1416	1431	21231	21246	GAGATAAAGCTGCCTG	$G_{ks}A_{ks}G_{ds}A_{ds}T_{ds}A_{ds}A_{ds}G_{ds}^m$ $C_{ds}T_{ds}G_{ks}^mC_{es}^mC_{ks}T_{es}G_k$	40	1592
1341566	1426	1441	21241	21256	TGTCCAGGTTGAGATA	$T_{ks}G_{ks}T_{ds}^mC_{ds}^mC_{ds}A_{ds}G_{ds}T$ $dsT_{ds}C_{ds}A_{ks}G_{es}A_{ks}T_{es}A_k$	48	191
1341574	1615	1630	21430	21445	GTCACCTTTCATAATG	$G_{ks}T_{ks}^mC_{ds}A_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ds}$ $T_{ds}^mC_{ds}A_{ds}T_{ks}A_{es}A_{ks}T_{es}G_k$	4	1907
1341577	1705	1720	21520	21535	TGGGACCAAGGATATA	$T_{ks}G_{ks}G_{ds}G_{ds}A_{ds}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}G_{ds}A_{ks}T_{es}A_{ks}T_{es}A_k$	30	1674
1341583	2194	2209	22009	22024	TTAGTCTTGATGTAGT	$T_{ks}T_{ks}A_{ds}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}G_{ks}T_{es}A_{ks}G_{es}T_k$	36	1366
1341589	N/A	N/A	11858	11873	GATTAAGCAGAATTGT	$G_{ks}A_{ks}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}^mC_{ds}A_d$ $s^sG_{ds}A_{ds}A_{ks}T_{es}T_{ks}G_{es}T_k$	31	1156
1341591	N/A	N/A	11543	11558	TGACAATGGTTGCCAC	$T_{ks}G_{ks}A_{ds}^mC_{ds}A_{ds}A_{ds}T_{ds}G_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ks}^mC_{es}^mC_{ks}A_{es}^mC_k$	38	1233
1341595	2350	2365	22165	22180	ACTACAAGAGGTTATT	$A_{ks}^mC_{ks}T_{ds}A_{ds}^mC_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}G_{ds}T_{ks}T_{es}A_{ks}T_{es}T_k$	42	899
1341600	N/A	N/A	12416	12431	ACGGTGATCAAATGTA	$A_{ks}^mC_{ks}G_{ds}G_{ds}T_{ds}G_{ds}A_{ds}T_{ds}^m$ $C_{ds}A_{ds}A_{ds}A_{ks}T_{es}G_{ks}T_{es}A_k$	33	2205
1341601	N/A	N/A	16218	16233	TATAGACTGGGTAGGA	$T_{ks}A_{ks}T_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}G_d$ $s^sG_{ds}G_{ds}T_{ks}A_{es}G_{ks}G_{es}A_k$	38	2808
1341607	N/A	N/A	12426	12441	CAGGTTGGTAACGGTG	$^mC_{ks}A_{ks}G_{ds}G_{ds}T_{ds}G_{ds}G_{ds}T_d$ $s^sA_{ds}A_{ds}^mC_{ks}G_{es}G_{ks}T_{es}G_k$	49	2729

TABLE 41

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 2 Stop Site	SEQ ID NO.: 3 Start Site	SEQ ID NO.: 4 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO.
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G_{ds}T_{ds}$ $sP_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_{ks}$	93	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_{ks}$	88	1526
1341165	608	623	N/A	N/A	ACAATATGGGATGAGG	$A_{ks}{}^mC_{ks}A_{ds}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}G_{ds}$ $G_{ds}A_{ds}T_{ks}G_{es}A_{ks}G_{es}G_{ks}$	51	1579
1341170	810	825	15691	15706	TATTGGTAAGTATTCC	$T_{ks}A_{ks}T_{ds}T_{ds}G_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}$ $sT_{ds}A_{ks}{}^mC_{es}T_{ks}{}^mC_{ks}$	36	1272
1341174	798	813	15679	15694	TTCCATCTATCAGACT	$T_{ks}T_{ks}{}^mC_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}$ $sT_{ds}{}^mC_{ds}A_{ks}G_{es}A_{ks}{}^mC_{es}T_{ks}$	33	571
1341177	696	711	12114	12129	GACATGAGGTTTTGAT	$G_{ks}A_{ks}{}^mC_{ds}A_{ds}T_{ds}G_{ds}A_{ds}G_{ds}G_{ds}$ $T_{ds}T_{ds}T_{ks}T_{es}G_{ks}{}^mC_{es}T_{ks}$	55	647
1341181	1078	1093	20893	20908	CTAGTGCCAAACCAAT	${}^mC_{ks}T_{ks}A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}$ $dsA_{ds}{}^mC_{ks}{}^mC_{ks}C_{es}A_{ks}A_{es}T_{ks}$	47	1276
1341185	1082	1097	20897	20912	GCTGCTAGTGCCAAAC	$G_{ks}{}^mC_{ks}T_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}{}^mC_{ds}{}^mC_{ks}A_{es}A_{ks}A_{es}{}^mC_{ks}$	62	1587
1341195	1180	1195	20995	21010	GTTTTTAAGAGCATG	$G_{ks}T_{ks}T_{ds}T_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_{ds}$ $G_{ds}G_{ks}{}^mC_{es}A_{ks}T_{es}G_{ks}$	79	1199
1341197	1086	1101	20901	20916	GACTGCTGCTAGTGCC	$G_{ks}A_{ks}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}$ $dsT_{ds}{}^mC_{ks}G_{es}G_{ks}{}^mC_{es}{}^mC_{ks}$	35	1899
1341201	1314	1329	21129	21144	CAGTCTTAAACCTTCC	${}^mC_{ks}A_{ks}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ks}T_{es}T_{ks}{}^mC_{es}{}^mC_{ks}$	51	30
1341206	1256	1271	21071	21086	GCTTTTGTCACCTTT	$G_{ks}{}^mC_{ks}T_{ds}T_{ds}T_{ds}T_{ds}G_{ds}T_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}{}^mC_{ks}{}^mC_{es}T_{ks}T_{es}T_{ks}$	75	1744
1341207	1261	1276	21076	21091	AGGTAGCTTTTGTCCA	$A_{ks}G_{ks}G_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}$ $dsT_{ds}G_{ks}T_{es}{}^mC_{ks}{}^mC_{es}A_{ks}$	26	109
1341210	1326	1341	21141	21156	ATGCTACTTGAACAGT	$A_{ks}T_{ks}G_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}A_{ks}{}^mC_{es}A_{ks}G_{es}T_{ks}$	54	967
1341212	1322	1337	21137	21152	TACTTGAACAGTCTTA	$T_{ks}A_{ks}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}G_{ds}T_{ks}{}^mC_{es}T_{ks}T_{es}A_{ks}$	60	656
1341219	1318	1333	21133	21148	TGAACAGTCTTAAACC	$T_{ks}G_{ks}A_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}$ $T_{ds}T_{ds}A_{ks}A_{es}A_{ks}{}^mC_{es}{}^mC_{ks}$	34	344
1341225	1344	1359	21159	21174	GCATGGCTACAGATTG	$G_{ks}{}^mC_{ks}A_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}$ ${}^mC_{ds}A_{ds}G_{ks}A_{es}T_{ks}T_{es}G_{ks}$	41	267
1341231	1354	1369	21169	21184	ATATTCTGTGGCATGG	$A_{ks}T_{ks}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}$ $dsG_{ds}{}^mC_{ks}A_{es}T_{ks}G_{es}G_{ks}$	43	968
1341233	1358	1373	21173	21188	GTTGATATTCTGTGCG	$G_{ks}T_{ks}T_{ds}G_{ds}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}$ $dsT_{ds}G_{ks}T_{es}G_{ks}G_{es}{}^mC_{ks}$	69	1280
1341238	1349	1364	21164	21179	CTGTGGCATGGCTACA	${}^mC_{ks}T_{ks}G_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}$ $G_{ds}G_{ds}{}^mC_{ks}T_{es}A_{ks}{}^mC_{es}A_{ks}$	33	657
1341240	1395	1410	21210	21225	ACTTGATCTCTTAGCT	$A_{ks}{}^mC_{ks}T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}T_{ks}A_{es}G_{ks}{}^mC_{es}T_{ks}$	62	813
1341243	1389	1404	21204	21219	TCTCTTAGCTGTGCAC	$T_{ks}{}^mC_{ks}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}$ $sT_{ds}G_{ds}T_{ks}G_{es}{}^mC_{ks}A_{es}{}^mC_{ks}$	36	346
1341246	1367	1382	21182	21197	TGTGTTCTTGTGTGATA	$T_{ks}G_{ks}T_{ds}G_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}$ $dsT_{ds}T_{ks}G_{es}A_{ks}T_{es}A_{ks}$	64	1591
1341251	1429	1444	21244	21259	ATATGTCCAGGTTGAG	$A_{ks}T_{ks}A_{ds}T_{ds}G_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}G_{ds}T_{ks}T_{es}G_{ks}A_{es}G_{ks}$	28	425

TABLE 41 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341257	1418	1433	21233	21248	TTGAGATAAAGCTGCC	$T_{ks}T_{ks}G_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}A_{ds}A_{ds}$ $sG_{ds}{}^mC_{ks}T_{es}G_{ks}{}^mC_{es}{}^mC_k$	2	1670
1341259	1422	1437	21237	21252	CAGGTTGAGATAAAGC	${}^mC_{ks}A_{ks}G_{ds}G_{ds}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}$ $A_{ds}T_{ds}A_{ks}{}^mC_{es}A_{ks}G_{es}{}^mC_k$	53	1982
1341260	1491	1506	21306	21321	ACCGTTTTGGGCTAAT	$A_{ks}{}^mC_{ks}{}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}{}^mC_{ks}T_{es}A_{ks}A_{es}T_k$	76	1438
1341261	1495	1510	21310	21325	TTGCACCGTTTGGGCG	$T_{ks}T_{ks}G_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}G_{ds}T_d$ $sT_{ds}T_{ds}T_{ks}G_{es}G_{ks}G_{es}{}^mC_k$	26	1749
1341268	1466	1481	21281	21296	AGAGGCTAGGGAAATC	$A_{ks}G_{ks}A_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}G_{ds}$ $G_{ds}G_{ds}A_{ks}{}^mC_{es}A_{ks}T_{es}{}^mC_k$	59	1048
1341273	1499	1514	21314	21329	AGAGTTGCACCGTTTT	$A_{ks}G_{ks}A_{ds}G_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}{}^m$ $C_{ds}{}^mC_{ds}G_{ks}T_{es}T_{ks}T_{es}T_k$	80	34
1341279	1600	1615	21415	21430	GTATCTTATAAGACTA	$G_{ks}T_{ks}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}T_{ds}A$ $dsA_{ds}G_{ks}A_{es}{}^mC_{ks}T_{es}A_k$	61	1595
1341285	1624	1639	21439	21454	ATAGAGTCGGTCACCT	$A_{ks}T_{ks}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}$ $G_{ds}T_{ds}{}^mC_{ks}A_{es}{}^mC_{ks}{}^mC_{es}T_k$	56	584
1341288	1620	1635	21435	21450	AGTCGGTCACCTTTCA	$A_{ks}G_{ks}T_{ds}{}^mC_{ds}G_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}$ ${}^mC_{ds}{}^mC_{ds}T_{ks}T_{es}T_{ks}{}^mC_{es}A_k$	55	272
1341298	1712	1727	21527	21542	ACATCTCTGGGACCAA	$A_{ks}{}^mC_{ks}A_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_d$ $sG_{ds}G_{ds}{}^mC_{ks}{}^mC_{es}{}^mC_{ks}A_{es}A_k$	67	195
1341299	1628	1643	21443	21458	TAAATAGAGTCGGTC	$T_{ks}A_{ks}A_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}A_{ds}G$ $dsT_{ds}{}^mC_{ks}G_{es}G_{ks}{}^mC_k$	68	895
1341306	1721	1736	21536	21551	ATTGTCTAAACATCTC	$A_{ks}T_{ks}T_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}A_{ds}A$ $ds{}^mC_{ds}A_{ks}T_{es}{}^mC_{ks}T_{es}{}^mC_k$	62	585
1341308	1752	1767	21567	21582	TCCTGTGTAGCTTTA	$T_{ks}{}^mC_{ks}{}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}{}^mC_{ks}T_{es}T_{ks}T_{es}A_k$	79	1442
1341312	1784	1799	21599	21614	GTTTCTTATGTAATAG	$G_{ks}T_{ks}T_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}T_{ds}G$ $dsT_{ds}{}^mC_{ks}A_{es}T_{ks}A_{es}G_k$	21	897
1341317	1780	1795	21595	21610	CTTATGTAATAGCCAG	${}^mC_{ks}T_{ks}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}T$ $dsA_{ds}G_{ks}{}^mC_{es}{}^mC_{ks}A_{es}G_k$	61	664
1341320	2189	2204	22004	22019	CTTGATGTAGTGGGAG	${}^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}G$ $dsT_{ds}G_{ks}G_{es}G_{ks}A_{es}G_k$	20	976
1341324	2252	2267	22067	22082	AGAGGATAGTCCATGC	$A_{ks}G_{ks}A_{ds}G_{ds}G_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T$ $ds{}^mC_{ds}{}^mC_{ks}A_{es}T_{ks}G_{es}{}^mC_k$	70	354
1341330	N/A	N/A	9334	9349	CGATCCTATATACATC	${}^mC_{ks}G_{ks}A_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}A_{ds}T_d$ $sA_{ds}T_{ds}{}^mC_{ks}{}^mC_{es}A_{ks}T_{es}{}^mC_k$	29	2788
1341331	N/A	N/A	9516	9531	ACACTAATATTGAGGC	$A_{ks}{}^mC_{ks}A_{ds}{}^mC_{ds}T_{ds}A_{ds}T_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ks}A_{es}G_{ks}G_{es}{}^mC_k$	55	1928
1341336	2353	2368	22168	22183	ATAACTACAAGAGGTT	$A_{ks}T_{ks}A_{ds}A_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ks}G_{es}G_{ks}T_{es}T_k$	46	1133
1341340	N/A	N/A	11546	11561	TCTTGACAATGGTTGC	$T_{ks}{}^mC_{ks}T_{ds}T_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}A_{ds}$ $T_{ds}G_{ds}G_{ks}T_{es}T_{ks}G_{es}{}^mC_k$	37	2199
1341341	N/A	N/A	9968	9983	AGTAGTGTAAGCTGAG	$A_{ks}G_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}T_{ds}A_{ds}A_d$ $sG_{ds}{}^mC_{ks}T_{es}G_{ks}A_{es}G_k$	52	2155
1341349	N/A	N/A	9603	9618	CACATATGTGAAGAGC	${}^mC_{ks}A_{ks}{}^mC_{ds}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}$ $G_{ds}A_{ds}A_{ks}G_{es}A_{ks}G_{es}{}^mC_k$	53	760
1341352	N/A	N/A	15498	15513	TGCTGCATTAATGCCA	$T_{ks}G_{ks}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}A_{ds}T_{ks}G_{es}{}^mC_{ks}{}^mC_{es}A_k$	30	2485

TABLE 41 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound 1	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341579	1771	1786	21586	21601	TAGCCAGTACAGTTCC	$T_{ks}A_{ks}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}$ ${}^mC_{ds}A_{ds}G_{ks}T_{es}{}^mC_{es}{}^mC_k$	57	1987
1341581	1716	1731	21531	21546	CTAACATCTCTGGGA	${}^mC_{ks}T_{ks}A_{ds}A_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ks}G_{es}G_{ks}G_{es}A_k$	29	2259
1341582	2196	2211	22011	22026	GATTAGTCTTGATGTA	$G_{ks}A_{ks}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ks}T_{es}G_{ks}T_{es}A_k$	56	1521
1341584	1776	1791	21591	21606	TGTAATAGCCAGTACA	$T_{ks}G_{ks}T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ks}T_{es}A_{ks}{}^mC_{es}A_k$	45	352
1341592	N/A	N/A	11578	11593	GTTATTCTCTTGACAA	$G_{ks}T_{ks}T_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ks}A_{es}{}^mC_{ks}A_{es}A_k$	27	2575
1341602	N/A	N/A	16292	16307	TTAGTTAGAATAGTCT	$T_{ks}T_{ks}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_{ds}A_d$ $sT_{ds}A_{ks}G_{es}T_{ks}{}^mC_{es}T_k$	53	1714
1341617	N/A	N/A	17995	18010	TGAAGTTAGTATAGTT	$T_{ks}G_{ks}A_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_d$ $sA_{ds}T_{ks}A_{es}G_{ks}T_{es}T_k$	46	1405

TABLE 42

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound 1	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G_{ds}T_d$ $sA_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_k$	84	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_k$	83	1526
1340984	1497	1512	21312	21327	AGTTGCACCGTTTTGG	$A_{ks}G_{ks}T_{ks}T_{ds}G_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}$ $dsG_{ds}T_{ds}T_{ds}T_{ks}T_{ks}G_{ks}G_e$	43	1905
1340987	1489	1504	21304	21319	CGTTTTGGGCTAATGA	${}^mC_{ks}G_{ks}T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}A_{ks}T_{ks}G_{ks}A_e$	52	1282
1340990	1618	1633	21433	21448	TCGGTCACCTTTTCATA	$T_{ks}{}^mC_{ks}G_{ks}G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}{}^mC_{ks}A_{ks}T_{ks}A_e$	72	116
1340997	1622	1637	21437	21452	AGAGTCGGTCACCTTT	$A_{ks}G_{ks}A_{ks}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}T_{ks}T_{ks}T_e$	66	428
1341001	1626	1641	21441	21456	AAATAGAGTCGGTCAC	$A_{ks}A_{ks}A_{ks}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ds}T_{ks}{}^mC_{ks}A_{ks}{}^mC_e$	48	740
1341005	1630	1645	21445	21460	TTTAAATAGAGTCGG	$T_{ks}T_{ks}T_{ks}A_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_d$ $sA_{ds}G_{ds}T_{ks}{}^mC_{ks}G_{ks}G_e$	58	1051
1341018	1710	1725	21525	21540	ATCTCTGGGACCAAGG	$A_{ks}T_{ks}{}^mC_{ks}T_{ds}{}^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}A_{ks}G_{ks}G_e$	53	37
1341019	1714	1729	21529	21544	AAACATCTCTGGGACC	$A_{ks}A_{ks}A_{ks}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ks}A_{ks}{}^mC_{es}{}^mC_e$	48	2129
1341026	1768	1783	21583	21598	CCAGTACAGTTTCCTTT	${}^mC_{ks}{}^mC_{ks}A_{ks}G_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}T_{ks}T_e$	76	1753
1341029	1750	1765	21565	21580	CTGTGTTAGCTTTAAT	${}^mC_{ks}T_{ks}G_{ks}T_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ks}A_{ks}A_{ks}T_e$	50	1286
1341031	2193	2208	22008	22023	TAGTCTTGATGTAGTG	$T_{ks}A_{ks}G_{ks}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ks}G_{ks}T_{ks}G_e$	61	1288

TABLE 42 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341036	1778	1793	21593	21608	TATGTAATAGCCAGTA	$T_{ks}A_{ks}T_{ks}G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}$ $s^mC_{ds}^mC_{ds}A_{ks}G_{ks}T_{ks}A_e$	82	508
1341038	1782	1797	21597	21612	TTCTTATGTAATAGCC	$T_{ks}T_{ks}^mC_{ks}T_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}$ $dsA_{ds}T_{ds}A_{ks}G_{ks}^mC_{ks}^mC_e$	73	819
1341047	2247	2262	22062	22077	ATAGTCCATGCAAAAG	$A_{ks}T_{ks}A_{ks}G_{ds}T_{ds}^mC_{ds}^mC_{ds}A_{ds}T_{ds}$ $G_{ds}^mC_{ds}A_{ds}A_{ks}A_{ks}G_e$	29	2436
1341058	N/A	N/A	9330	9345	CCTATATACATCCAAG	$^mC_{ks}^mC_{ks}T_{ks}A_{ds}T_{ds}A_{ds}T_{ds}A_{ds}^mC_{ds}$ $s^mP_{ds}T_{ds}^mC_{ds}^mC_{ks}A_{ks}G_e$	63	1771
1341063	1082	1097	20897	20912	GCTGCTAGTGCCAAAC	$G_{ks}^mC_{ks}T_{ks}G_{ds}^mC_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}^mC_{ds}^mC_{ds}A_{ks}A_{ks}A_{ks}^mC_e$	62	1587
1341064	N/A	N/A	11855	11870	TAAGCAGAATTGTGAA	$T_{ks}A_{ks}A_{ks}G_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ds}T_{ks}G_{ks}A_{ks}A_e$	52	2361
1341067	N/A	N/A	11571	11586	TCTTGACAATGGTTGA	$T_{ks}^mC_{ks}T_{ks}T_{ds}G_{ds}^mC_{ds}A_{ds}A_{ds}$ $T_{ds}G_{ds}G_{ds}T_{ks}T_{ks}A_e$	61	2440
1341069	N/A	N/A	9964	9979	GTGTAAGCTGAGAGTT	$G_{ks}T_{ks}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}A_{ks}G_{ks}T_{ks}T_e$	45	2760
1341070	N/A	N/A	14199	14214	TATTATGTGATTGAGT	$T_{ks}A_{ks}T_{ks}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}G_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ks}A_{ks}G_{ks}T_e$	51	73
1341073	N/A	N/A	15498	15513	TGCTGCATTAATGCCA	$T_{ks}G_{ks}^mC_{ks}T_{ds}G_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}A_{ds}T_{ds}T_{ks}^mC_{ks}^mC_{ks}A_e$	6	2485
1341079	N/A	N/A	14918	14933	TAATACATTAGCAAGC	$T_{ks}A_{ks}A_{ks}T_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}A_{ds}$ $dsG_{ds}^mC_{ds}A_{ks}A_{ks}G_{ks}^mC_e$	35	2266
1341082	N/A	N/A	16380	16395	CCAACCAAACTTCCAG	$^mC_{ks}^mC_{ks}A_{ks}A_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ds}$ $A_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ks}^mC_{ks}A_{ks}G_e$	47	2539
1341083	1078	1093	20893	20908	CTAGTGCCAAACCAAT	$^mC_{ks}T_{ks}A_{ks}G_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}$ $dsA_{ds}^mC_{ds}^mC_{ks}A_{ks}A_{ks}T_e$	61	1276
1341085	N/A	N/A	16287	16302	TAGAATAGTCTTCAGC	$T_{ks}A_{ks}G_{ks}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}^m$ $C_{ds}T_{ds}T_{ds}^mC_{ks}A_{ks}G_{ks}^mC_e$	50	1403
1341090	N/A	N/A	18697	18712	CAAAGACTTTGAGAC	$^mC_{ks}A_{ks}A_{ks}A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}$ $s^mT_{ds}T_{ds}G_{ds}A_{ks}G_{ks}^mC_e$	13	2186
1341095	N/A	N/A	18099	18114	CTGAAATTGATACACC	$^mC_{ks}T_{ks}G_{ks}A_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}A_{ds}^mC_{ks}^mC_{ks}^mC_e$	75	393
1341098	N/A	N/A	17992	18007	AGTTAGTATAGTTATC	$A_{ks}G_{ks}T_{ks}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}T_{ds}A_{ds}$ $sG_{ds}T_{ds}T_{ks}A_{ks}T_{ks}^mC_e$	47	1249
1341107	N/A	N/A	19907	19922	CTTAAGATACCCAGGT	$^mC_{ks}T_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}^m$ $C_{ds}^mC_{ds}^mC_{ks}G_{ks}G_{ks}T_e$	45	241
1341108	N/A	N/A	19911	19926	GCATCTTAAGATACCC	$G_{ks}^mC_{ks}A_{ks}T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}A_{ks}^mC_{ks}^mC_{ks}^mC_e$	84	553
1341110	N/A	N/A	20383	20398	GTAATCACAAGTAAGG	$G_{ks}T_{ks}A_{ks}A_{ds}T_{ds}^mC_{ds}A_{ds}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}T_{ds}A_{ks}A_{ks}G_{ks}G_e$	56	1411
1341116	N/A	N/A	20194	20209	TGCAAGTTATTAGAAG	$T_{ks}T_{ks}G_{ks}^mC_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}T_{ds}$ $dsT_{ds}A_{ds}G_{ks}A_{ks}A_{ks}G_e$	10	2123
1341122	1086	1101	20901	20916	GACTGCTGCTAGTGCC	$G_{ks}A_{ks}^mC_{ks}T_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}^mC_{ds}$ $dsT_{ds}A_{ds}G_{ds}T_{ks}G_{ks}^mC_{ks}^mC_e$	49	1899
1341123	N/A	N/A	12801	12816	AGGCTGTAGAGTCAG	$A_{ks}G_{ks}G_{ks}^mC_{ds}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}$ $G_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}^mC_{ks}G_e$	49	2809
1341125	N/A	N/A	12503	12518	TTGCAGAAATTCACCT	$T_{ks}T_{ks}G_{ks}^mC_{ds}A_{ds}G_{ds}A_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}^mC_{ds}A_{ks}^mC_{ks}^mC_{ks}T_e$	29	2528

TABLE 42 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341134	N/A	N/A	13954	13969	CCTCTAAGTTAGCCCC	${}^mC_{ks}{}^mC_{ks}T_{ks}{}^mC_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_d$ $sT_{ds}A_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_{ks}{}^mC_e$	16	1084
1341136	N/A	N/A	14168	14183	ATAGCAAGCCAACAGA	$A_{ks}T_{ks}A_{ks}G_{ds}{}^mC_{ds}A_{ds}A_{ds}G_{ds}{}^mC_d$ $s{}^mC_{ds}A_{ds}A_{ds}{}^mC_{ks}A_{ks}G_{ks}A_e$	30	2633
1341144	1258	1273	21073	21088	TAGCTTTTGTCCACCT	$T_{ks}A_{ks}G_{ks}{}^mC_{ds}T_{ds}T_{ds}T_{ds}G_{ds}T$ $ds{}^mC_{ds}{}^mC_{ds}A_{ks}{}^mC_{ks}{}^mC_{ks}T_e$	65	1900
1341146	1200	1215	21015	21030	ATGTTTATGTAAGCAC	$A_{ks}T_{ks}G_{ks}T_{ds}T_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ks}{}^mC_{ks}A_{ks}{}^mC_e$	73	2622
1341148	N/A	N/A	14192	14207	TGATTGAGTTCTCCAC	$T_{ks}G_{ks}A_{ks}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}T_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}{}^mC_e$	30	2116
1341149	606	621	10492	10507	AATATGGGATGAGGTA	$A_{ks}A_{ks}T_{ks}A_{ds}T_{ds}G_{ds}G_{ds}G_{ds}A_{ds}T_d$ $sG_{ds}A_{ds}G_{ks}G_{ks}T_{ks}A_e$	47	1424
1341155	1312	1327	21127	21142	GTCCTAAACCTTCCCT	$G_{ks}T_{ks}{}^mC_{ks}T_{ds}T_{ds}A_{ds}A_{ds}A_{ds}{}^mC_{ds}$ ${}^mC_{ds}T_{ds}T_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_{ks}T_e$	50	1901
1341157	1316	1331	21131	21146	AACAGTCTTAAACCTT	$A_{ks}A_{ks}{}^mC_{ks}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $A_{ds}A_{ds}A_{ds}{}^mC_{ks}{}^mC_{ks}T_{ks}T_e$	57	188
1341161	1322	1337	21137	21152	TACTTGAACAGTCTTA	$T_{ks}A_{ks}{}^mC_{ks}T_{ds}T_{ds}G_{ds}A_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}G_{ds}T_{ds}{}^mC_{ks}T_{ks}A_e$	65	656
1341162	1333	1348	21148	21163	GATTGGAATGCTACTT	$G_{ks}A_{ks}T_{ks}T_{ds}G_{ds}G_{ds}A_{ds}A_{ks}T_{ds}G_d$ $s{}^mC_{ds}T_{ds}A_{ks}{}^mC_{ks}T_{ks}T_e$	59	1435
1341169	1326	1341	21141	21156	ATGCTACTTGAACAGT	$A_{ks}T_{ks}G_{ks}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}A_{ds}{}^mC_{ks}A_{ks}G_{ks}T_e$	50	967
1341220	1342	1357	21157	21172	ATGGCTACAGATTGGA	$A_{ks}T_{ks}G_{ks}G_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}T_{ks}G_{ks}A_e$	65	111
1341283	1356	1371	21171	21186	TGATATTCTGTGGCAT	$T_{ks}G_{ks}A_{ks}T_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}G$ $dsT_{ds}G_{ds}G_{ks}{}^mC_{ks}A_{ks}T_e$	74	1124
1341309	1352	1367	21167	21182	ATTCTGTGGCATGGCT	$A_{ks}T_{ks}{}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}G_{ds}{}^m$ $C_{ds}A_{ds}T_{ds}G_{ks}G_{ks}{}^mC_{ks}T_e$	68	2328
1341366	1364	1379	21179	21194	GTTCTTGTGTGATATTC	$G_{ks}T_{ks}T_{ks}{}^mC_{ds}T_{ds}T_{ds}G_{ds}T_{ds}T_{ds}G$ $dsA_{ds}T_{ds}A_{ks}T_{ks}{}^mC_e$	69	1436
1341420	1418	1433	21233	21248	TTGAGATAAAGCTGCC	$T_{ks}T_{ks}G_{ks}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}A_{ks}A_d$ $sG_{ds}{}^mC_{ds}T_{ks}G_{ks}{}^mC_{ks}{}^mC_e$	43	1670
1341422	1389	1404	21204	21219	TCTCTTAGCTGTGCAC	$T_{ks}{}^mC_{ks}T_{ks}{}^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_d$ $sT_{ds}G_{ds}T_{ds}G_{ks}{}^mC_{ks}A_{ks}{}^mC_e$	69	346
1341423	1422	1437	21237	21252	CAGGTTGAGATAAAGC	${}^mC_{ks}A_{ks}G_{ks}G_{ds}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}$ $A_{ds}T_{ds}A_{ds}A_{ks}A_{ks}G_{ks}{}^mC_e$	63	1982
1341430	1493	1508	21308	21323	GCACCGTTTTGGGCTA	$G_{ks}{}^mC_{ks}A_{ks}{}^mC_{ds}{}^mC_{ds}G_{ds}T_{ds}T_{ds}T_d$ $sT_{ds}G_{ds}G_{ds}G_{ks}{}^mC_{ks}T_{ks}A_e$	45	1593
1341438	1504	1519	21319	21334	AGAATAGAGTTGCACC	$A_{ks}G_{ks}A_{ks}A_{ds}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T$ $dsT_{ds}G_{ds}{}^mC_{ks}A_{ks}{}^mC_{ks}{}^mC_e$	58	426
1341440	1461	1476	21276	21291	CTAGGGAAATCTTTCA	${}^mC_{ks}T_{ks}A_{ks}G_{ds}G_{ds}G_{ds}A_{ds}A_{ds}A_{ds}$ $T_{ds}{}^mC_{ds}T_{ds}T_{ks}T_{ks}{}^mC_{ks}A_e$	55	659
1341443	1614	1629	21429	21444	TCACCTTTCATAATGT	$T_{ks}{}^mC_{ks}A_{ks}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}{}^m$ $C_{ds}A_{ds}T_{ds}A_{ks}A_{ks}T_{ks}G_{ks}T_e$	41	2109
1341444	805	820	15686	15701	GTAAGTATTCCATCTA	$G_{ks}T_{ks}A_{ks}A_{ds}G_{ds}T_{ds}A_{ds}T_{ds}T_{ds}{}^mC$ $ds{}^mC_{ds}A_{ds}T_{ks}{}^mC_{ks}T_{ks}A_e$	39	1038
1341445	613	628	N/A	N/A	CTGGAACAATATGGGA	${}^mC_{ks}T_{ks}G_{ks}G_{ds}A_{ds}A_{ds}{}^mC_{ds}A_{ds}A_d$ $sT_{ds}A_{ds}T_{ds}G_{ks}G_{ks}A_e$	36	1969

TABLE 42 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% in- hibition	SEQ ID NO
1341452	1718	1733	21533	21548	GTCTAAACATCTCTGG	$G_{ks}T_{ks}^mC_{ks}T_{ds}^mA_{ds}A_{ds}^mC_{ds}^mA_{ds}$ $G_{ds}T_{ds}^mC_{ks}^mG_{ks}T_e$	49	429
1341453	1774	1789	21589	21604	TAATAGCCAGTACAGT	$T_{ks}A_{ks}A_{ks}T_{ds}^mG_{ds}^mC_{ds}^mC_{ds}^mA_{ds}$ $G_{ds}T_{ds}^mC_{ks}^mG_{ks}T_e$	64	196
1341459	1074	1089	20889	20904	TGCCAAACCAATGTTT	$T_{ks}G_{ks}^mC_{ks}^mC_{ds}^mA_{ds}A_{ds}^mC_{ds}^m$ $C_{ds}^mA_{ds}T_{ds}G_{ks}T_{ks}T_e$	43	2673
1341461	N/A	N/A	9511	9526	AATATTGAGGCACTGG	$A_{ks}A_{ks}T_{ks}A_{ds}T_{ds}G_{ds}A_{ds}G_{ds}G_d$ $s^mC_{ds}^mC_{ks}^mT_{ks}^mG_{ks}G_e$	36	1538
1341462	N/A	N/A	9598	9613	ATGTGAAGAGCTGGTA	$A_{ks}T_{ks}G_{ks}T_{ds}G_{ds}A_{ks}A_{ds}G_{ds}A_{ds}G$ $ds^mC_{ds}T_{ds}G_{ks}G_{ks}T_{ks}A_e$	56	527
1341464	2257	2272	22072	22087	AAACAAGAGGATATGC	$A_{ks}A_{ks}A_{ks}^mC_{ds}^mA_{ds}A_{ds}G_{ds}A_{ds}G_{ds}$ $G_{ds}A_{ks}T_{ds}A_{ks}G_{ks}T_{ks}^mC_e$	45	744
1341465	2186	2201	22001	22016	GATGTAGTGGGAGTCG	$G_{ks}A_{ks}T_{ks}G_{ds}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}G_d$ $s^mG_{ds}A_{ds}G_{ks}T_{ks}^mC_{ks}G_e$	42	743
1341470	N/A	N/A	12416	12431	ACGGTGATCAAATGTA	$A_{ks}^mC_{ks}G_{ks}G_{ds}T_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}$ $A_{ds}A_{ds}A_{ds}T_{ks}G_{ks}T_{ks}A_e$	41	2205
1341472	N/A	N/A	12426	12441	CAGGGTGGTAACGGTG	$^mC_{ks}A_{ks}G_{ks}G_{ds}T_{ds}G_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}^mC_{ds}G_{ks}G_{ks}T_{ks}G_e$	39	2729
1341478	N/A	N/A	17997	18012	TATGAAGTTAGTATAG	$T_{ks}A_{ks}T_{ks}G_{ds}A_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_d$ $s^mG_{ds}T_{ds}A_{ks}T_{ks}A_{ks}G_e$	13	1560
1341480	N/A	N/A	16134	16149	GATGCTCCATAATAAT	$G_{ks}A_{ks}T_{ks}G_{ds}^mC_{ds}T_{ds}^mC_{ds}^mA$ $dsT_{ds}A_{ds}A_{ds}T_{ks}A_{ks}T_e$	15	2049
1341486	N/A	N/A	16124	16139	AATAATAGCTCTATTG	$A_{ks}A_{ks}T_{ks}A_{ds}T_{ds}A_{ds}^mC_{ds}$ $T_{ds}^mC_{ds}T_{ds}A_{ks}T_{ks}G_e$	3	2410
1341494	N/A	N/A	19915	19930	GAGTGCATCTTAAGAT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}^mC_{ds}A_{ds}T_{ds}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ks}G_{ks}A_{ks}T_e$	34	2256
1341496	N/A	N/A	17987	18002	GTATAGTTATCTTCTC	$G_{ks}T_{ks}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_{ks}T_{ds}$ $^mC_{ds}T_{ds}T_{ks}^mC_{ks}T_{ks}^mC_e$	68	937
1341499	N/A	N/A	20489	20504	CATAACTTCATAGTGG	$^mC_{ks}A_{ks}T_{ks}A_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}^mC_d$ $s^mA_{ds}T_{ds}A_{ds}G_{ks}T_{ks}G_{ks}G_e$	16	866
1341501	N/A	N/A	20109	20124	AAGACATCTAGCCTG	$A_{ks}A_{ks}G_{ks}A_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}$ $T_{ds}A_{ds}G_{ds}^mC_{ks}^mC_{ks}T_{ks}G_e$	40	2744
1341506	N/A	N/A	20119	20134	TTACAGCATCAAGACA	$T_{ks}A_{ks}^mC_{ks}^mA_{ds}G_{ds}^mC_{ds}A_{ds}T_{ds}$ $^mC_{ds}A_{ds}A_{ds}G_{ks}A_{ks}^mC_{ks}A_e$	44	2409
1341517	N/A	N/A	20479	20494	TAGTGGACTTCATTAG	$T_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}T$ $ds^mC_{ds}A_{ds}T_{ks}^mC_{ks}A_{ks}G_e$	22	2351

TABLE 43

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% in- hibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G$ $dsT_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	88	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ks}dsT_{ds}^m$ $C_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}A_{ks}G_{ks}T_k$	84	1526

TABLE 43 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1340982	1490	1505	21305	21320	CCGTTTTGGGCTAATG	${}^mC_{ks}{}^mC_{ks}G_{ks}T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ks}A_{ks}T_{ks}G_e$	69	1360
1340988	1494	1509	21309	21324	TGCACCGTTTTGGGCT	$T_{ks}G_{ks}{}^mC_{ks}A_{ds}{}^mC_{ds}{}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ds}G_{ds}G_{ks}G_{ks}{}^mC_{ks}T_e$	43	1671
1340993	1619	1634	21434	21449	GTCGGTCACCTTTTCAT	$G_{ks}T_{ks}{}^mC_{ks}G_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ks}{}^mC_{ks}A_{ks}T_e$	68	194
1340994	1598	1613	21413	21428	ATCTTATAAGACTATA	$A_{ks}T_{ks}{}^mC_{ks}T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}{}^mC_{ds}T_{ks}A_{ks}T_{ks}A_e$	53	2661
1341006	1627	1642	21442	21457	AAAATAGAGTCGGTCA	$A_{ks}A_{ks}A_{ks}A_{ds}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ks}T_{ks}{}^mC_{ks}A_e$	65	817
1341008	808	823	15689	15704	TTGGTAAGTATTCAT	$T_{ks}T_{ks}G_{ks}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}T_e$	28	2750
1341009	1623	1638	21438	21453	TAGAGTCGGTCACCTT	$T_{ks}A_{ks}C_{ks}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ks}T_{ks}T_e$	59	506
1341012	1719	1734	21534	21549	TGTCTAAACATCTCTG	$T_{ks}G_{ks}T_{ks}{}^mC_{ds}T_{ds}A_{ds}A_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ks}{}^mC_{ks}T_{ks}G_e$	69	507
1341014	1711	1726	21526	21541	CATCTCTGGGACCAAG	${}^mC_{ks}A_{ks}T_{ks}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}A_{ks}G_e$	42	117
1341016	1715	1730	21530	21545	TAAACATCTCTGGGAC	$T_{ks}A_{ks}A_{ks}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}G_{ks}A_{ks}{}^mC_e$	32	351
1341020	1017	1032	20832	20847	CAGCATTGATTCGAAA	${}^mC_{ks}A_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}G_{ks}A_{ks}A_{ks}G_e$	53	263
1341025	1775	1790	21590	21605	GTAATAGCCAGTACAG	$G_{ks}T_{ks}A_{ks}A_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ks}{}^mC_{ks}A_{ks}G_e$	62	274
1341027	1751	1766	21566	21581	CCTGTGTAGCTTTAA	${}^mC_{ks}{}^mC_{ks}T_{ks}G_{ds}T_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ks}T_{ks}A_{ks}A_e$	66	1364
1341028	1770	1785	21585	21600	AGCCAGTACAGTTCCT	$A_{ks}G_{ks}{}^mC_{ks}{}^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}T_{ks}{}^mC_{ks}{}^mC_{ks}T_e$	59	1909
1341032	1779	1794	21594	21609	TTATGTAATAGCCAGT	$T_{ks}T_{ks}A_{ks}T_{ds}G_{ds}T_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}G_{ds}{}^mC_{ds}{}^mC_{ks}G_{ks}T_e$	77	586
1341035	2187	2202	22002	22017	TGATGTAGTGGGAGTC	$T_{ks}G_{ks}A_{ks}T_{ds}G_{ds}T_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ds}G_{ds}A_{ks}G_{ks}T_{ks}{}^mC_e$	77	820
1341037	1783	1798	21598	21613	TTTCTTATGTAATAGC	$T_{ks}T_{ks}T_{ks}{}^mC_{ds}T_{ds}T_{ds}A_{ks}T_{ds}G_{ds}T_{ds}T_{ds}A_{ds}T_{ks}A_{ks}G_{ks}{}^mC_e$	60	2128
1341041	1079	1094	20894	20909	GCTAGTGCCAAACCAG	$G_{ks}{}^mC_{ks}T_{ks}A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}A_e$	70	1354
1341046	2250	2265	22065	22080	AGGATAGTCCATGCAA	$A_{ks}G_{ks}G_{ks}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ks}{}^mC_{ks}A_{ks}A_e$	61	2690
1341050	N/A	N/A	9332	9347	ATCCTATATACATCCA	$A_{ks}T_{ks}{}^mC_{ks}{}^mC_{ds}T_{ds}A_{ds}T_{ds}A_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	75	1927
1341052	N/A	N/A	9966	9981	TAGTGAAGCTGAGAG	$T_{ks}A_{ks}G_{ks}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ds}G_{ds}A_{ks}G_{ks}S_{ks}G_e$	82	1852
1341054	N/A	N/A	9601	9616	CATATGTGAAGAGCTG	${}^mC_{ks}A_{ks}T_{ks}A_{ds}T_{ds}G_{ds}T_{ds}G_{ds}A_{ds}A_{ds}G_{ds}A_{ds}G_{ks}{}^mC_{ks}T_{ks}G_e$	50	683
1341055	N/A	N/A	9514	9529	ACTAATATTGAGGCAC	$A_{ks}{}^mC_{ks}T_{ks}A_{ds}T_{ds}T_{ds}A_{ds}T_{ds}T_{ds}G_{ds}A_{ds}G_{ks}{}^mC_{ks}A_{ks}{}^mC_e$	62	1772
1341061	N/A	N/A	12419	12434	GTAACGGTGATCAAAT	$G_{ks}T_{ks}A_{ks}A_{ds}{}^mC_{ds}G_{ds}G_{ds}T_{ds}G_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ks}A_{ks}A_{ks}T_e$	32	2345

TABLE 43 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341062	N/A	N/A	11548	11563	TCTCTTGACAATGGTT	$T_{ks}^{mC_{ks}}T_{ks}^{mC_{ds}}T_{ds}T_{ds}G_{ds}A_{ds}$ $mC_{ds}A_{ds}A_{ds}T_{ds}G_{ks}G_{ks}T_{ks}T_e$	84 100	1389
1341080	N/A	N/A	16289	16304	GTTAGAATAGTCTTCA	$G_{ks}T_{ks}T_{ks}A_{ds}G_{ds}A_{ds}A_{ds}T_{ds}A_{ds}$ $dsG_{ds}T_{ds}^{mC_{ds}}T_{ks}T_{ks}^{mC_{ks}}A_e$	71	1481
1341087	N/A	N/A	16382	16397	TCCCAACCAAACCTCC	$T_{ks}^{mC_{ks}}C_{ks}^{mC_{ds}}A_{ds}A_{ds}^{mC_{ds}}$ $mC_{ds}A_{ds}A_{ds}A_{ds}^{mC_{ds}}T_{ks}T_{ks}^{mC_{ks}}mC_e$	24	2026
1341088	N/A	N/A	16127	16142	CATAATAATAGTCTCTA	$mC_{ks}A_{ks}T_{ks}A_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}^{mC_{ds}}T_{ks}^{mC_{ks}}T_{ks}A_e$	55	2511
1341093	1083	1098	20898	20913	TGCTGCTAGTGCCAAA	$T_{ks}G_{ks}^{mC_{ks}}T_{ds}G_{ds}^{mC_{ds}}T_{ds}A_{ds}$ $G_{ds}T_{ds}G_{ds}^{mC_{ds}}mC_{ks}A_{ks}A_{ks}A_e$	56	1665
1341097	N/A	N/A	17989	18004	TAGTATAGTTATCTTCT	$T_{ks}A_{ks}G_{ks}T_{ks}A_{ds}T_{ds}A_{ds}G_{ds}T_d$ $s^T_{ds}A_{ds}T_{ds}^{mC_{ds}}T_{ks}T_{ks}^{mC_{ks}}C_e$	28	1093
1341099	N/A	N/A	17994	18009	GAAGTTAGTATAGTTA	$G_{ks}A_{ks}A_{ks}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}$ $dsA_{ds}T_{ds}A_{ds}G_{ks}T_{ks}T_{ks}A_e$	75	1327
1341100	N/A	N/A	19908	19923	TCTTAAGATACCCAGG	$T_{ks}^{mC_{ks}}T_{ks}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}^{mC_{ds}}mC_{ds}^{mC_{ks}}A_{ks}G_{ks}G_e$	48	319
1341105	N/A	N/A	19912	19927	TGCATCTTAAGATACC	$T_{ks}G_{ks}^{mC_{ks}}A_{ds}T_{ds}^{mC_{ds}}T_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{ks}^{mC_{ks}}mC_e$	37	2050
1341113	N/A	N/A	20385	20400	TAGTAATCACAAGTAA	$T_{ks}A_{ks}G_{ks}T_{ks}A_{ds}A_{ds}T_{ds}^{mC_{ds}}$ $A_{ds}^{mC_{ds}}A_{ds}A_{ds}G_{ks}T_{ks}A_{ks}A_e$	65	1489
1341114	N/A	N/A	20112	20127	ATCAAGACATTCTAGC	$AT_{ks}^{mC_{ks}}A_{ds}A_{ds}G_{ds}A_{ds}^{mC_{ds}}$ $s^T_{ds}T_{ds}T_{ds}^{mC_{ds}}mC_{ds}T_{ks}A_{ks}G_{ks}^{mC_e}$	52	1488
1341119	1088	1103	20903	20918	TTGACTGCTGCTAGTG	$T_{ks}T_{ks}G_{ks}A_{ds}^{mC_{ds}}T_{ds}G_{ds}^{mC_{ds}}$ $T_{ds}G_{ds}^{mC_{ds}}mC_{ds}T_{ds}A_{ks}G_{ks}T_{ks}G_e$	50	28
1341127	N/A	N/A	20482	20497	TCATAGTGGACTTCAT	$T_{ks}^{mC_{ks}}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}$ $G_{ds}A_{ds}^{mC_{ds}}mC_{ds}T_{ds}T_{ks}^{mC_{ks}}T_e$	54	633
1341130	N/A	N/A	13956	13971	TACCTCTAAGTTAGCC	$T_{ks}A_{ks}^{mC_{ks}}mC_{ds}T_{ds}^{mC_{ds}}T_{ds}A_{ds}$ $dsA_{ds}G_{ds}T_{ds}^{mC_{ds}}A_{ks}G_{ks}^{mC_{ks}}mC_e$	30	2249
1341132	N/A	N/A	14194	14209	TGTGATTGAGTTCTCC	$T_{ks}G_{ks}T_{ks}G_{ds}A_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $sG_{ds}T_{ds}^{mC_{ds}}mC_{ks}T_{ks}^{mC_{ks}}mC_e$	52	2020
1341133	N/A	N/A	14170	14185	GAATAGCAAGCCAACA	$G_{ks}A_{ks}A_{ks}T_{ds}A_{ds}G_{ds}^{mC_{ds}}$ $A_{ds}G_{ds}^{mC_{ds}}mC_{ds}A_{ks}A_{ks}^{mC_{ks}}mC_e$	30	1708
1341141	1259	1274	21074	21089	GTAGCTTTTGTCCACC	$G_{ks}T_{ks}A_{ks}G_{ds}^{mC_{ds}}T_{ds}T_{ds}T_{ds}T_{ds}$ $dsG_{ds}T_{ds}^{mC_{ds}}mC_{ds}mC_{ks}A_{ks}^{mC_{ks}}mC_e$	67	1978
1341147	1202	1217	21017	21032	GTATGTTTATGTAAAGC	$G_{ks}T_{ks}A_{ks}T_{ds}G_{ds}T_{ds}T_{ds}T_{ds}A_{ds}$ $s^T_{ds}G_{ds}T_{ds}A_{ks}A_{ks}G_{ks}^{mC_e}$	71	1511
1341152	1317	1332	21132	21147	GAACAGTCTTAAACCT	$G_{ks}A_{ks}A_{ks}^{mC_{ds}}A_{ds}G_{ds}T_{ds}^{mC_{ds}}$ $s^T_{ds}T_{ds}T_{ds}A_{ds}A_{ks}A_{ks}^{mC_{ks}}mC_{ks}T_e$	64	266
1341159	1313	1328	21128	21143	AGTCTTAAACCTTCCC	$A_{ks}G_{ks}T_{ks}^{mC_{ds}}T_{ds}T_{ds}A_{ds}A_{ds}$ $A_{ds}^{mC_{ds}}mC_{ds}T_{ds}T_{ks}^{mC_{ks}}mC_{ks}mC_e$	61	1979
1341163	1327	1342	21142	21157	AATGCTACTTGAACAGC	$A_{ks}A_{ks}T_{ks}G_{ds}^{mC_{ds}}T_{ds}A_{ds}^{mC_{ds}}$ $s^T_{ds}T_{ds}G_{ds}A_{ds}A_{ks}^{mC_{ks}}A_{ks}G_e$	50	1045
1341168	1323	1338	21138	21153	CTACTTGAACAGTCTT	$mC_{ks}T_{ks}A_{ks}^{mC_{ds}}T_{ds}T_{ds}G_{ds}A_{ds}$ $A_{ds}^{mC_{ds}}mC_{ds}A_{ds}G_{ds}T_{ks}^{mC_{ks}}T_{ks}T_e$	71	734
1341205	1335	1350	21150	21165	CAGATTGGAATGCTAC	$mC_{ks}A_{ks}G_{ks}A_{ds}T_{ds}T_{ds}G_{ds}G_{ds}$ $A_{ds}A_{ds}T_{ds}G_{ds}^{mC_{ds}}T_{ks}A_{ks}^{mC_{ks}}mC_e$	61	1590
1341216	1344	1359	21159	21174	GCATGGCTACAGATTG	$G_{ks}^{mC_{ks}}A_{ks}T_{ds}G_{ds}G_{ds}^{mC_{ds}}T_d$ $s^T_{ds}A_{ds}^{mC_{ds}}A_{ds}G_{ds}A_{ks}T_{ks}T_{ks}G_e$	67	267

TABLE 43 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341223	696	711	12114	12129	GACATGAGGTTTGTAT	$G_{ks}A_{ks}^mC_{ks}A_{ds}T_{ds}G_{ds}A_{ds}G_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}$	55	647
1341244	1353	1368	21168	21183	TATTCTGTGCGATGCG	$T_{ks}A_{ks}T_{ks}T_{ks}T_{ks}T_{ks}T_{ks}T_{ks}T_{ks}T_{ks}$ $dsG_{ds}^mC_{ds}A_{ds}T_{ks}G_{ks}G_{ks}^mC_e$	74	890
1341335	1357	1372	21172	21187	TTGATATTCTGTGGCA	$T_{ks}T_{ks}G_{ks}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}^mC$ $dsT_{ds}G_{ds}T_{ds}G_{ks}G_{ks}^mC_{ks}^mC_e$	76	1202
1341353	1366	1381	21181	21196	GTGTTCTTGTGTATAT	$G_{ks}T_{ks}G_{ks}T_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}G$ $dsT_{ds}T_{ds}G_{ds}T_{ks}T_{ks}A_{ks}T_e$	66	1514
1341413	1391	1406	21206	21221	GATCTCTTAGCTGTGC	$G_{ks}A_{ks}T_{ks}^mC_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ks}T_{ks}G_{ks}^mC_e$	70	502
1341424	794	809	15675	15690	ATCTATCAGACTTCTT	$A_{ks}T_{ks}^mC_{ks}T_{ds}A_{ds}T_{ds}^mC_{ds}A_{ds}$ $G_{ds}A_{ds}^mC_{ds}T_{ds}T_{ks}^mC_{ks}T_{ks}T_e$	51	2728
1341425	1423	1438	21238	21253	CCAGGTTGAGATAAG	$^mC_{ks}^mC_{ks}A_{ks}G_{ds}G_{ds}T_{ds}T_{ds}G_d$ $s^A_{ds}G_{ds}T_{ds}T_{ds}T_{ds}A_{ks}A_{ks}G_e$	51	33
1341433	1464	1479	21279	21294	AGGCTAGGGAATCTT	$A_{ks}G_{ks}G_{ks}^mC_{ds}T_{ds}A_{ds}G_{ds}G_{ds}$ $G_{ds}A_{ds}A_{ds}A_{ds}T_{ks}^mC_{ks}T_{ks}T_e$	55	892
1341434	1419	1434	21234	21249	GTTGAGATAAAGCTGC	$G_{ks}T_{ks}T_{ks}G_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A$ $dsA_{ds}T_{ds}G_{ds}^mC_{ks}T_{ks}G_{ks}^mC_e$	59	1748
1341439	1615	1630	21430	21445	GTCACCTTTTATAATG	$G_{ks}T_{ks}^mC_{ks}A_{ds}^mC_{ds}^mC_{ds}T_{ds}T$ $dsT_{ds}^mC_{ds}T_{ds}T_{ds}T_{ks}A_{ks}T_{ks}G_e$	21	1907
1341442	1498	1513	21313	21328	GAGTTGCACCGTTTGT	$G_{ks}A_{ks}G_{ks}T_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}G_{ds}T_{ds}T_{ks}T_{ks}T_{ks}G_e$	62	1983
1341448	1705	1720	21520	21535	TGGGACCAAGGATATA	$T_{ks}G_{ks}G_{ks}G_{ds}A_{ds}^mC_{ds}^mC_{ds}A_d$ $s^A_{ds}G_{ds}G_{ds}A_{ds}T_{ks}A_{ks}T_{ks}A_e$	55	1674
1341458	2194	2209	22009	22024	TTAGTCTTGATGTAGT	$T_{ks}T_{ks}A_{ks}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}G$ $dsA_{ds}T_{ds}G_{ds}T_{ks}A_{ks}G_{ks}T_e$	61	1366
1341463	2350	2365	22165	22180	ACTACAAGAGGTTATT	$A_{ks}^mC_{ks}T_{ks}A_{ds}^mC_{ds}A_{ds}A_{ds}G_d$ $s^A_{ds}G_{ds}G_{ds}T_{ds}T_{ks}A_{ks}T_{ks}T_e$	58	899
1341468	N/A	N/A	14204	14219	CAGATTATTATGTGAT	$^mC_{ks}A_{ks}G_{ks}A_{ds}T_{ds}T_{ds}A_{ds}T_{ds}$ $T_{ds}A_{ds}T_{ds}G_{ds}T_{ks}G_{ks}A_{ks}T_e$	38	2723
1341471	1075	1090	20890	20905	GTGCCAAACCAATGTT	$G_{ks}T_{ks}G_{ks}^mC_{ds}^mC_{ds}A_{ds}A_{ds}A_d$ $s^mC_{ds}^mC_{ds}A_{ds}A_{ds}T_{ks}G_{ks}T_{ks}T_e$	56	1042
1341473	N/A	N/A	11858	11873	GATTAAGCAGAATGTG	$G_{ks}A_{ks}T_{ks}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $A_{ds}G_{ds}A_{ds}T_{ks}T_{ks}T_{ks}T_e$	28	1156
1341475	N/A	N/A	11543	11558	TGACAATGGTTGCCAC	$T_{ks}G_{ks}A_{ks}^mC_{ds}A_{ds}T_{ds}G_{ds}$ $G_{ds}T_{ds}T_{ds}G_{ds}^mC_{ks}^mC_{ks}^mC_e$	45	1233
1341485	N/A	N/A	16218	16233	TATAGACTGGGTAGGA	$T_{ks}A_{ks}T_{ks}A_{ds}G_{ds}^mC_{ds}T_{ds}$ $G_{ds}G_{ds}G_{ds}T_{ds}A_{ks}G_{ks}G_{ks}T_e$	55	2808
1341487	N/A	N/A	15491	15506	TTAATGCCACCCTACC	$T_{ks}T_{ks}A_{ks}A_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}^mC_{ds}T_{ks}A_{ks}^mC_{ks}^mC_e$	20	2304
1341489	N/A	N/A	18092	18107	TGATACACCAATGCAG	$T_{ks}G_{ks}A_{ks}T_{ds}A_{ds}^mC_{ds}A_{ds}^mC_d$ $s^mC_{ds}A_{ds}A_{ds}T_{ds}G_{ks}^mC_{ks}A_{ks}G_e$	58	2370
1341490	N/A	N/A	19916	19931	GGAGTGCATCTTAAGA	$G_{ks}G_{ks}A_{ks}G_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}$ $T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ks}G_{ks}A_e$	40	2810
1341491	N/A	N/A	19904	19919	AAGATACCCAGGTTGC	$A_{ks}A_{ks}G_{ks}A_{ds}T_{ds}A_{ds}^mC_{ds}^mC_d$ $s^mC_{ds}A_{ds}G_{ds}G_{ds}T_{ks}T_{ks}G_{ks}^mC_e$	69	2712
1341497	N/A	N/A	18690	18705	CTTTGAGACTCTTGTT	$^mC_{ks}T_{ks}T_{ks}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}^m$ $C_{ds}T_{ds}^mC_{ds}T_{ds}T_{ks}T_{ks}T_e$	57	2583

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2							
Compound Number	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 3	SEQ ID NO: 4	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
	Start Site	Stop Site	Start Site	Stop Site			
1341502	N/A	N/A	20187	20202	TATTAGAAGTCAGCCC T _{ks} A _{ks} T _{ks} T _{ks} A _{ds} G _{ds} A _{ds} A _{ds} G _{ds} T _{ds} ^m C _{ds} A _{ds} G _{ks} ^m C _{ks} ^m C _e	32	164
1341504	N/A	N/A	20197	20212	ACATTGCAGTTATTAG A _{ks} ^m C _{ks} A _{ks} T _{ds} T _{ds} G _{ds} ^m C _{ds} A _d _s G _{ds} T _{ds} T _{ds} A _{ds} T _{ks} T _{ks} A _{ks} G _e	52	2309
1341512	N/A	N/A	12794	12809	AAGAGTCAGTATCCTC A _{ks} A _{ks} G _{ks} A _{ds} G _{ds} T _{ds} ^m C _{ds} A _{ds} G _{ds} T _{ds} A _{ds} T _{ds} ^m C _{ks} ^m C _{ks} T _{ks} ^m C _e	67	2572
1341514	N/A	N/A	12496	12511	AATTCACCTTGACTAA A _{ks} A _{ks} T _{ks} T _{ks} T _{ds} ^m C _{ds} A _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ds} G _{ds} A _{ds} ^m C _{ks} A _{ks} A _e	16	2144
1341515	N/A	N/A	12804	12819	CTCAGGCTGTAAGAGT ^m C _{ks} T _{ks} ^m C _{ks} A _{ds} G _{ds} G _{ds} ^m C _{ds} T _{ds} G _{ds} T _{ds} A _{ds} A _{ds} G _{ks} A _{ks} G _{ks} T _e	21	2240

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
	SEQ ID NO:	SEQ ID NO:	SEQ ID NO:	SEQ ID NO:					
Compound	1 Start	1 Stop	2 Start	2 Stop	Sequence	Chemistry	Notation	% in- hibition	SEQ ID NO
Number	Site	Site	Site	Site	(5' to 3')		(5' to 3')		
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	A _{ks} G _{ks} T _{ks} A _{ds} G _{ds} A _{ds} T _{ds} G _{ds} G _{ds}	T _{ds} A _{ds} A _{ds} G _{ds} T _{ks} ^m C _{ks} A _k	79	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTACAGT	G _{ks} A _{ks} G _{ks} T _{ks} G _{ds} A _{ds} A _{ds} T _{ds} ^m C _d	s _A _{ds} T _{ds} T _{ds} ^m C _d s _A _{ks} G _{ks} T _k	83	1526
1340981	1495	1510	21310	21325	TTGCACCGTTTTTGGGC	T _{ks} T _{ks} G _{ks} ^m C _{ks} A _{ds} ^m C _{ds} ^m C _{ds} G _{ds}	T _{ds} T _{ds} T _{ds} T _{ds} T _{ds} G _{ks} G _{ks} ^m C _e	50	1749
1340983	1499	1514	21314	21329	AGAGTTGCACCGTTTTT	A _{ks} G _{ks} A _{ks} G _{ds} T _{ds} T _{ds} G _{ds} ^m C _{ds} A _d	s ^m C _{ds} ^m C _{ds} G _{ds} T _{ks} T _{ks} T _{ks} T _e	78	34
1340991	810	825	15691	15706	TATTGGTAAGTATTCC	T _{ks} A _{ks} T _{ks} T _{ks} G _{ds} G _{ds} T _{ds} A _{ds} A _{ds}	G _{ds} T _{ds} A _{ds} T _{ks} T _{ks} ^m C _{ks} ^m C _e	48	1272
1340995	1620	1635	21435	21450	AGTCGGTCACCTTTCA	A _{ks} G _{ks} T _{ks} ^m C _{ds} G _{ds} G _{ds} T _{ds} ^m C _{ds}	A _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ks} ^m C _{ks} A _e	75	272
1340998	1600	1615	21415	21430	GTATCTTATAAGACTA	G _{ks} T _{ks} A _{ks} T _{ks} ^m C _{ds} T _{ds} T _{ds} A _{ds} T _{ds}	A _{ds} A _{ds} G _{ds} A _{ks} ^m C _{ks} T _{ks} A _e	72	1595
1341000	1624	1639	21439	21454	ATAGAGTCGGTCACCT	A _{ks} T _{ks} A _{ks} G _{ds} A _{ds} G _{ds} T _{ds} ^m C _d I _d G _d	s _G _{ds} T _{ds} ^m C _{ds} A _{ks} ^m C _{ks} ^m C _{ks} T _e	67	584
1341002	1712	1727	21527	21542	ACATCTCTGGGACCAA	A _{ks} ^m C _{ks} A _{ks} T _{ds} ^m C _{ds} T _{ds} ^m C _{ds} T _{ds}	G _{ds} G _{ds} G _{ds} A _{ds} ^m C _{ks} ^m C _{ks} A _e	65	195
1341004	1628	1643	21443	21458	TAAAATAGAGTCGGTC	T _{ks} A _{ks} A _{ks} A _{ds} A _{ds} T _{ds} A _{ds} G _{ds} A _{ds}	G _{ds} T _{ds} ^m C _{ds} G _{ks} G _{ks} T _{ks} ^m C _e	74	895
1341011	1752	1767	21567	21582	TCCTGTGTTAGCTTTA	T _{ks} ^m C _{ks} ^m C _{ks} T _{ds} G _{ds} T _{ds} G _{ds} T _{ds} T	ds ^m A _{ds} G _{ds} ^m C _{ds} T _{ks} T _{ks} T _{ks} A _e	75	1442
1341013	1721	1736	21536	21551	ATTGTCTAAACATCTC	A _{ks} T _{ks} T _{ks} T _{ks} G _{ds} T _{ds} ^m C _{ds} T _{ds} A _{ds} A _{ds}	A _{ds} ^m C _{ds} A _{ds} T _{ks} ^m C _{ks} T _{ks} ^m C _e	71	585
1341017	1019	1034	20834	20849	TGCAGCATTGATTCGA	T _{ks} G _{ks} ^m C _{ks} A _{ds} G _{ds} ^m C _{ds} A _{ds} T _{ds} T	ds _G _{ds} A _{ds} T _{ds} T _{ks} ^m C _{ks} G _{ks} A _e	52	419
1341030	1784	1799	21599	21614	GTTTCTTATGTAATAG	G _{ks} T _{ks} T _{ks} T _{ks} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ks} T _{ds}	G _{ds} T _{ds} A _{ds} A _{ds} T _{ks} A _{ks} G _e	44	897
1341033	2189	2204	22004	22019	CTTGATGTAGTGGGAG	^m C _{ks} T _{ks} T _{ks} G _{ds} A _{ds} T _{ds} G _{ds} T _{ds} A _{ds}	G _{ds} T _{ds} G _{ds} G _{ks} G _{ks} A _{ks} G _e	62	976

TABLE 44 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341039	1780	1795	21595	21610	CTTATGTAATAGCCAG	${}^mC_{ks}T_{ks}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}G_e$	70	664
1341042	N/A	N/A	9334	9349	CGATCCTATATACATC	${}^mC_{ks}G_{ks}A_{ks}T_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}A_{ds}$ $T_{ds}A_{ds}T_{ds}A_{ds}C_{ks}A_{ks}T_{ks}{}^mC_e$	29	2788
1341043	2252	2267	22067	22082	AGAGGATAGTCCATGC	$A_{ks}G_{ks}A_{ks}G_{ds}G_{ds}A_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}T_{ks}G_{ks}{}^mC_e$	78	354
1341044	2353	2368	22168	22183	ATAACTACAAGAGTT	$A_{ks}T_{ks}A_{ks}A_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}G_{ks}G_{ks}T_{ks}T_e$	67	1133
1341053	N/A	N/A	9968	9983	AGTAGTGTAAGCTGAG	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}T_{ds}G_{ds}T_{ds}A_{ds}$ $A_{ds}G_{ds}{}^mC_{ds}T_{ks}G_{ks}A_{ks}G_e$	71	2155
1341056	N/A	N/A	9516	9531	ACACTAATATTGAGGC	$A_{ks}{}^mC_{ks}A_{ks}{}^mC_{ds}T_{ds}A_{ds}T_{ds}$ $A_{ds}T_{ds}T_{ds}G_{ds}A_{ks}G_{ks}{}^mC_e$	68	1928
1341057	N/A	N/A	9603	9618	CACATATGTGAAGAGC	${}^mC_{ks}A_{ks}{}^mC_{ks}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}$ $dsG_{ds}A_{ds}A_{ds}G_{ks}A_{ks}G_{ks}{}^mC_e$	74	760
1341060	N/A	N/A	12421	12436	TGGTAACGGTGATCAA	$T_{ks}G_{ks}G_{ks}T_{ds}A_{ds}A_{ds}{}^mC_{ds}G_{ds}G_d$ $sT_{ds}G_{ds}A_{ds}T_{ks}{}^mC_{ks}A_{ks}A_e$	35	67
1341068	N/A	N/A	11546	11561	TCTTGACAATGGTTGC	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}A_{ds}$ $dsT_{ds}G_{ds}G_{ds}T_{ks}T_{ks}G_{ks}{}^mC_e$	77	2199
1341071	N/A	N/A	15494	15509	GCATTAATGCCACCCT	$G_{ks}{}^mC_{ks}A_{ks}T_{ds}T_{ds}A_{ds}A_{ds}T_{ds}G_d$ $s{}^mC_{ds}{}^mC_{ds}A_{ds}{}^mC_{ks}{}^mC_{ks}T_e$	51	1012
1341075	608	623	N/A	N/A	ACAATATGGGATGAGG	$A_{ks}{}^mC_{ks}A_{ks}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}G_d$ $sG_{ds}A_{ds}T_{ds}G_{ks}A_{ks}G_{ks}G_e$	76	1579
1341076	N/A	N/A	14914	14929	ACATTAGCAAGCTAAG	$A_{ks}{}^mC_{ks}A_{ks}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}$ $A_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ks}A_{ks}A_{ks}G_e$	41	2065
1341077	1080	1095	20895	20910	TGCTAGTGCCAAACCA	$T_{ks}G_{ks}{}^mC_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ds}$ $ds{}^mC_{ds}A_{ds}A_{ds}A_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	73	1432
1341081	N/A	N/A	16223	16238	CTTTTATAGACTGGG	${}^mC_{ks}T_{ks}T_{ks}T_{ds}T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}A_{ds}{}^mC_{ds}T_{ks}G_{ks}G_{ks}G_e$	44	857
1341084	N/A	N/A	16384	16399	AATCCCAACCAACTT	$A_{ks}A_{ks}T_{ks}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}{}^mC_{ks}T_{ks}T_e$	13	2755
1341089	N/A	N/A	16129	16144	TCCATAATAATAGCTC	$T_{ks}{}^mC_{ks}{}^mC_{ks}A_{ks}T_{ds}A_{ds}T_{ds}A_{ds}$ $dsA_{ds}T_{ds}A_{ds}G_{ks}{}^mC_{ks}T_{ks}{}^mC_e$	66	546
1341092	N/A	N/A	18095	18110	AATTGATACACCAATG	$A_{ks}A_{ks}T_{ks}T_{ds}G_{ds}A_{ds}T_{ds}A_{ds}{}^mC_d$ $sA_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ks}T_{ks}G_e$	53	2525
1341096	N/A	N/A	17990	18005	TTAGTATAGTTATCTT	$T_{ks}T_{ks}A_{ks}G_{ds}T_{ds}A_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ks}{}^mC_{ks}T_{ks}T_e$	45	2366
1341101	N/A	N/A	19909	19924	ATCTTAAGATACCCAG	$A_{ks}T_{ks}{}^mC_{ks}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_d$ $sT_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}G_e$	71	397
1341102	N/A	N/A	18693	18708	AGACTTTGAGACTCTT	$A_{ks}G_{ks}A_{ks}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_d$ $sG_{ds}A_{ds}{}^mC_{ds}T_{ks}{}^mC_{ks}T_{ks}T_e$	41	2804
1341103	N/A	N/A	19913	19928	GTGCATCTTAAGATAC	$G_{ks}T_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T$ $dsA_{ds}A_{ds}G_{ds}A_{ks}T_{ks}A_{ks}{}^mC_e$	29	2806
1341115	N/A	N/A	20190	20205	AGTTATTAGAAGTCAG	$A_{ks}G_{ks}T_{ks}T_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_{ks}G_e$	56	2746
1341118	N/A	N/A	20114	20129	GCATCAAGACATTCTA	$G_{ks}{}^mC_{ks}A_{ks}T_{ds}{}^mC_{ds}A_{ds}A_{ds}G_{ds}$ $A_{ds}{}^mC_{ds}A_{ds}T_{ds}T_{ks}{}^mC_{ks}T_{ks}A_e$	84	1565
1341120	1084	1099	20899	20914	CTGCTGCTAGTGCCAA	${}^mC_{ks}T_{ks}G_{ks}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}A_e$	66	1743

TABLE 44 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
SEQ ID NO: Compound 1 Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% in- hibition	SEQ ID NO
1341121	N/A	N/A	12499	12514	AGAAATTCACCTTGAC	$A_{ks}G_{ks}A_{ks}A_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ds}$ $s{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ks}G_{ks}A_{ks}{}^mC_e$	7	2284
1341126	N/A	N/A	20484	20499	CTTCATAGTGGACTTC	${}^mC_kT_{ks}T_{ks}{}^mC_{ds}A_{ds}T_{ds}A_{ds}G_{ds}T$ $dsG_{ds}G_{ds}A_{ds}{}^mC_{ks}T_{ks}T_{ks}{}^mC_e$	57	711
1341131	N/A	N/A	14196	14211	TATGTGATTGAGTTCT	$T_{ks}A_{ks}T_{ks}G_{ds}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}T_{ks}T_{ks}{}^mC_{ks}T_e$	43	2367
1341137	N/A	N/A	12797	12812	TGTAAGAGTCAGTATC	$T_{ks}G_{ks}T_{ks}A_{ds}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}$ ${}^mC_{ds}A_{ds}G_{ds}T_{ks}A_{ks}T_{ks}{}^mC_e$	46	2652
1341138	N/A	N/A	14172	14187	TTGAATAGCAAGCCAA	$T_{ks}T_{ks}G_{ks}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}{}^mC_d$ $s{}^mC_{ds}A_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}A_e$	33	1864
1341143	1261	1276	21076	21091	AGGTAGCTTTTGTCCA	$A_{ks}G_{ks}G_{ks}T_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ds}T_d$ $s{}^mC_{ds}T_{ds}G_{ds}T_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	58	109
1341145	1256	1271	21071	21086	GCTTTTGTCCACCTTT	$G_{ks}{}^mC_{ks}T_{ks}T_{ds}T_{ds}T_{ds}G_{ds}T_{ds}{}^mC$ $ds{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}T_{ks}T_e$	81	1744
1341150	1318	1333	21133	21148	TGAACAGTCTTAAACC	$T_{ks}G_{ks}A_{ks}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}{}^m$ $C_{ds}T_{ds}T_{ds}A_{ds}A_{ks}A_{ks}{}^mC_{ks}{}^mC_e$	61	344
1341154	1314	1329	21129	21144	CAGTCTTAAACCTTCC	${}^mC_{ks}A_{ks}G_{ks}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}A$ $dsA_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}T_{ks}{}^mC_{ks}{}^mC_e$	74	30
1341160	698	713	12116	12131	GAGACATGAGGTTTTC	$G_{ks}A_{ks}G_{ks}A_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}A$ $dsG_{ds}G_{ds}T_{ds}T_{ks}T_{ks}T_{ks}G_e$	61	725
1341166	1324	1339	21139	21154	GCTACTTGAACAGTCT	$G_{ks}{}^mC_{ks}T_{ks}A_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A$ $dsA_{ds}{}^mC_{ds}A_{ds}G_{ks}T_{ks}{}^mC_{ks}T_e$	60	811
1341184	1329	1344	21144	21159	GGAATGCTACTTGAAC	$G_{ks}G_{ks}A_{ks}A_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}A_d$ $s{}^mC_{ds}T_{ds}T_{ds}G_{ks}A_{ks}A_{ks}{}^mC_e$	68	1201
1341194	1337	1352	21152	21167	TACAGATTGGAATGCT	$T_{ks}A_{ks}{}^mC_{ks}A_{ds}G_{ds}A_{ds}T_{ds}T_{ds}G_d$ $sG_{ds}A_{ds}A_{ds}T_{ks}G_{ks}{}^mC_{ks}T_e$	66	1746
1341266	1349	1364	21164	21179	CTGTGGCATGGCTACA	${}^mC_{ks}T_{ks}G_{ks}T_{ds}G_{ds}G_{ds}{}^mC_{ds}A_{ds}T$ $dsG_{ds}G_{ds}{}^mC_{ds}T_{ks}A_{ks}{}^mC_{ks}A_e$	58	657
1341295	1354	1369	21169	21184	ATATTCTGTGGCATGG	$A_{ks}T_{ks}A_{ks}T_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ds}T_{ds}$ $G_{ds}G_{ds}{}^mC_{ds}A_{ks}T_{ks}G_{ks}G_e$	58	968
1341318	1358	1373	21173	21188	GTTGATATTCTGTGTC	$G_{ks}T_{ks}T_{ks}G_{ds}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}{}^m$ $C_{ds}T_{ds}G_{ds}T_{ks}G_{ks}G_{ks}{}^mC_e$	73	1280
1341392	1367	1382	21182	21197	TGTGTTCTTGTGTGATA	$T_{ks}G_{ks}T_{ks}G_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}T_{ds}T_{ds}G_{ks}A_{ks}T_{ks}A_e$	65	1591
1341400	796	811	15677	15692	CCATCTATCAGACTTC	${}^mC_{ks}{}^mC_{ks}A_{ks}T_{ds}{}^mC_{ds}T_{ds}A_{ds}T_{ds}$ ${}^mC_{ds}A_{ds}G_{ds}A_{ds}{}^mC_{ks}T_{ks}T_{ks}{}^mC_e$	60	415
1341421	1420	1435	21235	21250	GGTTGAGATAAAGCTG	$G_{ks}G_{ks}T_{ks}T_{ds}G_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}A_{ds}G_{ks}{}^mC_{ks}T_{ks}G_e$	67	1826
1341426	1393	1408	21208	21223	TTGATCTCTTAGCTGT	$T_{ks}T_{ks}G_{ks}A_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T$ $dsT_{ds}A_{ds}G_{ds}{}^mC_{ks}T_{ks}G_{ks}T_e$	64	658
1341431	1429	1444	21244	21259	ATATGTCCAGTTGAG	$A_{ks}T_{ks}A_{ks}T_{ds}G_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}A$ $dsG_{ds}G_{ds}T_{ds}T_{ks}G_{ks}A_{ks}G_e$	71	425
1341432	1491	1506	21306	21321	ACCGTTTGGGCTAAT	$A_{ks}{}^mC_{ks}{}^mC_{ks}G_{ds}T_{ds}T_{ds}T_{ds}G$ $dsG_{ds}G_{ds}{}^mC_{ds}T_{ks}A_{ks}A_{ks}T_e$	73	1438
1341435	1466	1481	21281	21296	AGAGGCTAGGGAAATC	$A_{ks}G_{ks}A_{ks}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}G$ $dsG_{ds}G_{ds}A_{ds}A_{ks}A_{ks}T_{ks}{}^mC_e$	68	1048
1341441	1616	1631	21431	21446	GGTCACCTTTCATAAT	$G_{ks}G_{ks}T_{ks}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}$ $T_{ds}T_{ds}{}^mC_{ds}A_{ds}T_{ks}A_{ks}A_{ks}T_e$	49	1985

TABLE 44 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341449	1771	1786	21586	21601	TAGCCAGTACAGTTCC	$T_{ks}A_{ks}G_{ks}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}{}^mC_{ds}G_{ds}T_{ks}T_{ks}{}^mC_{ks}{}^mC_e$	58	1987
1341451	1776	1791	21591	21606	TGTAATAGCCAGTACA	$T_{ks}G_{ks}T_{ks}A_{ds}A_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ks}A_{ks}{}^mC_{ks}{}^mC_e$	34	352
1341456	1707	1722	21522	21537	TCTGGGACCAAGGATA	$T_{ks}{}^mC_{ks}T_{ks}G_{ds}G_{ds}G_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_{ks}{}^mC_{ks}{}^mC_e$	57	1830
1341457	1716	1731	21531	21546	CTAAACATCTCTGGGA	${}^mC_{ks}T_{ks}A_{ks}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}C_{ds}T_{ds}G_{ks}G_{ks}G_{ks}A_e$	57	2259
1341466	2196	2211	22011	22026	GATTAGTCTTGATGTA	$G_{ks}A_{ks}T_{ks}T_{ks}A_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}T_{ks}G_{ks}A_e$	48	1521
1341474	N/A	N/A	11578	11593	GTTATTCTCTTGACAA	$G_{ks}T_{ks}T_{ks}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_e$	35	2575
1341482	N/A	N/A	16292	16307	TTAGTTAGAATAGTCT	$T_{ks}T_{ks}A_{ks}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ks}T_{ks}{}^mC_{ks}T_e$	67	1714
1341488	N/A	N/A	19905	19920	TAAGATACCCAGGTTG	$T_{ks}A_{ks}A_{ks}G_{ds}A_{ds}T_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_{ks}G_e$	58	2790
1341492	1076	1091	20891	20906	AGTGCCAAACCAATGT	$A_{ks}G_{ks}T_{ks}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ks}T_{ks}G_{ks}T_e$	60	1120
1341495	N/A	N/A	17995	18010	TGAAGTTAGTATAGTT	$T_{ks}G_{ks}A_{ks}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}T_{ds}A_{ks}G_{ks}T_{ks}T_e$	70	1405
1341498	N/A	N/A	19999	20014	CAGTTTTCCTCATGAT	${}^mC_{ks}A_{ks}G_{ks}T_{ds}T_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}C_{ds}A_{ds}T_{ks}G_{ks}A_{ks}T_e$	65	2331
1341500	N/A	N/A	20378	20393	CACAAGTAAGGTAAAG	${}^mC_{ks}A_{ks}{}^mC_{ks}A_{ds}A_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}G_{ds}T_{ds}A_{ks}A_{ks}A_{ks}G_e$	55	2448
1341503	N/A	N/A	20388	20403	CTATAGTAATCACAAAG	${}^mC_{ks}T_{ks}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}A_{ks}G_e$	53	1644
1341510	1091	1106	20906	20921	CGTTTGACTGCTGCTA	${}^mC_{ks}G_{ks}T_{ks}T_{ds}T_{ds}G_{ds}A_{ds}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}G_{ks}{}^mC_{ks}T_{ks}A_e$	75	264
1341513	N/A	N/A	13959	13974	GTCTACCTCTAAGTTA	$G_{ks}T_{ks}{}^mC_{ks}T_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}A_{ds}G_{ks}T_{ks}T_{ks}A_e$	41	2363
1341516	N/A	N/A	13949	13964	AAGTTAGCCCCAGGA	$A_{ks}A_{ks}G_{ks}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}G_{ks}G_{ks}A_e$	23	2811

TABLE 45

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_k$	79	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_k$	80	1526
1340980	610	625	N/A	N/A	GAACAATATGGGATGA	$G_{ks}A_{ks}A_{ks}{}^mC_{ds}A_{ds}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}G_{ds}G_{ds}A_{ks}T_{ks}G_{ks}A_e$	52	1735
1340985	1492	1507	21307	21322	CACCGTTTTGGGCTAA	${}^mC_{ks}A_{ks}{}^mC_{ks}{}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}G_{ds}{}^mC_{ks}T_{ks}A_{ks}A_e$	71	1516

TABLE 45 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341078	N/A	N/A	14916	14931	ATACATTAGCAAGCTA	$A_{ks}T_{ks}A_{ks}{}^mC_{ds}A_{ds}T_{ds}T_{ds}A_{ds}G_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}G_{ks}{}^mC_{ks}T_{ks}A_e$	48	309
1341086	N/A	N/A	16131	16146	GCTCCATAATAATAGC	$G_{ks}{}^mC_{ks}T_{ks}{}^mC_{ds}{}^mC_{ds}A_{ds}T_{ds}A_{ds}A_{ds}$ ${}_{ds}T_{ds}{}_{ds}A_{ds}{}_{ds}T_{ks}A_{ks}G_{ks}{}^mC_e$	8	2665
1341091	N/A	N/A	17991	18006	GTTAGTATAGTTATCT	$G_{ks}T_{ks}T_{ks}A_{ds}G_{ds}T_{ds}A_{ds}T_{ds}A_{ds}G_{ds}$ ${}_{ds}T_{ds}{}_{ds}A_{ks}{}_{ds}T_{ks}{}^mC_{ks}T_e$	71	1171
1341094	N/A	N/A	18097	18112	GAAATTGATACACCAA	$G_{ks}A_{ks}A_{ks}A_{ds}T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}A_{ds}$ ${}_{ds}{}^mC_{ds}A_{ds}{}^mC_{ks}{}^mC_{ks}A_{ks}A_e$	64	315
1341104	N/A	N/A	19910	19925	CATCTTAAGATACCCA	${}^mC_{ks}A_{ks}T_{ks}{}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}$ $A_{ds}T_{ds}A_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	76	475
1341106	N/A	N/A	18695	18710	AAAGACTTTGAGACTC	$A_{ks}A_{ks}A_{ks}G_{ds}A_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}A_{ks}{}^mC_{ks}T_{ks}{}^mC_e$	67	1250
1341109	1085	1100	20900	20915	ACTGCTGCTAGTGCCA	$A_{ks}{}^mC_{ks}T_{ks}G_{ds}{}^mC_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}$ ${}_{ds}A_{ds}G_{ds}T_{ds}G_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	57	1821
1341111	N/A	N/A	20116	20131	CAGCATCAAGACATTC	${}^mC_{ks}A_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ks}T_{ks}T_{ks}$	70	2191
1341112	N/A	N/A	20381	20396	AATCACAAGTAAGGTA	$A_{ks}A_{ks}T_{ks}{}^mC_{ds}A_{ds}{}^mC_{ds}A_{ds}A_{ds}G_{ds}$ ${}_{ds}T_{ds}A_{ds}{}_{ds}G_{ks}G_{ks}T_{ks}A_e$	63	1333
1341117	N/A	N/A	20192	20207	GCAGTTATTAGAAGTC	$G_{ks}{}^mC_{ks}A_{ks}G_{ds}T_{ds}T_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}A_{ds}A_{ks}G_{ks}T_{ks}{}^mC_e$	67	398
1341124	N/A	N/A	12501	12516	GCAGAAATTCACCTTG	$G_{ks}{}^mC_{ks}A_{ks}G_{ds}A_{ds}A_{ds}A_{ds}T_{ds}T_{ds}$ ${}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}T_{ks}G_e$	62	1002
1341128	N/A	N/A	20486	20501	AACTTCATAGTGGACT	$A_{ks}A_{ks}{}^mC_{ks}T_{ds}T_{ds}{}^mC_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}G_{ds}G_{ks}A_{ks}{}^mC_{ks}T_e$	70	2555
1341129	N/A	N/A	20422	20437	ACAGTAAAAATTATGCC	$A_{ks}{}^mC_{ks}A_{ks}G_{ds}T_{ds}A_{ds}A_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ks}G_{ks}{}^mC_{ks}{}^mC_e$	62	477
1341135	N/A	N/A	13952	13967	TCTAAGTTAGCCCCCA	$T_{ks}{}^mC_{ks}T_{ks}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}$ $G_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_{ks}A_e$	0	928
1341139	N/A	N/A	12799	12814	GCTGTAAGAGTCAGTA	$G_{ks}{}^mC_{ks}T_{ks}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}$ $G_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_{ks}A_e$	52	693
1341140	1180	1195	20995	21010	GTTTTTAAGAGGCATG	$G_{ks}T_{ks}T_{ks}T_{ds}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}$ ${}_{ds}G_{ds}G_{ds}{}^mC_{ks}{}_{ks}T_{ks}G_e$	78	1199
1341142	1257	1272	21072	21087	AGCTTTTGTCCACCTT	$A_{ks}G_{ks}{}^mC_{ks}T_{ds}T_{ds}T_{ds}T_{ds}G_{ds}T_{ds}$ $C_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}T_{ks}{}^mC_e$	69	1822
1341151	1315	1330	21130	21145	ACAGTCTTAAACCTTC	$A_{ks}{}^mC_{ks}A_{ks}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}$ $A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}{}^mC_e$	73	110
1341153	1325	1340	21140	21155	TGCTACTTGAACAGTC	$T_{ks}C_{ks}{}^mC_{ks}T_{ds}A_{ds}{}^mC_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}A_{ds}{}^mC_{ds}A_{ks}G_{ks}T_{ks}{}^mC_e$	68	889
1341156	1311	1326	21126	21141	TCTTAAACCTTCCCTG	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}$ ${}_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{ks}T_{ks}G_e$	38	2160
1341158	700	715	12118	12133	CAGAGACATGAGGTTT	${}^mC_{ks}A_{ks}G_{ks}A_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ds}$ ${}_{ds}G_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_{ks}T_e$	45	2319
1341167	1321	1336	21136	21151	ACTTGAACAGTCTTAA	$A_{ks}{}^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}A_{ds}{}^mC_{ds}A_{ds}$ ${}_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ks}T_{ks}A_{ks}A_e$	60	578
1341173	1331	1346	21146	21161	TTGGAATGCTACTTGA	$T_{ks}T_{ks}G_{ks}G_{ds}A_{ds}A_{ds}T_{ds}G_{ds}{}^mC_{ds}$ $T_{ds}A_{ds}{}^mC_{ds}T_{ks}T_{ks}G_{ks}A_e$	56	1279
1341232	1340	1355	21155	21170	GGCTACAGATTGGAAT	$G_{ks}G_{ks}{}^mC_{ks}T_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}A_{ds}$ ${}_{ds}T_{ds}T_{ds}G_{ds}G_{ks}A_{ks}A_{ks}T_e$	28	1980

TABLE 45 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
SEQ ID NO: Compound 1 Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% in- hibition	SEQ ID NO
1341255	1351	1366	21166	21181	TTCTGTGGCATGGCTA	$T_{ks}T_{ks}^mC_{ks}T_{ds}G_{ds}T_{ds}G_{ds}G_{ds}^mC_{ds}$ $A_{ds}T_{ds}G_{ds}G_{ks}^mC_{ks}T_{ks}A_e$	59	812
1341271	1355	1370	21170	21185	GATATTCTGTGGCATG	$G_{ks}A_{ks}T_{ks}T_{ks}A_{ds}T_{ds}^mC_{ds}T_{ds}G_{ds}$ $T_{ds}G_{ds}G_{ds}^mC_{ks}A_{ks}T_{ks}G_e$	71	1046
1341321	1359	1374	21174	21189	TGTTGATATTCTGTGG	$T_{ks}G_{ks}T_{ks}T_{ks}G_{ds}A_{ds}T_{ds}A_{ds}T_{ds}T_d$ $s^mC_{ds}T_{ds}G_{ks}T_{ks}G_{ks}G_e$	64	1358
1341323	798	813	15679	15694	TTCCATCTATCAGACT	$T_{ks}T_{ks}^mC_{ks}^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}A$ $dsT_{ds}^mC_{ds}A_{ds}G_{ks}A_{ks}^mC_{ks}T_e$	43	571
1341427	1395	1410	21210	21225	ACTTGATCTCTTAGCT	$A_{ks}^mC_{ks}T_{ks}T_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}T_{ds}T_{ds}A_{ks}G_{ks}^mC_{ks}T_e$	59	813
1341428	1387	1402	21202	21217	TCTTAGCTGTGCACTC	$T_{ks}^mC_{ks}T_{ks}T_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}$ $T_{ds}G_{ds}^mC_{ds}A_{ks}^mC_{ks}T_{ks}^mC_e$	52	190
1341436	1421	1436	21236	21251	AGGTTGAGATAAAGCT	$A_{ks}G_{ks}G_{ks}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}T$ $dsA_{ds}^mC_{ks}A_{ks}G_{ks}^mC_{ks}T_e$	71	1904
1341437	1431	1446	21246	21261	AAATATGTCCAGGTG	$A_{ks}A_{ks}A_{ks}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_{ks}G_e$	74	581
1341446	1488	1503	21303	21318	GTTTGGGCTAATGAA	$G_{ks}T_{ks}T_{ks}T_{ds}T_{ds}G_{ds}G_{ds}G_{ds}^mC_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ks}G_{ks}A_{ks}A_e$	47	1204
1341447	1496	1511	21311	21326	GTTGCACCGTTTGGG	$G_{ks}T_{ks}G_{ds}^mC_{ds}A_{ds}^mC_{ds}^mC_{ds}G$ $dsT_{ds}T_{ds}T_{ds}T_{ks}G_{ks}G_{ks}G_e$	45	1827
1341450	1726	1741	21541	21556	CTAAATGTCTAAAC	$^mC_{ks}T_{ks}A_{ks}A_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $T_{ds}^mC_{ds}T_{ds}A_{ks}A_{ks}A_{ks}^mC_e$	7	2590
1341454	1022	1037	20837	20852	CTTTCAGCATTGATT	$^mC_{ks}T_{ks}T_{ks}T_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}G_{ds}^mC$ $dsA_{ds}T_{ds}T_{ds}G_{ks}A_{ks}T_{ks}T_e$	56	2463
1341455	1717	1732	21532	21547	TCTAAACATCTCTGGG	$T_{ks}^mC_{ks}T_{ks}A_{ds}A_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}$ $^mC_{ds}T_{ds}^mC_{ds}T_{ks}G_{ks}G_{ks}G_e$	42	2287
1341460	N/A	N/A	9337	9352	CCTCGATCCTATATAC	$^mC_{ks}^mC_{ks}T_{ks}^mC_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}T_{ds}A_{ds}T_{ds}A_{ks}T_{ks}A_{ks}^mC_e$	11	2812
1341467	2184	2199	21999	22014	TGTAGTGGGAGTCGGA	$T_{ks}G_{ks}T_{ks}A_{ds}G_{ds}T_{ds}G_{ds}G_{ds}A$ $dsG_{ds}T_{ds}^mC_{ks}G_{ks}A_e$	53	587
1341469	N/A	N/A	9971	9986	TTGAGTAGTGTAAGCT	$T_{ks}T_{ks}G_{ks}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}G$ $dsT_{ds}^mC_{ks}A_{ks}G_{ks}^mC_{ks}T_e$	40	2346
1341476	N/A	N/A	11568	11583	TGACAATGGTTGATAG	$T_{ks}G_{ks}A_{ks}^mC_{ds}A_{ds}A_{ds}T_{ds}G_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ks}T_{ks}A_{ks}G_e$	38	1700
1341477	N/A	N/A	9961	9976	TAAGCTGAGAGTTCTA	$T_{ks}A_{ks}A_{ks}G_{ds}^mC_{ds}T_{ds}G_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ks}^mC_{ks}T_{ks}A_e$	48	2499
1341479	N/A	N/A	17986	18001	TATAGTTATCTTCTCA	$T_{ks}A_{ks}T_{ks}A_{ds}G_{ds}T_{ds}T_{ds}T_{ds}^m$ $C_{ds}T_{ds}T_{ds}^mC_{ks}T_{ks}^mC_{ks}A_e$	59	2295
1341481	N/A	N/A	17996	18011	ATGAAGTTAGTATAGT	$A_{ks}T_{ks}G_{ks}A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G$ $dsT_{ds}A_{ds}T_{ks}A_{ks}G_{ks}T_e$	61	1483
1341483	N/A	N/A	16282	16297	TAGTCTTCAGCAAAGT	$T_{ks}A_{ks}G_{ks}T_{ds}^mC_{ds}T_{ds}^mC_{ds}A_{ds}$ $G_{ds}^mC_{ds}A_{ds}A_{ks}G_{ks}T_e$	36	2632
1341484	N/A	N/A	16377	16392	ACCAAACCTCCAGCAG	$A_{ks}^mC_{ks}^mC_{ks}A_{ds}A_{ds}A_{ds}^mC_{ds}T_{ds}T$ $ds^mC_{ds}^mC_{ds}A_{ds}G_{ks}^mC_{ks}A_{ks}G_e$	36	1948
1341493	N/A	N/A	19914	19929	AGTGCATCTTAAGATA	$A_{ks}G_{ks}T_{ks}G_{ds}^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ks}A_{ks}T_{ks}A_e$	12	2130
1341505	N/A	N/A	19906	19921	TTAAGATACCCAGGTT	$T_{ks}T_{ks}A_{ks}A_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}G_{ks}T_{ks}T_e$	51	163

TABLE 45 -continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2							
SEQ ID NO:	SEQ ID NO:	SEQ ID NO:	SEQ ID NO:	Sequence	Chemistry Notation	% inhibition	SEQ ID NO
Compound 1	Start 1	Stop 2	Start 2	Stop 2	(5' to 3')		
Number	Site	Site	Site	Site	(5' to 3')		
1341507	N/A	N/A	20009	20024	AGAGAGTAATCAGTTT A _{ks} G _{ks} A _{ks} G _{ds} A _{ds} G _{ds} T _{ds} A _{ds} A _{ds} T _{ds} ^m C _{ds} A _{ds} G _{ks} T _{ks} T _{ks} T _e	51	2651
1341509	N/A	N/A	14165	14180	GCAAGCCAACAGAGAG G _{ks} ^m C _{ks} A _{ks} A _{ds} G _{ds} ^m C _{ds} ^m C _{ds} A _{ds} A _{ds} ^m C _{ds} A _{ds} G _{ds} G _{ks} A _{ks} G _e	38	2556
1341511	N/A	N/A	14175	14190	GCCTTGAAATAGCAAGC G _{ks} ^m C _{ks} ^m C _{ks} T _{ds} T _{ds} G _{ds} A _{ds} A _{ds} T _{ds} A _{ds} G _{ds} ^m C _{ds} A _{ks} A _{ks} G _{ks} ^m C _e	38	2063

TABLE 46

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	SEQ ID NO: 2	Sequence (5' to 3')	Chemistry Notation	%	SEQ ID
	Start Site	Stop Site	Start Site	Stop Site		(5' to 3')	inhibition	NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	A _{ks} G _{ks} T _{ks} A _{ds} G _{ds} A _{ds} T _{ds} G _{ds} G _{ds} T _{ds} A _{ds} A _{ds} G _{ds} T _{ks} ^m C _{ks} A _k	79	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	G _{ks} A _{ks} G _{ks} T _{ds} G _{ds} A _{ds} A _{ds} T _{ds} ^m C _{ds} A _{ds} T _{ds} T _{ds} ^m C _{ds} A _{ks} G _{ks} T _k	87	1526
1341648	1251	1266	21066	21081	TGTCCACCTTTAAATG	T _{ks} G _{ks} T _{ks} ^m C _{ds} C _{ys} A _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ds} T _{ds} A _{ds} A _{ds} A _{ks} T _{ks} G _k	17	1666
1341654	1317	1332	21132	21147	GAACAGTCTTAAACCT	G _{ks} A _{ks} A _{ks} ^m C _{ds} A _{ys} G _{ds} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ds} A _{ds} A _{ds} ^m C _{ks} ^m C _{ks} T _k	75	266
1341658	1313	1328	21128	21143	AGTCUTAAACCTTCCC	A _{ks} G _{ks} T _{ks} ^m C _{ds} U _{ys} T _{ds} A _{ds} A _{ds} A _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ds} ^m C _{ks} ^m C _{ks} ^m C _k	52	2813
1341661	1321	1336	21136	21151	ACTTGAACAGTCTTAA	A _{ks} ^m C _{ks} T _{ks} T _{ds} G _{ys} A _{ds} A _{ds} ^m C _{ds} A _{ds} G _{ds} T _{ds} ^m C _{ds} T _{ds} T _{ks} A _{ks} A _k	29	578
1341663	1308	1323	21123	21138	TAAACCTTCCCTGTGT	T _{ks} A _{ks} A _{ks} A _{ds} C _{ys} ^m C _{ds} T _{ds} T _{ds} ^m C _{ds} ^m C _{ds} ^m C _{ds} T _{ds} G _{ds} T _{ks} G _{ks} T _k	46	1745
1341668	1335	1350	21150	21165	CAGAUTGGAATGCTAC	^m C _{ks} A _{ks} G _{ks} A _{ds} U _{ys} T _{ds} G _{ds} G _{ds} A _{ds} A _{ds} T _{ds} G _{ds} ^m C _{ds} T _{ks} A _{ks} ^m C _k	70	2814
1341672	1329	1344	21144	21159	GGAAUGCTACTTGAAC	G _{ks} G _{ks} A _{ks} A _{ds} U _{ys} G _{ds} ^m C _{ds} T _{ds} A _{ds} ^m C _{ds} T _{ds} T _{ds} G _{ds} A _{ks} A _{ks} ^m C _k	62	2815
1341677	1325	1340	21140	21155	TGCTACTTGAACAGTC	T _{ks} G _{ks} ^m C _{ks} T _{ds} A _{ys} ^m C _{ds} T _{ds} T _{ds} G _{ds} A _{ds} A _{ds} ^m C _{ds} A _{ds} G _{ks} T _{ks} ^m C _k	65	889
1341685	1342	1357	21157	21172	ATGGCTACAGATTGGA	A _{ks} T _{ks} G _{ks} G _{ds} C _{ys} T _{ds} A _{ds} ^m C _{ds} A _{ds} G _{ds} A _{ds} T _{ds} T _{ds} T _{ds} G _{ks} G _{ks} A _k	44	111
1341691	1357	1372	21172	21187	TTGAUATTCTGTGGCA	T _{ks} T _{ks} G _{ks} A _{ds} U _{ys} A _{ds} T _{ds} T _{ds} ^m C _{ds} T _{ds} G _{ds} T _{ds} G _{ds} G _{ks} ^m C _{ks} A _k	60	2816
1341692	1348	1363	21163	21178	TGTGGCATGGCTACAG	T _{ks} G _{ks} T _{ks} G _{ds} G _{ys} ^m C _{ds} A _{ds} T _{ds} G _{ds} G _{ds} ^m C _{ds} T _{ds} A _{ds} ^m C _{ks} A _{ks} G _k	5	579
1341697	1353	1368	21168	21183	TATTCTGTGGCATGGC	T _{ks} A _{ks} T _{ks} T _{ds} C _{ys} T _{ds} G _{ds} T _{ds} G _{ds} G _{ds} ^m C _{ds} A _{ds} T _{ds} G _{ks} G _{ks} ^m C _k	55	890
1341698	1366	1381	21181	21196	GTGTUCTTGTTGATAT	G _{ks} T _{ks} G _{ks} T _{ds} U _{ys} ^m C _{ds} T _{ds} T _{ds} G _{ds} T _{ds} T _{ds} G _{ds} A _{ds} T _{ks} A _{ks} T _k	52	2817
1341702	1394	1409	21209	21224	CTTGATCTCTTAGCTG	^m C _{ks} T _{ks} T _{ks} G _{ds} A _{ys} T _{ds} ^m C _{ds} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ds} G _{ds} ^m C _{ks} T _{ks} G _k	32	736
1341703	1388	1403	21203	21218	CTCTUAGCTGTGCACT	^m C _{ks} T _{ks} ^m C _{ks} T _{ds} U _{ys} A _{ds} G _{ds} ^m C _{ds} T _{ds} G _{ds} T _{ds} G _{ds} ^m C _{ds} A _{ks} ^m C _{ks} T _k	67	2818

TABLE 46-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341713	1421	1436	21236	21251	AGGTUGAGATAAAGCT	$A_{ks}G_{ks}G_{ks}T_{ds}U_{ys}G_{ds}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}A_{ds}A_{ds}G_{ks}C_{ks}T_k$	58	2819
1341716	696	711	12114	12129	GACAUGAGGTTTTGAT	$G_{ks}A_{ks}^mC_{ks}A_{ds}U_{ys}G_{ds}A_{ds}G_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}T_{ds}G_{ks}A_{ks}T_k$	34	2820
1341719	1416	1431	21231	21246	GAGAUAAGCTGCCTG	$G_{ks}A_{ks}G_{ks}A_{ds}U_{ys}A_{ds}A_{ds}A_{ds}G_{ds}$ $^mC_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ks}T_{ks}G_k$	39	2821
1341726	1426	1441	21241	21256	TGTCACGTTGAGATA	$T_{ks}G_{ks}T_{ks}^mC_{ds}C_{ys}A_{ds}G_{ds}G_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ds}A_{ks}T_{ks}A_k$	41	191
1341727	1464	1479	21279	21294	AGGCUAGGGAATCTT	$A_{ks}G_{ks}G_{ks}^mC_{ds}U_{ys}A_{ds}G_{ds}G_{ds}$ $G_{ds}A_{ds}A_{ds}A_{ds}T_{ds}^mC_{ks}T_{ks}T_k$	41	2822
1341729	1494	1509	21309	21324	TGCACCGTTTTGGGCT	$T_{ks}G_{ks}^mC_{ks}A_{ds}C_{ys}^mC_{ds}G_{ds}T_{ds}$ $T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}G_{ks}^mC_{ks}T_k$	55	1671
1341734	1490	1505	21305	21320	CCGTUTTGGGCTAATG	$^mC_{ks}^mC_{ks}G_{ks}T_{ds}U_{ys}T_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}^mC_{ds}T_{ds}A_{ds}A_{ks}T_{ks}G_k$	51	2823
1341739	1623	1638	21438	21453	TAGAGTCGGTCACCTT	$T_{ks}A_{ks}G_{ks}A_{ds}G_{ys}T_{ds}^mC_{ds}G_{ds}G_{ds}$ $T_{ds}^mC_{ds}A_{ds}^mC_{ds}^mC_{ks}T_{ks}T_k$	55	506
1341743	1598	1613	21413	21428	ATCTUATAAGACTATA	$A_{ks}T_{ks}^mC_{ks}T_{ds}U_{ys}A_{ds}T_{ds}A_{ds}A_{ds}$ $G_{ds}A_{ds}^mC_{ds}T_{ds}A_{ks}T_{ks}A_k$	35	2824
1341745	1498	1513	21313	21328	GAGTUGCACCGTTTTTG	$G_{ks}A_{ks}G_{ks}T_{ds}U_{ys}G_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ks}T_{ks}G_k$	63	2825
1341753	1615	1630	21430	21445	GTCACCTTTCATAATG	$G_{ks}T_{ks}^mC_{ks}A_{ds}C_{ys}^mC_{ds}T_{ds}T_{ds}$ $T_{ds}^mC_{ds}A_{ds}T_{ds}A_{ks}T_{ks}G_k$	40	1907
1341755	1619	1634	21434	21449	GTCGGTCACCTTTCAT	$G_{ks}T_{ks}^mC_{ks}G_{ds}G_{ds}T_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}^mC_{ks}A_{ks}T_k$	66	194
1341758	1627	1642	21442	21457	AAAAUAGAGTCGGTCA	$A_{ks}A_{ks}A_{ks}A_{ds}U_{ys}A_{ds}G_{ds}A_{ds}G_{ds}$ $T_{ds}^mC_{ds}G_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	62	2826
1341759	1711	1726	21526	21541	CATCUCTGGGACCAAG	$^mC_{ks}A_{ks}T_{ks}^mC_{ds}U_{ys}^mC_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}A_{ds}^mC_{ds}^mC_{ds}A_{ks}G_k$	42	2827
1341763	1715	1730	21530	21545	TAAACATCTCTGGGAC	$T_{ks}A_{ks}A_{ks}A_{ds}C_{ys}A_{ds}T_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ks}^mT_k$	49	351
1341774	1705	1720	21520	21535	TGGGACCAAGGATATA	$T_{ks}G_{ks}G_{ks}G_{ds}A_{ys}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}G_{ds}T_{ds}A_{ks}T_{ks}A_k$	51	1674
1341777	796	811	15677	15692	CCATCTATCAGACTTC	$^mC_{ks}^mC_{ks}A_{ks}T_{ds}C_{ys}T_{ds}A_{ds}T_{ds}$ $^mC_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ks}T_{ks}^mC_k$	56	415
1341780	1770	1785	21585	21600	AGCCAGTACAGTTCCT	$A_{ks}G_{ks}^mC_{ks}^mC_{ds}A_{ys}G_{ds}T_{ds}A_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}T_{ds}^mC_{ks}^mC_{ks}T_k$	65	1909
1341785	1751	1766	21566	21581	CCTGUGTTAGCTTTAA	$^mC_{ks}^mC_{ks}T_{ks}G_{ds}U_{ys}G_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}T_{ds}T_{ks}A_{ks}A_k$	28	2828
1341786	1719	1734	21534	21549	TGTCUAAACATCTCTG	$T_{ks}G_{ks}T_{ks}^mC_{ds}U_{ys}A_{ds}A_{ds}A_{ds}$ $^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ks}T_{ks}G_k$	44	2829
1341788	1783	1798	21598	21613	TTTCUTATGTAATAGC	$T_{ks}T_{ks}T_{ks}^mC_{ds}U_{ys}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ds}A_{ks}G_{ks}^mC_k$	50	2830
1341793	1775	1790	21590	21605	GTAUAGCCAGTACAG	$G_{ks}T_{ks}A_{ks}A_{ds}U_{ys}A_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}^mC_{ks}A_{ks}G_k$	62	2831
1341796	1779	1794	21594	21609	TTATGTAATAGCCAGT	$T_{ks}T_{ks}A_{ks}T_{ds}G_{ys}T_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ks}G_{ks}T_k$	56	586
1341800	2187	2202	22002	22017	TGATGTAGTGGGAGTC	$T_{ks}G_{ks}A_{ks}T_{ds}G_{ys}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}G_{ds}G_{ds}A_{ds}G_{ks}T_{ks}^mC_k$	43	820

TABLE 46-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341805	2194	2209	22009	22024	TTAGUCTTGATGTAGT	$T_{ks}T_{ks}A_{ks}G_{ds}U_{ys}{}^mC_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}G_{ds}T_{ds}A_{ks}G_{ks}T_k$	50	2832
1341809	2350	2365	22165	22180	ACTACAAGAGGTTATT	$A_{ks}{}^mC_{ks}T_{ks}A_{ds}C_{ys}A_{ds}A_{ds}G_{ds}A_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}A_{ks}T_{ks}T_k$	29	899
1341810	2250	2265	22065	22080	AGGAUAGTCCATGCAA	$A_{ks}G_{ks}G_{ks}A_{ds}U_{ys}A_{ds}G_{ds}T_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}{}^mC_{ks}A_{ks}A_k$	55	2833
1341822	N/A	N/A	9514	9529	ACTAATATTGAGGCAC	$A_{ks}{}^mC_{ks}T_{ks}A_{ds}A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}G_{ds}{}^mC_{ks}A_{ks}C_k$	42	1772
1341825	N/A	N/A	9332	9347	ATCCUATATACATCCA	$A_{ks}T_{ks}{}^mC_{ks}{}^mC_{ds}U_{ys}A_{ds}T_{ds}A_{ds}$ $T_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ks}{}^mC_{ks}A_k$	64	2834
1341828	N/A	N/A	9601	9616	CATAUGTGAAGAGCTG	${}^mC_{ks}A_{ks}T_{ks}A_{ds}U_{ys}G_{ds}T_{ds}G_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}{}^mC_{ks}T_{ks}G_k$	39	2835
1341831	N/A	N/A	11578	11593	GTTAUTCTCTTGACAA	$G_{ks}T_{ks}T_{ks}A_{ds}U_{ys}T_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ds}{}^mC_{ks}A_{ks}A_k$	29	2836
1341833	N/A	N/A	9966	9981	TAGTGTAAAGCTGAGAG	$T_{ks}A_{ks}G_{ks}T_{ds}G_{ys}T_{ds}A_{ds}A_{ds}G_{ds}$ ${}^mC_{ds}T_{ds}G_{ds}A_{ds}G_{ks}A_{ks}G_k$	31	1852
1341836	808	823	15689	15704	TTGGUAGTATTCAT	$T_{ks}T_{ks}G_{ks}G_{ds}U_{ys}A_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ks}A_{ks}T_k$	13	2837
1341839	N/A	N/A	11546	11561	TCTTGACAATGGTTGC	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}G_{ys}A_{ds}{}^mC_{ds}A_{ds}A_{ds}$ $T_{ds}G_{ds}G_{ds}T_{ds}T_{ks}G_{ks}{}^mC_k$	67	2199
1341850	N/A	N/A	14197	14212	TTATGTGATTGAGTTC	$T_{ks}T_{ks}A_{ks}T_{ds}G_{ys}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}T_{ks}T_{ks}{}^mC_k$	35	2431
1341856	N/A	N/A	12419	12434	GTAACGGTGATCAAAT	$G_{ks}T_{ks}A_{ks}A_{ds}C_{ys}G_{ds}G_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}{}^mC_{ds}A_{ds}A_{ks}A_{ks}T_k$	21	2345
1341863	N/A	N/A	15496	15511	CTGCATTAATGCCACC	${}^mC_{ks}T_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}{}^mC_{ks}{}^mC_k$	52	1168
1341864	1017	1032	20832	20847	CAGCATTGATTGAAAT	${}^mC_{ks}A_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}G_{ds}A_{ks}A_{ks}A_k$	37	263
1341867	N/A	N/A	14916	14931	ATACATTAGCAAGCTA	$A_{ks}T_{ks}A_{ks}{}^mC_{ds}A_{ds}T_{ds}T_{ds}A_{ds}G_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}G_{ds}{}^mC_{ks}T_{ks}A_k$	36	309
1341871	N/A	N/A	16289	16304	GTTAGAATAGTCTTCA	$G_{ks}T_{ks}T_{ks}A_{ds}G_{ys}A_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ks}T_{ks}A_k$	42	1481
1341873	N/A	N/A	16218	16233	TATAGACTGGGTAGGA	$T_{ks}A_{ks}T_{ks}A_{ds}G_{ys}A_{ds}{}^mC_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}T_{ds}A_{ds}G_{ks}G_{ks}A_k$	30	2808
1341876	N/A	N/A	16127	16142	CATAATAATAGCTCTA	${}^mC_{ks}A_{ks}T_{ks}A_{ds}A_{ds}T_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}{}^mC_{ds}T_{ds}{}^mC_{ks}T_{ks}A_k$	33	2511
1341878	N/A	N/A	16382	16397	TCCCAACCAAACCTTCC	$T_{ks}{}^mC_{ks}{}^mC_{ks}{}^mC_{ds}A_{ds}A_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}{}^mC_{ds}T_{ks}{}^mC_{ks}{}^mC_k$	43	2026
1341885	N/A	N/A	17994	18009	GAAGUTAGTATAGTTA	$G_{ks}A_{ks}A_{ks}G_{ds}U_{ys}T_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ds}T_{ds}A_{ds}G_{ds}T_{ks}T_{ks}A_k$	43	2838
1341892	1083	1098	20898	20913	TGCTGCTAGTGCCAAA	$T_{ks}G_{ks}{}^mC_{ks}T_{ds}G_{ys}{}^mC_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}A_k$	50	1665
1341893	N/A	N/A	17989	18004	TAGTATAGTTATCTTC	$T_{ks}A_{ks}G_{ks}T_{ds}A_{ys}T_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ks}T_{ks}{}^mC_k$	13	1093
1341898	N/A	N/A	19912	19927	TGCAUCTTAAGATAACC	$T_{ks}G_{ks}{}^mC_{ks}A_{ds}U_{ys}{}^mC_{ds}T_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A_{ks}{}^mC_{ks}{}^mC_k$	51	2839
1341903	N/A	N/A	18690	18705	CTTTGAGACTCTTGTT	${}^mC_{ks}T_{ks}T_{ks}T_{ds}G_{ys}A_{ds}G_{ds}A_{ds}$ ${}^mC_{ds}T_{ds}{}^mC_{ds}T_{ds}G_{ks}T_{ks}T_k$	15	2583

TABLE 46-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341905	N/A	N/A	18092	18107	TGATACACCAATGCAG	$T_{ks}G_{ks}A_{ks}T_{ds}A_{ys}{}^mC_{ds}A_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}T_{ds}G_{ds}{}^mC_{ks}A_{ks}G_k$	38	2370
1341910	N/A	N/A	19904	19919	AAGAUACCCAGGTTCG	$A_{ks}A_{ks}G_{ks}A_{ds}U_{ys}A_{ds}{}^mC_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}G_{ds}G_{ds}T_{ds}T_{ks}G_{ks}{}^mC_k$	37	2840
1341913	608	623	N/A	N/A	ACAAUATGGGATGAGG	$A_{ks}{}^mC_{ks}A_{ks}A_{ds}U_{ys}A_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}A_{ds}T_{ds}G_{ds}A_{ks}G_kG_k$	70	2841
1341916	N/A	N/A	19908	19923	TCTTAAGATACCCAGG	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}A_{ys}A_{ds}G_{ds}A_{ds}T_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}A_{ks}G_{ks}G_k$	51	319
1341918	N/A	N/A	19916	19931	GGAGUGCATCTTAAGA	$G_{ks}G_{ks}A_{ks}G_{ds}U_{ys}G_{ds}{}^mC_{ds}A_{ds}$ $T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ks}G_{ks}A_k$	32	2842
1341920	N/A	N/A	20112	20127	ATCAAGACATTCTAGC	$A_{ks}T_{ks}{}^mC_{ks}A_{ks}A_{ys}G_{ds}A_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ks}G_{ks}{}^mC_k$	37	1488
1341927	1079	1094	20894	20909	GCTAGTGCCAAACCAA	$G_{ks}{}^mC_{ks}T_{ks}A_{ds}G_{ys}T_{ds}G_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}A_{ks}A_k$	65	1354
1341929	N/A	N/A	20385	20400	TAGTAATCACAGTAA	$T_{ks}A_{ks}G_{ks}T_{ds}A_{ys}A_kT_{ds}{}^mC_{ds}A_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}T_{ds}T_{ks}A_{ks}A_k$	38	1489
1341931	N/A	N/A	20197	20212	ACATUGCAGTTATTAG	$A_{ks}{}^mC_{ks}A_{ks}T_{ds}U_{ys}G_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}A_{ds}T_{ds}T_{ks}A_{ks}G_k$	26	2843
1341936	N/A	N/A	20187	20202	TATTAGAAGTCAGCCC	$T_{ks}A_{ks}T_{ks}T_{ds}A_{ys}G_{ds}A_{ds}G_{ds}$ $T_{ds}{}^mC_{ds}A_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}{}^mC_k$	29	164
1341940	N/A	N/A	20482	20497	TCATAGTGGACTTCAT	$T_{ks}{}^mC_{ks}A_{ks}T_{ds}A_{ys}G_{ds}T_{ds}G_{ds}G_{ds}$ $A_{ds}{}^mC_{ds}T_{ds}T_{ds}{}^mC_{ks}A_{ks}T_k$	44	633
1341946	1075	1090	20890	20905	GTGCCAAACCAATGTT	$G_{ks}T_{ks}G_{ks}{}^mC_{ds}C_{ys}A_{ds}A_{ds}A_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}T_{ds}G_{ks}T_{ks}T_k$	30	1042
1341951	N/A	N/A	12801	12816	AGGCUGTAAGAGTCAG	$A_{ks}G_{ks}G_{ks}{}^mC_{ds}U_{ys}G_{ds}T_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ks}A_{ks}G_k$	52	2844
1341955	N/A	N/A	12496	12511	AATTCACCTTGACTAA	$A_{ks}A_{ks}T_{ks}T_{ds}C_{ys}A_{ds}{}^mC_{ds}{}^mC_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ds}{}^mC_{ds}T_{ks}A_{ks}A_k$	0	2144

TABLE 47

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_k$	86	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_k$	87	1526
1341649	1314	1329	21129	21144	CAGTCTTAAACCTTCC	${}^mC_{ks}A_{ks}G_{ks}T_{ds}C_{ys}T_{ds}T_{ds}A_{ds}A_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ks}{}^mC_{ks}{}^mC_k$	59	30
1341652	1318	1333	21133	21148	TGAACAGTCTTAAACC	$T_{ks}G_{ks}A_{ks}A_{ds}C_{ys}A_{ds}G_{ds}T_{ds}{}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ds}A_{ks}{}^mC_k{}^mC_k$	57	344
1341655	1256	1271	21071	21086	GCTTUTGTCCACCTTT	$G_{ks}{}^mC_{ks}T_{ks}T_{ds}U_{ys}T_{ds}G_{ds}T_{ds}{}^mC_{ds}$ ${}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}T_{ks}T_k$	75	2845
1341666	1309	1324	21124	21139	TTAAACCTTCCCTGTG	$T_{ks}T_{ks}A_{ks}A_{ds}A_{ys}{}^mC_{ds}{}^mC_{ds}T_{ds}$ $T_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}G_{ks}T_{ks}G_k$	23	2807

TABLE 47-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO	
1341669	1326	1341	21141	21156	ATGCUACTTGAACAGT	$A_{ks}T_{ks}G_{ks}^mG_{ds}U_{ys}A_{ds}^mC_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}A_{ds}^mC_{ds}A_{ks}G_{ks}T_{ks}$	46	2846	
1341671	1330	1345	21145	21160	TGGAATGCTACTTGAA	$T_{ks}G_{ks}G_{ks}A_{ks}A_{ds}T_{ds}G_{ds}^mC_{ds}T_{ds}$ $A_{ds}^mC_{ds}T_{ds}T_{ds}G_{ks}A_{ks}A_{ks}$	79	2203	
1341673	1336	1351	21151	21166	ACAGATTGGAATGCTA	$A_{ks}^mC_{ks}A_{ks}G_{ds}A_{ys}T_{ds}T_{ds}G_{ds}G_{ds}$ $A_{ds}A_{ds}T_{ds}G_{ds}^mC_{ks}T_{ks}A_{ks}$	70	1668	
1341674	1322	1337	21137	21152	TACTUGAACAGTCTTA	$T_{ks}A_{ks}^mC_{ks}T_{ds}U_{ys}G_{ds}A_{ds}A_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}^mC_{ds}T_{ks}T_{ks}A_{ks}$	38	2847	
1341686	1344	1359	21159	21174	GCATGGCTACAGATTG	$G_{ks}^mC_{ks}A_{ks}T_{ds}G_{ys}G_{ds}^mC_{ds}T_{ds}$ $A_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}T_{ks}T_{ks}G_{ks}$	76	267	
1341689	1354	1369	21169	21184	ATATUCTGTGGCATGG	$A_{ks}T_{ks}A_{ks}T_{ds}U_{ys}^mC_{ds}T_{ds}G_{ds}T_{ds}$ $G_{ds}G_{ds}^mC_{ds}A_{ds}T_{ks}G_{ks}G_{ks}$	63	2848	
1341693	1358	1373	21173	21188	GTTGATATTCTGTGGC	$G_{ks}T_{ks}T_{ks}G_{ds}A_{ys}T_{ds}A_{ds}T_{ds}T_{ds}$ $^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ks}G_{ks}^mC_{ks}$	68	1280	
1341694	1349	1364	21164	21179	CTGTGGCATGGCTACA	$^mC_{ks}T_{ks}G_{ks}T_{ds}G_{ys}G_{ds}^mC_{ds}A_{ds}$ $T_{ds}G_{ds}G_{ds}^mC_{ds}T_{ds}A_{ks}^mC_{ks}A_{ks}$	35	657	
1341701	1389	1404	21204	21219	TCTCUTAGCTGTGCAC	$T_{ks}^mC_{ks}T_{ks}^mC_{ds}U_{ys}T_{ds}A_{ds}G_{ds}$ $^mC_{ds}T_{ds}T_{ds}T_{ds}G_{ds}^mC_{ks}A_{ks}^mC_{ks}$	49	2849	
1341706	1367	1382	21182	21197	TGTGUTCTTGTTGATA	$T_{ks}G_{ks}T_{ks}G_{ds}U_{ys}T_{ds}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}T_{ds}T_{ds}G_{ds}A_{ks}T_{ks}A_{ks}$	41	2850	
1341707	698	713	12116	12131	GAGACATGAGGTTTTTG	$G_{ks}A_{ks}G_{ks}A_{ds}C_{ys}A_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}T_{ks}T_{ks}G_{ks}$	61	725	
1341708	1395	1410	21210	21225	ACTTGATCTCTTAGCT	$A_{ks}^mC_{ks}T_{ks}T_{ds}G_{ys}A_{ds}T_{ds}^mC_{ds}$ $T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ks}^mC_{ks}T_{ks}$	39	813	
1341709	1418	1433	21233	21248	TTGAGATAAAGCTGCC	$T_{ks}T_{ks}G_{ks}A_{ds}G_{ys}A_{ds}T_{ds}A_{ds}A_{ds}$ $A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ks}^mC_{ks}^mC_{ks}$	51	1670	
1341717	1422	1437	21237	21252	CAGGUTGAGATAAAGC	$^mC_{ks}A_{ks}G_{ks}G_{ds}U_{ys}T_{ds}G_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}A_{ds}A_{ks}P_{ks}G_{ks}^mC_{ks}$	55	2851	
1341722	1466	1481	21281	21296	AGAGGCTAGGGAATC	$A_{ks}G_{ks}A_{ks}G_{ds}G_{ys}^mC_{ds}T_{ds}A_{ds}$ $G_{ds}G_{ds}G_{ds}A_{ds}A_{ds}A_{ks}T_{ks}^mC_{ks}$	48	1048	
1341723	1429	1444	21244	21259	ATATGTCCAGGTTGAG	$A_{ks}T_{ks}A_{ks}T_{ds}G_{ys}T_{ds}^mC_{ds}^mC_{ds}$ $A_{ds}G_{ds}G_{ds}T_{ds}T_{ds}G_{ks}A_{ks}G_{ks}$	48	425	
1341733	1491	1506	21306	21321	ACCGUTTTGGGCTAAT	$A_{ks}^mC_{ks}^mC_{ks}G_{ds}U_{ys}T_{ds}T_{ds}T_{ds}$ $G_{ds}G_{ds}G_{ds}^mC_{ds}T_{ds}A_{ks}A_{ks}T_{ks}$	65	2852	
1341737	1495	1510	21310	21325	TTGCACCGTTTGGGC	$T_{ks}T_{ks}G_{ks}^mC_{ds}A_{ys}^mC_{ds}^mC_{ds}G_{ds}$ $T_{ds}T_{ds}T_{ds}T_{ds}G_{ds}G_{ks}G_{ks}^mC_{ks}$	50	1749	
1341740	1499	1514	21314	21329	AGAGUTGCACCGTTTT	$A_{ks}G_{ks}A_{ks}G_{ds}U_{ys}T_{ds}G_{ds}^mC_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}G_{ds}T_{ds}T_{ks}T_{ks}T_{ks}$	68	2853	
1341741	1624	1639	21439	21454	ATAGAGTCGGTCACCT	$A_{ks}T_{ks}A_{ks}G_{ds}A_{ys}G_{ds}T_{ds}^mC_{ds}G_{ds}$ $G_{ds}T_{ds}^mC_{ds}A_{ds}^mC_{ks}^mC_{ks}T_{ks}$	50	584	
1341742	1600	1615	21415	21430	GTATCTTATAAGACTA	$G_{ks}T_{ks}A_{ks}T_{ds}C_{ys}T_{ds}T_{ds}A_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}^mC_{ks}T_{ks}A_{ks}$	79	1595	
1341751	1616	1631	21431	21446	GGTCACCTTTTCATAAT	$G_{ks}G_{ks}T_{ks}^mC_{ds}A_{ys}^mC_{ds}^mC_{ds}T_{ds}$ $T_{ds}T_{ds}^mC_{ds}A_{ds}T_{ds}A_{ks}P_{ks}T_{ks}$	48	1985	
1341756	798	813	15679	15694	TTCCATCTATCAGACT	$T_{ks}T_{ks}^mC_{ks}^mC_{ds}A_{ys}T_{ds}^mC_{ds}T_{ds}$ $A_{ds}T_{ds}^mC_{ds}A_{ds}G_{ds}A_{ks}^mC_{ks}T_{ks}$	38	571	
1341757	1620	1635	21435	21450	AGTCGGTCACCTTTCA	$A_{ks}G_{ks}T_{ks}^mC_{ds}G_{ys}G_{ds}T_{ds}^mC_{ds}$ $A_{ys}^mC_{ks}^mC_{ds}T_{ks}T_{ks}T_{ks}^mC_{ks}A_{ks}$	61	272	

TABLE 47-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341760	1712	1727	21527	21542	ACATCTCTGGGACCAA	$A_{ks}^m C_{ks} A_{ks} T_{ds} C_{ys} T_{ds}^m C_{ds} T_{ds}$ $G_{ds} G_{ds} G_{ds} A_{ds}^m C_{ds}^m C_{ks} A_{ks} A_k$	68	195
1341765	1628	1643	21443	21458	TAAATAGAGTCGGTC	$T_{ks} A_{ks} A_{ks} A_{ds} A_{ys} T_{ds} A_{ds} G_{ds} A_{ds}$ $G_{ds} T_{ds}^m C_{ds} G_{ds} G_{ks} T_{ks}^m C_k$	52	895
1341770	1721	1736	21536	21551	ATTGUCTAAACATCTC	$A_{ks} T_{ks} T_{ks} G_{ds} U_{ys}^m C_{ds} T_{ds} A_{ds} A_{ds}$ $A_{ds}^m C_{ds} A_{ds} T_{ds}^m C_{ks} T_{ks}^m C_k$	73	2854
1341771	1716	1731	21531	21546	CTAAACATCTCTGGGA	$^m C_{ks} T_{ks} A_{ks} A_{ds} A_{ys}^m C_{ds} A_{ds} T_{ds}$ $^m C_{ds} T_{ds}^m C_{ds} T_{ds} G_{ds} G_{ks} G_{ks} A_k$	32	2259
1341776	1707	1722	21522	21537	TCTGGGACCAAGGATA	$T_{ks}^m C_{ks} T_{ks} G_{ds} G_{ys} G_{ds} A_{ds}^m C_{ds}$ $^m C_{ds} A_{ds} A_{ds} G_{ds} G_{ds} A_{ks} T_{ks} A_k$	31	1830
1341779	1752	1767	21567	21582	TCCTGTGTAGCTTTA	$T_{ks}^m C_{ks}^m C_{ks} T_{ds} G_{ys} T_{ds} G_{ds} T_{ds}$ $T_{ds} A_{ds} G_{ds}^m C_{ds} T_{ds} T_{ks} T_{ks} A_k$	72	1442
1341784	1771	1786	21586	21601	TAGCCAGTACAGTTCC	$T_{ks} A_{ks} G_{ks}^m C_{ds} C_{ys} A_{ds} G_{ds} T_{ds} A_{ds}$ $^m C_{ds} A_{ds} G_{ds} T_{ds} T_{ks}^m C_{ks}^m C_k$	70	1987
1341791	1784	1799	21599	21614	GTTTCTTATGTAATAG	$G_{ks} T_{ks} T_{ks} T_{ds} C_{ys} T_{ds} T_{ds} A_{ds} T_{ds}$ $G_{ds} T_{ds} A_{ds} A_{ds} T_{ks} A_{ks} G_k$	25	897
1341795	1776	1791	21591	21606	TGTAATAGCCAGTACA	$T_{ks} G_{ks} T_{ks} A_{ds} A_{ys} T_{ds} A_{ds} G_{ds}^m C_{ds}$ $^m C_{ds} A_{ds} G_{ds} T_{ds} A_{ks}^m C_{ks} A_k$	40	352
1341797	1780	1795	21595	21610	CTTAUGTAATAGCCAG	$^m C_{ks} T_{ks} T_{ks} A_{ds} U_{ys} G_{ds} T_{ds} A_{ds} A_{ds}$ $T_{ds} A_{ds} G_{ds}^m C_{ds}^m C_{ks} A_{ks} G_k$	75	2855
1341799	2196	2211	22011	22026	GATTAGTCTTGATGTA	$G_{ks} A_{ks} T_{ks} T_{ds} A_{ys} G_{ds} T_{ds} m G_{ds} T_{ds}$ $T_{ds} G_{ds} A_{ds} T_{ds} G_{ks} T_{ks} A_k$	45	1521
1341804	2189	2204	22004	22019	CTTGATGTAGTGGGAG	$^m G_{ks} T_{ks} T_{ks} G_{ds} A_{ys} T_{ds} G_{ds} T_{ds} A_{ds}$ $G_{ds} T_{ds} G_{ds} G_{ds} G_{ks} A_{ks} G_k$	67	976
1341808	2353	2368	22168	22183	ATAACTACAAGAGGTT	$A_{ks} T_{ks} A_{ks} A_{ds} C_{ys} T_{ds} A_{ds}^m C_{ds} A_{ds}$ $A_{ds} G_{ds} A_{ds} G_{ds} T_{ks} T_k$	61	1133
1341813	2252	2267	22067	22082	AGAGGATAGTCCATGC	$A_{ks} G_{ks} A_{ks} G_{ds} G_{ys} A_{ds} T_{ds} A_{ds} G_{ds}$ $T_{ds}^m C_{ds}^m C_{ds} A_{ds} T_{ks} G_{ks}^m C_k$	53	354
1341818	610	625	N/A	N/A	GAACAATATGGGATGA	$G_{ks} A_{ks} A_{ks}^m C_{ds} A_{ys} A_{ds} T_{ds} A_{ds} T_{ds}$ $G_{ds} G_{ds} G_{ds} A_{ds}^m C_{ks} G_{ks} A_k$	50	1735
1341823	N/A	N/A	9516	9531	ACACUAATATTGAGGC	$A_{ks}^m C_{ks} A_{ks}^m C_{ds} U_{ys} A_{ds} A_{ds} T_{ds}$ $A_{ds} T_{ds} T_{ds} G_{ds} A_{ks} G_{ks}^m C_k$	43	2856
1341824	810	825	15691	15706	TATTGGTAAGTATTCC	$T_{ks} A_{ks} T_{ks} T_{ds} G_{ys} G_{ds} T_{ds} A_{ds} A_{ds}$ $G_{ds} T_{ds} A_{ds} T_{ds} T_{ks}^m C_{ks}^m C_k$	41	1272
1341826	N/A	N/A	9334	9349	CGATCCTATATACATC	$^m C_{ks} G_{ks} A_{ks} T_{ds} C_{ys}^m C_{ds} T_{ds} A_{ds}$ $T_{ds} A_{ds} T_{ds} A_{ds}^m C_{ds} A_{ks} T_{ks}^m C_k$	28	2788
1341835	N/A	N/A	9968	9983	AGTAGTGTAAGCTGAG	$A_{ks} G_{ks} T_{ks} A_{ds} C_{ys} T_{ds} G_{ds} T_{ds} A_{ds}$ $A_{ds} G_{ds}^m C_{ds} T_{ds} G_{ks} A_{ks} G_k$	68	2155
1341837	N/A	N/A	9603	9618	CACAUATGTGAAGAGC	$^m C_k S A_{ks}^m C_{ks} A_{ds} U_{ys} A_{ds} T_{ds} G_{ds}$ $T_{ds} G_{ds} A_{ds} A_{ds} G_{ds} A_{ks} G_{ks}^m C_k$	65	2857
1341843	N/A	N/A	11568	11583	TGACAATGGTTGATAG	$T_{ks} G_{ks} A_{ks}^m C_{ds} A_{ys} A_{ds} T_{ds} G_{ds} G_{ds}$ $T_{ds} T_{ds} G_{ds} A_{ds} T_{ks} A_{ks} G_k$	42	1700
1341844	N/A	N/A	11853	11868	AGCAGAATTGTGAACG	$A_{ks} G_{ks}^m C_{ks} A_{ds} G_{ys} A_{ds} T_{ds} T_{ds}$ $G_{ds} T_{ds} G_{ds} A_{ds} A_{ks}^m C_{ks} G_k$	51	844
1341848	N/A	N/A	12421	12436	TGGTAACGGTGATCAA	$T_{ks} G_{ks} G_{ks} T_{ds} A_{ys} A_{ds}^m C_{ds} G_{ds} G_{ds}$ $T_{ds} G_{ds} A_{ds} T_{ds}^m C_{ks} A_{ks} A_k$	41	67
1341854	N/A	N/A	14199	14214	TATTATGTGATTGAGT	$T_{ks} A_{ks} T_{ks} T_{ds} A_{ys} T_{ds} G_{ds} T_{ds} G_{ds}$ $A_{ds} T_{ds} T_{ds} G_{ds} A_{ks} G_{ks} T_k$	28	73

TABLE 47-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO.: 1 Start Site	SEQ ID NO.: 1 Stop Site	SEQ ID NO.: 2 Start Site	SEQ ID NO.: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341858	N/A	N/A	14918	14933	TAATACATTAGCAAGC	$T_{ks}A_{ks}A_{ks}T_{ds}A_{ys}^mC_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}A_{ds}A_{ks}G_{ks}C_k$	26	2266
1341865	1019	1034	20834	20849	TGCAGCATTGATTTCGA	$T_{ks}G_{ks}^mC_{ks}A_{ds}G_{ys}^mC_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}^mC_{ks}G_{ks}A_k$	65	419
1341866	N/A	N/A	15498	15513	TGCTGCATTAATGCCA	$T_{ks}G_{ks}^mC_{ks}T_{ds}G_{ys}^mC_{ds}A_{ds}T_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ds}G_{ds}^mC_{ks}^mC_{ks}A_k$	32	2485
1341869	N/A	N/A	16129	16144	TCCAUAATAATAGCTC	$T_{ks}^mC_{ks}^mC_{ks}A_{ds}U_{ys}A_{ds}A_{ds}T_{ds}$ $A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}^mC_{ks}^mC_{ks}C_k$	58	2858
1341872	N/A	N/A	16292	16307	TTAGUTAGAATAGTCT	$T_{ks}T_{ks}A_{ks}G_{ds}U_{ys}T_{ds}A_{ds}G_{ds}A_{ds}$ $A_{ds}T_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}T_k$	35	2859
1341875	N/A	N/A	16223	16238	CTTTUTATAGACTGGG	$^mC_{ks}T_{ks}T_{ks}T_{ds}U_{ys}T_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}A_{ds}^mC_{ds}T_{ds}G_{ks}G_{ks}G_k$	34	2860
1341879	N/A	N/A	17990	18005	TTAGUATAGTTATCTT	$T_{ks}T_{ks}A_{ks}G_{ds}U_{ys}A_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ds}^mC_{ks}T_{ks}T_k$	22	2861
1341881	N/A	N/A	17995	18010	TGAAGTTAGTATAGTT	$T_{ks}G_{ks}A_{ks}A_{ds}G_{ys}T_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}A_{ds}T_{ds}A_{ds}G_{ks}T_{ks}T_k$	62	1405
1341884	N/A	N/A	16384	16399	AATCCCAACCAAACTT	$A_{ks}A_{ks}T_{ks}m_{ds}C_{ys}m_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}A_{ds}^mC_{ks}T_{ks}^mT_k$	7	2755
1341888	N/A	N/A	18095	18110	AATTGATACACCAATG	$A_{ks}A_{ks}T_{ks}T_{ds}G_{ys}A_{ds}T_{ds}A_{ds}^mC_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}T_{ks}G_k$	40	2525
1341897	1084	1099	20899	20914	CTGCUGCTAGTGCCAA	$^mC_{ks}T_{ks}G_{ks}^mC_{ds}U_{ys}G_{ds}^mC_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ds}G_{ds}^mC_{ks}^mC_{ks}A_{ks}$	66	2862
1341902	N/A	N/A	19913	19928	GTGCATCTTAAGATAC	$G_{ks}T_{ks}G_{ks}^mC_{ds}A_{ys}T_{ds}^mC_{ds}T_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{ks}^mC_k$	40	2806
1341906	N/A	N/A	18693	18708	AGACUTTGAAGTCTT	$A_{ks}G_{ks}A_{ks}^mC_{ds}U_{ys}T_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}A_{ds}^mC_{ds}T_{ds}^mC_{ks}T_{ks}T_k$	48	2863
1341909	N/A	N/A	19909	19924	ATCTUAAGATACCCAG	$A_{ks}T_{ks}^mC_{ks}T_{ds}U_{ys}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}^mC_{ds}^mC_{ds}^mC_{ks}A_{ks}G_k$	55	2864
1341911	1080	1095	20895	20910	TGCTAGTGCCAAACCA	$T_{ks}G_{ks}^mC_{ks}T_{ds}A_{ys}G_{ds}T_{ds}G_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}A_{ds}A_{ds}^mC_{ks}^mC_{ks}A_k$	80	1432
1341912	N/A	N/A	19905	19920	TAAGATACCCAGGTG	$T_{ks}A_{ks}A_{ks}G_{ds}A_{ys}T_{ds}A_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}G_{ds}T_{ks}T_{ks}G_k$	11	2790
1341922	N/A	N/A	20114	20129	GCATCAAGACATTCTA	$G_{ks}^mC_{ks}A_{ks}T_{ds}C_{ys}A_{ds}G_{ds}$ $A_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ks}T_{ks}A_k$	51	1565
1341926	N/A	N/A	19999	20014	CAGTUTTCCTCATGAT	$^mC_{ks}A_{ks}G_{ks}T_{ds}U_{ys}T_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}T_{ds}^mC_{ds}A_{ds}T_{ds}G_{ks}A_{ks}T_k$	0	2865
1341932	1076	1091	20891	20906	AGTGCCAAACCAATGT	$A_{ks}G_{ks}T_{ks}G_{ds}C_{ys}C_{ds}A_{ds}A_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}A_{ds}T_{ks}G_{ks}T_k$	24	1120
1341933	N/A	N/A	20388	20403	CTATAGTAATCACAAG	$^mC_{ks}T_{ks}A_{ks}T_{ds}A_{ys}G_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}^mC_{ds}A_{ds}^mC_{ds}A_{ks}G_k$	37	1644
1341935	N/A	N/A	20190	20205	AGTTATTAGAAGTCAG	$A_{ks}G_{ks}T_{ks}T_{ds}A_{ys}T_{ds}T_{ds}A_{ds}G_{ds}$ $A_{ds}A_{ds}G_{ds}T_{ds}^mC_{ks}A_{ks}G_k$	25	2746
1341938	N/A	N/A	20378	20393	CACAAGTAAGGTAAAG	$^mC_{ks}A_{ks}^mC_{ks}A_{ds}A_{ys}G_{ds}T_{ds}A_{ds}$ $A_{ds}G_{ds}G_{ds}T_{ds}A_{ds}A_{ks}A_{ks}G_k$	43	2448
1341944	N/A	N/A	20484	20499	CTTCATAGTGGACTTC	$^mC_{ks}T_{ks}T_{ks}^mC_{ds}A_{ys}T_{ds}A_{ds}G_{ds}$ $T_{ds}G_{ds}G_{ds}A_{ds}^mC_{ds}T_{ks}T_{ks}^mC_k$	35	711

TABLE 47-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341952	N/A	N/A	12804	12819	CTCAGGCTGTAAGAGT	${}^m\text{C}_{ks}\text{T}_{ks}{}^m\text{C}_{ks}\text{A}_{ds}\text{G}_{ys}\text{G}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}$ $\text{G}_{ds}\text{T}_{ds}{}^m\text{A}_{ds}\text{A}_{ds}\text{G}_{ds}\text{A}_{ks}\text{G}_{ks}\text{T}_k$	46	2240
1341956	N/A	N/A	12499	12514	AGAAATTCACCTTGAC	$\text{A}_{ks}\text{G}_{ks}{}^m\text{A}_{ks}{}^m\text{A}_{ds}\text{T}_{ds}{}^m\text{T}_{ds}{}^m\text{C}_{ds}\text{A}_{ds}$ ${}^m\text{C}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}\text{T}_{ds}\text{G}_{ks}\text{A}_{ks}{}^m\text{C}_k$	9	2284

TABLE 48

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$\text{A}_{ks}\text{G}_{ks}\text{T}_{ks}\text{A}_{ds}\text{G}_{ds}\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}$ $\text{T}_{ds}\text{A}_{ds}\text{A}_{ds}\text{G}_{ds}\text{T}_{ks}{}^m\text{C}_{ks}\text{A}_k$	82	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCACT	$\text{G}_{ks}\text{A}_{ks}\text{G}_{ks}\text{T}_{ds}\text{G}_{ds}\text{A}_{ds}\text{T}_{ds}{}^m\text{C}_{ds}$ $\text{A}_{ds}\text{T}_{ds}\text{T}_{ds}{}^m\text{C}_{ds}\text{A}_{ks}\text{G}_{ks}\text{T}_k$	86	1526
1341651	1315	1330	21130	21145	ACAGUCTTAAACCTTC	$\text{A}_{ks}{}^m\text{C}_{ks}\text{A}_{ks}\text{G}_{ds}\text{U}_{ys}{}^m\text{C}_{ds}\text{T}_{ds}\text{T}_{ds}$ $\text{A}_{ds}\text{A}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}{}^m\text{C}_{ds}\text{T}_{ks}\text{T}_{ks}{}^m\text{C}_k$	81	2866
1341656	1257	1272	21072	21087	AGCTUTTGTCCACCTT	$\text{A}_{ks}\text{G}_{ks}{}^m\text{C}_{ks}\text{T}_{ds}\text{U}_{ys}\text{T}_{ds}\text{T}_{ds}\text{G}_{ds}\text{T}_{ds}$ ${}^m\text{C}_{ds}{}^m\text{C}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}{}^m\text{C}_{ks}\text{T}_{ks}\text{T}_k$	72	2867
1341657	613	628	N/A	N/A	CTGGAACAATATGGGA	${}^m\text{C}_{ks}\text{T}_{ks}\text{G}_{ks}\text{G}_{ds}\text{A}_{ys}{}^m\text{C}_{ds}\text{A}_{ds}$ $\text{A}_{ds}\text{T}_{ds}\text{A}_{ds}\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}\text{G}_{ks}\text{A}_k$	45	1969
1341659	1327	1342	21142	21157	AATGCTACTTGAACAG	$\text{A}_{ks}\text{A}_{ks}\text{T}_{ks}\text{G}_{ds}\text{C}_{ys}\text{T}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}$ $\text{T}_{ds}\text{G}_{ds}\text{A}_{ds}\text{A}_{ds}{}^m\text{C}_{ks}\text{A}_{ks}\text{G}_k$	34	1045
1341662	1319	1334	21134	21149	TTGAACAGTCTTAAAC	$\text{T}_{ks}\text{T}_{ks}\text{G}_{ks}\text{A}_{ds}\text{A}_{ys}{}^m\text{C}_{ds}\text{A}_{ds}\text{G}_{ds}\text{T}_{ds}$ ${}^m\text{C}_{ds}\text{T}_{ds}\text{T}_{ds}\text{A}_{ds}\text{A}_{ks}{}^m\text{C}_k$	11	422
1341665	1310	1325	21125	21140	CTTAAACCTTCCCTGT	${}^m\text{C}_{ks}\text{T}_{ks}\text{T}_{ks}\text{A}_{ds}\text{A}_{ys}\text{A}_{ds}{}^m\text{C}_{ds}{}^m\text{C}_{ds}$ $\text{T}_{ds}\text{T}_{ds}{}^m\text{C}_{ds}{}^m\text{C}_{ds}{}^m\text{C}_{ds}\text{T}_{ks}\text{G}_{ks}\text{T}_k$	24	1823
1341675	1323	1338	21138	21153	CTACUTGAACAGTCTT	${}^m\text{C}_{ks}\text{T}_{ks}\text{A}_{ks}{}^m\text{C}_{ds}\text{U}_{ys}\text{T}_{ds}\text{G}_{ds}\text{A}_{ds}$ $\text{A}_{ds}{}^m\text{C}_{ds}\text{A}_{ds}\text{G}_{ds}\text{T}_{ds}{}^m\text{C}_{ks}\text{T}_{ks}\text{T}_k$	79	2868
1341679	1355	1370	21170	21185	GATAUTCTGTGGCATG	$\text{GSA}_{ks}\text{T}_{ks}\text{A}_{ds}\text{U}_{ys}\text{T}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}\text{G}_{ds}$ $\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}{}^m\text{C}_{ds}\text{A}_{ks}\text{T}_{ks}\text{G}_k$	60	2869
1341681	1337	1352	21152	21167	TACAGATTGGAATGCT	$\text{T}_{ks}\text{A}_{ks}{}^m\text{C}_{ks}\text{A}_{ds}\text{G}_{ys}\text{A}_{ds}\text{T}_{ds}\text{T}_{ds}\text{G}_{ds}$ $\text{G}_{ds}\text{A}_{ds}\text{A}_{ds}\text{T}_{ds}\text{G}_{ks}{}^m\text{C}_{ks}\text{T}_k$	60	1746
1341682	700	715	12118	12133	CAGAGACATGAGGTTT	${}^m\text{C}_{ks}\text{A}_{ks}\text{G}_{ks}\text{A}_{ds}\text{G}_{ys}\text{A}_{ds}{}^m\text{C}_{ds}\text{A}_{ds}$ $\text{T}_{ds}\text{G}_{ds}\text{A}_{ds}\text{G}_{ds}\text{G}_{ds}\text{T}_{ks}\text{T}_{ks}\text{T}_k$	61	2319
1341684	1331	1346	21146	21161	TTGGAATGCTACTTGA	$\text{T}_{ks}\text{T}_{ks}\text{G}_{ks}\text{G}_{ds}\text{A}_{ys}\text{A}_{ds}\text{T}_{ds}\text{G}_{ds}{}^m\text{C}_{ds}$ $\text{T}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}\text{T}_{ks}\text{G}_{ks}\text{A}_k$	65	1279
1341687	1346	1361	21161	21176	TGGCATGGCTACAGAT	$\text{T}_{ks}\text{G}_{ks}\text{G}_{ks}{}^m\text{C}_{ds}\text{A}_{ys}\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}$ ${}^m\text{C}_{ds}\text{T}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}\text{A}_{ds}\text{G}_{ks}\text{A}_{ks}\text{T}_k$	57	423
1341688	1359	1374	21174	21189	TGTTGATATTCTGTGG	$\text{T}_{ks}\text{G}_{ks}\text{T}_{ks}\text{T}_{ds}\text{G}_{ys}\text{A}_{ds}\text{T}_{ds}\text{A}_{ds}\text{T}_{ds}$ $\text{T}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}\text{G}_{ds}\text{T}_{ks}\text{G}_{ks}\text{G}_k$	48	1358
1341695	1351	1366	21166	21181	TTCTGTGGCATGGCTA	$\text{T}_{ks}\text{T}_{ks}{}^m\text{C}_{ks}\text{T}_{ds}\text{G}_{ys}\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}{}^m\text{C}_{ds}$ $\text{A}_{ds}\text{T}_{ds}\text{G}_{ds}\text{G}_{ds}\text{mG}_{ks}\text{T}_{ks}\text{A}_k$	65	812
1341700	1384	1399	21199	21214	TAGCUGTGCACTCATT	$\text{T}_{ks}\text{A}_{ks}\text{G}_{ks}{}^m\text{C}_{ds}\text{U}_{ys}\text{G}_{ds}\text{T}_{ds}\text{G}_{ds}$ ${}^m\text{C}_{ds}\text{A}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}{}^m\text{C}_{ds}\text{A}_{ks}\text{T}_{ks}\text{T}_k$	60	2870
1341704	1391	1406	21206	21221	GATCUCTTAGCTGTGC	$\text{G}_{ks}\text{A}_{ks}\text{T}_{ks}{}^m\text{C}_{ds}\text{U}_{ys}{}^m\text{C}_{ds}\text{T}_{ds}\text{T}_{ds}$ $\text{A}_{ds}\text{G}_{ds}{}^m\text{C}_{ds}\text{T}_{ds}\text{G}_{ds}\text{T}_{ks}\text{G}_{ks}{}^m\text{C}_k$	65	2871

TABLE 48-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341710	1419	1434	21234	21249	GTTGAGATAAAGCTGC	G _{ks} T _{ks} T _{ks} G _{ds} A _{ys} G _{ds} A _{ds} T _{ds} A _{ds} A _{ds} A _{ds} G _{ds} ^m C _{ds} T _{ks} G _{ks} ^m C _k	51	1748
1341712	1423	1438	21238	21253	CCAGGTTGAGATAAAG	^m C _{ks} ^m C _{ks} A _{ks} G _{ds} G _{ys} T _{ds} T _{ds} G _{ds} A _{ds} G _{ds} A _{ds} T _{ds} A _{ds} A _{ks} A _{ks} G _k	49	33
1341715	1398	1413	21213	21228	GAAACTTGATCTCTTA	G _{ks} A _{ks} A _{ks} A _{ds} C _{ys} T _{ds} T _{ds} G _{ds} A _{ds} T _{ds} ^m C _{ds} T _{ds} ^m C _{ds} T _{ks} T _{ks} A _k	59	2717
1341720	1496	1511	21311	21326	GTTGCACCGTTTTGGG	G _{ks} T _{ks} T _{ks} G _{ds} C _{ys} A _{ds} ^m C _{ds} ^m C _{ds} G _{ds} T _{ds} T _{ds} T _{ds} T _{ds} T _{ds} G _{ks} G _{ks} G _k	38	1827
1341724	1431	1446	21246	21261	AAATATGTCCAGGTTG	A _{ks} A _{ks} A _{ks} T _{ds} A _{ys} T _{ds} G _{ds} T _{ds} ^m C _{ds} ^m C _{ds} A _{ds} G _{ds} G _{ds} T _{ks} T _{ks} G _k	63	581
1341730	1488	1503	21303	21318	GTTTUGGGCTAATGAA	G _{ks} T _{ks} T _{ks} T _{ds} U _{ys} G _{ds} G _{ds} ^m C _{ds} T _{ds} A _{ds} A _{ds} T _{ds} G _{ks} A _{ks} A _k	30	2872
1341732	801	816	15682	15697	GTATUCCATCTATCAG	G _{ks} T _{ks} A _{ks} T _{ds} U _{ys} ^m C _{ds} ^m C _{ds} A _{ds} T _{ds} ^m C _{ds} T _{ds} A _{ds} T _{ds} ^m C _{ks} A _{ks} G _k	41	2873
1341735	1492	1507	21307	21322	CACCGTTTTGGGCTAA	^m C _{ks} A _{ks} ^m C _{ks} ^m C _{ks} ^m C _{ds} G _{ys} T _{ds} T _{ds} T _{ds} T _{ds} G _{ds} G _{ds} G _{ds} ^m C _{ds} T _{ks} A _{ks} A _k	82	1516
1341746	1501	1516	21316	21331	ATAGAGTTGCACCGTT	A _{ks} T _{ks} A _{ks} G _{ds} A _{ys} G _{ds} T _{ds} T _{ds} G _{ds} ^m C _{ds} A _{ds} ^m C _{ds} ^m C _{ds} G _{ks} T _{ks} T _k	59	192
1341747	1602	1617	21417	21432	ATGTATCTTATAAGAC	A _{ks} T _{ks} G _{ks} T _{ds} A _{ys} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ds} T _{ds} A _{ds} T _{ds} G _{ks} A _{ks} ^m C _k	3	1751
1341749	1621	1636	21436	21451	GAGTCGGTCACCTTTC	G _{ks} A _{ks} G _{ks} T _{ds} C _{ys} G _{ds} G _{ds} T _{ds} ^m C _{ds} A _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ks} ^m C _k	55	350
1341750	1625	1640	21440	21455	AATAGAGTCGGTCACC	A _{ks} A _{ks} T _{ks} A _{ds} G _{ys} A _{ds} G _{ds} T _{ds} ^m C _{ds} G _{ds} G _{ds} T _{ds} ^m C _{ds} A _{ks} ^m C _{ks} ^m C _k	57	662
1341752	1617	1632	21432	21447	CGGTCACCTTTCATAA	^m C _{ks} G _{ks} G _{ks} T _{ds} C _{ys} A _{ys} ^m C _{ds} ^m C _{ds} T _{ds} T _{ds} T _{ds} ^m C _{ds} A _{ds} T _{ks} A _{ks} A _k	42	36
1341762	1713	1728	21528	21543	AACAUCTCTGGGACCA	A _{ks} A _{ks} ^m C _{ks} A _{ds} U _{ys} ^m C _{ds} T _{ds} ^m C _{ds} T _{ds} G _{ds} G _{ds} A _{ds} ^m C _{ks} ^m C _{ks} A _k	64	2874
1341766	1629	1644	21444	21459	TTAAAATAGAGTCGGT	T _{ks} T _{ks} A _{ks} A _{ds} A _{ys} T _{ds} A _{ds} G _{ds} A _{ds} T _{ds} T _{ds} ^m C _{ds} G _{ks} G _{ks} T _k	70	973
1341769	1726	1741	21541	21556	CTAAAATTGTCTAAAC	^m C _{ks} T _{ks} A _{ks} A _{ds} A _{ys} T _{ds} T _{ds} G _{ds} T _{ds} ^m C _{ds} T _{ds} A _{ds} A _{ks} A _{ks} ^m C _k	13	2590
1341772	1717	1732	21532	21547	TCTAAACATCTCTGGG	T _{ks} ^m C _{ks} T _{ks} A _{ds} A _{ys} ^m C _{ds} A _{ds} T _{ds} ^m C _{ds} T _{ds} ^m C _{ds} T _{ds} G _{ks} G _{ks} G _k	36	2287
1341775	1708	1723	21523	21538	CTCTGGGACCAAGGAT	^m C _{ks} T _{ks} ^m C _{ks} T _{ds} G _{ys} G _{ds} G _{ds} A _{ds} ^m C _{ds} ^m C _{ds} A _{ds} A _{ds} G _{ds} G _{ks} A _{ks} T _k	43	1908
1341778	1773	1788	21588	21603	AATAGCCAGTACAGTT	A _{ks} A _{ks} T _{ks} A _{ds} G _{ys} ^m C _{ds} ^m C _{ds} A _{ds} G _{ds} T _{ds} A _{ds} ^m C _{ds} A _{ds} G _{ks} T _{ks} T _k	66	118
1341782	1766	1781	21581	21596	AGTACAGTTCCTTTTC	A _{ks} G _{ks} T _{ks} A _{ds} C _{ys} A _{ds} G _{ds} T _{ds} T _{ds} ^m C _{ds} ^m C _{ds} T _{ds} T _{ds} T _{ks} T _{ks} ^m C _k	56	1597
1341790	791	806	15672	15687	TATCAGACTTCTTACG	T _{ks} A _{ks} T _{ks} ^m C _{ds} A _{ys} G _{ds} ^m C _{ds} T _{ds} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ks} ^m C _{ks} G _k	27	103
1341794	1777	1792	21592	21607	ATGTAATAGCCAGTAC	A _{ks} T _{ks} G _{ks} T _{ds} A _{ys} T _{ds} A _{ds} G _{ds} ^m C _{ds} ^m C _{ds} A _{ds} G _{ds} T _{ks} A _{ks} ^m C _k	42	430
1341798	1781	1796	21596	21611	TCTTATGTAATAGCCA	T _{ks} ^m C _{ks} T _{ks} T _{ds} A _{ys} T _{ds} G _{ds} T _{ds} A _{ds} A _{ds} T _{ds} A _{ds} G _{ds} ^m C _{ks} ^m C _{ks} A _k	64	742
1341802	2191	2206	22006	22021	GTCTUGATGTAGTGGG	G _{ks} T _{ks} ^m C _{ks} T _{ds} U _{ys} G _{ds} A _{ds} T _{ds} G _{ds} T _{ds} A _{ds} G _{ds} T _{ds} G _{ks} G _{ks} G _k	63	2875

TABLE 48-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341803	2184	2199	21999	22014	TGTAGTGGGAGTCGGA	$T_{ks}G_{ks}T_{ks}A_{ds}G_{ys}T_{ds}G_{ds}G_{ds}$ $A_{ds}G_{ds}T_{ds}{}^mC_{ds}G_{ks}G_{ks}A_{ks}$	50	587
1341811	2355	2370	22170	22185	TTATACTACAAGAGG	$T_{ks}T_{ks}A_{ks}T_{ds}A_{ys}A_{ds}{}^mC_{ds}T_{ds}A_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}G_{ds}A_{ks}G_{ks}G_{ks}$	58	1289
1341812	812	827	15693	15708	CTTAUTGGTAAGTATT	${}^mC_{ks}T_{ks}T_{ks}A_{ds}U_{ys}T_{ds}G_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}T_{ds}A_{ks}T_{ks}T_{ks}$	0	2876
1341814	2254	2269	22069	22084	CAAGAGGATAGTCCAT	${}^mC_{ks}A_{ks}A_{ks}G_{ds}A_{ys}G_{ds}G_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ds}{}^mC_{ds}{}^mC_{ks}A_{ks}T_{ks}$	60	510
1341815	2245	2260	22060	22075	AGTCCATGCAAAAGCA	$A_{ks}G_{ks}T_{ks}{}^mC_{ds}C_{ys}A_{ds}T_{ds}G_{ds}$ ${}^mC_{ds}A_{ds}A_{ds}A_{ds}G_{ks}{}^mC_{ks}A_{ks}$	73	120
1341819	N/A	N/A	9337	9352	CCTCGATCCTATATAC	${}^mC_{ks}{}^mC_{ks}T_{ks}{}^mC_{ds}G_{ys}A_{ds}T_{ds}{}^mC_{ds}$ ${}^mC_{ds}T_{ds}A_{ds}T_{ds}A_{ds}T_{ks}A_{ks}{}^mC_{ks}$	9	2812
1341821	N/A	N/A	9518	9533	CTACACTAATATTGAG	${}^mC_{ks}T_{ks}A_{ks}{}^mC_{ds}A_{ys}{}^mC_{ds}T_{ds}A_{ds}$ $A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}G_{ks}A_{ks}G_{ks}$	29	59
1341829	N/A	N/A	9961	9976	TAAGCTGAGAGTTCTA	$T_{ks}A_{ks}A_{ks}G_{ds}C_{ys}T_{ds}G_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ds}{}^mC_{ks}T_{ks}A_{ks}$	27	2499
1341834	N/A	N/A	9971	9986	TTGAGTAGTGTAAAGCT	$T_{ks}T_{ks}G_{ks}A_{ds}G_{ys}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}G_{ks}{}^mC_{ks}T_{ks}$	35	2346
1341838	1022	1037	20837	20852	CTTGCAGCATTGATT	${}^mC_{ks}T_{ks}T_{ks}T_{ds}G_{ys}{}^mC_{ds}A_{ds}G_{ds}$ ${}^mC_{ds}A_{ds}T_{ds}T_{ds}G_{ds}A_{ks}T_{ks}T_{ks}$	19	2463
1341841	N/A	N/A	11571	11586	TCTTGACAATGGTTGA	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}G_{ys}A_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}T_{ds}G_{ds}T_{ds}T_{ds}G_{ks}A_{ks}A_{ks}$	56	2440
1341846	N/A	N/A	11855	11870	TAAGCAGAATTGTGAA	$T_{ks}A_{ks}A_{ks}G_{ds}C_{ys}A_{ds}G_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ds}T_{ds}G_{ks}A_{ks}A_{ks}$	26	2361
1341849	N/A	N/A	14194	14209	TGTGATTGAGTTCTCC	$T_{ks}G_{ks}T_{ks}G_{ds}A_{ys}T_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ks}{}^mC_{ks}{}^mC_{ks}$	34	2020
1341852	N/A	N/A	12423	12438	GGTGGTAACGGTGATC	$G_{ks}G_{ks}T_{ks}G_{ds}G_{ys}T_{ds}A_{ds}A_{ds}{}^mC_{ds}$ $G_{ds}G_{ds}T_{ds}G_{ds}A_{ks}T_{ks}{}^mC_{ks}$	35	2591
1341857	N/A	N/A	14204	14219	CAGAUTATTATGTGAT	${}^mC_{ks}A_{ks}G_{ks}A_{ds}U_{ys}T_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}T_{ds}G_{ds}T_{ds}G_{ks}T_{ks}T_{ks}$	32	2877
1341859	N/A	N/A	15491	15506	TTAAGCCACCCCTACC	$T_{ks}T_{ks}A_{ks}A_{ds}U_{ys}G_{ds}{}^mC_{ds}{}^mC_{ds}$ $A_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}A_{ks}{}^mC_{ks}{}^mC_{ks}$	8	2878
1341868	N/A	N/A	16282	16297	TAGTCTTCAGCAAAGT	$T_{ks}A_{ks}G_{ks}T_{ds}C_{ys}T_{ds}T_{ds}{}^mC_{ds}A_{ds}$ $G_{ds}{}^mC_{ds}A_{ds}A_{ds}A_{ks}G_{ks}T_{ks}$	41	2632
1341874	N/A	N/A	16131	16146	GCTCCATAATAATAGC	$G_{ks}{}^mC_{ks}T_{ks}{}^mC_{ds}C_{ys}A_{ds}T_{ds}A_{ds}$ $A_{ds}T_{ds}A_{ds}T_{ds}T_{ds}G_{ks}G_{ks}{}^mC_{ks}$	16	2665
1341880	N/A	N/A	17991	18006	GTTAGTATAGTTATCT	$G_{ks}T_{ks}T_{ks}A_{ds}G_{ys}T_{ds}A_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}T_{ks}{}^mC_{ks}T_{ks}$	52	1171
1341883	N/A	N/A	17996	18011	ATGAAGTTAGTATAGT	$A_{ks}T_{ks}G_{ks}A_{ds}A_{ys}G_{ds}T_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}A_{ds}T_{ds}A_{ks}G_{ks}T_{ks}$	38	1483
1341886	N/A	N/A	16377	16392	ACCAAACCTCCAGCAG	$A_{ks}{}^mC_{ks}{}^mC_{ks}A_{ys}A_{ds}{}^mC_{ds}T_{ds}$ $T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}{}^mC_{ks}A_{ks}G_{ks}$	54	1948
1341891	N/A	N/A	18097	18112	GAAAUTGATACACCAA	$G_{ks}A_{ks}A_{ks}A_{ds}U_{ys}T_{ds}G_{ds}A_{ds}T_{ds}$ $A_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}A_{ks}A_{ks}$	58	2879
1341894	N/A	N/A	17986	18001	TATAGTTATCTTCTCA	$T_{ks}A_{ks}T_{ks}A_{ds}G_{ys}T_{ds}T_{ds}A_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ks}{}^mC_{ks}A_{ks}$	33	2295
1341899	N/A	N/A	18695	18710	AAAGACTTTGAGACTC	$A_{ks}A_{ks}A_{ks}G_{ds}A_{ys}{}^mC_{ds}T_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}A_{ds}{}^mC_{ks}T_{ks}{}^mC_{ks}$	76	1250

TABLE 48-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341901	N/A	N/A	19914	19929	AGTGCATCTTAAGATA	$A_{ks}G_{ks}T_{ks}G_{ds}C_{ys}A_{ds}T_{ds}{}^mC_{ds}T_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}A_{ks}T_{ks}A_k$	17	2130
1341904	1081	1096	20896	20911	CTGCUAGTGCCAAACC	${}^mC_{ks}T_{ks}G_{ks}{}^mC_{ds}U_{ys}A_{ds}G_{ds}T_{ds}$ $G_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ks}{}^mC_{ks}{}^mC_k$	67	2880
1341915	N/A	N/A	19906	19921	TTAAGATACCCAGGTT	$T_{ks}T_{ks}A_{ks}A_{ds}G_{ys}A_{ds}T_{ds}A_{ds}{}^mC_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_k$	30	163
1341917	N/A	N/A	19910	19925	CATCUTAAGATACCCA	${}^mC_{ks}A_{ks}T_{ks}{}^mC_{ds}U_{ys}T_{ds}A_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}A_{ds}{}^mC_{ds}{}^mC_{ks}{}^mC_{ks}A_k$	56	2881
1341921	N/A	N/A	20116	20131	CAGCATCAAGACATTC	${}^mC_{ks}A_{ks}G_{ks}{}^mC_{ds}A_{ys}T_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ks}T_{ks}{}^mC_k$	52	2191
1341925	N/A	N/A	20009	20024	AGAGAGTAATCAGTTT	$A_{ks}G_{ks}A_{ks}G_{ds}A_{ys}G_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ks}T_{ks}T_k$	25	2651
1341934	N/A	N/A	20192	20207	GCAGUTATTAGAAGTC	$G_{ks}{}^mC_{ks}A_{ks}G_{ds}U_{ys}T_{ds}A_{ds}T_{ds}T_{ds}$ $A_{ds}G_{ds}A_{ds}A_{ds}G_{ks}T_{ks}{}^mC_k$	59	2882
1341937	1077	1092	20892	20907	TAGTGCCAAACCAATG	$T_{ks}A_{ks}G_{ks}T_{ds}G_{ys}{}^mC_{ds}{}^mC_{ds}A_{ds}$ $A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ks}T_{ks}G_k$	55	1198
1341939	1085	1100	20900	20915	ACTGCTGCTAGTGCCA	$A_{ks}{}^mC_{ks}T_{ks}G_{ds}C_{ys}T_{ds}G_{ds}{}^mC_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ks}{}^mC_{ks}A_k$	64	1821
1341943	N/A	N/A	20486	20501	AACTUCATAGTGGACT	$A_{ks}A_{ks}{}^mC_{ks}T_{ds}U_{ys}{}^mC_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ds}G_{ds}G_{ds}A_{ks}{}^mC_{ks}T_k$	38	2883
1341945	N/A	N/A	20422	20437	ACAGUAAAATTATGCC	$A_{ks}{}^mC_{ks}A_{ks}G_{ds}U_{ys}A_{ds}A_{ds}A_{ds}$ $A_{ds}T_{ds}T_{ds}A_{ds}T_{ds}G_{ks}{}^mC_{ks}{}^mC_k$	24	2884
1341947	N/A	N/A	20381	20396	AATCACAAGTAAGGTA	$A_{ks}A_{ks}T_{ks}{}^mC_{ds}A_{ys}{}^mC_{ds}A_{ds}A_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}G_{ks}T_{ks}A_k$	31	1333
1341954	N/A	N/A	12501	12516	GCAGAAATTCACCTTG	$G_{ks}{}^mC_{ks}A_{ks}G_{ds}A_{ys}A_{ds}A_{ds}T_{ds}T_{ds}$ ${}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}T_{ks}G_k$	54	1002

TABLE 49

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}{}^mC_{ks}A_k$	86	121
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCAGT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}{}^mC_{ds}$ $A_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{ks}T_k$	82	1526
1341508	N/A	N/A	14189	14204	TTGAGTTCTCCACTGC	$T_{ks}T_{ks}G_{ks}A_{ds}G_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}$ ${}^mC_{ds}{}^mC_{ds}A_{ds}{}^mC_{ks}T_{ks}G_{ks}{}^mC_e$	26	2885
1341518	1251	1266	21066	21081	TGTCCACCTTTAAATG	$T_{ks}G_{ks}T_{ks}{}^mC_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ks}A_{ks}T_{ks}G_e$	28	1666
1341519	1310	1325	21125	21140	CTTAAACCTTCCCTGT	${}^mC_{ks}T_{ks}T_{ks}A_{ds}A_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}$ $T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ks}T_{ks}G_{ks}T_e$	34	1823
1341520	1328	1343	21143	21158	GAATGCTACTTGAACA	$G_{ks}A_{ks}A_{ks}T_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}$ ${}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ks}{}^mC_{ks}A_e$	56	1123
1341521	1330	1345	21145	21160	TGGAATGCTACTTGAA	$T_{ks}G_{ks}G_{ks}A_{ds}A_{ds}T_{ds}G_{ds}{}^mC_{ds}T_{ds}$ $A_{ds}{}^mC_{ds}T_{ds}T_{ks}G_{ks}A_{ks}A_e$	57	2203

TABLE 49-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO	
1341522	1320	1335	21135	21150	CTTGAACAGTCTTAAA	${}^mC_{ks}T_{ks}T_{ks}G_{ds}A_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ks}A_{ks}A_{ks}A_e$	43	500	
1341523	1309	1324	21124	21139	TTAAACCTTCCCTGTG	$T_{ks}T_{ks}A_{ks}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}G_{ks}T_{ks}G_e$	18	2807	
1341524	1319	1334	21134	21149	TTGAACAGTCTTAAAC	$T_{ks}T_{ks}G_{ks}A_{ds}A_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}A_{ks}A_{ks}A_{ks}{}^mC_e$	35	422	
1341525	1264	1279	21079	21094	GGGAGGTAGCTTTTGT	$G_{ks}G_{ks}G_{ks}A_{ds}G_{ds}G_{ds}T_{ds}A_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}T_{ks}T_{ks}G_{ks}T_e$	40	343	
1341526	1308	1323	21123	21138	TAAACCTTCCCTGTGT	$T_{ks}A_{ks}A_{ks}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}G_{ks}T_{ks}T_e$	23	1745	
1341527	1175	1190	20990	21005	TAAGAGGCATGAAAGG	$T_{ks}A_{ks}A_{ks}G_{ds}A_{ds}G_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}A_{ds}A_{ks}A_{ks}G_e$	36	2751	
1341530	693	708	12111	12126	ATGAGGTTTTGATACC	$A_{ks}T_{ks}G_{ks}G_{ks}A_{ds}G_{ds}G_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}A_{ks}{}^mC_{ks}{}^mC_e$	20	569	
1341531	1336	1351	21151	21166	ACAGATTGGAATGCTA	$A_{ks}{}^mC_{ks}A_{ks}G_{ds}A_{ds}T_{ds}T_{ds}G_{ds}G_{ds}A_{ds}A_{ds}T_{ds}G_{ks}{}^mC_{ks}T_{ks}A_e$	68	1668	
1341557	1347	1362	21162	21177	GTGGCATGGCTACAGA	$G_{ks}T_{ks}G_{ks}G_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ks}A_{ks}G_{ks}A_e$	38	501	
1341561	1346	1361	21161	21176	TGGCATGGCTACAGAT	$T_{ks}G_{ks}G_{ks}{}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ks}G_{ks}T_{ks}T_e$	68	423	
1341573	1348	1363	21163	21178	TGTGGCATGGCTACAG	$T_{ks}G_{ks}T_{ks}G_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ks}{}^mC_{ks}A_{ks}G_e$	25	579	
1341576	801	816	15682	15697	GTATTCCATCTATCAG	$G_{ks}T_{ks}A_{ks}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}T_{ks}{}^mC_{ks}A_{ks}G_e$	51	727	
1341608	N/A	N/A	19914	19929	AGTGCATCTTAAGATA	$A_{ks}G_{ks}T_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}A_{ks}G_{es}A_{es}T_{es}A_k$	27	2130	
1341610	N/A	N/A	19915	19930	GAGTGCATCTTAAGAT	$G_{ks}A_{ks}G_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ks}A_{ks}G_{es}G_{es}T_k$	27	2256	
1341611	N/A	N/A	18690	18705	CTTTGAGACTCTTGTT	${}^mC_{ks}T_{ks}T_{ds}T_{ds}G_{ds}A_{ds}G_{ks}A_{ds}{}^mC_{ds}T_{ds}{}^mC_{ds}T_{ks}T_{es}G_{ks}T_{es}T_k$	42	2583	
1341612	N/A	N/A	19904	19919	AAGATACCCAGGTGTC	$A_{ks}A_{ks}G_{ds}A_{ds}T_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}G_{ds}G_{ks}T_{es}T_{ks}G_{es}{}^mC_k$	59	2712	
1341613	N/A	N/A	18092	18107	TGATACACCAATGCAG	$T_{ks}G_{ks}A_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}T_{ks}T_{es}{}^mC_{ks}A_{es}G_k$	62	2370	
1341618	N/A	N/A	19906	19921	TTAAGATACCCAGGTT	$T_{ks}T_{ks}A_{ds}A_{ds}G_{ds}A_{ds}T_{ds}T_{ds}T_{ds}{}^mC_{ds}{}^mC_{ds}{}^mC_{ds}A_{ds}A_{ds}G_{es}G_{es}T_{es}T_k$	48	163	
1341619	N/A	N/A	20388	20403	CTATAGTAATCACAAG	${}^mC_{ks}T_{ks}A_{ds}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ks}{}^mC_{es}A_{ks}A_{es}G_k$	36	1644	
1341620	N/A	N/A	20187	20202	TATTAGAAGTCAGCCC	$T_{ks}A_{ks}T_{ds}T_{ds}A_{ds}G_{ds}A_{ds}A_{ds}G_{ds}T_{ds}{}^mC_{ds}A_{ks}G_{es}{}^mC_{ks}{}^mC_{es}{}^mC_k$	36	164	
1341621	N/A	N/A	20109	20124	AAGACATTCTAGCCTG	$A_{ks}A_{ks}G_{ds}A_{ds}{}^mC_{ds}A_{ds}T_{ds}T_{ds}{}^mC_{ds}T_{ds}A_{ds}G_{ks}{}^mC_{es}{}^mC_{ks}T_{es}G_k$	25	2744	
1341622	N/A	N/A	20197	20212	ACATTGCAGTTATTAG	$A_{ks}{}^mC_{ks}A_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ks}T_{es}T_{ks}A_{es}G_k$	46	2309	
1341623	N/A	N/A	20119	20134	TTACAGCATCAAGACA	$T_{ks}T_{ks}A_{ds}{}^mC_{ds}A_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}A_{ds}A_{ks}G_{es}A_{ks}{}^mC_{es}A_k$	22	2409	
1341624	N/A	N/A	19916	19931	GGAGTGCATCTTAAGA	$G_{ks}G_{ks}A_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ks}A_{es}A_{ks}G_{es}A_k$	34	2810	

TABLE 49-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO	
1341625	N/A	N/A	20009	20024	AGAGAGTAATCAGTTT	$A_{ks}G_{ks}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}^mC_{ds}A_{ks}G_{es}T_{ks}T_{es}T_k$	52	2651	
1341626	N/A	N/A	19905	19920	TAAGATACCCAGGTTG	$T_{ks}A_{ks}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}G_{ks}G_{es}T_{ks}T_{es}G_k$	30	2790	
1341627	N/A	N/A	19999	20014	CAGTTTTCCTCATGAT	$^mC_{ks}A_{ks}G_{ds}T_{ds}T_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}T_{ds}^mC_{ds}A_{ks}T_{es}G_{ks}A_{es}T_k$	59	2331	
1341628	N/A	N/A	20378	20393	CACAAGTAAGGTAAG	$^mC_{ks}A_{ks}^mC_{ds}A_{ds}A_{ds}G_{ds}T_{ds}A_{ds}$ $A_{ds}G_{ds}G_{ds}T_{ks}A_{es}A_{ks}A_{es}G_k$	38	2448	
1341629	N/A	N/A	12804	12819	CTCAGGCTGTAAGAGT	$^mC_{ks}T_{ks}^mC_{ds}A_{ds}G_{ds}G_{ds}^mC_{ds}T_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ks}G_{es}G_{ks}G_{es}T_k$	10	2240	
1341630	N/A	N/A	14165	14180	GCAAGCCAACAGAGAG	$G_{ks}^mC_{ks}A_{ds}A_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}^mC_{ds}A_{ds}G_{ks}A_{es}G_{ks}A_{es}G_k$	12	2556	
1341631	N/A	N/A	14175	14190	GCCTTGAATAGCAAGC	$G_{ks}^mC_{ks}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}^mC_{ks}A_{es}A_{ks}G_{es}^mC_k$	16	2063	
1341632	N/A	N/A	13959	13974	GTCTACCTCTAAGTTA	$G_{ks}T_{ks}^mC_{ds}T_{ds}A_{ds}^mC_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}T_{ds}A_{ds}A_{ks}G_{es}T_{ks}T_{es}A_k$	15	2363	
1341633	N/A	N/A	13949	13964	AAGTTAGCCCCAGGA	$A_{ks}A_{ks}G_{ds}T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}^mC_{ks}A_{es}G_{ks}G_{es}A_k$	12	2811	
1341634	N/A	N/A	12496	12511	AATTCACCTTGACTAA	$A_{ks}A_{ks}T_{ds}T_{ds}^mC_{ds}A_{ds}^mC_{ds}^mC_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ks}^mC_{es}T_{ks}A_{es}A_k$	9	2144	
1341635	N/A	N/A	20479	20494	TAGTGACTTCATTAG	$T_{ks}A_{ks}G_{ds}T_{ds}G_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}$ $T_{ds}^mC_{ds}A_{ks}T_{es}T_{ks}A_{es}G_k$	41	2351	
1341636	N/A	N/A	20489	20504	CATAACTTCATAGTGG	$^mC_{ks}A_{ks}T_{ds}A_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}$ $^mC_{ds}A_{ds}T_{ds}A_{ks}G_{es}T_{ks}G_{es}G_k$	15	866	
1341637	N/A	N/A	12794	12809	AAGAGTCAGTATCCTC	$A_{ks}A_{ks}G_{ds}A_{ds}G_{ds}T_{ds}^mC_{ds}A_{ds}G_{ds}$ $T_{ds}A_{ds}T_{ks}^mC_{es}^mC_{ks}T_{es}^mC_k$	14	2572	
1341638	1394	1409	21209	21224	CTTGATCTCTTAGCTG	$^mC_{ks}T_{ks}T_{ks}G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ks}^mC_{ks}T_{ks}G_e$	36	736	
1341639	1416	1431	21231	21246	GAGATAAAGCTGCCTG	$G_{ks}A_{ks}G_{ks}A_{ds}T_{ds}A_{ds}A_{ds}G_{ds}$ $^mC_{ds}T_{ds}G_{ds}^mC_{ks}^mC_{ks}T_{ks}G_e$	55	1592	
1341640	1426	1441	21241	21256	TGTCCAGGTTGAGATA	$T_{ks}G_{ks}T_{ks}^mC_{ds}^mC_{ds}A_{ds}G_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ds}G_{ks}A_{ks}T_{ks}A_e$	45	191	
1341641	1415	1430	21230	21245	AGATAAAGCTGCCTGC	$A_{ks}G_{ks}A_{ks}T_{ds}A_{ds}A_{ds}G_{ds}^mC_{ds}$ $T_{ds}G_{ds}^mC_{ds}^mC_{ks}T_{ks}G_{ks}^mC_e$	44	1515	
1341642	1425	1440	21240	21255	GTCCAGGTTGAGATAA	$G_{ks}T_{ks}^mC_{ks}^mC_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ds}A_{ks}T_{ks}A_{ks}A_e$	52	113	
1341643	791	806	15672	15687	TATCAGACTTCTTACG	$T_{ks}A_{ks}T_{ks}^mC_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}$ $T_{ds}T_{ds}^mC_{ds}T_{ds}T_{ks}A_{ks}^mC_{ks}G_e$	28	103	
1341644	1398	1413	21213	21228	GAAACTTGATCTCTTA	$G_{ks}A_{ks}A_{ks}A_{ds}^mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $T_{ds}^mC_{ds}T_{ds}^mC_{ks}T_{ks}T_{ks}A_e$	79	2717	
1341645	1388	1403	21203	21218	CTCTTAGCTGTGCACT	$^mC_{ks}T_{ks}^mC_{ks}T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $T_{ds}G_{ds}T_{ds}G_{ds}^mC_{ks}A_{ks}^mC_{ks}T_e$	66	268	
1341646	1384	1399	21199	21214	TAGCTGTGCACTCATT	$T_{ks}A_{ks}G_{ks}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}$ $^mC_{ds}A_{ds}^mC_{ds}T_{ds}^mC_{ks}A_{ks}T_{ks}T_e$	66	1981	
1341647	N/A	N/A	14189	14204	TTGAGTTCTCCACTGC	$T_{ks}T_{ks}G_{ds}A_{ds}G_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}^mC_{ds}A_{ks}^mC_{es}T_{ks}G_{es}^mC_k$	20	2885	
1341949	N/A	N/A	12799	12814	GCTGUAAGAGTCAGTA	$G_{ks}^mC_{ks}T_{ks}G_{ds}U_{ys}A_{ds}A_{ds}G_{ds}A_{ds}$ $G_{ds}T_{ds}^mC_{ds}A_{ds}G_{ks}T_{ks}A_k$	28	2886	

TABLE 49-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO	
1341950	1088	1103	20903	20918	TTGACTGCTGCTAGTG	$T_{ks}T_{ks}G_{ks}A_{ds}C_{ys}T_{ds}G_{ds}^mC_{ds}T_{ds}$ $G_{ds}^mC_{ds}T_{ds}A_{ds}G_{ks}T_{ks}G_k$	40	28	
1341958	N/A	N/A	12797	12812	TGTAAGAGTCAGTATC	$T_{ks}G_{ks}T_{ks}A_{ds}A_{ys}G_{ds}A_{ds}G_{ds}T_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ks}T_{ks}^mC_k$	29	2652	
1341959	1091	1106	20906	20921	CGTTUGACTGCTGCTA	$^mC_{ks}G_{ks}T_{ks}T_{ds}U_{ys}G_{ds}A_{ds}^mC_{ds}$ $T_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}^mC_{ks}T_{ks}A_k$	54	2887	
1341960	N/A	N/A	14172	14187	TTGAATAGCAAGCCAA	$T_{ks}T_{ks}G_{ks}A_{ds}A_{ys}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $A_{ds}A_{ds}G_{ds}^mC_{ds}^mC_{ks}A_k$	30	1864	
1341961	N/A	N/A	14175	14190	GCCTUGAATAGCAAGC	$G_{ks}^mC_{ks}^mC_{ks}T_{ds}U_{ys}G_{ds}A_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}^mC_{ds}A_{ds}A_{ks}^mC_k$	12	2888	
1341962	N/A	N/A	13949	13964	AAGTUAGCCCCCAGGA	$A_{ks}A_{ks}G_{ks}T_{ds}U_{ys}A_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}^mC_{ds}^mC_{ds}G_{ks}G_{ks}A_k$	0	2889	
1341963	N/A	N/A	13952	13967	TCTAAGTTAGCCCCCA	$T_{ks}^mC_{ks}T_{ks}A_{ds}A_{ys}G_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}^mC_{ds}^mC_{ds}^mC_{ks}^mC_{ks}A_k$	7	928	
1341964	N/A	N/A	13956	13971	TACCUCTAAGTTAGCC	$T_{ks}A_{ks}T_{ks}^mC_{ds}U_{ys}^mC_{ds}T_{ds}A_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ds}A_{ds}G_{ks}^mC_{ks}^mC_k$	13	2890	
1341965	N/A	N/A	13954	13969	CCTCUAAGTTAGCCCC	$^mC_{ks}^mC_{ks}T_{ks}^mC_{ds}U_{ys}A_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}^mC_{ks}^mC_{ks}^mC_k$	8	2891	
1341966	N/A	N/A	13959	13974	GTCTACCTCTAAGTTA	$G_{ks}T_{ks}^mC_{ks}T_{ds}A_{ys}^mC_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}T_{ks}A_k$	15	2363	
1341967	N/A	N/A	12794	12809	AAGAGTCAGTATCCTC	$A_{ks}A_{ks}G_{ks}A_{ds}G_{ys}T_{ds}^mC_{ds}A_{ds}G_{ds}$ $T_{ds}A_{ds}T_{ds}^mC_{ds}^mC_{ks}T_{ks}^mC_k$	55	2572	
1341968	1202	1217	21017	21032	GTATGTTTATGTAAGC	$G_{ks}T_{ks}A_{ks}T_{ds}G_{ys}T_{ds}T_{ds}T_{ds}A_{ds}$ $T_{ds}G_{ds}T_{ds}A_{ds}A_{ks}G_{ks}^mC_k$	66	1511	
1341969	1200	1215	21015	21030	ATGTUTATGTAAGCAC	$A_{ks}T_{ks}G_{ks}T_{ds}U_{ys}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}^mC_{ks}A_{ks}^mC_k$	45	2892	
1341970	N/A	N/A	14168	14183	ATAGCAAGCCAACAGA	$A_{ks}T_{ks}A_{ks}G_{ds}C_{ys}A_{ds}A_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}A_{ds}^mC_{ds}A_{ks}G_{ks}A_k$	1	2633	
1341971	606	621	10492	10507	AATAUGGGATGAGGTA	$A_{ks}A_{ks}T_{ks}A_{ds}U_{ys}G_{ds}G_{ds}G_{ds}A_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ds}G_{ks}T_{ks}A_k$	38	2893	
1341972	1175	1190	20990	21005	TAAGAGGCATGAAAGG	$T_{ks}A_{ks}A_{ks}G_{ds}G_{ds}A_{ys}G_{ds}G_{ds}^mC_{ds}A_{ds}$ $T_{ds}G_{ds}A_{ds}A_{ds}A_{ks}G_{ks}G_k$	50	2751	
1341973	1180	1195	20995	21010	GTTTUTAAGAGGCATG	$G_{ks}T_{ks}T_{ks}T_{ds}U_{ys}T_{ds}A_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}G_{ds}^mC_{ds}A_{ks}T_{ks}G_k$	38	2894	
1341974	N/A	N/A	14165	14180	GCAAGCCAACAGAGAG	$G_{ks}^mC_{ks}A_{ks}A_{ds}G_{ys}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}G_{ks}A_{ks}G_k$	26	2556	
1341975	N/A	N/A	14170	14185	GAATAGCAAGCCAACA	$G_{ks}A_{ks}A_{ks}T_{ds}A_{ys}G_{ds}^mC_{ds}A_{ds}A_{ds}$ $G_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}^mC_{ks}A_k$	23	1708	
1341976	N/A	N/A	14192	14207	TGATUGAGTTCTCCAC	$T_{ks}G_{ks}A_{ks}T_{ds}U_{ys}G_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}^mC_{ds}T_{ds}^mC_{ds}^mC_{ks}^mC_k$	4	2895	
1341977	N/A	N/A	14189	14204	TTGAGTTCTCCACTGC	$T_{ks}T_{ks}G_{ks}A_{ds}G_{ys}T_{ds}T_{ds}^mC_{ds}T_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}^mC_{ds}T_{ks}G_{ks}^mC_k$	18	2885	
1341978	1261	1276	21076	21091	AGGTAGCTTTTGTCCA	$A_{ks}G_{ks}G_{ks}T_{ds}A_{ys}G_{ds}^mC_{ds}T_{ds}T_{ds}$ $T_{ds}T_{ds}G_{ds}T_{ds}^mC_{ks}^mC_{ks}A_k$	59	109	
1341979	1259	1274	21074	21089	GTAGCTTTTGTCCACC	$G_{ks}T_{ks}A_{ks}G_{ds}C_{ys}T_{ds}T_{ds}T_{ds}T_{ds}$ $G_{ds}T_{ds}^mC_{ds}^mC_{ds}A_{ks}^mC_{ks}^mC_k$	64	1978	
1341980	1258	1273	21073	21088	TAGCUTTTGTCCACCT	$T_{ks}A_{ks}G_{ks}^mC_{ds}U_{ys}T_{ds}T_{ds}T_{ds}G_{ds}$ $T_{ds}^mC_{ds}^mC_{ds}A_{ds}^mC_{ks}^mC_{ks}T_k$	67	2896	

431

Example 4: Design of and Antisense Inhibition by Human HSD17B13 Modified Oligonucleotides in HepRG Cells

Modified oligonucleotides complementary to an HSD17B13 nucleic acid were synthesized and tested for their effect on HSD17B13 RNA levels in vitro, specifies the chemistry of each modified oligonucleotide; wherein subscript 'd' represents a 2'-β-D-deoxyribosyl sugar moiety, subscript 'e' represents a 2'-MOE sugar moiety, subscript 'y' represents a 2'-O-methyl sugar moiety, subscript 'k' represents a cEt modified sugar moiety, subscript 's' represents a phosphorothioate internucleoside linkage, and superscript 'in' before the cytosine residue represents a 5-methyl cytosine. "Start site" indicates the 5'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. "Stop site" indicates the 3'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. As shown in the Tables below, the modified oligonucleotide are complementary to either the human HSD17B13 mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_178135.4) or to the human

432

HSD17B13 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000) or to both. 'N/A' indicates that the modified oligonucleotide is not complementary to that particular target sequence with 100% complementarity.

The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with 1,000 nM of modified oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and HSD17B13 RNA levels were measured by quantitative real-time RTPCR. Human primer probe set RTS43553 was used to measure RNA levels. HSD17B3 RNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of HSD17B13 relative to untreated control cells. As used herein, a value of '0' indicates that treatment with the modified oligonucleotide did not inhibit HSD17B13 mRNA levels.

TABLE 50

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO	
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	81	121	
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCACT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}A_{ks}G_{ks}T_k$	83	1526	
1340072	N/A	N/A	19913	19928	GTGCATCTTAAGATAC	$G_{ks}T_{ks}G_{ks}^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{ks}^mC_k$	47	2806	
1340095	N/A	N/A	19395	19410	ATAAGAGATGAGTAGG	$A_{ks}T_{ks}A_{ks}A_{ds}G_{ds}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}A_{ds}G_{ds}T_{ds}A_{ks}G_{ks}G_k$	56	2897	
1340097	N/A	N/A	14604	14619	TGCGAGAAATCTAAGA	$T_{ks}G_{ks}^mC_{ks}G_{ds}A_{ds}G_{ds}A_{ds}A_{ds}P_{ds}A_{ds}T_{dsin}C_{ds}T_{ds}A_{ds}A_{ks}G_{ks}A_k$	50	2898	
1340102	N/A	N/A	10521	10536	TATAACATGGCTGGCA	$T_{ks}A_{ks}T_{ks}A_{ks}A_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}G_{ks}^mC_{ks}A_k$	33	2899	
1340109	1432	1447	21247	21262	AAAATATGTCCAGGTT	$A_{ks}A_{ks}A_{ks}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}T_{ds}^mC_{ds}A_{ds}G_{ds}G_{ks}T_{ks}T_k$	74	2900	
1340134	N/A	N/A	9967	9982	GTAAGTGAAGCTGAGA	$G_{ks}T_{ks}A_{ks}G_{ds}T_{ds}G_{ds}T_{ds}A_{ds}P_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}A_{ks}G_{ks}A_k$	81	2901	
1340137	N/A	N/A	18139	18154	ATGTTATCTCAAGTCA	$A_{ks}T_{ks}G_{ks}T_{ds}T_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ds}A_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	79	2902	
1340138	1632	1647	21447	21462	GATTTAAATAGAGTC	$G_{ks}A_{ks}T_{ks}T_{ds}T_{ds}A_{ds}A_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ds}A_{ds}G_{ks}T_{ks}^mC_k$	46	2903	
1340155	N/A	N/A	9442	9457	CTCCACAAAACCTAAC	$^mC_{ks}T_{ks}^mC_{ks}^mC_{ds}^mC_{ds}A_{ds}^mC_{ds}A_{ds}A_{ds}A_{ds}A_{ds}^mC_{ds}T_{ds}A_{ks}A_{ks}^mC_k$	13	2904	

TABLE 50-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1340172	N/A	N/A	11623	11638	TTCTGTAGGACTCTGC	$T_{ks}T_{ks}^mC_{ks}T_{ds}G_{ds}T_{ds}A_{ds}$ $G_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}^mC_{ds}$ $T_{ks}G_{ks}^mC_k$	72	2905
1340180	N/A	N/A	20417	20432	AAAATTATGCCTTGTG	$A_{ks}A_{ks}A_{ks}A_{ds}T_{ds}T_{ds}A_{ds}$ $T_{ds}G_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ds}$ $TG_{ks}T_{ks}G_k$	50	2906
1340182	N/A	N/A	14915	14930	TACATTAGCAAGCTAA	$T_{ks}A_{ks}^mC_{ks}A_{ds}T_{ds}T_{ds}A_{ds}$ $G_{ds}^mC_{ds}A_{ds}A_{ds}G_{ds}^mC_{ds}$ $T_{ks}A_{ks}A_k$	29	2907
1340213	N/A	N/A	20593	20608	GCTGAGAGTTATCTGG	$G_{ks}^mC_{ks}T_{ks}G_{ds}A_{ds}G_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}A_{ds}^mC_{ds}T_{ks}$ $G_{ks}G_k$	65	2908
1340249	N/A	N/A	18037	18052	TCATAGTTTATATGGA	$T_{ks}^mC_{ks}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}T_{ds}G_{ks}$ $G_{ks}A_k$	44	2909
1340263	N/A	N/A	13609	13624	TTGAAGACAATACAGG	$T_{ks}T_{ks}G_{ks}A_{ds}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}A_{ds}A_{ds}T_{ds}A_{ds}^mC_{ds}$ $A_{ks}G_{ks}G_k$	63	2910
1340282	N/A	N/A	19535	19550	ATAGGCTAAAATGGTC	$A_{ks}T_{ks}A_{ks}G_{ds}G_{ds}^mC_{ds}T_{ds}$ $A_{ds}A_{ds}A_{ds}A_{ds}T_{ds}T_{ds}G_{ds}G_{ks}$ $T_{ks}^mC_k$	61	2911
1340287	N/A	N/A	18696	18711	AAAAGACTTTGAGACT	$A_{ks}A_{ks}A_{ks}A_{ds}G_{ds}A_{ds}^mC_{ds}$ $T_{ds}T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}A_{ks}$ $^mC_{ks}T_k$	32	2912
1340288	N/A	N/A	11115	11130	ACAGAGGAGTTTGCAG	$A_{ks}^mC_{ks}A_{ks}G_{ds}A_{ds}G_{ds}G_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ds}T_{ds}G_{ds}^mC_{ks}$ $A_{ks}G_k$	37	2913
1340289	1309	1324	21124	21139	TTAAACCTTCCTGTG	$T_{ks}T_{ks}A_{ks}A_{ds}A_{ds}^mC_{ds}$ $^mC_{ds}T_{ds}T_{ds}T_{ds}^mC_{ds}^mC_{ds}^mC_{ds}$ $T_{ds}G_{ks}T_{ks}G_k$	22	2807
1340295	1178	1193	20993	21008	TTTTAAGAGGCATGAA	$T_{ks}T_{ks}T_{ks}T_{ds}A_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}G_{ds}^mC_{ds}A_{ds}T_{ds}G_{ks}$ $A_{ks}A_k$	53	2914
1340301	N/A	N/A	20113	20128	CATCAAGACATTCTAG	$^mC_{ks}A_{ks}T_{ks}^mC_{ds}A_{ds}A_{ds}$ $G_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}$ $^mC_{ds}T_{ks}A_{ks}G_k$	49	2915
1340306	N/A	N/A	19732	19747	AACTGAAGGCTCTGAGC	$A_{ks}A_{ks}^mC_{ks}T_{ds}G_{ds}A_{ds}A_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}G_{ds}A_{ks}$ $G_{ks}^mC_k$	55	2916
1340315	N/A	N/A	14284	14299	TGAATGTAAAGGCTGG	$T_{ks}G_{ks}A_{ks}A_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}A_{ds}G_{ds}G_{ds}^mC_{ds}T_{ks}$ $G_{ks}G_k$	38	2917
1340333	1764	1779	21579	21594	TACAGTTCCTTTTCTCT	$T_{ks}A_{ks}^mC_{ks}A_{ds}G_{ds}T_{ds}T_{ds}$ $^mC_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}T_{ds}$ $^mC_{ks}^mC_{ks}T_k$	68	2918
1340340	N/A	N/A	6913	6928	TCGGGAAGTTTAGACA	$T_{ks}^mC_{ks}G_{ks}G_{ds}G_{ds}A_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_{ks}$ $^mC_{ks}A_k$	43	2919
1340341	N/A	N/A	6330	6345	GACTTTCATAGGGAGA	$G_{ks}A_{ks}^mC_{ks}T_{ds}T_{ds}T_{ds}$ $^mC_{ds}A_{ds}T_{ds}A_{ds}G_{ds}G_{ds}$ $A_{ks}G_{ks}A_k$	72	2920
1340346	N/A	N/A	11510	11525	ATTATGAGGATCTGGA	$A_{ks}T_{ks}A_{ks}A_{ds}T_{ds}G_{ds}A_{ds}$ $G_{ds}G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}G_{ks}$ $G_{ks}A_k$	49	2921

TABLE 50-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1340359	N/A	N/A	13949	13964	AAGTTAGCCCCAGGA	$A_{ks}A_{ks}G_{ks}T_{ds}T_{ds}A_{ds}G_{ds}$ $mC_{ds}mC_{ds}mC_{ds}mC_{ds}mC_{ds}$ $A_{ds}G_{ks}G_{ks}A_k$	36	2811
1340362	N/A	N/A	6734	6749	GATTACCCCTGGCTTT	$G_{ks}A_{ks}T_{ks}T_{ds}A_{ds}mC_{ds}$ $mC_{ds}mC_{ds}mC_{ds}T_{ds}G_{ds}G_{ds}$ $mC_{ds}T_{ks}T_{ks}T_k$	31	2922
1340391	N/A	N/A	12204	12219	TATCATACCACATACC	$T_{ks}A_{ks}T_{ks}mC_{ds}A_{ds}T_{ds}A_{ds}$ $mC_{ds}mC_{ds}A_{ds}mC_{ds}A_{ds}T_{ds}$ $A_{ks}mC_{ks}mC_k$	24	2923
1340393	N/A	N/A	5744	5759	CACTGATTTAGTTGGT	$mC_{ks}A_{ks}mC_{ks}T_{ds}G_{ds}A_{ds}$ $T_{ds}T_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_{ds}T_{ds}$ $G_{ks}G_{ks}T_k$	66	2924
1340399	N/A	N/A	15419	15434	CCAGTAGGTGTGTTTC	$mC_{ks}mC_{ks}A_{ks}G_{ds}T_{ds}A_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}G_{ds}T_{ds}G_{ds}T_{ds}$ $T_{ks}T_{ks}mC_k$	55	2925
1340403	N/A	N/A	5673	5688	GGTCAGTAGAGAGCAT	$G_{ks}G_{ks}T_{ks}mC_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}A_{ds}G_{ds}mC_{ks}$ $A_{ks}T_k$	79	2926
1340409	683	698	12101	12116	GATACCAGTTTTTCCC	$G_{ks}A_{ks}T_{ks}A_{ds}mC_{ds}mC_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ds}T_{ds}T_{ds}T_{ds}$ $mC_{ks}mC_{ks}mC_k$	75	2927
1340411	N/A	N/A	9206	9221	AGATGTACTAGACAA	$A_{ks}G_{ks}A_{ks}T_{ds}G_{ds}T_{ds}A_{ds}$ $mC_{ds}A_{ds}mC_{ds}T_{ds}G_{ds}A_{ds}$ $mC_{ks}A_{ks}A_k$	66	2928
1340420	N/A	N/A	20193	20208	TGCAGTTATTAGAAGT	$T_{ks}G_{ks}mC_{ks}A_{ds}G_{ds}T_{ds}T_{ds}$ $A_{ds}T_{ds}T_{ds}T_{ds}A_{ds}G_{ds}A_{ks}T_{ks}$ $G_{ks}T_k$	46	2929
1340444	N/A	N/A	8963	8978	CTCCGTTATAAGTTTC	$mC_{ks}T_{ks}mC_{ks}mC_{ds}G_{ds}T_{ds}$ $T_{ds}A_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ks}T_{ks}mC_k$	59	2930
1340456	N/A	N/A	16218	16233	TATAGACTGGGTAGGA	$T_{ks}A_{ks}T_{ks}A_{ds}G_{ds}A_{ds}mC_{ds}$ $T_{ds}G_{ds}G_{ds}G_{ds}T_{ds}A_{ds}G_{ks}$ $G_{ks}A_k$	61	2808
1340460	N/A	N/A	9775	9790	ACTAGAACTCCCAACC	$A_{ks}mC_{ks}T_{ks}A_{ds}G_{ds}A_{ds}$ $mC_{ds}T_{ds}mC_{ds}mC_{ds}mC_{ds}A_{ds}$ $A_{ks}mC_{ks}mC_k$	20	2931
1340467	N/A	N/A	8129	8144	TTAGAGTGCTTAGTTC	$T_{ks}T_{ks}A_{ks}G_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}mC_{ds}T_{ds}T_{ds}A_{ds}G_{ds}T_{ks}$ $T_{ks}mC_k$	58	2932
1340473	N/A	N/A	3608	3623	TTAGCAGCTTGGAAG	$T_{ks}T_{ks}T_{ks}A_{ds}G_{ds}mC_{ds}A_{ds}$ $G_{ds}mC_{ds}T_{ds}T_{ds}G_{ds}A_{ks}$ $A_{ks}G_k$	43	2933
1340474	N/A	N/A	16425	16440	TGATATGTCAATACTC	$T_{ks}G_{ks}A_{ks}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}mC_{ds}A_{ds}A_{ds}T_{ds}A_{ds}$ $mC_{ks}T_{ds}mC_k$	57	2934
1340490	N/A	N/A	10124	10139	GGATATGTCATCTAAA	$G_{ks}G_{ks}A_{ks}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}mC_{ds}A_{ds}T_{ds}mC_{ds}T_{ds}$ $A_{ks}A_{ks}A_k$	68	2935
1340494	N/A	N/A	5896	5911	GTCCTTTGTATTTTCGC	$G_{ks}T_{ks}mC_{ks}mC_{ds}T_{ds}T_{ds}$ $T_{ds}G_{ds}T_{ds}A_{ds}T_{ds}T_{ds}T_{ds}$ $mC_{ks}G_{ks}mC_k$	45	2936
1340520	N/A	N/A	3701	3716	ATTTTCAGATCCCGTT	$A_{ks}T_{ks}T_{ks}T_{ds}T_{ds}mC_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}mC_{ds}mC_{ds}$ $G_{ks}T_{ks}T_k$	67	2937

TABLE 50-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1340530	N/A	N/A	3490	3505	TACAGACTAAGGGACC	$T_{ks}A_{ks}^mC_{ks}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}T_{ds}A_{ds}A_{ds}G_{ds}G_{ds}G_{ds}$ $A_{ks}^mC_{ks}^mC_k$	68	2938
1340531	N/A	N/A	9337	9352	CCTCGATCCTATATAC	$^mC_{ks}^mC_{ks}T_{ks}^mC_{ds}G_{ds}A_{ds}$ $T_{ds}^mC_{ds}^mC_{ds}T_{ds}A_{ds}T_{ds}$ $A_{ds}T_{ks}A_{ks}^mC_k$	31	2812
1340532	N/A	N/A	12801	12816	AGGCTGTAAGAGTCAG	$A_{ks}G_{ks}G_{ks}^mC_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}G_{ds}A_{ds}G_{ds}T_{ds}^mC_{ks}$ $A_{ks}G_k$	74	2809
1340565	N/A	N/A	5844	5859	CCCCAAACATGGATGT	$^mC_{ks}^mC_{ks}^mC_{ks}^mC_{ds}A_{ds}A_{ds}$ $A_{ds}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ds}A_{ds}$ $T_{ks}G_{ks}T_k$	62	2939
1340586	811	826	15692	15707	TTATTGGTAAGTATTC	$T_{ks}T_{ks}A_{ks}T_{ds}T_{ds}G_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}T_{ds}A_{ds}T_{ks}$ $T_{ks}^mC_k$	17	2940
1340599	N/A	N/A	6599	6614	GGCTACTTTCAAACCT	$G_{ks}G_{ks}^mC_{ks}T_{ks}A_{ds}^mC_{ds}$ $T_{ds}T_{ds}T_{ds}^mC_{ds}A_{ds}A_{ds}$ $^mC_{ks}^mC_{ks}T_k$	54	2941
1340600	N/A	N/A	8538	8553	GTTTTGCAAGTTTATC	$G_{ks}T_{ks}T_{ks}T_{ds}T_{ds}G_{ds}^mC_{ds}$ $A_{ds}A_{ds}G_{ds}T_{ds}T_{ds}T_{ds}A_{ks}$ $T_{ks}^mC_k$	73	2942
1340612	N/A	N/A	6450	6465	AGTAGTAATTCTAAAC	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}T_{ds}A_{ds}$ $A_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}A_{ds}A_{ks}$ $A_{ks}^mC_k$	23	2943
1340625	N/A	N/A	7527	7542	TCCGAAAAAAGTGGAG	$T_{ks}^mC_{ks}^mC_{ks}G_{ds}A_{ds}A_{ds}$ $A_{ds}A_{ds}A_{ds}G_{ds}G_{ds}T_{ds}G_{ds}$ $G_{ks}A_{ks}G_k$	12	2944
1340627	N/A	N/A	10292	10307	GAGAGAACTTATACAA	$G_{ks}A_{ks}G_{ks}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}T_{ds}T_{ds}A_{ds}T_{ds}A_{ds}$ $^mC_{ks}A_{ks}A_k$	37	2945
1340629	77	92	3171	3186	CAGAAGGATTCTAGG	$^mC_{ks}A_{ks}G_{ks}A_{ds}A_{ds}G_{ds}G_{ds}$ $A_{ds}T_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}A_{ks}$ $G_{ks}G_k$	43	2946
1340633	N/A	N/A	4662	4677	CCCGCCCTTAAGTCAT	$^mC_{ks}^mC_{ks}^mC_{ks}G_{ds}^mC_{ds}$ $^mC_{ds}^mC_{ds}T_{ds}T_{ds}A_{ds}A_{ds}$ $G_{ds}T_{ds}^mC_{ks}A_{ks}T_k$	45	2947
1340643	N/A	N/A	5277	5292	GCTAGACAATTGCAAA	$G_{ks}^mC_{ks}T_{ks}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}A_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $^mC_{ds}A_{ks}A_{ks}A_k$	77	2948
1340686	N/A	N/A	5189	5204	TTCCCAACGCAACAGT	$T_{ks}T_{ks}^mC_{ks}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}^mC_{ds}G_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}A_{ks}G_{ks}T_k$	60	2949
1340694	N/A	N/A	5648	5663	GAGCATTCATCAGATG	$G_{ks}A_{ks}G_{ks}^mC_{ds}A_{ds}T_{ds}T_{ds}$ $^mC_{ds}A_{ds}T_{ds}^mC_{ds}A_{ds}G_{ds}$ $A_{ks}T_{ks}G_k$	64	2950
1340698	N/A	N/A	6087	6102	AGTTTTACCTCAGGT	$A_{ks}G_{ks}T_{ks}T_{ds}T_{ds}T_{ds}^mC_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}T_{ds}^mC_{ds}A_{ds}$ $G_{ks}G_{ks}T_k$	82	2951
1340709	N/A	N/A	9116	9131	AGACAGACCAAGTAGC	$A_{ks}G_{ks}A_{ks}^mC_{ds}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}G_{ds}T_{ds}$ $A_{ks}G_{ks}^mC_k$	21	2952
1340719	N/A	N/A	9944	9959	ACCACACTAATGAATC	$A_{ks}^mC_{ks}^mC_{ks}A_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}T_{ds}A_{ds}A_{ds}T_{ds}G_{ds}A_{ds}$ $A_{ks}T_{ks}^mC_k$	54	2953

TABLE 50-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1340726	N/A	N/A	6262	6277	TGAGATGGGCAAGGCC	$T_{ks}G_{ks}A_{ks}G_{ds}A_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}^mC_{ds}A_{ds}A_{ds}G_{ds}G_{ks}$ $^mC_{ks}^mC_k$	53	2954
1340728	N/A	N/A	8645	8660	GGCAAGACAGACTGTT	$G_{ks}G_{ks}^mC_{ks}A_{ds}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}$ $G_{ks}T_{ks}T_k$	75	2955
1340729	N/A	N/A	14191	14206	GATTGAGTTCTCCTACT	$A_{ks}A_{ks}T_{ks}T_{ks}T_{ds}G_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ds}^mC_{ds}$ $A_{ks}^mC_{ks}T_k$	45	2956
1340739	N/A	N/A	16288	16303	TTAGAATAGTCTTCAG	$T_{ks}T_{ks}A_{ks}G_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}$ $^mC_{ks}A_{ks}G_k$	52	2957
1340752	N/A	N/A	9590	9605	AGCTGGTAAAGGTAAAG	$A_{ks}G_{ks}^mC_{ks}T_{ds}G_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}A_{ds}G_{ds}G_{ds}T_{ds}A_{ks}$ $A_{ks}G_k$	55	2958
1340755	N/A	N/A	3995	4010	ATGTAGTGTTTACAAG	$A_{ks}T_{ks}G_{ks}T_{ds}A_{ds}G_{ds}T_{ds}$ $G_{ds}T_{ds}T_{ds}T_{ds}A_{ds}^mC_{ds}A_{ks}$ $A_{ks}G_k$	58	2959
1340769	N/A	N/A	12666	12681	GTTAACCTGCAGCAGA	$G_{ks}T_{ks}T_{ks}A_{ds}A_{ds}^mC_{ds}$ $^mC_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}G_{ds}$ $^mC_{ds}A_{ks}G_{ks}A_k$	51	2960
1340791	N/A	N/A	17953	17968	TAGAACTGCTCCTCA	$T_{ks}A_{ks}G_{ks}A_{ds}A_{ds}^mC_{ds}$ $T_{ds}G_{ds}^mC_{ds}T_{ds}^mC_{ds}^mC_{ds}$ $T_{ks}^mC_{ks}A_k$	32	2961
1340795	N/A	N/A	7789	7804	GTGACATAACTACTTC	$G_{ks}T_{ks}G_{ks}A_{ds}^mC_{ds}A_{ds}T_{ds}$ $A_{ds}A_{ds}^mC_{ds}T_{ds}A_{ds}^mC_{ds}$ $T_{ks}T_{ks}^mC_k$	68	2962
1340801	N/A	N/A	4246	4261	TAAGTGTAAAGACAT	$T_{ks}A_{ks}A_{ks}G_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}G_{ds}^mC_{ks}$ $A_{ks}T_k$	63	2963
1340810	N/A	N/A	12480	12495	GGAGCTAGATACTCAA	$G_{ks}G_{ks}A_{ks}G_{ds}^mC_{ds}T_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}A_{ds}^mC_{ds}T_{ds}$ $^mC_{ks}A_{ks}A_k$	42	2964
1340819	N/A	N/A	18530	18545	TCCTACAGTGCTTAGT	$T_{ks}^mC_{ks}^mC_{ks}T_{ds}A_{ds}^mC_{ds}$ $A_{ds}G_{ds}T_{ds}G_{ds}^mC_{ds}T_{ds}T_{ds}$ $A_{ks}G_{ks}T_k$	44	2965
1340907	N/A	N/A	5121	5136	AAGGAGCCAGTTATGA	$A_{ks}A_{ks}G_{ks}G_{ds}A_{ds}G_{ds}$ $^mC_{ds}^mC_{ds}A_{ds}G_{ds}T_{ds}T_{ds}$ $A_{ds}T_{ks}G_{ks}A_k$	57	2966
1340917	N/A	N/A	9666	9681	ATGCAGCTATAGGTAG	$A_{ks}T_{ks}G_{ks}^mC_{ds}A_{ds}G_{ds}$ $^mC_{ds}T_{ds}A_{ds}T_{ds}A_{ds}G_{ds}G_{ds}$ $T_{ks}A_{ks}G_k$	41	2967

TABLE 51

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1245927	N/A	N/A	3433	3448	AGTAGATGGTAAAGTCA	$A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}$ $G_{ds}G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}T_{ks}$ $^mC_{ks}A_k$	77	121

TABLE 51-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1246023	N/A	N/A	5046	5061	GAGTGAATCATTCACT	$G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}T_{ds}$ $T_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}$ $A_{ks}G_{ks}T_k$	81	1526
1341164	610	625	N/A	N/A	GAACAATATGGGATGA	$G_{ks}A_{ks}A_{ds}^mC_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}T_{ds}G_{ds}G_{ds}G_{ks}A_{es}T_{ks}$ $G_{es}A_k$	49	1735
1341172	812	827	15693	15708	CTTATTGGTAAGTATT	$T_{ks}T_{ks}T_{ds}A_{ds}T_{ds}T_{ds}G_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}G_{ks}T_{es}A_{ks}$ $T_{es}T_k$	9	1350
1341178	698	713	12116	12131	GAGACATGAGGTTTTG	$G_{ks}A_{ks}G_{ds}A_{ds}^mC_{ds}A_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}G_{ds}T_{ks}T_{es}T_{ks}$ $T_{es}G_k$	49	725
1341186	1079	1094	20894	20909	GCTAGTGCCAAACCAA	$G_{ks}^mC_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}$ $T_{ds}^mC_{ds}A_{ds}A_{ds}A_{ks}^mC_{es}$ $T_{ks}A_{es}A_k$	62	1354
1341188	1017	1032	20832	20847	CAGCATTGATTCGAAA	$T_{ks}A_{ks}G_{ds}^mC_{ds}A_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}^mC_{ks}G_{es}$ $A_{ks}A_{es}A_k$	58	263
1341192	1200	1215	21015	21030	ATGTTTATGTAAGCAC	$A_{ks}T_{ks}G_{ds}T_{ds}T_{ds}A_{ds}$ $T_{ds}G_{ds}T_{ds}A_{ks}G_{es}^mC_{ks}$ $A_{es}^mC_k$	61	2622
1341196	1088	1103	20903	20918	TTGACTGCTGCTAGTG	$T_{ks}T_{ks}G_{ds}A_{ds}^mC_{ds}T_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ds}^mC_{ds}T_{ks}A_{es}$ $G_{ks}T_{es}G_k$	29	28
1341199	1083	1098	20898	20913	TGCTGCTAGTGCCAAA	$T_{ks}G_{ks}^mC_{ds}T_{ds}G_{ds}^mC_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}^mC_{ks}$ $T_{es}A_{ks}A_{es}A_k$	40	1665
1341203	1311	1326	21126	21141	TCTTAAACCTTCCTTG	$T_{ks}^mC_{ks}T_{ds}T_{ds}A_{ds}A_{ds}$ $T_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ks}^mC_{es}$ $T_{ks}T_{es}G_k$	24	2160
1341208	1315	1330	21130	21145	ACAGTCTTAAACCTTC	$A_{ks}^mC_{ks}A_{ds}G_{ds}T_{ds}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ds}^mC_{ks}$ $T_{es}T_{ks}T_{es}^mC_k$	46	110
1341209	1257	1272	21072	21087	AGCTTTTGTCACCTT	$A_{ks}G_{ks}^mC_{ds}T_{ds}T_{ds}T_{ds}$ $G_{ds}T_{ds}^mC_{ds}^mC_{ds}A_{ks}^mC_{es}$ $T_{ks}T_{es}T_k$	62	1822
1341215	1323	1338	21138	21153	CTACTTGAACAGTCTT	$T_{ks}T_{ks}A_{ds}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}A_{ds}^mC_{ds}A_{ds}G_{ks}T_{es}$ $T_{ks}T_{es}T_k$	47	734
1341221	1327	1342	21142	21157	AATGCTACTTGAAACAG	$A_{ks}A_{ks}T_{ds}G_{ds}^mC_{ds}T_{ds}A_{ds}$ $T_{ds}T_{ds}T_{ds}G_{ds}A_{ks}A_{es}$ $T_{ks}A_{es}G_k$	39	1045
1341224	1337	1352	21152	21167	TACAGATTGGAATGCT	$T_{ks}A_{ks}^mC_{ds}A_{ds}G_{ds}A_{ds}T_{ds}$ $T_{ds}G_{ds}G_{ds}A_{ds}A_{ks}T_{es}G_{ks}$ $T_{es}T_k$	35	1746
1341229	1331	1346	21146	21161	TTGGAATGCTACTTGA	$T_{ks}T_{ks}G_{ds}G_{ds}A_{ds}A_{ds}T_{ds}$ $G_{ds}^mC_{ds}T_{ds}A_{ds}^mC_{ks}T_{es}$ $T_{ks}G_{es}A_k$	54	1279
1341230	1351	1366	21166	21181	TTCTGTGGCATGGCTA	$T_{ks}T_{ks}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}$ $G_{ds}^mC_{ds}T_{ds}T_{ds}G_{ks}G_{es}$ $T_{ks}T_{es}A_k$	50	812
1341235	1355	1370	21170	21185	GATATTCTGTGGCATG	$G_{ks}A_{ks}T_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}$ $T_{ds}G_{ds}T_{ds}G_{ds}G_{ks}^mC_{es}A_{ks}$ $T_{es}G_k$	52	1046

TABLE 51-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341245	1391	1406	21206	21221	GATCTCTTAGCTGTGC	$G_{ks}A_{ks}T_{ds}^mC_{ds}T_{ds}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}G_{ds}^mC_{ds}T_{ks}G_{es}$ $T_{ks}G_{es}^mC_k$	45	502
1341249	1359	1374	21174	21189	TGTTGATATTCTGTGG	$T_{ks}G_{ks}T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}$ $A_{ds}T_{ds}T_{ds}^mC_{ds}T_{ks}G_{es}T_{ks}$ $G_{es}G_k$	54	1358
1341252	1431	1446	21246	21261	AAATATGTCCAGGTTG	$A_{ks}A_{ks}A_{ds}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}^mC_{ds}^mC_{ds}A_{ds}G_{ks}G_{es}$ $T_{ks}T_{es}G_k$	61	581
1341253	1419	1434	21234	21249	GTTGAGATAAAGCTGC	$G_{ks}T_{ks}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}A_{ds}A_{ds}G_{ks}^mC_{es}T_{ks}$ $G_{es}^mC_k$	40	1748
1341256	1423	1438	21238	21253	CCAGGTTGAGATAAAG	$^mC_{ks}^mC_{ks}A_{ds}G_{ds}G_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{es}$ $A_{ks}A_{es}G_k$	55	33
1341264	1492	1507	21307	21322	CACCGTTTTGGGCTAA	$^mC_{ks}A_{ks}^mC_{ds}^mC_{ds}G_{ds}T_{ds}$ $T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}G_{ks}^mC_{es}$ $T_{ks}A_{es}A_k$	42	1516
1341270	1602	1617	21417	21432	ATGTATCTTATAAGAC	$A_{ks}T_{ks}G_{ds}T_{ds}A_{ds}T_{ds}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ds}A_{ks}A_{es}G_{ks}$ $A_{es}^mC_k$	18	1751
1341272	1501	1516	21316	21331	ATAGAGTTGCACCGTT	$A_{ks}T_{ks}A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}G_{ds}^mC_{ds}A_{ds}^mC_{ks}^mC_{es}$ $G_{ks}T_{es}T_k$	45	192
1341276	1617	1632	21432	21447	CGGTCACCTTTCATAA	$^mC_{ks}G_{ks}G_{ds}T_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}^mC_{ks}$ $A_{es}T_{ks}A_{es}A_k$	70	36
1341278	1621	1636	21436	21451	GAGTCGGTCACCTTTC	$G_{ks}A_{ks}G_{ds}T_{ds}^mC_{ds}G_{ds}G_{ds}$ $T_{ds}^mC_{ds}A_{ds}^mC_{ds}^mC_{ks}T_{es}$ $T_{ks}T_{es}^mC_k$	62	350
1341282	1625	1640	21440	21455	AATAGAGTCGGTCACC	$A_{ks}A_{ks}T_{ds}A_{ds}G_{ds}A_{ds}G_{ds}$ $T_{ds}^mC_{ds}G_{ds}G_{ds}T_{ks}^mC_{es}$ $A_{ks}^mC_{es}^mC_k$	13	662
1341289	1629	1644	21444	21459	TTAAATAGAGTCGGT	$T_{ks}T_{ks}A_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}T_{ks}^mC_{es}G_{ks}$ $G_{es}T_k$	51	973
1341293	1713	1728	21528	21543	AACATCTCTGGGACCA	$A_{ks}A_{ks}^mC_{ds}A_{ds}T_{ds}^mC_{ds}$ $T_{ds}^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ks}A_{es}$ $^mC_{ks}^mC_{es}A_k$	36	273
1341294	1708	1723	21523	21538	CTCTGGGACCAAGGAT	$^mC_{ks}T_{ks}^mC_{ds}T_{ds}G_{ds}G_{ds}$ $G_{ds}A_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}$ $G_{es}G_{ks}A_{es}T_k$	39	1908
1341301	1777	1792	21592	21607	ATGTAATAGCCAGTAC	$A_{ks}T_{ks}G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ks}G_{es}$ $T_{ks}A_{es}^mC_k$	47	430
1341304	1766	1781	21581	21596	AGTACAGTTCCTTTTC	$A_{ks}G_{ks}T_{ds}A_{ds}^mC_{ds}A_{ds}G_{ds}$ $T_{ds}T_{ds}^mC_{ds}^mC_{ds}T_{ks}T_{es}$ $T_{ks}T_{es}^mC_k$	40	1597
1341311	1773	1788	21588	21603	AATAGCCAGTACAGTT	$A_{ks}A_{ks}T_{ds}A_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}^mC_{ks}$ $A_{es}G_{ks}T_{es}T_k$	36	118
1341319	1781	1796	21596	21611	TCTTATGTAATAGCCA	$T_{ks}^mC_{ks}T_{ds}T_{ds}A_{ds}T_{ds}G_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ds}A_{ks}G_{es}^mC_{ks}$ $^mC_{es}A_k$	65	742

TABLE 51-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2				
Compound Number	Start Site	Stop Site	Start Site	Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341322	2254	2269	22069	22084	CAAGAGGATAGTCCAT	$mC_{ks}A_{ks}A_{ds}G_{ds}A_{ds}G_{ds}G_{ds}$ $A_{ds}T_{ds}A_{ds}G_{ds}T_{ks}mC_{es}$ $mC_{ks}A_{es}T_k$	22	510
1341326	2245	2260	22060	22075	AGTCCATGCAAAAGCA	$A_{ks}G_{ks}T_{ds}mC_{ds}mC_{ds}A_{ds}$ $T_{ds}G_{ds}mC_{ds}A_{ds}A_{ds}A_{ks}A_{es}$ $G_{ks}mC_{es}A_{sk}$	52	120
1341327	2191	2206	22006	22021	GTCTTGATGTAGTGGG	$G_{ks}T_{ks}mC_{ds}T_{ds}T_{ds}G_{ds}A_{ds}$ $T_{ds}G_{ds}T_{ds}A_{ds}G_{ks}T_{es}G_{ks}$ $G_{es}G_k$	18	1132
1341333	N/A	N/A	9518	9533	CTACACTAATATTGAG	$mC_{ks}T_{ks}A_{ds}mC_{ds}A_{ds}mC_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}T_{ks}T_{es}$ $G_{ks}A_{es}G_k$	5	59
1341338	2355	2370	22170	22185	TTATACTACAAGAGG	$T_{ks}T_{ks}A_{ds}T_{ds}A_{ds}A_{ds}mC_{ds}$ $T_{ds}A_{ds}mC_{ds}A_{ds}A_{ks}G_{es}A_{ks}$ $G_{es}G_k$	55	1289
1341345	N/A	N/A	11853	11868	AGCAGAATTGTGAACG	$A_{ks}G_{ks}mC_{ds}A_{ds}G_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}G_{ds}T_{ds}G_{ks}A_{es}A_{ks}$ $mC_{es}G_k$	31	844
1341357	N/A	N/A	12421	12436	TGGTAACGGTGATCAA	$T_{ks}G_{ks}G_{ds}T_{ds}A_{ds}A_{ds}mC_{ds}$ $G_{ds}G_{ds}T_{ds}G_{ds}A_{ks}T_{es}mC_{ks}$ $A_{es}A_k$	26	67
1341368	N/A	N/A	16131	16146	GCTCCATAATAATAGC	$G_{ks}mC_{ks}T_{ds}mC_{ds}mC_{ds}A_{ds}$ $T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}A_{ks}T_{es}$ $A_{ks}G_{es}mC_k$	15	2665
1341374	N/A	N/A	18099	18114	CTGAAATTGATACACC	$mC_{ks}T_{ks}G_{ds}A_{ds}A_{ds}A_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}T_{ds}A_{ks}mC_{es}A_{ks}$ $mC_{es}mC_k$	45	393
1341378	N/A	N/A	17991	18006	GTTAGTATAGTTATCT	$G_{ks}T_{ks}T_{ds}A_{ds}G_{ds}T_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}T_{ks}A_{es}T_{ks}$ $mC_{es}T_k$	51	1171
1341381	N/A	N/A	19911	19926	GCATCTTAAGATACCC	$G_{ks}mC_{ks}A_{ds}T_{ds}mC_{ds}T_{ds}$ $T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{es}$ $mC_kmC_{es}mC_{sk}$	41	553
1341385	N/A	N/A	19907	19922	CTTAAGATACCCAGGT	$mC_{ks}T_{ks}T_{ds}A_{ds}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}mC_{ds}mC_{ds}mC_{ks}A_{es}$ $G_{ks}G_{es}T_k$	34	241
1341390	N/A	N/A	20114	20129	GCATCAAGACATTCTA	$G_{ks}mC_{ks}A_{ds}T_{ds}mC_{ds}A_{ds}$ $A_{ds}G_{ds}A_{ds}mC_{ds}A_{ds}T_{ks}T_{es}$ $mC_{ks}T_{es}A_k$	43	1565
1341398	N/A	N/A	20194	20209	TTGCAGTTATTAGAAG	$T_{ks}T_{ks}G_{ds}mC_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}A_{ds}T_{ds}T_{ds}A_{ks}G_{es}A_{ks}$ $A_{es}G_k$	27	2123
1341403	N/A	N/A	12801	12816	AGGCTGTAAGAGTCAG	$A_{ks}G_{ks}G_{ds}mC_{ds}T_{ds}G_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}G_{ks}T_{es}mC_{ks}$ $A_{es}G_k$	38	2809
1341405	N/A	N/A	12501	12516	GCAGAAATTCACCTTG	$G_{ks}mC_{ks}A_{ds}G_{ds}A_{ds}A_{ds}A_{ds}$ $T_{ds}T_{ds}mC_{ds}A_{ds}mC_{ks}mC_{es}$ $T_{ks}T_{es}G_k$	53	1002
1341409	N/A	N/A	20422	20437	ACAGTAAAATTATGCC	$A_{ks}mC_{ks}A_{ds}G_{ds}T_{ds}A_{ds}A_{ds}$ $A_{ds}A_{ds}T_{ds}T_{ds}A_{ks}T_{es}G_{ks}$ $mC_{es}mC_k$	28	477
1341411	N/A	N/A	14194	14209	TGTGATTGAGTTCTCC	$T_{ks}G_{ks}T_{ds}G_{ds}A_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}G_{ds}T_{ds}T_{ks}mC_{es}T_{ks}$ $mC_{es}mC_k$	11	2020

TABLE 51-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341414	N/A	N/A	14168	14183	ATAGCAAGCCAACAGA	$A_{ks}T_{ks}A_{ds}G_{ds}^mC_{ds}A_{ds}A_{ds}$ $G_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}^mC_{es}$ $A_{ks}G_{es}A_k$	23	2633
1341528	1075	1090	20890	20905	GTGCCAAACCAATGTT	$G_{ks}T_{ks}G_{ds}^mC_{ds}^mC_{ds}A_{ds}$ $A_{ds}A_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}$ $T_{es}G_{ks}T_{es}T_k$	38	1042
1341536	801	816	15682	15697	GTATTCCATCTATCAG	$G_{ks}T_{ks}A_{ds}T_{ds}T_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}T_{ds}A_{ks}$ $T_{es}^mC_{ks}A_{es}G_k$	48	727
1341537	791	806	15672	15687	TATCAGACTTCTTACG	$T_{ks}A_{ks}T_{ds}^mC_{ds}A_{ds}G_{ds}A_{ds}$ $^mC_{ds}T_{ds}T_{ds}^mC_{ds}T_{ks}T_{es}$ $A_{ks}^mC_{es}G_k$	32	103
1341542	1319	1334	21134	21149	TTGAACAGTCTTAAAC	$T_{ks}T_{ks}G_{ds}A_{ds}A_{ds}^mC_{ds}A_{ds}$ $G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ks}A_{es}A_{ks}$ $A_{es}^mC_k$	37	422
1341544	1264	1279	21079	21094	GGGAGGTAGCTTTTGT	$G_kC_kG_kG_{ds}A_{ds}G_{ds}G_{ds}$ $T_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}T_{ks}T_{es}$ $T_{ks}G_{es}T_k$	66	343
1341548	1384	1399	21199	21214	TAGCTGTGCACTCATT	$T_{ks}A_{ks}G_{ds}^mC_{ds}T_{ds}G_{ds}T_{ds}$ $G_{ds}^mC_{ds}A_{ds}^mC_{ds}T_{ks}^mC_{es}$ $A_{ks}T_{es}T_k$	53	1981
1341550	1398	1413	21213	21228	GAAACTTGATCTCTTA	$G_{ks}A_{ks}A_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}$ $G_{ds}A_{ds}T_{ds}^mC_{ds}T_{ks}^mC_{es}$ $T_{ks}T_{es}A_k$	54	2717
1341553	1346	1361	21161	21176	TGGCATGGCTACAGAT	$T_{ks}G_{ks}G_{ds}^mC_{ds}A_{ds}T_{ds}G_{ds}$ $G_{ds}^mC_{ds}T_{ds}A_{ds}^mC_{ks}A_{es}$ $G_{ks}A_{es}T_k$	58	423
1341559	1488	1503	21303	21318	GTTTGGGCTAATGAA	$G_{ks}T_{ks}T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}$ $G_{ds}^mC_{ds}T_{ds}A_{ds}A_{ks}T_{es}G_{ks}$ $A_{es}A_k$	42	1204
1341562	1496	1511	21311	21326	GTTGCACCGTTTGGG	$G_{ks}T_{ks}T_{ds}G_{ds}^mC_{ds}A_{ds}$ $^mC_{ds}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ks}$ $T_{es}G_{ks}G_{es}G_k$	35	1827
1341571	1717	1732	21532	21547	TCTAAACATCTCTGGG	$T_{ks}^mC_{ks}T_{ds}A_{ds}A_{ds}A_{ds}$ $^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ks}$ $T_{es}G_{ks}G_{es}G_k$	21	2287
1341586	2184	2199	21999	22014	TGTAGTGGGAGTCGGA	$T_{ks}G_{ks}T_{ds}A_{ds}G_{ds}T_{ds}G_{ds}$ $G_{ds}G_{ds}A_{ds}G_{ds}T_{ks}^mC_{es}G_{ks}$ $G_{es}A_k$	27	587
1341587	1726	1741	21541	21556	CTAAAATTGTCTAAAC	$^mC_{ks}T_{ks}A_{ds}A_{ds}A_{ds}T_{ds}$ $T_{ds}G_{ds}T_{ds}^mC_{ds}T_{ks}A_{es}A_{ks}$ $A_{es}^mC_k$	19	2590
1341590	N/A	N/A	9337	9352	CCTCGATCCTATATAC	$^mC_{ks}^mC_{ks}T_{ds}^mC_{ds}G_{ds}A_{ds}$ $T_{ds}^mC_{ds}^mC_{ds}T_{ds}A_{ds}T_{ks}$ $A_{es}T_{ks}A_{es}^mC_k$	3	2812
1341593	N/A	N/A	9961	9976	TAAGCTGAGAGTTCTA	$T_{ks}A_{ks}A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}$ $T_{es}A_k$	29	2499
1341594	N/A	N/A	11568	11583	TGACAATGGTTGATAG	$T_{ks}G_{ks}A_{ds}^mC_{ds}A_{ds}A_{ds}T_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}G_{ks}A_{es}T_{ks}$ $A_{es}G_k$	37	1700
1341597	N/A	N/A	9971	9986	TTGAGTAGTGTAAAGCT	$T_{ks}T_{ks}G_{ds}A_{ds}G_{ds}T_{ds}A_{ds}$ $G_{ds}T_{ds}G_{ds}T_{ds}A_{ks}A_{es}G_{ks}$ $^mC_{es}T_k$	29	2346

TABLE 51-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound	SEQ ID NO: 1 Start	SEQ ID NO: 1 Stop	SEQ ID NO: 2 Start	SEQ ID NO: 2 Stop	Sequence Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341598	N/A	N/A	16282	16297	TAGTCTTCAGCAAAGT	$T_{ks}A_{ks}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}$ $^mC_{ds}A_{ds}G_{ds}^mC_{ds}A_{ks}A_{es}$ $A_{ks}G_{es}T_k$	35	2632
1341599	N/A	N/A	16377	16392	ACCAAACCTTCAGCAG	$A_{ks}^mC_{ks}^mC_{ds}A_{ds}A_{ds}A_{ds}$ $^mC_{ds}T_{ds}T_{ds}^mC_{ds}^mC_{ds}A_{ks}$ $G_{es}^mC_{ks}A_{es}G_k$	32	1948
1341605	N/A	N/A	15491	15506	TTAATGCCACCCCTACC	$T_{ks}T_{ks}A_{ds}A_{ds}T_{ds}G_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}^mC_{ds}^mC_{ds}^mC_{ks}T_{es}$ $A_{ks}^mC_{es}^mC_k$	11	2304
1341606	N/A	N/A	14204	14219	CAGATTATTATGTGAT	$^mC_{ks}A_{ks}G_{ds}A_{ds}T_{ds}T_{ds}A_{ds}$ $T_{ds}T_{ds}A_{ds}T_{ds}G_{ks}T_{es}G_{ks}$ $A_{es}T_k$	31	2723
1341609	N/A	N/A	17986	18001	TATAGTTATCTTCTCA	$T_{ks}A_{ks}T_{ds}A_{ds}G_{ds}T_{ds}T_{ds}$ $A_{ds}T_{ds}^mC_{ds}T_{ds}T_{ks}^mC_{es}^m$ $T_{ks}C_{es}A_k$	45	2295
1341616	N/A	N/A	17996	18011	ATGAAGTTAGTATAGT	$A_{ks}T_{ks}G_{ds}A_{ds}A_{ds}G_{ds}T_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}A_{ks}T_{es}A_{ks}$ $G_{es}T_k$	34	1483

TABLE 52

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence	Chemistry Notation	%	SEQ	
Number	Start	Site	Stop	Site	Stop	Site (5' to 3')	inhibition	ID NO	
1245927	N/A		N/A	3433	3448	AGTAGATGGTAAGTCA $A_{ks}G_{ks}T_{ks}A_{ds}G_{ds}A_{ds}T_{ds}G_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}G_{ds}G_{ds}T_{ks}^mC_{ks}A_k$	79	121	
1246023	N/A		N/A	5046	5061	GAGTGAATCATTTCAGT $G_{ks}A_{ks}G_{ks}T_{ds}G_{ds}A_{ds}A_{ds}T_{ds}$ $^mC_{ds}A_{ds}T_{ds}T_{ds}^mC_{ds}A_{ks}G_{ks}$ T_k	89	1526	
1341650	1264		1279	21079	21094	GGGAGGTAGCTTTTGT $G_{ks}G_{ks}G_{ks}A_{ds}G_{ys}G_{ds}T_{ds}A_{ds}$ $G_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}T_{ks}G_{ks}T_k$	17	343	
1341653	1316		1331	21131	21146	AACAGTCTTAAACCTT $A_{ks}A_{ks}^mC_{ks}A_{ds}G_{ys}T_{ds}^mC_{ds}$ $T_{ds}T_{ds}A_{ds}A_{ds}A_{ds}^mC_{ds}^mC_{ks}$ $T_{ks}T_k$	75	188	
1341660	1328		1343	21143	21158	GAATGCTACTTGAACA $G_{ks}A_{ks}A_{ks}T_{ds}G_{ys}^mC_{ds}T_{ds}$ $A_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}G_{ds}A_{ds}A_{ks}$ $^mC_{ks}A_k$	48	1123	
1341664	1320		1335	21135	21150	CTTGAACAGTCTTAAA $^mC_{ks}T_{ks}T_{ks}G_{ds}A_{ys}A_{ds}^mC_{ds}$ $A_{ds}G_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}A_{ks}$ $A_{ks}A_k$	49	500	
1341667	1312		1327	21127	21142	GTCTUAAACCTTCCCT $G_{ks}T_{ks}^mC_{ks}T_{ds}U_{ys}A_{ds}A_{ds}$ $A_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ds}$ $^mC_{ks}^mC_{ks}T_k$	35	2968	
1341670	1333		1348	21148	21163	GATTGGAATGCTACTT $G_{ks}A_{ks}T_{ks}T_{ds}G_{ys}G_{ds}A_{ds}A_{ds}$ $T_{ds}G_{ds}^mC_{ds}T_{ds}A_{ds}^mC_{ks}T_{ks}$ T_k	48	1435	
1341676	1324		1339	21139	21154	GCTACTTGAACAGTCT $G_{ks}^mC_{ks}T_{ks}A_{ds}C_{ys}T_{ds}T_{ds}$ $G_{ds}A_{ds}A_{ds}^mC_{ds}A_{ds}G_{ds}T_{ks}$ $^mC_{ks}T_k$	82	811	

TABLE 52-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341678	1356	1371	21171	21186	TGATATTCTGTGGCAT	$T_{ks}G_{ks}A_{ks}T_{ds}A_{ys}T_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ks}$ $A_{ks}T_k$	61	1124
1341680	1347	1362	21162	21177	GTGGCATGGCTACAGA	$G_{iss}T_{ks}G_{iss}G_{ds}C_{ys}A_{ds}T_{ds}$ $G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}{}^mC_{ds}A_{ks}$ $G_{ks}A_k$	49	501
1341683	1340	1355	21155	21170	GGCTACAGATTGGAAT	$G_{ks}G_{ks}{}^mC_{ks}T_{ds}A_{ys}{}^mC_{ds}A_{ds}$ $G_{ds}A_{ds}T_{ds}T_{ds}G_{ds}G_{ds}A_{ks}A_{ks}$ T_k	40	1980
1341690	1364	1379	21179	21194	GTTCTGTTGATATTC	$G_{ks}T_{ks}T_{ks}{}^mC_{ds}U_{ys}T_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ds}T_{ds}A_{ds}T_{ks}T_{ks}$ mC_k	70	2969
1341696	1352	1367	21167	21182	ATTCTGTTGGCATGGCT	$A_{ks}T_{ks}T_{ks}{}^mC_{ds}U_{ys}G_{ds}T_{ds}$ $G_{ds}G_{ds}{}^mC_{ds}A_{ds}T_{ds}G_{ds}G_{ks}$ ${}^mC_{ks}T_k$	59	2970
1341699	1387	1402	21202	21217	TCTTAGCTGTGCACCTC	$T_{ks}{}^mC_{ks}T_{ks}T_{ds}A_{ys}{}^mC_{ds}$ $T_{ds}G_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ds}{}^mC_{ks}$ $T_{ks}{}^mC_k$	71	190
1341705	1393	1408	21208	21223	TTGAUCTCTTAGCTGT	$T_{ks}T_{ks}G_{ks}A_{ds}U_{ys}{}^mC_{ds}T_{ds}$ ${}^mC_{ds}T_{ds}T_{ds}A_{ds}G_{ds}{}^mC_{ds}T_{ks}$ $G_{ks}T_k$	50	2971
1341711	1420	1435	21235	21250	GGTTGAGATAAAGCTG	$G_{ks}G_{ks}T_{ks}T_{ds}G_{ys}A_{ds}G_{ds}A_{ds}$ $T_{ds}A_{ds}A_{ds}A_{ds}G_{ds}{}^mC_{ks}T_{ks}G_k$	57	1826
1341714	1425	1440	21240	21255	GTCAGGTTGAGATAA	$G_{ks}T_{ks}{}^mC_{ks}{}^mC_{ds}A_{ys}G_{ds}G_{ds}$ $T_{ds}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}T_{ks}A_{ks}$ A_k	68	113
1341718	1461	1476	21276	21291	CTAGGGAAATCTTTCA	${}^mC_{ks}T_{ks}A_{ks}G_{ds}G_{ys}G_{ds}A_{ds}$ $A_{ds}A_{ds}T_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ks}$ ${}^mC_{ks}A_k$	19	659
1341721	693	708	12111	12126	ATGAGGTTTTGATACC	$A_{ks}T_{ks}G_{ks}A_{ds}G_{ys}G_{ds}T_{ds}T_{ds}$ $T_{ds}T_{ds}G_{ds}T_{ds}T_{ds}A_{ks}{}^mC_{ks}$ mC_k	49	569
1341725	1415	1430	21230	21245	AGATAAAGCTGCCTGC	$A_{ks}G_{ks}A_{ks}T_{ks}A_{ys}A_{ds}G_{ds}$ ${}^mC_{ds}{}^mC_{ds}G_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}G_{ks}$ mC_k	35	1515
1341728	1489	1504	21304	21319	CGTTUTGGGCTAATGA	${}^mC_{ks}G_{ks}T_{ks}T_{ds}U_{ys}T_{ds}G_{ds}$ $G_{ds}G_{ds}{}^mC_{ds}T_{ds}A_{ds}A_{ds}T_{ks}$ $G_{ks}A_k$	34	2972
1341731	1497	1512	21312	21327	AGTTGCACCGTTTTGG	$A_{ks}G_{ks}T_{ks}T_{ds}G_{ys}{}^mC_{ds}A_{ds}$ ${}^mC_{ds}{}^mC_{ds}G_{ds}T_{ds}T_{ds}T_{ds}T_{ks}$ $G_{ks}G_k$	38	1905
1341736	1493	1508	21308	21323	GCACCGTTTTGGGCTA	$G_{ks}{}^mC_{ks}A_{ks}{}^mC_{ds}C_{ys}G_{ds}T_{ds}$ $T_{ds}T_{ds}T_{ds}G_{ds}G_{ds}{}^mC_{ks}$ $T_{ks}A_k$	67	1593
1341738	1622	1637	21437	21452	AGAGUCGGTCACCTTT	$A_{ks}G_{ks}A_{ks}G_{ds}U_{ys}{}^mC_{ds}G_{ds}$ $G_{ds}T_{ds}{}^mC_{ds}A_{ds}{}^mC_{ds}{}^mC_{ds}T_{ks}$ $T_{ks}T_k$	55	2973
1341744	1504	1519	21319	21334	AGAAUAGAGTTGCACC	$A_{ks}G_{ks}A_{ks}A_{ds}U_{ys}A_{ds}G_{ds}A_{ds}$ $G_{ds}T_{ds}T_{ds}G_{ds}{}^mC_{ds}A_{ks}{}^mC_{ks}$ mC_k	49	2974
1341748	1614	1629	21429	21444	TCACCTTTCATAATGT	$T_{ks}{}^mC_{ks}A_{ks}{}^mC_{ds}C_{ys}T_{ds}T_{ds}T_{ds}$ ${}^mC_{ds}A_{ds}T_{ds}A_{ds}A_{ds}T_{ks}G_{ks}$ T_k	47	2109
1341754	1618	1633	21433	21448	TCGGUCACCTTTCATA	$T_{ks}{}^mC_{ks}G_{ks}G_{ds}U_{ys}{}^mC_{ds}A_{ds}$ ${}^mC_{ds}{}^mC_{ds}T_{ds}T_{ds}T_{ds}{}^mC_{ds}A_{ks}$ $T_{ks}A_k$	21	2975

TABLE 52-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound	SEQ ID	SEQ ID	SEQ ID	SEQ ID	Sequence	Chemistry Notation	% inhibition	SEQ ID	
Number	Start	Site	Stop	Site					
									NO
1341761	1630	1645	21445	21460	TTTAAATAGAGTCGG	$T_{ks}T_{ks}T_{ks}A_{ds}A_{ys}A_{ds}T_{ds}$ $A_{ds}G_{ds}A_{ds}G_{ds}T_{ds}^mC_{ks}G_{ks}G_k$	77	1051	
1341764	1626	1641	21441	21456	AAATAGAGTCGGTCAC	$A_{ks}A_{ks}A_{ks}T_{ds}A_{ys}G_{ds}A_{ds}G_{ds}$ $T_{ds}^mC_{ds}G_{ds}G_{ds}T_{ds}^mC_{ks}A_{ks}$ mC_k	63	740	
1341767	1714	1729	21529	21544	AAACATCTCTGGGACC	$A_{ks}A_{ks}A_{ks}^mC_{ds}A_{ys}T_{ds}^mC_{ds}$ $T_{ds}^mC_{ds}T_{ds}G_{ds}G_{ds}G_{ds}A_{ks}$ $^mC_{ks}^mC_k$	66	2129	
1341768	1710	1725	21525	21540	ATCTCTGGGACCAAGG	$A_{ks}T_{ks}^mC_{ks}T_{ds}C_{ys}T_{ds}G_{ds}$ $G_{ds}G_{ds}A_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}$ $G_{ks}G_k$	52	37	
1341773	1718	1733	21533	21548	GTCTAAACATCTCTGG	$G_{ks}T_{ks}^mC_{ks}T_{ds}A_{ys}A_{ds}A_{ds}$ $^mC_{ds}A_{ds}T_{ds}^mC_{ds}T_{ds}^mC_{ds}T_{ks}$ $G_{ks}G_k$	47	429	
1341781	1768	1783	21583	21598	CCAGUACAGTTCCTTT	$^mC_{ks}^mC_{ks}A_{ks}G_{ds}U_{ys}A_{ds}^mC_{ds}$ $A_{ds}G_{ds}T_{ds}T_{ds}^mC_{ds}^mC_{ds}T_{ks}$ $T_{ks}T_k$	51	2976	
1341783	1750	1765	21565	21580	CTGTGTTAGCTTTAAT	$^mC_{ks}T_{ks}G_{ks}T_{ds}G_{ys}T_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}T_{ds}T_{ds}T_{ds}A_{ks}$ $A_{ks}T_k$	41	1286	
1341787	794	809	15675	15690	ATCTATCAGACTTCTT	$A_{ks}T_{ks}^mC_{ks}T_{ds}A_{ys}T_{ds}^mC_{ds}$ $A_{ds}G_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ks}$ $T_{ks}T_k$	34	2728	
1341789	1778	1793	21593	21608	TATGUAATAGCCAGTA	$T_{ks}A_{ks}T_{ks}G_{ds}U_{ys}A_{ds}A_{ds}T_{ds}$ $A_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}G_{ks}T_{ks}$ A_k	43	2977	
1341792	1774	1789	21589	21604	TAATAGCCAGTACAGT	$T_{ks}A_{ks}A_{ks}T_{ds}A_{ys}G_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}G_{ds}T_{ds}A_{ds}^mC_{ds}A_{ks}$ $G_{ks}T_k$	51	196	
1341801	2186	2201	22001	22016	GATGUAGTGGGAGTCG	$G_{ks}A_{ks}T_{ks}G_{ds}U_{ys}A_{ds}G_{ds}T_{ds}$ $G_{ds}G_{ds}G_{ds}A_{ds}G_{ds}T_{ks}^mC_{ks}G_k$	35	2978	
1341806	2193	2208	22008	22023	TAGTCTTGATGTAGTG	$T_{ks}A_{ks}G_{ks}T_{ds}C_{ys}T_{ds}T_{ds}G_{ds}$ $A_{ds}T_{ds}G_{ds}T_{ds}A_{ds}G_{ks}T_{ks}G_k$	56	1288	
1341807	1782	1797	21597	21612	TTCTUATGTAATAGCC	$T_{ks}T_{ks}^mC_{ks}T_{ds}U_{ys}A_{ds}T_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}T_{ds}A_{ds}G_{ks}$ $^mC_{ks}^mC_k$	66	2979	
1341816	2257	2272	22072	22087	AAACAAGAGGATAGTC	$A_{ks}A_{ks}A_{ks}^mC_{ds}A_{ys}A_{ds}G_{ds}$ $A_{ds}G_{ds}G_{ds}A_{ds}T_{ds}A_{ds}G_{ks}T_{ks}$ mC_k	53	744	
1341817	2247	2262	22062	22077	ATAGUCCATGCAAAAG	$A_{ks}T_{ks}A_{ks}G_{ds}U_{ys}^mC_{ds}^mC_{ds}$ $A_{ds}T_{ds}G_{ds}^mC_{ds}A_{ds}A_{ks}$ $A_{ks}G_k$	57	2980	
1341820	N/A	N/A	9511	9526	AATAUTGAGGCACTGG	$A_{ks}A_{ks}T_{ks}A_{ds}U_{ys}T_{ds}G_{ds}A_{ds}$ $G_{ds}G_{ds}^mC_{ds}A_{ds}^mC_{ds}T_{ks}G_{ks}$ G_k	48	2981	
1341827	N/A	N/A	9330	9345	CCTAUATACATCCAAG	$^mC_{ks}^mC_{ks}T_{ks}A_{ds}U_{ys}A_{ds}T_{ds}$ $A_{ds}^mC_{ds}A_{ds}T_{ds}^mC_{ds}^mC_{ds}A_{ks}$ $A_{ks}G_k$	39	2982	
1341830	N/A	N/A	9598	9613	ATGTGAAGAGCTGGTA	$A_{ks}T_{ks}G_{ks}T_{ds}G_{ys}A_{ds}A_{ds}G_{ds}$ $A_{ds}G_{ds}^mC_{ds}T_{ds}G_{ds}G_{ks}T_{ks}A_k$	26	527	
1341832	N/A	N/A	9964	9979	GTGTAAGCTGAGAGTT	$G_{ks}T_{ks}G_{ks}T_{ds}A_{ys}A_{ds}G_{ds}$ $^mC_{ds}T_{ds}G_{ds}A_{ds}G_{ds}A_{ds}G_{ks}$ $T_{ks}T_k$	55	2760	

TABLE 52-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2								
Compound Number	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	Sequence (5' to 3')	Chemistry Notation (5' to 3')	% inhibition	SEQ ID NO
1341840	N/A	N/A	11543	11558	TGACAATGGTTGCCAC	$T_{ks}G_{ks}A_{ks}^mC_{ds}A_{ys}A_{ds}T_{ds}$ $G_{ds}G_{ds}T_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ks}$ $A_{ks}^mC_k$	55	1233
1341842	N/A	N/A	11548 11573	11563 11588	TCTCUTGACAATGGTT	$T_{ks}^mC_{ks}T_{ks}^mC_{ds}U_{ys}T_{ds}G_{ds}$ $A_{ds}^mC_{ds}A_{ds}T_{ds}T_{ds}G_{ds}G_{ks}$ $T_{ks}T_k$	62	2983
1341845	N/A	N/A	11858	11873	GATTAAGCAGAATTGT	$G_{ks}A_{ks}T_{ks}T_{ds}A_{ys}A_{ds}G_{ds}$ $^mC_{ds}A_{ds}G_{ds}A_{ds}A_{ds}T_{ds}T_{ks}$ $G_{ks}T_k$	28	1156
1341847	805	820	15686	15701	GTAAGTATTCATCTA	$G_{ks}T_{ks}A_{ks}A_{ds}G_{ys}T_{ds}A_{ds}T_{ds}$ $T_{ds}^mC_{ds}^mC_{ds}A_{ds}T_{ds}^mC_{ks}T_{ks}$ A_k	43	1038
1341851	N/A	N/A	14196	14211	TATGUGATTGAGTTCT	$T_{ks}A_{ks}T_{ks}G_{ds}U_{ys}G_{ds}A_{ds}T_{ds}$ $T_{ds}G_{ds}A_{ds}G_{ds}T_{ds}T_{ks}^mC_{ks}T_k$	21	2984
1341853	N/A	N/A	12416	12431	ACGGUGATCAAATGTA	$A_{ks}^mC_{ks}G_{ks}G_{ds}U_{ys}G_{ds}A_{ds}T_{ds}$ $^mC_{ds}A_{ds}A_{ds}A_{ds}T_{ds}G_{ks}T_{ks}$ A_k	18	2985
1341855	N/A	N/A	12426	12441	CAGGGTGGTAACGGTG	$^mC_{ks}A_{ks}G_{ks}G_{ds}G_{ys}T_{ds}G_{ds}$ $G_{ds}T_{ds}A_{ds}A_{ds}^mC_{ds}G_{ds}G_{ks}$ $T_{ks}G_k$	29	2729
1341860	N/A	N/A	16134	16149	GATGCTCCATAATAAT	$G_{ks}A_{ks}T_{ks}G_{ds}C_{ys}T_{ds}^mC_{ds}$ $^mC_{ds}A_{ds}T_{ds}A_{ds}A_{ds}T_{ds}A_{ks}$ $A_{ks}T_k$	24	2049
1341861	N/A	N/A	14914	14929	ACATUAGCAAGCTAAG	$A_{ks}^mC_{ks}A_{ks}T_{ds}U_{ys}A_{ds}G_{ds}$ $^mC_{ds}A_{ds}A_{ds}G_{ds}^mC_{ds}T_{ds}A_{ks}$ $A_{ks}G_k$	23	2986
1341862	N/A	N/A	15494	15509	GCATUAATGCCACCCT	$G_{ks}^mC_{ks}A_{ks}T_{ds}U_{ys}A_{ds}A_{ds}$ $T_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}^mC_{ds}$ $^mC_{ks}^mC_{ks}T_k$	32	2987
1341870	N/A	N/A	16287	16302	TAGAATAGTCTTCAGC	$T_{ks}A_{ks}G_{ks}A_{ds}T_{ds}A_{ds}G_{ds}$ $T_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ds}A_{ks}G_{ks}$ mC_k	30	1403
1341877	N/A	N/A	16124	16139	AATAATAGCTCTATTG	$A_{ks}A_{ks}T_{ks}A_{ds}A_{ys}T_{ds}A_{ds}G_{ds}$ $^mC_{ds}^mC_{ds}^mC_{ds}T_{ds}T_{ds}T_{ks}T_{ks}$ G_k	16	2410
1341882	N/A	N/A	17992	18007	AGTTAGTATAGTTATC	$AG_{ks}T_{ks}T_{ds}A_{ys}G_{ds}T_{ds}A_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}T_{ds}A_{ks}T_{ks}^mC_k$	42	1249
1341887	N/A	N/A	16380	16395	CCAACCAAACTTCCAG	$^mC_{ks}^mC_{ks}A_{ks}A_{ds}C_{ys}^mC_{ds}A_{ds}$ $A_{ds}A_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ds}^mC_{ks}$ $A_{ks}G_k$	30	2539
1341889	N/A	N/A	17987	18002	GTATAGTTATCTTCTC	$G_{ks}T_{ks}A_{ks}T_{ds}A_{ys}G_{ds}T_{ds}T_{ds}$ $A_{ds}T_{ds}^mC_{ds}T_{ds}T_{ds}^mC_{ks}T_{ks}$ mC_k	54	937
1341890	1082	1097	20897	20912	GCTGCTAGTGCCAAAC	$G_{ks}^mC_{ks}T_{ks}G_{ds}C_{ys}T_{ds}A_{ds}$ $G_{ds}T_{ds}G_{ds}^mC_{ds}^mC_{ds}A_{ds}A_{ks}$ $A_{ks}^mC_k$	39	1587
1341895	N/A	N/A	18099	18114	CTGAAATTGATACACC	$^mC_{ks}T_{ks}G_{ks}A_{ds}A_{ys}T_{ds}$ $T_{ds}G_{ds}A_{ds}T_{ds}A_{ds}^mC_{ds}A_{ks}$ $^mC_{ks}^mC_k$	71	393
1341896	N/A	N/A	17997	18012	TATGAAGTTAGTATAG	$T_{ks}A_{ks}T_{ks}G_{ds}A_{ys}G_{ds}T_{ds}$ $T_{ds}A_{ds}G_{ds}T_{ds}T_{ks}A_{ks}G_k$	27	1560
1341900	N/A	N/A	19911	19926	GCATCTTAAGATACCC	$G_{ks}^mC_{ks}A_{ks}T_{ds}C_{ys}T_{ds}T_{ds}$ $A_{ds}A_{ds}G_{ds}A_{ds}T_{ds}A_{ds}^mC_{ks}$ $^mC_{ks}^mC_k$	65	553

TABLE 52-continued

Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2									
Compound	SEQ ID NO: 1	SEQ ID NO: 1	SEQ ID NO: 2	SEQ ID NO: 2	Sequence	Chemistry Notation	% inhibition	SEQ ID	
Number	Start	Site	Stop	Site	Start Site	Stop Site		NO	
1341907	N/A	N/A	18697	18712	CAAAAGACTTTGAGAC	^m C _{ks} A _{ks} A _{ks} A _{ks} A _{ys} G _{ds} A _{ds} ^m C _{ds} T _{ds} T _{ds} T _{ds} G _{ds} A _{ds} G _{ks} A _{ks} ^m C _k	25	2186	
1341908	N/A	N/A	19915	19930	GAGTGCATCTTAAGAT	G _{ks} A _{ks} G _{ks} T _{ds} G _{ys} ^m C _{ds} A _{ds} T _{ds} ^m C _{ds} T _{ds} T _{ds} A _{ds} A _{ds} G _{ks} A _{ks} T _k	7	2256	
1341914	N/A	N/A	19907	19922	CTTAAGATACCCAGGT	^m C _{ks} T _{ks} T _{ks} A _{ks} A _{ys} G _{ds} A _{ds} T _{ds} A _{ds} ^m C _{ds} ^m C _{ds} ^m C _{ds} A _{ds} G _{ks} G _{ks} T _k	50	241	
1341919	N/A	N/A	20109	20124	AAGACATTCTAGCCTG	A _{ks} A _{ks} G _{ks} A _{ds} C _{ys} A _{ds} T _{ds} T _{ds} ^m C _{ds} T _{ds} A _{ds} G _{ds} ^m C _{ds} ^m C _{ks} T _{ks} G _k	54	2744	
1341923	N/A	N/A	20119	20134	TTACAGCATCAAGACA	T _{ks} T _{ks} A _{ks} ^m C _{ds} A _{ys} G _{ds} ^m C _{ds} A _{ds} T _{ds} ^m C _{ds} A _{ds} A _{ds} G _{ds} A _{ks} ^m C _{ks} A _k	46	2409	
1341924	1078	1093	20893	20908	CTAGUGCCAAACCAAT	^m C _{ks} T _{ks} A _{ks} G _{ds} U _{ys} G _{ds} ^m C _{ds} ^m C _{ds} A _{ds} A _{ds} A _{ds} ^m C _{ds} ^m C _{ds} A _{ks} A _{ks} T _k	45	2988	
1341928	N/A	N/A	20194	20209	TTGCAGTTATTAGAAG	T _{ks} T _{ks} G _{ks} ^m C _{ds} A _{ys} G _{ds} T _{ds} T _{ds} A _{ds} T _{ds} T _{ds} A _{ds} G _{ds} A _{ks} A _{ks} G _k	17	2123	
1341930	N/A	N/A	20383	20398	GTAAUCACAAGTAAGG	G _{ks} T _{ks} A _{ks} A _{ds} U _{ys} ^m C _{ds} A _{ds} ^m C _{ds} A _{ds} A _{ds} G _{ds} T _{ds} A _{ds} A _{ks} G _{ks} G _k	20	2989	
1341941	N/A	N/A	20489	20504	CATAACTTCATAGTGG	^m C _{ks} A _{ks} T _{ks} A _{ds} A _{ys} ^m C _{ds} T _{ds} T _{ds} ^m C _{ds} A _{ds} T _{ds} A _{ds} G _{ds} T _{ks} G _{ks} G _k	38	866	
1341942	1074	1089	20889	20904	TGCCAAACCAATGTTT	T _{ks} G _{ks} ^m C _{ks} ^m C _{ds} A _{ys} A _{ds} A _{ds} ^m C _{ds} ^m C _{ds} A _{ds} A _{ds} T _{ds} G _{ds} T _{ks} T _{ks} T _k	35	2673	
1341948	N/A	N/A	12503	12518	TTGCAGAAATTCACCT	T _{ks} T _{ks} G _{ks} ^m C _{ds} A _{ys} G _{ds} A _{ds} A _{ds} A _{ds} T _{ds} T _{ds} ^m C _{ds} A _{ds} ^m C _{ks} ^m C _{ks} T _k	29	2528	
1341953	1086	1101	20901	20916	GACTGCTGCTAGTGCC	G _{ks} A _{ks} ^m C _{ks} T _{ds} G _{ys} ^m C _{ds} T _{ds} G _{ds} ^m C _{ds} T _{ds} A _{ds} G _{ds} T _{ds} G _{ks} ^m C _{ks} ^m C _k	60	1899	
1341957	N/A	N/A	20479	20494	TAGTGGACTTCATTAG	T _{ks} A _{ks} G _{ks} T _{ds} G _{ys} G _{ds} A _{ds} ^m C _{ds} T _{ds} T _{ds} ^m C _{ds} A _{ds} T _{ds} T _{ks} A _{ks} G _k	34	2351	

Example 5: Dose-Dependent Inhibition of Human HSD17B13 in HepaRG Cells by Modified Oligonucleotides

Certain modified oligonucleotides described in the studies above exhibiting significant in vitro inhibition of HSD17B13 RNA were selected and tested at various doses in HepaRG cells.

The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000-35,000 cells per well were transfected using electroporation with modified oligonucleotides diluted to different concentrations

as specified in the tables below. After a treatment period of approximately 24 hours, HSD17B13 RNA levels were measured as previously described using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total RNA content, as measured by human GAPDH. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells. Data indicated as "N.D." (no data) means that no data is available for that treatment with that compound.

The half maximal inhibitory concentration (IC₅₀) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

459

TABLE 53

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245094	0	5	75	68	N.D.	0.3
1245641	5	16	52	75	80	0.4
1245744	5	11	40	81	87	0.4
1245822	0	0	57	78	93	0.1
1245901	1	14	38	55	78	0.8
1245927	8	42	76	83	92	0.2
1245952	25	34	51	70	86	0.2
1245979	4	0	38	64	75	0.8
1246005	1	14	60	64	78	0.5
1246134	0	50	75	79	N.D.	0.2
1246135	10	10	42	68	88	0.5
1246161	18	20	63	75	86	0.3
1246550	14	28	58	76	88	0.3
1246706	60	69	68	76	N.D.	0.02

TABLE 54

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245096	3	17	70	85	N.D.	0.2
1245097	4	40	75	80	76	0.2
1245565	2	0	26	56	65	1.5
1245798	16	37	57	82	90	0.2
1245825	0	11	44	64	81	0.6
1245851	0	9	45	69	84	0.6
1245927	0	25	56	79	82	0.4
1245928	7	42	67	89	95	0.2
1245980	12	43	94	84	N.D.	0.1
1246059	1	15	44	68	81	0.5
1246084	4	18	69	89	93	0.2
1246110	10	41	71	90	N.D.	0.1
1246111	0	22	44	79	83	0.4
1246162	33	31	71	88	94	0.1
1246163	1	17	42	73	92	0.4

TABLE 55

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	19 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245098	25	27	71	80	71	0.2
1245463	0	0	20	63	78	0.9
1245696	15	30	52	74	89	0.3
1245800	0	30	55	82	82	0.3
1245904	12	19	51	73	97	0.3
1245927	3	11	48	68	76	0.6
1245930	7	26	58	77	89	0.3
1245957	0	0	54	75	86	0.5
1245982	7	25	45	78	92	0.3
1245983	0	18	42	67	90	0.5
1246035	0	2	51	73	75	0.6
1246113	12	0	47	75	84	0.6
1247023	0	23	46	75	90	0.4
1247048	9	22	41	74	84	0.4
1247075	0	0	4	50	68	2.0

460

TABLE 56

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245075	20	50	62	75	86	0.1
1245829	18	33	58	83	94	0.2
1245855	16	21	59	78	89	0.3
1245927	0	21	54	69	76	0.5
1245932	22	24	56	90	94	0.2
1245933	23	24	63	76	89	0.2
1245958	12	31	43	87	84	0.3
1245959	1	10	46	65	90	0.5
1245984	9	37	62	76	93	0.2
1246036	15	33	65	80	95	0.2
1246063	16	29	53	84	85	0.2
1246115	0	38	66	88	92	0.2
1246140	20	2	54	75	86	0.5
1246192	0	17	52	72	89	0.5
1247024	16	45	82	89	97	0.1

TABLE 57

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245076	0	0	32	66	83	0.6
1245415	0	0	22	65	88	0.9
1245649	8	23	49	79	97	0.3
1245857	0	2	35	74	91	0.6
1245927	4	25	57	82	86	0.3
1245960	14	34	54	78	88	0.2
1245986	2	2	30	78	95	0.4
1245987	26	32	58	82	91	0.2
1246013	0	0	27	68	83	0.8
1246064	19	36	63	74	92	0.2
1246116	17	41	75	92	95	0.1
1246169	0	0	30	55	86	1.0
1246195	0	0	26	63	82	0.9
1246610	5	0	2	57	82	1.3
1247078	3	4	28	66	93	0.6

TABLE 58

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245105	0	0	24	61	88	0.8
1245651	3	22	49	78	89	0.3
1245676	0	0	34	67	91	0.7
1245806	28	27	51	69	91	0.2
1245807	1	9	47	80	90	0.4
1245858	16	39	66	86	96	0.2
1245927	0	22	46	79	89	0.4
1245988	8	26	40	65	85	0.5
1246066	26	41	63	86	94	0.1
1246067	0	0	40	51	85	0.9
1246093	0	0	24	64	86	0.7
1246144	5	30	43	78	86	0.3
1246145	0	3	40	72	85	0.6
1246248	28	32	62	86	92	0.1
1247029	0	0	11	36	73	1.9

461

TABLE 59

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245080	7	13	39	70	87	0.5
1245081	4	28	44	72	85	0.4
1245106	0	28	46	80	81	0.4
1245809	0	0	1	48	77	1.6
1245886	0	4	21	62	80	0.9
1245927	16	29	64	84	90	0.2
1245964	16	22	48	72	95	0.3
1245990	15	4	29	61	86	0.7
1245991	0	9	26	63	83	0.8
1246068	14	31	49	73	92	0.3
1246095	10	16	46	69	88	0.4
1246121	0	18	37	76	86	0.5
1246173	3	26	58	83	89	0.3
1246562	5	19	26	70	88	0.5

TABLE 60

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245212	3	1	43	63	88	0.6
1245680	0	0	22	64	82	0.9
1245681	0	17	58	80	94	0.3
1245888	17	22	41	72	87	0.4
1245927	0	16	52	78	93	0.4
1245967	0	9	41	62	89	0.6
1246019	17	26	62	92	91	0.2
1246070	7	31	46	74	94	0.3
1246123	0	16	37	58	83	0.7
1246226	12	30	56	86	89	0.2
1246227	0	6	0	55	78	1.4
1246824	0	0	17	40	81	1.5
1246902	0	17	40	74	86	0.5
1246903	35	35	60	68	84	0.2
1246929	0	18	40	65	87	0.6

TABLE 61

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245085	18	21	49	79	95	0.3
1245162	19	5	41	78	97	0.4
1245579	0	8	39	65	87	0.6
1245656	20	33	47	80	83	0.3
1245682	23	42	53	87	93	0.2
1245838	9	0	28	77	92	0.5
1245916	1	25	26	60	82	0.7
1245927	14	36	76	77	86	0.2
1245968	24	43	69	88	88	0.1
1245969	0	9	32	67	74	0.9
1246124	7	12	31	76	90	0.5
1246203	0	0	21	58	87	1.0
1246489	2	14	44	86	86	0.4
1246853	0	0	16	52	77	1.3
1247061	0	12	32	70	93	0.5

462

TABLE 62

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245580	3	9	49	58	89	0.5
1245607	0	16	33	77	87	0.5
1245684	35	51	77	92	96	0.1
1245685	13	25	59	80	93	0.2
1245736	0	0	45	81	95	0.6
1245737	13	23	53	83	94	0.3
1245919	0	0	0	72	90	1.0
1245927	0	9	37	84	77	0.4
1245996	29	53	74	91	97	0.1
1245997	16	31	54	79	90	0.2
1246022	49	54	80	93	94	0.0
1246023	0	40	89	86	95	0.2
1246048	12	0	73	73	97	0.3
1246074	7	43	73	84	94	0.2
1246075	0	0	41	70	88	0.7

TABLE 63

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245141	12	30	57	84	90	0.2
1245374	29	7	43	65	85	0.4
1245660	7	12	36	73	94	0.4
1245738	36	38	61	80	88	0.1
1245790	50	61	68	87	95	<0.2
1245927	0	23	58	85	90	0.3
1245972	24	39	58	79	90	0.2
1245973	0	26	41	66	75	0.5
1245998	12	10	43	78	89	0.4
1245999	0	19	60	87	88	0.3
1246025	35	54	84	94	89	0.0
1246051	4	26	54	78	92	0.3
1246128	37	38	58	84	92	0.1
1246701	0	4	12	20	40	5.0
1246909	0	17	27	66	86	0.7

TABLE 64

Multi-dose assay of modified oligonucleotides in HepaRG cells						
% inhibition HSD17B13						
ION No.	20 nM	78 nM	313 nM	1250 nM	5000 nM	IC ₅₀ (μM)
1245142	0	9	40	64	78	0.8
1245246	15	32	33	74	91	0.3
1245610	28	44	61	82	97	0.1
1245715	0	0	37	65	86	0.9
1245844	38	37	53	75	89	0.1
1245927	11	27	64	75	81	0.3
1245975	0	11	48	71	82	0.5
1246027	0	2	15	71	85	0.8
1246052	36	41	57	79	85	0.1
1246053	0	0	9	56	74	1.5
1246104	21	40	64	77	91	0.2
1246105	0	0	3	47	73	1.8
1246130	35	13	54	78	91	0.2
1246157	44	37	62	78	91	0.1
1246391	0	0	2	48	83	1.5

463

TABLE 65

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1245092	9	20	39	68	89	0.4
1245639	0	0	21	65	85	0.8
1245716	20	27	40	69	86	0.4
1245927	0	0	25	74	64	1.2
1245976	9	29	57	77	86	0.3
1245977	0	21	44	76	94	0.4
1246002	38	51	64	79	91	0.1
1246028	0	0	19	56	80	1.2
1246029	0	18	50	73	86	0.4
1246081	5	19	48	75	88	0.4
1246185	13	15	43	62	79	0.6
1246314	3	8	36	72	92	0.5
1246340	28	32	37	58	85	0.4
1246705	1	14	20	42	70	1.8
1246861	0	0	7	25	63	5.0

TABLE 66

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	9	21	57	83	87	0.3
1340071	0	0	0	56	88	1.3
1340085	0	0	27	68	91	0.8
1340154	0	20	48	80	90	0.4
1340360	0	0	0	51	42	3.8
1340385	4	35	69	82	72	0.3
1340448	0	0	27	78	74	0.9
1340543	5	15	62	78	86	0.3
1340546	0	12	40	65	56	1.2
1340549	0	4	50	78	92	0.4
1340601	0	0	31	54	81	1.1
1340684	0	0	31	58	84	0.9
1340704	0	8	25	69	83	0.8
1340762	0	13	41	71	79	0.7
1340878	1	1	30	79	85	0.6

TABLE 67

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	20	55	82	88	0.4
1340206	0	22	55	71	90	0.4
1340208	12	12	51	69	92	0.4
1340226	0	0	43	73	96	0.6
1340260	0	0	33	56	78	1.0
1340268	0	0	35	55	90	0.9
1340334	0	5	30	67	86	0.8
1340397	0	0	49	73	88	0.7
1340555	1	13	41	63	76	0.7
1340585	0	13	47	77	74	0.6
1340603	18	16	47	69	85	0.4
1340647	14	35	76	83	94	0.2
1340789	10	14	32	73	83	0.5
1340834	14	18	48	74	79	0.4
1340895	0	19	32	65	87	0.6

464

TABLE 68

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	25	47	76	85	0.4
1340168	6	0	32	80	90	0.6
1340401	4	11	50	78	79	0.4
1340449	26	8	59	81	93	0.3
1340548	0	17	51	78	92	0.4
1340626	0	5	45	80	90	0.5
1340646	6	31	57	67	58	0.6
1340653	2	0	38	69	77	0.8
1340780	6	23	36	73	83	0.5
1340820	20	35	55	84	88	0.2
1340821	0	0	21	40	60	2.7
1340847	9	20	45	60	88	0.5
1340857	0	0	11	46	66	2.3
1340859	14	32	47	71	84	0.3
1340908	16	33	60	84	92	0.2

TABLE 69

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	16	42	80	87	0.5
1340153	0	18	38	75	82	0.5
1340227	0	21	57	70	90	0.4
1340435	14	27	57	78	90	0.3
1340441	6	14	32	70	74	0.7
1340493	13	23	54	70	88	0.3
1340510	11	13	27	66	87	0.6
1340511	0	0	7	52	74	1.6
1340572	19	23	53	75	89	0.3
1340630	20	25	42	75	87	0.3
1340670	16	24	51	74	93	0.3
1340782	0	26	54	77	92	0.3
1340840	0	3	23	43	67	1.9
1340848	0	0	30	71	91	0.7
1340861	22	34	47	74	90	0.2

TABLE 70

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	3	22	55	79	84	0.3
1340159	10	13	49	83	80	0.4
1340481	0	0	34	64	82	0.8
1340502	0	4	37	71	87	0.6
1340564	5	10	34	72	85	0.5
1340635	7	21	40	79	82	0.4
1340649	0	0	42	69	89	0.6
1340672	0	20	47	72	76	0.6
1340743	12	18	46	80	90	0.3
1340779	5	7	39	68	40	0.6
1340793	3	24	66	86	91	0.2
1340796	0	19	61	88	84	0.3
1340799	16	34	72	86	89	0.2
1340827	16	31	48	72	81	0.3
1340881	10	28	52	59	79	0.4

465

TABLE 71

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	20	42	58	79	0.7
1341183	11	18	27	53	57	2.0
1341190	0	3	29	53	48	2.4
1341193	11	27	53	63	68	0.5
1341195	15	13	12	61	71	1.2
1341239	16	0	27	62	83	0.7
1341262	0	4	36	52	78	0.9
1341273	6	18	38	63	81	0.6
1341303	29	21	44	61	90	0.3
1341313	9	33	58	67	85	0.3
1341315	8	28	52	79	82	0.3
1341342	24	25	37	51	81	0.6
1341348	15	25	43	71	70	0.5
1341391	0	2	35	49	70	1.3
1341547	13	2	40	61	83	0.6

TABLE 72

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	19	1	38	77	89	0.4
1341026	10	20	43	73	83	0.4
1341032	0	13	14	74	92	0.6
1341035	5	0	36	62	74	1.0
1341036	23	28	53	78	93	0.2
1341038	17	25	57	76	85	0.3
1341052	6	13	52	69	79	0.5
1341062	0	23	47	66	79	0.5
1341095	0	17	51	61	81	0.6
1341108	28	34	58	81	83	0.2
1341146	6	0	39	66	70	0.9
1341206	28	25	45	81	87	0.2
1341260	23	24	37	66	78	0.5
1341283	8	23	40	67	90	0.4
1341308	18	29	54	71	86	0.3

TABLE 73

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	0	47	66	89	0.6
1340983	0	0	26	55	45	3.1
1340995	1	17	37	66	62	0.9
1341011	0	0	34	50	81	1.0
1341043	0	3	40	50	44	2.2
1341050	0	7	39	55	76	0.9
1341057	6	9	39	57	78	0.8
1341068	0	0	11	46	79	1.5
1341075	0	0	0	39	55	3.1
1341118	0	8	13	31	74	2.2
1341145	0	17	37	69	83	0.6
1341154	0	0	47	43	79	1.0
1341244	10	0	21	55	68	1.5
1341335	0	0	25	51	55	2.4
1341510	4	17	34	57	66	1.0

466

TABLE 74

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	1	20	59	67	88	0.4
1340992	0	0	0	51	76	1.8
1341004	0	12	31	55	77	0.9
1341022	15	18	25	65	84	0.6
1341048	1	15	43	74	85	0.5
1341077	8	16	38	54	71	0.9
1341104	0	23	38	64	79	0.6
1341140	12	17	41	64	82	0.5
1341151	13	31	52	71	80	0.3
1341271	8	0	43	65	80	0.7
1341318	23	17	49	61	55	1.0
1341436	0	23	46	64	77	0.6
1341437	3	0	16	60	88	0.9
1341654	0	24	53	66	69	0.6
1341668	7	25	42	56	82	0.6

TABLE 75

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	2	24	40	70	92	0.4
1341651	19	35	41	67	85	0.3
1341655	0	0	13	70	89	0.9
1341671	13	27	35	66	91	0.4
1341675	9	22	46	77	76	0.4
1341686	2	2	31	61	78	0.9
1341703	16	14	33	52	58	1.8
1341735	12	32	54	71	84	0.3
1341742	13	26	39	66	84	0.4
1341797	0	0	34	65	91	0.6
1341815	0	21	20	65	75	0.9
1341839	25	16	39	60	87	0.5
1341899	14	28	42	66	77	0.5
1341911	7	29	46	67	90	0.4
1341913	0	2	0	48	59	3.2

TABLE 76

Multi-dose assay of modified oligonucleotides in HepaRG cells						
ION No.	% inhibition HSD17B13					IC ₅₀ (μM)
	20 nM	78 nM	313 nM	1250 nM	5000 nM	
1246023	0	7	18	81	92	0.6
1341531	18	21	38	62	76	0.6
1341561	26	22	39	75	70	0.4
1341644	0	11	60	82	90	0.4
1341645	0	20	0	39	79	1.8
1341646	3	0	8	57	76	1.3
1341968	0	0	10	72	90	0.8
1341980	8	9	26	54	78	1.0

Example 6: Design of Human HSD17B13 Modified Oligonucleotides

Certain modified oligonucleotides described above were further modified by conjugating at the 5'-end with a LICA-1 moiety. The resulting compounds and their unconjugated counterparts are listed in the table below. The compounds

467

have identical sequence and oligonucleotide chemistry as the “parent compounds” described above.

TABLE 77

Design of 5'-LICA-1-conjugated modified oligonucleotides targeting human HSD17B13			
GalNac Conjugated Compound No.	Parent Compound No.	Sequence (5' to 3')	SEQ ID NO
1310535	1246163	CAACTATGTAATAGGC	286
1371966	1245806	AAAATAGAGTCGGTCA	817
1371968	1246068	AGCATAGTTACCAGA	983
1371969	1246115	AAATTATACATTTCCG	596
1371971	1246063	GTTTTATACTGAGCT	594
1371973	1246019	GAATTTTGTAGGTTGG	1215
1371974	1245987	TCTCAAGGAGTACTTC	747
1371976	1245952	GTCATAAAAATCGCTG	43
1371978	1245930	AAGATAAGTAGATGGT	355
1371979	1245972	GCACTAAAGGTTTCTG	1602
1371981	1245980	ATACTAATGTCCAAGG	201
1373086	1245649	CTACTTGAACAGTCTT	734
1373087	1246903	AGTTAGTATAGTTATC	1249
1373089	1246134	CTCATTTATGTACCAA	51
1373090	1246162	ACTATGTAATAGGCAG	208
1373097	1245984	ACAATTTTCCCAATCC	513
1373098	1246022	AGTGAATCATTAGTA	1449
1373099	1245996	CATTGCAATTTCTTCA	1448
1373101	1245790	GTATCTTATAAGACTA	1595
1394140	1341675	CTACUTGAACAGTCTT	2868
1394141	1340820	GTAGATGGTAAGTCAA	2567
1394142	1341283	TGATATTCTGTGGCAT	1124
1394143	1341313	TATGTAATAGCCAGTA	508
1394144	1341742	GTATCTTATAAGACTA	1595
1394145	1341038	TTCTTATGTAATAGCC	819
1394146	1341735	CACCGTTTTGGGCTAA	1516
1394147	1341032	TTATGTAATAGCCAGT	586
1394148	1341108	GCATCTTAAGATACCC	553
1394149	1340435	TAGAACTGTGTGCTT	2535
1394150	1340908	TAGAGTGAATCATTCA	2420
1394151	1340208	GAGTAGTGTAAGCTGA	2204
1394152	1340782	GAGATTTGAAGGTTAG	2561
1371970	1246157	GCAGTAAAGTGCTAAG	1843
1371972	1246025	AGCCTTAGAGTGAATC	1682

468

TABLE 77-continued

Design of 5'-LICA-1-conjugated modified oligonucleotides targeting human HSD17B13			
GalNac Conjugated Compound No.	Parent Compound No.	Sequence (5' to 3')	SEQ ID NO
1373094	1246036	TGAAAACCTCAGCCAGC	515
1373096	1245968	GTAATTTTCAGATCCC	1291
1373100	1245682	GTGATATTCTGTGGC	1280

Example 7: Dose-Dependent Ex-Vivo Inhibition of Human HSD17B13 in Transgenic Mouse Hepatocytes

The conjugated modified oligonucleotides described above were tested for inhibition of HSD17B13 RNA at various doses in primary mouse hepatocytes extracted from transgenic mice.

A transgenic mouse model was developed in-house using a Fosmid containing the human HSD17B13 gene (NCBI Clone DB ID: ABC8-43206400A 10). The clone was digested at NotI restriction sites (2 different sites) to produce a fragment containing 19,118 bp of the human HSD17B13 gene, including 12,175 bp upstream and 7,553 bp downstream regions. That gene fragment was introduced into fertilized eggs from C57BL/6NTac strain mice by pronuclear injection to produce huHSD 17469 and 17470 founder lines. Transgenic mice derived from Founder Line 17469 was used in the experiments described herein. Human HSD17B13 RNA expression is found in the liver in this model. Mice were sacrificed and hepatocytes were collected for use in experiments testing conjugated modified oligonucleotides.

The conjugated modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Primary mouse transgenic hepatocytes described above were plated at a density of 20,000 cells per well and were treated by free uptake with conjugated modified oligonucleotides diluted to different concentrations as specified in the tables below. After an overnight incubation, HSD17B13 RNA levels were measured as previously described herein using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total GAPDH content. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells. The half maximal inhibitory concentration (IC₅₀) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

TABLE 78

Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes							
% Inhibition							
ION No.	1.95 nM	7.81 nM	31.25 nM	125 nM	500 nM	2000 nM	IC ₅₀ (nM)
1310535	38	53	80	87	85	90	3.2
1371966	41	52	87	88	91	89	4.1

TABLE 78-continued

Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes							
ION No.	% Inhibition						IC ₅₀ (nM)
	1.95 nM	7.81 nM	31.25 nM	125 nM	500 nM	2000 nM	
1371968	39	56	84	94	91	90	4.1
1371969	23	60	81	88	88	92	5.2
1371971	6	45	54	70	85	76	34.1
1371973	38	61	84	93	94	96	2.2
1371974	4	31	65	74	74	82	38.6
1371976	16	62	79	87	90	93	7.3
1371978	0	39	84	92	89	88	18.4
1371979	10	38	70	82	85	87	21.0
1371981	33	68	86	94	93	95	1.9

TABLE 79

Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes							
ION No.	% Inhibition						IC ₅₀ (nM)
	1.95 nM	7.81 nM	31.25 nM	125 nM	500 nM	2000 nM	
1373086	24	49	78	83	86	84	8.9
1373087	23	20	67	70	64	65	22.6
1373089	36	62	84	92	91	92	4.0
1373090	43	49	69	81	80	82	5.4
1373097	41	64	84	89	92	91	1.3
1373098	43	69	80	87	89	89	2.3
1373099	19	37	68	84	83	86	13.8
1373101	9	41	66	83	85	92	20.5
1394140	5	42	80	78	85	88	19.6
1394142	5	49	73	87	89	91	16.4
1371966	4	55	77	88	88	87	11.5

TABLE 80

Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes							
ION No.	% Inhibition						IC ₅₀ (nM)
	1.95 nM	7.81 nM	31.25 nM	125 nM	500 nM	2000 nM	
1394143	19	45	80	83	89	89	10.6
1394144	16	21	54	70	75	80	48.1
1394145	37	44	83	91	88	88	6.2
1394146	3	30	67	74	82	77	38.2
1394147	30	71	85	94	94	93	4.0
1394148	41	80	88	90	87	91	1.7
1394149	1	26	56	72	78	82	50.9
1394150	14	45	80	84	88	85	12.8
1394151	0	37	63	81	83	85	42.5
1394152	8	35	63	82	81	87	28.1
1371966	0	61	72	82	89	88	16.7

Example 8: Tolerability of Modified
Oligonucleotides Complementary to Human
HSD17B13 in CD-1 Mice

CD-1 mice are a multipurpose mouse model frequently utilized for safety and efficacy testing. In this study, CD-1 mice were treated with conjugated modified oligonucleotides selected from studies described above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

Groups of 6- to 8-week-old male CD-1 mice were injected subcutaneously once a week for six weeks (for a total of 7 treatments) with 25 mg/kg of conjugated modified oligonucleotides. One group of male CD-1 mice was injected with PBS. Mice were euthanized 48 hours following the final administration.

Study 1

Plasma Chemistry Markers

To evaluate the effect of conjugated modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

TABLE 81

Plasma chemistry markers in male CD-1 mice						
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)	
PBS	2.5	25	52	0.2	23	
1310535	2.6	32	44	0.2	24	
1371966	2.7	31	40	0.2	23	
1371968	2.7	26	42	0.2	23	
1371969	2.6	28	59	0.2	24	
1371970	2.6	277	225	0.2	24	
1371971	2.8	41	74	0.2	24	
1371972	2.4	810	568	0.2	24	
1371973	2.3	50	78	0.2	22	
1371974	2.7	60	61	0.2	25	
1371976	2.7	32	61	0.2	22	
1371978	2.7	51	59	0.2	21	
1371979	2.7	32	70	0.2	23	

Body and Organ Weights

Body weights of CD-1 mice were measured at days 1 and 44, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

TABLE 82

Body and organ weights (in grams)					
ION No.	Body Weight (g)		Liver (g)	Kidney (g)	Spleen (g)
	Day 1	Day 44			
PBS	34	42	2.4	0.5	0.1
1310535	33	42	2.4	0.6	0.1
1371966	32	39	2.2	0.6	0.1
1371968	30	38	2.1	0.5	0.1
1371969	33	43	2.4	0.5	0.1
1371970	29	36	2.1	0.5	0.1
1371971	33	41	2.5	0.6	0.2
1371972	33	40	3.1	0.7	0.2
1371973	31	38	1.7	0.5	0.1
1371974	34	40	2.4	0.6	0.1
1371976	32	40	2.4	0.6	0.1
1371978	32	38	2.4	0.5	0.1
1371979	33	38	2.3	0.6	0.1

Hematology Assays

Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (MIT), and platelet counts (PLT). The results are presented in the table below.

471

TABLE 83

Blood cell counts in male CD-1 mice		
ION NO.	HCT (%)	PLT (10 ⁹ /L)
PBS	46	1155
1310535	50	1389
1371966	46	1415
1371968	47	1372
1371969	45	1059
1371970	48	1395
1371971	46	1244
1371972	44	1360
1371973	47	968
1371974	45	1362
1371976	49	1199
1371978	47	1417
1371979	48	1190

Study 2

Plasma Chemistry Markers

To evaluate the effect of modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

TABLE 84

Plasma chemistry markers in male CD-1 mice					
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)
PBS	2.8	36	57	0.11	19
1371981	2.7	26	39	0.14	24
1373086	2.6	39	59	0.12	24
1373087	2.6	31	54	0.13	23
1373089	2.5	107	88	0.12	24
1373090	2.5	27	45	0.11	25
1373094	2.8	687	532	0.15	23
1373096	2.7	57	87	0.13	23
1373097	2.4	67	71	0.13	23
1373098	2.5	67	73	0.12	24
1373099	2.7	43	75	0.12	22
1373100	2.7	373	365	0.12	25
1373101	2.5	153	179	0.14	23

Body and Organ Weights

Body weights of CD-1 mice were measured at days 1 and 42, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

TABLE 85

Body and organ weights (in grams)					
ION	Body Weight (g)		Liver	Kidney	Spleen
No.	Day 1	Day 42	(g)	(g)	(g)
PBS	27	38	1.9	0.6	0.1
1371981	27	39	2.5	0.6	0.1
1373086	27	38	2.1	0.5	0.1
1373087	27	40	2.4	0.6	0.1
1373089	26	39	2.8	0.5	0.1
1373090	27	38	2.4	0.5	0.2
1373094	28	41	2.9	0.6	0.2
1373096	27	42	2.5	0.6	0.2
1373097	27	42	2.52	0.49	0.17

472

TABLE 85-continued

Body and organ weights (in grams)					
ION	Body Weight (g)		Liver	Kidney	Spleen
No.	Day 1	Day 42	(g)	(g)	(g)
1373098	27	39	2.48	0.51	0.13
1373099	27	39	2.44	0.61	0.14
1373100	26	37	2.11	0.52	0.16
1373101	29	45	2.68	0.63	0.20

Hematology Assays

Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

TABLE 86

Blood cell counts in male CD-1 mice					
ION NO.	HCT (%)	PLT (10 ⁹ /μL)	NEU (%)	RET (%)	LYM (%)
PBS	45	1150	19	3	77
1371981	46	1398	18	3	76
1373086	44	1415	17	3	74
1373087	42	1343	21	3	73
1373089	43	1216	21	3	72
1373090	42	1332	21	3	52
1373094	44	1307	24	3	68
1373096	47	1161	19	3	69
1373097	42	1394	25	3	67
1373098	44	1432	24	4	68
1373099	44	1201	20	3	75
1373100	41	1031	16	3	77
1373101	45	1236	28	4	65

Study 3

Plasma Chemistry Markers

To evaluate the effect of modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TBIL) and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

TABLE 87

Plasma chemistry markers in male CD-1 mice						
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)	TBIL (mg/dL)
PBS	2.8	25	56	0.1	26	0.2
1394140	2.9	38	72	0.2	24	0.2
1394142	2.5	67	92	0.1	23	0.2
1394143	2.7	30	55	0.1	22	0.2
1394144	2.7	37	69	0.1	24	0.2
1394145	2.7	50	67	0.1	25	0.1
1394146	2.8	142	203	0.1	24	0.2
1394147	2.5	63	80	0.1	25	0.1
1394148	3.1	219	193	0.1	23	0.2
1394149	2.9	746	757	0.2	25	0.3
1394150	2.7	608	451	0.1	22	0.2

Body and Organ Weights

Body weights of CD-1 mice were measured at days 1 and 39, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

473

TABLE 88

Body and organ weights (in grams)					
ION No.	Body Weight (g)		Liver (g)	Kidney (g)	Spleen (g)
	Day 1	Day 39			
PBS	29	36	1.8	0.5	0.1
1394140	28	36	2.0	0.5	0.1
1394142	30	38	1.8	0.5	0.2
1394143	29	37	2.1	0.5	0.1
1394144	29	37	2.1	0.5	0.1
1394145	30	38	2.4	0.5	0.2
1394146	29	37	2.1	0.5	0.1
1394147	31	38	2.5	0.5	0.2
1394148	29	37	2.6	0.5	0.2
1394149	27	35	2.3	0.4	0.1
1394150	27	35	2.3	0.5	0.2

Hematology Assays

Blood obtained from mouse groups at week 6 were sent to IDEXX BioResearch for measurement of hematocrit (HCT), platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

TABLE 89

Blood cell counts in male CD-1 mice					
ION NO.	HCT (%)	PLT (10 ³ /μL)	NEU (%)	RET (%)	LYM (%)
PBS	53	1271	17	3	74
1394140	52	1221	20	3	72
1394142	50	1249	28	3	64
1394143	50	1165	17	3	76
1394144	49	1090	18	3	75
1394145	52	1292	21	3	73
1394146	50	1382	27	3	66
1394147	52	1343	22	3	69
1394148	52	1286	27	3	64
1394149	46	1613	26	3	66
1394150	47	1444	13	2	79

Example 9: Tolerability of Modified
Oligonucleotides Targeting Human HSD17B13 in
Sprague-Dawley Rats

Sprague-Dawley rats are a multipurpose model used for safety and efficacy evaluations. In this study, the rats were treated with Ionis conjugated modified oligonucleotides from the studies described in the Examples above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

Male Sprague-Dawley rats were maintained on a 12-hour light/dark cycle and fed ad libitum with Purina normal rat chow. Groups of 4 Sprague-Dawley rats each were weekly injected subcutaneously with 25 mg/kg of conjugated modified oligonucleotide for 6 weeks (total 7 doses). 72 hours after the last dose, rats were euthanized; and organs, urine and plasma were harvested for further analysis.

Study 1

Plasma Chemistry Markers

To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented

474

in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

TABLE 90

Plasma chemistry markers in Sprague-Dawley rats					
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)
PBS	3.4	34	53	0.3	13
1310535	3.2	35	68	0.3	14
1371966	3.5	33	66	0.3	16
1371968	3.8	36	57	0.4	16
1371969	3.0	25	68	0.3	16
1371970	3.4	48	101	0.3	15
1371971	3.6	34	72	0.3	16
1371972	3.5	51	76	0.4	15
1371973	3.6	36	62	0.3	14
1371974	3.3	37	76	0.4	15
1371976	3.4	31	55	0.3	15
1371978	3.4	37	67	0.3	15
1371979	3.5	117	122	0.3	12

Hematology Assays

Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT). The results are presented in the table below.

TABLE 91

Blood Cell Count in Sprague-Dawley Rats		
ION NO.	HCT (%)	PLT (10 ⁹ /L)
PBS	47	849
1310535	43	954
1371966	46	844
1371968	47	1010
1371969	43	810
1371970	48	1016
1371971	46	893
1371972	46	1019
1371973	44	857
1371974	46	930
1371976	45	843
1371978	45	900
1371979	47	1077

Kidney Function

To evaluate the effect of the conjugated modified oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of total protein to creatinine (P/C ratio) are presented in the table below.

TABLE 92

Total protein to creatinine ratio in Sprague-Dawley rats	
ION NO.	Urine P/C Ratio
PBS	0.5
1310535	0.4
1371966	0.8
1371968	0.7
1371969	0.6
1371970	0.6
1371971	1.1
1371972	0.4

475

TABLE 92-continued

Total protein to creatinine ratio in Sprague-Dawley rats	
ION NO.	Urine P/C Ratio
1371973	0.7
1371974	0.8
1371976	0.7
1371978	0.5
1371979	0.8

Body and Organ Weights

Body weights of Sprague-Dawley rats were measured at days 1 and 40, and the average body weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

TABLE 93

Body and Organ weights In Sprague-Dawley rats (g)					
ION No.	Body Weight (g)		Liver (g)	Kidney (g)	Spleen (g)
	Day 1	Day 40			
PBS	140	464	18	3.2	1.1
1310535	152	476	20	3.7	1.5
1371966	150	437	17	3.1	1.0
1371968	147	419	15	2.9	1.0
1371969	151	423	16	3.5	2.0
1371970	151	440	16	2.9	1.1
1371971	138	406	17	3.1	1.4
1371972	144	424	12	2.7	1.0
1371973	141	401	14	3.0	1.3
1371974	143	443	16	3.0	1.1
1371976	126	412	16	2.8	1.0
1371978	155	452	17	3.3	1.3
1371979	148	458	19	3.1	1.3

Study 2

Plasma Chemistry Markers

To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

TABLE 94

Plasma chemistry markers in Sprague-Dawley rats					
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)
PBS	3.6	33	56	0.3	16
1371981	3.7	35	57	0.3	17
1373086	3.7	35	87	0.4	19
1373087	3.3	33	64	0.4	18
1373089	3.7	92	129	0.3	18
1373090	3.5	38	72	0.4	18
1373094	3.7	187	311	0.4	16
1373096	3.3	69	171	0.3	18
1373097	3.2	40	86	0.3	17
1373098	3.8	37	74	0.3	17
1373099	3.6	33	60	0.4	16

476

TABLE 94-continued

Plasma chemistry markers in Sprague-Dawley rats					
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)
1373100	3.1	39	115	0.3	20
1373101	3.4	37	80	0.3	14

Hematology Assays

Blood obtained from mouse groups at week 6 were sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

TABLE 95

Blood Cell Count in Sprague-Dawley Rats					
ION NO.	HCT (%)	PLT (10 ³ /μL)	NEU (%)	RET (%)	LYM (%)
PBS	51	709	14	3	72
1371981	52	769	11	2	84
1373086	47	921	8	2	88
1373087	50	764	6	3	89
1373089	49	1217	7	3	88
1373090	50	766	10	4	85
1373094	53	943	14	2	81
1373096	61	376	6	3	89
1373097	49	746	5	3	90
1373098	49	929	10	3	85
1373099	47	717	8	3	85
1373100	42	823	6	4	87
1373101	48	693	7	3	87

Kidney Function

To evaluate the effect of Ionis oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of total protein to creatinine (P/C ratio) are presented in the table below.

TABLE 96

Total protein to creatinine ratio in Sprague-Dawley rats	
ION NO.	Urine P/C Ratio
PBS	1.0
1371981	1.7
1373086	1.5
1373087	1.7
1373089	1.5
1373090	1.3
1373094	3.3
1373096	1.5
1373097	1.3
1373098	1.2
1373099	1.9
1373100	0.9
1373101	1.7

Body and Organ Weights

Body weights of Sprague-Dawley rats were measured at days 1 and 44, and the average body weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

477

TABLE 97

Body and Organ weights In Sprague-Dawley rats (g)					
ION	Body Weight (g)		Liver	Kidney	Spleen
No.	Day 1	Day 44	(g)	(g)	(g)
PBS	146	458	19	3.1	1.0
1371981	141	424	21	3.4	1.1
1373086	151	434	19	3.0	1.8
1373087	143	433	19	3.3	1.3
1373089	149	431	21	3.5	2.1
1373090	142	453	19	3.2	1.3
1373094	142	449	19	3.2	1.4
1373096	149	460	23	3.6	3.2
1373097	136	400	17	3.3	2.2
1373098	130	389	20	2.9	1.5
1373099	152	421	16	2.9	1.6
1373100	146	453	19	4.2	2.3
1373101	140	412	17	2.9	1.5

Study 3

Plasma Chemistry Markers

To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, total bilirubin (TBIL) and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

TABLE 98

Plasma chemistry markers in Sprague-Dawley rats						
ION NO.	Albumin (g/dL)	ALT (IU/L)	AST (IU/L)	CREA (mg/dL)	BUN (mg/dL)	TBIL (mg/dL)
PBS	3.2	41	62	0.3	18	0.2
1394140	3.6	48	123	0.4	20	0.3
1394142	3.0	57	136	0.4	23	0.2
1394143	3.3	122	201	0.3	19	0.4
1394144	3.2	38	73	0.4	18	0.1
1394145	3.6	112	187	0.4	17	0.2
1394146	3.4	54	98	0.4	20	0.2
1394147	3.1	53	110	0.3	18	0.2
1394148	3.5	45	57	0.4	19	0.3
1394149	3.7	94	102	0.5	27	0.3
1394150	3.8	60	83	0.4	19	0.2

Hematology Assays

Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT1), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte count (LYM). The results are presented in the table below.

TABLE 99

Blood Cell Count in Sprague-Dawley Rats					
ION NO.	HCT (%)	PLT (K/ μ L)	NEU (%)	RET (%)	LYM (%)
PBS	52	795	12	3	84
1394140	45	876	8	3	88
1394142	47	768	4	3	90
1394143	59	1277	11	2	80
1394144	51	574	6	3	87

478

TABLE 99-continued

Blood Cell Count in Sprague-Dawley Rats					
ION NO.	HCT (%)	PLT (K/ μ L)	NEU (%)	RET (%)	LYM (%)
1394145	51	806	12	2	82
1394146	48	1052	11	2	81
1394147	49	922	8	3	85
1394148	49	1134	12	3	84
1394149	46	1816	14	3	80
1394150	52	1094	12	3	83

Kidney Function

To evaluate the effect of Ionis oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of total protein to creatinine (P/C ratio) are presented in the table below.

TABLE 100

Total protein to creatinine ratio in Sprague-Dawley rats	
ION NO.	Urine P/C Ratio
PBS	0.8
1394140	1.6
1394142	1.1
1394143	1.3
1394144	1.3
1394145	1.1
1394146	1.8
1394147	1.5
1394148	1.0
1394149	1.1
1394150	1.9

Body and Organ Weights

Body weights of Sprague-Dawley rats were measured at days 1 and 42, and the average body weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

TABLE 101

Body and Organ weights In Sprague-Dawley rats (g)					
ION NO.	Body Weight (g)		Liver (g)	Kidney (g)	Spleen (g)
PBS	270	503	19	3.0	0.9
1394140	263	440	19	3.0	1.5
1394142	258	370	13	2.9	1.5
1394143	257	393	16	2.9	1.1
1394144	254	461	18	3.2	1.5
1394145	266	475	21	3.4	1.4
1394146	268	444	17	3.1	1.7
1394147	258	401	17	3.2	2.0
1394148	267	447	21	3.0	1.5
1394149	261	422	20	5.8	1.4
1394150	251	423	18	3.0	1.2

479

Example 10: Measurement of Viscosity of Modified
Oligonucleotides Targeting Human HSD17B13

The viscosity of select conjugated modified oligonucleotides from the studies described above was measured.

Conjugated modified oligonucleotides (within a range of 32-38 mg) were placed into separate glass vials; approximately 100 μ L of water was added to each vial, and the conjugated modified oligonucleotides were dissolved into solution by heating the vials to 55° C. A 75 μ L portion of the pre-heated sample was pipetted to a micro-viscometer (PAC Cambridge Viscosity Viscometer). The temperature of the micro-viscometer was set to 25° C. and the viscosity of the sample was measured. The 75 μ L portion of each sample was then combined with the remaining portion of the original sample, which was then diluted for UV reading at 260 nM (Cary UV instrument). The data is presented in the Table below. Compounds having viscosity of less than 40 cP in this assay are generally suitable for use in therapy.

TABLE 102

Viscosity of modified oligonucleotides			
Compound ID	Concentration by weight (mg/mL)	Concentration by UV (mg/mL)	Viscosity (cP)
1310535	300	206	6
1371966	300	223	10

480

TABLE 102-continued

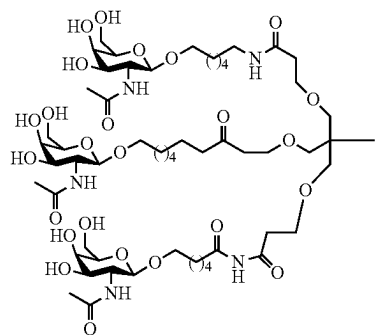
Viscosity of modified oligonucleotides			
Compound ID	Concentration by weight (mg/mL)	Concentration by UV (mg/mL)	Viscosity (cP)
1371974	300	207	8
1371976	300	254	64
1371978	300	219	7
1371979	300	244	10
1373086	300	215	7
1373087	300	219	9
1373090	300	191	8
1373097	300	222	6
1373098	300	203	5
1373099	300	242	6
1373101	300	205	11
1394140	300	197	10
1394141	300	225	9
1394142	300	271	24
1394143	300	250	11
1394144	300	257	32
1394145	300	263	10
1394146	300	289	11
1394147	300	246	15
1394148	300	205	9
1394149	300	304	14
1394150	300	202	16
1394151	300	223	8
1394152	300	225	24
1371976	200	151	16

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12409188B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

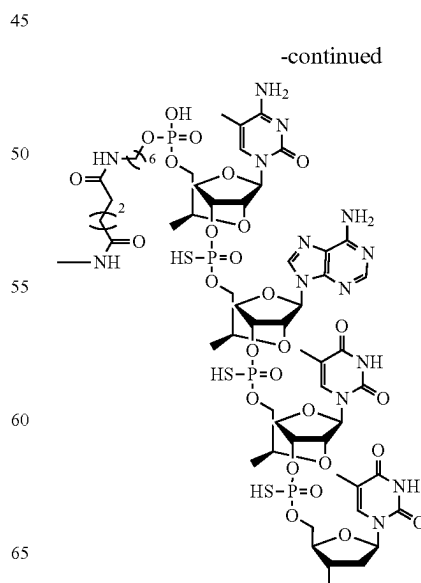
What is claimed is:

1. An oligomeric compound, according to the following chemical structure:



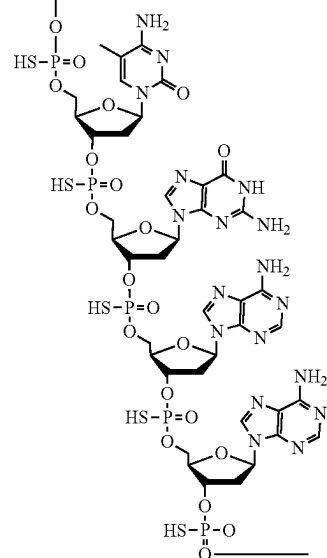
(SEQ ID NO: 3003)

-continued



481

-continued



5

10

15

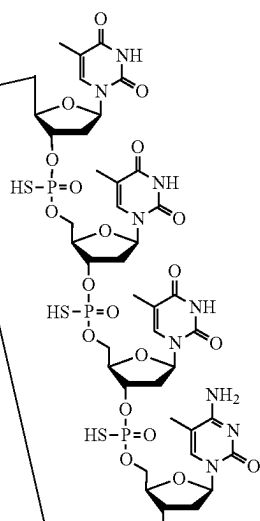
20

25

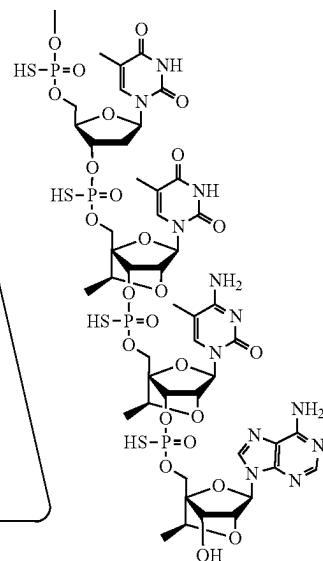
30

35

40

**482**

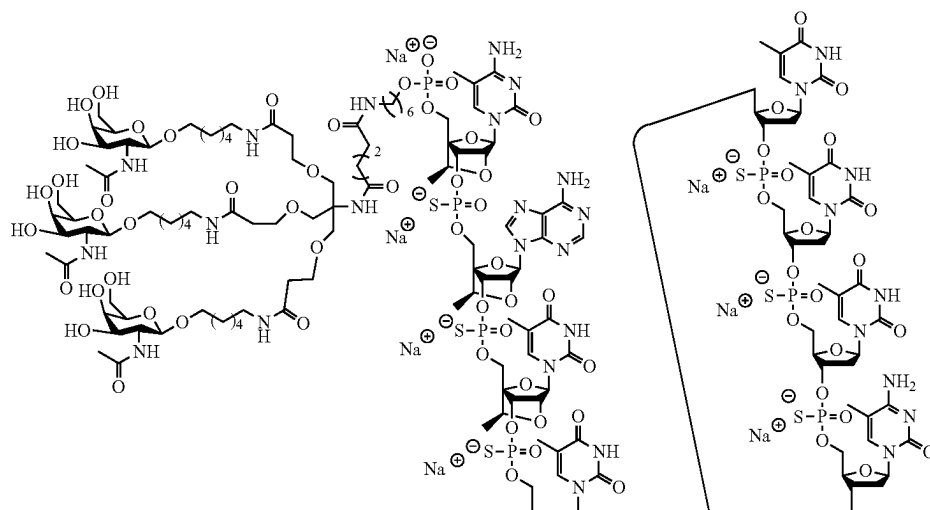
-continued



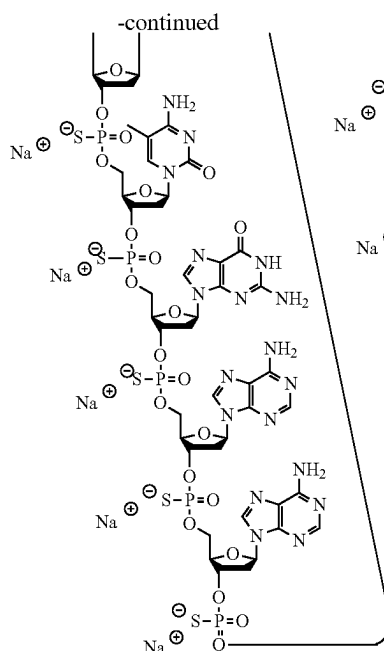
or a pharmaceutically acceptable salt thereof.

2. An oligomeric compound, according to the following chemical structure:

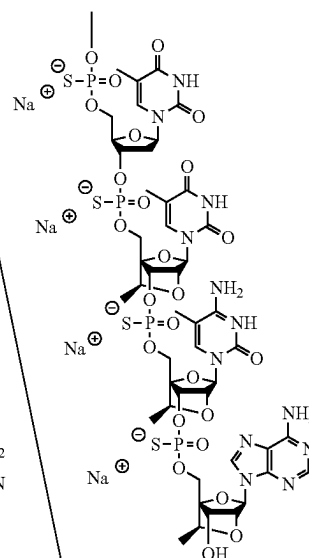
(SEQ ID NO. 3003)



483



484



3. An oligomeric compound comprising a modified oligonucleotide consisting of 16 linked nucleosides having a nucleobase sequence SEQ ID NO: 3003, wherein the modified oligonucleotide comprises:

a gap segment consisting of ten linked deoxynucleosides;
a 5' wing segment consisting of three linked nucleosides;
and

a 3' wing segment consisting of three linked nucleosides;
wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein the 5' wing segment and the 3' wing segment comprise cEt sugars; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

4. The oligomeric compound of claim 1, which is a sodium salt or a potassium salt.

5. A pharmaceutical composition comprising the oligomeric compound of claim 1 and a pharmaceutically acceptable diluent.

6. The pharmaceutical composition of claim 5, wherein the pharmaceutically acceptable diluent is water.

7. A pharmaceutical composition comprising the oligomeric compound of claim 2 and a pharmaceutically acceptable diluent.

8. The pharmaceutical composition of claim 7, wherein the pharmaceutically acceptable diluent is water.

9. A pharmaceutical composition comprising the oligomeric compound of claim 3 and a pharmaceutically acceptable diluent.

10. The pharmaceutical composition of claim 9, wherein the pharmaceutically acceptable diluent is water.

11. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 1, thereby treating the disease.

12. The method of claim 11, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

13. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 2, thereby treating the disease.

14. The method of claim 13, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

15. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 3, thereby treating the disease.

16. The method of claim 15, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

* * * * *